THE STACKCALC LIBRARY • VOLUME II

THINKING IN THE STACK

The Art of Calculating Without Parentheses

BOOKS & OPEN-SOURCE INSTRUMENTS IN THE STACKCALC LIBRARY

Volume I — From Blocks to the Stack (Grades 3–6 / Ages 8–12)
How Hands-On Blocks Make Numbers Click
ISBN 979-8-16-593829-0 • Place Value Blocks, Prime Factor Blocks & 84mm Fraction Towers

Volume II — Thinking in the Stack (High School, University & Practicing Engineers)
The Art of Calculating Without Parentheses
Reverse Polish Notation, Historical Derivations, Numerical Methods & 100 Canonical STEM Equations

Software Applications & Open-Source Hardware (stackcalc.io)
• **StackCalc iOS, iPadOS & watchOS Applications:** Four-register RPN calculators for phone, tablet, or wrist.
• **OpenSCAD & KiCad Hardware Files:** Printable SC-32 calculator enclosure, tactile keycap matrix, and PCB schematics.

THE STACKCALC LIBRARY • VOLUME II

THINKING IN THE STACK

The Art of Calculating Without Parentheses

Reverse Polish Notation, Historical Derivations, Numerical Methods, and Applied Engineering on the **StackCalc SC-32**

StackCalc
stackcalc.io
First Edition

THINKING IN THE STACK: The Art of Calculating Without Parentheses
Copyright © 2026 by **StackCalc** (https://stackcalc.io). Some rights reserved.
Published by **StackCalc**.
ISBN-13: 979-8-9914832-2-3 (Hardcover Edition) | **First Edition**

Open-Education License (CC BY-NC-SA 4.0): University instructors, classroom teachers, and students are expressly permitted to reproduce tables, 4-register stack traces, and worked engineering examples for non-commercial educational instruction under Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International.

Trademark Notice: *StackCalc*, *SC-32*, and *SC-6* are trademarks of StackCalc. *Hewlett-Packard*, *IEEE*, and *Gridfinity* are trademarks of their respective owners, none of whom endorse or sponsor this publication.

Print & Typography Colophon: Typeset in Computer Modern Unicode (CMU Serif), SpaceMono, STIX Math, and Terminus LCD on 6.00 × 9.00 in (152.4 × 228.6 mm) archival white paper in high-contrast monochrome ink for trade paperback and hardcover editions.

For every engineer, scientist, and student who has ever wanted to see inside the equation—and for the pioneers of parenthesis-free calculation from Warsaw to Palo Alto.

"Civilization advances by extending the number of important operations which we can perform without thinking about them."

— **ALFRED NORTH WHITEHEAD, *An Introduction to Mathematics* (1911)**

Table of Contents

Prologue: Writing Equations for Space, Solving Them in Time 1

The Typesetter's Bargain and the Inside-Out Mind 1

From Warsaw to Palo Alto: The Four-Register Lineage 3

Evaluating Nested Equations Across the Four Registers 5

Chapter 1: Slide Rules, Quartz Oscillators & Pocket Instrument Design . . 9

1.1 Bill Hewlett's Shirt Pocket and the Death of the Slide Rule 9

1.2 The Corvallis Lineage: Designing an Instrument for the Hand and Eye 12

1.3 Scientific Notation, Orders of Magnitude, and the 32,768 Hz Quartz Heartbeat 15

1.4 Daylight Optics, Tactile Mechanics, and the Laws of Ohm and Joule 19

1.5 Instrument Diagnostics, Optics & Power-Law Calculations 23

Chapter 2: The Architecture of Inside-Out Thinking 30

2.1 George Miller's Four Chunks and Dijkstra's Railway Yard 30

2.2 Unpeeling Rational Expressions and Harmonic Work Rates 31

2.3 Continued Fractions, Zu Chongzhi's Rational Pi, and Viète's Nested Radicals 33

2.4 Top-of-Stack Replication in Register T and Named Memory 35

2.5 Four-Register Stack Mechanics & Memory Architecture 38

Chapter 3: Logarithms, Power Laws & Exponential Scaling 46

3.1 John Napier's Logarithms and the Weber-Fechner Law 46

3.2 Power Laws, Biological Scaling & The Universal Harmonic Operator 49

3.3 Leonhard Euler's Natural Base e, Carbon-14 & Cooling Coffee 52

3.4 Factorials, Combinatorics & Extending n! to Fractional Numbers 54

3.5 Evaluating Powers, Logarithms & Exponential Laws 56

Chapter 4: Circular Trigonometry, Vector Projections & Hyperbolic Functions 65

4.1 Eratosthenes' Shadow, Angular Measure, and the Unit Circle 65

4.2 Triangles, Crosswinds & Resolving 2D Vectors 68

4.3 Hanging Chains, Catenary Arches, and Hyperbolic Functions 74

4.4 Circular, Vector & Hyperbolic Projections 76

Chapter 5: Complex Numbers, Euler's Identity & Alternating-Current Phasors 83

5.1 Multiplying by i as a Quarter-Turn in the Plane 83

5.2 Euler's Formula and De Moivre's Roots of Unity 85

5.3 Charles Proteus Steinmetz and the War of the Currents 89

5.4 Gaussian Complex Pairs & Alternating-Current Phasors 93

Chapter 6: The Square-Cube Law, Solid Geometry & Three-Dimensional Space 102

6.1 Galileo's Giant and the Square-Cube Law 102

6.2 Heron's Triangle, Ramanujan's Ellipse & Archimedes' Cylinder 103

6.3 Navigating 3D Space: Euclidean Distance, Centroids & Rotational Inertia 110

6.4 Solid Mensuration, Weighted Centroids & Moments of Inertia 113

Chapter 7: Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder 122

7.1 Al-Khwarizmi's Squares and the 1535 Contest in Milan 122

7.2 Unpeeling Polynomials from the Inside Out: Horner's Scheme 127

7.3 Tangent Instabilities in Newton's Method and Brent's Hybrid Algorithm 128

7.4 Horner Evaluation, Cardano's Cubic Solver & Brent's Root-Finder 133

Chapter 8: Archimedes' Balance Beam, Central Differences & Romberg Quadrature 142

8.1 Symmetric Difference Quotients and Numerical Derivatives 142

8.2 Parabolic Quadrature and the Composite Trapezoidal Rule 144

8.3 The 4/3 Minus 1/3 Extrapolation: Richardson & Romberg Integration 146

8.4 Non-Elementary Integrals: Rockets, Dams & Large-Amplitude Pendulums 149

8.5 Central Differences & Adaptive Romberg Quadrature 150

Chapter 9: Least Squares, Small Samples & the Geometry of Uncertainty 157

9.1 The Lost Planet Ceres, Galton's Quincunx & Bessel's n − 1 Correction 157

9.2 Regression to the Mean, Least-Squares Lines & Linearized Curves 160

9.3 Student's t-Distribution and Small-Sample Inference 163

9.4 Statistical Accumulators, Probability Distributions & Regression 165

Chapter 10: Boolean Logic, Two's Complement & the 36-Bit Word 174

10.1 Leibniz's Medallion, Boole's Logic & Claude Shannon's 1937 Master's Thesis 174

10.2 Positional Radices and the 36-Bit Word Architecture 175

10.3 Mechanical Odometers, Two's Complement Arithmetic & Bitwise Masks 177

10.4 Combinatorics (PROB), Cryptographic Keyspaces, and the Birthday Collision Paradox 181

10.5 36-Bit Radix Representations, Boolean Masks & Bitwise Arithmetic 182

Chapter 11: Compound Interest, Amortization & the Time Value of Money 190

11.1 Fibonacci's Merchants, Percentage Asymmetry & The Rule of 72 190

11.2 The Five-Variable Balance Law and Mortgage Amortization 192

11.3 Valuing Future Cash Streams: Bond Pricing, Net Present Value & Internal Rate of Return 195

11.4 Percentages, Mortgage Amortization & Discounted Cash Flows 196

Chapter 12: Analytic Geometry, Composite Areas & Integral Mensuration 204

12.1 René Descartes's Coordinate Plane, Gauss's Shoelace Theorem & Analytic Loci 204

12.2 Composite Shapes, Cutout Apertures & Surface Area Takeoffs in STAT Mode 206

12.3 Tying Definite Integration ($\int$) to Planar Area, Cross-Sections & Surface Area of Revolution 210

12.4 Worked Stack Traces: Composite Area Ledgers, Curve Profiles & Integral Mensuration 212

Chapter 13: Coupled Laws of Classical and Modern Engineering 217

13.1 Bending Light and Stacking Tolerances: Compound Microscopes & Pythagorean Error 217

13.2 Osborne Reynolds's Dye Ribbon, Continuity & Leonhard Euler's Buckling Column . . . 219

13.3 Sadi Carnot's Heat Engine Limit & Counter-Flow Heat Exchangers 221

13.4 Acoustic Reverberation, Rocket Staging & Geostationary Orbits 223

13.5 Multi-Domain Engineering Calculations 225

Chapter 14: Fermi Estimation, Algebraic Symmetry & High-Speed Calculation 230

14.1 Enrico Fermi's Torn Paper Scraps and the Art of Back-of-the-Envelope Estimation . . 230

14.2 Structural Algebraic Symmetries and Mental Calculation 232

14.3 Reading Giant Expressions from the Inside Out: Visual Chunking & Horner Rhythm . . 234

14.4 High-Speed Estimation & Multi-Stage Evaluations 237

14.5 Epilogue: The Mind and the Instrument 243

Appendix A: Formal Stack-Lift State Machine 244

A.1 State-Machine Architecture & Register Transitions 244

A.2 Shifted Operations & Formal Register Delta Algebra 247

A.3 Comprehensive Stack Lift Traces 250

Appendix B: Error Catalog & Singularity Analysis 255

B.1 The Non-Destructive Error Recovery Principle & Domain Singularities 255

B.2 Statistical, Iterative Solver & Financial Singularities 258

B.3 Memory, Quadrature & Hardware Diagnostic Recovery Protocols 261

Appendix C: Physical Keycap Directory & Engineering Architecture . . . 264

C.1 Physical Keypad Layout & the 43-Key Matrix 264

C.2 Architectural Refinements, Dynamic LFU Menus & Complete HP-32SII / HP-15C / HP-12C Comparison 270

C.3 Open-Source Hardware & Firmware Architecture 275

Appendix D: Mathematical Constants, SI Units & STEM Equations . . . 277

D.1 Mathematical Constants, BIPM SI Prefixes & Exact Unit Conversions 277

D.2 Fundamental Equations of Science and Engineering 279

D.3 Stack Optimization Idioms, Solver Templates & Boundary Guards 311

Appendix E: Primary Sources & Annotated Bibliography 313

E.1 Primary Historical & Technical Sources 313

E.2 Thematic Reading Pathways 320

Prologue: Writing Equations for Space, Solving Them in Time

The Typesetter's Bargain and the Inside-Out Mind

In the autumn of 1924, in a quiet lecture hall at the University of Warsaw, the Polish logician and philosopher Jan Łukasiewicz stood before a blackboard covered in logic formulas and examined a foundational anomaly of mathematical notation *(Łukasiewicz, 1924)*: the ubiquity of parentheses.

Every high-school student who has ever stared at a page of Algebra II or Precalculus knows the feeling of parenthesis vertigo. You look at an equation on a printed page:

$$x = \sqrt{7 + \sqrt{13 + \sqrt{29 + \sqrt{41}}}} \tag{Eq. 0.1}$$

or a compound rational fraction from physics or chemistry:

$$Q = \frac{12.4 \times 8.6}{3.2 + 4.9} \tag{Eq. 0.2}$$

On paper, your eyes naturally read from left to right and from the outside in. In this radical pyramid, the first symbol you encounter on the left is the outer square-root radical √..., followed by 7 +, followed by a second radical, a third radical, and finally, at the far right end of the line, the number 41.

The moment you sit down to *solve* that equation, the outer square root cannot be evaluated first because you cannot take the square root of a sum that has not been added yet. To find x, evaluation must ignore the left-to-right order of the ink and travel straight to the innermost number, 41, extracting its square root first, adding 29, taking the next square root, adding 13, extracting that root, and finally adding 7 before one last square root completes the chain from right to left.

Writing mathematics from the outside in while every human mind and every computer processor on Earth can only evaluate it from the inside out is a historical artifact of the printing press.

Space Versus Time: Sixteenth-Century Typography and Infix Notation

For thousands of years, from Babylonian clay scribes and Greek geometers to medieval Arab algebraists, mathematics was written out in spoken sentences: *"Take forty-one, extract its root, add twenty-nine to what results..."* Spoken language unfolds in *time*. Physical measurement and construction follow the same chronological order, gathering materials first and applying the tool second: a carpenter aligns two boards on a bench before joining them, a chemist weighs two reagents on a balance before combining them in a beaker, and an action in speech or calculation cannot occur until its objects exist.

In the sixteenth and early seventeenth centuries, however, Renaissance mathematicians—including Robert Recorde (who introduced the equals sign = in 1557 in *The Whetstone of Witte*), François Viète, and Thomas Harriot—working with European printers and typesetters faced an economic problem. Movable lead type was expensive, and

rag paper was costly. Printers wanted to pack an entire finished mathematical thought onto a single static horizontal line of type. To fit a multi-step timeline onto a flat two-dimensional line, they wedged action symbols (**+**, **-**, **×**, **÷**) *between* numbers in a convention called *infix notation*, and invented parentheses **( )**, brackets **[ ]**, and horizontal fraction bars as punctuation fences to warn the eye: *"Do not read this line from left to right; jump inside the deepest fence first."*

That 16th-century typesetter's bargain was well suited to static paper, where the eye can scan back and forth across a page. It became a cognitive obstacle the moment 20th-century engineers copied paper notation onto electronic calculator keyboards.

When you type an infix string like **((12.4 × 8.6) ÷ (3.2 + 4.9)) =** into an algebraic pocket or smartphone calculator, you are forced to feed the machine in the exact order of the printed ink rather than the order of evaluation. Halfway through the formula, when you press **÷ (**, the calculation stalls because the denominator **(3.2 + 4.9)** does not yet exist. As cognitive psychologist George A. Miller demonstrated in his 1956 study of human working memory *(Miller, 1956; Sweller, 1988)*, the mind holds only four to seven active items at once. Hiding intermediate numbers behind a single-line display while counting nested parentheses **((...) / (...))** diverts working memory from the physical structure of the problem to clerical punctuation *(Strogatz, 2012; Strogatz, 2019; Ellenberg, 2014)*.

Jan Łukasiewicz's Breakthrough and the Four Visible Registers

In 1924, Jan Łukasiewicz proved a foundational theorem *(Łukasiewicz, 1924)*: if operators are placed consistently before or after their operands, parentheses disappear from mathematical notation. No matter how deeply an equation is nested, the order in which numbers combine is unambiguous without a single bracket or precedence rule.

Three decades later, Australian philosopher and computer pioneer Charles Hamblin (1957) *(Hamblin, 1957)* paired the postfix (*Reverse Polish*) form of Łukasiewicz's parenthesis-free notation, in which operands precede their operator, with a last-in, first-out hardware memory structure: an operational *stack*.

On a four-register operational stack, four horizontal display lines are arranged from bottom to top as Registers *X*, *Y*, *Z*, and *T*. Register *X* at the bottom of the screen serves as the active register where keyed digits appear and where every unary or binary operation writes its immediate result. When a first operand is completed in Register *X* and the central **ENTER** bar is pressed, that value copies upward into Register *Y*, preparing Register *X* to receive the second operand. Whenever a calculation spans two or three separate branches, such as the numerator and denominator of a rational expression, the completed subtotal from the first branch lifts into Register *Z* (and Register *T*) to remain visible while the second branch is evaluated across Registers *Y* and *X*. As soon as that second branch reduces to a single value in Register *X*, the waiting subtotal in Register *Z* drops back into Register *Y* so the two operands align for the final binary operation.

When equations are read from their innermost operands outward, expressions from disparate branches of science reveal shared algebraic skeletons *(HP, 1982)*. Wiring electrical resistors in parallel, focusing light through a thin optical lens, and averaging a round-trip velocity all reduce to the same binary harmonic operation $x \parallel y = (x^{-1} + y^{-1})^{-1}$ *(Horowitz & Hill, 2015)*, and multi-layer formulas resolve into a direct sequence of two-register reductions from the innermost core outward.

From Warsaw to Palo Alto: The Four-Register Lineage

Łukasiewicz's 1924 logic notation and Hamblin's 1957 hardware stack, formalized for digital computer evaluation by Arthur Burks, Don Warren, and Jesse Wright in 1954 *(Burks et al., 1954)*, became the native operating language of scientific handhelds in 1972, when William Hewlett, David Cochran, and the Palo Alto engineering team *(Whitney, Rodé & Tung, 1972; Cochran, 1972)* introduced the HP-35, the world's first pocket scientific calculator, supplanting the slide rule with a 4-register operational stack (X, Y, Z, T). Over the next two decades, Hewlett-Packard's Corvallis division steadily expanded that four-register architecture: in 1979–1982, William M. Kahan and the HP-34C / HP-15C Voyager team *(HP, 1982; Kahan, 1979)* brought Richard Brent's hybrid root-finding algorithm (**SOLVE**), adaptive Romberg quadrature (**∫f(x)**), and complex-number arithmetic into a shirt-pocket instrument.

In 1991, the HP-32SII Pioneer (succeeding the 1988 HP-32S) *(HP, 1991)* paired those same Brent **SOLVE** and Romberg **∫** engines with 26 named **A**–**Z** variables, a symbolic Equation Bank (**EQN**), exact fraction reduction, and eight SI/Imperial unit conversions. Whereas 1970s–1990s calculators showed only a single line (**Register X**) through a narrow display window, the StackCalc SC-32 puts all four stack registers (T, Z, Y, X) in plain sight at once on a 4-line transflective LCD, paired with a 6-key dynamic softkey bar, RPN equation storage, closed-form quadratic/cubic [→] [POLY], five-variable [←] [TVM] finance, and dedicated parallel-harmonic ([x ∥ y]), absolute-value ([ABS]), and quotient-remainder ([→] [÷R]) keys.

While the SC-32 defaults to the classic 4-register stack (**4-LVL**) for direct keystroke parity with the HP-35, HP-15C, and HP-32SII, its [→] [FLAGS] menu also allows expanding the operational stack to 8 registers (**8-LVL**, providing a full 4-level complex stack when each complex number occupies two registers) or an unbounded 999-level LIFO stack (**INF**) for deep multi-branch networks. Because the operational rules of Registers X, Y, Z, and T remain constant across this lineage, every four-register postfix recurrence and keystroke sequence applies directly across any four-register RPN instrument as well as the free, open-source browser simulator at **stackcalc.io**:

Architectural Feature	HP-35 (1972)	HP-15C Voyager (1982)	HP-32SII Pioneer (1991)	StackCalc SC-32 (Present)
Processor Architecture	56-bit serial PMOS (10-digit BCD)	56-bit Nut CMOS (10-digit BCD)	64-bit Saturn CMOS (12-digit BCD)	64-bit IEEE-754 double-precision math engine
Display & Stack View	1-line 15-char red LED (**X** only)	1-line 10-digit 7-seg LCD (**X** only)	1-line 12-char dot-matrix LCD (**X** only)	4-line 132×65 transflective LCD (T, Z, Y, X visible)
Operational Stack	4-level (X, Y, Z, T)	4-level (X, Y, Z, T)	4-level (X, Y, Z, T)	Selectable **4-LVL**, **8-LVL**, or **INF** (999-level)
Dynamic Softkeys	None	None	Remaps top row (**√x**..**Σ+**) in menus	Dedicated 6-key LFU softkey strip
Variables & Equations	1 storage register	Numbered **R0**–**R9** / **LBL 0..9**	Named **A**–**Z** & Infix **EQN** (**(**, **)**, **=**)	Named **A**–**Z** (89 slots) & RPN [→] [EQN] Bank
Numerical Solver	None	Brent–Kahan hybrid (via **LBL**)	Brent–Kahan hybrid (via **EQN** / **LBL**)	Brent–Kahan [→] [SOLVE] (10^{-12}) + Closed-Form [→] [POLY]
Quadrature Engine	None	Adaptive Romberg (via **LBL**)	Adaptive Romberg (via **EQN** / **LBL**)	Adaptive Romberg Richardson extrapolation (10^{-12})

Architectural Feature	HP-35 (1972)	HP-15C Voyager (1982)	HP-32SII Pioneer (1991)	StackCalc SC-32 (Present)
Complex Numbers	None	Parallel imaginary stack (**I** / **Re↔Im**)	4-register (Y,X) & (T,Z) via **CMPLX**	4-line visible (T,Z,Y,X) [↶] [CMPLX] + complex [x‖y] & [↷] [POLY]
Financial (TVM)	None	None (HP-12C required)	None (HP-12C required)	Built-in 5-variable [↶] [TVM] (**N**, **I%**, **PV**, **PMT**, **FV**)
Parallel & Integer Math	Manual reciprocal ($1/x$)	Manual reciprocal ($1/x$)	Fractions **a b/c**; **PARTS** softkeys	Direct [x‖y], [ABS] & [↷] [÷R] keys (**GCD**/**LCM** in **PARTS** menu)
Exam Compliance & Base Word	Non-programmable	Programmable (10-digit)	Programmable (32-bit unsigned **BASE**)	Non-graphing scientific + proctored [EXAM] lockdown & 36-bit signed **BASE**
Non-Volatile Memory	None (volatile RAM)	CMOS continuous memory	CMOS continuous memory	Continuous non-volatile flash memory

Chapter	Title & Scope	Mathematical & Engineering Landmarks
Chapter 1	**Slide Rules, Quartz Oscillators & Pocket Instrument Design**	Bill Hewlett's 1971 shirt-pocket specification; scientific notation ($m \times 10^n$); 32,768 Hz (2^{15} Hz) quartz oscillators; daylight optics & Ohm's/Joule's laws
Chapter 2	**The Architecture of Inside-Out Thinking**	George Miller's working-memory capacity; Dijkstra's Shunting-Yard algorithm; reciprocal rate combination; continued fractions ($\pi \approx 355/113$, Golden Ratio φ) & Viète's nested radicals
Chapter 3	**Logarithms, Power Laws & Exponential Scaling**	John Napier (1614) & Leonhard Euler (e); Weber-Fechner scales (decibels, Richter, pH); Henderson-Hasselbalch buffers & blood-gas physiology; aviation pressure & density altitude; Carbon-14 decay & Lanczos Gamma (70!)
Chapter 4	**Circular Trigonometry, Vector Projections & Hyperbolic Functions**	Eratosthenes's 240 BC meridian arc; surveying DMS traverses & circular curves ([↶] [▸HR], [↷] [▸y,x], [Σ+]); pilot E6B wind triangles & endurance ([↷] [▸HMS]); forestry clinometers, machinist sine bars & catenaries ($\cosh x$)
Chapter 5	**Complex Numbers, Euler's Identity & Alternating-Current Phasors**	Orthogonal 90° rotation by $i = \sqrt{(-1)}$; Euler's identity ($e^{i\pi} + 1 = 0$); Charles Proteus Steinmetz's AC impedance phasors; polyphase networks & loudspeaker crossovers
Chapter 6	**The Square-Cube Law, Solid Geometry & Three-Dimensional Space**	Galileo's 1638 Square-Cube Law; Archimedes's sphere-and-cylinder ratio; Heron's formula; Cavalieri's principle; 3D print slicing, Center of Mass & Moment of Inertia (I)
Chapter 7	**Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder**	Al-Khwarizmi's completing the square (AD 820); Cardano's cubic ([↷] [POLY]); exact weak-acid & van der Waals chemical equilibria; Colebrook-White pipe flow & Richard Brent's [↷] [SOLVE] algorithm
Chapter 8	**Archimedes' Balance Beam, Central Differences & Romberg Quadrature**	Symmetric central-difference derivatives; Werner Romberg's $(4/3)T_{\text{fine}}$ - $(1/3)T_{\text{coarse}}$ Richardson extrapolation ([↷] [∫]); pharmacokinetic AUC bioavailability, stormwater hydrographs & rocket staging
Chapter 9	**Least Squares, Small Samples & the Geometry of Uncertainty**	Gauss's 1801 Least Squares; Bessel's n - 1 variance; Beer-Lambert UV-Vis spectrophotometric calibration ([↷] [L.R.] → [x̂]); Six Sigma CNC capability (C_{pk}) & Gosset's t-distribution
Chapter 10	**Boolean Logic, Two's Complement & the 36-Bit Word**	Leibniz's 1703 binary arithmetic; Claude Shannon's 1937 switching algebra; modular ring arithmetic & Two's Complement signed integers; 36-bit words, IPv4 subnet masks & RGB bitfield packing
Chapter 11	**Compound Interest, Amortization & the Time Value of Money**	Fibonacci's 1202 *Liber Abaci*; geometric return asymmetry; the cash-flow sign convention; 30-year mortgage amortization dynamics; bond pricing, **NPV** & **IRR**

Chapter	Title & Scope	Mathematical & Engineering Landmarks
Chapter 12	**Analytic Geometry, Composite Areas & Integral Mensuration**	Descartes's 1637 coordinate plane; Gauss's Shoelace polygon theorem; composite section & cutout takeoffs ([Σ+], [↶] [Σ−], [x⇄w]); definite-integral area ([↷] [∫]) & surfaces of revolution
Chapter 13	**Coupled Laws of Classical and Modern Engineering**	Compound microscope optics; Root-Sum-Square (RSS) mechanical tolerance stacking; Reynolds pipe flow; Euler column buckling; Carnot heat exchangers; Breguet aircraft range, Sabine acoustics & geostationary orbits
Chapter 14	**Fermi Estimation, Algebraic Symmetry & High-Speed Calculation**	Enrico Fermi's 1945 order-of-magnitude estimation; dimensional bounds checking; algebraic difference of squares (48×52); half-integer squaring; single-entry [↶] [LAST x] Horner evaluation
Appendices A–E	**Formal Reference Compendium**	**App. A:** Stack-Lift State Machine • **App. B:** 12 Error Codes • **App. C:** 43-Key Directory & Instrument Architecture • **App. D:** Constants & 100 STEM Equations • **App. E:** Annotated Bibliography

Evaluating Nested Equations Across the Four Registers



Figure: The StackCalc SC-32 handheld RPN scientific computer: monolithic chassis, tactile TPU keypad, and daylight-reflective vector display.

On the StackCalc SC-32 display glass, four live registers sit stacked vertically from Register T at the top down to Register X at the bottom, without a single **(**, **)**, or **=** key on the bezel. At power-on, the instrument initializes in decimal floating-point format ([DEC], [ALL]) and degree angle mode ([DEG]).

Chaining Measurements and Eliminating Parentheses

Three foundational operations—[ENTER], *automatic stack lift*, and *two-branch drops*—govern how operands move across the four visible registers:

1. **Adding Two Numbers (14 and 6):** Keying **14** enters the first value into Register X. Pressing [ENTER] marks that first number complete and lifts **14** into Register Y. Keying **6** then places the second value into Register X so that **14** (in Y) and **6** (in X) are visible simultaneously. Pressing [+] drops Register Y and combines it with Register X to yield **20**.

2. **Automatic Stack Lift (Chaining Without Pressing [ENTER] Again):** To multiply three physical dimensions in sequence, such as the volumetric melt flow rate $Q = w \times h \times v$ of a 3D printer hotend extruding a $w = 0.45$ mm bead width at a $h = 0.20$ mm layer height and $v = 180$ mm/s printhead speed, keying **0.45** [ENTER] **0.20** [×] computes the nominal rectangular cross-sectional area of the bead, **0.09** mm^2, in

Register *X*. Multiplying that **0.09** by **180** requires no second [ENTER]. Because **0.09** is a completed result, touching **1** to begin typing **180** triggers *automatic stack lift*, which moves **0.09** into Register *Y*. Keying **180** and pressing [×] yields **16.2** mm^3/s in Register *X*.

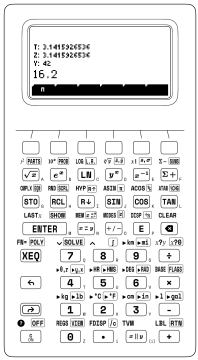


Figure: Chaining three measurements on the live 4-register stack: evaluating the volumetric melt flow rate $Q = w \times h \times v = 0.45\ mm \times 0.20\ mm \times 180\ mm/s = 16.2\ mm^3/s$ of a 3D printer hotend.

3. **Untangling a Two-Branch Fraction Across Registers *Z* and *Y*:** A two-branch rational expression such as dividing a hydraulic pump's output capacity by the combined cross-sectional area of two parallel intake manifolds (3.2 cm^2 and 4.9 cm^2),

$$Q = \frac{12.4 \times 8.6}{3.2 + 4.9} \quad \text{[L/min]} \qquad \textbf{(Eq. 0.3)}$$

requires no parentheses on the SC-32 because Registers *Y* and *Z* stage intermediate subtotals above the active register. Once the numerator $12.4 \times 8.6 = 106.64$ is evaluated in Register *X*, that completed subtotal is armed for stack lift. Typing the first denominator term (**3.2**) lifts 106.64 into Register *Y*, and pressing [ENTER] to separate **3.2** from **4.9** shifts 106.64 into Register *Z* while the two intake areas occupy Registers *Y* and *X*. When [+] collapses $3.2 + 4.9$ into 8.1000 cm^2 in Register *X*, the waiting numerator drops from Register *Z* back into Register *Y* so the two completed branches align for [÷] (**12.4** [ENTER] **8.6** [×] **3.2** [ENTER] **4.9** [+] [÷] → **13.165432099 L/min**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
12.4	0.0000	0.0000	0.0000	12.4000	Keying first numerator factor
[ENTER]	0.0000	0.0000	12.4000	12.4000	Stack lift: pushes 12.4 into *Y*
8.6	0.0000	0.0000	12.4000	8.6000	Entering second factor into *X*
[×]	0.0000	0.0000	0.0000	106.6400	Numerator product: 12.4×8.6
3.2	0.0000	0.0000	106.6400	3.2000	Automatic stack lift: numerator preserved in *Y*
[ENTER]	0.0000	106.6400	3.2000	3.2000	Numerator elevated to *Z*; 3.2 in *Y*
4.9	0.0000	106.6400	3.2000	4.9000	Loading second denominator term
[+]	0.0000	0.0000	106.6400	8.1000	Denominator summed in *X*; numerator drops to *Y*
[÷]	0.0000	0.0000	0.0000	**13.1654**	Final quotient 106.64 / 8.1

Parallel Harmonics, Nernst Potentials, and Nested Radicals

Three electrical pathways wired in parallel ($R_1 = 100\ \Omega$, $R_2 = 220\ \Omega$, $R_3 = 470\ \Omega$) obey the reciprocal conductance relation:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \qquad \textbf{(Eq. 0.4)}$$

Parallel resistors combine by reciprocals because all three branches share the same voltage while their electrical conductances ($G_i = 1/R_i$, in Siemens) add directly. On a traditional RPN keypad, you invert each resistance into a conductance in place with the unary $[x^{-1}]$ key, accumulate the running sum across Register Y and Register X, and invert the total conductance back into Ohms (**100** $[x^{-1}]$ **220** $[x^{-1}]$ $[+]$ **470** $[x^{-1}]$ $[+]$ $[x^{-1}]$ → **59.9768 Ω**).

More directly, the SC-32's dedicated parallel-harmonic key $[x\|y]$ ($x \parallel y = yx \,/\, (y{+}x)$) reduces that reciprocal sum-and-invert sequence to an associative fold across the lower two registers: pressing $[x\|y]$ collapses the first two branches in Y and X into their combined equivalent resistance and arms stack lift, so typing the third resistor pushes the two-branch subtotal into Register Y for a second parallel fold (**100** $[\text{ENTER}]$ **220** $[x\|y]$ **470** $[x\|y]$ → **59.9768 Ω**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
100 $[\text{ENTER}]$	0.0000	0.0000	100.0000	100.0000	Load $R_1 = 100\ \Omega$ into Y and X; stack lift disabled
220	0.0000	0.0000	100.0000	220.0000	Overwrite Register X with second branch $R_2 = 220\ \Omega$
$[x\|y]$	0.0000	0.0000	0.0000	68.7500	Collapse Y and X into $100 \parallel 220 = 68.75\ \Omega$; stack lift armed
470	0.0000	0.0000	68.7500	470.0000	Auto-lift 68.75 Ω to Y and load third branch $R_3 = 470\ \Omega$ in X
$[x\|y]$	0.0000	0.0000	0.0000	**59.9768**	Collapse Y and X into 3-branch equivalent $R_{\text{eq}} \approx 59.9768\ \Omega$

Across that entire three-branch reduction, Registers Z and T remain at zero because each harmonic fold immediately collapses Y and X back into a single number in Register X. That same two-register economy carries straight over to geometrical optics. In Carl Friedrich Gauss's real-is-positive thin-lens equation ($1/f = 1/d_o + (1/d_i) \Rightarrow f = d_o \parallel d_i$), a converging glass lens bends diverging wavefronts from an object distance $d_o = 45$ cm into converging wavefronts focused at an image distance $d_i = 90$ cm by adding their reciprocal curvatures ($1/45\ \text{cm}^{-1}$ and $1/90\ \text{cm}^{-1}$). Loading $d_o = 45$ cm into Register Y and $d_i = 90$ cm into Register X and pressing $[x\|y]$ (**45** $[\text{ENTER}]$ **90** $[x\|y]$) collapses the two distances directly into the lens's focal length $f =$ **30.0000 cm** (+3.33 diopters) in Register X, while Registers Z and T remain at zero.

Inside-out evaluation also flattens multi-layer transcendental formulas so they barely need upper stack registers at all. Like a chemical waterfall that pushes harder when the upper supply reservoir is full and the lower outflow pool is nearly empty, a Daniell copper-zinc battery at 298.15 K ($n = 2$ moles of electrons per mole of reaction, thermal voltage $RT/F = 0.025693$ V, and standard potential $E^\circ = 1.100$ V) with a diluted zinc-ion anode (0.015 M, activity $a_{\text{Zn}^{2+}} = 0.015$ relative to $c^\circ = 1$ M) and concentrated copper-ion cathode (0.850 M, effective activity $a_{\text{Cu}^{2+}} = 0.850$) obeys Walther Nernst's 1889 equation:

$$E = E^\circ - \frac{RT}{nF}\ln Q = 1.100 - \frac{0.025693}{2}\ln\left(\frac{0.015}{0.850}\right) \qquad \textbf{(Eq. 0.5)}$$

Rather than typing from left to right and stalling after 1.100 -, inside-out evaluation begins with the reaction quotient $Q = 0.015 \,/\, 0.850 = 0.017647$. Once that ratio sits in Register X, the next three physical transformations happen entirely inside Register X without lifting the stack: [LN] converts the concentration ratio in place into $\ln Q = -4.0372$, multiplying by 0.025693 V scales it into single-electron Volts, and dividing by 2 leaves the Nernst overpotential -0.05186 V in Register X while Y, Z, and T remain at zero. Entering the standard potential 1.100 V then lifts -0.05186 V into Register Y, where [x⇄y] (the Register Swap key) orders the two terms for subtraction (**0.015** [ENTER] **0.850** [÷] [LN] **0.025693** [×] **2** [÷] **1.100** [x⇄y] [-] → **1.15186 V**).

The clearest demonstration of this single-register collapse is the four-level nested radical pyramid from the opening equation:

$$x = \sqrt{7 + \sqrt{13 + \sqrt{29 + \sqrt{41}}}} \qquad \textbf{(Eq. 0.6)}$$

On an algebraic keypad, typing that expression from the outside in requires opening four nested levels of parentheses while three unfinished additions hang suspended in memory. On the SC-32, walking from the innermost radicand outward via **41** [√x] extracts $\sqrt{41} =$ **6.4031242374** in place in Register X. Typing 29 lifts 6.4031 into Register Y for an instant before addition and [√x] collapse $29 + 6.4031 = 35.4031$ into $\sqrt{(35.4031)} \approx 5.9501$ in Register X. Adding 13 and extracting the root collapses $\sqrt{(18.9501)}$ into 4.3532, and adding 7 before one final square root (**41** [√x] **29** [+] [√x] **13** [+] [√x] **7** [+] [√x]) resolves the four-level radical to **3.3694459184** with Registers Y, Z, and T resting at zero at the close of every layer.

Across the chapters that follow, this same inside-out discipline extends across the sciences and field professions: from electrical, mechanical, civil, and aerospace engineering to land surveying and geodesy, cockpit air navigation, analytical and physical chemistry, clinical pharmacokinetics, precision machining, timber forestry, and actuarial finance—each executed directly on the live stack.

Chapter 1: Slide Rules, Quartz Oscillators & Pocket Instrument Design

1.1 Bill Hewlett's Shirt Pocket and the Death of the Slide Rule

1.1.1 January 1972: Thirty-Five Keys That Ended Three Centuries of Wood and Ivory

In January 1971 in Palo Alto, California, William R. Hewlett *(Whitney, Rodé & Tung, 1972)*, co-founder of Hewlett-Packard, stood in his laboratory beside Dave Packard, independent calculator designer Thomas Osborne, and algorithmic architect David S. Cochran *(Cochran, 1972)*. Sitting on the wooden laboratory bench in front of them was the Hewlett-Packard 9100A: a forty-pound desktop electronic calculator the size of an electric typewriter, priced at nearly five thousand dollars, which sat in NASA mission-control rooms, nuclear physics laboratories, and university engineering departments.

Three years earlier, in 1968, Hewlett-Packard had launched the 9100A, born after Thomas Osborne walked into HP in December 1964 carrying a floating-point prototype he had built at home in his spare time after leaving the University of California at Berkeley. Every major adding-machine company in America had rejected Osborne's prototype because it used floating-point scientific notation and Reverse Polish logic instead of a mechanical paper-tape crank. Yet the moment Bill Hewlett pressed its keys, he recognized the future of scientific computing. That prototype became the forty-pound HP 9100A, the first machine ever called a "personal computer" in print.

Now, in 1971, Hewlett pointed to the forty-pound desktop machine, then patted the breast pocket of his white cotton dress shirt. He issued a challenge that has become legend in the history of engineering:

Shrink the mathematical brain of that forty-pound desktop computer—trigonometry, logarithms, exponentials, square roots, and ten-digit floating-point range—into a battery-powered instrument small enough to slip inside my shirt pocket.

Before a single silicon transistor was laid out, Hewlett's industrial designers took a brass ruler to his dress shirts, setting the target envelope at $3.2 \times 5.8 \times 1.0$ inches *(Whitney, Rodé & Tung, 1972)*.

An engineering classroom or aerospace drafting hall in 1970 illustrates the scale of Hewlett's shirt-pocket challenge. For more than three hundred and fifty years, ever since the English mathematician William Oughtred placed two of John Napier's logarithmic scales side by side in 1622 so one wooden ruler could slide past the other, every suspension bridge, steam locomotive, transatlantic liner, jet turbine, and Apollo lunar trajectory had been designed with a *slide rule*: a slipstick of boxwood, bamboo, or engraved yellow aluminum.

When Apollo 11 descended toward the Sea of Tranquility in July 1969, Buzz Aldrin and Neil Armstrong carried a six-inch aluminum Pickett N600-ES slide rule in their spacesuit pockets as their backup flight computer. Nine months later, in April 1970, when an oxygen tank exploded aboard Apollo 13 two hundred thousand miles from Earth and forced the crew to power down the Command Module's electronic guidance computer to save every amp-hour of battery power, NASA flight engineers in Houston and Astronaut Jack Swigert

in the frigid cabin pulled out their Pickett slide rules to verify the emergency engine-burn attitude and free-return trajectory around the Moon.

The slide rule was a compact logarithmic analog instrument that taught generations of engineers, chemists, and aviators to develop an instinctive feel for numbers and orders of magnitude—whether on a ten-inch Hemmi bamboo desk rule in a chemical laboratory or on naval aviator Philip Dalton's circular aluminum E6B dead-reckoning computer ("whiz wheel") spun one-handed in a turbulent cockpit. Even so, analog scales suffered from two physical limitations that every twentieth-century practitioner knew firsthand. No matter how sharply an observer squinted through the glass hairline cursor, a ten-inch desk slide rule, and certainly a six-inch pocket rule or circular flight computer, could only be read to *three significant figures* (roughly 0.1% accuracy).

On a wooden C/D scale, 1,495 and 1,496 shared the same engraved line. Surveying a property boundary to the millimeter with a brass transit or calculating an orbital trajectory correction where the fourth and fifth digits mattered required setting the slide rule aside and opening a thick volume of seven-place Peters trigonometric or logarithmic tables (or cranking a mechanical Curta drum) to add columns of digits by hand.

Furthermore, a slide rule could neither add nor subtract, and its scales carried no record of the decimal point. Because sliding two logarithmic scales past one another adds their lengths ($\log A + \log B = \log(A \times B)$), the rule multiplied and divided rapidly, yet a simple sum such as $14.2 + 6.8$ in the middle of a formula required computing the addition with pencil and paper and re-entering the sum under the cursor hairline. Worse still, the logarithmic markings on a slide rule look identical for 0.0025, 0.25, 2.5, 250, and 25,000,000. The engineer had to track the order of magnitude mentally after every slide, where a single misplaced decimal point meant a bridge girder ten times too thin or ten times too heavy.

Many marketing executives inside Hewlett-Packard believed Hewlett's shirt-pocket challenge was financially reckless. Their market research reports warned that no engineer would pay $395 (nearly half the price of a used car in 1972) to replace a $20 bamboo slide rule. Meanwhile, the earliest pocket electronic calculators of 1970 could barely add, subtract, multiply, and divide four-function grocery totals, and their power-hungry red LED numerals drained nickel-cadmium batteries in less than an hour.

More daunting still was the silicon memory barrier. To fit logarithmic, trigonometric, exponential, and power algorithms onto three tiny 256-word read-only memory chips, totaling 768 ten-bit words of firmware, David Cochran and Thomas Osborne realized they could not spend scarce silicon transistors on an algebraic syntax parser to count open and closed parentheses `( ( ... ) )` or stack up pending operators *(Cochran, 1972)*. Instead, they modeled the calculator directly on the chronological order of human calculation:

An engineer or a student builds every calculation sequentially in time.

Evaluating $(14 + 6) \times (9 - 4)$ mentally never stalls halfway through waiting for a closing parenthesis: the first pair sums to $14 + 6 = 20$, that intermediate value is held in working memory while the second pair subtracts to $9 - 4 = 5$, and the two subtotals multiply to $20 \times 5 = 100$.

Drawing on Jan Łukasiewicz's parenthesis-free logic and Charles Hamblin's push-down store, Cochran and Osborne built the *four-register operational stack (X, Y, Z, T)*. Hewlett bypassed the skeptical market research, ordered production tooling on his own authority, and unveiled the HP-35 in January 1972 (beginning commercial shipments on February 1, 1972, and named for its 35 keys). Expecting to sell perhaps ten thousand units in the first year, Hewlett-Packard sold over 100,000 HP-35 calculators in year one and 300,000 before

its successor arrived. Within forty-eight months, Keuffel & Esser and Pickett shut down their slide-rule factories permanently; the three-hundred-and-fifty-year reign of the wooden slipstick had ended in a shirt pocket.

1.1.2 Instruments Versus Gadgets

The HP-35 *(Cochran, 1972)* and its Corvallis descendants—the expandable alphanumeric HP-41C *(Musch, Wong & Conklin, 1980)*, the horizontally sculpted HP-15C Voyager *(HP, 1982; Evett et al., 1983)*, and the alphanumeric HP-32SII Pioneer *(HP, 1991)* championed by the classic RPN engineering community *(Nelson, 1974; Hamrick, 1996)*—established the architecture of the pocket scientific instrument. Unlike a smartphone glass screen crowded with pop-up notifications and flat virtual buttons that demand visual attention for every tap, a dedicated scientific instrument is organized around visible working memory, tactile three-plane key geometry, and instant-on non-volatile state retention. Four horizontal registers (T, Z, Y, X) remain in plain sight at all times, relieving biological working memory from counting brackets or recording scratchpad fractions *(Miller, 1956; Sweller, 1988)*:

Stack Register	Display Readout	Architectural Role
T	`0.000000000000`	**Top Register & Constant Source:** Holds upper subtotals; automatically copies its value downward whenever the stack drops
Z	`0.000000000000`	**Upper Holding Register:** Holds earlier subtotals during multi-part formulas without needing memory registers
Y	`0.000000000000`	**Pending Operand Register:** Holds the first operand waiting to combine with X ($Y + X$, $Y - X$, $Y \times X$, $Y \div X$)
X	`0.000000000000`	**Active Register:** Bottom display line where digits appear as you type and where every immediate result lands

Rather than hiding scientific functions inside nested computer menus, the 43 sculpted Shore 90A TPU keys of the StackCalc SC-32 distribute 129 functions across three physical planes. Direct unshifted presses invoke the digits **0**–**9**, [ENTER], the four arithmetic keys [+] through [÷], stack and storage controls [x⇄y], [R↓], [STO], and [RCL], and primary scientific operations including [SIN], [COS], [TAN], [LN], [e^x], [$\sqrt{x}$], [y^x], [$1/x$], [%], and [E].

Pressing Left Shift [↰] (**Row 5, Col 0**) activates upper-left inverse and structural operations, including [↰] [ASIN], [↰] [ACOS], [↰] [ATAN], [↰] [LOG], [↰] [10^x], [↰] [x^2], [↰] [x≤y], [↰] [x!], [↰] [CMPLX], [↰] [TVM], [↰] [▸θ,r], and [↰] [LASTx]. Pressing Right Shift [↱] (**Row 6, Col 0**) activates upper-right numerical engines and inspectors, including [↱] [SOLVE], [↱] [∫], [↱] [POLY], [↱] [EQN], [↱] [÷R], [↱] [▸y,x], and [↱] [SHOW]. Touching [ON] wakes the processor in under five milliseconds with the operational stack, named variables A... Z, and stored equations preserved in non-volatile flash memory.

The physical geometry of a dedicated keypad also guides the hand when visual attention remains on a circuit board, a telescope, or an examination sheet. Stepped wedge keycaps feature a sloped top surface and chamfered edges so the thumb can sense column boundaries by touch. Beneath each keycap, a mechanical dome switch builds resistance up to an actuation peak of roughly 1.6 Newtons ($\approx$ 160 grams-force) before snapping downward onto gold-plated contacts, which provides crisp physical confirmation that a digit or operator has registered. Across the bezel, primary operations occupy the raised key face, paired inverse operations sit above and to the left for [↰], and numerical solvers, definite integrals, and coordinate conversions sit above and to the right for [↱].



Figure: The StackCalc SC-32 precision enclosure, high-contrast transflective LCD, and tactile key switches

Both the physical StackCalc SC-32 faceplate editions and the companion software app support ten interchangeable faceplate palettes: **`Pure Black`**, **`Retro Futurism`**, **`Pioneer Brown`**, **`Pioneer Black`**, **`Stealth`**, **`Supernova`**, **`Deep Space`**, **`Voyager`**, **`Synthwave`**, and **`Arcade`**. Because shift legend accent colors adapt dynamically to the active faceplate theme, shifted operations are denoted independent of color by the directional hardware shift symbols ↖ (**`Row 5, Col 0`**) for upper-left legends and ↗ (**`Row 6, Col 0`**) for upper-right legends.

1.2 The Corvallis Lineage: Designing an Instrument for the Hand and Eye

1.2.1 The Dvorak Principle and One-Handed Portrait Ergonomics

In 1932, educational psychologist August Dvorak and his brother-in-law William Dealey at the University of Washington pointed a motion-picture camera at the hands of dozens of professional typists. Typists suffered rapid finger fatigue because Christopher Latham Sholes's 1870s QWERTY typewriter had scattered the most common letters of the English language into outlying corners to keep nineteenth-century iron typebars from jamming, so a typist's fingers had to leap across rows for nearly seventy percent of every sentence. Dvorak's insight was direct and enduring: an instrument designed for human thought should place the actions performed ninety-nine percent of the time directly under the resting fingertips.

Over the past three decades, that human-factors wisdom was quietly abandoned in scientific computing. Smartphones, desktop computer terminals, and modern graphing calculators are well suited to writing long programs or browsing encyclopedic trees of rarely used settings that no human could ever memorize. Yet when calculator manufacturers began imitating computer terminals, they swept everyday mathematical tools—absolute value, integer remainders, rounding, unit conversions, parallel impedance, and polynomial root solvers—off the physical keypad and buried them three or four layers deep inside scrolling on-screen submenus.

A nested menu tree exacts an immediate cost to concentration during a live calculation. In the middle of analyzing an AC filter network, a thermodynamic cycle, or a timed examination problem, pausing a mathematical derivation to press a menu button, tap arrow keys across three tabs, and scroll down a list of text labels just to take an absolute

value or convert inches to millimeters breaks cognitive focus.

The StackCalc family was built from the opposite principle, applying Dvorak's ergonomic rule to scientific calculators: leave long, scrolling option lists to phones and computer terminals. On a dedicated mathematical instrument, ninety-nine percent of daily work should require zero menus, and every high-frequency operation belongs on a direct, tactile keystroke that the fingers can reach by muscle memory alone.

That single design rule shapes both instruments in the StackCalc family. On the elementary StackCalc SC-6, dedicated physical keys for prime factorization, greatest common factors, integer quotient-and-remainder division ([↗] [÷R]), exact fraction simplification, and 2D/3D geometry let young learners explore numbers with their hands instead of hunting through menus. On the scientific StackCalc SC-32, that same Dvorak discipline shapes the classic Corvallis RPN lineage.

Among veteran Hewlett-Packard engineers and calculator designers, the architectural choice between the 1991 HP-32SII Pioneer *(HP, 1991)* and the 1982 HP-15C Voyager *(HP, 1982)* turns on ergonomics and symbolic clarity.

Among collectors, the horizontally oriented HP-15C Voyager is rightly admired as a classic of twentieth-century silicon engineering, bringing numerical root-finding (**SOLVE**), numerical quadrature (**∫f(x)**), complex-number arithmetic, and matrix operations into a brushed-metal shirt-pocket slab. Yet comparing both calculators side by side at an electronics test bench, at a property-corner tripod, in an aircraft cockpit, at a chemistry titration burette, or during a timed NCEES licensing examination reveals why the 1991 HP-32SII Pioneer provides the stronger physical and symbolic foundation.

In practical field and laboratory work, one hand is almost always occupied holding an oscilloscope probe, a surveyor's plumbing pole, an aircraft yoke, a volumetric pipette, or a machinist's Vernier caliper. The HP-15C's wide landscape chassis was sculpted for two-handed thumb typing or flat desk tapping, whereas the HP-32SII's vertical portrait footprint places the entire digit pad, the double-wide [ENTER] bar, and the arithmetic column inside the natural sweep of a single thumb.

That ergonomic advantage is paired with a structural difference in how the two machines represent human thought. Because the 1982 HP-15C relied on a seven-segment numeric LCD that could not render alphabetical letters, physical constants had to be stored inside anonymous numbered registers (**R0** through **R9** and **.0** through **.9**) and **SOLVE** or **∫f(x)** subroutines had to be recorded under numeric labels (**LBL 0**) read back as row-column keycodes (**001- 43 21 0**). Halfway through an orbital mechanics or chemistry derivation, the user had to remember whether the molar gas constant sat in **R3** or **R7**, and finding a typo in a ten-step formula meant single-stepping through numeric codes.

The HP-32SII swept that bookkeeping away by introducing 26 named variables (**A**–**Z**) that activate whenever [STO] or [RCL] is pressed, turning the register bank into self-documenting algebra (**G** for gravity, **L** for inductance, **C** for capacitance), alongside a dedicated Equation Bank ([↗] [EQN]) for [XEQ], [↗] [SOLVE], and [↗] [∫]. Yet the 1991 HP-32SII introduced one modal compromise: its **EQN** editor required algebraic infix syntax (**=**, **(**, **)**), forcing RPN users to switch mental models whenever storing a formula. On the SC-32, [↗] [EQN] remains pure postfix RPN, so the exact keystroke sequence tested interactively on the four-register stack is recorded directly for evaluation, root-finding, and integration.

Furthermore, whereas the HP-15C lacked physical unit conversions and fraction arithmetic, the HP-32SII placed direct SI ↔ Imperial unit conversions right on the digit keys, expanded on the SC-32 to ten conversions (**km ↔ mi** on [9], **kg ↔ lb** on [1], **°C ↔ °F** on [2], **cm**

↔ **in** on [3], and **1** ↔ **gal** on [-], with Metric always on [↶] and Imperial always on [↷]) alongside native rational fractions ([↶] [FDISP] and [↷] [/c]) for machinist blueprints.

At the same time, an engineer should not have to carry an HP-32SII for named equations, unit conversions, and numerical root finding, a separate tool for closed-form polynomial roots, and an HP-12C for financial cash flows. The 1991 HP-32SII had already inherited William Kahan's hybrid Brent–Dekker **SOLVE** root-finder, adaptive Romberg ∫ quadrature, and four-register **CMPLX** arithmetic from the 1979 HP-34C and 1982 HP-15C Voyager, upgrading **SOLVE** and ∫ to operate directly on symbolic equations in **EQN**. By applying the Dvorak principle to the Pioneer keypad, the SC-32 completes that synthesis on dedicated shifted keys.

From the 1981 HP-12C Voyager, pressing [↶] [TVM] opens a 5-variable Time Value of Money engine across the top softkeys (**N**, **I%**, **PV**, **PMT**, **FV**, and **END/BEG**), complete with one-tap store-or-solve semantics and standard cash-flow sign conventions, plus William Kahan's base-preserving [↷] [%] and [↷] [%CHG] operators.

Whereas both the HP-15C and HP-32SII required running **SOLVE** iteratively one real root at a time (or writing a synthetic-division subroutine) to factor a polynomial, pressing [↷] [POLY] on the SC-32 launches a dedicated Quadratic & Cubic Polynomial Root Workspace with softkeys **A**, **B**, **C**, **D**, **EXEC**, and **CLR**. This workspace solves $Ax^2 + Bx + C = 0$ and $Ax^3 + Bx^2 + Cx + D = 0$ analytically in closed form, including complex conjugate root pairs, and writes all roots into indirect registers **R34**–**R39**, while [↶] [CMPLX] on Left-Shift [STO] brings the HP-32SII's four-register Gaussian Complex phasor engine onto a four-line display where both complex operands (T, Z) and (Y, X) are visible at once.

1.2.2 Four Visible Registers and the Dedicated Softkey Strip

Preserving the HP-32SII's portrait keyboard architecture also offered an opportunity to address three hardware constraints of the 1991 instrument. First, because of 1991 LCD driver limits, the HP-32SII displayed only a single 12-character line (**Register X**), requiring Registers Y, Z, and T to be tracked mentally; moreover, lacking a dedicated softkey row, opening a submenu (**PARTS**, **PROB**, **SUMS**) temporarily remapped the top row of fixed scientific keys ([$\sqrt{x}$], [e^x], [LN], [y^x], [x^{-1}], [Σ+]). On the SC-32, the 132 × 65 transflective display keeps all four stack registers (T, Z, Y, X) visible simultaneously, and a dedicated 6-key Dynamic Softkey Row ([LFU0]–[LFU5]) sits directly beneath the glass, displaying six dynamically tracked functions in normal operation and menu labels inside submenus while leaving the fixed scientific keys untouched.

Second, the 1991 HP-32SII mixed two distinct notations: on the live stack it operated in RPN, yet entering **EQN** mode required algebraic infix formulas with parentheses. To support infix entry, three right-shifted slots on Row 3 (**R3C3**–**R3C5**) were assigned to an open parenthesis **(**, a close parenthesis **)**, and an equals sign **=**.

On the SC-32, [↷] [EQN] mode operates in side-effect-free RPN. Formulas are entered using the same postfix sequence as the live stack, and the relational tests [↷] [x?0] and [↶] [x?y] can be embedded inside an equation as 1/0 *Iverson Boolean indicators* to model piecewise engineering functions. Reclaiming **(**, **)**, and **=** on Row 3 places Absolute Value [ABS] (**|x|**) on Right-Shift [+/−] and Euclidean Quotient-Remainder Division [↷] [÷R] on Right-Shift [E] (leaving Right-Shift [⌫] unassigned). Meanwhile, just as on the HP-32SII where number-decomposition utilities reside in the **PARTS** softkey menu (**Right-Shift** [$\sqrt{x}$]), the SC-32's 6-softkey **PARTS** menu retains **IP**, **FP**, and **ABS** on **LFU0**–**LFU2** and adds Signum (**SGN**), Greatest Common Divisor (**GCD**), and Least Common Multiple (**LCM**) on **LFU3**–**LFU5** (whereas the elementary SC-6 places **GCF** and **LCM** on dedicated physical keys).

Third, on the 1991 HP-32SII, key **Row 7, Col 3** was dedicated to **R/S** (Run/Stop, with **PRGM** and **PSE** on its shifts) for keystroke-loop programming, while **INPUT** sat on Left-Shift [0]. Because named RPN equations ([→] [EQN]), [XEQ], [→] [SOLVE], and [→] [∫] replace line-number programming, the SC-32 assigns Left-Shift [0] to the full-screen [←] [REGS] viewer and assigns **Row 7, Col 3** to three core analytical operations.

Its primary tap is the universal parallel-harmonic operator [x‖y], which computes $Y \times X \,/\, (Y + X)$ in a single keystroke for parallel resistors, series capacitors, thin lenses, reduced masses, and complex AC impedances. Its left shift launches the 5-variable [←] [TVM] financial solver (**N**, **I%**, **PV**, **PMT**, **FV**, and **END/BEG**), bringing the actuarial engine of the HP-12C into the scientific keypad while keeping the physical instrument strictly non-graphing for standardized examination compliance. Finally, its Alpha label **(i)** works with the index pointer **i** on [.] and the full-screen [←] [REGS] inspector on [0] to provide indirect addressing across all 89 memory slots.

For users accustomed to Corvallis RPN, zero retraining is required: in the default [4-LVL] stack mode, every [ENTER] lift disable, top-of-stack T-replication, [←] [LASTx] capture, storage-register arithmetic (**STO + A**), and flag test behaves identically to the 1991 HP-32SII, while the [→] [FLAGS] menu expands the stack to [8-LVL] or [INF] (unlimited LIFO depth) or locks the instrument into proctored [EXAM] mode whenever a task demands it.

1.3 Scientific Notation, Orders of Magnitude, and the 32,768 Hz Quartz Heartbeat

A pocket scientific calculator represented a decisive leap over both a four-function adding machine and a wooden slide rule because physics, chemistry, and engineering do not live solely in the everyday neighborhood between 1 and 1,000.

In the third century BCE, the Greek mathematician and physicist Archimedes of Syracuse wrote a treatise to King Gelon II titled *The Sand Reckoner* (*Psammites*). Critics had claimed that the number of grains of sand on the Sicilian seashore was infinite, or at least larger than any number that could ever be named. Archimedes saw that Greek numerals only had words up to a *myriad* ($10{,}000 = 10^4$). By inventing a hierarchy of powers, grouping myriads of myriads (10^8) as his "first order" and raising them to higher powers, Archimedes proved that even if the entire spherical cosmos of Aristarchus from Earth to the fixed stars were filled with fine beach sand, the total number of sand grains could be written in just a few words: 10^{63}.

Archimedes had formulated the central principle of *scientific notation*: to reason across vast scales without drowning in zeros, a notation must separate how precisely a quantity is known from how large or small that quantity is.

1.3.1 Separating Shape from Scale: Mantissa (m) and Exponent (10^n)

In chemistry, a single mole of carbon-12 atoms (12 grams) contains Avogadro's number of atoms, 602,214,076,000,000,000,000,000. In atomic optics, a red helium-neon laser beam has a vacuum wavelength of 0.0000006328 meters. Writing twenty-three trailing zeros on a chalkboard or counting six leading zeros after a decimal point blurs the eye within seconds, and a single missed zero throws a bridge, orbit, or circuit off by a factor of ten.

To represent numbers spanning forty orders of magnitude from the subatomic to the astronomical, scientific notation factors every real number x into two components:

$$x = m \times 10^n \qquad (1 \le |m| < 10) \tag{Eq. 1.1}$$

The mantissa (or significand) m contains the meaningful measured digits, or *significant figures*, normalized so that a single non-zero digit sits to the left of the decimal point (for example, 6.02214076 or 6.328). The mantissa records the *shape* and *precision* of the measurement, which is the exact portion that William Oughtred's seventeenth-century slide rule displayed along its engraved wooden scale. Paired with the mantissa, the integer exponent n specifies the *order of magnitude*, counting how many powers of ten to scale upward for $n > 0$ or downward for $n < 0$. Stepping up from kilometers at 10^3 to megameters at 10^6, or down through millimeters at 10^{-3}, red-blood-cell micrometers at 10^{-6}, silicon-lattice nanometers at 10^{-9}, and quartz-crystal picofarads at 10^{-12}, compares physical scales across the universe at a glance.

In electrical, mechanical, and optical work, powers of ten that are multiples of three ($10^{\pm 3k}$) match the International System of Units (SI) metric prefixes. For that reason, the SC-32 provides both Scientific [SCI] display mode, where $1 \leq |m| < 10$, and Engineering [ENG] display mode, where the exponent n always locks to a multiple of 3 for direct reading in kilo-, mega-, milli-, micro-, nano-, or pico- units:

Power of Ten	SI Prefix	Symbol	Physical Scale Example in Instrument Design
10^{9}	giga-	G	GPS L1 satellite carrier (1.57542 GHz = 1.57542×10^{9} Hz) & glass-polymer modulus (3.5 GPa)
10^{6}	mega-	M	High-speed Pierce quartz oscillator (12.000 MHz = 1.2×10^{7} Hz)
10^{3}	kilo-	k	RC reset pull-up resistor (100 kΩ) & laser repetition rate (35.0 kHz)
10^{0}	(unit)	—	Coin-cell electromotive force (3.0 V) & drop test height (1.20 m)
10^{-3}	milli-	m	Millisecond display refresh window & solver burst current (18.2 mA)
10^{-6}	micro-	μ	Beer-Lambert optical absorption skin & decoupling capacitance (0.10 μF)
10^{-9}	nano-	n	Sub-nanohenry package lead inductance & optical shutter window (100 ns)
10^{-12}	pico-	p	Quartz crystal load capacitor (29.0 pF = 2.9×10^{-11} F)

When two numbers in scientific notation are multiplied or divided, their mantissas (m_1, m_2) multiply or divide while their orders of magnitude (n_1, n_2) add or subtract:

$$(m_1 \times 10^{n_1}) \times (m_2 \times 10^{n_2}) = (m_1 \times m_2) \times 10^{n_1 + n_2} \quad \textbf{(Eq. 1.2)}$$

$$\frac{m_1 \times 10^{n_1}}{m_2 \times 10^{n_2}} = \left(\frac{m_1}{m_2}\right) \times 10^{n_1 - n_2} \quad \textbf{(Eq. 1.3)}$$

A frequent error when entering scientific notation on a calculator is treating 3.0×10^{8} as two separate numbers rather than a *single* quantity: a number written as 3.0×10^{8} is one atomic value, not an arithmetic multiplication problem between 3.0 and 10^{8}.

Take the transit time required for a laser pulse traveling at $c = 3.0 \times 10^{8}$ m/s to cross a $d = 1.5$ m optical bench:

$$t = \frac{1.5}{3.0 \times 10^{8}} \quad \textbf{(Eq. 1.4)}$$

If a student using an ordinary calculator types **`1.5 ÷ 3.0 × 10 ^ 8`** as separate arithmetic operations, the calculator divides 1.5 by 3.0 first to get 0.5 and then *multiplies* that quotient by 10^{8}. Instead of 5.0×10^{-9} seconds (five nanoseconds), the screen returns 50,000,000 seconds, or nineteen months: an error of sixteen orders of magnitude (10^{16}).

That is why Bill Hewlett's 1972 HP-35 introduced a dedicated Exponent Entry key, labeled [E] on the SC-32. Keying **3**, pressing [E], and keying **8**—or keying **6.328** [E] **7** [+/−] for 6.328×10^{-7}—does not issue an arithmetic multiplication command; it binds the scale factor 10^{8} or 10^{-7} to the mantissa inside the entry buffer so the entire quantity moves through the stack as a single atomic number. A direct corollary governs pure powers of ten: because [E] already means *"× 10to the power of"*, a pure power of ten such as 10^{-7} F (1×10^{-7} F) or 10^{6} Hz (1×10^{6} Hz) is entered as **1** [E] **7** [+/−] or **1** [E] **6**. Keying **10** [E] **6** means $10 \times 10^{6} = 10^{7}$, a 10× order-of-magnitude error.

Equally important, changing how a number is *displayed* on the screen by pressing [↶] [DISP] and choosing Fixed-Point [FIX], Scientific [SCI], Engineering [ENG], or Full Precision [ALL] never truncates or rounds the underlying value in memory. Internally, every stack register (X, Y, Z, T) and every variable (A... Z) holds full IEEE 754 double precision (53 bits of binary significand, or roughly 15 to 17 decimal digits), and pressing [↷] [SHOW] at any moment reveals the unrounded mantissa of Register X.

1.3.2 The Quartz Tuning Fork and $2^{15} = 32{,}768$ Hz

Beyond representing microscopic and astronomical numbers, a scientific instrument must also measure physical time.

For centuries, timekeeping depended on swinging pendulums (discovered by Galileo and perfected by Christiaan Huygens in 1656) or coiled hairsprings (invented by Robert Hooke and Huygens for marine chronometers). Yet even the finest mechanical pocket watch drifted by several seconds a day as temperature changed the length of its balance wheel.

In 1880, two French physicists, the brothers Jacques and Pierre Curie, discovered an electromechanical coupling inside crystals of quartz (SiO_2, silicon dioxide), tourmaline, and Rochelle salt, which they named *piezoelectricity* (from the Greek *piezein*, "to squeeze"). When a quartz crystal is compressed or bent mechanically, the positive silicon ions and negative oxygen ions in its hexagonal lattice shift slightly apart and generate an electrical potential across its faces; conversely, when an alternating voltage is applied across metal electrodes plated onto the quartz wafer, the crystal physically expands, contracts, and flexes like a microscopic tuning fork.

Because a single crystal of quartz is an almost perfectly elastic atomic lattice with virtually zero internal friction in a vacuum, a quartz tuning fork rings thousands of times longer than a steel tuning fork. In 1927 at Bell Telephone Laboratories, Warren Marrison used a vacuum-tube amplifier to keep a quartz crystal vibrating continuously, building the world's first quartz clock. Four decades later, Seiko and Swiss watchmakers laser-trimmed a quartz tuning fork just four millimeters long to fit inside a wristwatch.

The specification sheet for almost any digital wristwatch, spacecraft timer, or the real-time clock inside the SC-32 lists a characteristic vibration frequency:

$$f_{\mathrm{RTC}} = 32{,}768 \text{ Hz} \qquad \textbf{(Eq. 1.5)}$$

Electrical engineers chose 32,768 vibrations per second rather than a round decimal number like 10,000 Hz or 50,000 Hz for a binary reason.

In physics, the *frequency* f (in Hertz, cycles per second) and the *period* T (in seconds per cycle) are exact *reciprocals* of one another:

$$T = \frac{1}{f} \quad \text{and} \quad f = \frac{1}{T} \qquad \textbf{(Eq. 1.6)}$$

Each single swing of a 32,768 Hz quartz tuning fork takes a period of:

$$T = \frac{1}{32,768\ \text{Hz}} \approx 0.000030517578\ \text{s} = 30.5176\ \text{microseconds} \quad \textbf{(Eq. 1.7)}$$

A digital circuit counts those 30.5-microsecond vibrations to advance the clock by 1 second (1 Hz) through binary frequency division.

In binary electronics, the simplest, lowest-power circuit element is a *toggle flip-flop*, a two-transistor gate that changes state on every second incoming pulse and divides the frequency in half (÷ 2). Chaining fifteen divide-by-two flip-flops in series halves the frequency at every stage:

$$32,768 \xrightarrow{\div 2} 16,384 \xrightarrow{\div 2} 8,192 \xrightarrow{\div 2} 4,096 \xrightarrow{\div 2} \cdots \xrightarrow{\div 2} 4 \xrightarrow{\div 2} 2 \xrightarrow{\div 2} 1\ \text{pulse per second} \quad \textbf{(Eq. 1.8)}$$

Because $2^{15} = 32,768$, fifteen ultra-low-power halving stages reduce 32,768 Hz down to 1.000000 Hz without a single decimal comparator. Engineers selected $2^{15} = 32,768$ Hz rather than $2^{14} = 16,384$ Hz because the upper limit of human hearing is roughly 20,000 Hz. A 16,384 Hz crystal would emit a faint, high-pitched whine near the ear, whereas 32,768 Hz is the first power of two that is ultrasonic and completely silent.

1.3.3 High-Speed Pierce Oscillators, Thermal Drift, and Parasitic Resonance

While a 32,768 Hz tuning-fork crystal keeps human calendar time during deep sleep on nanoamperes of current, waking the processor to evaluate a Romberg definite integral or a closed-form cubic root sweep in milliseconds requires a much faster heartbeat: a 12.000 MHz (1.2×10^7 Hz) Pierce quartz crystal oscillator (named after Harvard physicist George Washington Pierce, who invented the single-stage crystal feedback loop in 1923), whose period is $T = 1 / (1.2 \times 10^7\ \text{Hz}) = 83.33$ nanoseconds.

A thin slice of quartz locks onto a single frequency because, through the piezoelectric effect, a vibrating crystal acts as a microscopic mechanical resonator coupled to an electrical circuit: its physical mass behaves like an inductor L, its elastic stiffness behaves like a spring or small series capacitor C, and an inverter gate sustains oscillation on every cycle.

First, just as adding a small calibration weight to a pendulum bob fine-tunes its oscillation rate, a Pierce quartz wafer requires a specific effective *load capacitance* C_L across its terminals to resonate at its nominal factory frequency. On the circuit board, two symmetric ceramic capacitors ($C_1 = C_2 = C$) sit at the input and output pins of the crystal, wired in series with each other through the ground plane, while the copper PCB traces and amplifier pins contribute an unavoidable parallel stray capacitance C_{stray}. The series combination of two equal capacitors $C_1 = C_2 = C$ equals $C/2$ because placing two identical dielectric gaps in series doubles the effective plate separation across the alternating current loop and halves the net capacitance according to the harmonic product-over-sum law (the operation $C_1 \parallel C_2$ computed on the keypad by [x‖y]):

$$C_{\text{series}} = C_1 \parallel C_2 = \frac{C_1 C_2}{C_1 + C_2} = \frac{C^2}{2C} = \frac{C}{2} \quad \textbf{(Eq. 1.9)}$$

Adding the parallel stray capacitance C_{stray} of the copper traces gives the total effective load capacitance seen by the vibrating quartz wafer, which inverts to specify each external board capacitor:

$$C_L = \frac{C}{2} + C_{\text{stray}} \Rightarrow C = 2(C_L - C_{\text{stray}}) \quad \textbf{(Eq. 1.10)}$$

Rather than guessing capacitor values on a breadboard, an engineer subtracts the measured board stray capacitance C_{stray} from the crystal datasheet's target C_L on the stack and doubles that net difference to select the symmetric ceramic pair.

Second, even crystalline quartz expands and changes elastic stiffness slightly with temperature. Ordinary X-cut or Y-cut quartz wafers, as well as the 32.768 kHz tuning-fork crystals inside wristwatches, drift parabolically with temperature and lose speed whenever they warm above or cool below room temperature. In 1934, Lack, Willard, and Fair at Bell Laboratories discovered that by slicing a synthetic quartz boule along a crystallographic plane inclined at 35° 15' to the optical *Z*-axis, known as the *AT-cut* orientation, the positive and negative elastic temperature coefficients of the crystal lattice cancel each other out. At the room-temperature inflection point ($T_0 = 25.0$°C) of a zero-slope cut, both the first-order (linear) and second-order (quadratic) temperature derivatives vanish identically, and only a cubic restoring term governed by a third-order constant k remains:

$$\frac{\Delta f}{f_0} = k(T - T_0)^3 \qquad \textbf{(Eq. 1.11)}$$

Because the temperature excursion $\Delta T = T - T_0$ is cubed, small room-temperature fluctuations around 25.0°C produce negligible frequency error, while sub-zero winter excursions can be modeled on the stack by cubing ΔT with [y^x] and scaling by k (entered as an atomic scientific constant via [E] and [+/−]).

Third, beside the processor's power pins sit ceramic *bypass capacitors* of capacitance C that act as local charge reservoirs, shunting high-frequency transistor switching noise to ground. In an ideal circuit diagram, a capacitor's opposition to alternating current, its capacitive reactance $X_C = 1 / 2\pi f C$, drops toward zero as frequency f climbs toward infinity. In a physical component, however, every surface-mount capacitor has metal end-caps and microscopic internal plates that form a small current loop with *equivalent series inductance* L_{ESL}, whose inductive reactance $X_L = 2\pi f L_{\text{ESL}}$ *grows* linearly with frequency. At the exact frequency where the falling capacitive reactance X_C equals the rising inductive reactance X_L, the two reactances cancel completely and the capacitor reaches its *self-resonant frequency (f_{SRF})*:

$$\frac{1}{2\pi f_{\text{SRF}} C} = 2\pi f_{\text{SRF}} L_{\text{ESL}} \Rightarrow f_{\text{SRF}} = \frac{1}{2\pi\sqrt{L_{\text{ESL}} C}} \qquad \textbf{(Eq. 1.12)}$$

Above f_{SRF}, inductive reactance dominates and the capacitor behaves like an inductor. The SC-32 evaluates f_{SRF} from the inside out in Register *X*: first L_{ESL} and C are entered with [E] and multiplied together, then [$\sqrt{x}$] extracts the characteristic electromagnetic time scale $\tau_{LC} = \sqrt{(L_{\text{ESL}}\ C)}$ in place, then **2** and [↻] [π] scale τ_{LC} into the full radian period $T_0 = 2\pi\sqrt{(L_{\text{ESL}}\ C)}$, and finally a single press of the unary reciprocal key [x^{-1}] inverts that period into the crossover frequency $f_{\text{SRF}} = 1/T_0$.

1.4 Daylight Optics, Tactile Mechanics, and the Laws of Ohm and Joule

Early 1970s red-LED calculators drained their batteries in forty-five minutes, whereas a modern pocket scientific instrument operates for months or years on a single lithium coin cell. That difference in endurance follows from the physics of reflective optics, electrochemical internal resistance, and continuum mechanics.

1.4.1 Reflective Daylight Optics and Exponential Light Absorption

Direct noon sunlight washes out a smartphone screen because an emissive display must project internal backlight photons against the 100,000-lux irradiance of the sky, expending hundreds of milliwatts of battery power to remain legible. A dedicated scientific instrument such as the SC-32 instead employs a *transflective liquid-crystal display (LCD)* that modulates ambient light much like ink on paper.

Room light or sunlight enters through a front linear polarizer and a layer of twisted-nematic liquid-crystal molecules, reflects off a brushed metallic backing behind the glass, and returns to the observer. Where a pixel is energized by an electric field, the rod-shaped liquid-crystal molecules untwist and align with the field so the crossed polarizer extinguishes the reflected ray and forms an ink-black numeral; where a pixel remains unenergized, the incident polarization twists smoothly along the molecular helix and reflects back through the front glass. Brighter outdoor illumination sharpens the contrast while the panel draws a fraction of a milliwatt.

Maintaining that optical clarity across rapid keystroke entry and outdoor temperature extremes couples digital bus timing with liquid-crystal thermodynamics. Refreshing all four visible registers (132×65 pixels, organized into N_{cmd} command bytes plus N_{col} pixel-column bytes) without perceptual tearing requires streaming the frame buffer over a serial peripheral interface (SPI) bus of clock frequency f_{SPI} within a small fraction of the human eye's persistence-of-vision window ($\sim$ 16 ms at 60 Hz), governed by the transfer duration $T_{\text{frame}} = 8(N_{\text{cmd}} + N_{\text{col}})/f_{\text{SPI}}$. Completing that serial commit in roughly a millisecond locks the updated registers onto the glass and returns the bus to zero-current sleep before a pressed keycap rebounds.

Once the frame is latched, the nematic liquid-crystal layer must maintain uniform optical contrast across temperature excursions. Much like cold honey on a winter morning, liquid-crystal molecules stiffen as temperature drops: their rotational viscosity $\gamma_1(T)$ increases, resisting untwisting against their elastic restoring forces unless the driving electric field is strengthened. Applying a negative linear temperature coefficient $dV/dT < 0$ around a room-temperature charge-pump reference V_0 at T_0,

$$V(T) = V_0 + \frac{dV}{dT}(T - T_0) \qquad \textbf{(Eq. 1.13)}$$

raises the LCD bias rail automatically in freezing air so numerals neither ghost nor fade outdoors.

Surrounding that reflective glass, the dark matte polymer shell is compounded with carbon-black pigment so stray sunlight and near-infrared radiation cannot penetrate the walls and scatter across the optical cavity. In the eighteenth and nineteenth centuries, Pierre Bouguer, Johann Heinrich Lambert, and August Beer established why radiation attenuates exponentially rather than linearly inside an absorbing medium: if the first thin lamina absorbs half of the incoming photons, the second identical lamina absorbs half of the remaining half, which reduces transmitted intensity to one-quarter and then one-eighth. Integrating the differential loss $dI/dz = -\alpha\, I$ across a constant linear attenuation coefficient α yields the Beer–Lambert decay law $I(z) = I_0\, e^{-\alpha z}$, which inverts via the natural logarithm ([LN], acting in place on Register X) to specify the wall thickness z required to reduce transmitted stray light to a target fraction I/I_0:

$$I(z) = I_0 e^{-\alpha z} \Rightarrow z = -\frac{\ln(I/I_0)}{\alpha} \qquad \textbf{(Eq. 1.14)}$$

Because α is large in carbon-loaded polymers, ninety percent of incident near-infrared light is extinguished within a surface skin thinner than a human hair. That shallow optical absorption skin also enables permanent keycap legending. During manufacturing, a Q-switched near-infrared fiber laser operating at average power P_{avg} and pulse repetition frequency f_{rep} stores optical energy inside its resonator between openings, delivering $E_{\text{pulse}} = P_{\text{avg}}/f_{\text{rep}}$ per pulse. Releasing that microjoule packet across a nanosecond optical shutter window τ concentrates an instantaneous peak power of

$$P_{\text{peak}} = \frac{E_{\text{pulse}}}{\tau} = \frac{P_{\text{avg}}}{f_{\text{rep}}\tau} \qquad \textbf{(Eq. 1.15)}$$

onto a microscopic focal spot of radius r and area $A = \pi\ r^2$, where the resulting optical fluence $F = E_{\text{pulse}}/(\pi\ r^2)$ nucleates gas micro-bubbles inside the polymer matrix and forms a wear-proof legend flush with the keycap surface.

1.4.2 Ohm's Law, Joule's Law, and Coin-Cell Internal Resistance

Inside the battery compartment sits a CR2032 lithium coin cell (3.0 V nominal electromotive force, 225 mAh capacity). Power delivery from that electrochemical cell to the processor and keypad rests on two relationships established in 1827 by Georg Simon Ohm—who used bismuth-copper thermocouples dipped in boiling water and crushed ice to construct a stable voltage source—and in 1841 by James Prescott Joule:

$$V = IR \quad (\text{Ohm's Law}) \qquad \text{and} \qquad P = VI = I^2R = \frac{V^2}{R} \quad (\text{Joule's Power Law}) \qquad \textbf{(Eq. 1.16)}$$

Here V is potential difference in Volts (Joules per Coulomb), current I in Amperes measures charge flow in Coulombs per second, resistance R in Ohms (Ω) measures opposition to current, and thermal power P in Watts measures heat dissipation in Joules per second. Substituting Ohm's Law $V = IR$ into $P = VI$ yields the quadratic heating law $P = I^2\ R$, under which doubling the current through a conductor or battery multiplies resistive heat dissipation by four ($2^2 = 4$).

Whereas an ideal Thévenin voltage source maintains a constant open-circuit electromotive force E at any current, a physical lithium-manganese-dioxide (Li/MnO_2) coin cell relies on Li^+ ions migrating through an organic electrolyte between the metallic lithium anode and the porous MnO_2 cathode. That ionic transport resistance acts as an internal series resistance R_{int} inside the cell, causing the terminal voltage delivered to the circuit board to droop under load according to $V_{\text{term}}(I) = E - I\ R_{\text{int}}$.

Because probing E directly under load is impossible, internal resistance is determined by measuring terminal voltage across two operating states, such as low-current idle refresh (I_1, V_1) and a high-current numerical solver burst (I_2, V_2). Subtracting the two linear terminal equations eliminates the open-circuit potential E and isolates the dynamic internal resistance, along with the internal Joule heating dissipated during peak computation:

$$R_{\text{int}} = \frac{V_1 - V_2}{I_2 - I_1}, \qquad P_{\text{loss}} = I_{\text{burst}}^2 R_{\text{int}} \qquad \textbf{(Eq. 1.17)}$$

Over a full day of use, the instrument cycles among discrete operating states—solver bursts (I_1, t_1), interactive key entry (I_2, t_2), standby display hold (I_3, t_3), and quartz-timed deep sleep (I_4, t_4)—so total daily charge consumption in milliampere-hours (1 mAh = 3.6 Coulombs) is the piecewise sum $Q_{\text{daily}} = \sum_{k=1}^{4} I_k\ t_k$, and cell service life follows from the quotient $t_{\text{life}} = Q_{\text{cell}}\ /\ Q_{\text{daily}}$.

Ohm's Law and Kirchhoff's loop equations likewise govern the digital interface between the mechanical key switches and the processor logic rails. When a thumb press snaps a gold-plated tactile dome onto its PCB contact ring, elastic momentum causes the metal surfaces to rebound microscopically for a bounce duration t_{bounce} before settling into steady contact; registering the keystroke on the initial falling edge and enforcing a firmware lockout window T_{lockout} establishes a dimensionless debounce safety factor $M = T_{\text{lockout}} \,/\, t_{\text{bounce}}$.

Within the multiplexed row-and-column key matrix, placing a one-way electrical check-valve (diode) in series with each switch prevents a sneak path from registering a phantom fourth key when three keys in an L-corner are pressed simultaneously. Selecting a metal-semiconductor Schottky barrier diode with a low forward voltage drop V_F rather than a diffused silicon p-n junction diode preserves the logic-low noise margin $V_{NM,L} = V_{IL,\text{max}} - V_F$ below the microcontroller's maximum logic-low threshold $V_{IL,\text{max}}$, while transient voltage suppressor (TVS) diodes clamp electrostatic human-body-model discharges ($I_{\text{peak}} = V_{\text{ESD}} \,/\, R_{\text{HBM}}$).

Finally, when a fresh coin cell first contacts the battery clips, supply voltage sputters for a few milliseconds before settling. Acting like a tiny 10-millisecond hourglass that holds the processor still until the power rail is rock-steady, an external pull-up resistor R charges a timing capacitor C (time constant $\tau = RC$) toward the rail V_{dd} until the capacitor potential crosses the Schmitt-trigger release threshold V_{th}. Integrating Kirchhoff's loop equation $V_{\text{dd}} - IR - Q/C = 0$ shows that the uncharged voltage fraction $(1 - V(t)/V_{\text{dd}})$ decays exponentially as $e^{-t/RC}$:

$$V(t) = V_{\text{dd}}(1 - e^{-t/RC}) \Rightarrow t = -RC \ln\left(1 - \frac{V_{\text{th}}}{V_{\text{dd}}}\right) \quad \textbf{(Eq. 1.18)}$$

1.4.3 Mechanical Permanence, Thermal Fatigue, and the Cubic Law of Stiffness ($I = bh^3/12$)

The structural permanence of a pocket instrument across decades of service rests on four continuum mechanics relationships governing its fasteners, solder joints, perimeter elastomer, and internal ribbing. Rather than driving steel machine screws into bare plastic bosses that strip after repeated battery changes, heat-set knurled brass inserts of outer diameter D and embedment length L anchor into the glass-reinforced chassis across a cylindrical shear cylinder of area $A_{\text{shear}} = \pi \, D \, L$, resisting pull-out up to $F_{\text{pull}} = \pi \, D \, L \, \tau$, where τ is the shear yield strength of the surrounding polymer matrix.

Inside the enclosure, carrying the instrument repeatedly between winter air and a heated laboratory subjects the printed circuit assembly to cyclic temperature excursions ΔT. Because rigid FR-4 fiberglass and copper component leads expand at different thermal rates, each excursion imposes plastic shear strain across the lead-free solder fillets; under the Coffin–Manson inverse power law with fatigue exponent m and material constant C, the expected thermal cycle life scales as $N_f = C \,/\, (\Delta T)^m$. Similarly, if the instrument drops from height h onto a rigid floor while its perimeter TPU bumper compresses over a stopping stroke Δy, equating gravitational potential energy mgh to average deceleration work $F_{\text{avg}} \Delta y = m \, a_{\text{avg}} \Delta y$ cancels both mass m and gravitational acceleration g to show that peak shock deceleration in multiples of gravity depends strictly on the geometric ratio $G = a_{\text{avg}}/g = h \,/\, \Delta y$.

Finally, keypad rigidity under thumb pressure follows the beam theory formulated in the eighteenth century by Leonhard Euler and Daniel Bernoulli. When a concentrated thumb load F acts at the midpoint of a structural rib of span L, width b, vertical depth h, and

Young's elastic modulus E, the upper fibers compress while the lower fibers stretch, generating an internal restoring moment proportional to the rectangular *area moment of inertia* I:

$$I = \frac{bh^3}{12}, \qquad \delta = \frac{FL^3}{48EI} \tag{Eq. 1.19}$$

Because width b enters linearly whereas vertical thickness h enters cubed (h^3), doubling the vertical depth of an internal chassis rib or battery-bay cantilever latch multiplies its flexural stiffness eightfold ($2^3 = 8$), suppressing keypad sponge without adding bulk.



Figure: Precision dovetail service bay and cantilever snap-latch geometry

1.5 Instrument Diagnostics, Optics & Power-Law Calculations

With the physical models of the oscillator, display, power rail, and chassis established, the four-register stack can audit the exact design parameters of the SC-32 hardware. Across these evaluations, every formula follows one of two register patterns: *single-register cascades*, in which sequential products, quotients, and unary transformations such as $F = \pi\ D\ L\ \tau$ or $f_{\mathrm{SRF}} = 1\ /\ 2\pi\sqrt{LC}$ collapse back into Register X at each step and call on Register Y only momentarily while Registers Z and T remain at zero; and *multi-register staging*, in which two-branch expressions such as $R_{\mathrm{int}} = (V_1 - V_2)\ /\ (I_2 - I_1)$ or $\delta = FL^3/48EI$ evaluate the first branch in Register X and shift it upward into Registers Y, Z, or T while the second branch is assembled across the lower registers.

1.5.1 Mechanical Fasteners, Impact Kinematics & Switch Resilience

Single-branch physical laws execute without leaving intermediate values parked in Registers Z or T because each dyadic multiplication or division immediately folds Register Y and Register X into a single updated quantity in the active register. For the ohmic series resistance $R = \rho\ L\ /\ w\ t$ of a 1-oz copper PCB power rail ($\rho = 1.68 \times 10^{-5}\ \Omega\cdot\mathrm{mm}$) routed across length $L = 45.0$ mm with trace width $w = 0.25$ mm and foil thickness $t = 0.035$ mm: keying **0.0000168** [ENTER] **45** [×] forms the $\rho\ L = 7.56 \times 10^{-4}\ \Omega\cdot\mathrm{mm}^2$ numerator in Register X. Because [×] arms automatic stack lift, typing **0.25** [÷] **0.035** [÷] divides sequentially by width and thickness in place to land on 0.0864 Ω (86.4 mΩ) without a second [ENTER] keystroke.

Applying that same single-register economy to switch contact debounce, fastener shear, and freefall impact quantifies the mechanical and electrostatic safety margins of the keypad and enclosure:

1. **Switch Debounce & Electrostatic Discharge Protection ($M = T_{\text{lockout}} / t_{\text{bounce}}$, $I_{\text{peak}} = V_0 \ / \ R_{\text{HBM}}$):** For gold-plated dome contacts that rebound elastically for $t_{\text{bounce}} = 4.8$ ms inside a $T_{\text{lockout}} = 20.0$ ms firmware refractory window, dividing 20.0 ms in Register Y by 4.8 ms in Register X verifies a $>4\times$ debounce margin. When an 8,000 V winter-carpet electrostatic spark discharges across the standardized human-body-model skin resistance ($R_{\text{HBM}} = 1{,}500\ \Omega$), dividing 8,000 V by 1,500 Ω yields the peak surge current that the board's TVS diodes shunt to ground:

   - **Debounce Safety Margin:** `20` [ENTER] `4.8` [÷] → **4.1667**× safety factor.
   - **ESD Peak Clamping Current:** `8000` [ENTER] `1500` [÷] → **5.3333** A shunted to ground.

2. **Heat-Set Brass Insert Pull-Out Strength ($F_{\text{pull}} = \pi\ D\ L\ \tau$):** Whereas a small $D = 2.6$ mm, $L = 4.0$ mm insert in unfilled ABS ($\tau = 30$ MPa) shears at 980 N, the SC-32 chassis anchors its fasteners with $D = 3.6$ mm heavy-knurl brass inserts embedded across $L = 5.0$ mm in glass-reinforced polymer ($\tau = 42.0$ N/mm^2). On the stack, multiply diameter D by π to form the 11.31 mm cylindrical perimeter, scale by the 5.0 mm embedment depth to sweep out shear area $A_{\text{shear}} = 56.55$ mm^2, and multiply by $\tau = 42.0$ N/mm^2 while Registers Z and T remain at zero:

   - **Ultimate Pull-Out Force:** `3.6` [↻] [π] [×] `5.0` [×] `42.0` [×] → **2375.04 N** (≈ 242 kgf).

3. **Freefall Impact & Elastomeric Shock Deceleration ($G = h \ / \ \Delta y$):** When the instrument slips off a $h = 1.20$ m bench onto concrete and its perimeter TPU bumper compresses across a stopping stroke of $\Delta y = 2.2$ mm $= 0.0022$ m (or 0.0015 m on an 800 G corner strike), dividing drop height h in Register Y by cushion stroke Δy in Register X gives the average deceleration in G's directly. (As an independent kinematic check on a $h = 1.50$ m drop with $\Delta y = 0.0018$ m, Torricelli's law $v = \sqrt{2gh}$ gives an impact speed of **5.4240 m/s**; squaring in place with [↶] [x²] to 29.4200 m^2/s^2 and dividing by $2\ g\ \Delta y$ confirms an **833.3333 *G*** shock pulse.)

   - **Bench Drop Shock Pulse (1.20 m onto 2.2 mm bumper):** `1.20` [ENTER] `0.0022` [÷] → **545.45 *G***.

1.5.2 High-Speed Clocks, Parasitic Inductance & Multi-Register Beam Flexure

Stepping from mechanical geometry to high-frequency timing and structural flexure engages the Scientific Exponent key [E], the dyadic Power key [y^x], and the unary Reciprocal key [x^{-1}] across nineteen orders of magnitude:

1. **Pierce Oscillator Load Tuning ($C = 2(C_L - C_{\text{stray}})$):** For the 12.000 MHz Pierce quartz crystal to resonate at its calibrated factory frequency, its terminals require a net series load capacitance of $C_L = 18.0$ pF. With PCB traces and amplifier pins contributing $C_{\text{stray}} = 3.5$ pF in parallel, start inside the parentheses by subtracting 3.5 pF from 18.0 pF in Register X (→ 14.5 pF), then double that difference to specify each symmetric series capacitor:

- **External Load Capacitor Value:** `18` [ENTER] `3.5` [-] `2` [×] → **29.0000** pF per capacitor.

2. **AT-Cut Cubic Thermal Drift ($\Delta f / f_0 = k\Delta T^3$):** When that AT-cut quartz wafer moves from its T_0 = 25.0°C inflection point to a -15.0°C winter field site (ΔT = -40 K, cubic lattice constant $k = 0.035 \times 10^{-9}$ K^{-3}), cubing ΔT = -40 K with [y^x] forms -64,000 K^3 in Register X. Multiplying by k (entered via [E]) yields a fractional drift of $\Delta f / f_0$ = **-2.2400 × 10⁻⁶** (-2.24 ppm). Scaling by the 12.0 × 10^6 Hz carrier gives a shift of **-26.8800 Hz**, or if operated continuously over 86,400 s/day, a timekeeping drift of **-0.1935 s/day**.

3. **SPI Display Refresh Duty Cycle ($T_{\text{frame}} = 1065 \times 8 / (8.0 \times 10^6)$):** A complete 132 × 65-pixel screen refresh transmits N_{cmd} = 9 command bytes plus N_{col} = 1,056 column bytes (1,065 bytes total). Multiplying 1,065 by 8 bits/byte forms the 8,520-bit payload in Register X; dividing by the 8.0 × 10^6 bits/s SPI clock (entered via [E]) yields 0.001065 s, and scaling by 1,000 converts the transfer window into milliseconds (at one keystroke per second, the display bus sleeps for 99.89% of every second):

- **Frame Commit Duration:** `1065` [ENTER] `8` [×] `8` [E] `6` [÷] `1000` [×] → **1.065 ms**.

4. **Q-Switched Laser Keycap Foaming ($P_{\text{peak}} = P_{\text{avg}} / f_{\text{rep}}\tau$):** Dividing a 1064 nm fiber laser's P_{avg} = 1.20 W continuous beam power by its f_{rep} = 35,000 Hz repetition rate isolates the single-pulse energy E_{pulse} = 34.29 μJ in Register X, and dividing by the $\tau = 1.0 \times 10^{-7}$ s (100 ns) shutter width reveals the peak optical power delivered to the keycap. (Dividing 34.29 μJ in Register Y by the spot area $\pi r^2 = \pi(0.00225\text{ cm})^2 = 1.5904 \times 10^{-5}$ cm^2 formed in Register X via [↶] [x^2] and [↷] [π] confirms a foaming fluence of **2.1558 J/cm²**.)

- **Peak Optical Pulse Power:** `1.20` [ENTER] `35000` [÷] `1` [E] `7` [+/−] [÷] → **342.85714286** Watts.

5. **Bypass Capacitor Self-Resonant Frequency ($f_{\text{SRF}} = 1 / 2\pi\sqrt{LC}$):** For a C = 100 nF (1.0 × 10^{-7} F) decoupling capacitor whose package terminals contribute equivalent series inductance L_{ESL} = 0.85 nH (8.5 × 10^{-10} H, or 0.65 nH in reverse-geometry packages where f_{SRF} reaches 19.7 MHz), multiply L by C in Register X to form LC = 8.50 × 10^{-17} s^2, press [$\sqrt{x}$] to extract $\sqrt{LC}$ = 9.22 ns in place, multiply by 2π for the radian period (57.93 ns), and tap [x^{-1}] before dividing by 10^6 for Megahertz:

- **Self-Resonant Crossover Frequency:** `8.5` [E] `10` [+/−] [ENTER] `1` [E] `7` [+/−] [×] [$\sqrt{x}$] `2` [×] [↷] [π] [×] [x^{-1}] `1` [E] `6` [÷] → **17.262777335 MHz**.

6. **Coffin-Manson Solder-Joint Fatigue Life ($N_f = C / (\Delta T)^m$):** Under cyclic thermal excursions of ΔT = 55 K between winter air and an indoor bench (m = 2.2, C = 1.8 × 10^7 for ductile lead-free SAC305 fillets on FR-4), inside-out evaluation computes the strain denominator $(\Delta T)^{2.2} = 55^{2.2}$ = 6,742.14 first in Register X via [y^x]. Entering the fatigue constant 1.8 × 10^7 lifts 6,742.14 into Register Y, where a single tap of [$x \leftrightarrow y$] orders numerator and denominator for division:

– **Thermal Shock Fatigue Life:** 55 [ENTER] 2.2 [y^x] 1.8 [E] 7 [$x\rightleftarrows y$] [÷] → **2,669.774547** extreme thermal cycles.

7. **Two-Branch Structural Flexure: Euler-Bernoulli Beam Deflection ($\delta = FL^3/48EI$ where $I = bh^3/12$):** Evaluating both Euler–Bernoulli equations in a single pass engages all four registers: first computing the cross-sectional Area Moment of Inertia $I = bh^3/12$ of an internal chassis stiffening rib (b = 15.0 mm, h = 2.5 mm), and then dividing the cubic bending load by that stiffness under a F = 3.92 N center thumb press across a L = 60.0 mm span (E = 3,500 N/mm^2). Once I = 19.53125 mm^4 is formed on Row 2, entering F, L, and the exponent 3 pushes I up to Register T, where *T-register replication* automatically copies 19.53125 mm^4 back down into Y as the load numerator FL^3 = 846,720 N·mm^3 and stiffness factor $48E$ = 168,000 N/mm^2 collapse across the lower registers (**15** [ENTER] **2.5** [ENTER] **3** [y^x] [×] **12** [÷] **3.92** [ENTER] **60** [ENTER] **3** [y^x] [×] **48** [ENTER] **3500** [×] [÷] [$x\rightleftarrows y$] [÷] → **0.2580 mm**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
15 [ENTER] **2.5** [ENTER] **3** [y^x]	0.0000	0.0000	15.0000	15.6250	Cube vertical rib thickness $h^3 = 2.5^3 = 15.625$ mm^3
[×] **12** [÷]	0.0000	0.0000	0.0000	19.5312	Area moment of inertia $I = bh^3/12 = 19.53125$ mm^4
3.92 [ENTER] **60** [ENTER] **3** [y^x]	19.5312	19.5312	3.9200	216000.00	Cube span $L^3 = 60^3$ mm^3 (I lifts to T and replicates into Z)
[×]	19.5312	19.5312	19.5312	846720.00	Bending load $F\,L^3$ (N·mm^3); T-replication brings I down to Y
48 [ENTER] **3500** [×]	19.5312	19.5312	846720.00	168000.00	Stiffness denominator factor $48\,E$ = 168,000 N/mm^2
[÷]	19.5312	19.5312	19.5312	5.0400	Quotient $F\,L^3$ / $(48\,E)$ = 5.0400 mm^5; I waits in Y
[$x\rightleftarrows y$] [÷]	19.5312	19.5312	19.5312	**0.2580**	Swap I into X and divide to resolve center flexure δ in mm

*(Applying that identical four-register sequence to the battery-door snap-latch cantilever clip of width b = 4.0 mm, thickness h = 1.5 mm, and length L = 12.0 mm under a F = 6.0 N thumb press—forming $I = bh^3/12$ = 1.125 mm^4 first so it rides in Register T while the cantilever load FL^3 = 10,368 N·mm^3 is divided by $3E$ = 10,500 N/mm^2 and swapped via [$x\rightleftarrows y$]—confirms an elastic latch engagement stroke of $\delta = FL^3/(3EI)$ = **0.8777 mm**.)*

1.5.3 Hardware Diagnostics, LASTX Recovery & Euclidean Checksums (÷R)

Three diagnostic traces demonstrate how [↶] [LASTx], signed temperature offsets ([+/−]), and Euclidean quotient-remainder division ([↻] [÷R]) audit digital logic levels and non-volatile flash integrity:

1. **Schottky Diode Anti-Ghosting Margin & LASTX Comparison ($V_{NM,L} = V_{IL,\max}$ - V_F):** Because a pulled-low key matrix column must read below the microcontroller's maximum logic-low input threshold ($V_{IL,\max}$ = 0.80 V) after dropping across the series isolation diode's forward barrier V_F, placing a metal-semiconductor Schottky diode (V_F = 0.32 V) at each switch intersection preserves a 0.48 V noise margin. Because subtraction saves 0.32 V inside [↶] [LASTx], recalling [↶] [LASTx] and pressing [+] to restore the 0.80 V threshold before subtracting 0.72 V shows immediately why an ordinary silicon p-n junction diode (V_F = 0.72 V) would compress the margin to 0.0800 V:

- **Schottky Logic-Low Noise Margin:** `0.80` [ENTER] `0.32` [-] → **0.4800 V**.

2. **Cold-Weather Liquid-Crystal Viscosity Compensation ($V(T) = V_0 + (dV/dT)(T - T_0)$):** When ambient temperature drops from room reference (T_0 = 25.0°C, V_0 = 9.00 V) to freezing (T = 0.0°C) along a charge-pump compensation slope of dV/dT = -0.0125 V/°C, start inside the parentheses by subtracting 0 - 25 to form the signed thermal excursion ΔT = -25.0°C in Register X, multiply by -0.0125 V/°C (negated via [+/−]) to form the +0.3125 V boost in place, and add the 9.00 V baseline:

   - **Freezing-Air Contrast Rail:** `0` [ENTER] `25` [-] `0.0125` [+/−] [×] `9.00` [+] → **9.3125** V.

3. **Simultaneous Two-Register Output in the FNV-1a Flash Checksum ([↷] [÷R]):** Acting as a digital tamper-seal that verifies no stored variable or formula was corrupted while the calculator slept, the firmware audits flash memory on wake-up using the 32-bit FNV-1a (Fowler–Noll–Vo) hash, XOR-folding each stored byte into a running integer state and multiplying by the FNV prime 16,777,619 modulo 2^{32} = 4,294,967,296. Unlike standard arithmetic operators that collapse two registers into one, the Euclidean division key [↷] [÷R] consumes the dividend in Y and divisor in X and writes two simultaneous results back onto the stack: the 32-bit wrapped hash remainder ($Y \bmod X$ = 3.3562 × 10^9) in Register Y and the truncated integer overflow quotient ($\lfloor Y / X \rfloor$ = **8,461,672**) in Register X (`2166136200` [ENTER] `16777619` [×] `4294967296` [↷] [÷R]):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`2166136200` [ENTER]	0.0000	0.0000	2.1661 × 10^9	2.1661 × 10^9	Load XOR-accumulated hash state
`16777619` [×]	0.0000	0.0000	0.0000	3.6343 × 10^{16}	Multiply by 32-bit FNV prime
`4294967296` [↷] [÷R]	0.0000	0.0000	3.3562 × 10^9	**8461672.00**	Divide with remainder: Y=32-bit wrapped hash, X=quotient

1.5.4 Battery Electrochemistry, Optical Extinction & RC Reset Timing

Electrochemical load testing, Beer–Lambert optical attenuation, and Kirchhoff RC charging complete the hardware audit across the CR2032 cell, the carbon-loaded enclosure, and the power-on reset supervisor:

1. **Four-State Daily Charge Accumulation ($Q_{\text{daily}} = \sum I_k\ t_k$):** Across an accelerated 24-hour factory torture-test profile—0.5 h of intensive solver bursts at 18.2 mA, 1.5 h of interactive typing at 4.8 mA, 2.0 h of standby pause at 0.85 mA, and 20.0 h of deep sleep at 0.0118 mA—each state's charge folds into the running sum in Register X as soon as its product is formed (9.10 + 7.20 → 16.30 + 1.70 → 18.00 + 0.236 → 18.236 mAh/day). Entering the CR2032 cell's 225 mAh capacity and swapping via [x⇄y] divides capacity by daily load without touching Register T:

   - **Continuous Bench Endurance:** `18.2` [ENTER] `0.5` [×] `4.8` [ENTER] `1.5` [×] [+] `0.85` [ENTER] `2.0` [×] [+] `0.0118` [ENTER] `20.0` [×] [+] `225` [x⇄y] [÷] → **12.3382** days of continuous bench testing.

2. **Coin-Cell Dynamic Internal Resistance (R_{int} = (V_1 - V_2) / (I_2 - I_1)) & Joule Heating:** Stepping from idle display refresh (I_1 = 0.0048 A, V_1 = 2.940 V) into a root-finding burst (I_2 = 0.0182 A, V_2 = 2.685 V) evaluates the two-load Thévenin resistance across Registers Y and X: subtract ΔV = 2.940 - 2.685 = 0.2550 V, form ΔI = 0.0182 - 0.0048 = 0.0134 A in X while ΔV rides to Z and drops to Y, and divide ΔV / ΔI (confirming internal Joule heating $P_{loss} = I_{burst}^2 R_{int}$ = **6.3035 mW** via [↶] [LASTx]):

   - **Internal Dynamic Resistance:** **2.940** [ENTER] **2.685** [-] **0.0182** [ENTER] **0.0048** [-] [÷] → R_{int} = **19.0299 Ω**.

3. **Beer-Lambert Optical Extinction Depth (z = -ln(I/I_0)/α):** To find the carbon-black polymer wall depth z required to extinguish 90% of stray 1064 nm near-infrared light (α = 450 cm^{-1}, transmitted fraction I/I_0 = 0.10), the evaluation stays entirely inside Register X: apply [LN] to 0.10 (→ -2.3026 nepers), divide by α = 450 cm^{-1}, negate via [+/−], and scale by 10,000 μm/cm:

   - **90% Optical Absorption Depth:** **0.10** [LN] **450** [÷] [+/−] **10000** [×] → **51.1686** μm.

4. **Three-Register RC Power-On Reset Lockout Time (t = -RC ln(1 - V_{th}/V_{dd})):** When the power rail V_{dd} = 3.30 V charges the reset timing capacitor (C = 0.10 μF = 1.0 × 10^{-7} F) through a pull-up resistor (R = 100 kΩ = 100,000 Ω) toward the microcontroller's V_{th} = 2.10 V release threshold, this seven-step sequence (**2.10** [ENTER] **3.30** [÷] [+/−] **1** [+] [LN] **100000** [ENTER] **1** [E] **7** [+/−] [×] [×] [+/−] **1000** [×] → **10.116009117 ms**) first forms the uncharged voltage headroom 1 - V_{th}/V_{dd} = 0.3636 and applies [LN] in place to obtain -1.0116 in Register X. Entering R and C across the lower registers to form $\tau = RC$ = 0.0100 s lifts -1.0116 to Register Z and drops it back to Register Y for multiplication:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
2.10 [ENTER] **3.30** [÷]	0.0000	0.0000	0.0000	0.6364	Normalized threshold V_{th} / V_{dd}
[+/−] **1** [+]	0.0000	0.0000	0.0000	0.3636	Uncharged voltage headroom (1 - V_{th}/V_{dd})
[LN]	0.0000	0.0000	0.0000	-1.0116	Natural logarithm of voltage headroom
100000 [ENTER]	0.0000	-1.0116	100000.0000	100000.00	Pull-up resistance R = 100 kΩ
1 [E] **7** [+/−] [×]	0.0000	0.0000	-1.0116	0.0100	Circuit time constant RC = 0.01 s
[×]	0.0000	0.0000	0.0000	-0.010116	Negative elapsed delay in seconds
[+/−] **1000** [×]	0.0000	0.0000	0.0000	**10.1160**	POR reset lockout time in milliseconds

5. **Aviation Preflight Fuel-Load Audit & Machinist Blueprint Conversions ([↶] [▸kg], [↶] [▸l], [↶] [▸°C], [↷] [▸in]):** Because all ten SI ↔ Imperial unit conversions sit directly on the shifted digit pad ([↶] for Metric output, [↷] for Imperial output) and act as unary in-place transformations on Register X, field practitioners never open a conversion menu. When a pilot on a hot ramp (86.0°F) loads 52.0 US gallons of 100LL avgas (at the standard FAA nominal weight of 6.00 lb/gal)

and needs the metric mass (**kg**), metric volume (**L**), and Celsius outside air temperature (**°C**) for an international weight-and-balance manifest—followed by a machinist converting a 25.4 cm gauge span to inches (**in**)—all four converted figures stack cleanly across Registers (T, Z, Y, X):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
52 ENTER 6 ×	0.0000	0.0000	0.0000	312.0000	Avgas payload weight 52 gal × 6.0 lb/gal = 312.0 lb
↶ ▸kg	0.0000	0.0000	0.0000	141.5208	Convert fuel payload mass in place to 141.5208 kg
52 ↶ ▸l	0.0000	0.0000	141.5208	196.8414	Convert 52 gal to 196.8414 L; fuel mass in kg lifts to Y
86 ↶ ▸°C	0.0000	141.5208	196.8414	30.0000	Convert ramp OAT 86°F to 30.0000°C
25.4 ↷ ▸in	141.5208	196.8414	30.0000	**10.0000**	Convert 25.4 cm span to 10.0000 in; all 4 metrics visible

Every precision scientific instrument must reckon with its own physical parasitics: contact rebound in mechanical switches, ionic impedance inside electrochemical cells, stray capacitance and package inductance along copper traces, and photon attenuation inside polymer walls. Expressing those constraints in postfix order turns the four-register stack into a diagnostic instrument for the machine itself; complete KiCad schematics, PCB layouts, and CAD models are archived at **stackcalc.io/hardware** and on the **StackCalc Hackaday Project**.

Chapter 2: The Architecture of Inside-Out Thinking

2.1 George Miller's Four Chunks and Dijkstra's Railway Yard

2.1.1 The Four-Chunk Limit of Human Working Memory

A human being can glance at a four-item grocery list—*milk, apples, coffee, bread*—and walk through a supermarket without writing anything down, yet stall when staring at a four-layer algebraic fraction on a mathematics examination.

In 1956, Harvard cognitive psychologist George A. Miller published a foundational study on the limits of human information-processing capacity *(Miller, 1956)* and showed that *working memory*, the mental scratchpad where numbers and intermediate relationships are held during active reasoning, spans only about seven items. Subsequent experiments by cognitive scientists Nelson Cowan and John Sweller *(Sweller, 1988)* refined Miller's bound further: when a person is actively transforming quantities rather than reciting a static digit string, biological working memory holds only about four independent chunks at one time.

Evaluating a two-branch rational expression from left to right on paper or on an ordinary infix calculator exposes this bottleneck immediately:

$$U = \frac{(a \cdot b) + (c \cdot d)}{e - f} \tag{Eq. 2.1}$$

Reading that equation strictly from left to right, in the order sixteenth-century typesetters printed it, consumes all four working-memory slots before arithmetic even begins. The opening bracket marks the numerator of a compound fraction in the first mental slot, while a second nested parenthesis opens to multiply $a \cdot b$ in the second slot. Next, the central + sign must be deferred in the third slot because the second product $(c \cdot d)$ has not yet been evaluated, and the division bar ÷ (must likewise wait in the fourth slot until the denominator $(e - f)$ is complete. Before a single challenging operation has taken place, working memory is saturated with *suspended syntax*: open parentheses and deferred operators waiting for their right-hand operands.

Evaluating that same fraction from the inside out eliminates the suspended syntax altogether: multiplying $a \cdot b$ forms a completed product P_1 in one register; multiplying $c \cdot d$ forms P_2 and adds the two products into the completed numerator $N = P_1 + P_2$, which immediately frees the second register; and subtracting $D = e - f$ prepares the denominator for the final division $N \div D$. At no point does the mind hold more than three numbers simultaneously, and at no point does it track an unclosed parenthesis.

2.1.2 Edsger Dijkstra's 1961 Railway Siding

Humans are not the only ones impeded by left-to-right infix notation. Digital processors cannot evaluate infix equations in left-to-right order either.

In 1960 and 1961, Dutch computer scientist Edsger W. Dijkstra *(Dijkstra, 1961)* and his colleague Jaap Zonneveld were writing the first ALGOL 60 compiler on an Electrologica X1 computer at the Mathematical Centre in Amsterdam. Faced with teaching a digital machine with limited magnetic-core memory to parse formulas such as $A + B \times C$ in a

single left-to-right pass, Dijkstra noted that a processor cannot add A and B the moment it encounters the **+** sign, because **×** carries higher operator precedence (**PEMDAS**). The compiler must shunt the **+** sign onto a temporary waiting track, much like freight cars in a railway switchyard, until $B \times C$ has been coupled together.

Dijkstra named this translation method the *Shunting-Yard Algorithm*. As an incoming sequence of tokens (**a**, **×**, **b**, **+**, **c**, **×**, **d**) arrives from left to right along a main track, each numeric operand passes directly to the output queue, whereas each operator (**×**, **+**, **÷**) or left parenthesis **(** is diverted onto a last-in, first-out operator siding to wait until its right-hand operands have passed. Only when an operation of equal or higher precedence completes, or when a closing right parenthesis **)** arrives to clear the siding, do the waiting operators roll back onto the main output track behind their operands:

$$\text{Infix Input: } (a \times b) + (c \times d) \xrightarrow{\text{Shunting Yard}} \text{Postfix Output: } a\ b\ \times\ c\ d\ \times\ + \qquad \textbf{(Eq. 2.2)}$$

In the postfix sequence emerging from Dijkstra's shunting yard on the right, every parenthesis has been discarded, producing the chronological postfix order developed from Łukasiewicz *(Łukasiewicz, 1924)* and Burks, Warren, and Wright *(Burks et al., 1954)* to Charles Hamblin's hardware push-down stack *(Hamblin, 1957)*. Whenever an equation is entered into Python, Java, C++, JavaScript, or an algebraic pocket calculator followed by **=**, an internal shunting-yard parser translates the left-to-right infix text into postfix stack instructions before any arithmetic takes place, a mechanism that powers the Java Virtual Machine (JVM), WebAssembly (Wasm), Adobe PostScript, and spacecraft Forth navigation computers *(Knuth, 1997; HP, 1982)*.

When Bill Hewlett and David Cochran designed the HP-35 in 1972 *(Whitney, Rodé & Tung, 1972; Cochran, 1972)*, they joined George Miller's cognitive limit with Edsger Dijkstra's compiler architecture. Rather than requiring the user to hold open parentheses in working memory while an internal parser translates infix keystrokes back into a postfix stack, they exposed the four-register stack (X, Y, Z, T) directly on the instrument so that every intermediate subtotal appears as soon as it is formed.

2.2 Unpeeling Rational Expressions and Harmonic Work Rates

Four visible registers (X, Y, Z, T) evaluate any rational expression in algebra or engineering through the paired mechanics of stack lift and stack drop.

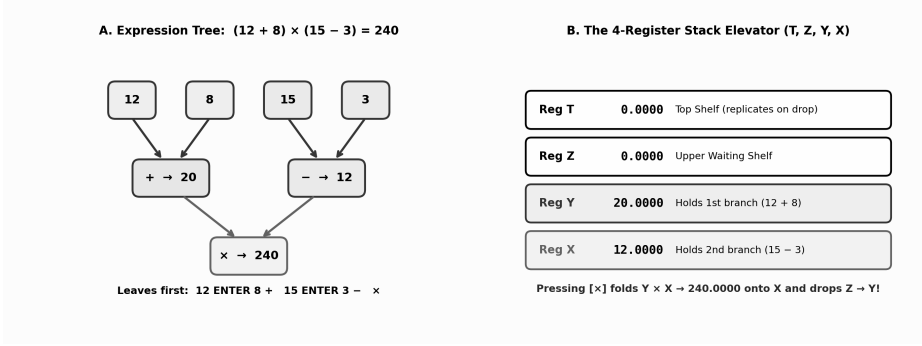


Figure: Inside-out expression tree evaluation paired with the four-register operational stack (T, Z, Y, X).

2.2.1 Stack-Lift State Transitions: Neutral After ENTER, Armed After Operators

Across the four vertical registers, every stack lift shifts subtotals upward toward Register T, and every dyadic (two-operand) operation drops waiting subtotals downward toward Register X:

Stack Register	Architectural Role	Movement on Lift (**ENTER** / Armed Digit)	Movement on Dyadic Drop (+, -, ×, ÷, y^x, x‖y)
Register T (Line 4)	**Top Register & Constant Source**	Receives Z_{old} (T_{old} shifts out in **4-LVL**)	Copies T_{old} downward into Z while remaining in T
Register Z (Line 3)	**Upper Holding Register**	Receives Y_{old} as a second branch starts	Drops Z_{old} down into Y, ready to combine with X
Register Y (Line 2)	**Pending Operand Register**	Receives X_{old} (**ENTER** disables lift; armed digit lifts)	Consumed with X_{old}; receives Z_{old} from Register Z
Register X (Line 1)	**Active Register**	Holds copy on **ENTER** (overwritten by next digit)	Receives new result $f(Y_{\text{old}}, X_{\text{old}})$ and arms stack lift

To govern how newly keyed numbers enter Register X, the stack-lift state machine transitions between two states: *Neutral* (**Stack Lift = FALSE**) after **ENTER**, and *Armed* (**Stack Lift = TRUE**) after any operator or function. To multiply a by b, the user keys a into Register X and presses **ENTER** to terminate digit entry for the first operand. Pressing **ENTER** copies a upward into Register Y ($Y = a$) while a remains displayed in Register X ($X = a$), and sets the stack-lift state to neutral (**Stack Lift = FALSE**). Because Register X still holds that duplicated copy of a, the next number keyed (b) overwrites Register X in place so that $Y = a$ and $X = b$ sit ready for multiplication.

Pressing **×** then drops Register Y (a) and combines it with Register X (b) to form the product $P_1 = a\ b$ in Register X. Because P_1 is a completed result rather than a temporary duplicate from **ENTER**, every arithmetic operator or mathematical function sets the stack-lift state to armed (**Stack Lift = TRUE**). When the first digit of the next operand (c) is keyed, **ENTER** is not pressed: the armed stack automatically lifts P_1 into Register Y so c can occupy Register X. Pressing **ENTER** after c and keying d shifts the completed first branch (P_1) into Register Z while c and d occupy Registers Y and X. When **×** multiplies $P_2 = c\ d$ in Register X, the upper registers drop by one level and return P_1 from Register Z into Register Y, where **+** sums $P_1 + P_2$ into the complete numerator N. Evaluating the denominator $D = e$ - f across the lower registers while N waits in the upper registers then sets up the final division $N \div D$.

2.2.2 Harmonic Work Rates and Parallel Reciprocals

This two-branch stack discipline clarifies a classic problem in algebra and hydraulics. Consider a civil engineer testing the emergency water supply for a town reservoir fed by two independent conduits discharging into the open basin: Pipe A, the main valley aqueduct, fills the empty reservoir alone in 100 minutes, whereas Pipe B, an auxiliary river pump, fills the same reservoir alone in 220 minutes. When the valves on both Pipe A and Pipe B are opened simultaneously, the two conduits work in parallel to fill the basin.

A common algebraic mistake is to average the two elapsed times ($(100 + 220)/2 = 160$ minutes) or add them (320 minutes). Physical reasoning rules out both results immediately *(Strogatz, 2012; Ellenberg, 2014)*: if Pipe A alone fills the reservoir in 100 minutes, opening a second pipe alongside it cannot make the filling take longer than 100 minutes. Two conduits operating together must finish in less time than the faster single conduit ($T_{\text{together}} < 100$ minutes), because parallel conduits add their *volumetric flow rates* rather than their

elapsed times.

Examining the flow rate per minute resolves the relationship directly. Because Pipe A fills 1 reservoir in 100 minutes, in a single minute it fills a fraction of the basin equal to its reciprocal, $R_A = 1/100 = 0.0100$ reservoirs per minute. Because Pipe B fills 1 reservoir in 220 minutes, in a single minute it contributes $R_B = (1/220) \approx 0.004545$ reservoirs per minute. With both valves open simultaneously, the per-minute flow rates sum directly to $R_{\text{total}} = R_A + R_B = 1/100 + (1/220) \approx 0.014545$ reservoirs per minute, and the total time T_{together} required to fill 1 whole reservoir is the reciprocal of that combined rate:

$$T_{\text{together}} = \frac{1}{\frac{1}{100} + \frac{1}{220}} = \frac{100 \times 220}{100 + 220} = \frac{22{,}000}{320} = \mathbf{68.75}\ \text{minutes} \qquad \textbf{(Eq. 2.3)}$$

When the town opens two additional backup conduits—Pipe C, requiring 470 minutes alone, and Pipe D, requiring 1,000 minutes alone—those four time constants (100, 220, 470, and 1,000) obey the exact same harmonic reciprocal-sum law as 100 Ω, 220 Ω, and 470 Ω electrical resistors wired in parallel. Each additional pipe contributes its reciprocal rate, 1/470 and 1/1000, which shortens the combined filling time from 100 minutes for Pipe A alone to 68.75 minutes for Pipes A and B, 59.98 minutes with Pipe C added, and **56.58 minutes** with all four pipes open:

$$T_{\text{all four}} = \left(\frac{1}{100} + \frac{1}{220} + \frac{1}{470} + \frac{1}{1000}\right)^{-1} = 56.5831\ \text{minutes} \qquad \textbf{(Eq. 2.4)}$$

Every harmonic system follows this three-step sequence—invert into rates, sum the rates, and invert back into the net quantity—or folds each additional branch into the running total with the associative product-over-sum operator $T_A\ T_B\ /\ (T_A + T_B)$.

2.3 Continued Fractions, Zu Chongzhi's Rational Pi, and Viète's Nested Radicals

Alternating the reciprocal operator $(1/x)$ with integer addition generates a fundamental structure in number theory known as a *continued fraction*. When Christiaan Huygens designed the brass gear trains of his astronomical planetarium in 1682 to match the orbital periods of Saturn and Earth, he expressed the target ratio as a four-tier continued fraction:

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3}}} \qquad \textbf{(Eq. 2.5)}$$

When read from the top down, a continued fraction presents a deeply nested hierarchy in which each denominator contains another rational expression. Keying that expression from the top down into an algebraic calculator (`a0 + 1 / ( a1 + 1 / ( a2 + 1 / a3 ) )`) requires opening a new parenthesis level at every tier and closing all of them at the end.

Evaluating that same structure from the innermost denominator upward eliminates every parenthesis. The calculation begins at the deepest partial denominator a_3 and inverts it in place into $r_3 = 1/a_3$. Adding the next tier's integer a_2 forms the intermediate denominator $d_2 = a_2 + r_3$, and inverting that sum in place ($r_2 = 1/d_2$) before adding a_1 yields $d_1 = a_1 + r_2$. One final inversion ($r_1 = 1/d_1$) followed by addition of the leading integer a_0 produces the convergent $x = a_0 + r_1$. No two branches ever compete for attention simultaneously; every stage alternates between a unary reciprocal and an addition in Register X.

2.3.1 Zu Chongzhi's Rational Approximation π ≈ 355/113

Mathematicians from ancient Greece and fifth-century China to Christiaan Huygens and Srinivasa Ramanujan prized continued fractions because they generate the best possible rational approximations to any irrational number.

Applying Euclidean reciprocal extraction to π = 3.14159265358979... exposes each convergent in succession. Subtracting the integer part 3 isolates the fractional remainder 0.14159265..., whose reciprocal (1 / 0.14159265...) is 7.0625133.... Because 7.0625 lies close to the integer 7, truncating the expansion at the first tier yields Archimedes' classical approximation:

$$\pi \approx 3 + \frac{1}{7} = \frac{22}{7} \approx 3.142857 \quad \text{(accurate to 2 decimal places)} \tag{Eq. 2.6}$$

Subtracting 7 from 7.0625133... isolates the next remainder 0.0625133..., and inverting it (1 / 0.0625133...) produces 15.996594..., which lies within 0.0035 of the integer 16 (or 15 + 1/1). Truncating immediately before the large subsequent partial denominator (292) produces a four-tier continued fraction:

$$\pi \approx 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1}}} = 3 + \cfrac{1}{7 + \cfrac{1}{16}} = 3 + \frac{16}{113} = \mathbf{\frac{355}{113}} \approx \mathbf{3.14159292} \tag{Eq. 2.7}$$

The rational approximation 355/113 is easily recalled by writing the first three odd integers twice (**`113355`**) and dividing the second triplet by the first (**`355 / 113`**). Using only three-digit integers, 355/113 matches π to six decimal places, corresponding to an error of less than one-millionth of an inch per foot. Discovered in AD 480 by the Chinese mathematician and astronomer Zu Chongzhi (who called it the *Milü*, or close ratio), this convergent remained the most accurate rational approximation to π known anywhere in the world for more than a millennium.

2.3.2 The Golden Ratio φ and Fibonacci Phyllotaxis

Because large partial denominators like 292 inside a continued fraction indicate that an irrational number can be closely approximated by rationals, the irrational number that is hardest to approximate with fractions is the one whose partial denominators are as small as possible. That extremal number is the continued fraction whose partial denominators equal **`1`** at every tier:

$$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}} \tag{Eq. 2.8}$$

In that infinite expansion of 1s, the sub-fraction beneath the uppermost division bar is identical to the entire continued fraction φ itself, satisfying the quadratic self-similarity equation:

$$\varphi = 1 + \frac{1}{\varphi} \Rightarrow \varphi^2 - \varphi - 1 = 0 \Rightarrow \varphi = \frac{1 + \sqrt{5}}{2} \approx \mathbf{1.6180339887} \tag{Eq. 2.9}$$

Evaluating this expansion from the bottom up by alternating **`1/x`** and **`+ 1`** generates the sequence of rational convergents step by step: the first stage gives 1 + 1/1 = 2/1 = 2.0000, the second gives 1 + 1/2 = 3/2 = 1.5000, the third gives 1 + 2/3 = (5/3) ≈ 1.6667, the fourth gives 1 + 3/5 = 8/5 = 1.6000, and subsequent iterations trace 13/8 = 1.6250 → (21/13) → (34/21) → (55/34) → (89/55) ≈ **1.618182**. Every convergent is the ratio of two consecutive *Fibonacci numbers* (1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89...).

This number-theoretic extremal property governs botanical phyllotaxis. Across the center of a sunflower seed head (*capitulum*), a daisy, or a pine cone, the florets arrange themselves into two intersecting families of opposing spirals (*parastichies*), typically 34 spirals curving clockwise and 55 curving counterclockwise (or 55 and 89 on larger heads). As each new seed bud (*primordium*) emerges at the central growing tip (*meristem apex*), it turns away from its predecessor by the golden angle ($360° / \varphi^2 \approx 137.5°$). Were the divergence angle a simple fraction of a circle such as 1/4 or 22/7, the seeds would line up along 4 or 7 radial spokes separated by wasted gaps. Because φ has only 1s in its continued-fraction expansion, successive buds never align along radial spokes, packing the growing disk at uniform density.

2.3.3 François Viète's 1593 Nested Square-Root Cascade

In 1593, the French mathematician and royal cryptographer François Viète published the first exact analytical formula for π in European mathematics. Viète expressed the area of a circle through an infinite product of nested square roots. Extending Archimedes' method of exhaustion, he inscribed a square (4 sides) inside a unit circle of radius $r = 1$ and repeatedly doubled the number of polygon sides to an octagon (8 sides), a 16-gon, a 32-gon, and a 64-gon.

Each bisection of a central angle ϑ applies the half-angle cosine identity, enclosing the previous cosine expression inside an outer radical:

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\cos\theta}{2}} = \frac{\sqrt{2+2\cos\theta}}{2} \qquad \textbf{(Eq. 2.10)}$$

Starting at a right angle $\vartheta_1 = 90°$ ($\pi/2$ radians, where $\cos 90° = 0$), successive bisections build a tower of nested radicals: the first bisection at 45° (an octagon) gives $2\cos(45°) = \sqrt{(2 + 2\cos(90°))} = \sqrt{2}$; the second bisection at 22.5° (a 16-gon) gives $2\cos(22.5°) = \sqrt{(2 + \sqrt{2})}$; the third bisection at 11.25° (a 32-gon) gives $2\cos(11.25°) = \sqrt{(2 + \sqrt{(2 + \sqrt{2})})}$; and the fourth bisection at 5.625° (a 64-gon) reaches $2\cos(5.625°) = \sqrt{(2 + \sqrt{(2 + \sqrt{(2 + \sqrt{2})})})}$.

Comparing the area of each inscribed 2^n-gon to the area of the circumscribed circle, Viète proved that the ratio $2/\pi$ is the infinite product of those nested cosine factors:

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2+\sqrt{2}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \cdot \frac{\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}}}{2} \cdots \qquad \textbf{(Eq. 2.11)}$$

Like continued fractions, nested radical expressions—whether Viète's polygon product, a radical quotient such as $\sqrt{((128.5 + \sqrt{(48.2)}) / (19.4 - \sqrt{(7.6)}))}$, or a four-tier continued radical $\sqrt{(3 + 5 / \sqrt{(7 + 11 / \sqrt{(13 + 17)})})}$—require deep parenthesis nesting when written from the outside in, yet evaluate directly in Register *X* when executed from the innermost radicand outward *(Cochran, 1972)*.

2.4 Top-of-Stack Replication in Register T and Named Memory

At the top of the four-register stack, Register *T* (Line 4) implements a retention rule that governs what happens when two operands in Registers *Y* and *X* combine and the upper registers drop by one level.

2.4.1 T-Register Duplication and Iterative Constants

If Register T cleared to zero every time the stack dropped, three consecutive dyadic operations would leave the upper registers empty. Instead, the architecture of the classic HP-32SII and HP-15C gives Register T a structural property called *T-Register Duplication*:

$$X_{new} \leftarrow Y_{old}\ [\text{op}]\ X_{old},\ Y_{new} \leftarrow Z_{old},\ Z_{new} \leftarrow T_{old},\ T_{new} \leftarrow T_{old} \quad \text{(Copied downward while remaining in } T\text{)} \qquad \textbf{(Eq. 2.12)}$$

Whenever a dyadic operation drops the stack, Register T copies its value downward into Register Z while retaining its own value in T. That single hardware rule supports three standard numerical idioms. First, in one-touch compound growth, filling all four registers with a growth factor $(1 + r)$ and keying the initial principal P_0 into Register X allows every tap of [×] to advance the balance by one period ($P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow ...$), because Register T feeds an endless stream of $(1 + r)$ multipliers down into Register Y.

Second, in repeated blueprint scaling, loading a unit conversion factor k across the upper registers makes it possible to convert an entire table of machinist dimensions by keying each measurement, multiplying, and rolling the preserved constant k back into position ([R↓]) for the next entry. Third, in fixed-point continued-fraction iteration, filling all four registers with a constant integer a turns Register T into a steady source of a's for continued-fraction expansions $(a + 1/(a + 1/...))$, including the Fibonacci convergents of the Golden Ratio φ.

2.4.2 Register Permutations, LASTX Recovery, and Storage Arithmetic

To reorder operands and manage intermediate constants without re-entry, the four stack registers are paired with in-place permutation keys, an automatic **LASTX** shadow register, and 26 lettered storage registers (**A**–**Z**) with indirect addressing.

The permutation keys [x⇄y], [R↓], and [↗] [R↑] rearrange values already on the stack. Pressing [x⇄y] exchanges Registers Y and X whenever a denominator or subtrahend is evaluated before its numerator or minuend: if a complicated denominator D is computed first in Register X and a simple numerator N is keyed next (lifting D into Register Y), pressing [÷] immediately would evaluate $D \div N$ rather than $N \div D$, whereas a single tap of [x⇄y] places $Y = N$ and $X = D$ so [÷] computes $N \div D$. The same swap restores operand order when a complicated subtrahend B is evaluated before A in A - B. Meanwhile, [R↓] (Roll Down) and [↗] [R↑] (Roll Up) cycle all four registers in a closed ring ($X \leftarrow Y \leftarrow Z \leftarrow T \leftarrow X$) without losing a single digit:

$$\text{Roll Down (R}\downarrow\text{)}: (X, Y, Z, T) \leftarrow (Y, Z, T, X), \qquad \text{Roll Up (R}\uparrow\text{)}: (X, Y, Z, T) \leftarrow (T, X, Y, Z) \qquad \textbf{(Eq. 2.13)}$$

Because four consecutive rolls return every register to its original position, an engineer can load four thermocouple readings onto the stack, cycle through them with [R↓] to inspect each value at full precision, and then press [+] [+] [+] **4** [÷] to compute their four-point ensemble mean.

Alongside the four visible registers, every operation that modifies Register X automatically saves the previous occupant of Register X inside the shadow register [↶] [LASTx]. If an unintended operator or function key is pressed, [↶] [LASTx] recovers the input operand immediately. Whenever a formula uses the same variable x or a shared divisor twice—as in the optical diffraction sinc function $\sin \vartheta/\vartheta$, the symmetric rational function $(x^2 - 1) / (x^2 + 1)$, the quadratic roots $(-b \pm \sqrt{\Delta}) / 2a$, or Arthur Kennelly's 1899 three-phase Delta-to-Wye ($\Delta \rightarrow$ Y) impedance transformation—[↶] [LASTx] acts as an automatic fifth register.

Kennelly's $\Delta \rightarrow$ Y transformation illustrates how [↰] [LASTx] eliminates redundant summation: when three resistors R_{12}, R_{23}, R_{31} are wired in a closed delta (Δ) inside a Wheatstone bridge or a three-phase industrial motor starter, neither series nor parallel rules apply in isolation. Converting the delta into an equivalent three-branch star (Y) requires dividing each adjacent pair product by the shared perimeter sum $S = R_{12} + R_{23} + R_{31}$:

$$R_a = \frac{R_{12}R_{31}}{R_{12}+R_{23}+R_{31}}, \qquad R_b = \frac{R_{12}R_{23}}{R_{12}+R_{23}+R_{31}}, \qquad R_c = \frac{R_{23}R_{31}}{R_{12}+R_{23}+R_{31}} \tag{Eq. 2.14}$$

Because any division $N \div S$ automatically captures its divisor S inside **LASTX**, evaluating the first wye branch R_a latches the perimeter sum S inside **LASTX** as a byproduct of the division, so R_b and R_c can be evaluated next by recalling [↰] [LASTx] instead of re-summing the three resistors.

When all four stack registers (T, Z, Y, X) hold live subtotals and Register X must be combined with a constant stored in one of the 26 lettered variables **A**–**Z**, a standard [RCL] **A** would lift the stack by one level and push the occupant of Register T off the top in **4-LVL** mode. To preserve all four stack levels, the SC-32 provides two-way in-place storage arithmetic that bypasses stack lift altogether. *Recall Arithmetic* ([RCL] followed by **+**, **-**, **×**, or **÷** and **A**) updates Register X in place as $X \leftarrow X$ op A while leaving Registers Y, Z, and T untouched. Conversely, *Store Arithmetic* ([STO] followed by **+**, **-**, **×**, or **÷** and **A**) updates Variable **A** in place as $A \leftarrow A$ op X.

In 1949, Tom Kilburn, Geoff Tootill, and Max Newman at the University of Manchester introduced the B-line index register on the Manchester Mark 1 and showed that instead of hard-wiring a fixed memory address (**A**, **B**, **C**) into an instruction sequence, a processor can hold an integer address k inside a dedicated *index register (***i***)* and use *indirect addressing (***(i)***)* to reference whichever memory slot resides at address k.

On the SC-32, when [STO] or [RCL] activates the **A–Z** Alpha layer, the decimal point key [•] carries the Alpha label **i** (the index register that holds a "room number" 1 to 89) and the neighboring key [$x \| y$] carries the Alpha label **(i)** (the indirect pointer that opens whichever door **i** currently points to, raising **INVALID DATA** on non-integer or out-of-bounds addresses). Storing 2 into **i** sets the pointer to slot 2 (**B**), so [RCL] **(i)** recalls Variable **B**, [STO] [+] **(i)** adds Register X into Variable **B**, and incrementing **i** by 1 steps the pointer across all 89 slots of the calculator's unified memory map:

Pointer Range $k \in (1, \ldots, 89)$	Target Memory Bank	Addressed Register / Content	Access Permissions
1 ... 26	**Lettered Variables (A–Z)**	**1** = **A**, **2** = **B**, **3** = **C**, ..., **26** = **Z**	Full Read / Write (**STO (i)**, **RCL (i)**, **x↔? (i)**)
27	**Index Register (i)**	**27** = Index register **i** itself	Full Read / Write
28 ... 33	**Statistical Accumulators**	**28** = $\sum n$, **29** = $\sum x$, **30** = $\sum y$, **31** = $\sum x^2$, **32** = $\sum y^2$, **33** = $\sum xy$	Read / Write via **(i)**
34 ... 39	**POLY Solver Roots**	**R34**–**R39** (Real & imaginary parts of roots r_1, r_2, r_3)	Read via **RCL (i)**
40 ... 89	**Statistical Data Points**	**DATA0** (**40**) through **DATA49** (**89**)	Read via **RCL (i)**
$k < 1$ **or** $k > 89$	**Out of Bounds**	Displays **INVALID DATA** (Code 02)	N/A (Halts without stack corruption)

Finally, the division key [÷] carries two six-operator relational test menus: left-shifted [↰] [x?y] compares Register X against Register Y (**x=y**, **x≠y**, **x>y**, **x<y**, **x≥y**, **x≤y**), while

right-shifted [→] [x?0] compares Register X against zero (**x=0**, **x≠0**, **x>0**, **x<0**, **x≥0**, **x≤0**). On the live stack, any test flashes **YES** or **NO** non-destructively; inside the Equation Editor [→] [EQN], each test acts as Kenneth Iverson's Boolean indicator bracket $[P] \in (1, 0)$—a digital valve that evaluates to **1.0** when true and **0.0** when false to switch piecewise formula branches on or off. Internally, the SC-32 computes in 64-bit IEEE-754 double precision (53 significand bits $\approx$ 15.95 decimal digits) with a 12-digit guard-rounded display across [←] [DISP] (**ALL**, **FIX**, **SCI**, **ENG**) and 15-digit inspection via [→] [SHOW].

2.4.3 Selectable Stack Depth: Classic 4-LVL, Extended 8-LVL & Unbounded INF

In classic RPN calculator design, two distinct stack architectures emerged: the 1972 Corvallis four-register stack (**HP-35**, **HP-15C**, **HP-32SII**), where a fixed ceiling at Register T duplicates downward on every dyadic drop to serve as an automatic constant source (**k ENTER ENTER ENTER**), and the 1987 RPL unbounded stack (**HP-28C**, **HP-48SX**), where an expandable Last-In, First-Out (**LIFO**) heap holds arbitrarily many subtotals without dropping upper values off the ceiling or duplicating the top register on drop. Through the [→] [FLAGS] menu (**STACK** → **4-LVL**, **8-LVL**, or **INF**), the SC-32 supports both paradigms alongside an intermediate 8-register mode, and pressing [←] [REGS] (**Left-Shift** [0]) inspects registers **R5**–**R8** or deeper **INF** stack levels above the four-line T, Z, Y, X viewport:

Mode ID	Architecture Name	Accessible Registers	Stack Drop Behavior	Recommended Application Zone
4-LVL	**Classic 4-Level Stack (Default)**	X, Y, Z, T	Top-of-stack T-duplication on drop; full HP-32SII / HP-15C parity	Everyday engineering, Horner recurrences, constant scaling
8-LVL	**Extended 8-Level Stack**	R_1 ... R_8 (with R_1–R_4 mapped to X, Y, Z, T)	Holds up to 8 real operands (or 4 complex pairs) before R_8 replication	Multi-branch complex phasor algebra, 5-to-8-register expressions
INF	**Unbounded Infinite Stack**	Dynamic heap allocation (up to 999 levels)	True LIFO stack; dyadic drop never duplicates; pulls from history	5+ branch parallel networks, long product-sum series

2.5 Four-Register Stack Mechanics & Memory Architecture

Two transition rules govern every calculation on the stack: pressing [ENTER] sets stack lift to neutral so the next number keyed overwrites Register X, while any dyadic operator ([+], [-], [×], [÷]) drops the upper registers by one level, replicates Register T downward, and re-arms stack lift.



Figure: Real-time 4-register operational stack (T, Z, Y, X) displaying intermediate calculation states without deferred parentheses.

2.5.1 Two-Branch Rational Fractions & Combined Reservoir Flow Rates

1. **Weighted Span Rational Fraction (((45 × 3) + (18 × 4)) / (15 - 3)):** Tracing every individual key press on its own row illustrates how [ENTER] disables stack lift so **3**, **4**, and **3** overwrite Register X in place, how [×] and [-] drop the upper stack and re-arm stack lift, and how keying **18** and **15** automatically lifts completed subtotals into Register Y by position alone:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
45	0.0000	0.0000	0.0000	45.0000	Enter numerator term 1
[ENTER]	0.0000	0.0000	45.0000	45.0000	Stack lift disabled; X copied to Y
3	0.0000	0.0000	45.0000	3.0000	Overwrites X directly
[×]	0.0000	0.0000	0.0000	135.0000	45 × 3 = 135; Stack lift armed
18	0.0000	0.0000	135.0000	18.0000	Stack lifts; 135 preserved in Y
[ENTER]	0.0000	135.0000	18.0000	18.0000	Stack lift disabled; 135 pushed to Z
4	0.0000	135.0000	18.0000	4.0000	Overwrites X directly
[×]	0.0000	0.0000	135.0000	72.0000	18 × 4 = 72; Z drops to Y
[+]	0.0000	0.0000	0.0000	207.0000	135 + 72 = 207 (Numerator complete)
15	0.0000	0.0000	207.0000	15.0000	Stack lifts; 207 preserved in Y
[ENTER]	0.0000	207.0000	15.0000	15.0000	Stack lift disabled; 207 pushed to Z
3	0.0000	207.0000	15.0000	3.0000	Overwrites X directly
[-]	0.0000	0.0000	207.0000	12.0000	15 - 3 = 12; Z drops to Y
[÷]	0.0000	0.0000	0.0000	**17.2500**	207 ÷ 12 = 17.25 (Final result)

2. **Two-Pass Heat Exchanger Thermal Balance (U = ((18.5 × 4.2) + (7.1 × 9.6)) / (14.3 - 5.8)):** For a two-pass heat-exchanger balance $U = ((18.5 \times 4.2) + (7.1 \times 9.6)) / (14.3 - 5.8)$, grouping each logical phrase (**18.5 ENTER 4.2 ×**) onto a single row highlights the physical subtotals across Registers Z, Y, and X. Executing **18.5** [ENTER] **4.2** [×] **7.1** [ENTER] **9.6** [×] [+] **14.3** [ENTER] **5.8** [-] [÷] resolves the overall heat transfer coefficient to **17.16 W/m²K**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
18.5 [ENTER] **4.2** [×]	0.0000	0.0000	0.0000	77.7000	First numerator term (18.5 × 4.2)
7.1 [ENTER] **9.6** [×]	0.0000	0.0000	77.7000	68.1600	Second numerator term; 77.7 held in Y
[+]	0.0000	0.0000	0.0000	145.8600	Total numerator in X (77.7 + 68.16)
14.3 [ENTER] **5.8** [-]	0.0000	0.0000	145.8600	8.5000	Denominator (14.3 - 5.8 = 8.5) in X
[÷]	0.0000	0.0000	0.0000	**17.1600**	Final heat transfer coefficient

3. **Combined Reservoir Pipe Flow Rates (T = 100 ∥ 220 ∥ 470 ∥ 1000 min):** Compare the two-branch thermal fraction, which required Register Z to hold the finished numerator while the denominator formed, with the four-aqueduct town reservoir network (100, 220, 470, and 1,000 min). Because the parallel-harmonic operator [x∥y] is associative, all four pipe times never need to occupy T, Z, Y, and X simultaneously.

Keying the initial pair **100** [ENTER] **220** [x∥y] collapses 100 min in Register Y and 220 min in Register X directly into their two-pipe harmonic time (68.7500 min) in Register X,

returning Register Y to zero and arming stack lift. Opening Pipe C (**470** [x⇌y] → 59.9768 min) and Pipe D (**1000** [x⇌y]) in the full sequence **100** [ENTER] **220** [x⇌y] **470** [x⇌y] **1000** [x⇌y] then folds each additional aqueduct into that running subtotal in Register X (**68.7500** → **59.9768** → **56.583123564 minutes**) while Registers Z and T remain at zero from start to finish.

2.5.2 Register T Constant Replication, Cyclic Roll, and LASTX Recovery

1. **Five-Year Compounding via Register T Duplication ($A_n = 2500 \times 1.065^n$):** Priming all four levels with the growth factor 1.065 and tapping [×] five times via **1.065** [ENTER] [ENTER] [ENTER] **2500** [×] [×] [×] [×] [×] compounds \$2,500 to **3,425.2167** at Year 5:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1.065 [ENTER]	0.0000	0.0000	1.0650	1.0650	Load annual compounding factor $1 + r = 1.065$
[ENTER] [ENTER]	1.0650	1.0650	1.0650	1.0650	Replicate multiplier across all 4 stack levels
2500 [×]	1.0650	1.0650	1.0650	2,662.5000	Enter principal $P = \$2500$; Year 1 balance in X
[×]	1.0650	1.0650	1.0650	2,835.5625	T duplicates down; Year 2 balance in X
[×]	1.0650	1.0650	1.0650	3,019.8741	Year 3 balance compounded via T-replication
[×]	1.0650	1.0650	1.0650	3,216.1659	Year 4 balance compounded
[×]	1.0650	1.0650	1.0650	**3,425.2167**	Year 5 portfolio balance without re-entry

2. **Blueprint Inch-to-Millimeter Conversion Table ($x_{\text{mm}} = 25.4\ x_{\text{in}}$):** Flooding the stack with **25.4** locks the conversion factor in Register T:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
25.4 [ENTER]	0.0000	0.0000	25.4000	25.4000	Enter conversion constant 25.4 mm/in
[ENTER] [ENTER]	25.4000	25.4000	25.4000	25.4000	Flood entire stack with 25.4 conversion factor
0.375 [×]	25.4000	25.4000	25.4000	9.5250	Convert 0.375 in → 9.5250 mm; T replicates
[R↓] **1.125** [×]	25.4000	25.4000	25.4000	28.5750	Roll down 25.4 into place; convert 1.125 in → 28.5750 mm
[R↓] **2.875** [×]	25.4000	25.4000	25.4000	73.0250	Roll down 25.4 into place; convert 2.875 in → 73.0250 mm
[R↓] **5.500** [×]	25.4000	25.4000	25.4000	**139.7000**	Roll down 25.4 into place; convert 5.500 in → 139.7000 mm

3. **Cyclic 4-Register Carousel ([R↓]):** When logging four thermocouple channels along a heat-exchanger manifold ($T_1 = 21.5$°C, $T_2 = 24.8$°C, $T_3 = 28.2$°C, $T_4 = 31.0$°C) with all four temperatures loaded onto the stack ($T = 21.5$, $Z = 24.8$, $Y = 28.2$, $X = 31.0$°C), pressing [R↓] four consecutive times steps through every sensor channel in Register X and restores the stack to its exact starting configuration:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
21.5 ENTER **24.8** ENTER	0.0000	21.5000	24.8000	24.8000	Load sensor channels T_1 and T_2 into stack
28.2 ENTER **31.0**	21.5000	24.8000	28.2000	31.0000	Load channels T_3 and T_4; all 4 channels primed
R↓	31.0000	21.5000	24.8000	28.2000	1st roll down: channel $T_3 = 28.2°C$ in X
R↓	28.2000	31.0000	21.5000	24.8000	2nd roll down: channel $T_2 = 24.8°C$ in X
R↓	24.8000	28.2000	31.0000	21.5000	3rd roll down: channel $T_1 = 21.5°C$ in X
R↓	21.5000	24.8000	28.2000	**31.0000**	4th roll down: returns full cycle to $T_4 = 31.0°C$

4. **Optical Diffraction Sinc Function via ↶ LASTx ($\text{sinc}\vartheta = \sin\vartheta/\vartheta$):** An expression like $\text{sinc}\ x = \sin\ x\ /\ x$, in which the independent variable x appears in both the numerator and the denominator, never requires ENTER or the upper stack registers because ↶ LASTx acts as an automatic fifth shadow register. In radian mode (RAD), entering 0.42 into Register X and pressing SIN replaces 0.420000 in Register X with $\sin(0.42) = 0.407760$ while automatically preserving the pre-operator argument 0.420000 inside the shadow register.

Recalling that saved argument lifts 0.407760 into Register Y and restores 0.420000 into Register X, so a single division resolves $\sin(0.42)/0.42 =$ **0.970858** while Registers Z and T stay empty. Applying that same four-stroke sequence to Fraunhofer single-slit diffraction at $\vartheta = 0.085$ radians via RAD **0.085** SIN ↶ LASTx ÷ resolves the central diffraction lobe amplitude to **0.9987962683** without keying the angle twice (switch back to DEG afterward).

5. **Three-Phase Delta-to-Wye ($\Delta \rightarrow$ Y) Transformation ($R_a = R_{12}R_{31}\ /\ (R_{12}+R_{23}+R_{31})$):** That same shadow-register behavior applies directly to dyadic subtraction and division, where ↶ LASTx captures the *second operand* (X_{old}) and supports relative-change calculations (Y - X) / X via subtraction followed immediately by division by ↶ LASTx, or reuse of a shared denominator across multiple branches. In Arthur Kennelly's three-phase $\Delta \rightarrow$ Y transformation for a closed Δ loop with $R_{12} =$ 30 Ω, $R_{23} =$ 45 Ω, $R_{31} =$ 60 Ω, all three equivalent star branches share the identical perimeter denominator $S = R_{12} + R_{23} + R_{31} =$ 135 Ω.

Evaluating **30** ENTER **60** × **30** ENTER **45** + **60** + ÷ computes the first star branch R_a while priming ↶ LASTx for the remaining two branches: first multiplying $R_{12}R_{31} = 30 \times 60 =$ 1,800 Ω^2 in Register X, then accumulating the perimeter sum $S = 30 + 45 + 60 =$ 135 Ω in Register X while 1,800 Ω^2 shifts up to Register Z and drops back to Register Y, and finally dividing to resolve $R_a =$ 1,800 / 135 = **13.3333 Ω** in Register X while automatically latching the shared perimeter divisor 135.0000 Ω inside ↶ LASTx.

The sum 30 + 45 + 60 never needs to be keyed again: recalling 135 Ω from ↶ LASTx *before* forming each remaining pair product ($R_{12}R_{23} =$ 1,350 Ω^2 and $R_{23}R_{31} =$ 2,700 Ω^2, since × overwrites **LASTX** whereas the subsequent ÷ re-latches 135 Ω back into **LASTX**) and swapping via x⇄y for division aligns all three equivalent wye impedances ($Z =$ 13.3333 Ω, $Y =$ 10.0000 Ω, $X =$ 20.0000 Ω) on the stack.

Twelve-digit base-10 guard rounding likewise suppresses repeating-binary residuals on the display: whereas raw 64-bit binary floating-point arithmetic evaluates 0.1 + 0.2 as

0.3000000000000000444..., executing **0.1** ENTER **0.2** + displays exact **0.3** on screen while holding ↷ SHOW exposes the full 15-digit internal significand whenever numerical precision is being audited.

2.5.3 Continued Fractions, Nested Radicals & Recall Arithmetic

1. **Symmetric Rational Function ($f(x) = (x^2 - 1) / (x^2 + 1)$ at $x = 3.75$):** Computing $x^2 = 14.0625$ once and duplicating it across Z and Y via **3.75** ↶ x² ENTER ENTER **1** - x⇄y **1** + ÷ produces **0.867219917**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3.75 ↶ x²	0.0000	0.0000	0.0000	14.0625	Compute $x^2 = 14.0625$
ENTER ENTER	0.0000	14.0625	14.0625	14.0625	Duplicate x^2 into Y and Z
1 -	0.0000	0.0000	14.0625	13.0625	Numerator x^2 - 1; 14.0625 drops to Y
x⇄y	0.0000	0.0000	13.0625	14.0625	Swap x^2 into X, numerator into Y
1 +	0.0000	0.0000	13.0625	15.0625	Denominator $x^2 + 1 = 15.0625$
÷	0.0000	0.0000	0.0000	**0.867220**	Final quotient 13.0625 / 15.0625

2. **Evaluating a Four-Level Continued Fraction ($x = 1 + 1 / (2 + 1 / (3 + 1/4))$), Zu Chongzhi's $\pi \approx 355/113$, and the Golden Ratio φ:** Evaluating the four-level continued fraction $1 + 1 / (2 + 1 / (3 + 1/4))$ confirms that Registers Z and T remain empty from start to finish: the unary reciprocal key x^{-1} inverts each completed denominator in place inside Register X, and adding the next tier's integer immediately collapses Y and X back into a single number in Register X.

Starting at the innermost denominator, keying **4** x^{-1} **3** + inverts 4 into $1/4 = 0.2500$ in Register X and adds 3 to form the intermediate denominator 3.2500 (13/4) with Register Y cleared to zero. Continuing upward in the full sequence **4** x^{-1} **3** + x^{-1} **2** + x^{-1} **1** + alternates x^{-1} and + directly in Register X—inverting 13/4 into 4/13 ≈ 0.3077, adding 2 to reach 30/13 ≈ 2.3077, inverting to 13/30 ≈ 0.4333, and adding 1 to land on **1.4333333333** (43/30) while Registers Y, Z, and T remain at zero.

Applying this identical inside-out cadence to Archimedes' $\pi \approx 3 + 1/7 = 22/7$ or Zu Chongzhi's *Milü* $\pi \approx 3 + 1 / (7 + 1 / (15 + 1/1))$ (starting at the deepest 1, inverting and adding 15, inverting and adding 7, and inverting and adding 3) yields **3.14159292035**, where toggling fraction display mode ↶ FDISP reveals the exact reduced rational **3 16/113** (355/113). Likewise, filling all four stack registers with 1 via ENTER uses Register T's constant replication to supply an endless stream of 1's, so repeating the two-keystroke sequence x^{-1} + steps through the Fibonacci convergents (2.0000 → 1.5000 → 1.6667 → 1.6000 → 1.6250 → 1.6154 → ...) to the Golden Ratio $\varphi = $ **1.61803398875**.

3. **Nested Radical Quotient ($x = \sqrt{(128.5 + \sqrt{48.2})} / (19.4 - \sqrt{7.6})$):** When an expression branches into an independent radical numerator and radical denominator, Registers Z, Y, and X coordinate across three levels. Evaluating the numerator first via **48.2** √x **128.5** + (→ 135.44262) and then assembling the denominator via **19.4** ENTER **7.6** √x - (→ 16.64319) pushes the completed numerator up to Register Z while 19.4 and $\sqrt{7.6} \approx 2.7568$ occupy Y and X, before - drops 135.4426 back to Register Y. Chaining both branches and extracting the outer square root in **48.2** √x **128.5** + **19.4** ENTER **7.6** √x - ÷ √x then yields **2.8527215801**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
48.2 [√x] **128.5** [+]	0.0000	0.0000	0.0000	135.4426	Numerator: √(48.2) + 128.5 ≈ 135.4426
19.4 [ENTER] **7.6** [√x]	0.0000	135.4426	19.4000	2.7568	Start denominator: enter 19.4; compute √(7.6)
[-]	0.0000	0.0000	135.4426	16.6432	Denominator: 19.4 - √(7.6) ≈ 16.6432
[÷]	0.0000	0.0000	0.0000	8.1380	Divide numerator by denominator; radicand ratio
[√x]	0.0000	0.0000	0.0000	**2.8527**	Extract outer square root; final radical quotient

4. **Four-Level Continued Radical ($x = \sqrt{3 + 5 / \sqrt{7 + 11 / \sqrt{13 + 17}}}$):** Contrast that two-branch radical fraction with a four-level continued radical that mixes square roots and fraction bars at every tier. Working outward from √(13 + 17) via **13** [ENTER] **17** [+] [√x] **11** [x⇄y] [÷] **7** [+] [√x] **5** [x⇄y] [÷] **3** [+] [√x] never touches Registers Z or T because each inner radical becomes the *divisor* of the tier immediately above it, and swapping via [x⇄y] before division collapses each tier in place.

Specifically, summing 13 + 17 and pressing [√x] extracts the deepest root √30 ≈ 5.4772 in Register X. Entering the numerator 11 next lifts 5.4772 into Register Y, where [x⇄y] and division place 11 / √30 ≈ 2.0083 back in Register X with Register Y cleared to zero. Adding 7 and extracting the root forms the middle radical √(9.0083) ≈ 3.0014 in place; dividing 5 by that root yields 1.6659; and adding 3 before one final square root extracts √(4.6659) ≈ **2.1601** (2.160069) while Registers Z and T remain at zero.

5. **Preserving a Full 4-Register Stack with Recall Arithmetic ([RCL] [×] A):** With **2.5** stored in variable **A** (**2.5** [STO] **A**) and all four stack registers (T, Z, Y, X) occupied by active subtotals (**10** [ENTER] **20** [ENTER] **30** [ENTER] **40**, placing $T = 10$, $Z = 20$, $Y = 30$, $X = 40$), a standard [RCL] **A** [×] would lift the stack ($T \leftarrow 20$, $Z \leftarrow 30$, $Y \leftarrow 40$, $X \leftarrow 2.5$), discard 10 from the top of Register T across successive operations, and leave $T = 20$, $Z = 20$, $Y = 30$, $X = 100$. Keying [RCL] [×] **A** instead scales Register X to $40 \times 2.5 = 100$ in place while preserving $T = 10$, $Z = 20$, $Y = 30$ intact:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
2.5 [STO] **A**	0.0000	0.0000	0.0000	2.5000	Store scale factor 2.5 in variable **A**; lift armed
10 [ENTER] **20** [ENTER]	2.5000	10.0000	20.0000	20.0000	Load first two subtotals (10 and 20) onto stack
30 [ENTER] **40**	10.0000	20.0000	30.0000	40.0000	All 4 registers (T, Z, Y, X) are now completely full
[RCL] [×] **A**	**10.0000**	**20.0000**	**30.0000**	**100.0000**	In-place $X \leftarrow 40 \times 2.5$ (**LASTX** = 40.0000, **A** = 2.5000); T = **10 is preserved**

6. **Indirect Register Addressing & Pointer Arithmetic via `i` and (i) ([STO] `i`, [RCL] (i), [STO] [+] (i)):** Store calibration offset 120.0 in Variable **A** at slot $k = 1$ and 250.0 in Variable **B** at slot $k = 2$. Loading $k = 1$ into index register **i** via **1** [STO] **i**—tapping [•] for Alpha **i**—lets [RCL] **(i)** on [x⇆y] dereference slot 1 to return 120.0. Incrementing the pointer in place with **1** [STO] [+] **i** advances **i** to slot 2, where **35** [STO] [+] **(i)** and [RCL] **(i)** update and recall Variable **B** as 250 + 35 = 285.0:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
120 STO **A**	0.0000	0.0000	0.0000	120.0000	Store 120.0 in slot 1 (**A**)
250 STO **B**	0.0000	0.0000	120.0000	250.0000	Store 250.0 in slot 2 (**B**)
1 STO **i**	0.0000	120.0000	250.0000	1.0000	Point index register **i** (•) to slot $k = 1$ (**A**)
RCL **(i)**	120.0000	250.0000	1.0000	120.0000	Indirect recall **RCL (i)** ($x \leftrightarrow y$) fetches **A** = 120.0
1 STO + **i**	250.0000	1.0000	120.0000	1.0000	Increment pointer **i** in place to $k = 2$ (**B**)
35 STO + **(i)**	1.0000	120.0000	1.0000	35.0000	Indirect store-add updates **B** ← 250 + 35 = 285
RCL **(i)**	120.0000	1.0000	35.0000	**285.0000**	Indirect recall fetches updated slot 2 (**B** = **285.0**)

2.5.4 Multi-Register Engineering Synthesis

Across scientific and engineering domains, algebraic formulas fall into three recurring stack-utilization patterns determined by the depth of their expression trees:

Inside-Out Radical & Reactance Scaling (Two Active Registers, Z and T Free)

Whenever an equation multiplies an outer physical scale by a dimensionless inner radical, starting inside the radical resolves the entire formula across Registers Y and X without touching Registers Z or T:

- **Hohmann Orbital Transfer ($\Delta v_1 = \sqrt{(\mu/r_1)}(\sqrt{(2r_2 \,/\, (r_1{+}r_2))} - 1)$):** To boost a spacecraft from low Earth parking orbit (r_1 = 6,678 km) to geostationary altitude (r_2 = 42,164 km, with gravitational parameter μ = 398,600 km^3/s^2), start inside the dimensionless radius ratio $2r_2 \,/\, (r_1{+}r_2)$ = 84,328/48,842 = 1.72655. Extracting $\sqrt{x}$ ($\rightarrow$ 1.31398) and subtracting 1 isolates the fractional velocity boost 0.31398 in Register X; forming the circular orbital speed $\sqrt{(\mu/r_1)}$ = 7.72584 km/s in Register X while 0.31398 waits in Register Y and multiplying delivers the perigee burn Δv_1:
  - **Perigee Velocity Impulse:** **2** ENTER **42164** × **6678** ENTER **42164** + ÷ $\sqrt{x}$ **1** − **398600** ENTER **6678** ÷ $\sqrt{x}$ × $\rightarrow \Delta v_1 =$ **2.425767684 km/s**.

- **First-Order High-Pass RC Filter Gain ($|H| = X \,/\, \sqrt{(1{+}X^2)}$, where $X = 2\pi f R C$):** For an audio network with R = 10,000 Ω and $C = 15 \times 10^{-9}$ F at f = 2,500 Hz, first form the dimensionless frequency ratio $X \approx$ 2.35619 in Register X and press ENTER once to copy 2.35619 into Register Y. Transforming Register X in place via ↶ x^2, **+ 1**, and $\sqrt{x}$ produces the denominator $\sqrt{(1 + X^2)} \approx$ 2.55962 directly below the waiting numerator in Y:
  - **High-Pass Gain Evaluation:** **2** ↷ π × **2500** × **10000** × **15** E **9** +/− × ENTER ↶ x^2 **1** + $\sqrt{x}$ ÷ $\rightarrow |H| =$ **0.9205252585**.

Two-Branch Difference Over Reciprocal or Series Sum (Registers Z, Y, X in Concert)

Whenever a physical driving difference ($\Delta(T^4)$ or ΔT) sits above a multi-term sum of thermal resistances, evaluate the numerator first so it shifts up to Register Z while the denominator terms combine in Registers Y and X:

- **Stefan-Boltzmann Radiative Heat Shield ($q = \sigma(T_1^4 - T_2^4) / (1 / \varepsilon_1 + 1 / \varepsilon_2 - 1)$):** Between spacecraft shields at $T_1 = 800$ K ($\varepsilon_1 = 0.85$) and $T_2 = 500$ K ($\varepsilon_2 = 0.70$, $\sigma = 5.67037 \times 10^{-8}$ W/m^{2}K^4), scaling $800^4 - 500^4 = 3.4710 \times 10^{11}$ K^4 by σ places radiant flux 19,681.8543 W/m^2 in Register X. Next, $[x^{-1}]$ inverts ε_1 (1.1765) and ε_2 (1.4286) in place while the numerator waits in Z; when their sum minus 1 collapses to 1.6050 in X, the numerator drops from Z to Y for division:
  - **Radiative Heat Flux: `800` [ENTER] `4` [y^x] `500` [ENTER] `4` [y^x] [-] `5.67037` [E] `8` [+/-] [×] `0.85` [x^{-1}] `0.70` [x^{-1}] [+] `1` [-] [÷] → q = 12,262.51653 W/m^2.**

- **Composite Series Wall Conduction ($q = \Delta T / (R_1 + R_2 + R_3)$):** Across an envelope supporting $\Delta T = 22.0$ K over air film ($R_1 = 0.08$), fiberglass ($R_2 = 2.20$), and sheathing ($R_3 = 0.15$ m^{2}K/W), hold $\Delta T = 22.0$ K in Register Y while summing $R_{\text{total}} = 2.4300$ m^{2}K/W in Register X, then divide:
  - **Conductive Heat Flux: `0.08` [ENTER] `2.20` [+] `0.15` [+] `22.0` [$x \rightleftarrows y$] [÷] → q = 9.0535 W/m^2.**

Single-Register Multiplicative & Additive Cascades (Register X Alone)

When a formula is a chain of factors, divisors, and in-place squares—or a running sum divided by a count—each step folds into Register X so Z and T remain at zero:

- **Darcy-Weisbach Pipe Head Loss ($h_f = f(L/D)v^2/2g$):** Pumping water at $v = 2.4$ m/s through an $L = 250$ m main of diameter $D = 0.30$ m ($f = 0.022$, $g = 9.80665$ m/s^2) forms $f\,L\,/\,D = 18.3333$, squares $v^2 = 5.76$ m^2/s^2 in place ($\rightarrow 105.60$ m^2/s^2), and divides by 2 and g:
  - **Frictional Head Loss: `0.022` [ENTER] `250` [×] `0.30` [÷] `2.4` [↶] [x^2] [×] `2` [÷] `9.80665` [÷] → h_f = 5.3841 m.**

- **Cantilever Steel Girder End Slope ($\vartheta = PL^2/2EI$):** Applying an end load $P = 12{,}000$ N to a girder of span $L = 3.5$ m and flexural rigidity $EI = 4.2 \times 10^6$ N·m^2 squares $L^2 = 12.25$ m^2 in place, multiplies by P (147,000 N·m^2), divides by 2 and EI (0.0175 rad), and scales by 1,000:
  - **Girder Tip Deflection Angle: `12000` [ENTER] `3.5` [↶] [x^2] [×] `2` [ENTER] `4.2` [E] `6` [×] [÷] `1000` [×] → 17.5 mrad.**

- **Three-Phase Feeder Mean Phase Loading ($I_{\text{avg}} = (I_a + I_b + I_c) / 3$):** Across unbalanced feeder RMS ammeter readings $I_a = 120$ A, $I_b = 85$ A, $I_c = 95$ A, summing $120 + 85 + 95 = 300.0$ A in Register X and dividing by 3 isolates the mean phase loading without touching the upper stack:
  - **Mean Phase Current: `120` [ENTER] `85` [+] `95` [+] `3` [÷] → I_{avg} = 100.00 A.**

Across every algebraic structure—single-register cascades, two-register harmonic folds, three-register rational quotients, and four-register Horner recurrences—the operational stack aligns the sequence of keystrokes with the dependency order of the quantities themselves. Once operands and subtotals remain visible across X, Y, Z, and T, parentheses disappear because every intermediate magnitude is formed and inspected in the exact order that nature combines it.

Chapter 3: Logarithms, Power Laws & Exponential Scaling

3.1 John Napier's Logarithms and the Weber-Fechner Law

3.1.1 Inside Merchiston Castle: Turning Products into Sums

At Merchiston Castle near Edinburgh in the opening years of the seventeenth century, Scottish mathematician John Napier (1550–1617) sought a method to relieve the arithmetic bottleneck of observational astronomy.

Across Europe, astronomers had just inherited the richest body of observational data in history. On the island of Hven in Denmark, Tycho Brahe had spent thirty years measuring the nightly positions of Mars and the stars to within one arcminute. In Prague, Tycho's assistant Johannes Kepler was working to establish that the planets orbit the Sun in ellipses rather than circles. Yet every time Kepler tested an elliptical orbit against Tycho's observations, he had to multiply and divide eight-digit trigonometric chords by hand across nearly a thousand folio pages of dense arithmetic. A single carried-digit error could invalidate months of calculation.

Napier set out to construct a mapping in mathematics that converts multiplication into addition, division into subtraction, and square or cube root extraction into division by 2 or 3. Aligning a row of integer exponents above the corresponding powers of two and powers of ten reveals the underlying isomorphism:

Step Sequence	0	1	2	3	4	5	6	7
Power of 2 (2^n)	1	2	4	8	16	32	64	128
Power of 10 (10^n)	1	10	100	1,000	10,000	100,000	10^6	10^7

To multiply 8 × 16 in the middle row, compare the exponents directly above them: 8 corresponds to exponent 3 (2^3), and 16 corresponds to exponent 4 (2^4). Adding their exponents (3 + 4 = 7) and reading the entry below step 7 gives 128 (8 × 16 = 128). That correspondence holds because an exponent counts how many times the base is multiplied by itself:

$$b^x \cdot b^y = b^{x+y}, \qquad \frac{b^x}{b^y} = b^{x-y}, \qquad (b^x)^n = b^{nx} \tag{Eq. 3.1}$$

Napier spent twenty years interpolating the continuous values between the integers through a kinematic model of two points moving simultaneously along straight lines. The first point (the logarithm track) travels along an infinite straight line at a uniform velocity v_0, covering equal *arithmetic* increments of distance in equal intervals of time (0, Δx, $2\Delta x$, $3\Delta x$, ...). Simultaneously, the second point (the number track) starts at distance L from a terminal mark and moves toward it at a speed proportional to the distance remaining, covering equal *geometric* ratios in equal intervals of time (L, $L\,r$, $L\,r^2$, $L\,r^3$, ...).

By pairing the arithmetic position of the first point with the geometric position of the second point at every instant, Napier established a continuous correspondence between addition and multiplication. In 1614, three years before his death, he published *Mirifici Logarithmorum Canonis Descriptio* (*Description of the Wonderful Rule of Logarithms*) *(Napier, 1614)*. There he coined the word *logarithm* from the Greek *logos* ("ratio") and

arithmos ("number") to signify the number that counts ratios. A year later, London geometry professor Henry Briggs traveled to Edinburgh and worked with Napier to normalize logarithms to base 10, where $\log_{10} 1 = 0$, $\log_{10} 10 = 1$, $\log_{10} 100 = 2$, and $\log_{10} 1000 = 3$.

For any two positive numbers a and b, Napier and Briggs's logarithmic function satisfies three algebraic identities:

$$\log(a \cdot b) = \log a + \log b, \qquad \log\left(\frac{a}{b}\right) = \log a - \log b, \qquad \log(y^x) = x \log y \qquad \textbf{(Eq. 3.2)}$$

Rather than spending twenty minutes multiplying two eight-digit numbers by hand, an astronomer could look up their two logarithms in Briggs's table, add them in seconds, and read the antilogarithm matching that sum. Extracting a cube root ($\sqrt[3]{y} = y^{1/3}$), which previously required hours of digit-by-digit trial division, reduced to looking up log y and dividing by 3. When Kepler received Napier's treatise in 1617, he dedicated his *Ephemerides* to Napier and used logarithmic tables to establish his Third Law of Planetary Motion ($T^2 = a^3$). Two centuries later, Pierre-Simon Laplace summarized the historical impact of the invention: *"Logarithms, by shortening the labors, doubled the life of the astronomer."*

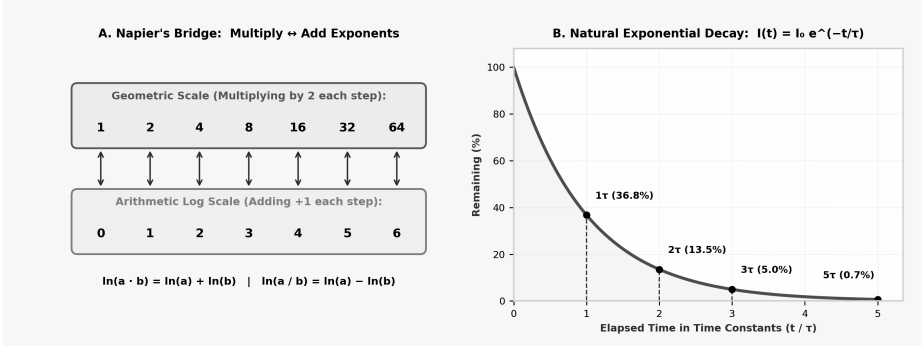


Figure: John Napier's logarithmic bridge between geometric multiplication and arithmetic addition, paired with natural exponential decay across characteristic time constants.

How does a pocket instrument find any logarithm to 16 digits without storing Henry Briggs's thousand-page book in memory? Following the classical algorithm of William J. Cody and William Waite *(Cody & Waite, 1980)*, the SC-32 strips off powers of two—instant in binary floating-point—splitting any positive number $x = m \cdot 2^k$ into its normalized binary significand $m \in [1/\sqrt{2}, \sqrt{2})$ near 1 and its exact integer binary exponent k:

$$\ln(m \cdot 2^k) = \ln(m) + k \ln(2) \qquad \textbf{(Eq. 3.3)}$$

Because m is confined to a narrow interval around 1, the natural logarithm $\ln(m)$ can be expanded in terms of the rapidly converging rational transformation $z = (m - 1) / (m + 1)$:

$$\ln(m) = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \frac{z^7}{7} + \ldots\right), \qquad \text{where } z = \frac{m-1}{m+1} \qquad \textbf{(Eq. 3.4)}$$

which converges to 16 decimal digits in a small number of odd powers.

3.1.2 The Weber-Fechner Law of Logarithmic Perception

Logarithms extend well beyond seventeenth-century astronomical tables. In the nineteenth century, German physiologists Ernst Heinrich Weber and Gustav Fechner established a quantitative law of psychophysics: human sensory systems perceive physical stimulus intensity on a logarithmic scale. In a dark room illuminated by a single candle, introducing a second candle doubles the photon flux and produces an unmistakable increase in perceived brightness. In a hall already illuminated by 100 candles, introducing a 101st candle changes the stimulus by only one percent, and the increment is imperceptible. Sensory receptors respond not to the absolute difference (+1 candle), but to the relative ratio (new intensity / old intensity). Each tenfold multiplication of physical energy produces one equal perceptual step.

Four standard measurement scales in acoustics, chemistry, seismology, and astrophysics apply this principle directly through the base-10 logarithm key [↰] [LOG] and its inverse [↰] [10ˣ], each acting in place on Register X once the underlying physical ratio has been formed across Registers Y and X. In acoustics, the faintest pressure fluctuation detectable by the human eardrum has a root-mean-square amplitude of $p_0 = 20$ μPa (2.0×10^{-5} Pa), whereas a nearby jet engine reaches 20 Pa. Between those extremes, acoustic pressure spans a million-fold range (10^6:1) and acoustic intensity $I \propto p^2$ spans twelve orders of magnitude (10^{12}:1). Along the inner ear, the tapered basilar membrane of the mammalian cochlea maps those twelve orders of magnitude onto the 0 to 120 dB Sound Pressure Level scale:

$$L_p = 20\log_{10}\left(\frac{p_{\text{rms}}}{p_0}\right) \text{ dB SPL} \qquad \textbf{(Eq. 3.5)}$$

Evaluating this expression from the inside out places the measured sound pressure p_{rms} in Register Y and the reference threshold $p_0 = 2.0 \times 10^{-5}$ Pa in Register X so [÷] forms the dimensionless pressure ratio in Register X. Pressing [↰] [LOG] transforms that ratio in place into base-10 orders of magnitude (Bels), and multiplying by 20 converts the result into decibels (**dB SPL**). That same 60 dB million-fold energy decay threshold defines the Sabine reverberation time T_{60} in architectural acoustics.

In aqueous and clinical chemistry, the molar concentration of hydronium ions $[H_3O^+]$ spans fourteen orders of magnitude from strong mineral acid (10^0 M) to concentrated alkali (10^{-14} M). In 1909, Danish biochemist Søren Peder Lauritz Sørensen at the Carlsberg Laboratory compressed those exponents onto a linear scale by defining pH $= -\log_{10} a_{H_3O^+} \approx -\log_{10}[H_3O^+]$, inverted in two keystrokes on Register X via [+/−] [↰] [10ˣ] ($[H_3O^+] = 10^{-pH}$). When a laboratory buffer or human arterial blood plasma equilibrates a weak acid ([HA] or dissolved $\propto P_{CO_2}$) with its conjugate base ($[A^-]$ or $[HCO_3^-]$), the *Henderson-Hasselbalch equation* relates the equilibrium pH to the acid dissociation constant pK_a:

$$\text{pH} = \text{p}K_a + \log_{10}\left(\frac{[\text{A}^-]}{[\text{HA}]}\right) \qquad \textbf{(Eq. 3.6)}$$

On the stack, dividing the conjugate base concentration in Register Y by the weak acid concentration in Register X, pressing [↰] [LOG] in place, and adding pK_a resolves the buffer pH without intermediate storage.

Two further logarithmic scales govern geophysics and observational astronomy. In 1935, Caltech seismologist Charles F. Richter represented the wide dynamic range of ground motion recorded on Wood-Anderson torsion seismographs by defining local earthquake magnitude as $M_L = \log_{10}(A/A_0)$ (extended to modern Hiroo Kanamori Moment Magnitude M_w): each integer step up the magnitude scale corresponds to a tenfold increase in ground-motion amplitude and a $10^{1.5} = 10\sqrt{10} \approx$ **31.62**× increase in released elastic wave

energy.

Similarly, around 130 BC the Greek astronomer Hipparchus classified naked-eye stars from First Magnitude (brightest) to Sixth Magnitude (faintest). In 1856, English astronomer Norman Robert Pogson formalized Hipparchus's catalog through the logarithmic relation $m_1 - m_2 = -2.5 \log_{10}(F_1/F_2)$, establishing that a 5-magnitude span corresponds to a $100\times$ (10^2) ratio in photon flux and that each single magnitude step represents a flux ratio of $100^{1/5} = 10^{0.4} \approx \mathbf{2.5119}\times$.

3.2 Power Laws, Biological Scaling & The Universal Harmonic Operator

Once Napier established that $\log(y^x) = x \log y$, raising any positive real number y to an arbitrary fractional or irrational exponent x followed directly from the composition of the logarithm and exponential:

$$y^x = e^{x \ln y} \quad \text{and} \quad \sqrt[x]{y} = y^{1/x} \tag{Eq. 3.7}$$

On the SC-32, powers, roots, and reciprocals divide into two operational classes: *in-place unary keys* ([↶] [x²], [√x], [1/x]) that transform Register X in place without disturbing values held in Registers Y, Z, or T, and *two-register dyadic keys* ([yˣ], [↶] [ˣ√y], [x ∥ y]) that combine Register Y with Register X:

Mathematical Function	Physical Keycap	Arity	Exact Mathematical Definition	Register Stack Effect
Square	[↶] [x²]	Unary	$f(x) = x^2$	$X \leftarrow X^2$
Square Root	[√x]	Unary	$f(x) = \sqrt{x},\ x \geq 0$	$X \leftarrow \sqrt{X}$
Reciprocal	[1/x]	Unary	$f(x) = 1/x,\ x \neq 0$	$X \leftarrow 1/X$
General Power	[yˣ]	Dyadic	$f(y, x) = y^x = \exp(x \ln y)$	$X \leftarrow Y^X$ (Dyadic drop)
Arbitrary Root	[↶] [ˣ√y]	Dyadic	$f(y, x) = \sqrt[x]{y} = y^{1/x}$	$X \leftarrow Y^{1/X}$ (Dyadic drop)
Parallel Harmonic	[x ∥ y]	Dyadic	$f(x, y) = x \cdot y \,/\, (x + y)$	$X \leftarrow X \cdot Y \,/\, (X + Y)$ (Dyadic drop)

3.2.1 Kepler's Planets, Kleiber's Mice and Whales, and Relativistic Muons

Fractional exponents such as $y^{2/3}$, $y^{3/4}$, $y^{1.32}$, and $(1 - \beta^2)^{-1/2}$ govern scaling laws across celestial mechanics, physiology, and special relativity. In 1619, Johannes Kepler established that the square of a planet's orbital period T (in Earth years) equals the cube of its semi-major axis a (in Astronomical Units, where 1 AU is the mean Earth-Sun distance). Sixty-eight years later, Isaac Newton derived $T^2 \propto a^3$ from universal gravitation: equating gravitational attraction $G\,M_\odot\,m\,/\,a^2$ with centripetal force $m\,v^2\,/\,a = m(2\pi\,a\,/\,T)^2\,/\,a$ yields $T^2 = (4\pi^2\,/\,G\,M_\odot)\,a^3$. Taking the cube root of both sides expresses $T^2 = a^3$ as a $2/3$ fractional power law:

$$a = (T^2)^{1/3} = \sqrt[3]{T^2} = T^{2/3} \tag{Eq. 3.8}$$

This fractional exponent links the unary and dyadic power keys on the keypad: squaring an orbital period T in place in Register X with the unary key [↶] [x²] and extracting the third root with the dyadic key [↶] [ˣ√y] (or raising T in Register Y directly to the $2/3$ power in Register X with [yˣ]) converts observed sidereal period directly into heliocentric distance.

A second fractional power law governs comparative physiology. In 1932, Swiss agricultural biologist Max Kleiber plotted basal metabolic rate B (in Watts or kilocalories per day) against body mass M across homeothermic mammals ranging from a 20-gram mouse to a 4,000-kilogram elephant (200,000× more massive). Rather than scaling linearly with mass (M^1) or with Euclidean exterior surface area ($M^{2/3} \approx M^{0.67}$), metabolic rate across organisms from unicellular microbes to cetaceans follows the three-quarters power of mass:

$$B = B_0 M^{3/4} = B_0 M^{0.75} \qquad \textbf{(Eq. 3.9)}$$

As physicists Geoffrey West, James Brown, and Brian Enquist demonstrated, this 3/4 exponent emerges because biological vascular and bronchial trees are space-filling fractal networks that minimize hydrodynamic pumping work across three spatial dimensions, introducing an effective fourth scaling dimension (3 / (3 + 1) = 3/4). Because basal metabolism scales as $M^{0.75}$, an organism 10,000× heavier (10^4) requires only $(10^4)^{0.75} = 10^3$ = 1,000× the caloric intake, while its heart rate scales as $M^{-1/4}$ (10× slower), separating the 800-beat-per-minute pulse of a shrew from the sub-10-beat-per-minute pulse of a blue whale.

A negative half-integer power law governs relativistic particle kinematics. In Earth's stratosphere, 15 kilometers (15,000 meters) above sea level, primary cosmic rays collide with atmospheric nuclei and produce unstable secondary *muons*. In its rest frame, a muon decays with a mean proper lifetime of $\tau_0 = 2.2$ microseconds (2.2×10^{-6} s). Even for a muon descending at 85% of the speed of light ($v = 0.85c \approx 2.55 \times 10^8$ m/s), Galilean kinematics predicts a mean propagation distance of less than six hundred meters before decay:

$$d_{\text{classical}} = v \cdot \tau_0 = (2.55 \times 10^8 \text{ m/s})(2.2 \times 10^{-6} \text{ s}) \approx \mathbf{560} \text{ meters} \qquad \textbf{(Eq. 3.10)}$$

Without relativistic time dilation, nearly every stratospheric muon would decay more than 14 kilometers above the surface, yet sea-level scintillation detectors record a steady flux of cosmic-ray muons. Under Albert Einstein's 1905 Special Theory of Relativity, invariance of the Minkowski spacetime interval $c^2\,\Delta\tau^2 = c^2\,\Delta t^2 - \Delta x^2$ across inertial frames dilates the laboratory lifetime $\Delta t = \gamma\,\Delta\tau$ according to Hendrik Lorentz's factor γ:

$$\gamma = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} = \left(1 - \beta^2\right)^{-1/2} \qquad \textbf{(Eq. 3.11)}$$

Structurally, γ evaluates as a pure unary chain from the dimensionless velocity ratio $\beta = v/c$ outward: squaring β in place with [↶] [x²], subtracting from 1 to form the proper-time factor $1 - \beta^2$, extracting the length-contraction factor $\sqrt{(1 - \beta^2)}$ with [√x], and inverting with [x^{-1}] yields the laboratory time-dilation factor while automatically preserving the length-contraction ratio $1/\gamma$ inside [↶] [LASTx]. At $v = 0.999c$, γ exceeds 22 and enables stratospheric muons to reach sea-level detectors.

These three laws illustrate why the SC-32 separates *in-place unary keys* ([↶] [x²], [√x], [x^{-1}]) from *two-register dyadic keys* ([y^x], [↶] [x√y], [x‖y]). When a power or root acts on a single physical quantity already in Register X, a unary key transforms that register without [ENTER] and without disturbing values in the upper registers—whether squaring the speed of light c in Einstein's gravitational redshift $z \approx GM/Rc^2$ for light escaping a white dwarf star, squaring molar volume V in Johannes Diderik van der Waals's equation of state $P = nRT / (V - nb) - an^2/V^2$, or chaining [√x] and [$x^{-1}$] to compute the electromagnetic skin depth $\delta = 1 / \sqrt{(\pi\, f\, \mu\, \sigma)}$ of alternating current in a copper conductor.

Conversely, whenever an equation pairs a base waiting in Register Y with a general fractional exponent or root index formed in Register X—as in polytropic gas compression work ($W = (P_1\ V_1\ /\ (n\ -\ 1))[(V_1/V_2)^{n\ -\ 1}\ -\ 1]$), Weibull wind-speed distributions, or solar air-mass attenuation ($I = I_0 \cdot 0.7^{(\mathrm{AM}^{0.678})}$)—pressing the dyadic key [y^x] or [↶] [$\sqrt[x]{y}$] combines Registers Y and X into Y^X or $\sqrt[X]{Y}$ and drops the upper stack by one level. When a rational power or decimal result needs to be inspected as a reduced fraction a/b or mixed number $n(a/b)$, toggling fraction display mode ([↶] [FDISP]) formats Register X as a best rational convergent via Euclid's continued-fraction / Farey algorithm (or to a fixed denominator $1/c$ for $c \in [2, 4095]$ popped from Register X via [↷] [1/c]), while retaining all 53 bits of floating-point precision internally.

3.2.2 The Five-Domain Universal Harmonic Operator (x‖y)

Extending the parallel reciprocal law ($1/T = 1/T_1 + 1/T_2$), the SC-32 provides a dedicated primary hardware keycap labeled [$x \| y$] that evaluates the product-over-sum harmonic combination:

$$x \| y = \frac{x \cdot y}{x+y} = \left(\frac{1}{x} + \frac{1}{y}\right)^{-1} \qquad \textbf{(Eq. 3.12)}$$

in a single keystroke across five distinct domains of physics and engineering:

Physical Domain	What Combines via $x \parallel y = x\ y\ /\ (x + y)$	Why the Reciprocal-Sum Law Appears
1. Parallel Electrical Resistors	$R_{\mathrm{eq}} = R_1 \parallel R_2 = R_1\ R_2\ /\ (R_1 + R_2)$	Both resistors share the same voltage V, while their currents ($I_i = V/R_i$) add.
2. Thin Optical Lenses & Mirrors	$f = d_o \parallel d_i = d_o\ d_i\ /\ (d_o + d_i)$	Light rays from object distance d_o bend by constant optical power $1/f$ to focus at image distance d_i ($1/f = 1/d_o + 1/d_i$).
3. Capacitors Wired in Series	$C_{\mathrm{eq}} = C_1 \parallel C_2 = C_1\ C_2\ /\ (C_1 + C_2)$	Series capacitor plates trap the same charge Q, while their voltage drops ($V_i = Q/C_i$) add.
4. Mechanical Springs in Series	$k_{\mathrm{eq}} = k_1 \parallel k_2 = k_1\ k_2\ /\ (k_1 + k_2)$	Two springs hooked end-to-end feel the same tension force F, while their stretches ($\Delta x_i = F/k_i$) add.
5. Round-Trip Average Speed	$v_{\mathrm{avg}} = 2(v_1 \parallel v_2) = 2\ v_1\ v_2\ /\ (v_1 + v_2)$	Driving D miles uphill at $v_1 = 30$ mph and D miles downhill at $v_2 = 60$ mph adds *times* ($D/v_1 + D/v_2$) to give $2(30 \parallel 60) =$ **40 mph** (not 45 mph).

All five systems reduce to $x \parallel y$ because each pairs a shared intensive variable with an additive extensive variable. In a *parallel resistor network*, Kirchhoff's current law requires that the total current $I_{\mathrm{total}} = V\ /\ R_{\mathrm{eq}}$ entering a node equal the sum of the branch currents $I_1 + I_2 = V/R_1 + V/R_2$; dividing out the common node voltage V gives $1\ /\ R_{\mathrm{eq}} = 1/R_1 + 1/R_2$. In *series capacitors*, the dual relationship holds: equal displacement charge $\pm\ Q$ is induced on each capacitor in the chain while the potential drops add ($V_{\mathrm{total}} = V_1 + V_2 \Rightarrow Q\ /\ C_{\mathrm{eq}} = Q/C_1 + Q/C_2$).

In *series Hookean springs*, both springs carry the same axial tension F while their elongations add ($\Delta x_{\mathrm{total}} = \Delta x_1 + \Delta x_2 \Rightarrow F\ /\ k_{\mathrm{eq}} = F/k_1 + F/k_2$), so the combined assembly exhibits a lower effective stiffness than either individual spring. Because $x \parallel y$ is both *commutative* ($x \parallel y = y \parallel x$) and *associative* ($(x \parallel y) \parallel z = x \parallel (y \parallel z)$), three, four, or more parallel branches fold sequentially on the stack with [$x \| y$] in the same manner as [+] or [×].

3.3 Leonhard Euler's Natural Base e, Carbon-14 & Cooling Coffee

Whereas Henry Briggs's base-10 logarithm ([↶] [LOG]) scales by powers of ten, continuous differential equations are governed by the natural base studied by Jacob Bernoulli in 1683 during his analysis of compound interest and designated e by Leonhard Euler in 1727:

$$e = \lim_{n\to\infty}\left(1+\tfrac{1}{n}\right)^n = 1+\tfrac{1}{1!}+\tfrac{1}{2!}+\tfrac{1}{3!}+\tfrac{1}{4!}+\ \ldots \approx \mathbf{2.718281828459} \qquad \textbf{(Eq. 3.13)}$$

Bernoulli identified this limit by analyzing an idealized bank paying 100% nominal annual interest on a principal of \$1.00 across progressively finer compounding intervals. Compounded once at year's end ($n = 1$), the balance reaches $(1 + 1)^1 = \$2.00$; compounded semiannually (50% every six months, $n = 2$), it reaches $(1 + 1/2)^2 = \$2.25$; compounded quarterly (25% every three months, $n = 4$), it reaches $(1 + 1/4)^4 \approx \$2.4414$. Increasing the frequency to monthly compounding ($n = 12$) yields $(1 + 1/12)^{12} \approx \$2.6130$, daily compounding ($n = 365$) yields $(1 + 1/365)^{365} \approx \$2.7146$, and compounding every second ($n = 31{,}536{,}000$)—where growth occurs continuously in direct proportion to the instantaneous balance—drives the terminal value to Euler's constant $e \approx \$2.718281828459...$

In mathematical analysis, e^x ([e^x]) and its inverse $\ln x$ ([LN]) form the *natural* exponential and logarithm because term-by-term differentiation of the factorial power series maps each term $x^k/k!$ to $k\,x^{k-1} / k! = x^{k-1}/(k\text{-}1)!$, reproducing the series unchanged:

$$\frac{d}{dx}\left(1+x+\tfrac{x^2}{2!}+\tfrac{x^3}{3!}+\tfrac{x^4}{4!}+\ \ldots\right) = 0+1+x+\tfrac{x^2}{2!}+\tfrac{x^3}{3!}+\ \ldots = e^x \qquad \textbf{(Eq. 3.14)}$$

The function e^x is the unique eigenfunction of the differential operator d/dx with unit eigenvalue: in everyday terms, e^x is the only curve in nature whose steepness (growth rate) at every moment equals its current height. Whenever a physical quantity grows or decays at a rate proportional to its instantaneous magnitude ($dN/dt = \pm\,\lambda\,N$), the solution is an exponential in base e.

3.3.1 Willard Libby's Carbon-14 Radiometric Dating

In 1949 at the University of Chicago, physical chemist Willard F. Libby demonstrated that natural logarithmic decay inversion ([LN]) could determine the age of organic archaeological specimens, a contribution recognized with the 1960 Nobel Prize in Chemistry.

Radiocarbon dating relies on an equilibrium between stratospheric production and biological uptake. High in the stratosphere, cosmic-ray neutrons continuously bombard nitrogen-14 nuclei ($^{14}\mathrm{N} + n \rightarrow {}^{14}\mathrm{C} + p$) and convert them into radioactive *carbon-14 (^{14}C)* at a nearly constant global rate. Those $^{14}\mathrm{C}$ atoms oxidize into carbon dioxide ($^{14}\mathrm{CO_2}$), mix throughout the atmosphere and oceans, and enter the biosphere through photosynthesis.

So long as a plant or animal remains alive and exchanges carbon with its environment, the ratio of radioactive $^{14}\mathrm{C}$ to stable carbon ($^{12}\mathrm{C}$) in its tissues remains locked to the atmospheric baseline ($N_0 \approx 1$ part per trillion). Once the organism dies—when a tree is felled for timber or flax is cut for linen—metabolic carbon exchange ceases, and the trapped $^{14}\mathrm{C}$ nuclei undergo beta decay back into nitrogen-14 with a physical Cambridge half-life of $t_{1/2} = 5{,}730$ years (refining Libby's original 5,568-year convention prior to tree-ring calibration):

$$N(t) = N_0\left(\tfrac{1}{2}\right)^{t/t_{1/2}} = N_0 e^{-\lambda t}, \qquad \text{where } \lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{5730} \approx 1.20968\times10^{-4}\ \mathrm{yr}^{-1} \qquad \textbf{(Eq. 3.15)}$$

Solving Libby's exponential decay law for elapsed time t isolates the surviving isotope fraction $N(t)/N_0$ inside the natural logarithm:

$$t = -\frac{\ln(N(t)/N_0)}{\lambda} = -\frac{\ln(N(t)/N_0) \times t_{1/2}}{\ln 2} \qquad \textbf{(Eq. 3.16)}$$

Because $\ln(N(t)/N_0)$ measures the number of elapsed natural e-folding lifetimes ($1/\lambda \approx$ 8,267 years), dividing by ln 2 ≈ 0.69315 converts those natural decay units into elapsed half-lives, and multiplying by $t_{1/2}$ = 5,730 years yields the age of the specimen. For Precambrian igneous rocks billions of years old, in which all original carbon-14 has decayed away, geochronologists apply the same [LN] inversion to Potassium-40 ($t_{1/2} = 1.248 \times 10^9$ yr) decaying into radiogenic Argon-40 trapped within crystalline mineral lattices, while the discrete compound-interest counterpart $t = \ln 2 \,/\, \ln(1+r)$ gives the exact doubling period of an accumulating reserve.

3.3.2 Newton's Law of Cooling and Thermal Relaxation

First-order exponential relaxation also governs macroscopic heat transfer. In 1701, Isaac Newton placed a heated iron block in a steady draft and observed that the rate dT/dt at which a warm body loses heat to its surroundings is proportional to the instantaneous temperature excess $\Delta T(t) = T(t) - T_{\text{room}}$ between the body and the ambient air:

$$\frac{d}{dt}(T(t) - T_{\text{room}}) = -\frac{1}{\tau}(T(t) - T_{\text{room}}) \qquad \textbf{(Eq. 3.17)}$$

When a vessel of liquid at 90°C sits in a 20°C room ($\Delta T_0 = 70$°C), thermal energy transfers rapidly across the boundary layer; once the liquid has cooled to 35°C ($\Delta T = 15$°C), the rate of heat loss drops nearly fivefold in proportion to the reduced temperature gradient. Integrating Newton's differential equation yields an exponential relaxation curve characterized by the thermal time constant τ:

$$\Delta T(t) = \Delta T_0 e^{-t/\tau} \Rightarrow t = -\tau \ln\left(\frac{\Delta T(t)}{\Delta T_0}\right) \qquad \textbf{(Eq. 3.18)}$$

The identical first-order differential equation governs an electrical RC network discharging through a resistor ($\tau = R\,C$), where each elapsed time constant τ reduces the remaining voltage or temperature excess to $1/e \approx 36.79\%$ of its initial value, and decaying to 10% of the initial disturbance requires -ln(0.10) = ln(10) ≈ 2.3026 time constants.

3.3.3 The Boltzmann Factor in Atmospheric Pressure and Thermal Radiation

Chemical reaction kinetics, reduced-pressure boiling points, barometric pressure at high altitude, and stellar blackbody radiation all depend on a single statistical-mechanical ratio: the competition between an activation or potential energy barrier (ΔE) and thermal agitation energy (RT or $k_B\,T$) inside the natural exponential ([e^x]).

In Svante Arrhenius's 1889 reaction-rate law ($k = A\,\exp(-E_a \,/\, RT)$), in solid-state interstitial lattice diffusion ($D = D_0\,\exp(-Q \,/\, RT)$), and in the Clausius-Clapeyron phase-equilibrium relation ($\ln(P_2/P_1) = -(\Delta H_{\text{vap}} \,/\, R)(1/T_2 - 1/T_1)$), molecules must surmount an energy threshold to react, hop between lattice sites, or escape from a liquid phase into vapor. On the stack, each Boltzmann factor is evaluated from the exponent outward: first form the dimensionless energy-to-temperature ratio $\Delta E \,/\, RT$ (or reciprocal-temperature difference $1/T_1 - 1/T_2$ via two applications of [x^{-1}] and [−]) in Register X, negate via [+/−] for a barrier, press [e^x] to obtain the Boltzmann probability factor in place, and multiply by the pre-exponential coefficient (A, D_0, or P_1).

The same exponential law governs planetary atmospheres. Whereas liquid water is nearly incompressible and exhibits a linear increase in hydrostatic pressure with depth, air is a compressible ideal gas whose density $\rho = PM/RT$ is proportional to local pressure P. Substituting ρ into the hydrostatic balance $dP = -\rho\, g\, dh$ yields the isothermal scale-height equation $dP/P = -Mg/RT_0\ dh$, in which gravitational potential energy Mgh competes against thermal energy RT_0 so that barometric pressure $P(h) = P_0\ e^{-Mgh/RT_0}$ decays exponentially with altitude h and drops to 60.7% of its sea-level value atop the 4,207-meter summit of Mauna Kea.

In solid-state physics and quantum electrodynamics, the Boltzmann factor appears inside the difference $(e^x - 1)$. Across a forward-biased silicon p-n junction ($I = I_s[\exp(V / nV_t) - 1]$) or intrinsic semiconductor bandgap ($n_i = B\ T^{3/2}\ \exp(-E_g / 2k_B\ T)$), electrostatic potential competes against thermal voltage $k_B\ T/q$. In December 1900, Max Planck placed that same $(e^x - 1)$ factor, with $x = hc / \lambda\ k_B\ T$, in the denominator of his blackbody spectral radiance law:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5\left(e^{\frac{hc}{\lambda k_B T}} - 1\right)} \qquad \textbf{(Eq. 3.19)}$$

resolving the classical ultraviolet divergence and introducing the quantum of action h. On the SC-32, evaluating $(e^x - 1)$ first in Register X ([e^x] **1** [-]) shifts that quantum denominator into Register Z while $2hc^2 / \lambda^5$ is formed across Registers Y and X, before [x⇄y] [÷] completes the quotient.

Numerical analysts distinguish two separate stages where direct exponential evaluation can lose precision: *inside* the [e^x] routine on large arguments, and *on the stack* when subtracting 1 from e^x on small arguments. As William J. Cody and William Waite showed *(Cody & Waite, 1980)*, a high-accuracy library routine must first reduce any argument x modulo ln 2 ($x = k \ln 2 + r$, where $|r| \le (1/2) \ln 2$) using split extended-precision constants $C_1 + C_2 = \ln 2$ before evaluating a minimax approximation on the remainder r, avoiding internal range-reduction cancellation.

Separately, evaluating $e^x - 1$ on the stack for a very small argument such as $x = 10^{-10}$ by pressing [e^x] and subtracting 1 loses trailing significand bits in floating-point representation: storing $1 + 10^{-10} + 1/2\ 10^{-20}$... in a 53-bit IEEE-754 word devotes the leading ten decimal digits of the significand to **1.000000000** and rounds away the $x^2/2$ curvature. For moderate exponents such as a forward-biased Shockley diode ($x = 17.95$) or visible-wavelength Planck radiation ($x = 4.98$), [e^x] minus 1 retains full precision; for small arguments $|x| \ll 10^{-5}$ (as in Rayleigh-Jeans radio-frequency astronomy), catastrophic cancellation on the stack is avoided by evaluating the two-term Taylor expansion $e^x - 1 \approx x(1 + x/2)$ or the identity $2\ e^{x/2}\sinh(x/2)$ in place.

3.4 Factorials, Combinatorics & Extending *n*! to Fractional Numbers

In combinatorics, the number of distinct top-to-bottom permutations of an ordinary deck of 52 playing cards follows directly from the multiplication principle. The first position can be filled by any of the 52 cards; for each of those 52 possibilities, the second position can be filled by any of the remaining 51 cards (52×51); and continuing down to the final card yields *52 factorial (52!)* distinct arrangements:

$$52! = 52 \times 51 \times 50 \times \ldots \times 3 \times 2 \times 1 \approx \mathbf{8.0658 \times 10^{67}} \qquad \textbf{(Eq. 3.20)}$$

That integer, exceeding 8×10^{67}, surpasses the total number of atoms in the Earth. Even a modest 16×16 fiber-optic crossbar switch admits $16! = 20{,}922{,}789{,}888{,}000$ (≈ 20.92 trillion) distinct one-to-one routing configurations.

3.4.1 Stirling's Asymptotic Formula and the Lanczos Gamma Approximation

On pocket scientific calculators of the 1970s and 1980s, keying **`69 x!`** succeeded ($\approx 1.7112 \times 10^{98}$), whereas keying **`70 x!`** halted with an **`OVERFLOW`** error because early BCD displays allocated only two decimal digits (**`E99`**) to the power-of-ten exponent and could not represent numbers $\geq 10^{100}$, while $70! \approx 1.197857 \times 10^{100}$ exceeds 10^{100}. Because the SC-32 uses a 64-bit IEEE-754 floating-point core supporting exponents up to $10^{\pm 308}$, its unary [↰] [x!] key evaluates 70! ($\rightarrow 1.1978571670 \times 10^{100}$) in place in Register X up to $170! \approx 7.2574 \times 10^{306}$ without overflow.

For analytical work in statistical mechanics, factorials are approximated asymptotically. In 1730, James Stirling and Abraham de Moivre proved that for large n, the factorial grows as $(n/e)^n$ scaled by $\sqrt{(2\pi\, n)}$:

$$n! \approx \sqrt{2\pi n}\,\left(\frac{n}{e}\right)^n \qquad \textbf{(Eq. 3.21)}$$

Even at $n = 10$ ($10! = 3{,}628{,}800$), Stirling's leading term yields 3,598,695.62, capturing over 99.17% of the exact integer product.

In 1729, Christian Goldbach posed to Leonhard Euler in St. Petersburg the problem of interpolating the discrete factorial $n!$ to non-integer arguments such as $(1/2)!$. Because a finite product of positive integers has no direct meaning at $n = 1/2$, Euler introduced a continuous improper integral, the *Gamma function*:

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t}\,dt = x\,\Gamma(x) \qquad \textbf{(Eq. 3.22)}$$

which agrees with the factorial at every non-negative integer ($n! = \Gamma(n+1)$) while extending analytically across real and complex arguments. Substituting $x = 1/2$ into Euler's integral and applying the change of variables $t = u^2$ transforms the expression into Carl Friedrich Gauss's error integral $\int_0^\infty e^{-u^2}\,du = \sqrt{\pi}/2$:

$$\left(\tfrac{1}{2}\right)! = \Gamma\left(\tfrac{3}{2}\right) = \frac{\sqrt{\pi}}{2} \approx \mathbf{0.886226925453} \qquad \textbf{(Eq. 3.23)}$$

Once $(1/2)! = \sqrt{\pi}/2$ is fixed, Euler's functional recurrence $(x+1)! = (x+1)\,x!$ determines the factorial of every half-integer by successive multiplication:

$$\left(\tfrac{3}{2}\right)! = \frac{3}{2}\frac{\sqrt{\pi}}{2} = \frac{3\sqrt{\pi}}{4}, \qquad \left(\tfrac{5}{2}\right)! = \frac{5}{2}\cdot\frac{3}{2}\frac{\sqrt{\pi}}{2} = \frac{15\sqrt{\pi}}{8} \approx 3.32335097045 \qquad \textbf{(Eq. 3.24)}$$

To evaluate [↰] [x!] for arbitrary real arguments in microseconds, the SC-32 firmware combines floating-point minimax range reduction *(Cody & Waite, 1980)* with Cornelius Lanczos's 1964 Chebyshev rational approximation:

$$\Gamma(z+1) = \sqrt{2\pi}\left(z+g+\tfrac{1}{2}\right)^{z+1/2} e^{-(z+g+1/2)} A_g(z) \qquad \textbf{(Eq. 3.25)}$$

This unified Gamma routine evaluates both discrete Stirling comparisons and continuous geometric formulas such as the volume of an n-dimensional unit ball, $V_n = \pi^{n/2}/(n/2)!$, which increases from $V_2 = \pi \approx 3.1416$ and $V_3 = (4/3)\pi \approx 4.1888$ to its counterintuitive

maximum across all Euclidean dimensions at $n = 5$ ($V_5 = \pi^{2.5}/(2.5)! = (8/15)\pi^2 \approx 5.2638$) before shrinking toward zero as $n \to \infty$ (because in higher dimensions, nearly all the volume of a cube hides in its distant corners while an inscribed unit ball wastes away).

3.4.2 Slicing Numbers into Parts (PARTS) and Formatting the Display (DISP)

Two additional softkey menus handle real-number decomposition and non-destructive display formatting. When an astronomical observation log records 14.736 decimal hours or a marine GPS receiver reports latitude 30.267153° N, a single floating-point value encodes integer units to the left of the radix point and fractional sub-units to the right. The [→] [PARTS] softkey menu (Right Shift [→] above [√x]) decomposes any value in Register X into its *Integer Part* **IP** and *Fractional Part* **FP** while preserving the original unsplit value in [↶] [LASTx]. The same softkey menu provides *Absolute Value* **ABS** for $|x|$ (complementing the change-sign key [+/−]), *Sign* **SGN** for $\pm$ 1 or 0, and the Euclidean *Greatest Common Divisor* [GCD] and *Least Common Multiple* [LCM]:

Menu Softkey	Operation Name	Mathematical Definition	Example on X = -14.7360
[IP]	Integer Part	$\mathrm{sgn}(x)\lfloor \lvert x\rvert \rfloor$ (truncation toward zero)	-14.0000
[FP]	Fractional Part	x - IP(x) (signed fractional remainder)	-0.7360
[ABS]	Absolute Value	$\lvert x\rvert = \sqrt{(x^2)}$ (non-negative magnitude)	+14.7360
[SGN]	Signum Function	+1.0 if $x > 0$, 0.0 if $x = 0$, -1.0 if $x < 0$	-1.0000

Because [IP] saves the original unsplit value inside [↶] [LASTx], extracting the integer part ([IP]) in Register X, recalling [↶] [LASTx] (which lifts the integer hours into Register Y), taking the fractional part ([FP]), and multiplying by 60 separates 14.736 hours into 14 hours in Register Y and 44.16 minutes in Register X.

Complementing [→] [PARTS], the [↶] [DISP] menu formats numbers rounded to fixed decimal places via [FIX] **n**, scientific notation via [SCI] **n**, engineering multiples-of-three exponents via [ENG] **n** for SI prefixes such as pF, µF, mA, kΩ, and MHz, or full 12-digit general precision via [ALL]. Switching the display format never alters or truncates the 64-bit value stored inside the stack registers. For the Beer-Lambert 90% optical absorption depth z = -ln(0.10)/α = 0.00005116856 m (51.169 µm) of the carbon-black polymer enclosure, the underlying register value renders across the four display modes as follows:

Display Mode	Key Sequence	Display Format	How 0.00005116856 m Appears
FIX 4	[↶] [DISP] [FIX] **4**	Fixed n digits after decimal point	**0.0001**
SCI 4	[↶] [DISP] [SCI] **4**	Scientific $m \times 10^k$ ($1 \le \lvert m\rvert < 10$)	**5.1169E-5**
ENG 4	[↶] [DISP] [ENG] **4**	Engineering $m \times 10^{3k}$ (SI prefixes)	**51.169E-6** (51.169 µm)
ALL	[↶] [DISP] [ALL]	Full 12-digit general representation	**5.11685576221E-5**

3.5 Evaluating Powers, Logarithms & Exponential Laws

Physical laws involving powers, logarithms, and exponentials obey a strict constraint of dimensional analysis: the argument of a logarithm or exponential cannot carry raw physical dimensions such as meters, seconds, or Pascals, because the Taylor series $e^x = 1 + x + x^2/2! + x^3/3! + \ldots$ would otherwise sum incompatible powers of a unit. Every transcendental expression therefore forms an inner dimensionless ratio—such as p_{rms}/p_0 in acoustics, $N(t)/N_0$ in radiometric dating, t/τ in thermal relaxation, or $\Delta E/(RT)$ in statistical mechanics—before applying the transcendental function and scaling by an outer

physical constant.

This structure maps directly onto the two operator arities of the keypad. Once two quantities of identical dimension are divided to form the dimensionless argument in Register X, an in-place unary key ([LN], [e^x], [↶] [LOG], [↶] [10^x], [$\sqrt{x}$], [↶] [x^2], [x^{-1}], [↶] [x!]) transforms Register X in place ($X \leftarrow f(X)$) without requiring [ENTER] and without disturbing any pre-exponential baseline or outer coefficient held in Registers Y, Z, and T. Conversely, the dyadic two-register keys ([y^x], [↶] [$\sqrt[x]{y}$], [$x \| y$]) combine a base, radicand, or branch impedance waiting in Register Y with the exponent, root index, or parallel partner in Register X, dropping the upper stack by one level.



*Figure: Acoustic sound pressure level 20 * log10(2 / 0.00002) = 100 dB evaluated from the inside out on the SC-32 stack.*

3.5.1 In-Place Squares, Reciprocal Roots & Harmonic Parallelism

Putting this unary-versus-dyadic distinction to work across gravitational physics, thermodynamics, electromagnetism, and relativistic kinematics resolves multi-term equations across the lower registers of the stack:

- **Einstein's White-Dwarf Gravitational Redshift ($z \approx GM/Rc^2$):** For a photon climbing out of the gravitational well of Sirius B, a solar-mass white dwarf ($M = 1.989 \times 10^{30}$ kg) packed into a sphere the size of the Earth ($R = 6.4 \times 10^6$ m, with $G = 6.6743 \times 10^{-11}$ N·m^2/kg^2 and $c = 2.99792 \times 10^8$ m/s), multiplying G by M on the stack establishes the stellar gravitational parameter $GM_\odot = 1.3275 \times 10^{20}$ m^3/s^2 in Register X, and dividing by the white dwarf's radius R yields the surface gravitational potential $\Phi = 2.0742 \times 10^{13}$ J/kg. When the speed of light c is keyed next, Φ lifts into Register Y; because the unary key [↶] [x^2] squares c in place inside Register X ($\rightarrow 8.9875 \times 10^{16}$ m^2/s^2) without disturbing Φ in Register Y, one division completes the ratio without touching Registers Z or T:
  - **Gravitational Redshift:** **6.6743** [E] **11** [+/−] [ENTER] **1.989** [E] **30** [×] **6.4** [E] **6** [÷] **2.99792** [E] **8** [↶] [x^2] [÷] $\rightarrow z =$ **0.0002307918**.
- **Van der Waals Real-Gas Equation of State ($P = nRT / (V - nb) - an^2/V^2$):** Contrast that single-branch ratio with Johannes Diderik van der Waals's 1873 equation of state, where two physical mechanisms compete: the repulsive kinetic pressure amplified by the finite volume nb of the molecules themselves, and the attractive intermolecular cohesion an^2/V^2 that pulls molecules back from the container walls.

For $n = 2.0$ mol ($n^2 = 4.0$ mol^2) compressed into a vessel of volume $V = 0.0050$ m^3 at $T = 320.0$ K ($R = 8.31446$ J/(mol·K), excluded molecular volume $b = 4.27 \times 10^{-5}$ m^3/mol, and cohesion constant $a = 0.364$ Pa·m^6/mol^2), assembling the two-part denominator ($V - nb$) pushes the thermal energy product $nRT = 5{,}321.2544$ J up to Register T, where T-replication preserves it across the upper registers while the kinetic pressure (1,082,744.15 Pa) and cohesive tension (58,240.00 Pa, simplified by squaring 0.0050 in place via [↶] [x^2])

meet in Registers Y and X (**2** [ENTER] **8.31446** [×] **320** [×] **0.0050** [ENTER] **2** [ENTER] **4.27** [E] **5** [+/−] [×] [-] [÷] **0.364** [ENTER] **4** [×] **0.0050** [↰] [x²] [÷] [-] → $P \approx$ **1,024,504.15 Pa**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
2 [ENTER] **8.31446** [×] **320** [×]	0.0000	0.0000	0.0000	5321.2544	Ideal kinetic energy numerator $n\ R\ T$
0.0050 [ENTER] **2** [ENTER] **4.27** [E] **5** [+/−] [×] [-] [÷]	5321.2544	5321.2544	5321.2544	1082744.1501	Divide by free volume (V - $n\ b$) for kinetic pressure term
0.364 [ENTER] **4** [×] **0.0050** [↰] [x²] [÷] [-]	5321.2544	5321.2544	5321.2544	**1024504.1501**	Subtract molecular attraction term $a\ n^2$ / V^2 for net pressure P (Pa)

Equally direct is the two-keystroke unary combination [√x] [x^{-1}], which turns any inverse square-root law ($1/\sqrt{u}$) into a single-register cascade in Register X without keying a numerator **1**:

- **RF Electromagnetic Skin Depth ($\delta = 1 / \sqrt{(\pi\ f\mu\sigma)}$):** When a $f = 100$ MHz (1.0×10^8 Hz) radio-frequency current flows through a copper conductor ($\sigma = 5.8 \times 10^7$ S/m, magnetic permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m), eddy currents crowd the alternating current into a thin outer skin. Multiply π f, μ_0, and σ directly in Register X to build the attenuation product $\pi\ f\mu_0\sigma = 2.2897 \times 10^{10}$ m^{-2}, then tap [√x] [x^{-1}] to extract the root (151,319.1 m^{-1}) and invert it in place into meters before scaling by 10^6 for micrometers:
  - **Copper Skin Depth:** [↻] [π] **1** [E] **8** [×] **4** [↻] [π] [×] **1** [E] **7** [+/−] [×] [×] **5.8** [E] **7** [×] [√x] [x^{-1}] **1** [E] **6** [×] → $\delta =$ **6.6085493101 μm**.

- **Hendrik Lorentz's Relativistic Time Dilation ($\gamma = 1 / \sqrt{(1 - (v/c)^2)}$) & Aviation True Airspeed ($V_{TAS} = V_{CAS} / \sqrt{\sigma}$):** Evaluating stratospheric muon time dilation at $\beta = v/c = 0.85$ demonstrates that same unary chain: square 0.85 in place with [↰] [x²] (→ 0.7225), negate and add 1 to form the proper-time variance $1 - \beta^2 = 0.2775$, and tap [√x] [x^{-1}] to extract the length-contraction factor $\sqrt{(1-\beta^2)} = 0.5268$ (saved in [↰] [LASTx]) and invert it into $\gamma = 1.8983$ (or $\gamma =$ **5.0252**× at $\beta = 0.98$). In subsonic mountain aviation, because a Pitot-static airspeed indicator senses dynamic pressure $q = (1/2)\rho\ V^2$, a pilot departing Leadville, Colorado (9,934 ft, where warm air reduces the density ratio to $\sigma = \rho/\rho_0 = 0.6944$) converts Calibrated Airspeed ($V_{CAS} =$ 120.0 kt) to True Airspeed via the identical cascade: keying **0.6944** [√x] [x^{-1}] **120** [×] reveals $V_{TAS} =$ **144.0000 kt** (20% faster through the airmass and over the runway in calm air).
  - **Muon Lorentz Factor ($\beta = 0.85$):** **0.85** [↰] [x²] **1** [-] [+/−] [√x] [x^{-1}] → $\gamma =$ **1.898315**.

The dyadic two-register keys ([y^x], [↶] [x⇆y], [x‖y]) combine Register Y and Register X with a one-level stack drop:

- **General Powers ([y^x]) & Polytropic Gas Compression ($W = (P_1\ V_1\ /\ (n - 1))[(V_1/V_2)^{n-1} - 1]$):** To raise a base $y = 2.5$ to an exponent $x = 3.8$, load the base into Register Y, key the exponent into Register X, and tap [y^x]. In a mechanical gas compressor operating across a volume ratio $V_1/V_2 = 4.0$ with polytropic exponent n - 1 = 0.32 and energetic pre-factor $P_1\ V_1\ /\ (n\text{-}1)$ = 15,000 J, start inside the brackets: raise $4.0^{0.32} = 1.5583$ with [y^x], subtract 1 to isolate the dimensionless work factor 0.5583, and multiply by 15,000 J:
  - **Pure Power Benchmark ($2.5^{3.8}$): `2.5` [ENTER] `3.8` [y^x] → 32.521609664.**
  - **Polytropic Compression Work: `4.0` [ENTER] `0.32` [y^x] `1` [-] `120000` [ENTER] `0.040` [×] `0.32` [÷] [×] → 8,374.93739 J.**

- **Kepler's Third Law ([↶] [x⇆y]) & Harmonic Round-Trip Speed ([x‖y]):** Applying Kepler's fractional-power law $a = \sqrt[3]{(T^2)}$ to Mars's sidereal period $T = 1.8808$ yr, square T in place with [↶] [x^2] (→ 3.5374 yr^2), enter the root index **3**, and tap the dyadic [↶] [x⇆y] key to recover Mars's semi-major axis a = **1.5237 AU**. Meanwhile, the round-trip mountain speed law (30 mph uphill and 60 mph downhill) collapses via [x‖y] and a factor of 2 into 2(30 ‖ 60) = **40.0000 mph**.

- **Weibull Wind-Turbine Distribution ($f(v) = (k/c)(v/c)^{k-1}\exp[-(v/c)^k]$) & Single-Pass Quadratic Sprints:** When an equation uses the *same* dimensionless ratio inside both a power law and a stretched exponential—as in Waloddi Weibull's wind-speed density for a turbine at $v = 8.5$ m/s, scale velocity $c = 9.0$ m/s, and shape parameter $k = 2.2$—the values 8.5 and 9.0 do not need to be keyed twice. Instead, form the normalized wind ratio $v/c = 0.9444$ once and press [ENTER] to duplicate it across the stack.

Raise the first copy to $k = 2.2$, negate, and fire [e^x] (→ 0.4140) while the saved copy of v/c = 0.9444 waits in Register Y; swap via [x⇄y] to raise 0.9444 to k - 1 = 1.2 (→ 0.9337), multiply the two factors together, and scale by $k/c = 2.2/9.0$. Similarly, in a single-pass quadratic evaluation on 2.4 x^2 + 7.8 x - 15.6 = 0, starting inside the discriminant radical $\sqrt{(b^2 - 4ac)} = \sqrt{(7.8^2 - 4(2.4)(-15.6))} = 14.5121$, subtracting $b = 7.8$, and dividing by $2a$ = 4.8 isolates the positive root in 13 keystrokes without touching a storage register:

- **Weibull Wind Probability Density: `8.5` [ENTER] `9.0` [÷] [ENTER] [ENTER] `2.2` [y^x] [+/-] [e^x] [x⇄y] `1.2` [y^x] [×] `2.2` [ENTER] `9.0` [÷] [×] → $f(v)$ = 0.094496 (m/s)$^{-1}$.**
- **Single-Pass Quadratic Positive Root: `7.8` [↶] [x^2] `4` [ENTER] `2.4` [×] `15.6` [+/-] [×] [-] [$\sqrt{x}$] `7.8` [-] `2` [÷] `2.4` [÷] → x_+ = 1.39834665561.**

3.5.2 Base-10 & Natural Logarithms (LOG, LN)

Because both [↶] [LOG] and [LN] are unary operators that transform Register X in place, logarithmic perception scales and radiometric dating laws evaluate from the physical concentration or activity ratio straight outward:

- **Weber-Fechner Acoustic Decibels & Sabine Reverberation ([↶] [LOG]):** For a stadium acoustic wave of root-mean-square pressure $p_{\text{rms}} = 1.45$ Pa referenced to the $p_0 = 2.0 \times 10^{-5}$ Pa cochlear hearing threshold, dividing p_{rms} by p_0 reveals that the pressure wave is 72,500× stronger than the threshold of hearing. Tapping [↶] [LOG] compresses that 72,500:1 ratio in place into 4.8603 Bels (powers of ten), and scaling by 20 yields the sound pressure level in decibels. In architectural acoustics, a 60 dB (10^6:1) drop in sound energy sets Wallace Clement Sabine's 0.161 s/m reverberation constant across a concert hall of volume $V = 4{,}500$ m^3 and acoustic absorption $A = 320.0$ Sabins:
  - **Stadium Sound Pressure Level:** `1.45` [ENTER] `2.0` [E] `5` [+/−] [÷] [↶] [LOG] `20` [×] → **97.206760131 dB SPL**.
  - **Sabine Concert Hall Reverberation (T_{60}):** `0.161` [ENTER] `4500` [×] `320` [÷] → T_{60} = **2.2641 s**.
- **Henderson-Hasselbalch Buffer Equilibrium & Arterial Blood-Gas Physiology (pH = pK$_a$ + log$_{10}$[A$^-$] / [HA]):** In analytical chemistry, when an acetic acid buffer (pK$_a$ = 4.76) pairs conjugate base [A$^-$] = 0.185 M with weak acid [HA] = 0.120 M, divide the molar ratio 0.185 / 0.120 = 1.5417 in Register X, tap [↶] [LOG] (→ +0.1880 pH units) in place, and add pK$_a$ = 4.76:
  - **Buffer Equilibrium pH:** `0.185` [ENTER] `0.120` [÷] [↶] [LOG] `4.76` [+] → **4.9479904824**.

In pulmonary and critical-care medicine, the carbonic acid–bicarbonate buffer maintains arterial blood plasma near pH 7.40 (pK$_a$ = 6.10, Henry's law CO_2 solubility α = 0.0301 mmol/(L·mmHg)):

$$\text{pH} = 6.10 + \log_{10}\left(\frac{[\text{HCO}_3^-]}{0.0301 \times P_{\text{CO}_2}}\right), \qquad [\text{H}_3\text{O}^+] = 10^{-\text{pH}} \tag{Eq. 3.26}$$

For an arterial blood-gas panel showing serum bicarbonate [HCO_3^-] = 24.0 mmol/L and partial pressure P_{CO_2} = 40.0 mmHg, evaluating the dissolved CO_2 denominator in Register X while [HCO_3^-] waits in Register Y yields pH = **7.3996**, and inverting Sørensen's scale in place via [+/−] [↶] [10^x] (scaled by 10^9 nmol/mol) reveals an effective free hydronium concentration of **39.8489 nmol/L**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`24` [ENTER]	0.0000	0.0000	24.0000	24.0000	Stage serum bicarbonate [HCO_3^-] = 24.0 mmol/L in Y
`0.0301` [ENTER] `40` [×]	0.0000	0.0000	24.0000	1.2040	Dissolved carbonic acid α P_{CO_2} = 1.2040 mmol/L
[÷] [↶] [LOG]	0.0000	0.0000	0.0000	1.2996	Base-to-acid ratio 19.9336 logged in place to +1.2996
`6.10` [+]	0.0000	0.0000	0.0000	7.3996	Arterial blood plasma pH = 6.10 + 1.2996 = 7.3996
[+/−] [↶] [10^x] `1` [E] `9` [×]	0.0000	0.0000	0.0000	**39.8489**	Inverse $10^{-\text{pH}} \times 10^9$: free [$H_3O^+$] (39.85 nmol/L)

- **Willard Libby's Carbon-14 & Potassium-Argon Radiometric Clocks (LN):** Evaluating Libby's radiocarbon inversion $t = -\ln(N/N_0)\ t_{1/2} / \ln 2$ for an Ice Age hearth bone retaining 23.4% of its original ^{14}C ($N/N_0 = 0.234$, $t_{1/2} = 5{,}730$ yr), press LN to transform 0.234 in place into -1.4524 natural decay units, multiply by 5,730 yr, divide by $\ln(2) = 0.6931$ (computed in Register X via **2** LN while the numerator waits in Register Y), and negate via +/− while Registers Z and T remain at zero. For Precambrian volcanic basalt billions of years old, potassium-argon dating ($t = (1/\lambda)\ \ln(1 + C(^{40}Ar\ /\ ^{40}K))$ with $^{40}Ar/^{40}K = 0.0850$, $C = 9.54$, $\lambda = 5.543 \times 10^{-10}\ yr^{-1}$) evaluates from the inner isotope ratio outward:

  - **Carbon-14 Ice Age Hearth Bone (23.4% remaining):** **0.234** LN **5730** × **2** LN ÷ +/− → **12,006.75 years** (12,006.7541 yr).
  - **Potassium-Argon Volcanic Core Age:** **9.54** ENTER **0.0850** × **1** + LN **5.543** E **10** +/− ÷ → **1.071304*E*9 years** (1.071 billion years).

Under the first-order relaxation law $t = -\tau\ \ln(\Delta T/\Delta T_0)$, a two-register logarithmic trace unfolds when an RC timing network with resistance $R = 150$ kΩ (150,000 Ω) and capacitance $C = 47$ μF (47×10^{-6} F) relaxes to 10% (0.10) of its initial step voltage (**150000** ENTER **47** E **6** +/− × **0.10** LN × +/− → **16.233224906 seconds**). Once R and C multiply into the characteristic time constant $\tau = 7.0500$ s in Register X, keying **0.10** and pressing LN lifts $\tau = 7.0500$ s into Register Y while transforming 0.10 in place into -2.302585 time constants in Register X:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
150000 ENTER	0.0000	0.0000	150000.000	150000.000	Load scale factor 150,000
47 E **6** +/− ×	0.0000	0.0000	0.0000	7.050000	Relaxation time constant τ = 7.05 s
0.10 LN	0.0000	0.0000	7.050000	-2.302585	ln(0.10) natural logarithm
× +/−	0.0000	0.0000	0.0000	**16.233225**	Relaxation time in seconds

Similarly, at an annual compound growth rate of $r = 7.2\%$, dividing $\ln(2) = 0.6931$ in Register Y by $\ln(1.072) = 0.0695$ in Register X (**2** LN **1.072** LN ÷) yields the logarithmic doubling time of **9.9696 years**, the exact benchmark behind the financial Rule of 72 (72 / 7.2 = 10.0 years).

3.5.3 Natural Exponentials (e^x) & Boltzmann's Universal Thermal Factor

Across physical chemistry, atmospheric physics, solid-state electronics, and quantum astrophysics, thermal processes share a single algebraic form: Ludwig Boltzmann's thermal activation factor $\exp(-\Delta E\ /\ k_B\ T)$ (or its molar counterpart $\exp(-\Delta E\ /\ RT)$), which compares a physical energy barrier ΔE against the thermal agitation energy $k_B\ T$ of the surrounding medium. Because e^x is a unary operator on Register X, every Boltzmann calculation on the SC-32 follows a single inside-out sequence: assemble the dimensionless energy-to-temperature ratio inside Register X first, fire e^x in place second, and scale by the physical pre-factor third:

- **Arrhenius Chemical Activation Barrier ($E_a = R \ln(k_2/k_1) \;/\; (1/T_1 - 1/T_2)$):** Warming a chemical reaction by fifty degrees from $T_1 = 300$ K ($k_1 = 3.25 \times 10^{-4}$ s^{-1}) to $T_2 = 350$ K ($k_2 = 2.85 \times 10^{-2}$ s^{-1}) accelerates the reaction rate nearly ninety-fold. To extract Svante Arrhenius's activation barrier, first form the reciprocal-temperature window $1/300 - 1/350 = 4.7619 \times 10^{-4}$ K^{-1} in Register X via two taps of the unary $[x^{-1}]$ key, then evaluate the scaled logarithmic rate ratio $R \ln(k_2/k_1)$ while 4.7619×10^{-4} K^{-1} waits in Register Y, and swap via $[x \leftrightarrow y]$ before dividing and scaling by 1,000 J/kJ to reveal an activation barrier of $E_a = \mathbf{78.1148\ kJ/mol}$:

  - **Reciprocal-Temperature Window ($1/T_1 - 1/T_2$):** **300** $[x^{-1}]$ **350** $[x^{-1}]$ $[-]$ → 4.7619×10^{-4} K^{-1}.
  - **Logarithmic Rate Numerator ($R \ln(k_2/k_1)$):** **2.85** $[E]$ **2** $[+/-]$ $[ENTER]$ **3.25** $[E]$ **4** $[+/-]$ $[\div]$ $[LN]$ **8.314462** $[\times]$ → **37.197524317 J/(mol·K)**.

- **Solid-State Atomic Diffusion ($D = D_0 \exp(-Q \;/\; RT)$):** When interstitial carbon atoms diffuse through austenite steel inside a $T = 1{,}200$ K heat-treating furnace ($Q = 148{,}000$ J/mol, $D_0 = 2.3 \times 10^{-5}$ m^2/s, $R = 8.31446$ J/(mol·K)), form the dimensionless barrier ratio $Q/(RT) = 14.8336$ in Register X, negate and fire $[e^x]$ to show that only 3.613×10^{-7} of lattice vibrations clear the energy jump, and scale by D_0:

  - **Carbon-in-Steel Diffusivity:** **148000** $[ENTER]$ **8.31446** $[ENTER]$ **1200** $[\times]$ $[\div]$ $[+/-]$ $[e^x]$ **2.3** $[E]$ **5** $[+/-]$ $[\times]$ → $D =$ **8.309585*E*-12 m^2/s**.

- **Clausius-Clapeyron Vapor Pressure ($P_2 = P_1 \exp[-(\Delta H_{vap} \;/\; R)(1/T_2 - 1/T_1)]$):** That identical Boltzmann factor explains why water boils at 60°C ($T_2 = 333.15$ K) inside a partial vacuum relative to its sea-level boiling point of 100°C ($T_1 = 373.15$ K, $P_1 = 101.325$ kPa, molar enthalpy of vaporization $\Delta H_{vap} = 40{,}660$ J/mol). Invert both temperatures with $[x^{-1}]$ and subtract to form $1/T_2 - 1/T_1 = 3.2176 \times 10^{-4}$ K^{-1} in Register X; multiply by $\Delta H_{vap}/R = 4{,}890.275$ K for the dimensionless exponent 1.5735; negate and fire $[e^x]$ to obtain the vapor-pressure fraction $P_2/P_1 = 0.2073$; and scale by $P_1 = 101.325$ kPa:

  - **Partial-Vacuum Vapor Pressure:** **333.15** $[x^{-1}]$ **373.15** $[x^{-1}]$ $[-]$ **40660** $[ENTER]$ **8.31446** $[\div]$ $[\times]$ $[+/-]$ $[e^x]$ **101.325** $[\times]$ → $P_2 =$ **21.0063 kPa**.

- **Isothermal Barometric Scale Height on Mauna Kea ($P = P_0 \exp(-Mgh \;/\; RT_0)$):** When a mountaineer climbs to the $h = 4{,}207$ m summit of Hawaii's Mauna Kea under the isothermal scale-height model ($M = 0.0289644$ kg/mol, $g = 9.80665$ m/s^2, $R = 8.314462$ J/(mol·K), reference temperature $T_0 = 288.15$ K, $P_0 = 1{,}013.25$ hPa), gravitational potential energy Mgh replaces the chemical barrier ΔH_{vap}. Forming the dimensionless scale-height ratio $\varkappa = Mgh/RT_0 = 0.4987$, firing $[e^x]$ on -$\varkappa$ (→ 0.6073), and scaling by $P_0 = 1{,}013.25$ hPa yields the summit pressure:

  - **Mauna Kea Summit Pressure:** **0.0289644** $[ENTER]$ **9.80665** $[\times]$ **4207** $[\times]$ **8.314462** $[\div]$ **288.15** $[\div]$ $[+/-]$ $[e^x]$ **1013.25** $[\times]$ → $P =$ **615.3203 hPa**.

- **Silicon Intrinsic Carrier Density ($n_i = B\ T^{3/2}\ \exp(-E_g\ /\ 2k_B\ T)$) & Shockley Diode ($I = I_s[\exp(V\ /\ nV_t)\ -\ 1]$):** Inside a crystal of pure silicon at room temperature $T = 300$ K ($k_B = 8.61733 \times 10^{-5}$ eV/K, $B = 5.23 \times 10^{15}$ cm^{-3}K$^{-3/2}$, bandgap $E_g = 1.12$ eV), dividing E_g by 2 $k_B\ T = 0.051704$ eV reveals a thermal barrier ratio of 21.6618; negating, firing [e^x] ($\rightarrow 3.9121 \times 10^{-10}$), and scaling by $T^{3/2} = 300^{1.5}$ and B gives the intrinsic carrier density without touching Registers Z or T. Conversely, applying a forward bias of $V = 0.58$ V across a Shockley silicon p-n junction ($n = 1.25$, $V_t = 0.025852$ V, $I_s = 1.8 \times 10^{-11}$ A) lowers the electrostatic barrier so the Boltzmann exponent $V/(nV_t) = 17.9483$ turns positive; firing [e^x], subtracting 1, and scaling by I_s gives the forward current:

  - **Silicon Intrinsic Carrier Density:** **1.12** [ENTER] **2** [ENTER] **8.61733** [E] **5** [+/−] [×] **300** [×] [÷] [+/−] [e^x] **300** [ENTER] **1.5** [y^x] [×] **5.23** [E] **15** [×] $\rightarrow n_i \approx$ **1.0631×10^{10} cm^{-3}**.
  - **Shockley Diode Forward Current:** **0.58** [ENTER] **1.25** [ENTER] **0.025852** [×] [÷] [e^x] **1** [-] **1.8** [E] **11** [+/−] [×] $\rightarrow I =$ **0.0011223527 A** (1.122 mA).

An exponential calculation recruits all four stack registers (T, Z, Y, X) when the quantum occupancy factor (e^x - 1) sits in the denominator of a multi-power physical law, as in Max Planck's 1900 blackbody spectral radiance law $B_\lambda(T) = c_1\ /\ \lambda^5(e^x - 1)$ at solar green wavelength $\lambda = 500$ nm $= 5.0 \times 10^{-7}$ m ($c_1 = 2hc^2 = 1.19104 \times 10^{-16}$ W·m^2/sr, dimensionless photon-to-thermal energy ratio $x = hc\ /\ \lambda\ k_B\ T = 4.98020$).

Evaluating the Bose-Einstein photon denominator ($e^{4.98020} - 1$) = 144.5035 first on Row 1 shifts it up into Registers Z and T while c_1 and the fifth-power wavelength $\lambda^5 = 3.1250 \times 10^{-32}$ m^5 are assembled in Registers Y and X, before dropping back into Register Y for a final [$x \rightleftarrows y$] [÷] (**4.98020** [e^x] **1** [-] **1.19104** [E] **16** [+/−] [ENTER] **5.0** [E] **7** [+/−] [ENTER] **5** [y^x] [÷] [$x \rightleftarrows y$] [÷] $\rightarrow B_\lambda \approx$ **2.6375×10^{13} W/(m^3·sr)**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
4.98020 [e^x] **1** [-]	0.0000	0.0000	0.0000	144.5035	Planck photon distribution denominator (e^x - 1)
1.19104 [E] **16** [+/−] [ENTER]	0.0000	144.5035	1.1910×10^{-16}	1.1910×10^{-16}	Load first radiation constant $2hc^2$
5.0 [E] **7** [+/−] [ENTER] **5** [y^x]	144.5035	144.5035	1.1910×10^{-16}	3.1250×10^{-32}	Raise wavelength to fifth power λ^5
[÷]	144.5035	144.5035	144.5035	3.8113×10^{15}	Divide radiation constant c_1 by λ^5
[$x \rightleftarrows y$] [÷]	144.5035	144.5035	144.5035	**2.6375×10^{13}**	Divide spectral power by photon denominator for radiance B_λ

3.5.4 Factorials (x!), Lanczos Gamma, and Domain Guard Verification

Across both discrete permutations and continuous integrals, the unary [↶] [x!] key invokes Lanczos's rational Gamma engine across both integers and fractions. For a 16-channel photonic crossbar switch, **16** [↶] [x!] evaluates 16! in place in Register X to display **2.092279*E*13** (20,922,789,888,000) routing states. Passing a half-integer to that same key (**0.5** [↶] [x!] → **0.886226925453**) confirms that Euler's fractional factorial $(1/2)! = \Gamma(3/2)$ matches Gauss's constant $\sqrt{\pi}/2$ ([↷] [π] [√x] **2** [÷] → **0.886226925453**) to all 12 digits.

That half-integer identity determines the five-dimensional unit hypersphere volume $V_5 = \pi^{2.5}/(2.5)! = (8/15)\pi^2$: raising π to the 2.5 power in Register Y and dividing by (2.5)! in Register X ([↷] [π] **2.5** [y^x] **2.5** [↶] [x!] [÷]) confirms $V_5 =$ **5.263789**. Finally, passing an out-of-bounds argument such as a negative number to [LN] (**5** [+/−] [LN]) triggers **INVALID DATA** (Code 02) without altering Registers Y, Z, or T, and tapping [C ON] clears the banner so the calculation resumes intact.

Chapter 4: Circular Trigonometry, Vector Projections & Hyperbolic Functions

4.1 Eratosthenes' Shadow, Angular Measure, and the Unit Circle

At local solar noon on the summer solstice of 240 BCE, Eratosthenes of Cyrene, chief librarian of the Great Library of Alexandria, measured the shadow of an upright bronze rod called a *gnomon* in a sunlit courtyard. As preserved in Cleomedes' astronomical treatise *The Heavens (Cleomedes, c. 1st C. CE / 2004)*, travelers had brought Eratosthenes an observation from the frontier garrison town of Syene (modern Aswan), some five thousand stadia up the Nile to the south: at midday on the summer solstice, vertical stone columns in Syene cast no shadow, and the reflection of the overhead sun shone directly onto the water at the bottom of a deep well.

At that same hour in Alexandria, Eratosthenes observed that his vertical bronze gnomon cast a distinct wedge of shadow across the courtyard paving stones. Measuring the ratio of the shadow's horizontal length to the rod's vertical height, he determined that the sun's rays met the upright rod at an angle equal to one-fiftieth of a full circle:

$$\theta = \frac{360^\circ}{50} = 7.2^\circ \tag{Eq. 4.1}$$

A sun standing at the zenith in Syene while inclined 7.2° south of the zenith in Alexandria could not be reconciled with both a flat Earth and a distant sun. Because the sun is sufficiently distant that its rays arrive across the Earth as parallel lines, only the curvature of the Earth could tilt Alexandria's vertical rod relative to Syene's vertical well. By Euclid's theorem on alternate interior angles of parallel lines cut by a transversal, the 7.2° shadow angle at the tip of the gnomon in Alexandria equaled the central angle subtended at the center of the Earth between Syene and Alexandria. Multiplying the five-thousand-stadia overland distance between the two cities by fifty ($50 \times 5{,}000 = 250{,}000$ stadia, or roughly 40,000 kilometers), Eratosthenes determined the circumference of the Earth to within a few percent of modern satellite geodesy from a vertical rod, its shadow, and circular geometry.

Two centuries after Eratosthenes, the astronomer Hipparchus of Nicaea addressed the general problem of translating any central angle on the celestial sphere into a straight-line distance. He inscribed triangles inside a circle of fixed radius and compiled the first table of *chords*, the direct precursor of the modern sine and cosine functions. Five hundred years later in India, astronomers led by Aryabhata halved Hipparchus's chord, observing that the half-chord of a double angle forms the vertical leg of a right triangle inscribed inside a circle of radius $r = 1$. They termed that half-chord *jya-ardha* ("half-bowstring"), which Arabic translators rendered as *jiba*, twelfth-century Latin translators in Toledo rendered as *sinus* ("bay" or "fold"), and modern mathematics shortened to *sine*.

When a circle of radius $r = 1$ is centered at the origin of an (x, y) coordinate plane—the *Unit Circle*—and a radial vector is rotated counterclockwise from the positive horizontal axis through an angle ϑ, the tip of that vector defines a right triangle of hypotenuse 1:

$$x = \cos\theta \quad \text{(Horizontal Projection)}, \qquad y = \sin\theta \quad \text{(Vertical Projection)} \tag{Eq. 4.2}$$

$$x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1 \tag{Eq. 4.3}$$

During the Middle Ages and the Renaissance, Islamic and European astronomers from al-Biruni to Regiomontanus and Johannes Kepler engraved this unit-circle geometry onto brass astrolabes equipped with a sighting arm (*alidade*) and an orthogonal shadow grid. Rather than computing series approximations by hand, a navigator could rotate the radius arm to an angle ϑ and read its horizontal and vertical projections off the engraved scales:

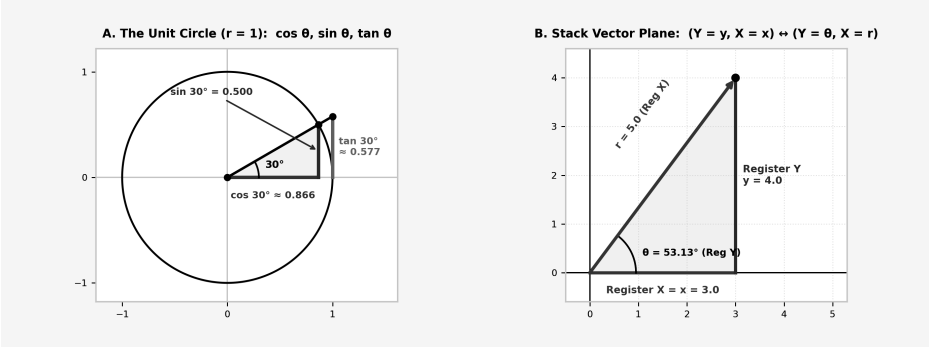


Figure: The Unit Circle ($r = 1$) right-triangle projections ($\cos\vartheta$, $\sin\vartheta$, $\tan\vartheta$) and the SC-32 (Y, X) stack vector plane.

Rotating the unit radius arm ($r = 1$) counterclockwise from 0° to 30° ($\pi/6$ rad) frames the right-triangle projections on the horizontal and vertical axes:

$$\cos(30^\circ) = \frac{\sqrt{3}}{2} \approx 0.866025, \qquad \sin(30^\circ) = \frac{1}{2} = 0.500000 \tag{Eq. 4.4}$$

Continuing the rotation through 45°, where $\cos 45^\circ = \sin 45^\circ = \sqrt{2}/2 \approx 0.707107$, and 60°, where $\cos 60^\circ = 1/2$ and $\sin 60^\circ = \sqrt{3}/2$, brings the radius vector to the vertical axis at 90°. There the horizontal projection vanishes to $\cos 90^\circ = 0$ while the vertical projection reaches $\sin 90^\circ = 1$. The third circular ratio, the *tangent* $\tan\vartheta = \sin\vartheta \,/\, \cos\vartheta$, takes its name from the Latin *tangens* ("touching"): it measures the vertical intercept cut off by the extended radius line along the vertical tangent line $x = 1$ at the right-hand boundary of the unit circle.

4.1.1 Degrees, Radians, and Gradians

Before evaluating a trigonometric function, the angular unit that subdivides a full revolution around the unit circle must be selected. The StackCalc SC-32 supports all three standard angular conventions, indicated continuously on the top status bar:

Angular Mode	LCD Annunciator	Full Circle Revolution	Right Angle	Primary Engineering Field
Degrees	**DEG** (Default)	360°	90°	Structural Civil Engineering, Geodesy, Avionics
Radians	**RAD**	$2\pi \approx 6.283185$ rad	$\pi/2 \approx 1.570796$ rad	Mathematical Physics, Calculus, Wave Mechanics
Gradians	**GRAD**	400 grad	100 grad	European Surveying, Mining Engineering

All three conventions remain in engineering use because each arose from a distinct structural symmetry. Four thousand years ago, Babylonian astronomers observed that the sun advances against the background zodiac by roughly one degree per day over a 360-day schematic year. In sexagesimal (base-60) arithmetic, 360 is a highly composite integer divisible without remainder by 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60,

72, 90, 120, and 180, so degrees (**DEG**) allow a machinist, surveyor, or navigator to subdivide a circle into halves, thirds, quarters, fifths, sixths, eighths, or twelfths using exact integers.

In mathematical analysis, dividing a circle into 360 parts is an external convention. Roger Cotes and Leonhard Euler defined radian measure (**RAD**) intrinsically from the circle itself by taking the unit angle ϑ as the central angle subtending an arc whose length s equals the radius r. Defining an angle as the dimensionless ratio of arc length to radius ($\vartheta = s \,/\, r$) makes one full revolution equal to 2π radians (≈ 6.283185 rad) and one radian equal to $180° \,/\, \pi \approx 57.29578°$. Only in radians does the small-angle limit $\lim_{\vartheta \to 0} (\sin \vartheta)/\vartheta$ equal 1, so that $(d/d\vartheta) \sin \vartheta = \cos \vartheta$ without introducing a factor of $\pi/180$ into differential equations of harmonic oscillation or wave propagation.

During the French Revolution, the geodesists who established the metric system introduced gradians (**GRAD**) to decimalize angular measure alongside length and mass. By defining a right angle as 100 gradians (*gons*) and a full circle as 400 gradians, they tied terrestrial coordinates directly to the meter: because the meter was defined as one ten-millionth of the meridian quadrant from the Equator to the North Pole (10,000 km per 100 grad), one gradian of latitude corresponds to 100 kilometers along a meridian and one centigrad (0.01 grad) corresponds to 1 meter.



Figure: StackCalc SC-32 Live 4-Level XYZT Stack Hardware Display.

The active angular mode determines how the SC-32 interprets circular inputs (**SIN**, **COS**, **TAN**) and what angular units it returns from inverse functions (**ASIN**, **ACOS**, **ATAN**) and rectangular-to-polar conversions ([←] [▸θ,r]). Opening [←] [MODES] ([←] + [+/−]) switches among [DEG], [RAD], and [GRAD] at any time. When a nautical chart or surveyor's plat specifies an angle in sexagesimal degrees-minutes-seconds **D.MMSS**—such as 35° 15' 36''—keyed as **35.1536**—pressing [←] [▸HR] (shifted above the **5** key) converts Register X in place into decimal degrees (**35.2600°**), while [→] [▸HMS] converts decimal degrees or hours back into **D.MMSS** format. Alongside them above the **6** key, [←] [▸DEG] and [→] [▸RAD] convert Register X between degrees and radians directly without changing the active angular mode.

Function	Dedicated Keystroke	Arity	Valid Domain	Output Range (in [DEG])
Sine	[SIN]	Unary	$(-\infty, +\infty)$	$[-1.0, +1.0]$
Cosine	[COS]	Unary	$(-\infty, +\infty)$	$[-1.0, +1.0]$
Tangent	[TAN]	Unary	$\vartheta \neq 90° + k \cdot 180°$	$(-\infty, +\infty)$
Arcsine	[←] [ASIN]	Unary	$[-1.0, +1.0]$	$[-90°, +90°]$
Arccosine	[←] [ACOS]	Unary	$[-1.0, +1.0]$	$[0°, 180°]$
Arctangent	[←] [ATAN]	Unary	$(-\infty, +\infty)$	$(-90°, +90°)$

4.1.2 Exact Axial Angle Clamping at 90°, 180°, and 270°

Evaluating cos(90°) in a generic floating-point environment frequently returns a non-zero residual such as $6.123233995736766 \times 10^{-17}$ rather than exact zero. Inside a conventional math library, an angle in degrees is first converted to radians by multiplying by π / 180, so 90° becomes a floating-point approximation to π / 2. Because π is transcendental, its binary expansion does not terminate; an IEEE-754 double-precision register truncates π/2 after 53 significand bits, producing a value that differs from mathematical π/2 by approximately 6.123234×10^{-17} radians.

Near $\vartheta = \pi/2$, the derivative of cos ϑ is -sin(π/2) = -1, so evaluating the cosine polynomial on that truncated binary approximation returns the residual 6.123234×10^{-17} instead of zero. In a structural stiffness matrix or coordinate transformation, a spurious non-zero entry of 6.12×10^{-17} can propagate into subsequent divisions or prevent zero-threshold tests from matching.

To prevent binary radian conversion from perturbing orthogonal geometry, the SC-32 performs *exact axial angle clamping* prior to polynomial evaluation: in Degrees (**DEG**) or Gradians (**GRAD**) mode, integer multiples of a right angle are reduced modulo a full circle so that sin(0°) = 0.0, cos(0°) = 1.0, sin(90°) = 1.0, cos(90°) = 0.0, sin(180°) = 0.0, cos(180°) = -1.0, sin(270°) = -1.0, and cos(270°) = 0.0 return exact IEEE-754 zero and unity values. Likewise, evaluating tan(90°) or tan(270°) traps the vertical asymptote with **INVALID DATA** (Code 02) rather than returning a 1.63×10^{16} reciprocal of rounding noise, and passing an argument $|x| > 1$ to [↖] [ASIN] or [↖] [ACOS] raises **INVALID DATA** (Code 02) while preserving the stack registers unchanged.

4.2 Triangles, Crosswinds & Resolving 2D Vectors

Across classical mechanics, civil surveying, geometrical optics, and air navigation, two-dimensional problems reduce to two reciprocal operations on the unit circle: resolving a directed magnitude into orthogonal components via cos ϑ and sin ϑ, or reconstructing a resultant magnitude and direction from orthogonal components.

4.2.1 Projecting Forces, Trajectories & Mountain Summits

In the *Two New Sciences* (1638) *(Galilei, 1638)*, Galileo Galilei demonstrated that the parabolic trajectory of a projectile in a uniform gravitational field (neglecting air resistance) is the superposition of two independent orthogonal motions. Horizontally, no net force acts after launch, so the horizontal velocity component $v_{0x} = v_0 \cos \vartheta$ remains constant downrange. Vertically, gravity g produces a constant downward acceleration that decelerates the initial vertical component $v_{0y} = v_0 \sin \vartheta$ until the projectile reaches its apex altitude $H_{\max}$, where the vertical kinetic energy $(1/2)mv_{0y}^2$ equals the gravitational potential energy $mgH_{\max}$:

$$H_{\max} = \frac{(v_0 \sin\theta)^2}{2g}, \qquad R = \frac{2v_{0x}v_{0y}}{g} = \frac{v_0^2 \sin(2\theta)}{g} \tag{Eq. 4.5}$$

By the double-angle identity 2 sin ϑ cos ϑ = sin(2ϑ), maximum horizontal range over level ground occurs when sin(2ϑ) = 1 (ϑ = 45°), and any two complementary launch angles summing to 90°, such as 30° and 60°, produce the same downrange distance R.

The same orthogonal resolution governs a mass m sliding down a plane inclined at angle ϑ with kinetic friction coefficient μ_k. Gravity acts vertically downward with magnitude mg, while the surface constrains motion along the incline. Resolving the gravitational vector into components parallel and normal to the surface shows that mg sin ϑ acts down the

slope while $mg\cos\vartheta$ acts perpendicular to the surface, producing a normal reaction $N = mg\cos\vartheta$ and an opposing Coulomb friction force $\mu_k N = \mu_k mg\cos\vartheta$:

$$a = g(\sin\theta - \mu_k \cos\theta) \qquad \textbf{(Eq. 4.6)}$$

Because the mass m cancels out of both terms, bodies of any mass sharing the same coefficient μ_k accelerate down the incline at the same rate, and at the critical *angle of repose* where $\sin\vartheta = \mu_k\cos\vartheta$ ($\tan\vartheta = \mu_k$), the downslope gravitational component and upslope friction balance to zero.

When an inaccessible summit cannot be plumbed directly, trigonometry determines its elevation from a surveyed horizontal baseline. During the Great Trigonometrical Survey of India in the 1850s, surveyors determined the altitude of Peak XV (Mount Everest) through *two-station optical leveling*: sighting the vertical elevation angle α to the summit from Station 1, advancing a measured horizontal baseline distance d toward the peak to Station 2, and sighting the steeper elevation angle β. Because the horizontal distances from the summit's plumb line to the two stations are $x_1 = H\cot\alpha$ and $x_2 = H\cot\beta$, their difference $x_1 - x_2 = d$ eliminates the unmeasured horizontal distance to the mountain's interior plumb line:

$$d = H(\cot\alpha - \cot\beta) \Rightarrow H = \frac{d}{\cot\alpha - \cot\beta} = \frac{d}{\frac{1}{\tan\alpha} - \frac{1}{\tan\beta}} \qquad \textbf{(Eq. 4.7)}$$

Because the cotangent $\cot\vartheta = 1/\tan\vartheta$ is the reciprocal of the tangent, dedicated keys for $\sec\vartheta$, $\csc\vartheta$, and $\cot\vartheta$ are unnecessary when x^{-1} sits on the primary keypad: inverting each tangent in place via TAN x^{-1} converts elevation angles directly into horizontal-to-vertical ratios.

When the horizontal baseline D to the base of a vertical target can be taped directly—as in *forestry timber cruising* with a handheld Suunto clinometer—sighting the inclination angle ϑ_{top} to the tree's crown leader and the depression angle ϑ_{base} to its root collar gives the total tree height $H = D(\tan\vartheta_{top} - \tan\vartheta_{base})$ (or merchantable bole height when sighted to the upper utilization limit). Conversely, in a *precision toolroom machine shop*, setting up a surface grinder with a hardened steel *sine bar* of center-to-center roller spacing L (5.0000 in or 10.0000 in) uses the vertical sine projection $h = L\sin\vartheta$ to determine the exact stack of gauge blocks wrung together beneath the raised roller to grind a specified taper angle ϑ.

Circular projection also governs the period of a simple pendulum: at finite angular amplitude ϑ_0, the exact gravitational restoring torque $\sin\vartheta$ is smaller than the linearized approximation ϑ, lengthening the small-angle period $T_0 = 2\pi\sqrt{(L/g)}$ according to the leading half-angle sine term $T \approx T_0[1 + (1/4)\sin^2(\vartheta_0/2)]$ (+3.66% at $\vartheta_0 = 45°$). Christiaan Huygens's 1656 horological analysis expresses the exact period through the complete elliptic integral of the first kind $K(\sin(\vartheta_0/2))$, which Romberg numerical integration (↗ ∫) evaluates directly across any release angle.

4.2.2 Reversing the Projection: Snell's Refraction, Brewster's Angle & the Inverse Tangent

Whereas direct trigonometric functions map an angle to a dimensionless ratio of lengths, *inverse trigonometric functions* (↖ ASIN, ↖ ACOS, ↖ ATAN) invert that mapping, recovering the geometric angle from a measured ratio of velocities, refractive indices, accelerations, or lengths.

In geometrical optics, a light ray crossing the interface between two transparent media of refractive indices n_1 and n_2 obeys the law of refraction first formulated in 984 CE in

Baghdad by Ibn Sahl (*On Burning Mirrors and Lenses*) and independently rediscovered in Europe by Thomas Harriot (1602) and Willebrord Snell (1621): across the boundary, the product of the refractive index and the sine of the ray angle from the normal remains invariant ($n_1 \sin \vartheta_1 = n_2 \sin \vartheta_2$). Pierre de Fermat later derived this invariance from the Principle of Least Time, given phase velocities $v_1 = c/n_1$ and $v_2 = c/n_2$. When light propagates inside a higher-index optical fiber core ($n_1 > n_2$), a ray meeting the cladding boundary refracts away from the normal; at the *critical angle* ϑ_c, the refracted ray grazes the interface at $\vartheta_2 = 90°$ ($\sin 90° = 1$), and [↰] [ASIN] recovers the confinement threshold:

$$\theta_c = \arcsin\left(\frac{n_2}{n_1}\right) \qquad \textbf{(Eq. 4.8)}$$

Rays incident at angles exceeding ϑ_c undergo lossless *total internal reflection*, confining optical signals along transoceanic glass fibers.

In 1815, Scottish physicist David Brewster determined the incidence angle ϑ_B at which light reflected from a dielectric surface becomes completely polarized parallel to the interface. Brewster found that complete polarization occurs when the reflected and refracted rays are mutually perpendicular ($\vartheta_B + \vartheta_2 = 90°$). Because $\sin(90° - \vartheta_B) = \cos \vartheta_B$, Snell's relation $n_1 \sin \vartheta_B = n_2 \sin \vartheta_2$ becomes $n_1 \sin \vartheta_B = n_2 \cos \vartheta_B$, expressing the refractive-index ratio as a tangent:

$$\tan \theta_B = \frac{\sin \theta_B}{\cos \theta_B} = \frac{n_2}{n_1} \Rightarrow \theta_B = \arctan\left(\frac{n_2}{n_1}\right) \qquad \textbf{(Eq. 4.9)}$$

Comparing these two optical laws illustrates how the three inverse keys partition the unit circle. Because neither the horizontal projection $\cos \vartheta$ nor the vertical projection $\sin \vartheta$ on the unit circle can exceed the unit hypotenuse 1, [↰] [ASIN] and [↰] [ACOS] are restricted to arguments within $[-1, +1]$, mapping onto principal ranges $[-90°, +90°]$ and $[0°, 180°]$. By contrast, the tangent ratio $\tan \vartheta = y/x$ spans all real numbers as a line approaches the vertical axis, so [↰] [ATAN] maps $(-\infty, +\infty)$ onto $(-90°, +90°)$ whenever two orthogonal components are divided.

4.2.3 Oblique Triangles and the Law of Cosines

In geodetic surveying and wave interference, triangles are rarely right-angled. When a surveyor occupies a corner station C and measures straight-line distances a and b to two boundary monuments separated by an included angle $C \neq 90°$, dropping an altitude h from the vertex opposite side a decomposes the oblique triangle into two right triangles. In the inner right triangle containing angle C, the vertical altitude has length $h = b \sin C$, while the horizontal projection of side b along side a has length $b \cos C$, so the remaining base of the outer right triangle is $(a - b \cos C)$. Applying the Pythagorean theorem to that outer right triangle and simplifying via $\cos^2 C + \sin^2 C = 1$ yields:

$$c^2 = (a - b\cos C)^2 + (b \sin C)^2 = a^2 + b^2(\cos^2 C + \sin^2 C) - 2ab\cos C \qquad \textbf{(Eq. 4.10)}$$

Extracting the square root gives the *Law of Cosines*:

$$c = \sqrt{a^2 + b^2 - 2ab\cos C} \qquad \textbf{(Eq. 4.11)}$$

Structurally, the Law of Cosines augments the Pythagorean sum $a^2 + b^2$ with the angular projection term $-2ab \cos C$, governing all planar triangles across the full range of angle C. When $C = 90°$, $\cos 90° = 0$ and the cross-term vanishes, recovering the Pythagorean theorem ($c = \sqrt{(a^2 + b^2)}$); when $C < 90°$ (an acute angle), $\cos C > 0$ reduces c below the right-triangle hypotenuse; and when $C > 90°$ (an obtuse angle), $\cos C < 0$ renders $-2ab \cos$

C positive and extends the opposite side c.

Replacing the interior angle C with the exterior supplement 180° - $\Delta\varphi$ reverses the sign of the cosine term and yields the law of *coherent wave interference*: when two acoustic or electromagnetic waves of amplitudes A_1 and A_2 arrive at a detector with phase difference $\Delta\varphi$, their superposed amplitude is the vector resultant of two phasors placed tip-to-tail:

$$A_{\text{resultant}} = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos(\Delta\phi)} \qquad \textbf{(Eq. 4.12)}$$

4.2.4 The (*Y*, *X*) Coordinate Plane: Polar ↔ Rectangular Vector Conversion

In scalar arithmetic, binary operators treat Registers Y and X as two scalar operands that collapse into a single result. In two-dimensional vector geometry, Registers Y and X operate together as a coordinate pair representing a point or vector in the plane.

Navigators, surveyors, and electrical engineers convert continuously between *polar coordinates* (r, ϑ) and *rectangular coordinates* (x, y) because field instruments measure in polar form while vector addition requires rectangular components. When a radar antenna tracks a target, a surveyor sights a monument with a theodolite, or an aircraft instrument panel displays airspeed and heading, the measurement consists of a radial magnitude r and a direction angle ϑ.

Yet as Simon Stevin demonstrated in 1586 through his *Wreath of Spheres* (*Clootcrans*) equilibrium proof establishing the parallelogram law of forces, non-collinear vectors cannot be summed by adding their scalar magnitudes $r_1 + r_2$. To combine an aircraft's airspeed vector with a crosswind, or to verify that a multi-leg property survey closes back onto its starting monument, each polar vector (r, ϑ) must first be resolved into its horizontal component ($x = r\cos\vartheta$) and vertical component ($y = r\sin\vartheta$), summed along each orthogonal axis ($\sum x$, $\sum y$), and converted back into a resultant magnitude r and direction ϑ.

On the SC-32 display, Register X occupies Line 1 and Register Y sits directly above it on Line 2. Mapping the horizontal component x (and radial magnitude r) to Register X and the vertical component y (and angle ϑ) to Register Y aligns the display registers with standard (x, y) notation.

The shifted legends above [1] indicate the destination registers directly: for Rectangular → Polar ([↖] [▸θ,r], Left Shift [↖] above [1]), the first symbol after the arrow (ϑ) is placed in Register Y and the second symbol (r) is placed in Register X, so keying **y** [ENTER] **x** [↖] [▸θ,r] transforms ($Y = y$, $X = x$) into ($Y = \vartheta$, $X = r$). Conversely, for Polar → Rectangular ([↗] [▸y,x], Right Shift [↗] above [1]), y is placed in Register Y and x is placed in Register X, so keying ϑ [ENTER] **r** [↗] [▸y,x] transforms ($Y = \vartheta$, $X = r$) into ($Y = y$, $X = x$) without disturbing Registers Z or T:

Transformation	Dedicated Keystroke	Stack Input (*Y*, *X*)	Stack Output (*Y*, *X*)	Mathematical Formula
Polar to Rectangular	[↗] [▸y,x]	$Y = \vartheta$ (Angle in active mode) • $X = r$ (Radius / Magnitude)	$Y = y = r\sin\vartheta$ • $X = x = r\cos\vartheta$	$(x, y) = (r\cos\vartheta, r\sin\vartheta)$
Rectangular to Polar	[↖] [▸θ,r]	$Y = y$ (Orthogonal vertical) • $X = x$ (Orthogonal horizontal)	$Y = \vartheta = \text{atan2}(y, x)$ • $X = r = \sqrt{(x^2 + y^2)}$	$(r, \vartheta) = (\sqrt{(x^2+y^2)}, \text{atan2}(y, x))$

Using [↶] [▸θ,r] avoids the quadrant ambiguity and axial singularity of dividing y / x and pressing [↶] [ATAN], which receives only the single scalar quotient y/x. In Quadrant III, where both x = -3 and y = -4 are negative, the quotient (-4)/(-3) = +1.3333 is positive, so $\arctan(y/x)$ returns +53.1301° in Quadrant I, an error of 180°; on the positive or negative vertical axis where $x = 0$ and $y = \pm 10$, dividing y/x halts with **DIVIDE BY 0**. By contrast, [↶] [▸θ,r] evaluates the four-quadrant atan2(y, x) function, inspecting the individual signs of y in Register Y and x in Register X across the full (-180°, +180°] circle:

Quadrant / Axis	Signs of (x, y)	Four-Quadrant atan2(y, x) Formula (in **DEG**)	Example Vector (x, y)	Output ($Y = \vartheta$, $X = r$)
Quadrant I	$x > 0$, $y \geq 0$	$\arctan(y/x)$	(+3, +4)	ϑ = +53.1301°, r = 5.0
Quadrant II	$x < 0$, $y \geq 0$	$\arctan(y/x)$ + 180°	(-3, +4)	ϑ = +126.8699°, r = 5.0
Quadrant III	$x < 0$, $y < 0$	$\arctan(y/x)$ - 180°	(-3, -4)	ϑ = -126.8699°, r = 5.0
Quadrant IV	$x > 0$, $y < 0$	$\arctan(y/x)$	(+3, -4)	ϑ = -53.1301°, r = 5.0
Vertical Axes	$x = 0$, $y \neq 0$	+90° if $y > 0$, -90° if $y < 0$	(0, +10)	ϑ = +90.0000°, r = 10.0

This three-stage sequence—project via [↷] [▸y,x], sum the orthogonal components, and recombine via [↶] [▸θ,r]—applies identically across marine, aeronautical, and structural vector mechanics. When a vessel steers across a tidal current or an aircraft flies at true airspeed $\boldsymbol{V}_{\text{air}}$ through a crosswind $\boldsymbol{V}_{\text{wind}}$, its ground track is the vector sum $\boldsymbol{V}_{\text{ground}} = \boldsymbol{V}_{\text{air}} + \boldsymbol{V}_{\text{wind}}$, just as two tension forces on a laboratory force table require an equilibrant $\boldsymbol{F}_3 = -(\boldsymbol{F}_1 + \boldsymbol{F}_2)$ for static equilibrium.

When a land surveyor or navigator records the distance r_i and *North Azimuth* α_i (measured clockwise from True North, where 0° = North and 90° = East) around a closed property boundary, reflecting the coordinate axes across the 45° Northeast diagonal swaps sine and cosine so that Northing latitude is $\Delta N_i = r_i \cos \alpha_i$ and Easting departure is $\Delta E_i = r_i \sin \alpha_i$:

$$\Sigma \Delta N = \sum r_i \cos \alpha_i = 0, \qquad \Sigma \Delta E = \sum r_i \sin \alpha_i = 0 \tag{Eq. 4.13}$$

Because [↷] [▸y,x] evaluates ($Y = r \sin \vartheta$, $X = r \cos \vartheta$), entering a Cartesian angle ϑ (counterclockwise from East) places ($Y = \Delta y$, $X = \Delta x$), whereas entering a surveyor's or pilot's North Azimuth α (clockwise from North) automatically places **Easting departure ($\boldsymbol{\Delta E} = \boldsymbol{r} \sin \boldsymbol{\alpha}$) in Register $\boldsymbol{Y}$** and **Northing latitude ($\boldsymbol{\Delta N} = \boldsymbol{r} \cos \boldsymbol{\alpha}$) in Register $\boldsymbol{X}$**—and pressing [↶] [▸θ,r] evaluates atan2(ΔE, ΔN) to recover the true North Azimuth directly! Chaining [↷] [▸y,x] [Σ+] after each course accumulates an entire multi-leg boundary traverse inside the statistical registers without occupying named storage variables.

4.2.5 Spherical Geodesy, Great-Circle Orthodromes & Avian Celestial Navigation

Over intercontinental distances, planar Cartesian trigonometry must be replaced by spherical geometry. On a spherical Earth of mean radius R_E = 6,371.0 km, the shortest path between two points at latitude/longitude coordinates (φ_1, λ_1) and (φ_2, λ_2) is not a constant-bearing rhumb line (*loxodrome*) on a Mercator projection, but a *great-circle orthodrome*: the arc formed on the surface by a plane passing through both points and the center of the Earth.

The *Spherical Law of Cosines* follows from three-dimensional unit-sphere coordinates. On a sphere of unit radius $r = 1$, a point at latitude φ and longitude λ has polar coordinate $z =$

sin φ and equatorial radius cos φ, which projects onto the equatorial (x, y) axes as $x = \cos \varphi \cos \lambda$ and $y = \cos \varphi \sin \lambda$:

$$\hat{u}_1 = [\cos\phi_1 \cos\lambda_1,\ \cos\phi_1 \sin\lambda_1,\ \sin\phi_1], \qquad \hat{u}_2 = [\cos\phi_2 \cos\lambda_2,\ \cos\phi_2 \sin\lambda_2,\ \sin\phi_2] \qquad \textbf{(Eq. 4.14)}$$

Evaluating the inner product $\hat{\mathbf{u}}_1 \cdot \hat{\mathbf{u}}_2 = \cos d$ between the two unit vectors and applying the cosine difference identity $\cos \lambda_1 \cos \lambda_2 + \sin \lambda_1 \sin \lambda_2 = \cos(\lambda_2 - \lambda_1)$ expresses the central geocentric angle d as the sum of the axial projection ($\sin \varphi_1 \sin \varphi_2$) and the equatorial projection ($\cos \varphi_1 \cos \varphi_2 \cos \Delta\lambda$):

$$\cos d = \sin\phi_1 \sin\phi_2 + \cos\phi_1 \cos\phi_2 \cos(\lambda_2 - \lambda_1), \qquad D = R_E \times d_{\text{rad}} \qquad \textbf{(Eq. 4.15)}$$

Because meridians converge toward the poles as $\cos \varphi$, the great-circle route between San Francisco ($\varphi_1 \approx 37.6°$ N) and Tokyo ($\varphi_2 \approx 35.8°$ N) arcs northward toward the Aleutian Islands near 50° N, where each degree of longitude spans a shorter arc than at 36° N and shortens the total path by hundreds of kilometers compared to sailing due west along a parallel of latitude.

The same Spherical Law of Cosines solves the *astronomical triangle* connecting the North Celestial Pole, an observer's zenith at latitude φ, and the sun at seasonal declination δ and hour angle ω (15° per hour from local solar noon), determining the solar horizon altitude α and azimuth γ_s:

$$\sin\alpha = \sin\phi \sin\delta + \cos\phi \cos\delta \cos\omega, \qquad \cos\gamma_s = \frac{\sin\alpha \sin\phi - \sin\delta}{\cos\alpha \cos\phi} \qquad \textbf{(Eq. 4.16)}$$

Migratory birds combine these same geometric relationships during transoceanic flight. Each autumn, the bar-tailed godwit (*Limosa lapponica*) departs Alaska's tidal flats and flies 11,000 kilometers non-stop across the Pacific Ocean to New Zealand over nine days. Because the sun's azimuth shifts at roughly 15° per hour, maintaining a solar heading requires time-compensated azimuth correction: the godwit couples its internal circadian rhythm with solar elevation and geomagnetic inclination cues to follow a great-circle orthodrome while compensating for crosswind drift (↻ ▸y,x vector addition).

Where satellite navigation requires sub-meter geodetic accuracy, the spherical model is refined to Isaac Newton's *oblate spheroid*. In the *Principia* (1687), Newton demonstrated that Earth's diurnal rotation causes the equatorial radius to exceed the polar radius by 21.385 km—a bulge large enough that if a smartphone GPS receiver treated Earth as a perfect sphere, its map pin at mid-latitudes would miss the street by nearly 20 kilometers. The World Geodetic System 1984 (WGS84) reference ellipsoid (equatorial semi-major axis a = 6,378,137.0 m, flattening $f = 1/298.257223563$, first eccentricity squared $e^2 = 2f - f^2 \approx 0.00669437999014$) defines the prime-vertical radius of curvature $N(\varphi)$ and Earth-Centered Earth-Fixed coordinates (X, Y, Z) at geodetic latitude φ, longitude λ, and ellipsoidal height h:

$$N(\phi) = \frac{a}{\sqrt{1 - e^2 \sin^2\phi}}, \qquad [X, Y, Z] = [(N(\phi) + h)\cos\phi\cos\lambda,\ (N(\phi) + h)\cos\phi\sin\lambda,\ (N(\phi)(1 - e^2) + h)\sin\phi] \qquad \textbf{(Eq. 4.17)}$$

4.3 Hanging Chains, Catenary Arches, and Hyperbolic Functions

4.3.1 The 1691 Bernoulli Contest, Catenary Chains & the Gateway Arch

A flexible cable or chain suspended between two supports assumes a distinct static equilibrium curve under its own gravitational weight. In the *Two New Sciences* (1638), Galileo Galilei conjectured that a hanging chain forms a quadratic parabola ($y = kx^2$). In 1646, seventeen-year-old Christiaan Huygens proved geometrically that the equilibrium curve of a uniform chain cannot be algebraic, so the analytical form of the curve remained an open problem.

In May 1690, Jakob Bernoulli published a challenge in the *Acta Eruditorum* calling for the exact equation of the hanging chain. Within a year, Christiaan Huygens, Gottfried Wilhelm Leibniz, and Johann Bernoulli solved the problem independently. They showed that a parabola arises only when weight is distributed uniformly per unit of *horizontal span dx*, as in a suspension bridge whose level roadway deck hangs from vertical suspenders spaced uniformly along the horizontal axis. In an unburdened flexible chain or overhead transmission line, weight is distributed uniformly per unit of *curved arc length* $ds = \sqrt{(1 + (dy/dx)^2)}\ dx$ along the conductor itself; near the towers where the inclination is steep, each unit of horizontal span carries a greater length and weight of cable than near the vertex.

Huygens, Leibniz, and Johann Bernoulli formulated the static equilibrium of a chain segment of arc length s measured from the lowest vertex (where the chain is horizontal under tension T_0) to a point (x, y) where the tangent makes angle ψ with the horizontal. Because no external horizontal forces act on the segment, the horizontal component of tension $T \cos \psi$ must equal the vertex tension T_0 everywhere along the span ($T \cos \psi = T_0$). Vertically, the upward tension component $T \sin \psi$ must balance the gravitational weight $w \cdot s$ of the arc length s suspended below (x, y), where w is the linear weight density in Newtons per meter ($T \sin \psi = w \cdot s$). Dividing the vertical balance by the horizontal balance eliminates the variable tension T and relates the local slope $dy/dx = \tan \psi$ to the arc length s:

$$\frac{dy}{dx} = \tan\psi = \frac{w \cdot s}{T_0} = \frac{s}{a}, \qquad \text{where } a = \frac{T_0}{w} \tag{Eq. 4.18}$$

Differentiating with respect to x yields the differential equation $d^2y / dx^2 = (1/a)ds/dx = (1/a)\sqrt{(1 + (dy/dx)^2)}$, whose solution is the symmetric combination of rising and decaying exponentials $e^{x/a}$ and $e^{-x/a}$. Huygens termed the curve the *catenary* (from the Latin *catena*, "chain"), corresponding to the *hyperbolic cosine*:

$$\cosh(u) = \frac{e^u + e^{-u}}{2}, \qquad \sinh(u) = \frac{e^u - e^{-u}}{2}, \qquad \tanh(u) = \frac{\sinh(u)}{\cosh(u)} = \frac{e^u - e^{-u}}{e^u + e^{-u}} \tag{Eq. 4.19}$$

With the origin placed at the lowest vertex of the span, the vertical sag $y(x)$ follows the hyperbolic cosine while the arc length $s(x)$ measured from the vertex to x follows the hyperbolic sine:

$$y(x) = a\left(\cosh\left(\frac{x}{a}\right) - 1\right), \qquad s(x) = a \sinh\left(\frac{x}{a}\right) \tag{Eq. 4.20}$$

These exponential combinations are termed *hyperbolic* functions and share keys with sin, cos, and tan because whereas the circular functions ($\cos u$, $\sin u$) parameterize the unit circle $x^2 + y^2 = 1$ (sweeping out a circular sector of area $u/2$), squaring and subtracting

cosh u and sinh u shows that (cosh u, sinh u) parameterize the right branch of the unit hyperbola x^2 - y^2 = 1 (sweeping out a hyperbolic sector of area $u/2$):

$$\cosh^2(u) - \sinh^2(u) = \left(\frac{e^u + e^{-u}}{2}\right)^2 - \left(\frac{e^u - e^{-u}}{2}\right)^2 = \frac{e^{2u} + 2 + e^{-2u} - (e^{2u} - 2 + e^{-2u})}{4} = 1 \qquad \textbf{(Eq. 4.21)}$$

Their derivatives parallel the circular functions without sign reversal on (d/du) cosh u:

$$\frac{d}{du}\sinh(u) = \cosh(u), \qquad \frac{d}{du}\cosh(u) = +\sinh(u), \qquad \frac{d}{du}\tanh(u) = \operatorname{sech}^2(u) = 1 - \tanh^2(u) \qquad \textbf{(Eq. 4.22)}$$

Inverting a hanging catenary curve exchanges pure axial tension for pure axial compression without bending moment. As Robert Hooke stated to the Royal Society in 1675, *"As hangs the flexible chain, so but inverted will stand the rigid arch,"* the structural principle applied by architect Eero Saarinen and engineer Hannskarl Bandel in the 192-meter stainless-steel catenary profile of the Gateway Arch in St. Louis.

4.3.2 Ocean Tsunami Dispersion, Relativistic Rapidity & The HYP Prefix

On the SC-32 keypad, the hyperbolic prefix [↖] [HYP] is located on Left Shift [↖] above [R↓]. Pressing [↖] [HYP] activates the transient **HYP** annunciator and pairs with the primary keys [SIN], [COS], and [TAN] for sinh(x), cosh(x), and tanh(x), or with their Left-Shifted counterparts [↖] [ASIN], [↖] [ACOS], and [↖] [ATAN] for the inverse hyperbolic functions (because hyperbolic arguments are dimensionless real numbers, [↖] [HYP] operations are independent of **DEG**/**RAD**/**GRAD** angular modes):

Function	Key Sequence	Mathematical Definition	Valid Domain
Hyperbolic Sine	[↖] [HYP] [SIN]	$\sinh(x) = (e^x - e^{-x}) / 2$	$(-\infty, +\infty)$
Hyperbolic Cosine	[↖] [HYP] [COS]	$\cosh(x) = (e^x + e^{-x}) / 2$	$(-\infty, +\infty)$
Hyperbolic Tangent	[↖] [HYP] [TAN]	$\tanh(x) = \sinh(x)/\cosh(x)$	$(-\infty, +\infty)$
Inverse Sinh	[↖] [HYP] [↖] [ASIN]	$\operatorname{arsinh}(x) = \ln(x + \sqrt{(x^2 + 1)})$	$(-\infty, +\infty)$
Inverse Cosh	[↖] [HYP] [↖] [ACOS]	$\operatorname{arcosh}(x) = \ln(x + \sqrt{(x^2 - 1)})$	$x \geq 1.0$
Inverse Tanh	[↖] [HYP] [↖] [ATAN]	$\operatorname{artanh}(x) = (1/2)\ln((1+x) / (1-x))$	$\lvert x \rvert < 1.0$

Beyond catenary cables and masonry arches, the hyperbolic tangent tanh(u) governs physical systems that transition smoothly between linear and saturated asymptotic regimes. When surface gravity waves of wavelength λ (wavenumber $k = 2\pi/\lambda$) propagate across an ocean of depth h, George Biddell Airy derived from potential-flow boundary conditions at the seabed and free surface that their phase velocity c_{wave} depends on the hyperbolic tangent of the dimensionless depth-to-wavelength product kh:

$$c_{\text{wave}} = \sqrt{\frac{g}{k}\tanh(kh)} \qquad \textbf{(Eq. 4.23)}$$

In deep water ($kh \gg 1$), tanh(kh) $\to$ 1, so long-wavelength swell travels faster than short-wavelength chop ($c_{\text{wave}} \approx \sqrt{(g/k)}$). In shallow water—or during a transoceanic tsunami whose wavelength far exceeds the ocean depth ($kh \ll 1$)—tanh(kh) $\approx kh$, the wavenumber k cancels out of the radical, and the wave propagates across the basin near the non-dispersive shallow-water speed $c_{\text{wave}} \approx \sqrt{gh}$.

The hyperbolic tangent also linearizes relativistic velocity addition. In Galilean kinematics, collinear velocities add linearly ($v_1 + v_2$), which would permit successive boosts to exceed the speed of light c. In 1905, Albert Einstein showed that Lorentz transformations combine normalized collinear velocities $\beta_i = v_i/c$ according to $\beta_{\text{total}} = (\beta_1 + \beta_2)/(1 + \beta_1\beta_2)$. Three years later, Hermann Minkowski identified that relation as the hyperbolic tangent addition

identity $\tanh(\vartheta_1 + \vartheta_2) = (\tanh\vartheta_1 + \tanh\vartheta_2)/(1 + \tanh\vartheta_1 \tanh\vartheta_2)$. Mapping each normalized velocity $\beta = v/c$ to its hyperbolic angle (*rapidity*) $\vartheta = \operatorname{artanh}(v/c)$ restores additive composition in Minkowski spacetime:

$$\theta_{\text{total}} = \theta_1 + \theta_2 \Rightarrow \frac{v_{\text{total}}}{c} = \tanh\left(\operatorname{artanh}\left(\frac{v_1}{c}\right) + \operatorname{artanh}\left(\frac{v_2}{c}\right)\right) \qquad \textbf{(Eq. 4.24)}$$

Because $\tanh(\vartheta)$ approaches 1.0 asymptotically as $\vartheta \to \infty$, arbitrarily many finite rapidity boosts sum without ever exceeding c.

4.4 Circular, Vector & Hyperbolic Projections

Applying the unit circle, the (Y, X) vector plane, and the unit hyperbola on the keypad relies on the distinction between single-register circular projections and two-register vector transformations.

First, every circular and hyperbolic key ([SIN], [COS], [TAN], [↶] [ASIN], [↶] [ACOS], [↶] [ATAN], and their [↶] [HYP] counterparts) acts as a unary operator on Register X in place and folds its argument via extended-precision Cody-Waite octant reduction without disturbing any physical baseline waiting in Registers Y, Z, or T. Second, the coordinate keys shifted above [1]—[↶] [▸θ,r] (y [ENTER] x [↶] [▸θ,r]) and [↷] [▸y,x] (ϑ [ENTER] r [↷] [▸y,x])—couple Registers Y and X as a simultaneous 2D vector pair whose rectangular output ($Y = y$, $X = x$) feeds directly into [Σ+].

4.4.1 Angular Mode Verification & Direct Circular Projections

Standard surveying, mechanical, and navigational projections operate in Degrees mode ([DEG]) unless radians are explicitly required. Before starting a trigonometric calculation, a two-keystroke check via **45** [SIN] confirms the active angular mode: in Degrees mode ([DEG]), $\sin(45°)$ displays the unit-diagonal projection **0.707106781187** ($\sqrt{2}/2$), whereas in radian mode ([RAD]), $\sin(45\text{ rad})$ returns **0.850903524534**.

Planar trigonometric projections execute across Registers X and Y alone while Registers Z and T remain at zero because every circular function key ([SIN], [COS], [TAN]) is a unary in-place operator: when an angle is keyed into Register X and [SIN] or [COS] is pressed, the angle is replaced in Register X by its dimensionless unit-circle projection without lifting or disturbing any value waiting in Register Y:

- **Galileo's Projectile Apex & Range ($H_{\max} = (v_0 \sin\vartheta)^2 / 2g$, $R = v_0^2 \sin(2\vartheta) / g$):** Under Galileo's orthogonal parabolic decomposition for an artillery shell leaving the muzzle at $v_0 = 85.0$ m/s inclined at $\vartheta = 38.0°$ ($g = 9.80665$ m/s^2), hold the muzzle speed 85.0 m/s in Register Y while transforming 38° in place in Register X into its vertical unit-circle projection $\sin(38°) = 0.615661$. Multiplying those two terms yields the initial vertical climb velocity $v_{0y} = 52.3312$ m/s, and squaring in place with [↶] [x²] produces $v_{0y}{}^2 = 2{,}738.5574$ m^2/s^2; dividing by $2g = 19.6133$ m/s^2 gives the apex altitude. For the horizontal downrange impact distance across level terrain, evaluating the double-angle factor $\sin(2\vartheta) = \sin(76°) = 0.9703$ and scaling by $v_0{}^2 / g$ resolves R across the lower two registers:
  - **Projectile Apex Altitude ($H_{\max}$): 85 [ENTER] 38 [SIN] [×] [↶] [x²] 2 [ENTER] 9.80665 [×] [÷] → 139.62755641 m.**
  - **Horizontal Downrange Distance (R): 85 [↶] [x²] 38 [ENTER] 2 [×] [SIN] [×] 9.80665 [÷] → 714.8604898 m.**

- **Inclined-Plane Friction & LASTX Angle Recovery ($a = g(\sin\vartheta - \mu_k \cos\vartheta)$):** Along an inclined ramp where a supply crate slides down a surface tilted at $\vartheta = 34.0°$ against a kinetic friction coefficient of $\mu_k = 0.22$ ($g = 9.80665$ m/s^2), two perpendicular projections of gravity compete along the incline: the downslope component $\sin(34°) = 0.5592$ g and the normal force $\cos(34°) = 0.8290$ g, which scales by $\mu_k = 0.22$ into an upslope friction drag of 0.1824 g. Place $\sin(34°) = 0.5592$ in Register Y, form the friction term $0.22\cos(34°) = 0.1824$ in Register X (recalling 34° from [↶] [LASTx] if desired so the angle is not keyed twice), subtract them into the net traction factor 0.3768, and multiply by g:

  - **Downhill Crate Acceleration:** **34** [SIN] **34** [COS] **0.22** [×] [-] **9.80665** [×] → $a = \mathbf{3.6952\ m/s^2}$.

- **Two-Station Optical Mountain Triangulation ($H = d\ /\ (\cot\alpha - \cot\beta)$):** Applying the Great Trigonometrical Survey baseline formula across a level baseline of $d = 500.0$ m—measuring elevation angle $\alpha = 24.5°$ at Station 1 and steeper angle $\beta = 31.2°$ at Station 2—invert $\tan(24.5°)$ via [TAN] [x^{-1}] into $\cot(24.5°) = 2.1943$ in Register Y, subtract $\cot(31.2°) = 1.6512$ in Register X to isolate the differential cotangent $\Delta\cot = 0.5431$ m/m, and use [$x\leftrightarrow y$] to divide the 500 m baseline by 0.5431:

  - **Triangulated Summit Height:** **24.5** [TAN] [x^{-1}] **31.2** [TAN] [x^{-1}] [-] **500** [$x\leftrightarrow y$] [÷] → $H = \mathbf{920.63\ meters}$.

- **Large-Amplitude Pendulum Period Dilation ($T \approx T_0[1 + (1/4)\sin^2(\vartheta_0/2)]$):** To evaluate Huygens's nonlinear period lengthening for a clock pendulum ($T_0 = 2.000$ s) released from a $\vartheta_0 = 45.0°$ amplitude, square the half-angle sine $\sin(22.5°)$ in place to $\sin^2(22.5°) = 0.1464$, scale by $1/4$ and add 1 to form the dilation factor 1.0366, and multiply by $T_0 = 2.0$ s:

  - **Stretched Pendulum Period:** **22.5** [SIN] [↶] [x^2] **0.25** [×] **1** [+] **2.0** [×] → $T = \mathbf{2.0732\ seconds}$ (+3.66%).

- **Multi-Tier Trigonometric Quotients & Nested Radical Cascades:** Compound trigonometric ratios and nested radicals such as:

$$x_1 = \frac{\sin(35°)\cos(42°) + \tan(18°)}{\sqrt{\sin^2(65°) + \cos^2(25°)}}, \qquad x_2 = \sqrt{\cos(28°) + \sqrt{\sin(48°) + \sqrt{\tan(15°)}}} \qquad \text{(Eq. 4.25)}$$

never touch Register T because every circular function, square, and square root transforms Register X in place. In x_1, the numerator folds into 0.7512 in Register Y while the denominator $\sqrt{(\sin^2(65°) + \cos^2(25°))} = 1.2817$ assembles in Register X. In the three-level nested radical x_2, working from the innermost root outward evaluates $\sqrt{(\tan(15°))} = 0.5176$, adds $\sin(48°)$ and takes [$\sqrt{x}$] (→ 1.1228), and adds $\cos(28°)$ before the final [$\sqrt{x}$]:

  - **Compound Trigonometric Quotient (x_1):** **35** [SIN] **42** [COS] [×] **18** [TAN] [+] **65** [SIN] [↶] [x^2] **25** [COS] [↶] [x^2] [+] [$\sqrt{x}$] [÷] → $\mathbf{0.5861}$.
  - **Nested Trigonometric Radical (x_2):** **15** [TAN] [$\sqrt{x}$] **48** [SIN] [+] [$\sqrt{x}$] **28** [COS] [+] [$\sqrt{x}$] → $\mathbf{1.4162603903}$.

4.4.2 Inverse Trigonometry & Oblique Triangles on the Stack

Just as [SIN], [COS], and [TAN] project an angle onto a dimensionless ratio in Register X, the inverse keys [↶] [ASIN], [↶] [ACOS], and [↶] [ATAN] recover the geometric angle from a physical ratio directly inside Register X:

- **Optical Fiber Critical Angle ([↶] [ASIN]) & Banked Highway Super-Elevation ([↶] [ATAN]):** Applying Ibn Sahl and Snell's total-internal-reflection law to laser light inside a silica optical fiber core ($n_1 = 1.480$) surrounded by glass cladding ($n_2 = 1.450$), divide n_2 / $n_1 = 0.9797$ and tap [↶] [ASIN] to find the critical trapping angle ϑ_c. Similarly, to bank a highway curve of radius $R = 350.0$ m for vehicles at $v = 100.0$ km/h (27.78 m/s, $g = 9.80665$ m/s^2), form the lateral acceleration ratio $\tan\vartheta = v^2/gR = 0.2248$ in Register X and tap [↶] [ATAN]:
  - **Optical Fiber Critical Angle ($\vartheta_c = \arcsin(n_2/n_1)$): `1.450` [ENTER] `1.480` [÷] [↶] [ASIN] → $\vartheta_c =$ 78.4441°.**
  - **Zero-Slip Highway Banking Angle ($\vartheta = \arctan(v^2/gR)$): `100` [ENTER] `3.6` [÷] [↶] [x²] `9.80665` [ENTER] `350` [×] [÷] [↶] [ATAN] → $\vartheta =$ 12.6698°.**

- **Four-Register Geocentric Satellite Look-Angle Elevation ($\tan E = (\cos\gamma - R_E/(R_E + h)) \,/\, \sin\gamma$):** Contrast that single-ratio highway turn with pointing a ground-station tracking dish at a low-Earth-orbit satellite at altitude $h = 850$ km above an Earth of radius $R_E = 6{,}371$ km and geocentric angular separation $\gamma = 18.0°$. This trace (`18` [COS] `6371` [ENTER] `6371` [ENTER] `850` [+] [÷] [-] `18` [SIN] [÷] [↶] [ATAN] → E = 12.546175175°) engages all four registers (T, Z, Y, X) because after $\cos(18°) = 0.9511$ is evaluated on Row 1, computing the Earth-to-orbit radius ratio 6371 / (6371 + 850) = 0.8823 requires two [ENTER] strokes that lift 0.9511 up into Register Z and Register T. When [+] and [÷] collapse that radius ratio into 0.8823, Register T's automatic replication copies 0.9511 back down into Register Y so [-] can form the net vertical clearance 0.9511 - 0.8823 = 0.0688 in Register X:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`18` [COS]	0.0000	0.0000	0.0000	0.9511	Cosine of angular distance cos γ
`6371` [ENTER] `6371` [ENTER] `850` [+] [÷] [-]	0.9511	0.9511	0.9511	0.0688	Subtract orbital radius ratio R_E / ($R_E + h$)
`18` [SIN] [÷]	0.9511	0.9511	0.9511	0.2225	Divide by sine sin γ for elevation tangent
[↶] [ATAN]	0.9511	0.9511	0.9511	**12.5462**	Extract ground satellite elevation angle E (degrees)

- **Keplerian True Anomaly, the Law of Cosines & Wave Interference:** Because a satellite in an elliptical orbit ($e = 0.240$) speeds up near perigee and slows down near apogee, Carl Friedrich Gauss's half-angle tangent transformation $\tan(\nu/2) = \sqrt{((1 + e) / (1 - e))}\ \tan(E/2)$ converts Kepler's uniform-circle *eccentric anomaly* $E = 40.0°$ into the *true polar anomaly* ν—the real sky angle from focus to satellite. Similarly, in oblique-triangle surveying and coherent wave superposition—a boundary corner with sides $a = 45.0$ m and $b = 62.0$ m separated by $C = 48.0°$ ($c = \sqrt{(a^2 + b^2 - 2ab \cos C)}$), or two coherent sound waves of amplitudes $A_1 = 2.40$ Pa and $A_2 = 3.10$ Pa interfering at phase angle $\Delta\varphi = 55.0°$ ($A_{\text{res}} = \sqrt{(A_1{}^2 + A_2{}^2 + 2A_1\ A_2 \cos \Delta\varphi)}$)—form the Pythagorean sum of squares in Register Y, build the $2ab \cos \vartheta$ cross-term in Register X, subtract or add, and press [√x]:

  - **Orbital True Anomaly (ν at $E = 40°$, $e = 0.240$):** `1.240` [ENTER] `0.760` [÷] [√x] `20` [TAN] [×] [↶] [ATAN] `2` [×] → $\nu = 49.8685°$.
  - **Law of Cosines Boundary Length (c):** `45` [↶] [x²] `62` [↶] [x²] [+] `2` [ENTER] `45` [×] `62` [×] `48` [COS] [×] [-] [√x] → $c = 46.208778566$ **m**.
  - **Coherent Acoustic Wave Superposition (A_{res}):** `2.4` [↶] [x²] `3.1` [↶] [x²] [+] `2` [ENTER] `2.4` [×] `3.1` [×] `55` [COS] [×] [+] [√x] → $A_{\text{res}} = 4.8893$ **Pa**.

4.4.3 Catenary Sag, Tsunami Wave Dispersion & Relativistic Rapidity on the Stack

Catenary sag, ocean wave dispersion, and relativistic velocity addition follow the same inside-out hyperbolic evaluation pattern. When a high-voltage transmission cable spans $L = 200$ meters, its half-span is $x = 100$ m; with catenary parameter $a = T_0 / w = 800$ meters, the midpoint sag $y = 800[\cosh(100/800) - 1]$ evaluates from the dimensionless ratio $x/a = 0.125$ outward ([↶] [HYP] [COS], subtract 1, and multiply by 800 m). In physical oceanography, Airy's wave dispersion equation $c_{\text{wave}} = \sqrt{((g/k) \tanh(kh))}$ across a $h = 4{,}000$ m ocean basin for a tsunami of wavelength $\lambda = 200{,}000$ m ($k = 2\pi/\lambda \approx 0.000031416$ m^{-1}, $kh = 0.12566$) applies [↶] [HYP] [TAN] to 0.12566, scales by g/k, and takes [√x] to yield a transoceanic wave speed of $c_{\text{wave}} \approx 197.53$ **m/s** (711.1 km/h).

For Einstein and Minkowski's relativistic velocity addition, an interstellar probe cruising at $v_1 = 0.75c$ relative to Earth that launches a probe forward at $v_2 = 0.60c$ relative to the ship converts both speeds to hyperbolic rapidities via [↶] [HYP] [↶] [ATAN], adds them on the stack ($\vartheta_{\text{total}} = 0.9730 + 0.6931 = 1.6661$), and maps the sum back via [↶] [HYP] [TAN]:

- **Catenary Transmission-Line Sag:** `100` [ENTER] `800` [÷] [↶] [HYP] [COS] `1` [-] `800` [×] → **6.2581422606 m**.
- **Relativistic Velocity Addition ($0.75c \oplus 0.60c$):** `0.75` [↶] [HYP] [↶] [ATAN] `0.60` [↶] [HYP] [↶] [ATAN] [+] [↶] [HYP] [TAN] → **0.9310344828 *c*** (93.10% of light speed).

4.4.4 2D Vector Projections, Wind Triangles, Traverses & 3D Space Frames

The ($Y = \vartheta$, $X = r$) ↔ ($Y = y$, $X = x$) coordinate keys operate directly on Registers Y and X. Keying a due-North vector of radius $r = 10$ at $\vartheta = 90°$ via `90` [ENTER] `10` [R↓] [▸y,x] verifies the register mapping: because [R↓] [▸y,x] takes Angle ϑ in Register Y and Radius r in Register X, the projection places `0.0000` in Register X ($x = 10 \cos 90°$) and `10.0000` in Register Y ($y = 10 \sin 90°$). Conversely, keying a 3-4-5 right triangle ($y = 4$ in Register Y, $x = 3$ in Register X) and pressing [↶] [▸θ,r] places the angle $\vartheta = 53.1301°$ in Register Y and

the hypotenuse $r = 5.0000$ in Register X:



Figure: Direct rectangular-to-polar vector conversion on the StackCalc SC-32 places the angle $\vartheta = 53.1301^\circ$ in Register Y and the radial magnitude $r = 5.0000$ in Register X.

Four applied vector calculations illustrate how [R↓] [▸y,x], [↖] [▸θ,r], [Σ+], and 3D dot products interact across the registers:

- **AC Electrical Power Triangle (S = 1,020 VA at φ = 32.0°):** When an inductive AC motor draws $I_{\text{rms}} = 8.50$ A from a $V_{\text{rms}} = 120.0$ V line at a lagging phase angle of $\varphi = 32.0^\circ$, total apparent power $S = V_{\text{rms}} I_{\text{rms}} = 1{,}020$ VA splits into horizontal real shaft power $P = S \cos\varphi$ (Watts) and vertical reactive magnetic power $Q = S \sin\varphi$ (VAR). Duplicating $S = 1{,}020$ VA across the stack and scaling by $\cos(32^\circ)$ and $\sin(32^\circ)$ (**120** [ENTER] **8.5** [×] [ENTER] [ENTER] **32** [COS] [×] [x⇄y] **32** [SIN] [×]) places P = **865.01 W** in Register Y and Q = **540.51764952 VAR** in Register X (or in a single coordinate stroke, keying **32** [ENTER] **1020** [R↓] [▸y,x] [x⇄y] resolves both legs onto (Y, X) simultaneously):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
120 [ENTER] **8.5** [×]	0.0000	0.0000	0.0000	1020.0000	Total apparent electrical power $S = V_{\text{rms}} I_{\text{rms}}$ (VA)
[ENTER] [ENTER] **32** [COS] [×]	0.0000	0.0000	1020.0000	865.0091	Multiply by power factor $\cos(32^\circ)$ for real power P (W)
[x⇄y] **32** [SIN] [×]	0.0000	0.0000	865.0091	**540.5176**	Multiply by reactive factor $\sin(32^\circ)$ for reactive power Q (VAR)

- **Aircraft Crosswind Vector Triangle ($\boldsymbol{V}_{\text{ground}} = \boldsymbol{V}_{\text{air}} + \boldsymbol{V}_{\text{wind}}$):** For an aircraft flying at true airspeed $V_{\text{air}} = 220.0$ kts on North-Azimuth heading $\vartheta_1 = 45.0^\circ$ with a 35.0-kt downwind drift vector toward azimuth $\vartheta_2 = 130.0^\circ$ (a wind from 310°), projecting both vectors via [R↓] [▸y,x] places Easting velocity in Y and Northing velocity in X:
  - **Airspeed Projection ($\vartheta_1 = 45^\circ$, $r_1 = 220$ kts):** **45** [ENTER] **220** [R↓] [▸y,x] → **155.56349186** ($V_{\text{air},\,E} = 155.5635$ kts in Y, $V_{\text{air},\,N} = 155.5635$ kts in X; store in **A** and **B**).
  - **Crosswind Projection ($\vartheta_2 = 130^\circ$, $r_2 = 35$ kts):** **130** [ENTER] **35** [R↓] [▸y,x] → places $V_{\text{wind},\,E} = 26.8116$ kts in Y and $V_{\text{wind},\,N}$ = **-22.49756634** kts in X.
  - **Component Summation & Polar Recovery ([↖] [▸θ,r]):** Summing components ($V_{\text{ground},\,E} = 182.3750$ kts in Y, $V_{\text{ground},\,N} = 133.0659$ kts in X) and pressing [↖] [▸θ,r] yields ground speed V_{ground} = **225.76 knots** in X and true compass track ϑ_{track} = **53.89°** in Y.

- **Four-Leg Land Survey Traverse ([→] [▸y,x] [Σ+]):** When walking a four-station closed-loop boundary around a rectangular parcel after clearing statistical memory ([CLΣ]), chaining [→] [▸y,x] [Σ+] projects each North-Azimuth course (α_i, r_i) into (ΔE_i, ΔN_i) and accumulates both coordinates straight into the statistical registers:
  - **Leg 1 (α = 60°, r = 100 m): 60 [ENTER] 100 [→] [▸y,x] [Σ+]** → projects (ΔE = 86.6025 m, ΔN = 50.0000 m) at station n = **1**.
  - **Leg 2 (α = 150°, r = 80 m): 150 [ENTER] 80 [→] [▸y,x] [Σ+]** → adds (ΔE = 40.0000 m, ΔN = -69.2820 m) at station n = **2**.
  - **Leg 3 (α = 240°, r = 100 m): 240 [ENTER] 100 [→] [▸y,x] [Σ+]** → adds (ΔE = -86.6025 m, ΔN = -50.0000 m) at station n = **3**.
  - **Leg 4 (α = 330°, r = 80 m): 330 [ENTER] 80 [→] [▸y,x] [Σ+]** → adds (ΔE = -40.0000 m, ΔN = 69.2820 m) and closes the loop at n = **4**, where [→] [SUMS] confirms zero closure error ($\sum x = \sum \Delta N$ = 0.0000 m, $\sum y = \sum \Delta E$ = 0.0000 m).

- **Surveyor DMS Bearings ([←] [▸HR]), Highway Curves, Pilot Endurance ([→] [▸HMS]) & CNC Bolt Circles:** In field surveying, air navigation, forestry, and toolroom machining, sexagesimal **D.MMSS** / **H.MMSS** conversions ([←] [▸HR] and [→] [▸HMS] above [5]) and polar-rectangular projections ([→] [▸y,x]) eliminate scratchpad conversion tables:
  - **DMS Theodolite Bearing Projection & Horizontal Highway Curve Stakeout:** When a total station records a North-Azimuth bearing of 42° 18' 36'' (**42.1836**) over a horizontal distance of 125.00 m, pressing [←] [▸HR] converts the sexagesimal angle in place to 42.3100°, and keying **125** [→] [▸y,x] resolves the leg into Easting departure ΔE = **84.1386 m** in Register Y and Northing latitude ΔN = **92.4392 m** in Register X. Similarly, staking out a horizontal circular highway curve of radius R = 250.00 m and central deflection angle Δ = 36° 24' 00'' (**36.24** [←] [▸HR] → 36.4000°) evaluates the long chord $L_c = 2R \sin(\Delta/2)$ = 500 sin(18.20°) = **156.1675 m** and middle ordinate $M = R(1 - \cos(\Delta/2))$ = **12.5028 m**.
  - **Pilot Cockpit Time-En-Route & Fuel Endurance ([→] [▸HMS]):** Flying a 385.0 NM cross-country leg at a groundspeed of 162.0 knots takes 385 / 162 = 2.3765 decimal hours. Duplicating that quotient with [ENTER] and pressing [→] [▸HMS] converts Register X into **2.2236** (2 hr 22 min 36 s en route), while swapping via [x⇄y] and multiplying the decimal hours by 14.8 gal/hr fuel flow yields **35.1728 gallons** burned with the **H.MMSS** ETA (**2.2236**) waiting in Register Y.
  - **Forestry Clinometer Tree Height & Machinist Sine-Bar / Bolt-Hole Circle Layout:** Sighting a Douglas fir from a horizontal tape distance D = 30.0 m with clinometer angles ϑ_{top} = +52.0° and ϑ_{base} = -8.0° (**52** [TAN] **8** [+/−] [TAN] [−] **30** [×]) yields a total height of **42.6143 m**. In a machine shop, grinding a 14° 15' 00'' (**14.15** [←] [▸HR] [SIN] **5** [×]) taper on a 5.0000-inch sine bar requires a **1.2308-inch** gauge-block stack (wrung from 0.1008'' + 0.130'' + 1.000'' blocks), while indexing the 60° hole on a R = 3.2500-inch bolt circle (**60** [ENTER] **3.25** [→] [▸y,x]) places both milling coordinates directly on (Y = **2.8146 in**, X = **1.6250 in**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
42.1836 [↶] [▸HR]	0.0000	0.0000	0.0000	42.3100	Convert DMS bearing 42° 18' 36'' to 42.3100°
125 [R↓] [▸y,x]	0.0000	0.0000	84.1386	92.4392	Project 125 m survey course to (Δy, Δx) (m)
385 [ENTER] **162** [÷]	84.1386	84.1386	92.4392	2.3765	Flight time 385 NM / 162 kts = 2.3765 hr
[ENTER] [R↓] [▸HMS]	84.1386	92.4392	2.3765	2.2236	Convert leg time to **H.MMSS** (2h 22m 36s)
[x⇄y] **14.8** [×]	92.4392	92.4392	2.2236	**35.1728**	Fuel burn (35.17 gal) in X; **H.MMSS** ETA in Y

- **Mohr's Circle Strain Tensor, 3D Space-Frame Mitre Joints & Transpacific Great Circles:** Multi-term directional cosines fold into Register X without overflowing the 4-level stack:
  - **Mohr's Circle Strain Transformation ($\varepsilon_{x'} = (\varepsilon_x + \varepsilon_y)/2 + ((\varepsilon_x - \varepsilon_y)/2)\cos(2\vartheta) + (\gamma_{xy}/2)\sin(2\vartheta)$):** For an aircraft wing rosette with $\varepsilon_x = 450$ με, $\varepsilon_y = 120$ με, and $\gamma_{xy} = 280$ με rotated by $\vartheta = 30.0°$ ($2\vartheta = 60.0°$), accumulate the mean strain 285.0 με, normal projection 82.5 με, and shear projection 121.24 με sequentially (**450** [ENTER] **120** [+] **2** [÷] **450** [ENTER] **120** [-] **2** [÷] **60** [COS] [×] [+] **280** [ENTER] **2** [÷] **60** [SIN] [×] [+] → $\varepsilon_{x'} \approx$ **488.74 με**; [↶] [▸θ,r] on 140 and 165 με extracts R_{Mohr} directly).
  - **3D Space-Frame Cauchy-Schwarz Bevel Angle ($\cos\vartheta = \boldsymbol{A}\cdot\boldsymbol{B} / |\boldsymbol{A}||\boldsymbol{B}|$):** For skew struts $\boldsymbol{A}$ = (4, 6, 8 m) ($|\boldsymbol{A}|^2 = 116$ m^2) and $\boldsymbol{B}$ = (2, -3, 5 m) ($|\boldsymbol{B}|^2 = 38$ m^2) with dot product $\boldsymbol{A}\cdot\boldsymbol{B} = 30.0$ m^2, divide 30.0 by √116 and √38 ([√x] [÷]) and apply [↶] [ACOS]: **30** [ENTER] **116** [√x] [÷] **38** [√x] [÷] [↶] [ACOS] → $\vartheta =$ **63.1371°**.
  - **Transpacific Great-Circle Orthodrome (San Francisco → Tokyo Narita):** Applying the Spherical Law of Cosines between SFO ($\varphi_1 = 37.6189°$ N, $\lambda_1 = -122.3750°$ W) and NRT ($\varphi_2 = 35.7720°$ N, $\lambda_2 = 140.3929°$ E, span $\Delta\lambda = 97.2321°$), form $\sin\varphi_1 \sin\varphi_2 = 0.356820$ (**37.6189** [SIN] **35.7720** [SIN] [×]), assemble $\cos\varphi_1 \cos\varphi_2 \cos\Delta\lambda = -0.080904$ (**37.6189** [COS] **35.7720** [COS] [×] **97.2321** [COS] [×]), sum into $\cos d = 0.275916$, and apply [↶] [ACOS] (→ $d = 73.9833°$, or **8,226.6 km** at R_E = 6,371 km).
  - **Heliostat Solar Concentrator Elevation ($\sin\alpha = \sin\varphi\sin\delta + \cos\varphi\cos\delta\cos\omega$):** At latitude $\varphi = 32.0°$ N, declination $\delta = 18.5°$, and hour angle $\omega = 45.0°$, summing the two spherical projections and pressing [↶] [ASIN] (**32** [SIN] **18.5** [SIN] [×] **32** [COS] **18.5** [COS] [×] **45** [COS] [×] [+] [↶] [ASIN]) locks onto $\alpha \approx$ **47.46°** (47.4611°).

From Eratosthenes's shadow angle at Alexandria to space-frame bevels and relativistic rapidities, circular and hyperbolic functions project directed quantities onto orthogonal axes, while [R↓] [▸y,x] and [↶] [▸θ,r] unify planar vectors, traverses, and spherical astronomy across (Y, X).

Chapter 5: Complex Numbers, Euler's Identity & Alternating-Current Phasors

5.1 Multiplying by i as a Quarter-Turn in the Plane

Few concepts in algebra have been more obscured by historical nomenclature than $\sqrt{(-1)}$. Because the square of any positive real number is positive ($+3 \times +3 = +9$) and the square of any negative real number is likewise positive ($-3 \times -3 = +9$), no point on the real number line has a negative square. When René Descartes encountered square roots of negative quantities in his 1637 treatise *La Géométrie (Descartes, 1637)*, he termed them *"imaginary,"* a label that persisted for four centuries even after complex numbers became central to mathematical physics.

Historically, $\sqrt{(-1)}$ arose not from quadratic equations, which simply lack real intersections when the parabola lies above the horizontal axis, but from cubic equations whose roots are real. In sixteenth-century Italy, when Scipione del Ferro, Niccolò Fontana (Tartaglia), and Gerolamo Cardano derived the general algebraic solution to the depressed cubic $x^3 = px + q$, they applied it to $x^3 = 15x + 4$. Direct substitution confirms that $x = 4$ is a positive real root ($4^3 = 64 = 15(4) + 4$). Yet substituting the coefficients of $x^3 - 15x - 4 = 0$ into Cardano's formula expressed that real integer as the sum of two radicals involving $\sqrt{(-121)}$:

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}} \tag{Eq. 5.1}$$

In his 1572 treatise *L'Algebra*, the Italian hydraulic engineer Rafael Bombelli demonstrated that treating $\sqrt{(-121)}$ as $11\sqrt{(-1)}$ and applying the ordinary distributive and associative laws of arithmetic resolves the expression consistently. Cubing $(2 + \sqrt{(-1)})$ with the rule $(\sqrt{(-1)})^2 = -1$ yields $(2 + \sqrt{(-1)})^3 = 2 + 11\sqrt{(-1)}$, and cubing the conjugate expression yields $(2 - \sqrt{(-1)})^3 = 2 - 11\sqrt{(-1)}$. Extracting the cube roots and summing the two branches cancels the opposite $+\sqrt{(-1)}$ and $-\sqrt{(-1)}$ terms:

$$(2 + \sqrt{-1}) + (2 - \sqrt{-1}) = 4 \tag{Eq. 5.2}$$

Passing from the real coefficients of the cubic to its real root $x = 4$ required algebraic trajectories through a second dimension.

5.1.1 Wallis, Wessel, Argand, and Gauss: The Perpendicular Direction

The quantity $\sqrt{(-1)}$ cannot lie on the real line because real numbers span only one dimension, and multiplication by a negative number is geometrically a rotation rather than a reflection. Beginning with John Wallis in 1685 and completed by Caspar Wessel (1797), Jean-Robert Argand (1806), and Carl Friedrich Gauss (1831), mathematicians established the geometric interpretation of complex arithmetic in the plane.

For a unit vector at the origin 0 directed along the positive real axis toward $+1$ (0°), multiplying by -1 preserves its unit length while rotating its heading by 180° to point along the negative real axis toward -1. Multiplying by -1 a second time rotates the vector by a further 180°, completing a 360° revolution back to $+1$:

$$(-1) \times (-1) = +1 \iff 180^\circ + 180^\circ = 360^\circ \tag{Eq. 5.3}$$

Interpreting multiplication by -1 as a 180° half-turn immediately defines $i = \sqrt{(-1)}$ as a 90° counterclockwise quarter-turn. Starting at $i^0 = +1$ along the positive real axis at 0°, a single multiplication by i rotates the vector 90° counterclockwise onto the perpendicular vertical axis at $+i$. A second multiplication ($i \times i = i^2 = -1$) rotates the vector by a second 90° quarter-turn (90° + 90° = 180°) onto the negative real axis at -1; two 90° rotations compose into a 180° reversal, which is the geometric meaning of $i^2 = -1$. A third quarter-turn ($i^3 = -i$) directs the vector along the negative vertical axis at 270°, and a fourth quarter-turn ($i^4 = +1$) completes the 360° cycle back to +1.

Gauss proposed replacing Descartes's term "imaginary" with *lateral unit*. A *complex number* $z = a + bi$ specifies a point or vector in the two-dimensional *Complex Plane* (*Argand Diagram*) through a displacement of a units along the horizontal Real axis and b units along the perpendicular vertical Imaginary axis.

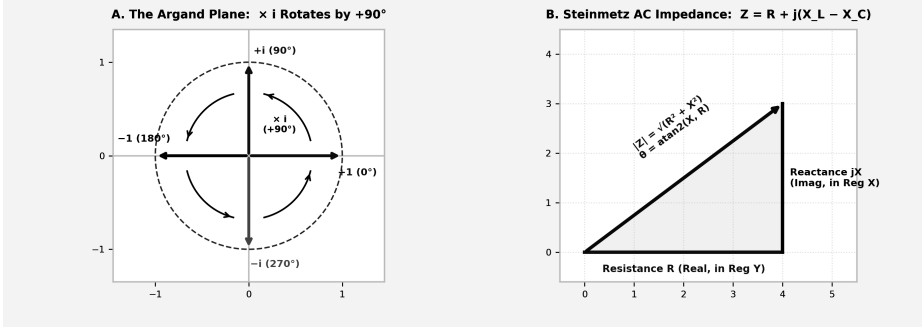


Figure: Multiplying by i as a 90-degree counterclockwise rotation in the Argand plane, paired with Steinmetz's AC impedance phasor triangle.

5.1.2 Unpeeling Complex Arithmetic and the Conjugate Mirror

Because a complex number $z = a + bi$ is an ordered pair on a two-dimensional plane, the SC-32 holds its horizontal Real component (a) in Register Y and its vertical Imaginary component (b) in Register X, while a second complex operand $z_2 = c + di$ entered into (Y, X) lifts the first operand into ($T = a$, $Z = b$). Pressing the transient [↖] [CMPLX] prefix ([↖] + [STO]) before [+], [−], [×], [÷], or [x^{-1}] executes the corresponding two-dimensional operation across the paired registers.

Under complex addition and subtraction ([↖] [CMPLX] [+] and [↖] [CMPLX] [−]), two complex vectors $z_1 = a + bi$ and $z_2 = c + di$ combine by head-to-tail translation, summing their horizontal components and vertical components independently:

$$(a+bi)+(c+di)=(a+c)+(b+d)i \qquad \textbf{(Eq. 5.4)}$$

Under complex multiplication ([↖] [CMPLX] [×]), expanding the binomial product $(a + bi)(c + di)$ with $i^2 = -1$ groups the two real terms ac and $i^2\ bd = -bd$ along the horizontal axis and the two cross-terms adi and bci along the vertical axis:

$$(a+bi)(c+di)=(ac-bd)+(ad+bc)i \qquad \textbf{(Eq. 5.5)}$$

Complex reciprocals ([↖] [CMPLX] [x^{-1}]) and division ([↖] [CMPLX] [÷]) rely on reflection across the real axis. Every complex number $z = c + di$ has a reflection twin called its *complex conjugate*, $z^* = c - di$, formed by negating the imaginary component with [+/−]. If z lies at angle $+\vartheta$ relative to the positive real axis, z^* lies at $-\vartheta$; multiplying z by its conjugate z^*

cancels their opposite angles (ϑ - ϑ = 0°) and yields the squared Euclidean modulus $|z|^2 = c^2 + d^2$ as a purely real number:

$$(c+di)(c-di) = c^2 - d^2 i^2 = c^2 + d^2 \quad \textbf{(Eq. 5.6)}$$

Multiplying the numerator and denominator of a complex quotient $(a + bi) / (c + di)$ by the denominator's conjugate c - di reduces the denominator to the real scalar $c^2 + d^2$, which is the algebraic reduction performed by [↶] [CMPLX] [x^{-1}] and [↶] [CMPLX] [÷]:

$$\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2} = \left(\frac{ac+bd}{c^2+d^2}\right) + \left(\frac{bc-ad}{c^2+d^2}\right)i \quad \textbf{(Eq. 5.7)}$$

5.1.3 Complex Eigenvalues in Damped Springs and Skyscrapers

This orthogonal separation between the horizontal Real axis and the vertical Imaginary axis characterizes linear damped oscillators across mechanical and structural engineering, from vehicle suspension struts to tuned mass dampers in tall buildings.

When a mass m coupled to a spring of stiffness k and a viscous damper of coefficient c is displaced from equilibrium, Newton's second law gives the homogeneous differential equation $mddotx + c\dot{x} + kx = 0$. Substituting an exponential trial solution $x(t) = e^{\lambda t}$ produces the quadratic characteristic equation $m\lambda^2 + c\lambda + k = 0$, whose roots are the system's *eigenvalues*:

$$\lambda = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} = -\alpha \pm i\omega_d \quad \textbf{(Eq. 5.8)}$$

Whenever the restoring stiffness exceeds the damping threshold ($k/m > (c/2m)^2$), the discriminant is negative and the eigenvalues form a complex conjugate pair $\lambda = -\alpha \pm i\omega_d$. The two axes of the complex plane separate the two physical processes acting simultaneously in the oscillator. Along the horizontal Real axis, the attenuation rate $-\alpha = -c/(2m)$ measures irreversible energy dissipation: viscous fluid forced through the damper orifices resists motion in phase with velocity and converts mechanical kinetic energy into heat, so the oscillation envelope decays exponentially as $e^{-\alpha t}$.

Along the vertical Imaginary axis, the damped natural frequency $\pm i\omega_d$ measures lossless quadrature energy exchange: the spring and mass dissipate no heat, exchanging kinetic energy in the moving mass ($(1/2)mv^2$, which peaks at the equilibrium crossing) with potential energy in the strained spring ($(1/2)kx^2$, which peaks at the turning points) 90° out of phase at angular frequency $\omega_d = \sqrt{(k/m - \alpha^2)}$.

In the undamped limit ($c \to 0$), the real part $-\alpha$ vanishes, the eigenvalues lie on the imaginary axis ($\lambda = \pm i\sqrt{(k/m)}$), and the oscillation persists at constant amplitude. At the critical damping threshold $c_{\text{crit}} = 2\sqrt{mk}$, the imaginary part ω_d vanishes and the displaced mass returns to equilibrium without oscillation.

5.2 Euler's Formula and De Moivre's Roots of Unity

Leonhard Euler's constant $e \approx 2.71828$ governs continuous exponential growth and decay along the real line, whereas $\cos\vartheta$ and $\sin\vartheta$ parameterize rotation around the unit circle. In his 1748 treatise *Introductio in analysin infinitorum* (building on Roger Cotes's 1714 logarithmic relation $i\vartheta = \ln(\cos\vartheta + i\sin\vartheta)$), Euler evaluated the exponential function e^z for an imaginary argument $z = i\vartheta$, establishing the fundamental identity of complex analysis:

$$e^{i\theta} = \cos\theta + i\sin\theta \quad \textbf{(Eq. 5.9)}$$

Prior to Euler's formulation, exponential functions (e^x, logarithms, and compound growth) and circular functions (cos ϑ, sin ϑ, and trigonometric tables) were treated as distinct families of transcendental functions. Euler's identity demonstrated that they are orthogonal cross-sections of a single holomorphic function e^z in the complex plane: along the horizontal real axis e^z scales exponentially in magnitude, whereas along the vertical imaginary axis e^z rotates at unit radius around the origin.

5.2.1 Continuous Imaginary Growth Around the Unit Circle

The unit-circle trajectory of $e^{i\vartheta}$ follows independently from differential kinematics and from power-series expansion.

In differential kinematics, when a state variable $z(t)$ evolves at a real growth rate k, its velocity vector $dz/dt = k \cdot z$ is collinear with the position vector z, driving the trajectory radially outward or inward along a straight ray from the origin. Replacing the real rate k with the imaginary rate $i\omega$ gives the differential equation:

$$\frac{dz}{dt} = i\omega z \quad \textbf{(Eq. 5.10)}$$

Because multiplication by i rotates any complex vector by 90° counterclockwise, the velocity vector dz/dt of $z(t) = e^{i\omega\ t}$ is at every instant perpendicular (90°) to the radius vector $z(t)$, identical to the tangential velocity of uniform circular motion. Because a velocity orthogonal to the radius performs zero radial work and changes the radial distance at rate $d/dt|z|^2 = 0$, the modulus remains invariant at $|e^{i\vartheta}| = 1$ for all real ϑ, while the point traces the unit circle with horizontal projection cos ϑ and vertical projection i sin ϑ.

Algebraically, substituting $x = i\vartheta$ into the Maclaurin series for e^x and reducing the powers $i^2 = -1$, $i^3 = -i$, and $i^4 = +1$ yields:

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots \quad \textbf{(Eq. 5.11)}$$

Separating the real even-power terms from the imaginary odd-power terms recovers the Maclaurin series for cosine and sine:

$$e^{i\theta} = \left(\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right)\right)_{[\cos\theta]} + i\left(\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)\right)_{[\sin\theta]} = \cos\theta + i\sin\theta \quad \textbf{(Eq. 5.12)}$$

Evaluating Euler's formula at a half-revolution ϑ = π radians (180°), where cos π = -1 and sin π = 0, gives $e^{i\pi} = -1$, or in homogeneous form:

$$e^{i\pi} + 1 = 0 \quad \textbf{(Eq. 5.13)}$$

connecting the additive identity 0, the multiplicative identity 1, the circle constant π, the base of natural logarithms e, and the imaginary unit i. Applying [↖] [CMPLX] [e^x] to $0 + i\pi$ evaluates Euler's formula directly on the stack, while [↖] [CMPLX] [LN] inverts the mapping to return the principal complex logarithm $\ln(-1) = 0 + i\pi$. At a quarter-turn ϑ = π/2, where $i = e^{i\pi/2}$, raising i to the i-th power via [↖] [CMPLX] [y^x] produces a purely real value: $i^i = (e^{i\pi/2})^i = e^{i^2\,\pi/2} = e^{-\pi/2} \approx 0.207880$.

5.2.2 Polar Form, Abraham de Moivre's Theorem, and the Roots of Unity

By Euler's formula, any nonzero complex number $z = x + iy$ can be expressed in *polar form* through its Euclidean modulus $r = |z| = \sqrt{(x^2 + y^2)}$ and its principal argument $\vartheta = \text{atan2}(y, x)$:

$$z = x + iy = r(\cos\theta + i\sin\theta) = re^{i\theta} = r\angle\theta \qquad \textbf{(Eq. 5.14)}$$

Because the coordinate conversion keys [↶] [▸θ,r] and [↷] [▸y,x] expect ($Y = y$, $X = x$), pairing them with [x⇄y] maps between Gaussian complex coordinates (Y = Real, X = Imag) and polar coordinates ($Y = \vartheta$, $X = r$) without computing the complementary angle arctan(Real/Imag) = 90° - ϑ.

In polar form $z = r\,e^{i\vartheta}$, multiplication, division, and exponentiation follow directly from the laws of exponents. Multiplying two complex numbers $z_1\ z_2 = (r_1\ e^{i\vartheta}1)(r_2\ e^{i\vartheta}2) = (r_1\ r_2)\ e^{i(\vartheta}1\ +\ \vartheta_2)$ multiplies their moduli and adds their phase angles; dividing two complex numbers $z_1\ /\ z_2 = (r_1\ /\ r_2)\ e^{i(\vartheta}1\ -\ \vartheta_2)$ divides their moduli and subtracts their phase angles. Raising $z = r\ e^{i\vartheta}$ to the n-th power ([↶] [CMPLX] [y^x]) yields *Abraham de Moivre's theorem*, $(r\ e^{i\vartheta})^n = r^n\ e^{in\vartheta}$:

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos(n\theta) + i\sin(n\theta)) \qquad \textbf{(Eq. 5.15)}$$

Geometrically, raising a complex number to the n-th power raises its modulus to r^n while multiplying its argument by n.

Inverting De Moivre's relation to extract an n-th root $\sqrt[n]{z}$ explains why an n-th degree polynomial equation has n roots in the complex plane. Because adding integer multiples of a full revolution $2k\pi$ (360° k) leaves a complex number unchanged ($\vartheta + 2k\pi$), dividing the angle by n produces n distinct roots spaced at equal angular intervals of $2\pi/n$ around a circle of radius $r^{1/n}$:

$$\sqrt[n]{z} = r^{1/n}\left[\cos\left(\frac{\theta+2k\pi}{n}\right) + i\sin\left(\frac{\theta+2k\pi}{n}\right)\right], \qquad k = 0, 1, \ldots, n-1 \qquad \textbf{(Eq. 5.16)}$$

Whereas early algebraists conjectured that extracting the square root of i ($\sqrt{i}$) might require a new class of numbers outside the complex plane, Gauss noted that $i = 1\angle$ 90° has modulus 1 and argument 90°: its principal square root ([↶] [CMPLX] [$\sqrt{x}$]) has modulus $\sqrt{1} = 1$ and half the angle, 45° ($\pi/4$ rad), which lies in the complex plane at cos 45° + i sin 45° = $\sqrt{2}/2 + i(\sqrt{2}/2) \approx 0.7071 + 0.7071i$.

Similarly, whereas the equation $z^3 = 1$ has only the single root 1 on the real line (and $z^3 =$ -8 has only -2), in the complex plane the three cube roots of unity ($z^3 = 1 = e^{i\cdot\ 2k\pi}$) lie 360° / 3 = 120° apart at the vertices of an equilateral triangle inscribed in the unit circle:

$$1 = 1\angle 0^\circ, \qquad a = 1\angle 120^\circ = -\tfrac{1}{2} + i\frac{\sqrt{3}}{2}, \qquad a^2 = 1\angle 240^\circ = -\tfrac{1}{2} - i\frac{\sqrt{3}}{2} \qquad \textbf{(Eq. 5.17)}$$

Because those three unit vectors are spaced symmetrically at 0°, 120°, and 240°, their vector sum vanishes identically ($1 + a + a^2 = 0$). This 120° three-phase symmetry is the foundational principle of polyphase AC power transmission (pioneered in 1888–1891 by Nikola Tesla and Mikhail Dolivo-Dobrovolsky, and generalized to unbalanced fault analysis through the operator $a = 1\angle$ 120° in Charles LeGeyt Fortescue's 1918 method of symmetrical components): in a balanced three-phase network with phase currents displaced by 0°, 120°, and 240°, the instantaneous return currents sum to $1 + a + a^2 = 0$ at the neutral node, so three conductors transmit polyphase power without a fourth return

conductor.

5.2.3 Beyond the Unit Circle: Analytic Continuation and Joukowsky Airfoils

Euler's exponential relations extend the circular trigonometric functions analytically to arbitrary complex arguments $z = x + iy$. For a real angle x, sin x and cos x are bounded within [-1, +1]; combining $e^{ix} = \cos x + i \sin x$ and $e^{-ix} = \cos x - i \sin x$ defines cosine and sine across the entire complex plane:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i} \tag{Eq. 5.18}$$

When a complex argument $z = x + iy$ is evaluated via [↶] [CMPLX] [SIN] or [↶] [CMPLX] [COS], multiplication of iy by i converts the imaginary coordinate into real exponential factors $e^{\mp y}$, coupling circular oscillation in x with hyperbolic scaling (cosh y, sinh y) in y:

$$\sin(x+iy) = \sin x \cosh y + i \cos x \sinh y, \qquad \cos(x+iy) = \cos x \cosh y - i \sin x \sinh y \tag{Eq. 5.19}$$

Even when cosh y drives the magnitudes of sin z and cos z well past [-1, +1], *analytic continuation* preserves the Pythagorean identity $\sin^2 z + \cos^2 z = 1$ across the complex plane. For multivalued inverse and fractional-power functions—the complex natural logarithm $\mathrm{Ln}(\boldsymbol{z}) = \ln|\boldsymbol{z}| + i\mathrm{Arg}(\boldsymbol{z})$, the complex square root $\sqrt{(\boldsymbol{z})} = \sqrt{(|\boldsymbol{z}|)}\ e^{i\mathrm{Arg}(\boldsymbol{z})/2}$, and general complex powers $\boldsymbol{z^w} = e^{\boldsymbol{w}\mathrm{Ln}(\boldsymbol{z})}$—the SC-32 follows William Kahan's 1987 specification (*"Branch Cuts for Complex Elementary Functions"*), placing a seam (branch cut) along the negative real axis (-∞, 0) just as cartographers place the International Date Line along the 180° meridian so every point has one unambiguous principal angle $\mathrm{Arg}(\boldsymbol{z}) \in (-\pi, \pi]$.

This convention guarantees that [↶] [CMPLX] [√x] always maps into the closed right half-plane $\mathrm{Re}(\sqrt{(\boldsymbol{z})}) \geq 0$ (with $\sqrt{(-4 + 0i)} = 0 + 2i$ on the upper boundary of the cut), while a Nyquist contour crossing the negative real axis (-∞, 0) exhibits a 2π phase transition between +π and -π.

Holomorphic complex functions also define *conformal mappings*: at every point where $f'(z) \neq 0$, a complex mapping $w = f(z)$ acts like an angle-preserving mathematical lens that transforms solutions of Laplace's equation ($\nabla^2\varphi = 0$) around a simple shape into exact solutions around a complex shape.

In 1910, before wind tunnels and digital computers existed, Russian aerodynamicist Nikolai Zhukovsky (Joukowsky) used conformal mapping to transform the easily solved airflow (with circulation Γ) around a circular cylinder into the airflow around a curved aircraft wing. Evaluating the sum of a complex coordinate z on an offset circle (passing through the critical point $z = +c$ and enclosing $z = -c$) and its scaled reciprocal c^2/z defines the *Joukowsky transformation*:

$$\zeta = z + \frac{c^2}{z} \tag{Eq. 5.20}$$

This rational complex map ($\boldsymbol{z} + c^2 \cdot 1/\boldsymbol{z}$) transforms the offset circle into an airfoil profile through three geometric effects: at the critical point $z = +c$, where the derivative $d\zeta/dz = 1 - c^2/z^2$ vanishes, the smooth circular boundary folds into a sharp trailing edge at $\zeta = +2c$; on the opposite side of the circle extending slightly beyond $z = -c$, the boundary rounds into a blunt leading edge; and displacing the circle's center slightly above the real axis introduces camber along the chord line. Because $\zeta = z + c^2/z$ is conformal everywhere exterior to the circle, it maps the analytical streamlines around the cylinder directly onto

the airflow around the cambered wing and forms the mathematical foundation of the *Kutta-Joukowsky Lift Theorem* ($L' = \rho_\infty\ V_\infty\ \Gamma$).

5.3 Charles Proteus Steinmetz and the War of the Currents

Complex arithmetic entered electrical power engineering in August 1893 at the International Electrical Congress in Chicago. During the preceding decade, the expansion of electrical distribution had centered on the competition between Thomas Edison's direct-current (DC) networks and George Westinghouse and Nikola Tesla's alternating-current (AC) polyphase networks. In a DC network, steady-state analysis required only scalar Ohm's law ($V = I \cdot R$), but because DC voltage could not be stepped up efficiently with static transformers, resistive line losses ($P_{\text{loss}} = I^2\ R$) restricted low-voltage DC distribution to short urban radii. Alternating current enabled high-voltage step-up and step-down transformation over long distances, but introduced phase shifts between voltage and current across reactive elements.

In an AC circuit operating at frequency $f = 60$ Hz (angular frequency $\omega = 2\pi\ f$), inductors and capacitors displace voltage and current by $\pm$ 90°. In an inductor (L, such as a transformer winding or motor coil), Faraday's law of induction states that induced voltage is proportional to the time derivative of current, $v_L(t) = L\ di/dt$; because $(d/dt)\ \sin(\omega\ t) = \omega\ \cos(\omega\ t) = \omega\ \sin(\omega\ t + 90°)$, inductor voltage peaks 90° ahead of current. Conversely, in a capacitor (C, such as parallel dielectric plates or underground cable insulation), charging current is proportional to the time derivative of voltage, $i_C(t) = C\ dv/dt$, so capacitor voltage lags 90° behind current. Calculating steady-state voltages and currents across multi-branch AC networks therefore required solving coupled systems of linear differential equations with sinusoidal forcing:

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = V_0\cos(\omega t + \phi) \qquad \textbf{(Eq. 5.21)}$$

At the 1893 Chicago Congress, Charles Proteus Steinmetz presented *"Complex Quantities and Their Use in Electrical Engineering" (Steinmetz, 1893)* and showed how Euler's complex exponential reduces steady-state AC differential equations to complex algebra.

5.3.1 Steinmetz's Phasor Transform: Turning Differential Equations into Algebra

Steinmetz observed that in a linear AC network driven at a single angular frequency ω, every steady-state voltage and current oscillates at that same angular frequency ω. Rather than carrying explicit $\cos(\omega\ t + \vartheta)$ time functions through differential equations, Steinmetz represented each sinusoid by a complex number—a *phasor*—scaled by its root-mean-square amplitude ($|\boldsymbol{V}| = V_0/\sqrt{2}$, eliminating the factor of 1/2 so AC power laws $P = |\boldsymbol{I}|^2\ R$ and $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{I}^*$ match DC Ohm's and Joule's laws) and initial phase angle relative to a common reference.

Differentiating a rotating complex exponential $e^{j\omega\ t}$ with respect to time (where electrical engineering denotes the imaginary unit by $j = \sqrt{(-1)}$ to reserve i for electrical current) multiplies the phasor by $j\omega$:

$$\frac{d}{dt}(e^{j\omega t}) = j\omega e^{j\omega t} \qquad \textbf{(Eq. 5.22)}$$

Time differentiation becomes algebraic multiplication by $j\omega$ (a +90° rotation scaled by ω), while time integration becomes division by $j\omega$ (a -90° rotation, since $1/j = -j$).

Under Steinmetz's transformation, the three passive circuit elements map onto the axes of the complex impedance plane. Resistors (R) lie along the positive horizontal Real axis as $\boldsymbol{Z}_R = R + j0\ \Omega$, introducing 0° phase shift and converting electrical energy irreversibly into Joule heat. Inductors (L) lie along the positive vertical Imaginary axis as $\boldsymbol{Z}_L = +j\ X_L = +j\omega\ L\ \Omega$, introducing a +90° voltage phase lead as magnetic field energy cycles losslessly between the coil and the source. Capacitors (C) lie along the negative vertical Imaginary axis as $\boldsymbol{Z}_C = -j\ X_C = -j(1\ /\ \omega\ C)\ \Omega$, introducing a -90° voltage phase lag as electrostatic field energy cycles losslessly between the plates and the source.

For a series RLC branch, the total *complex impedance* $\boldsymbol{Z}$ is the vector sum of real resistance and net imaginary reactance:

$$\mathbf{Z} = R + j(X_L - X_C) = R + j\left(\omega L - \frac{1}{\omega C}\right) \quad \textbf{(Eq. 5.23)}$$

Because inductive reactance points along $+j$ and capacitive reactance points along $-j$, at the angular frequency where $\omega_0\ L = 1\ /\ \omega_0\ C$—the *resonant frequency* $f_0 = 1\ /\ 2\pi\sqrt{LC}$—the opposing imaginary reactances cancel to zero and the net impedance reduces to the real resistance R.

With complex phasors, AC Ohm's law takes the multiplicative form $\boldsymbol{V} = \boldsymbol{I} \cdot \boldsymbol{Z}$, series impedances add directly ($\boldsymbol{Z}_{\text{series}} = \boldsymbol{Z}_1 + \boldsymbol{Z}_2$), and parallel impedances combine either by summing their reciprocal *complex admittances* ($\boldsymbol{Y} = 1/\boldsymbol{Z} = G + jB$, with conductance G and susceptance B in Siemens) or by applying the complex parallel operator [↶] [CMPLX] [x‖y] *(Horowitz & Hill, 2015)*:

$$\mathbf{Z}_{\text{parallel}} = \mathbf{Z}_1 \| \mathbf{Z}_2 = \frac{1}{\frac{1}{\mathbf{Z}_1} + \frac{1}{\mathbf{Z}_2}} = \frac{\mathbf{Z}_1 \mathbf{Z}_2}{\mathbf{Z}_1 + \mathbf{Z}_2} \quad \textbf{(Eq. 5.24)}$$

5.3.2 The Complex Power Triangle, Power Factor Correction, and Maximum Power Transfer

Steinmetz's complex representation also resolved an accounting anomaly in early AC power distribution: industrial plants containing large induction motors drew high line currents from utility feeders while registering lower real power consumption on wattmeters.

When an AC source delivers an RMS voltage phasor $\boldsymbol{V} = |\boldsymbol{V}|\angle\vartheta_v$ and an RMS current phasor $\boldsymbol{I} = |\boldsymbol{I}|\angle\vartheta_i$ to an inductive load $\boldsymbol{Z} = R + jX_L = |\boldsymbol{Z}|\angle\vartheta$ (where $\vartheta = \vartheta_v - \vartheta_i$ is the phase lag of current behind voltage), the phase-invariant power product requires complex conjugation. Multiplying $\boldsymbol{V}$ by $\boldsymbol{I}$ without conjugation sums their phase angles ($\vartheta_v + \vartheta_i$), which varies with the arbitrary choice of time origin $t = 0$. Defining *complex apparent power* $\boldsymbol{S}$ (in Volt-Amperes, VA) as the product of the voltage phasor $\boldsymbol{V}$ and the complex conjugate of the current phasor, $\boldsymbol{I}^* = |\boldsymbol{I}|\angle(-\vartheta_i)$, isolates the physical phase difference $\vartheta = \vartheta_v - \vartheta_i$:

$$\mathbf{S} = \mathbf{V} \cdot \mathbf{I}^* = |\mathbf{V}||\mathbf{I}|\angle(\theta_v - \theta_i) = |\mathbf{S}|\cos\theta + j|\mathbf{S}|\sin\theta = P + jQ \quad \textbf{(Eq. 5.25)}$$

This complex product defines the right-angled *Power Triangle* $\boldsymbol{S} = P + jQ$ in the complex plane. The horizontal real leg is the *real (active) power* $P = |\boldsymbol{V}||\boldsymbol{I}|\cos\vartheta = |\boldsymbol{I}|^2\ R$ (in Watts, W), representing the rate at which electrical energy is converted into mechanical shaft work—driving lathe spindles, hoists, and pumps—or dissipated as heat; the dimensionless cosine ratio $\text{PF} = \cos\vartheta = P\ /\ |\boldsymbol{S}|$ is the *power factor*.

The vertical imaginary leg is the *reactive (quadrature) power* $Q = |\boldsymbol{V}||\boldsymbol{I}|\sin\vartheta = |\boldsymbol{I}|^2\ X$ (in Volt-Amperes Reactive, VAR), measuring the peak rate of lossless energy oscillation twice

per AC cycle between the generator and the magnetic fields of inductive loads ($Q > 0$) or the electric fields of capacitive loads ($Q < 0$). Although reactive power performs zero net work over a complete cycle, utility transmission lines must carry the associated quadrature current, incurring real $I^2 R_{\text{line}}$ resistive losses in the distribution grid.

The hypotenuse of the triangle is the *apparent power* $|\boldsymbol{S}| = \sqrt{(P^2 + Q^2)} = |\boldsymbol{V}||\boldsymbol{I}|$ (in Volt-Amperes, VA); because substation transformers and alternators must be insulated for total voltage $|\boldsymbol{V}|$ and sized to conduct total current $|\boldsymbol{I}|$ regardless of phase angle, power equipment is rated in kVA or MVA rather than kilowatts. Industrial facilities eliminate quadrature line current through *power factor correction*. Because an induction motor draws lagging reactive power $+jQ_L$, connecting a shunt capacitor bank of equal and opposite leading reactive power $-jQ_C = -jQ_L$ in parallel across the load bus supplies the oscillating magnetic energy locally, canceling the imaginary component $j(Q_L - Q_C) = 0$ so the utility feeder sees a purely resistive load at unity power factor ($\cos 0^\circ = 1.000$).

In low-power RF communications, acoustic transducers, and instrumentation, the design objective shifts from minimizing transmission losses to extracting the maximum real power P_{load} from a source with fixed internal Thévenin impedance $\boldsymbol{Z}_{\text{source}} = R_s + jX_s$. Terminating the source with a load $\boldsymbol{Z}_{\text{load}} = R_L + jX_L$ produces loop current magnitude $|\boldsymbol{I}| = |\boldsymbol{V}_s| / \sqrt{((R_s + R_L)^2 + (X_s + X_L)^2)}$ and real load power:

$$P_{\text{load}} = |\mathbf{I}|^2 R_L = \frac{|\mathbf{V}_s|^2 R_L}{(R_s + R_L)^2 + (X_s + X_L)^2} \qquad \textbf{(Eq. 5.26)}$$

Because $(X_s + X_L)^2 \geq 0$ in the denominator, real power is maximized for any R_L by choosing the load reactance equal and opposite to the source reactance ($X_L = -X_s$), canceling the net series reactance to zero. Differentiating the remaining resistive ratio $R_L / (R_s + R_L)^2$ with respect to R_L shows that maximum power occurs when $R_L = R_s$. Combining both conditions ($R_L = R_s$ and $X_L = -X_s$) yields the *Complex Conjugate Maximum Power Transfer Theorem*:

$$\mathbf{Z}_{\text{load}} = R_s - jX_s = \mathbf{Z}_{\text{source}}^{*} \qquad \textbf{(Eq. 5.27)}$$

Maximum real power transfer occurs when the load impedance equals the complex conjugate of the source impedance.

5.3.3 Audio Filters, Linkwitz-Riley Crossovers, and Zobel Compensation

Complex impedance algebra also governs electroacoustic loudspeaker crossovers and impedance-compensation networks *(Horowitz & Hill, 2015)*. A moving-coil loudspeaker driver is not a pure 8 Ω resistance: the copper voice coil wound around the magnet pole piece introduces series inductance ($R + j\omega L$) that causes the driver impedance to rise at high audio frequencies. Flattening that inductive rise requires wiring a conjugate *Zobel compensation network* ($R - jX_C$) in parallel with the driver terminals. When two conjugate impedances $Z_1 = R + jX$ and $Z_2 = R - jX$ are connected in parallel, their opposite susceptances $\mp jX / (R^2 + X^2)$ sum to zero and leave a purely real conductance $2G = 2R / (R^2 + X^2)$ with equivalent resistance $Z_{\text{eq}} = (R^2 + X^2) / 2R$.

In multi-way loudspeaker design, dividing the audio spectrum between a low-frequency woofer and a high-frequency tweeter requires matching both amplitude and phase at the crossover frequency. In 1930, British engineer Stephen Butterworth derived the maximally flat second-order low-pass transfer function $H(s) = 1 / (s^2 + \sqrt{2}s + 1)$. At the normalized crossover frequency $s = j\omega/\omega_c = j$, the $s^2 = j^2 = -1$ term and the $+1$ constant in the

denominator cancel:

$$H_{\text{Butter2}}(j) = \frac{1}{(-1+1)+j\sqrt{2}} = \frac{1}{j\sqrt{2}} = -\frac{j}{\sqrt{2}} \approx 0 - 0.707107j \quad (-3.01\ \text{dB, phase } -90^\circ) \qquad \textbf{(Eq. 5.28)}$$

If a single second-order Butterworth low-pass filter feeds the woofer and its high-pass complement $s^2 / (s^2 + \sqrt{2}s + 1)$ feeds the tweeter, the low-pass output lags by -90° ($-j/\sqrt{2}$) at crossover while the high-pass output leads by +90° ($+j/\sqrt{2}$). Because -90° and +90° differ by 180°, the two acoustic outputs sum to zero at the crossover frequency when connected with identical polarity.

In 1976, Siegfried Linkwitz and Russ Riley resolved this phase opposition by cascading two second-order Butterworth sections in series to form a fourth-order *Linkwitz-Riley* (**LR4**) crossover, squaring the complex transfer function at the crossover frequency:

$$H_{\text{LR4}}(j) = [H_{\text{Butter2}}(j)]^2 = \left(-\frac{j}{\sqrt{2}}\right)^2 = \frac{j^2}{2} = -0.5000 + 0.0000j \quad (-6.02\ \text{dB, phase } -180^\circ) \qquad \textbf{(Eq. 5.29)}$$

At crossover, both the low-pass and high-pass outputs have magnitude $|H| = 0.5$ (-6.02 dB) and identical phase (-180°), so their acoustic outputs sum in phase to unit magnitude ($|-0.5 + (-0.5)| = 1.0$, or 0 dB all-pass response).

5.3.4 Coaxial Transmission Lines, Standing Waves, and Phillip Smith's Chart

At radio and microwave frequencies where the signal wavelength is comparable to or shorter than the interconnecting conductors, a transmission line acts as a distributed network of series impedance per unit length ($Z' = R' + j\omega L'$) and shunt admittance per unit length ($Y' = G' + j\omega C'$). Oliver Heaviside's telegrapher's equations combine these distributed parameters into the line's *characteristic impedance*:

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \qquad \textbf{(Eq. 5.30)}$$

Much like light partially reflecting off a pane of glass or an ocean wave rebounding off a seawall, when an electromagnetic wave propagating along a feedline of characteristic impedance $Z_0 = 50\ \Omega$ encounters an antenna whose complex impedance $\boldsymbol{Z}_L = R_L + jX_L$ differs from Z_0, part of the wave echoes backward down the cable and interferes with the forward wave to create stationary voltage peaks (*standing waves*). In 1939, Bell Telephone Laboratories engineer Phillip H. Smith observed that the complex *voltage reflection coefficient* $\boldsymbol{\Gamma}$ (denoted S_{11} in scattering-parameter measurements):

$$\boldsymbol{\Gamma} = \frac{\mathbf{Z}_L - Z_0}{\mathbf{Z}_L + Z_0} \qquad \textbf{(Eq. 5.31)}$$

maps the entire right half-plane of passive impedances ($R_L \geq 0$) onto the unit disk $|\boldsymbol{\Gamma}| \leq 1$, forming the *Smith Chart*. Subtracting Z_0 from the real part R_L while recovering Z_0 via [↶] [LASTx] constructs both $\boldsymbol{Z}_L - Z_0$ and $\boldsymbol{Z}_L + Z_0$ for [↶] [CMPLX] [÷]. Converting $\boldsymbol{\Gamma}$ to polar magnitude $|\boldsymbol{\Gamma}|$ via [x⇄y] [↶] [→θ,r] yields both the *return loss* (RL = -20 $\log_{10}|\boldsymbol{\Gamma}|$ in dB) and the *voltage standing wave ratio* (VSWR):

$$\text{VSWR} = \frac{1 + |\boldsymbol{\Gamma}|}{1 - |\boldsymbol{\Gamma}|} \qquad \textbf{(Eq. 5.32)}$$

To match a resistive antenna load Z_L to a transmission line Z_0 without reflection, an RF designer inserts a quarter-wavelength ($\lambda/4$) transformer section whose characteristic impedance is the geometric mean $Z_{\text{line}} = \sqrt{(Z_0\ Z_L)}$, which maps the input impedance to Z_{in}

$= Z_{\text{line}}^2 \,/\, Z_L = Z_0$ in exact analogy to a quarter-wave optical anti-reflection coating of refractive index $n_{\text{film}} = \sqrt{(n_{\text{air}}\ n_{\text{glass}})}$.

5.4 Gaussian Complex Pairs & Alternating-Current Phasors

Where the 1982 HP-15C hid imaginary components inside a separate parallel stack *(HP, 1982)* and the single-line 1991 HP-32SII stored (Y = Imag, X = Real) (so that $a + bi$ had to be keyed backward as **b ENTER a**), the four-line StackCalc SC-32 pairs (Y = Real, X = Imag) across the lower two registers of the display. Keying **a** [ENTER] **b** in natural left-to-right reading order places the horizontal Real part a on Line Y and the vertical Imaginary part b on Line X.

Pushing a second complex number $c + di$ onto the stack via **a** [ENTER] **b** [ENTER] **c** [ENTER] **d** lifts the first operand into Registers ($T = a$, $Z = b$) while the second operand occupies ($Y = c$, $X = d$), holding both complex vectors in plain sight simultaneously. Tapping the transient [↰] [CMPLX] prefix ([↰] + [STO]) before any arithmetic ([×], [÷], [+], [-], [x⇄y]), power, or trigonometric key executes the full two-dimensional operation across (T, Z) and (Y, X) and writes the result to (Y = Real, X = Imag) without locking the calculator into a separate complex mode.

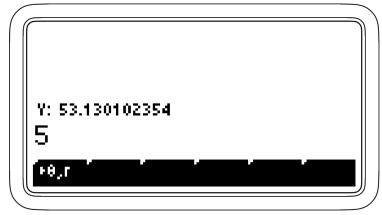


Figure: Impedance phasor 3 + j4 rectangular-to-polar conversion on the SC-32, resolving magnitude r = 5 in Register X and phase angle theta = 53.1301 deg in Register Y.

5.4.1 Pairing Two Complex Numbers Across Four Registers

Within each adjacent register pair, the upper line holds the real projection and the lower line holds the imaginary projection:

$$\mathbf{z} = Y + Xi \qquad \textbf{(Eq. 5.33)}$$

Register / Display	Complex Plane Component	Mathematical Notation	Display Appearance ($z = a + bi$)
Register *Y* (Line 2)	Real Part	Re(z)	a (Upper line on transflective LCD)
Register *X* (Line 1)	Imaginary Part	Im(z)	b (Active prompt line on transflective LCD)

Pressing [C ON] cancels the transient [↰] [CMPLX] prefix if tapped by mistake, and complex trigonometric and exponential operations ([↰] [CMPLX] [SIN], [COS], [TAN], [e^x], [LN], [$\sqrt{x}$]) always evaluate in natural analytic radians regardless of the scalar [DEG]/[RAD] angular setting. Internally, direct unscaled complex division $(a+bi) \,/\, (c+di) = ((ac+bd) + i(bc-ad)) \,/\, (c^2+d^2)$ fails whenever $|c|$ or $|d|$ exceeds $\sqrt{(\Omega_{\max})} \approx 1.34 \times 10^{154}$, because squaring $c^2 + d^2$ overflows to ∞ even when the quotient itself is near unity. To prevent premature overflow above 10^{154} or underflow below 10^{-154}, [↰] [CMPLX] [÷] and [↰] [CMPLX] [x^{-1}] implement Robert L. Smith's 1962 *Algorithm 116* scaling pivot:

$$\frac{a+bi}{c+di} = \frac{a+b(d/c)}{c+d(d/c)} + i\frac{b-a(d/c)}{c+d(d/c)} \text{if } |c| \geq |d|, \; [10pt] \frac{a(c/d)+b}{c(c/d)+d} + i\frac{b(c/d)-a}{c(c/d)+d} \text{if } |d| > |c|. \qquad \textbf{(Eq. 5.34)}$$

By always forming the smaller-over-larger ratio $r = d/c$ or $r = c/d$ so that $|r| \leq 1$, every intermediate product stays bounded within the native exponent range of the operands.

Similarly, [↶] [CMPLX] [$\sqrt{x}$] avoids catastrophic cancellation in $\sqrt{((|z| + a) / 2)}$ when $a < 0$ by evaluating the larger root component first from the scaled hypotenuse $|z| = \text{hypot}(a, b)$ and recovering the smaller component by division:

$$\sqrt{a+bi} = \sqrt{\frac{|z|+a}{2}} + i\frac{b}{2\sqrt{(|z|+a)/2}} \text{if } a \geq 0, \; [10pt] \frac{|b|}{2\sqrt{(|z|-a)/2}} + i\,\text{sgn}(b)\sqrt{\frac{|z|-a}{2}} \text{if } a < 0. \qquad \textbf{(Eq. 5.35)}$$

Because one complex number occupies two adjacent registers, the four-register stack holds two complete complex numbers simultaneously—$z_2 = T + Zi$ in the upper pair (T, Z) and $z_1 = Y + Xi$ in the lower pair (Y, X):

Operation Class	Key Sequence	Input Stack	Output in (Y, X)	Upper Stack (T, Z)
Complex Negation	[↶] [CMPLX] [+/−]	$z_1 = Y + Xi$	$-Y - Xi$	Preserved (T, Z unchanged)
Unary Complex Functions	[↶] [CMPLX] + [$\sqrt{x}$], [x^{-1}], [↶] [x^2], [e^x], [LN], [SIN], [COS], [TAN]	$z_1 = Y + Xi$	$f(z_1)$	Preserved (T, Z unchanged)
Binary Complex Arithmetic	[↶] [CMPLX] + [+], [−], [×], [÷], [y^x]	$z_2 = T + Zi$, $z_1 = Y + Xi$	z_2 op z_1	2-level drop ($T \to Z$, T)
Complex Parallel Impedance	[↶] [CMPLX] [$x \parallel y$]	$z_2 = T + Zi$, $z_1 = Y + Xi$	$z_2\, z_1 / (z_2 + z_1)$	2-level drop ($T \to Z$, T)

This register architecture highlights the distinction between *unary* and *dyadic* complex operations. Whereas the 1982 HP-15C stored imaginary components in a hidden parallel stack accessed via **Re↔Im**, and the 1-line HP-32SII required tapping **x↔y** to peek at the real part in Register Y, the SC-32's four-line LCD displays both $\text{Re}(z_1)$ in Register Y and $\text{Im}(z_1)$ in Register X simultaneously alongside $z_2 = T + Zi$ in (T, Z).

Because a single complex number $z_1 = Y + Xi$ lives entirely in the lower two registers of the stack, unary complex operators (such as [↶] [CMPLX] [e^x], **LN**, **√x**, **x²**, and **1/x**) transform **Register Y** (Real) and **Register X** (Imaginary) in place while **Register Z** and **Register T** remain untouched, so a second complex number can wait in the upper registers. Dyadic complex operators (such as [↶] [CMPLX] [×], **÷**, **y^x**, and **x‖y**), by contrast, engage all four registers simultaneously as two coupled two-dimensional Argand planes, combining $z_2 = T + Zi$ and $z_1 = Y + Xi$ into a single complex result in (Y, X).

Because each complex number consumes two real registers, a classic four-register stack (**4-LVL**) acts as a *two-level* complex stack (z_1 in Y, X and z_2 in T, Z). Whenever a multi-branch AC network requires holding three or four complex phasors in flight simultaneously, switching [↷] [FLAGS] → **STACK** → [8-LVL] expands the real stack to eight registers (R_1–R_8), transforming [↶] [CMPLX] mode into a true **four-level complex stack** (z_1 in Y, X; z_2 in T, Z; z_3 in R_6, R_5; and z_4 in R_8, R_7).

Five complex traces illustrate how (Y, X) and (T, Z) operate together:

- **Damped Mechanical Suspension Eigenvalues ($\lambda = -c/2m \pm i\sqrt{(k/m - (c/2m)^2)}$):** For an underdamped mechanical shock-absorber suspension ($m = 2$ kg, so $2m = 4$ kg; viscous damping $c = 8$ N·s/m; spring stiffness $k = 50$ N/m, so $\omega_0{}^2 = k/m = 25$ s^{-2}), the characteristic eigenvalues are $\lambda = -2.0 \pm i\sqrt{(25 - 4)} = -2.0 \pm i\sqrt{21} \approx -2.0 \pm 4.582576i$. Keying **8** [ENTER] **4** [÷] [+/−] evaluates the horizontal energy-dissipation rate $\sigma = -\alpha =$ **-2.0000 s^{-1}** in Register X. Keying **4** [ENTER] **25** [-] [+/−] then forms the net oscillatory variance $\omega_0{}^2 - \alpha^2 = 21.0000$ s^{-2} in Register X while lifting the real part $\sigma = -2.0000$ into Register Y, and pressing [$\sqrt{x}$] extracts the vertical damped ringing frequency $\omega_d = \sqrt{21} =$ **4.582576 rad/s** in Register X so the complete eigenvalue pair $Y = -2.000000$, $X = 4.582576$ occupies the lower two registers.

- **Verifying a Complex Quadratic Root ($z^2 - 4z + 13 = 0$ at $z_1 = 2 + 3i$):** As soon as we step from assembling a single eigenvalue to verifying a complex root inside $z^2 - 4z + 13 = 0$, both complex register pairs (T, Z) and (Y, X) come into play. Expanding $(2+3i)^2 - 4(2+3i) + 13$ algebraically gives $(-5 + 12i) - (8 + 12i) + 13 = 0 + 0i$. On the stack, squaring $2 + 3i$ in place via **2** [ENTER] **3** [↶] [CMPLX] [↶] [x^2] transforms (Y, X) into $z_1{}^2 = -5 + 12i$; keying the linear term **8** [ENTER] **12** then lifts $-5 + 12i$ into (T, Z) so [↶] [CMPLX] [-] and **13** [ENTER] **0** [↶] [CMPLX] [+] combine both 2D vectors across all four registers:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
2 [ENTER] **3**	0.0000	0.0000	2.0000	3.0000	Candidate complex root $z_1 = 2 + 3i$ loaded
[↶] [CMPLX] [↶] [x^2]	0.0000	0.0000	-5.0000	12.0000	Complex square $z_1{}^2 = -5 + 12i$ in place
8 [ENTER] **12** [↶] [CMPLX] [-]	-5.0000	-5.0000	-13.0000	0.0000	Subtract $4z_1 = 8 + 12i$
13 [ENTER] **0** [↶] [CMPLX] [+]	-13.0000	-13.0000	**0.0000**	**0.0000**	Add $13 + 0i$; residual = $0.0 + 0.0i$

- **Multiplying a Complex Number by Its Conjugate Twin ($z \cdot \bar{z} = (6 + 8i)(6 - 8i)$):** When we multiply $z = 6 + 8i$ in (T, Z) by its complex conjugate $z^* = 6 - 8i$ in (Y, X), their opposite phase angles cancel and leave the squared Euclidean norm $6^2 + 8^2 = 100 + 0i$:

  - **Conjugate Norm Product:** **6** [ENTER] **8** [ENTER] **6** [ENTER] **8** [+/−] [↶] [CMPLX] [×] [x⇄y] → ($Y = 100.0000$, $X =$ **100**).

- **Spacecraft Re-Entry Plasma Permittivity ($\varepsilon_r = 1 - \omega_p{}^2 / \omega(\omega - j\nu)$):** That conjugate mechanism powers [↶] [CMPLX] [x^{-1}] when modeling hypersonic re-entry plasma blackout. For normalized plasma frequency $\omega_p = 1$, wave frequency $\omega = 2$, and electron collision rate $\nu = 0.5$, the denominator is $\omega(\omega - j\nu) = 4.0 - 1.0i$. Inverting $4 - i$ in place on (Y, X) via [↶] [CMPLX] [x^{-1}], negating with [↶] [CMPLX] [+/−], and keying **1** [ENTER] **0** lifts $-0.235294 - 0.058824i$ into (T, Z) so [↶] [CMPLX] [+] resolves $\varepsilon_r =$ **$0.764706 - 0.058824i$**.

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
4 ENTER **1** +/−	0.0000	0.0000	4.0000	-1.0000	Scaled denominator z = 4.0 - 1.0i
↶ CMPLX x^{-1} ↶ CMPLX +/−	0.0000	0.0000	-0.235294	-0.058824	Negated reciprocal -1/(4-i)
1 ENTER **0**	-0.235294	-0.058824	1.0000	0.0000	Unit real reference 1.0 + 0i in (Y, X)
↶ CMPLX +	-0.235294	-0.235294	**0.764706**	**-0.058824**	Add unity: ε_r = 0.7647 - 0.0588i

- **Conformal Joukowsky Airfoil Map ($\zeta = z + 1/z$ at $z = 1.1 + 0.2i$):** Because unary complex operators leave (T, Z) undisturbed, keying **1.1** ENTER **0.2** ENTER **1.1** ENTER **0.2** loads z into both (T, Z) and (Y, X); pressing ↶ CMPLX x^{-1} inverts only the lower pair (Y, X) → 0.880000 - 0.160000i while the uninverted z = 1.1 + 0.2i waits in (T, Z), and pressing ↶ CMPLX + sums them to land at the cambered wing coordinate (Y = **1.980000**, X = **0.040000**).

5.4.2 Evaluating Euler's Identity on the Complex Stack

Because ↶ CMPLX e^x and ↶ CMPLX LN are unary complex functions, they transform a complex argument $z = x + iy$ in place across (Y, X) according to $e^{x+iy} = e^x \cos y + i\, e^x \sin y$ without disturbing Registers Z and T. With the SC-32 in Radians mode (RAD), we can verify Euler's formula $e^{i\pi}$ = -1 + 0i, Euler's identity $e^{i\pi} + 1 = 0$, and the principal complex logarithm Ln(-1) = 0 + $i\pi$ in three crisp steps:

- **Half-Turn Rotation ($e^{0 + i\pi}$):** Place the real growth rate $x = 0$ in Register Y and radian angle $y = \pi$ in Register X, then fire ↶ CMPLX e^x: **0** ↷ π ↶ CMPLX e^x → (Y = **-1.0000000000**, X = **0.0000000000**).

- **Euler's Five-Constant Identity ($e^{i\pi} + 1 = 0$):** Add the unit real vector +1 + 0i across all four registers: **0** ↷ π ↶ CMPLX e^x **1** ENTER **0** ↶ CMPLX + → **0** (Y = **0.0000000000**, X = **0.0000000000**).

- **Principal Complex Logarithm (Ln(-1)):** Evaluate $e^{i\pi}$ = -1 + 0i and immediately invert it in place via ↶ CMPLX LN onto the upper lip of William Kahan's branch cut: **0** ↷ π ↶ CMPLX e^x → (Y = **0.0000000000**, X = **3.1415926536**).

5.4.3 De Moivre Powers, Roots of Unity & Complex Trigonometry

The Euclidean vector keys ↶ →θ,r and ↷ →y,x expect vertical y in Register Y and horizontal x in Register X. Because Gaussian complex format places Real (x) in Register Y and Imaginary (y) in Register X, $x \rightleftarrows y$ is the one-keystroke bridge between Gaussian complex (Y = Real, X = Imag) and polar (ϑ, r) coordinates: pressing $x \rightleftarrows y$ ↶ →θ,r converts Complex (Y = Real, X = Imag) to Polar (Y = ϑ, X = r), and pressing ↷ →y,x $x \rightleftarrows y$ converts Polar (Y = ϑ, X = r) back to Complex (Y = Real, X = Imag). Swapping with $x \rightleftarrows y$ before ↶ →θ,r and after ↷ →y,x avoids the *complementary angle trap*, where pressing ↶ →θ,r on (Y = Real, X = Imag) without swapping first computes atan2(Real, Imag) = 90° - ϑ.

Converting $z = 3 + 4i$ to polar coordinates and back in [RAD] mode (where $\vartheta = 0.927295$ rad equals $\vartheta = 53.1301°$ in [DEG] mode) proceeds in four steps:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3 [ENTER] 4	0.0000	0.0000	3.0000	4.0000	Gaussian complex coordinates (Y=Real, X=Imag)
[x⇄y] [↶] [▸θ,r]	0.0000	0.0000	0.927295	**5.000000**	Polar conversion: $\vartheta = 0.9273$ rad (53.1301°) in Y, magnitude r in X
[↷] [▸y,x]	0.0000	0.0000	4.000000	**3.000000**	Rectangular conversion: horizontal x in X, vertical y in Y
[x⇄y]	0.0000	0.0000	**3.000000**	**4.000000**	Realign to Gaussian complex format: Real in Y, Imag in X

In Degrees mode ([DEG]), we can use [↷] [▸y,x] [x⇄y] to verify Fortescue's three-phase neutral cancellation $1 + a + a^2 = 0$ ($\rightarrow X =$ **0.000000**) for the cube roots of unity $a = 1\angle\ 120° =$ -0.5 + 0.866025i and $a^2 = 1\angle\ 240° =$ -0.5 - 0.866025i:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
120 [ENTER] 1 [↷] [▸y,x] [x⇄y]	0.0000	0.0000	-0.500000	0.866025	Polar to Gaussian: $a =$ -0.5 + 0.8660i
240 [ENTER] 1 [↷] [▸y,x] [x⇄y]	-0.500000	0.866025	-0.500000	-0.866025	Polar to Gaussian: $a^2 =$ -0.5 - 0.8660i
[↶] [CMPLX] [+]	-0.500000	-0.500000	-1.000000	0.000000	Complex sum: $a + a^2 =$ -1.0 + 0.0i
1 [ENTER] 0 [↶] [CMPLX] [+]	-1.000000	-1.000000	**0.000000**	**0.000000**	Add +1 reference: $1 + a + a^2 = 0 + 0i$

In Radians mode ([RAD]), [↶] [CMPLX] [√x] and [↶] [CMPLX] [y^x] evaluate De Moivre's fractional powers directly across the stack:

- **Principal Square Root of i ($\sqrt{(0 + 1i)} = \sqrt{2}/2 + (\sqrt{2}/2)i$):** Bisects the 90° ($\pi/2$ rad) angle to 45° in place on (Y, X): 0 [ENTER] 1 [↶] [CMPLX] [√x] $\rightarrow$ ($Y =$ **0.707107**, $X =$ **0.7071067812**).
- **Principal Cube Root of -8 ($8^{1/3}e^{i\pi/3}$):** Loading -8 + 0i into (T, Z) with a 6-digit decimal exponent 0.333333 + 0i in (Y, X) gives an approximate root, whereas forming the exact rational exponent 1/3 + 0i first lands on the exact principal branch root:
  - **6-Digit Decimal Exponent (0.333333):** 8 [+/−] [ENTER] 0 [ENTER] 0.333333333333 [ENTER] 0 [↶] [CMPLX] [y^x] [x⇄y] $\rightarrow X =$ **1**.
  - **Full-Precision Rational Exponent (1/3):** 8 [+/−] [ENTER] 0 [ENTER] 0.333333333333 [ENTER] 0 [↶] [CMPLX] [y^x] $\rightarrow$ ($Y =$ **1.000000**, $X =$ **1.7320508075**).

Even when both the base $z_2 = 1 + i$ in (T, Z) and the exponent $z_1 = 1 - i$ in (Y, X) are complex, pressing [↶] [CMPLX] [y^x] evaluates $(1 + i)^{(1 - i)} = e^{(1-i)\ \ln(1+i)} =$ **2.807879** + **1.317865**i (and keying 0 [ENTER] 1 [ENTER] 0 [ENTER] 1 [↶] [CMPLX] [y^x] confirms Euler's real-valued identity $i^i = e^{-\pi/2} =$ **0.207880** + **0.000000**i):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [ENTER] 1 [ENTER]	0.0000	1.0000	1.0000	1.0000	Base $z_2 = 1 + i$ loaded into (T, Z)
1 [ENTER] 1 [+/−]	1.0000	1.0000	1.0000	-1.0000	Exponent $z_1 = 1 - i$ loaded into (Y, X)
[↶] [CMPLX] [y^x]	1.0000	1.0000	**2.807879**	**1.317865**	General complex power ${z_2}^{z1}$ evaluated

To test analytic continuation of the unit-circle identity $\sin^2 z + \cos^2 z = 1$ far outside [-1, +1] at $z = 1 + 2i$, we first evaluate $\sin(1 + 2i) = \sin(1)\cosh(2) + i\cos(1)\sinh(2)$ and $\cos(1 + 2i) = \cos(1)\cosh(2) - i\sin(1)\sinh(2)$:

- **Complex Sine ($\sin(1 + 2i)$): 1 ENTER 2 ↶ CMPLX SIN** → (Y = **3.165779**, X = **1.9596010414**).
- **Complex Cosine ($\cos(1 + 2i)$): 1 ENTER 2 ↶ CMPLX COS** → (Y = **2.032723**, X = **-3.051897799**).

Because ↶ CMPLX SIN, ↶ CMPLX COS, and ↶ CMPLX ↶ x^2 are unary complex operators on (Y, X), evaluating both branches directly on the 4-register stack lifts $(\sin z)^2$ intact into (T, Z) while $(\cos z)^2$ forms in (Y, X), confirming $\sin^2(1+2i) + \cos^2(1+2i) = 1 + 0i$ without re-typing intermediate decimals:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
RAD 1 ENTER 2 ↶ CMPLX SIN	0.0000	0.0000	3.165779	1.959601	Evaluate $\sin(1+2i)$ in (Y, X)
↶ CMPLX ↶ x^2	0.0000	0.0000	6.182115	12.407147	Square complex sine $(\sin z)^2$ in place
1 ENTER 2 ↶ CMPLX COS	6.182115	12.407147	2.032723	-3.051898	$(\sin z)^2$ lifts to (T, Z); $\cos(1+2i)$ in (Y, X)
↶ CMPLX ↶ x^2	6.182115	12.407147	-5.182115	-12.407147	Square complex cosine $(\cos z)^2$ in place
↶ CMPLX +	6.182115	6.182115	**1.000000**	**0.000000**	Sum squares: $\sin^2 z + \cos^2 z = 1 + 0i$

5.4.4 AC Impedance, Crossovers & Transmission Lines

In steady-state AC circuit analysis, series impedances add directly across the complex stack while parallel branches combine through complex reciprocals or ↶ CMPLX $x \parallel y$. When two AC impedances sit in series—say an inductive branch $\boldsymbol{Z}_1 = 8.0 + 4.0i\ \Omega$ and a capacitive branch $\boldsymbol{Z}_2 = 4.0 - 8.0i\ \Omega$—their resistances and reactances add component by component (**4** ENTER **8** +/- ENTER **8** ENTER **4** ↶ CMPLX +), loading $\boldsymbol{Z}_2$ into (T, Z) and $\boldsymbol{Z}_1$ into (Y, X) before ↶ CMPLX +:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
4 ENTER	0.0000	0.0000	4.0000	4.0000	Real part of $\boldsymbol{Z}_2$ pushed
8 +/- ENTER	0.0000	4.0000	-8.0000	-8.0000	Imag part of $\boldsymbol{Z}_2$ pushed
8 ENTER	4.0000	-8.0000	8.0000	8.0000	Real part of $\boldsymbol{Z}_1$ pushed
4	**4.0000**	**-8.0000**	**8.0000**	**4.0000**	Complete 4-register setup
↶ CMPLX +	4.0000	4.0000	**12.000000**	**-4.000000**	$\boldsymbol{Z}_{\text{total}} = 12.0 - 4.0i\ \Omega$

When two branches are wired in parallel ($\boldsymbol{Z}_1 = 20 + 15i\ \Omega \parallel \boldsymbol{Z}_2 = 10 - 25i\ \Omega$), you can either fold both branches in a single dyadic stroke by loading them onto (T, Z) and (Y, X) and pressing ↶ CMPLX $x \parallel y$, or inspect their individual branch admittances $\boldsymbol{Y}_k = 1/\boldsymbol{Z}_k = G_k + jB_k$ along the way:

- **Invert Branch 1 ($\boldsymbol{Y}_1 = 1/\boldsymbol{Z}_1$): 20 ENTER 15** ↶ CMPLX $1/x$ → **-0.024** ($\boldsymbol{Y}_1 = 0.032000 - 0.024000i$ S).
- **Invert Branch 2 ($\boldsymbol{Y}_2 = 1/\boldsymbol{Z}_2$, while $\boldsymbol{Y}_1$ waits in T, Z): 10 ENTER 25** +/- ↶ CMPLX $1/x$ → **0.0344827586** ($\boldsymbol{Y}_2 = 0.013793 + 0.034483i$ S).

- **Sum Admittances & Re-Invert ($Z_{\text{parallel}} = (Y_1 + Y_2)^{-1}$): `0.045793` ENTER `0.010483` ↶ CMPLX x^{-1} $x \leftrightarrow y$ → `20.749994624`** ($Z_{\text{parallel}} = \mathbf{20.7516 - 4.7507}i\ \Omega$):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`20` ENTER `15` ↶ CMPLX x^{-1}	0.0000	0.0000	0.032000	-0.024000	Branch 1 admittance $\boldsymbol{Y}_1$ in (Y, X)
`10` ENTER `25` +/− ↶ CMPLX x^{-1}	0.032000	-0.024000	0.013793	0.034483	$\boldsymbol{Y}_1$ lifts to (T, Z); $\boldsymbol{Y}_2$ in (Y, X)
↶ CMPLX +	0.032000	0.032000	0.045793	0.010483	Sum admittances: $\boldsymbol{Y}_{\text{parallel}} = \boldsymbol{Y}_1 + \boldsymbol{Y}_2$
↶ CMPLX x^{-1}	0.032000	0.032000	**20.7516**	**-4.7507**	Invert admittance: $\boldsymbol{Z}_{\text{parallel}} = 20.75 - 4.75i\ \Omega$

In a Zobel conjugate loudspeaker compensation network ($Z_1 = 3 + 4i\ \Omega \parallel Z_2 = 3 - 4i\ \Omega$), where a series resistor-capacitor branch $3 - 4i\ \Omega$ is wired across an inductive voice coil $3 + 4i\ \Omega$, conjugate symmetry lets you solve the network entirely within the lower two stack registers without touching Z or T: invert $3 + 4i\ \Omega$ in place on (Y, X) into $Y_1 = 0.120000 - 0.160000i$ S, swap the real conductance $G = 0.120000$ S into Register X with $x \leftrightarrow y$ (since the conjugate susceptances $\mp\ 0.160000i$ S cancel identically), double G to 0.240000 S, and tap x^{-1}:

- **Zobel Compensated Loudspeaker Load: `3` ENTER `4` ↶ CMPLX x^{-1} $x \leftrightarrow y$ `2` × x^{-1} → 4.166667 Ω.**

When an AC network is not symmetric—such as a four-arm Thévenin AC impedance bridge $Z_{\text{th}} = (Z_A \parallel Z_C) + (Z_B \parallel Z_D)$ with $Z_A = 10 + 20i\ \Omega$, $Z_C = 20 + 0i\ \Omega$, $Z_B = 30 - 10i\ \Omega$, and $Z_D = 0 - 25i\ \Omega$—each parallel arm uses the dyadic complex harmonic operator ↶ CMPLX $x \parallel y$ across four real registers. Activating the 8-register stack (↷ FLAGS 8-LVL), providing a four-level complex stack) lets $Z_{AC} = Z_A \parallel Z_C$ ride intact into registers (R_6, R_5) while Z_B and Z_D are loaded, drop back into (T, Z) when ↶ CMPLX $x \parallel y$ resolves Z_{BD}, and combine directly via ↶ CMPLX + at full 16-digit internal precision without **STO**/**RCL** or re-typing:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
↷ FLAGS 8-LVL `10` ENTER `20` ENTER `20` ENTER `0`	10.0000	20.0000	20.0000	0.0000	Enable 8-level stack; load Z_A and Z_C
↶ CMPLX $x \parallel y$	0.0000	0.0000	10.7692	6.1538	Parallel branch $Z_{AC} = Z_A \parallel Z_C$ in (Y, X)
`30` ENTER `10` +/− ENTER `0` ENTER `25` +/−	30.0000	-10.0000	0.0000	-25.0000	Load Z_B, Z_D; Z_{AC} rides up to (R_6, R_5)
↶ CMPLX $x \parallel y$	10.7692	6.1538	8.8235	-14.7059	$Z_{BD} = Z_B \parallel Z_D$ in (Y, X); Z_{AC} drops to (T, Z)
↶ CMPLX +	0.0000	0.0000	**19.5928**	**-8.5520**	Sum branches: $Z_{\text{th}} = Z_{AC} + Z_{BD}$

In a second-order Butterworth low-pass section $H_{\text{Butter2}}(s) = 1 \,/\, (s^2 + \sqrt{2}s + 1)$, evaluating the normalized crossover frequency $s = j$ cancels $j^2 = -1$ against $+1$ in the denominator and reduces it to $B_2(j) = 0 + \sqrt{2}i$. Inverting that denominator vector in place on (Y, X) via ↶ CMPLX x^{-1} yields the Butterworth crossover response, and pressing ↶ CMPLX ↶ x^2 squares it in place to verify the fourth-order Linkwitz-Riley (**LR4**) loudspeaker crossover ($Y =$ **-0.500000**, $X =$ **0.000000**, or -6.02 dB at -180° phase):

- **2nd-Order Butterworth Crossover Response ($1/(j\sqrt{2})$):** `0` ENTER `1.414214` ↶ CMPLX x^{-1} → $H_{\text{Butter2}}(j)$ = **0.000000 - 0.707107*i*** (-3.01 dB, -90°).

Likewise, in an inverting operational-amplifier active RC stage where a virtual-ground summing node sets the closed-loop voltage gain to A_v = -$\boldsymbol{Z}_f$ / $\boldsymbol{Z}_{\text{in}}$, pairing feedback impedance $\boldsymbol{Z}_f$ = 5000 - 5000i Ω in (*T*, *Z*) with input impedance $\boldsymbol{Z}_{\text{in}}$ = 1000 - 1000i Ω in (*Y*, *X*) (`5000` ENTER `5000` +/- ENTER `1000` ENTER `1000` +/- ↶ CMPLX ÷ ↶ CMPLX +/- $x\leftrightarrow y$) engages all four registers during ↶ CMPLX ÷ and ↶ CMPLX +/- to determine A_v = **-5.000000** + **-5** i:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`5000` ENTER `5000` +/- ENTER	0.0000	5000.0000	-5000.0000	-5000.0000	Feedback impedance Z_f pushed
`1000` ENTER `1000` +/-	5000.0000	-5000.0000	1000.0000	-1000.0000	Input impedance Z_{in} in (*Y*, *X*)
↶ CMPLX ÷	5000.0000	5000.0000	5.000000	0.000000	Complex ratio Z_f / Z_{in} = 5.0 + 0i
↶ CMPLX +/-	5000.0000	5000.0000	**-5.000000**	**0.000000**	Negate gain: A_v = -5.0 + 0i

Finally, in RF transmission-line analysis, when a Z_0 = 50 Ω coaxial cable is terminated by a dipole antenna of impedance $\boldsymbol{Z}_L$ = 75 + 40i Ω, loading $\boldsymbol{Z}_L$ - Z_0 = 25 + 40i into (*T*, *Z*) and $\boldsymbol{Z}_L$ + Z_0 = 125 + 40i into (*Y*, *X*) lets ↶ CMPLX ÷ resolve the reflection coefficient $\boldsymbol{\Gamma}$ = 0.274311 + 0.232221i (|$\boldsymbol{\Gamma}$| = 0.359406, or |$\boldsymbol{\Gamma}$|^2 = 12.92% reflected power):

- **Voltage Standing Wave Ratio (VSWR = (1 + |Γ|) / (1 - |Γ|)):** `0.359406` ENTER ENTER `1` + $x\leftrightarrow y$ `1` $x\leftrightarrow y$ - ÷ → VSWR = **2.1221**.
- **Dipole Antenna Return Loss (RL = -20 $\log_{10}$|Γ|):** `0.359406` ↶ LOG `20` × +/- → RL = **8.89 dB** (8.8883 dB):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`25` ENTER `40` ENTER	0.0000	25.0000	40.0000	40.0000	Numerator Z_L - Z_0 = (75-50) + 40i pushed
`125` ENTER `40`	25.0000	40.0000	125.0000	40.0000	Denominator Z_L + Z_0 = (75+50) + 40i in (*Y*, *X*)
↶ CMPLX ÷	25.0000	25.0000	**0.274311**	**0.232221**	Reflection coefficient $\boldsymbol{\Gamma}$ = 0.2743 + 0.2322i
$x\leftrightarrow y$ ↶ R→P	25.0000	25.0000	0.702494	**0.359406**	Polar: Phase ϑ = 0.702494 rad (40.25°) in *Y*, magnitude |$\boldsymbol{\Gamma}$| = 0.3594 in *X*
ENTER ENTER `1` + $x\leftrightarrow y$ `1` $x\leftrightarrow y$ - ÷	0.702494	0.702494	0.702494	**2.1221**	VSWR = (1+|$\boldsymbol{\Gamma}$|)/(1-|$\boldsymbol{\Gamma}$|) = 2.1221

- **VNA Microwave Port Return Loss (S_{11} = 0.25 - 0.40i) & Quarter-Wave (λ/4) Matching:** For S_{11} = 0.25 - 0.40i ($|S_{11}|$ = 0.471699, or 22.25% reflected power), convert S_{11} on (Y, X) to polar magnitude via [x⇄y] [↶] [▸θ,r] and evaluate -20 $\log_{10}|S_{11}|$; to match a Z_L = 100 Ω load to Z_0 = 50 Ω via a quarter-wave line ($Z_{\text{line}} = \sqrt{(50 \times 100)}$ ≈ 70.7107 Ω), square Z_{line} and divide by Z_L:
  - **VNA Port Return Loss:** `0.25` [ENTER] `0.40` [+/−] [↶] [▸θ,r] [↶] [LOG] `20` [×] [+/−] → RL = **`6.5266998468`** dB.
  - **Quarter-Wave Transformer Input Impedance ($Z_{\text{in}} = Z_{\text{line}}^2 / Z_L$):** `70.7107` [↶] [x²] `100` [÷] → 50.0000 Ω (VSWR = 1.0000:1).

For a lossy coaxial line with series impedance Z' = 0.1 + 2.5i Ω/m in (T, Z) and shunt admittance Y' = 10^{-6} + $10^{-3}i$ S/m in (Y, X), [↶] [CMPLX] [÷] followed by [↶] [CMPLX] [√x] evaluates Oliver Heaviside's characteristic impedance $Z_0 = \sqrt{(Z'/Y')}$ = **50.0095 - 0.9748i Ω**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`0.1` [ENTER] `2.5` [ENTER]	0.0000	0.1000	2.5000	2.5000	Load series impedance Z' = 0.1 + 2.5i
`1` [E] `6` [+/−] [ENTER] `1` [E] `3` [+/−]	0.1000	2.5000	1.0000×10^{-6}	1.0000×10^{-3}	Load shunt admittance Y' into (Y, X)
[↶] [CMPLX] [÷]	0.1000	0.1000	2499.9975	-97.4999	Complex impedance ratio Z'/Y'
[↶] [CMPLX] [√x]	0.1000	0.1000	**50.0095**	**-0.9748**	Characteristic impedance $Z_0 = \sqrt{(Z'/Y')}$

Attempting complex division or [↶] [CMPLX] [x⁻¹] or [↶] [CMPLX] [LN] at the origin 0 + 0i halts non-destructively with **`DIVIDE BY 0`** (Code 01), while unsupported combinations such as [↶] [CMPLX] [↶] [ASIN] display **`INVALID DATA`** (Code 02) until cleared with [C ON]. By pairing two complete Gaussian planes across (T, Z) and (Y, X) on a four-line display, complex arithmetic replaces trigonometric differential equations with direct vector algebra across alternating-current networks, acoustic filters, and transmission lines.

Chapter 6: The Square-Cube Law, Solid Geometry & Three-Dimensional Space

6.1 Galileo's Giant and the Square-Cube Law

6.1.1 Bone Cross-Section Versus Body Mass

A common garden ant hoists a leaf ten times its own body weight, whereas a human athlete struggles to lift a barbell of a single body mass. If a human being were scaled up by a factor of ten into a sixty-foot giant with every proportion of bone, muscle, and sinew held geometrically identical, the giant's femurs would shatter the moment it tried to stand.

In 1638, while confined under house arrest at his villa in Arcetri outside Florence, Galileo Galilei explained this disparity in his final treatise, *Two New Sciences* (*Discorsi e dimostrazioni matematiche intorno a due nuove scienze*) *(Galilei, 1638).* Galileo demonstrated that nature is governed by a strict geometric asymmetry between area and volume, now known as the *Square-Cube Law.*

When any three-dimensional object, such as an ant, a cathedral column, a spherical fuel tank, or a silicon microprocessor, has every linear dimension (length, width, and height) scaled by a multiplier ℓ, its area and volume diverge in growth:

$$L \longrightarrow \ell L, \qquad A \propto \ell^{2}, \qquad V \propto \ell^{3}, \qquad \frac{A}{V} \propto \frac{\ell^{2}}{\ell^{3}} = \frac{1}{\ell} \tag{Eq. 6.1}$$

The tensile or compressive strength of a bone or steel column depends on its *cross-sectional area* ($A \propto \ell^{2}$), because that cross-section is where intermolecular bonds resist the load. Likewise, aerodynamic drag during a fall and the rate at which an organism or machine can shed heat, absorb oxygen across lung membranes, or radiate thermal energy depend on its *exposed surface area* ($A \propto \ell^{2}$). Yet the gravitational weight the skeleton must support, as well as the metabolic or electrical heat generated inside the body, depends on the number of cells or atoms inside its *enclosed volume* ($V \propto \ell^{3}$).

When a human being's height, width, and depth are multiplied by $\ell = 10$ to form a sixty-foot giant, the cross-sectional area of the leg bones increases by $10^{2} = 100$ times, whereas the volume and weight increase by $10^{3} = 1{,}000$ times. Every square centimeter of the giant's bone is subjected to ten times the compressive stress of a normal human femur and fractures under its own weight. Evolutionary biologist J. B. S. Haldane analyzed this biomechanical limit in his 1926 essay *"On Being the Right Size,"* calculating why the sixty-foot Giant Pope and Giant Pagan in *Pilgrim's Progress* would shatter their thighs at every step.

Conversely, as Haldane noted, a mouse or an ant dropped down a thousand-foot mine shaft walks away unharmed: because terminal velocity in air occurs where gravitational weight ($\propto \ell^{3}$) balances aerodynamic drag ($\propto A\, v^{2} \propto \ell^{2}\, v^{2}$), terminal impact speed scales as $v_{\text{term}} \propto \sqrt{(\ell^{3} \,/\, \ell^{2})} = \sqrt{\ell}$. To survive on land, large animals cannot be scaled-up small animals. As Galileo sketched in the margins of *Two New Sciences*, an elephant or a *Brachiosaurus* requires disproportionately thick, pillar-like legs. For the same reason, a civil engineer scaling a tabletop balsa-wood bridge model up by $\ell = 50$ across a river gorge must thicken the steel chords or switch to hollow box girders lest the full-scale span collapse under its own dead weight.

6.1.2 J. B. S. Haldane's Thermal Wall: Surface Area (L^2) vs. Volume (L^3)

This same $A/V \propto 1/\ell$ ratio governs thermal survival across biology and modern electronics. Inside a high-performance microprocessor, billions of nanometer-scale transistors switch at gigahertz frequencies and produce localized thermal flux exceeding 100 W/cm^2, a power density approaching the heat flux inside a nuclear reactor core. Under Jean-Baptiste Joseph Fourier's Law of Thermal Conduction and Isaac Newton's Law of Cooling *(Incropera et al., 2007)*, convective heat removal from a solid boundary to a surrounding fluid is proportional to the exposed surface area A_{surface}:

$$\dot{Q} = h \cdot A_{\text{surface}} \cdot (T_{\text{solid}} - T_{\text{fluid}}) \qquad \textbf{(Eq. 6.2)}$$

Because heat generation scales with volume (ℓ^3) while heat dissipation scales with surface area (ℓ^2), scaling up any heat-producing body creates a thermal bottleneck. In warm-blooded biology, this scaling contrasts metabolic torpor in hummingbirds with vascular radiators in elephants. A 3-gram ruby-throated hummingbird ($\ell \approx 2$ cm) has a large surface-area-to-volume ratio ($A/V \propto 1/\ell$). It loses body heat across its skin so rapidly relative to its small thermal mass that its heart must beat 1,200 times per minute by day, and it must enter deep hypothermic *torpor* at night to avoid starving before dawn.

Conversely, a 5,000-kilogram African elephant ($\ell \approx 200$ cm, a 100× linear scale-up over the hummingbird) generates $100^3 = 1{,}000{,}000\times$ the internal metabolic heat while possessing only $100^2 = 10{,}000\times$ the skin surface area. With 1/100th the relative cooling area, an elephant would overheat if evolution had not departed from geometric similarity by equipping it with thin, vascularized ear-fins that act as biological radiators.

A silicon CPU package faces the same thermal constraint, which engineers resolve with extruded pin-fin arrays. Leaving an aluminum heat spreader as a flat block provides minimal surface area, whereas machining or extruding the aluminum into dozens of thin parallel fins removes volume and mass while multiplying the exposed convective boundary area A_{surface} by 300% to 2,500%. This geometry overcomes Galileo's Square-Cube Law by folding additional surface area into the same footprint.

In real-world mechanical design, aerospace enclosures, heat sinks, and structural castings are never isolated cubes. They are bounded by dozens of planar walls, sloped frustum roofs, cylindrical bosses, and cutout apertures. On the SC-32, you audit a composite solid by using the statistical accumulator [Σ+] as a *surface-area and volume ledger*. First clear the statistical registers with [CLΣ] ([↶] + [Σ+]), then evaluate each rectangular wall, circular boss, or trapezoidal facet on Registers Y and X and add it with [Σ+], or deduct drilled holes and window cutouts with [↶] [Σ−]. Register X counts each finished face group (n = 1, 2, 3, ...) until [↷] [SUMS] ([↷] + [Σ+]) and softkey [Σx] recall the grand total.

6.2 Heron's Triangle, Ramanujan's Ellipse & Archimedes' Cylinder

To measure triangular land parcels, bolted pipe flanges, planetary oblateness, and curved pressure vessels, mathematicians over two millennia established a body of planar and solid mensuration formulas. Each formula encodes a geometric relation that reduces a multi-step measurement to a direct stack calculation.

6.2.1 Planar Primitives, Difference-of-Squares Rings & Silver-Ratio Octagons

Every three-dimensional cross-section or enclosure boundary begins with two-dimensional planar figures, ranging from rectangles, triangles, trapezoids, and circles to circular sectors, ellipses, parabolic arches, and cyclic quadrilaterals. On the four-register stack, *factoring differences of squares* ($R^2 - r^2 = (R - r)(R + r)$) and *exploiting geometric symmetry* reduce intricate planar sections to one-pass calculations that require no intermediate scratchpad storage.

The *circular annulus* is the flat ring formed by a pipe flange gasket, a thrust washer, or a hollow cylindrical wall bounded by outer radius R and inner bore radius r. Evaluating $\pi R^2 - \pi r^2$ term by term multiplies by π twice and loses significant digits when the wall is thin ($R \approx r$), because squaring two nearly equal radii and subtracting them cancels their leading digits. Factoring the difference of squares resolves both issues at once:

$$A = \pi(R^2 - r^2) = \pi(R - r)(R + r) = 2\pi\left(\frac{R + r}{2}\right)(R - r) \qquad \textbf{(Eq. 6.3)}$$

Geometrically, this factored form says that if you snip a circular ring and unroll it, its exact area equals its *average circumference* $2\pi R_{\text{mean}}$ multiplied by its *radial wall thickness* $t = R - r$. On the SC-32, you can either subtract the squares **R** [↶] [x²] **r** [↶] [x²] [-] [↷] [π] [×] or form $(R - r)$ in Register Y and $(R + r)$ in Register X before pressing [×] [↷] [π] [×] to preserve full precision on thin-walled aerospace tubing. Deferring the transcendental multiplier π until after the radial terms have been combined keeps the intermediate span exact and avoids scaling by π twice.

When an instrument cover or structural gusset is a *regular octagon* of side length s rather than a circle, its area follows directly from square dissection without a trigonometric cotangent lookup. A regular octagon is a bounding square of width $(1 + \sqrt{2})s$ that has had four 45°–45°–90° right-isosceles corner triangles of leg $s/\sqrt{2}$ clipped off its four corners. Each clipped corner has area $1/2(s/\sqrt{2})^2 = (1/4)s^2$, so all four corners together remove s^2 from the bounding square $(1 + \sqrt{2})^2 s^2 = (3 + 2\sqrt{2})s^2$:

$$A = (3 + 2\sqrt{2})s^2 - s^2 = 2(1 + \sqrt{2})s^2 = 2\delta_S s^2 \approx 4.828427\, s^2 \qquad \textbf{(Eq. 6.4)}$$

where $\delta_S = 1 + \sqrt{2} \approx 2.414214$ is the classical *silver ratio* (**2** [√x] **1** [+]), which satisfies $\delta_S^2 = 2\delta_S + 1$ and $1/\delta_S = \delta_S - 2 = \sqrt{2} - 1$.

6.2.2 Ramanujan's Ellipse Perimeter & Earth's Equatorial Bulge

One of the great surprises of classical geometry is the stark contrast between the *area* of an ellipse and its *perimeter*. Because an ellipse with semi-major axis a and semi-minor axis b is a circle of radius b stretched along one axis by the factor a/b, its area scales by that same factor: $A = \pi\, b^2 \cdot (a/b) = \pi\, a\, b$. Yet if you ask for the *perimeter* (circumference) of that same ellipse, no closed-form expression in elementary functions exists anywhere in mathematics. Setting up the arc-length integral $\int \sqrt{(dx^2 + dy^2)}$ leads to an irreducible integral that defied Leonhard Euler and Adrien-Marie Legendre and gave *elliptic integrals* their name.

In 1914, the Indian mathematician Srinivasa Ramanujan published a compact algebraic approximation for the perimeter of an ellipse. Ramanujan defined the dimensionless symmetric flattening parameter h:

$$h=\frac{(a-b)^2}{(a+b)^2}, \qquad P\approx\pi(a+b)\left[1+\frac{3h}{10+\sqrt{4-3h}}\right] \tag{Eq. 6.5}$$

The accuracy of Ramanujan's formula becomes clear when expanding the exact Gauss–Kummer complete elliptic integral for the perimeter of an ellipse as a power series in h: $P = \pi(a+b)[1 + (1/4)h + (1/64)h^2 + (1/256)h^3 + (25/16384)h^4 + (49/65536)h^5 + ...]$. Expanding Ramanujan's nested square root in a Taylor series, $3h / (10 + \sqrt{(4 - 3h)}) = (1/4)h + (1/64)h^2 + (1/256)h^3 + (25/16384)h^4 + (45/65536)h^5 + ...$, matches the exact Gauss–Kummer complete elliptic integral through the h^1, h^2, h^3, and h^4 terms ($(25/16384)h^4$), first differing at the h^5 coefficient ($(45/65536)h^5$ vs. $(49/65536)h^5$) by just $4/65536 = (1/16384)h^5$. Even for highly elongated orbits and elliptical machine cams, Ramanujan's formula is accurate to better than a part per million.

That axial difference (a - b) between semi-major and semi-minor axes also governs the shape of rotating planets. As Isaac Newton predicted in Book III of the *Principia* (1687) and Alexis Claude Clairaut formalized in *Théorie de la figure de la terre* (1743), because the Earth rotates once every 23 hours 56 minutes, centrifugal acceleration forces ocean and mantle outward at the equator and deforms our planet from a sphere into an *oblate spheroid* with equatorial semi-major axis a and polar semi-minor axis b. Geodesists express this polar compression by the dimensionless *flattening* f and its reciprocal $1/f$ (evaluated on the stack without re-typing a via **b** [ENTER] **a** [÷] **1** [x⇄y] [-] [x⁻¹]):

$$f=\frac{a-b}{a}=1-\frac{b}{a}, \qquad \frac{1}{f}=\frac{a}{a-b}=\frac{1}{1-b/a} \tag{Eq. 6.6}$$

Because the Mississippi River flows southward toward Earth's equatorial bulge, its mouth at the Gulf of Mexico sits roughly four miles *farther* from the center of the Earth than its headwaters at Lake Itasca.

6.2.3 Heron's Semiperimeter Formula & Right-Triangle Incircles

When a surveyor in the field measures the three boundary lengths (a, b, c) of a triangular parcel using a steel tape or laser rangefinder, finding the parcel's area from $A = (1/2)bh$ would normally require solving for an interior angle via the Law of Cosines and then taking a sine to find the perpendicular altitude h.

In the 1st century CE, Heron of Alexandria proved in his *Metrica* (a work rediscovered in a Constantinople manuscript in 1896, though Archimedes likely knew the theorem two centuries earlier) that you can bypass trigonometry completely. Define the *semiperimeter* s as half the triangle's perimeter:

$$s=\frac{a+b+c}{2}, \qquad A=\sqrt{s(s-a)(s-b)(s-c)} \tag{Eq. 6.7}$$

Storing the semiperimeter s in Variable **S** ([STO] **S**) lets you subtract each side in turn and multiply the three excess factors (s - a), (s - b), and (s - c) right on the stack before scaling by s and extracting [√x]. Better still, Heron's radicand contains a built-in physical diagnostic: by the Euclidean *triangle inequality*, the sum of any two sides of a real triangle must exceed the third ($a + b > c$, which means $s > c$ and (s - c) > 0). If a misread blueprint specifies three lengths that cannot possibly close into a triangle, such as 3, 4, and 9 where $s = 8$ and (s - 9) = -1 < 0, the factor (s - c) turns negative and [√x] flags an

impossible geometry (**INVALID DATA**).

The semiperimeter s also governs the inscribed circle of a triangle. Dropping an *incircle* of radius r inside any triangle so that it touches all three sides tangentially, and drawing lines from the incenter to the three vertices, partitions the triangle into three smaller triangles of heights r and bases a, b, c, whose areas sum to $(1/2)ar + (1/2)br + (1/2)cr = ((a+b+c)\ /\ 2)r = s \cdot r$. Thus, for *every* triangle:

$$A = r \cdot s \Rightarrow r = \frac{A}{s} \qquad \textbf{(Eq. 6.8)}$$

Once Heron's area A sits in Register X and the semiperimeter s still resides in Variable **S** (or in [↶] [LASTx] before the square root), a single recall and division ([RCL] **S** [÷]) delivers the exact inscribed circle radius $r = A/s$ of *any* triangle. When the triangle happens to be a *right triangle* with orthogonal legs a and b and hypotenuse $c = \sqrt{(a^2 + b^2)}$ (computed in one keystroke via [↶] [▸θ,r]), the tangent segments around the 90° corner simplify $r = A/s$ into the linear relation $r = (a + b - c)\ /\ 2$: because the two tangent segments at the right angle form a square of side r, the remaining segments $(a - r)$ and $(b - r)$ fold onto the hypotenuse so that $(a - r) + (b - r) = c$.

When a parcel's corners are specified by Cartesian survey coordinates (x_1, y_1) and (x_2, y_2) relative to an origin monument, Carl Friedrich Gauss's *Shoelace Theorem*—so named because multiplying diagonally across a two-column coordinate table ($x_1\ y_2$ downward-right minus $y_1\ x_2$ upward-right) criss-crosses like lacing a boot—gives the signed parallelogram area spanned by the two vectors as the 2×2 determinant $D = \det[\ x_1, y_1\ ;\ x_2, y_2\ ] = x_1\ y_2 - y_1\ x_2$ (with triangle area $A = 1/2|D|$). That determinant evaluates in seven RPN keystrokes without a single storage register via x_1 [ENTER] y_2 [×] y_1 [ENTER] x_2 [×] [−], and it likewise solves any 2×2 simultaneous linear system $a_1\ x + b_1\ y = c_1$, $a_2\ x + b_2\ y = c_2$ on the stack via Gabriel Cramer's Rule ($x = (c_1\ b_2 - b_1\ c_2)\ /\ (a_1\ b_2 - b_1\ a_2)$, $y = (a_1\ c_2 - c_1\ a_2)\ /\ (a_1\ b_2 - b_1\ a_2)$), where each coordinate is the ratio of a target parallelogram area to the base parallelogram area.

6.2.4 Cavalieri's Coin Stack, Archimedes' Inscribed Sphere and Cylinder & the Steinmetz Bicylinder

Centuries before Isaac Newton and Gottfried Wilhelm Leibniz codified integral calculus, geometers determined the exact volumes of cones, spheres, paraboloids, and intersecting pipes by comparing cross-sectional slices.

The foundational principle of solid mensuration was formalized in 1635 by Galileo's student Bonaventura Cavalieri, building on methods developed earlier by Archimedes of Syracuse and the 3rd-century Chinese mathematicians Liu Hui and Zu Chongzhi. If twenty coins are stacked into a vertical cylinder on a desk and then nudged sideways so the column leans over or curves into an S-shape, the total volume of the stack does not change by a single cubic millimeter. *Cavalieri's Principle* states that if two solids have the same total height, and if slicing them parallel to their bases at *every* height z reveals cross-sections of identical area $A(z)$ (or a constant ratio of areas), then their total volumes must be identical (or stand in that same ratio).

When the cross-sectional area $A(z)$ of a solid varies as any linear, quadratic, or cubic function of height z, English mathematician Thomas Simpson showed in 1743 that you can find the solid's exact volume from just three horizontal slices—the bottom base A_{bottom}, the mid-height slice A_{mid} (weighted by 4), and the top roof A_{top}—via *Simpson's Universal Prismoidal Formula*:

$$V = \frac{h}{6}(A_{\text{bottom}} + 4A_{\text{mid}} + A_{\text{top}}) \qquad \textbf{(Eq. 6.9)}$$

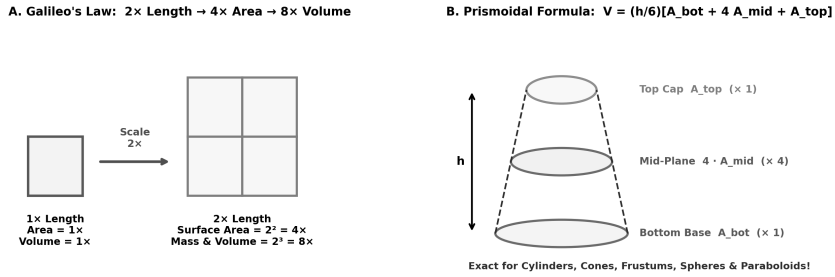


Figure: Galileo's square-cube scaling law and Simpson's universal prismoidal three-slice volume formula.

Cavalieri's Principle and Simpson's three-slice rule unify the classical volume formulas of solid geometry. When any pyramid or cone is sliced horizontally at distance z below its apex, its linear dimensions grow linearly with z/h, so its cross-sectional area grows *quadratically* as $A(z) = A_{\text{base}}(z/h)^2$. Because the average value of $(z/h)^2$ from 0 to h is $1/3$ (which also follows from dissecting a cube into three identical square pyramids meeting at one corner), *every* cone or pyramid encloses one-third the volume of its bounding prism or cylinder, $V = (1/3)A_{\text{base}}h$ (**3** ÷), whether its base is a circle, a square, an octagon, or an ellipse ($A_{\text{base}} = \pi\ ab$, where $V = (1/3)\pi\ abh$). Likewise, a *paraboloid of revolution* whose radius grows as $r(z) \propto \sqrt{z}$ has cross-sectional area $A(z) \propto z$, whose average from 0 to h is one-half ($V = (1/2)\pi\ r^2\ h$).

Above his war machines, his law of the lever, and his hydrostatics, Archimedes prized his proof in *On the Sphere and Cylinder* (c. 225 BCE): when a sphere of radius r is inscribed inside a right circular cylinder of radius r and height $h = 2r$, both the volume and the total surface area of the sphere equal two-thirds $(2/3)$ of the cylinder. Archimedes established this ratio by comparing horizontal slices. When the top hemisphere of radius r is placed beside a cylinder of radius r and height r that has had an inverted 45° cone ($r_{\text{cone}}(z) = z$) carved out of its center, and both solids are sliced at height z above the equator, the Pythagorean theorem shows that the hemisphere's slice is a disk of radius $\sqrt{(r^2 - z^2)}$ and area $\pi(r^2 - z^2)$.

Meanwhile, the punctured cylinder's slice is an annulus of outer radius r and inner hole radius z, whose area is also $\pi(r^2 - z^2)$. Because every horizontal slice matches identically, the hemisphere's volume equals the cylinder minus the cone: $\pi\ r^3 - (1/3)\pi\ r^3 = (2/3)\pi\ r^3$. Doubling for both hemispheres gives $V_{\text{sphere}} = (4/3)\pi\ r^3 = 2/3(2\pi\ r^3) = (2/3)\,V_{\text{cylinder}}$, while its surface area $A_{\text{sphere}} = 4\pi\ r^2$ is likewise $2/3$ of the cylinder's total area ($2\pi\ r^2 + 2\pi\ r(2r) = 6\pi\ r^2$). At Archimedes' request, a sphere inscribed inside a cylinder with the ratio 2:3 was carved onto his funerary monument in Syracuse, and it was by spotting that carving overgrown with brambles in 75 BCE that the Roman orator Cicero rediscovered Archimedes' forgotten grave.

When two equal circular cylinders of radius r intersect at a 90° right angle, as in a high-pressure refinery cross-joint or a Roman groin vault, the common volume forms a Steinmetz bicylinder. Both Archimedes (in *The Method*, preserved on the Archimedes Palimpsest) and Liu Hui (who named the intersection solid the *Mouhefanggai*, or "two square umbrellas") identified a geometric cancellation when a sphere of radius r is inscribed

inside the intersection of the two pipes. Slicing the bicylinder horizontally at height z, where both cylinders have half-width $\sqrt{(r^2 - z^2)}$, produces a *square* cross-section of side $2\sqrt{(r^2 - z^2)}$ and area $4(r^2 - z^2)$, whereas the inscribed sphere's slice is a *circle* of area $\pi(r^2 - z^2)$. At every slice from bottom to top, the bicylinder's square slice is $4/\pi$ times the sphere's circular slice, so by Cavalieri's Principle π cancels out of the volume completely:

$$V_{\text{bicylinder}} = \frac{4}{\pi}\left(\frac{4}{3}\pi r^3\right) = \frac{16}{3}r^3, \qquad A_{\text{bicylinder}} = 16r^2 \tag{Eq. 6.10}$$

Even though the solid is bounded entirely by curved cylindrical walls, neither its volume nor its surface area contains π.

When a cone or pyramid is sliced flat at height h before reaching its apex, with bottom radius R and top radius r, the resulting conical frustum has volume $V = (1/3)\pi\, h(R^2 + Rr + r^2)$, a formula already recorded for square pyramids in the 1850 BCE Moscow Mathematical Papyrus. On the SC-32, squaring R via **R** [↶] [x²] automatically saves R in [↶] [LASTx], so keying [↶] [LASTx] **r** [×] [↶] [LASTx] [↶] [x²] [+] [+] forms the three-term sum $(R^2 + Rr + r^2)$ while typing R and r only once each. Physically, the frustum's volume equals its height h multiplied by the *arithmetic average of three areas*: the bottom base $A_1 = \pi\, R^2$, the top base $A_2 = \pi\, r^2$, and their *geometric mean* $\sqrt{(A_1\, A_2)} = \pi\, Rr$.

The volume coefficient $(\sqrt{2}/12)a^3$ of a regular tetrahedron of edge length a follows from inscribing it inside a cube. Drawing the face diagonals that connect four alternating corners of a cube of edge length $c = a/\sqrt{2}$ (so each face diagonal has length $\sqrt{2}c = a$) constructs a regular tetrahedron of edge a inside the cube. The space outside the tetrahedron consists of four right-angled corner pyramids, each having three mutually perpendicular legs of length c and volume $1/3((1/2)c^2)c = (1/6)c^3$. Slicing off those four corners removes $4 \times 1/6 = 2/3$ of the cube, so the central tetrahedron retains $1 - 2/3 = 1/3$ of the cube's volume: $V = (1/3)c^3 = 1/3(a/\sqrt{2})^3 = (\sqrt{2}/12)a^3$.

In the volume ($V = 2\pi^2\, R\, r^2$) and surface area ($A = 4\pi^2\, R\, r$) of a *torus*—an O-ring gasket with major orbit radius R and minor tube cross-section radius r—the constant π appears *squared* ($\pi^2 \approx 9.8696$, evaluated via [↻] [π] [↶] [x²]) because two orthogonal circles are multiplied together. Around 340 CE, Pappus of Alexandria (rediscovered in 1640 by Swiss mathematician Paul Guldin in his *Centrobaryca*, 1635–1641) proved two theorems that link solid mensuration to the *centroid* (center of mass) of a planar figure. Under the First Theorem of Pappus-Guldinus for surface area of revolution, when a planar curve of arc length L is revolved around an external axis in its plane, the surface area swept out equals the curve's length L multiplied by the circumference of the circle $2\pi\, R_{\text{cm}}$ traveled by its centroid:

$$A = L \cdot (2\pi R_{\text{cm}}) \tag{Eq. 6.11}$$

For a torus, the generating circle has circumference $L = 2\pi\, r$ and its centroid travels a circle of length $2\pi\, R$, which yields $A = (2\pi\, r)(2\pi\, R) = 4\pi^2\, R\, r$, equivalent to slicing the O-ring open and unrolling it into a straight cylinder of length $2\pi\, R$. Under the Second Theorem of Pappus-Guldinus for volume of revolution, when a planar region of area A_{cross} is revolved around an external axis, the volume swept out equals the cross-sectional area A_{cross} multiplied by the distance $2\pi\, R_{\text{cm}}$ traveled by its centroid:

$$V = A_{\text{cross}} \cdot (2\pi R_{\text{cm}}) \tag{Eq. 6.12}$$

For the torus, a circular disk of area $A_{\text{cross}} = \pi\, r^2$ orbits a path of length $2\pi\, R$, which produces $V = (\pi\, r^2)(2\pi\, R) = 2\pi^2\, R\, r^2$ without evaluating an integral.

Whereas two-dimensional space admits infinitely many regular polygons, three-dimensional space permits only five regular convex polyhedra: the 4-faced tetrahedron, 6-faced cube, 8-faced octahedron, 12-faced dodecahedron, and 20-faced icosahedron. In 1750, Leonhard Euler discovered the topological invariant governing every convex polyhedron with V vertices, E edges, and F faces:

$$V - E + F = 2 \qquad \textbf{(Eq. 6.13)}$$

Stretching a polyhedron's wireframe onto a rubber sphere, puncturing one face, and flattening the remaining F - 1 faces onto a plane illustrates this invariant: removing each edge merges two faces (ΔE = -1, ΔF = -1, which preserves V - E + F), and contracting a tree of V vertices and V - 1 edges to a single point yields 1 - 0 + 1 = 2.

At every vertex of a convex polyhedron, at least 3 faces must meet, and their flat corner angles must sum to strictly *less than 360°* so the faces can fold into a 3D corner rather than lying flat. Equilateral triangles have 60° corners, so 3, 4, or 5 of them can fold around a vertex at 180°, 240°, or 300° to build the tetrahedron, octahedron, and icosahedron. Squares have 90° corners, so 3 meet per vertex at 270° to form the cube, while regular pentagons have 108° corners, so 3 meet per vertex at 324° to form the dodecahedron. At six sides, three regular hexagons meet at 3 × 120° = 360° and lie flat in a plane, which precludes any regular polyhedron with six or more sides per face.

6.2.5 Spherical Caps, Girard's Spherical Excess & Asteroid Power Means

Curved shells and planetary surfaces follow three additional mensuration laws for spherical caps, spherical triangles, and triaxial ellipsoids. In petrochemical refineries, liquefied gases are stored inside spherical *Hortonsphere* tanks of radius R because a sphere resists internal pressure with uniform membrane tension (half the hoop stress of a cylinder of equal radius). When an ultrasonic sensor measures liquid depth h at the bottom of the sphere, integrating horizontal disk slices of radius $r^2(z) = R^2 - (R\text{-}z)^2 = 2Rz - z^2$ from $z = 0$ to h gives the *spherical cap volume*:

$$V_{\text{cap}} = \int_0^h \pi(2Rz - z^2)\,dz = \pi\left(Rh^2 - \tfrac{1}{3}h^3\right) = \tfrac{1}{3}\pi h^2(3R - h) \qquad \textbf{(Eq. 6.14)}$$

Archimedes also proved that the curved surface area of any spherical cap or zone of thickness h is $A_{\text{zone}} = 2\pi\, R\, h$, which depends *only* on the slice thickness h and matches the lateral wall area punched out on the circumscribing cylinder.

On a flat plane, the interior angles of every triangle sum to 180° (π rad), whether the triangle is drawn on a postage stamp or across a field. On a curved sphere of radius R, however, great-circle paths bow outward, and the interior angles (α, β, γ) of a *spherical triangle always* sum to more than 180°. Starting at the North Pole, walking south to the Equator along the Prime Meridian, turning 90° west along the Equator to 90° W, and turning 90° north back to the North Pole traces a spherical triangle with three 90° right angles (α + β + γ = 270°) enclosing one-eighth of the globe ((1/8)· $4\pi\, R^2 = (\pi/2)R^2$).

In 1629, the Dutch-French mathematician Albert Girard proved how the angle sum and area are locked together. Each pair of intersecting great circles forms two crescent-shaped *spherical lunes* of area $A_{\text{lune}} = 2R^2\vartheta_{\text{rad}}$. By overlaying the three lune pairs covering the three sides of a spherical triangle—which cover the entire sphere once while triple-covering the triangle itself and its antipodal twin—Girard proved that the *spherical excess* E_{rad} = (α + β + γ) - π is proportional to the triangle's area:

$$E_{\text{rad}} = (\alpha^\circ + \beta^\circ + \gamma^\circ - 180^\circ)\frac{\pi}{180^\circ}, \qquad A_{\text{triangle}} = R^2 E_{\text{rad}} \tag{Eq. 6.15}$$

On a sphere, the physical area of a continent-sized survey network follows directly from three angles measured with a theodolite.

Just as the perimeter of a 2D ellipse has no elementary closed form, the surface area of a 3D *triaxial ellipsoid* with three unequal semi-axes (a, b, c)—shaped like a smooth, flattened river pebble and used as the standard model for elongated asteroids such as Eros or Toutatis—requires incomplete elliptic integrals. In 2004, Danish mathematician Knud Thomsen showed that taking a power mean of the three projected semi-axis products (ab, ac, bc) with exponent $p = 1.6075$ (reusing the exponent p across all three [y^x] terms via [↶] [LASTx] before inverting with [x^{-1}]):

$$A \approx 4\pi\left(\frac{(ab)^p + (ac)^p + (bc)^p}{3}\right)^{1/p} \qquad (p = 1.6075) \tag{Eq. 6.16}$$

approximates the exact elliptic surface integral to within 1.06% across all aspect ratios.

6.2.6 Shell Surface Area and Infill Volume in Additive Manufacturing

Surface area and enclosed volume govern mass estimation in Fused Deposition Modeling (FDM) additive manufacturing when slicing an engineering enclosure or structural bracket. Unlike a cast-iron or injection-molded part that is 100% solid throughout, a 3D-printed part—like the hollow trabecular wing bones of an albatross or eagle—is a *composite sandwich structure* combining a dense outer skin with a porous core.

During fabrication, the print nozzle traces N_{walls} solid perimeter loops (typically 3 loops at $w_{\text{line}} = 0.45$ mm extrusion width) together with solid top and bottom roof/floor layers of comparable thickness $t_{\text{shell}} \approx 1.35$ mm across the exterior boundary surface area A_{surface} to establish a thin-shell skin volume $V_{\text{shell}} \approx A_{\text{surface}} \cdot t_{\text{shell}}$. The enclosed interior volume $V_{\text{interior}} = V_{\text{total}} - V_{\text{shell}}$ is then filled not with solid polymer, but with a periodic minimal-surface *gyroid* or cubic lattice at $f_{\text{infill}} = 15\%$–25% relative density, contributing core polymer volume $V_{\text{core}} = V_{\text{interior}} \cdot f_{\text{infill}}$. Multiplying the combined shell and core polymer volumes by the filament density ρ (such as $\rho \approx 1.24$ g/cm^3 = 0.00124 g/mm^3 for PLA or 1.27 g/cm^3 for PETG) gives the mass of the finished part:

$$m_{\text{part}} = \rho(A_{\text{surface}} \cdot t_{\text{shell}} + V_{\text{interior}} \cdot f_{\text{infill}}) \tag{Eq. 6.17}$$

For thin-walled instrument enclosures and structural brackets, over 70% of the polymer mass resides in the outer perimeter skin ($A_{\text{surface}} \cdot t_{\text{shell}}$) rather than the sparse interior infill, so accurate mass estimation begins with a multi-face surface area takeoff via [Σ+] and [↷] [SUMS].

6.3 Navigating 3D Space: Euclidean Distance, Centroids & Rotational Inertia

Once the surface area and volume of a three-dimensional body are established, spatial analysis turns to Euclidean distance, center of mass, and rotational moment of inertia.

6.3.1 The 3D Pythagorean Theorem, Unwrapped Helical Springs & Direction Cosines

On a flat sheet of paper, the distance between two points follows the 2D Pythagorean theorem $r = \sqrt{(\Delta x^2 + \Delta y^2)}$. Stepping out of the flat plane by rising an elevation Δz perpendicular to the floor forms a second right triangle from the floor diagonal $r = \sqrt{(\Delta x^2 + \Delta y^2)}$ and the vertical rise Δz, whose hypotenuse is the straight-line *3D Euclidean distance*:

$$d = \sqrt{r^2 + \Delta z^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad \textbf{(Eq. 6.18)}$$

On the SC-32, you can evaluate this 3D distance either by summing the three squares ([↶] [x²]) or by chaining the polar key [↶] [▸θ,r] twice—first finding the diagonal shadow r_{xy} across the floor, and then tilting up from that floor diagonal by the vertical height Δz to reach the 3D corner: loading (Δz, Δy, Δx) onto Registers (Z, Y, X) and pressing [↶] [▸θ,r] [x⇄y] [R↓] [↶] [▸θ,r] collapses $\sqrt{(\Delta x^2 + \Delta y^2)}$ into r_{xy} on the first tap (with floor azimuth ϑ in Y) and combines r_{xy} with Δz into $d = \sqrt{(r_{xy}^2 + \Delta z^2)}$ on the second tap (with elevation angle $\varphi = \arctan(\Delta z \,/\, r_{xy})$ in Y) without squaring a single number.

That three-component Pythagorean sum also gives the small-taper active-coil wire blank length L required to wind a *tapered conical compression spring* with bottom radius R_1, top radius R_2, axial height h, and N active coils (prior to squared-and-ground end turns). Unrolling the conical helix resolves the wire path into three mutually perpendicular components: a circumferential wrap of N full turns at the mean radius $R_m = 1/2(R_1 + R_2)$ spanning horizontal distance $\Delta s_\vartheta = 2\pi\ N\ R_m$, an axial climb of vertical rise $\Delta z = h$, and a radial inward taper of spiral step $\Delta r = R_1 - R_2$. Because the circumferential, axial, and radial directions are locally orthogonal along the wire, the active wire length is their 3D Pythagorean norm:

$$L \approx \sqrt{(2\pi N R_m)^2 + h^2 + (R_1 - R_2)^2} \quad \textbf{(Eq. 6.19)}$$

Similarly, when a deep-space radar or robotic gimbal tracks a target at 3D displacement vector $\boldsymbol{V} = (\Delta x, \Delta y, \Delta z)$ at total range $d = |\boldsymbol{V}| = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$, dividing each coordinate by the range d produces the three *direction cosines* ($\cos\alpha$, $\cos\beta$, $\cos\gamma$) = (Δx / d, Δy / d, Δz / d). These direction cosines are the components of the 3D unit pointing vector $\hat{\mathbf{u}} = \boldsymbol{V}/d$ and obey the 3D Pythagorean identity:

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \quad \textbf{(Eq. 6.20)}$$

6.3.2 The Balance Point of a Composite Structure: Center of Mass

As Archimedes established in *On the Equilibrium of Planes*, the balance point of an asymmetric airplane wing, a welded I-beam girder, or a multi-stage rocket follows from static torque equilibrium. When a body is assembled from N components with masses (or cross-sectional areas) m_1, m_2, ..., m_N centered at positions x_1, x_2, ..., x_N, each piece exerts a gravitational torque (first moment) $m_k\ x_k$ about the origin. The composite *center of mass* (*centroid*) $\bar{\mathbf{x}}_{\text{cm}}$ is the exact location where a single support fulcrum balances all those torques, defined as the *mass-weighted mean* of the coordinates:

$$\bar{x}_{\text{cm}} = \frac{m_1 x_1 + m_2 x_2 + \ldots + m_N x_N}{m_1 + m_2 + \ldots + m_N} = \frac{\sum_{k=1}^{N} m_k x_k}{\sum_{k=1}^{N} m_k} \quad \textbf{(Eq. 6.21)}$$

The SC-32 models composite centroids through its statistical registers: when you place each component's mass (or area) m_k in Register Y and its coordinate x_k in Register X, pressing [Σ+] automatically accumulates the torque sum $\sum m_k x_k$ into $\sum xy$ and the total mass $\sum m_k$ into $\sum y$. Opening the [→] [x̄,ȳ] menu ([→] + [y^x]) and pressing the weighted-mean softkey [x̄w] then divides $\sum xy$ / $\sum y$ in a single keystroke to return $\bar{x}_{cm}$.

6.3.3 Rotational Moments of Inertia, Hollow Shafts & the Parallel-Axis Theorem

In linear motion, a 10-kilogram steel block requires the same force to accelerate whether it is shaped like a compact cannonball or a thin bicycle wheel. In *rotational motion*, however, shape and mass distribution determine the inertial resistance.

When a small mass m orbits an axle at radial distance r with angular velocity ω (rad/s), its linear speed is $v = r\omega$, so its kinetic energy is:

$$K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}m(r\omega)^2 = \tfrac{1}{2}(mr^2)\omega^2 \qquad \textbf{(Eq. 6.22)}$$

Summing $m\,r^2$ over every particle in a rigid body defines its *rotational moment of inertia* $I = \int r^2\,dm$, so that rotational kinetic energy is $K_{rot} = (1/2)\,I\,\omega^2$ and angular momentum is $L = I\,\omega$. Because the radial distance r is *squared* ([←] [x^2]), moving a kilogram of mass twice as far from the axis of rotation makes it *four times* harder to spin up or slow down.

That r^2 radial weighting governs conservation of angular momentum, flywheel shape factors, off-axis pivots, and hollow shaft stiffness. When a figure skater enters a spin with her arms and free leg extended outward, a large fraction of her mass sits at a substantial radius r from her vertical spin axis, which produces a large moment of inertia I_{open}. Pulling her arms and leg tightly against her chest reduces r by roughly a factor of two and cuts her moment of inertia $I_{closed} \propto r^2$ by a factor of three or four. Because the ice exerts negligible rotational friction torque, her angular momentum $L = I\omega$ is conserved, and reducing I to one-third forces her spin rate ω to triple.

For symmetric bodies of total mass M and outer radius R, the center-of-mass moment of inertia takes the canonical form $I_{cm} = k\,M\,R^2$, where the dimensionless shape factor k reflects how close the mass lies to the outer boundary: $I_{cm} = M\,R^2$ ($k = 1$, all mass at radius R) for a thin cylindrical hoop or rim, $I_{cm} = (1/2)\,M\,(r_1^2 + r_2^2)$ for a thick annular ring of radii r_1 and r_2, $I_{cm} = (1/2)\,M\,R^2$ ($k = 0.50$) for a solid uniform cylinder or disk, $I_{cm} = (2/3)\,M\,R^2$ ($k \approx 0.67$) for a thin spherical shell, $I_{cm} = (2/5)\,M\,R^2$ ($k = 0.40$) for a solid uniform sphere, and $I_{cm} = (1/12)\,M\,L^2$ for a slender rod of length L pivoted about its midpoint. Consequently, a mechanical punch-press flywheel concentrates nearly 80% of its cast iron into a thick outer rim supported by a thin inner web so that almost 90% of its mass acts at maximum r^2 radius.

When a robotic arm link or pendulum pivots about a shoulder joint displaced by a perpendicular distance d from its centroid rather than about its center of mass, Swiss geometer Jakob Steiner proved that one need not re-integrate $\int r^2\,dm$ from scratch. Because the cross-terms in $(r_{cm} + d)^2 = r_{cm}^2 + 2\,d\,r_{cm} + d^2$ integrate to zero around the center of mass ($\int r_{cm}\,dm = 0$), Jakob Steiner's *Parallel-Axis Theorem* states that the moment of inertia about *any* parallel axis at distance d equals the centroidal inertia plus the inertia of treating the entire body as a point mass at distance d:

$$I_{parallel} = I_{cm} + Md^2 \qquad \textbf{(Eq. 6.23)}$$

For a slender robotic arm link of mass m and length L, shifting the pivot from its midpoint ($I_{\text{cm}} = (1/12)mL^2$) to its shoulder joint at one end ($d = L/2$) adds $m(L/2)^2 = (1/4)mL^2 = 3\ I_{\text{cm}}$, which quadruples the rotational inertia to $I_{\text{pivot}} = (1/12)mL^2 + (3/12)mL^2 = (1/3)mL^2$.

When a solid steel shaft of radius R twists under engine torque, the metal along the central axis barely displaces ($r \approx 0$), so it carries negligible shear stress while adding dead weight. Because a circular cross-section's area grows as $dA = 2\pi\ r\ dr$, integrating $r^2\ dA$ from an inner bore radius R_i to an outer skin radius R_o shows that the geometric *area polar moment of inertia* J (mm^4 or m^4, as distinct from the 3D mass moment of inertia I in $\text{kg}\cdot\text{m}^2$, evaluated via **4** [y^x]) scales with the fourth power of the radius:

$$J = \int_{R_i}^{R_o} r^2 (2\pi r\, dr) = \frac{\pi}{2}\left(R_o^4 - R_i^4\right) \qquad \textbf{(Eq. 6.24)}$$

Because mass scales with cross-sectional area (R_o^2 - R_i^2) while polar stiffness scales with (R_o^4 - R_i^4) = (R_o^2 - R_i^2)(R_o^2 + R_i^2), boring out the central core where r is small removes dead weight while sacrificing very little torsional rigidity. For a hollow spherical pressure hull (such as a deep-ocean titanium bathysphere bounded by inner radius R_1 and outer radius R_2), subtracting the inner sphere from the outer sphere scales with the fifth power of radius (**5** [y^x]):

$$I_{\text{shell}} = \frac{2}{5} M \left(\frac{R_2^5 - R_1^5}{R_2^3 - R_1^3} \right) \qquad \textbf{(Eq. 6.25)}$$

6.4 Solid Mensuration, Weighted Centroids & Moments of Inertia

Evaluating planar, volumetric, and inertial formulas on the four-register stack separates into two operational patterns depending on whether the geometry is a single closed-form solid or a multi-part assembly.

For a single geometric primitive—such as the annular pipe flange, the silver-ratio octagon, Heron's triangle via [STO] **S**, the Moscow Papyrus frustum polynomial $R^2 + Rr + r^2$ via [↶] [LASTx], or the hollow drive shaft's fourth-power polar inertia $J = (\pi/2)(R_o^4$ - $R_i^4)$—factoring common multipliers (π, $(1/3)\pi\ h$, or $3\sqrt{3}/2$) to the outside allows the inner dimensional core to accumulate inside Register X; Register Y combines terms transiently while Registers Z and T remain untouched.

When auditing a composite structure assembled from multiple boundary facets or sub-masses, clearing the statistical registers with [CLΣ] ([↶] + [Σ+]) couples the live stack to the six statistical accumulators. Evaluating each face group in Register X and pressing [Σ+] (or [↶] [Σ−] for drilled cutouts) increments the item counter n in Register X and sums the net area or volume in $\sum x$ ([↷] [SUMS] → [Σx]), whereas staging each sub-mass m_k in Register Y and its coordinate y_k in Register X before [Σ+] prepares the first-moment ratio $\bar{\mathbf{y}} = \sum xy\ /\ \sum y$ for [↷] [$\bar{x}$,$\bar{y}$] → [x̄w].

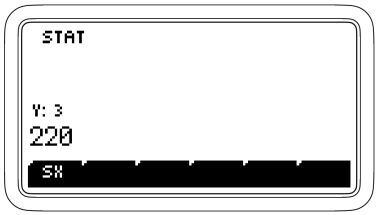


Figure: Surface area of 10x5x4 prism accumulated via Sigma+ → 220

6.4.1 2D Flanges, Silver-Ratio Octagons, Planetary Oblateness & Heron's Triangle

Geometric Region	Analytical Area Formula	Factored / Auxiliary Relation	Geometric Origin & Physical Meaning
Rectangle	$A = l \times w$	$P = 2(l + w)$	Cartesian product of orthogonal basis lengths.
Triangle (Base/Height)	$A = (1/2)\ b \times h$	$A = (1/2)ab \sin C$	Exactly half of the circumscribing parallelogram.
Circle	$A = \pi\ r^2$	$C = 2\pi\ r = dA/dr$	Integration of concentric thin rings ($dA = 2\pi\ r\ dr$).
Circular Sector	$A = (1/2)\ r^2\ \vartheta$ (ϑ in rad)	$s = r\vartheta$	Fractional angular wedge (ϑ / 2π) of full disk $\pi\ r^2$.
Circular Segment	$A = (1/2)\ r^2\ (\vartheta$ - $\sin \vartheta)$	$c = 2r \sin(\vartheta/2)$	Circular sector minus the isosceles chord triangle $(1/2)r^2 \sin \vartheta$.
Circular Annulus (Ring)	$A = \pi\ (R^2$ - $r^2)$	$A = \pi(R$ - $r)(R + r)$	Difference of concentric disks; mean circumference $2\pi((R+r)\ /\ 2)$ times wall thickness (R-r).
Trapezoid	$A = 1/2\ (a + b)\ h$	$m = (a + b)\ /\ 2$	Average of parallel base lengths multiplied by perpendicular altitude h.
Ellipse	$A = \pi\ a\ b$	$e = \sqrt{}(1$ - $(b/a)^2)$	Circle of radius b stretched along one axis by affine ratio a/b.
Parabolic Segment	$A = (2/3)\ b\ h$	$A = (4/3)\ A_{\text{triangle}}$	Archimedes' exhaustion proof: 2/3 of circumscribing box bh.
Regular n-gon	$A = (1/4)\ n\ s^2 \cot(\pi/n)$	$A = (1/2)\ P \cdot r_{\text{in}}$	Sum of n congruent isosceles triangles radiating from the center.
Cyclic Quadrilateral	$A = \sqrt{}((s$-$a)(s$-$b)(s$-$c)(s$-$d))$	$s = (a$+b+c+$d)\ /\ 2$	Brahmagupta's 628 CE theorem for any four-sided figure inscribed in a circle.

Applying this inside-out ordering to planar ring, polygon, and geodetic sections keeps every intermediate result within Registers Y and X:

- **Bolted ASME B16.5 Neoprene Pipe Flange Gasket ($A = \pi(R^2$ - $r^2)$):** On a high-pressure flange gasket with outer radius $R = 85.0$ mm and inner bore radius $r = 50.0$ mm, deferring the π multiplier avoids scaling both radii separately: square $R^2 = 7{,}225.0$ mm^2 in Register X, square $r^2 = 2{,}500.0$ mm^2 (which lifts R^2 into Register Y), subtract to collapse the net radial span $4{,}725.0$ mm^2 back into Register X, and multiply by [R↓] [π] last (**85** [↖] [x²] **50** [↖] [x²] [-] [R↓] [π] [×] → **14,844.02529**, or **14,844.03 mm²** $= 148.44$ cm^2; on thin-walled aerospace shells where $R \approx r$, forming (R - r) in Y and ($R + r$) in X before multiplying by π preserves full guard-digit precision).
- **Regular Octagonal Instrument Bezel ($A = 2(1 + \sqrt{}2)s^2$):** For an octagonal cover of side length $s = 65.0$ mm ($A = 2\delta_S\ s^2$), construct twice the silver ratio $2(1 + \sqrt{}2)$

$\approx$ 4.828427 in Register X first, then square the side length 65^2 = 4,225.0 mm^2 via [↶] [x²] (lifting $2\delta_S$ into Register Y) and multiply via **2** [√x] **1** [+] **2** [×] **65** [↶] [x²] [×] $\rightarrow$ **20,400.1046** (**20,400.10 mm²**).

- **Earth's Geodetic Reciprocal Flattening ($1/f = a/(a - b)$):** For the 3-decimal WGS-84 semi-axes a = 6,378.137 km and b = 6,356.752 km (6,356.7523142 km at full geodetic precision, where $1/f$ = **298.257223563**), subtracting b from a across (Y, X) isolates the equatorial bulge (a - b) = 21.385 km in Register X, dividing by a = 6,378.137 km forms $f \approx 0.00335286$, and inverting with [x^{-1}] (or evaluating $f = 1 - b/a$ in a single entry per axis) via **6378.137** [ENTER] **6356.752** [-] **6378.137** [÷] [x^{-1}] returns **298.2528**.

Heron's semiperimeter formula $A = \sqrt{(s(s\text{-}a)(s\text{-}b)(s\text{-}c))}$, which on paper appears to need four registers for s, (s-a), (s-b), and (s-c) simultaneously, likewise runs as a Register-X accumulator once the semiperimeter $s = (a+b+c)/2$ is stored in Variable **S** ([STO] **S**). On a 13–14–15 cm machinist's layout triangle, storing s = 21.0 cm in **S** lets you form (21 - 13) = 8, (21 - 14) = 7, and (21 - 15) = 6 cm sequentially in Register X, multiply them into 336 cm^3, and scale by s = 21 cm to reach 7,056 cm^4 before [√x] extracts A = 84.0 cm^2. On a surveyed triangular parcel with boundary lengths a = 7.0 m, b = 9.0 m, and c = 12.0 m, the stack sequence proceeds in three stages:

1. **Compute and store the semiperimeter s:** Sum the three boundary lengths and halve them to place $s = (7 + 9 + 12)/2$ = 14.0000 m in Register X, then save s in Variable **S** while keeping 14.0000 live on the stack via **7** [ENTER] **9** [+] **12** [+] **2** [÷] [STO] S $\rightarrow$ **14.0000 m**.

2. **Multiply the three (s - side) excess factors in place:** Subtract the first side via **7** [-] ($\rightarrow$ 7.0000 m), recall **S** and form the second factor via [RCL] **S** **9** [-] [×] ($\rightarrow$ 35.0000 m^2), and form the third factor via [RCL] **S** **12** [-] [×] to accumulate (s-a)(s-b)(s-c) = 7 × 5 × 2 = **70.0000 m³** in Register X.

3. **Scale by s, extract the root, and find the incircle radius:** Keying [RCL] **S** [×] [√x] multiplies 70.0000 × 14.0000 = 980.0000 m^4 and extracts the parcel area $A = \sqrt{980} = 14\sqrt{5}$ = **31.304952 m²**; dividing immediately by [RCL] **S** [÷] yields the inscribed circle radius $r = A/s$ = **2.236068 m** ($\sqrt{5}$ m).

For the right-angled structural channel with orthogonal legs a = 15.0 cm and b = 20.0 cm, evaluating the linear tangent relation $r = (a + b - c) / 2$ on (Y, X) begins by converting (15, 20) to polar form via **15** [ENTER] **20** [↶] [▸θ,r] to place the hypotenuse c = 25.0000 cm in Register X; forming the leg sum $a + b$ = 35.0000 cm via **15** [ENTER] **20** [+] lifts c into Register Y, and [x⇄y] [-] **2** [÷] returns the cooling-pipe inradius r = **5.0000 cm**.

6.4.2 Accumulating Composite Surface Areas via Σ+

When stepping from a single planar face to a three-dimensional enclosure or finned casting bounded by many polygon groups, the statistical accumulator [Σ+] replaces manual scratchpad variables with a running ledger:

1. Press [CLΣ] ([↶] + [Σ+]) to zero all six statistical registers ($\sum x = 0$, $n = 0$).

2. Evaluate the area A_1 of the first face (or symmetric opposing face pair) in Register X and press [Σ+], which adds A_1 into $\sum x$ and replaces Register X with the integer face-group counter n = 1.0.

3. To deduct a milled window aperture, cooling bore, or mistyped entry, evaluate its area in Register X and press [↶] [Σ−].

4. Press [↷] [SUMS] (Right Shift [↷] above [Σ+]) and tap softkey [Σx] ([LFU0]) to recall the net surface area $\sum x$ into Register X.

Applying this ledger to the StackCalc SC-32 rectangular chassis (L = 135 mm, W = 70 mm, H = 16 mm) for an additive-manufacturing shell takeoff, the six exterior walls group into three opposing pairs ($2 \times 135 \times 70 = 18{,}900$ mm^2, $2 \times 135 \times 16 = 4{,}320$ mm^2, and $2 \times 70 \times 16 = 2{,}240$ mm^2). After [CLΣ], each pair is evaluated on (Y, X) and committed to $\sum x$:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	Accumulator $\sum x$	What Happens on the Stack
135 [ENTER] 70 [×] 2 [×]	0.0	0.0	0.0	18900.0	0.0	Top/Bottom area pair evaluated (18,900.0 mm^2)
[Σ+]	0.0	0.0	0.0	**1.0**	**18900.0**	n=1 in Register X; first pair accumulated into $\sum x$
135 [ENTER] 16 [×] 2 [×]	0.0	0.0	0.0	4320.0	18900.0	Side walls area pair evaluated (4,320.0 mm^2)
[Σ+]	0.0	0.0	0.0	**2.0**	**23220.0**	n=2 in Register X; running area accumulated into $\sum x$
70 [ENTER] 16 [×] 2 [×]	0.0	0.0	0.0	2240.0	23220.0	End caps area pair evaluated (2,240.0 mm^2)
[Σ+]	0.0	0.0	0.0	**3.0**	**25460.0**	n=3 in Register X; all 6 faces accumulated into $\sum x$
[↷] [SUMS] → Σx	0.0	0.0	0.0	**25460.0000**	25460.0	Tap [Σx] ([LFU0]): total surface area recalled (25,460.0 mm^2)

The same [Σ+] accumulation tallies curved solids. For a cylindrical lithium battery compartment of radius r = 15 mm and height h = 65 mm, we sum the two circular end caps ($2\pi r^2 = 450\pi \approx 1{,}413.7167$ mm^2) and the unrolled lateral wall ($2\pi r h = 1{,}950\pi \approx 6{,}126.1057$ mm^2):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	Accumulator $\sum x$	What Happens on the Stack
15 [↶] [x²] [↷] [π] [×] 2 [×]	0.0	0.0	0.0	1413.716694	0.0	Two circular end caps $2\pi r^2$ evaluated in mm^2
[Σ+]	0.0	0.0	0.0	**1.0**	**1413.716694**	Accumulate end caps into $\sum x$; face-group count n=1 in X
2 [↷] [π] [×] 15 [×] 65 [×]	0.0	0.0	0.0	6126.105675	1413.716694	Cylindrical lateral wall $2\pi r h$ evaluated in mm^2
[Σ+]	0.0	0.0	0.0	**2.0**	**7539.822369**	Accumulate lateral wall into $\sum x$; face-group count n=2 in X
[↷] [SUMS] → Σx	0.0	0.0	0.0	**7539.822369**	7539.822369	Tap [Σx] ([LFU0]): total cylinder surface area recalled (7,539.82 mm^2)

Because [Σ+] increments the face counter n in Register X while building the running total inside $\sum x$, complex multi-step castings and thermal radiators can be audited by stepping around the blueprint and feeding each face group into [Σ+]:

- **Stepped Cast-Iron Machine Bracket (n = 6 face groups):** When a base block (120 × 80 × 30 mm) is surmounted by a centered boss (60 × 40 × 25 mm), subtracting the 60 × 40 mm boss footprint from the base roof and adding the boss roof back is redundant: viewed from above, the boss roof plus the surrounding ledge form one uninterrupted 120 × 80 = 9,600.0 mm^2 top projection. After [CLΣ], feed each group into [Σ+]: bottom base (120 × 80 = 9,600 mm^2, n=1), long base walls (2 × 120 × 30 = 7,200 mm^2, n=2), short base ends (2 × 80 × 30 = 4,800 mm^2, n=3), long boss walls (2 × 60 × 25 = 3,000 mm^2, n=4), short boss ends (2 × 40 × 25 = 2,000 mm^2, n=5), and top projection (120 × 80 = 9,600 mm^2, n=6). Pressing [R↓] [SUMS] and [Σx] recalls $\sum x$ = **36,200.0000 mm^2**.
- **8-Fin Extruded Aluminum CPU Heat Sink (n = 4 face groups):** A 100 × 100 mm square aluminum base plate has 10,000 mm^2 of top area (20,000 mm^2 counting both sides), whereas extruding 8 vertical cooling fins (100 mm long, 35 mm tall, 2 mm thick) out of its upper surface folds four convective face groups into the same footprint. After [CLΣ], accumulate each group with [Σ+]: the 16 fin sidewalls (16 × 100 × 35 = 56,000.0 mm^2, n=1), the 8 fin top tips (8 × 100 × 2 = 1,600.0 mm^2, n=2), the valley floor between fins (100 × 100 - 1,600 = 8,400.0 mm^2, n=3), and the bottom mounting plate (100 × 100 = 10,000.0 mm^2, n=4). Recalling [R↓] [SUMS] → [Σx] yields $\sum x$ = **76,000.0000 mm^2**—a **3.8**× total area gain (and a **6.6**× gain on the active air-cooled top face prior to applying the hyperbolic fin efficiency factor $\eta_f = \tanh(mL)/(mL)$).
- **Pyramidal Frustum Pavilion Roof:** When a park pavilion is capped by a four-sided hipped pyramidal frustum roof with a 12 m square eaves perimeter, a 4 m square skylight curb at the top, and an inclined slant height of s = 6.5 m, each sloping roof plane is a trapezoid of average width 1/2(12 + 4) = 8.0 m and area A_{face} = 8.0 × 6.5 = 52.0000 m^2. After [CLΣ], multiplying the single-slope trapezoidal area (12 + 4) / 2 × 6.5 = 52.0 m^2 by 4 in Register X and tapping [Σ+] logs all four identical roof slopes at once: $\sum x$ = **208.0000 m^2** (n = 1).
- **Galileo's Square Pyramid Surface-to-Volume Ratio (A/V):** On a regular right square pyramid of base side s = 12.0 mm and altitude h = 8.0 mm, the right triangle formed by h = 8.0 mm and base apothem $s/2$ = 6.0 mm has a 6–8–10 slant height of $l = \sqrt{(8^2 + 6^2)}$ = 10.0000 mm. Multiplying l = 10.0 mm by the four half-bases 4 × (1/2)s = 24.0 mm yields lateral area A_{lat} = 240.0000 mm^2, and adding the square base 12^2 = 144.0 mm^2 via [↰] [x²] and [+] gives total surface area A = 384.0000 mm^2 in Register X. Evaluating the volume $V = (1/3)s^2\,h$ = 1/3(144)(8) = 384.0000 mm^3 lifts A into Register Y while V lands in Register X, so pressing [÷] confirms A/V = **1.0000 mm^{-1}** (matching $V = (1/3)\,r_{\text{in}}\,A$ for inradius r_{in} = 3.0 mm).

6.4.3 3D Volumes, Conical Frustums, Spherical Caps & Girard's Excess

3D Solid Geometry	Analytical Volume Formula	Total External Surface Area Formula	Geometric Origin & Physical Meaning
Rectangular Prism	$V = l \cdot w \cdot h$	$A = 2(lw + lh + wh)$	Extrusion of rectangle lw along h; 6 orthogonal faces.
Right Circular Cylinder	$V = \pi\, r^2\, h$	$A = 2\pi\, r^2 + 2\pi\, r\, h$	Extrusion of circular disk $\pi\, r^2$ along normal axis h.

3D Solid Geometry	Analytical Volume Formula	Total External Surface Area Formula	Geometric Origin & Physical Meaning
Right Circular Cone	$V = (1/3)\ \pi\ r^2\ h$	$A = \pi\ r^2 + \pi\ r\ \sqrt{(r^2 + h^2)}$	Exactly 1/3 of bounding cylinder (Eudoxus & Archimedes).
Sphere	$V = (4/3)\ \pi\ r^3$	$A = 4\pi\ r^2 = dV/dr$	Exactly 2/3 of circumscribing cylinder in both V and A.
Torus (Major R, minor r)	$V = 2\pi^2\ R\ r^2$	$A = 4\pi^2\ R\ r$	Pappus-Guldinus: Centroid orbit ($2\pi\ R$) $\times$ disk area ($\pi\ r^2$) or circumference ($2\pi\ r$).
Frustum of Right Cone	$V = (1/3)\ \pi\ h\ (R^2 + Rr + r^2)$	$A = \pi(R^2 + r^2) + \pi(R + r)s$	Height h times average of top area, bottom area, and geometric mean $\pi\ Rr$.
Spherical Cap (depth h)	$V = (1/3)\ \pi\ h^2\ (3R - h)$	$A_{\text{dome}} = 2\pi\ R\ h$	Horizontal slice of sphere; curved dome area equals wall area of cylinder of height h.
Spherical Zone (2 planes)	$V = (\pi\ h\ /\ 6)(3r_1{}^2 + 3r_2{}^2 + h^2)$	$A_{\text{zone}} = 2\pi\ R\ h$	Slice between two parallel cutting planes through a sphere.
Triaxial Ellipsoid	$V = (4/3)\ \pi\ a\ b\ c$	$A \approx 4\pi\ (((ab)^p + (ac)^p + (bc)^p)\ /\ 3)^{1/p}$	3D affine stretch of unit sphere along semi-axes a, b, c ($p \approx 1.6075$).
Paraboloid of Revolution	$V = (1/2)\ \pi\ r^2\ h$	$A = (\pi\ r\ /\ 6h^2)\ [(r^2 + 4h^2)^{3/2} - r^3]$	Exactly 1/2 of its bounding cylinder (Archimedes' *On Conoids and Spheroids*).
Steinmetz Bicylinder	$V = (16/3)\ r^3$	$A = 16\ r^2$	Orthogonal intersection of two equal cylinders of radius r; π cancels completely.
Universal Prismoid	$V = (h/6)(A_{\text{top}} + 4A_{\text{mid}} + A_{\text{bot}})$	Sum of boundary faces	Exact for any solid whose cross-sectional area $A(z)$ is up to a cubic polynomial in z.
Regular Tetrahedron	$V = (\sqrt{2}/12)\ a^3$	$A = \sqrt{3}\ a^2$	4 equilateral triangular faces; inscribed inside a cube of edge $a/\sqrt{2}$.
Regular Octahedron	$V = (\sqrt{2}/3)\ a^3$	$A = 2\sqrt{3}\ a^2$	8 equilateral triangular faces; two square pyramids joined base-to-base.
Regular Dodecahedron	$V = ((15 + 7\sqrt{5})\ /\ 4)\ a^3$	$A = 3\sqrt{(25 + 10\sqrt{5})}\ a^2$	12 regular pentagonal faces governed by the golden ratio $\varphi = (1+\sqrt{5})\ /\ 2$.
Regular Icosahedron	$V = (5(3 + \sqrt{5})\ /\ 12)\ a^3$	$A = 5\sqrt{3}\ a^2$	20 equilateral triangular faces; dual polyhedron of the dodecahedron.

Each single-solid volume and curved-shell formula from Cavalieri's principle unfolds as a one-dimensional sequence inside Register X (using Register Y only transiently as terms combine), without populating Registers Z or T:

- **Aerodynamic Elliptic Nose Cone vs. the π-Free Steinmetz Bicylinder:** For an aerodynamic elliptic nose cone of semi-axes $a = 24.0$ mm and $b = 15.0$ mm and axial length $h = 50.0$ mm, Cavalieri's one-third cone factor ($V = (1/3)\pi\ abh = 6{,}000\pi$ mm^3) evaluates by forming the bounding box product $a \cdot b \cdot h = 18{,}000.0000$ mm^3 in Register X and scaling by $\pi\ /\ 3$ via [R↓] [π] to return **18,849.555922 mm³**. On the Steinmetz bicylinder (*Mouhefanggai*) where two equal circular pipes of radius $r = 15.0$ mm intersect at right angles, the $4/\pi$ slice ratio already canceled π, so evaluating $V = (16/3)r^3 = 16/3(3{,}375)$ via [y^x] delivers the rational volume **18,000.0 mm³**, while $A = 16r^2 = 16(225)$ via [←] [x^2] gives the curved intersection surface area **3,600.0 mm²**.
- **Egyptian Conical Frustum Grain Hopper via [←] [LASTx]:** Applying the Moscow Papyrus frustum polynomial $R^2 + Rr + r^2$ to a grain hopper of bottom radius $R = 1.80$ m, top throat radius $r = 0.40$ m, and height $h = 2.50$ m, squaring $R^2 = 3.24$ m^2 in Register X saves 1.8 in [←] [LASTx] to form the geometric-mean cross-product $Rr = 0.72$ m^2, which in turn saves 0.4 in [←] [LASTx] to square $r^2 = 0.16$ m^2. This sequence sums all three terms into $R^2 + Rr + r^2 = 4.1200$ m^2 while typing **1.8** and **0.4** only once each, before scaling by $2.5\pi/3$ via **1.8** [←] [x^2] **1.8** [ENTER] **0.4** [×] [+] **0.4** [←] [x^2]

[+] [R↓] [π] [×] **2.5** [×] **3** [÷] → **10.786134777** (**10.7861 m³**).

- **Hollow Hexagonal Brass Bar Stock:** On prismatic machine stock, a hollow hexagonal brass bar ($S = 20.0$ mm outer flat-to-corner side, $s = 8.0$ mm inner hexagonal bore side, and length $L = 150.0$ mm) pairs the difference of squares (S^2 - s^2) with the six-equilateral-triangle hexagon constant $C_6 = 3\sqrt{3}/2 \approx 2.598076$. Subtracting 20^2 - 8^2 = 336.0000 mm² in Register X, scaling by $3\sqrt{3}/2$ into A_{net} = 872.953607 mm², and extruding along $L = 150$ mm via **20** [↶] [x²] **8** [↶] [x²] [-] **3** [√x] [×] **3** [×] **2** [÷] **150** [×] yields **130,943.0411 mm³** (130.94 cm³; similarly, for a solid steel hex bolt head of side $s = 10.0$ mm, height $h = 12.0$ mm, and density $\rho = 7.85$ g/cm³, chaining [√x] [↶] [LASTx] [×] forms $3\sqrt{3}$ without re-typing **3** and yields a mass of **24.4739 grams**).

- **Inscribed Regular Tetrahedron:** For a regular tetrahedron of edge length $a = 50.0$ mm inscribed inside a cube of edge $a/\sqrt{2}$, multiplying the corner-sliced packing factor $\sqrt{2}/12 \approx 0.117851$ by a^3 = 125,000 mm³ evaluates in five steps via **2** [√x] **12** [÷] **50** [ENTER] **3** [y^x] [×] → **14,731.39127** (**14,731.39 mm³**).

- **Spherical Lune Canopy Gores & Albert Girard's Geodetic Excess:** On curved spherical shells, a parachute spherical lune gore panel on a hemispherical canopy of radius $R = 25.0$ cm spanning dihedral wedge angle $\vartheta = 36.0°$ covers $36°/360° = 0.1000$ of the full sphere area $4\pi R^2$ = 7,853.981634 cm², giving A_{lune} = **785.398163 cm²**. Across a geodetic triangulation network on the curved Earth (R = 6,371.0 km) with measured interior angles $\alpha = 82.5°$, $\beta = 68.2°$, and $\gamma = 44.7°$, summing the three theodolite angles to 195.4000° and subtracting 180° isolates Girard's spherical excess E = 15.4000° in Register X, which converts via × π / 180 into $E_{\text{rad}} \approx 0.268781$ rad (2.139% of the globe) and scales by $R^2 = 6{,}371^2$ = 40,589,641 km² to give $A = R^2 E_{\text{rad}}$ = **10,909,712.33 km²**.

- **Knud Thomsen's Triaxial Asteroid & Hortonsphere Spherical Caps:** Applying Thomsen's power mean ($p = 1.6075$) to a triaxial asteroid with semi-axes a = 15.0 km, b = 10.0 km, c = 6.0 km ($ab = 150$, $ac = 90$, $bc = 60$), raising $150^{1.6075}$ with [y^x] and recalling p from [↶] [LASTx] for 90^p and 60^p averages the three terms on (Y, X), raises that mean to $1/p$ via [x⁻¹] [y^x], and multiplies by 4π to yield **1,308.90 km²**. For a Hortonsphere tank of radius $R = 4.0$ m filled to depth $h = 1.5$ m, evaluating $V_{\text{cap}} = (1/3)\pi h^2(3R - h)$ forms $(3R - h)$ = 10.5 m in Register X, recalls $h = 1.5$ m from [↶] [LASTx] to form $h^2(3R - h)$ = 23.625 m³, and scales by π / 3 via **4** [ENTER] **3** [×] **1.5** [-] **1.5** [↶] [x²] [×] [R↓] [π] [×] **3** [÷] → **24.740042147** (**24.7400 m³**).

6.4.4 3D Helical Coils, Direction Cosines, Weighted Centroids & Moments of Inertia

In three-dimensional kinematics and structural mechanics, the stack alternates between single-register Pythagorean reductions and four-register composite audits.

For a tapered conical compression spring with base radius $R_1 = 30$ mm, top radius $R_2 = 10$ mm (mean radius $R_m = 20$ mm), axial height $h = 60$ mm, and $N = 6$ active coils, the three orthogonal displacement squares accumulate sequentially in Register X via $L = \sqrt{((2\pi\ N\ R_m)^2 + h^2 + (R_1 - R_2)^2)}$. Squaring the circumferential wrap $(2\pi\ N\ R_m)^2$ = (753.9822 mm)² = 568,489.2130 mm² via [↶] [x²] in Register X, adding the squared axial climb $h^2 = 60^2$ = 3,600.0000 mm² (→ 572,089.2130 mm²), and adding the squared radial inward taper $(R_1 - R_2)^2 = 20^2$ = 400.0000 mm² (→ 572,489.2130 mm²) before pressing [√x] yields a developed wire blank of **756.6302 mm**—showing that the axial climb and radial taper add 2.65 mm to the flat 753.98 mm circumferential wrap.

Similarly, when a deep-space radar tracks an ascending launch vehicle at 3D displacement vector $\boldsymbol{V}$ = (850, 1100, 550) m, computing the 3D Euclidean range $D = |\boldsymbol{V}|$ = 1,494.9916 m, storing it in Variable **D**, and dividing each Cartesian component in turn places all three direction cosines (cos α, cos β, cos γ) simultaneously across Registers (Z, Y, X) via **1494.9916** [STO] [D] **850** [RCL] [D] [÷] **1100** [RCL] [D] [÷] **550** [RCL] [D] [÷] → (Z = 0.568565, Y = 0.735790, X = **0.3678950437**).

Locating the composite centroid (and homogeneous elastic neutral axis) $\bar{y} = \sum m_i\, y_i \,/\, \sum m_i$ of an asymmetric welded steel girder built from three uniform-grade plates—a bottom flange of m_1 = 40 kg centered at height y_1 = 10 mm, a vertical web of m_2 = 60 kg at y_2 = 160 mm, and a top flange of m_3 = 30 kg at y_3 = 310 mm—engages all four stack registers alongside the statistical accumulators. After [CLΣ], loading each plate's mass m_i into Register Y and its centroid height y_i into Register X before pressing [Σ+] lifts the previous plate masses into Registers Z and T while Register X counts n = 1, 2, 3, and opening [↗] [x̄,ȳ] ([↗] + [yʷ]) to tap [x̄w] divides $\sum xy$ / $\sum y$:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.0	0.0	0.0	0.0	Clear all 6 statistical accumulators
40 [ENTER] **10** [Σ+]	0.0	0.0	40.0	1.0	Log bottom flange (m_1=40 kg, y_1=10 mm; n=1)
60 [ENTER] **160** [Σ+]	0.0	40.0	60.0	2.0	Log web section (m_2=60 kg, y_2=160 mm; n=2)
30 [ENTER] **310** [Σ+]	40.0	60.0	30.0	3.0	Log top flange (m_3=30 kg, y_3=310 mm; n=3)
[x̄w]	40.0	60.0	30.0	**148.461538**	Via [↗] [x̄,ȳ] → [x̄w]: weighted centroid $\bar{y}$ = 148.4615 mm

Evaluating rotational moments of inertia ($I = \int r^2\, dm$) maps the canonical shape factors and Steiner's parallel-axis transfer term directly onto the stack:

- **Composite Cast-Iron Punch-Press Flywheel:** On an 85-kg punch-press flywheel assembled from a thick outer rim (M_{rim} = 65 kg, inner radius r_1 = 0.28 m, outer radius r_2 = 0.35 m) and a solid web disk (M_{web} = 20 kg, radius R = 0.28 m), sum $r_1^2 + r_2^2$ = 0.200900 m² in Register X and scale by $(1/2)M_{\text{rim}}$ = 32.5 kg to store I_{rim} = 6.529250 kg·m² in **A**. Evaluating $I_{\text{web}} = (1/2)M_{\text{web}}R^2$ = 0.784000 kg·m² and recalling **A** to add them via **0.28** [↶] [x²] **0.35** [↶] [x²] [+] **65** [×] **2** [÷] [STO] **A** **0.28** [↶] [x²] **20** [×] **2** [÷] [RCL] **A** [+] → **7.31325** (I_{total} = **7.313250 kg·m²**) confirms that the rim supplies **89.3%** of the energy storage.
- **Jakob Steiner's Parallel-Axis Shift on a Robotic Arm Link:** Applying Steiner's parallel-axis theorem ($I_{\text{pivot}} = I_{\text{cm}} + md^2$) to the slender robotic arm link (m = 3.5 kg, length L = 0.80 m, shoulder offset $d = L/2$ = 0.40 m) executes in two stages across (Y, X). First evaluate the midpoint centroidal inertia $I_{\text{cm}} = (1/12)mL^2$ = 0.186667 kg·m² in Register X; then form the parallel-axis transfer term md^2 = 0.560000 kg·m² (= 3 I_{cm}) in Register X while I_{cm} waits in Register Y, and add them via **0.80** [↶] [x²] **3.5** [×] **12** [÷] **0.40** [↶] [x²] **3.5** [×] [+] → **0.7466666667** ($I_{\text{pivot}} = (1/3)mL^2$ = **0.746667 kg·m²**).

For the thick-walled titanium bathysphere pressure hull (R_2 = 1.10 m outer radius, R_1 = 0.98 m inner cabin radius, and total mass M = 7,347.74 kg for a Ti-6Al-4V shell of density ρ = 4,500 kg/m³ and shell volume $V_{\text{shell}} = (4/3)\pi(R_2^3 - R_1^3)$ = 1.63283 m³, where substituting $M = \rho\, V_{\text{shell}}$ cancels $(R_2^3 - R_1^3)$ to leave $I = (8/15)\pi\rho(R_2^5 - R_1^5)$), evaluating $I_{\text{shell}} = (2/5)M((R_2^5 - R_1^5) \,/\, (R_2^3 - R_1^3))$ engages all four stack registers as the fifth-power difference $(R_2^5 - R_1^5)$ = 0.706589 m⁵ rides up into Registers (T, Z, Y) while the third-power

difference ($R_2{}^3$ - $R_1{}^3$) = 0.389808 m^3 forms in Register X:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1.1 ENTER **5** y^x **0.98** ENTER **5** y^x -	0.0000	0.0000	0.0000	0.706589	Fifth-power difference ($R_2{}^5$ - $R_1{}^5$)
1.1 ENTER **3** y^x **0.98** ENTER **3** y^x -	0.706589	0.706589	0.706589	0.389808	Third-power difference ($R_2{}^3$ - $R_1{}^3$)
÷	0.706589	0.706589	0.706589	1.812660	Ratio ($R_2{}^5$ - $R_1{}^5$)/($R_2{}^3$ - $R_1{}^3$)
7347.74 × **2** × **5** ÷	0.706589	0.706589	0.706589	**5327.4160**	Mass moment of inertia I in kg·m^2

Finally, for a hollow steel drive shaft of outer radius R_o = 40 mm and inner bore radius R_i = 35 mm, the fourth-power polar moment of inertia $J = (\pi/2)(R_o{}^4 - R_i{}^4)$ runs directly on (Y, X) and reflects the algebraic factorization of $R_o{}^4$ - $R_i{}^4$ as a difference of squares of squares:

$$R_o^4 - R_i^4 = (R_o^2 - R_i^2)(R_o^2 + R_i^2) = (40^2 - 35^2)(40^2 + 35^2) = 375 \times 2{,}825 = 1{,}059{,}375 \text{ mm}^4 \qquad \textbf{(Eq. 6.26)}$$

Here the first factor ($R_o{}^2$ - $R_i{}^2$) = 375 mm^2 is proportional to the cross-sectional steel area (375 / 1,600 = 23.44% of a solid 40-mm bar), while ($R_o{}^2$ + $R_i{}^2$) = 2,825 mm^2 weights that steel by its squared radius near the outer wall. Keying **40** ENTER **4** y^x evaluates $R_o{}^4$ = 2,560,000.0000 mm^4 in Register X; keying **35** ENTER **4** y^x - subtracts the hollow core $R_i{}^4$ = 1,500,625.0000 mm^4 to leave $R_o{}^4$ - $R_i{}^4$ = 1,059,375.0000 mm^4; and scaling by R↓ π × **2** ÷ returns J = **1,664,071.7612 mm^4**—retaining 41.38% of the solid shaft's torsional rigidity with less than a quarter of the steel (just as boring out a shaft to R_i / R_o = 0.76 removes over half the mass while retaining 1 - 0.76^4 = **66.6%** of its torsional stiffness).

Across all composite geometric audits, clearing statistical memory with CLΣ isolates each new multi-face or centroid tally, Heron's radical guards against caliper misreads that violate the triangle inequality ($a + b \le c$) by signaling **INVALID DATA** (Code 02, cleared with ON), and halving blueprint diameters ($D \to R = D/2$) before squaring or cubing prevents four-fold and eight-fold scaling errors.

Chapter 7: Polynomial Evaluation, Cardano's Cubic & Brent's Root-Finder

7.1 Al-Khwarizmi's Squares and the 1535 Contest in Milan

Around the year 820 CE, in Baghdad's House of Wisdom (*Bayt al-Hikma*), the Persian mathematician and astronomer Muhammad ibn Musa al-Khwarizmi wrote the foundational treatise *Al-Kitāb al-mukhtaṣar fī ḥisāb al-jabr wa'l-muqābala* (*The Compendious Book on Calculation by Completion and Balancing*). Its title gave every language on Earth the word *algebra*, from *al-jabr* (restoring a subtracted term to the opposite side of an equation), while his own name, Latinized as *Algoritmi*, gave us the word *algorithm*.

When al-Khwarizmi set out to solve a quadratic equation such as his canonical benchmark:

$$x^2 + 10x = 39 \qquad \textbf{(Eq. 7.1)}$$

he did not reach for modern algebraic symbols, because plus signs, minus signs, and exponents would not be invented for another seven hundred years. Notably, al-Khwarizmi did not view x^2 as an abstract number multiplied by itself; he viewed x^2 as a *physical square* of side x, and $10x$ as the area of a rectangle.

To find the side x of a square when its area x^2 plus $10x$ equals 39, al-Khwarizmi constructed a geometric figure in four steps. He began with a central square of side x and area x^2, then split the linear term $10x$ in half into two equal rectangles of width 5 and length x (each having area $5x$) and attached them to two adjacent edges of the $x \times x$ square. The resulting L-shaped figure (a *gnomon*) forms a larger square of side $(x + 5)$ except that the upper-right corner where the two $5 \times x$ wings meet is missing a 5×5 corner square of area 25. Dropping a $5 \times 5 = 25$ square into that empty corner *completes the square*, and adding the same 25 to the 39 on the right balances the areas:

$$(x+5)^2 = 39 + 25 = 64 = 8^2 \Rightarrow x + 5 = 8 \Rightarrow \mathbf{x = 3} \qquad \textbf{(Eq. 7.2)}$$

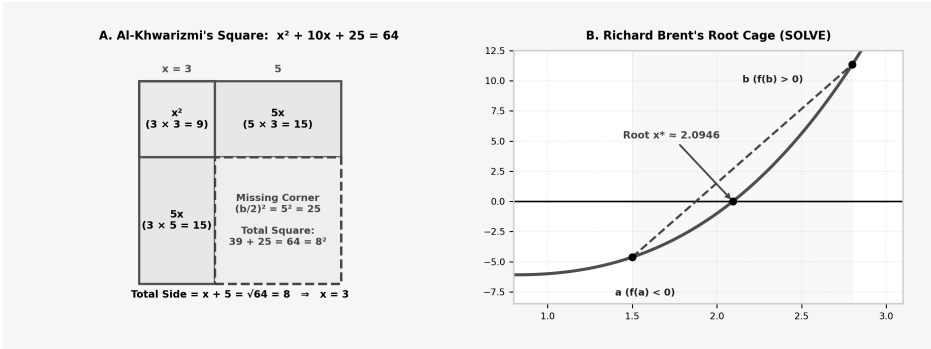


Figure: Al-Khwarizmi's geometric completion of the square (820 CE) and Richard Brent's root-hunting bracket cage (SOLVE).

Applying that same geometric completion to the general quadratic equation $a\,x^2 + b\,x + c = 0$—dividing by a, splitting the middle term $(b/a)x$ into two wings of width $b/2a$, and

filling the missing corner with $(b/2a)^2$—yields the universal *quadratic formula*:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \textbf{(Eq. 7.3)}$$

7.1.1 William Kahan's Warning: Catastrophic Cancellation in the Quadratic Formula

In finite-precision floating-point arithmetic, al-Khwarizmi's quadratic formula requires careful numerical handling. As Berkeley numerical analyst William Kahan demonstrated, the standard formula $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$ contains a numerical instability whenever b^2 is much larger than $4ac$ ($b^2 \gg |4ac|$).

When $b > 0$ and $4ac$ is tiny compared to b^2, the radical $\sqrt{(b^2 - 4ac)}$ is nearly equal to b. Evaluating the root with the plus sign ($-b + \sqrt{(b^2 - 4ac)}$) subtracts two nearly equal numbers and cancels up to half or more of the calculator's significant digits through *catastrophic cancellation*. As codified by William Kahan and in Press, Teukolsky, Vetterling, and Flannery's *Numerical Recipes*, avoiding this loss of significance requires never subtracting $\sqrt{(b^2 - 4ac)}$ from a number of the same sign. Instead, one forms the *Citardauq* ("Quadratic" spelled backward) intermediate quantity q by choosing the sign that *adds* the magnitudes together, and then extracts the larger root $x_1 = q/a$ and the smaller root $x_2 = c/q$ via François Viète's root-product identity $x_1\, x_2 = c/a$:

$$q = -\tfrac{1}{2}\left(b + \operatorname{sgn}(b)\sqrt{b^2 - 4ac}\right), \qquad x_1 = \frac{q}{a}, \qquad x_2 = \frac{c}{q} \qquad \textbf{(Eq. 7.4)}$$

Because q is formed by pure magnitude addition (**b** [↶] [x²] **4** [ENTER] **a** [×] **c** [×] [−] [√x] **|b|** [+] **2** [÷] with sign $-\operatorname{sgn}(b)$), both $x_1 = q/a$ and $x_2 = c/q$ retain every guard digit of precision even when b^2 exceeds $4ac$ by ten orders of magnitude.

Quadratic equations govern two classic inverse problems in mechanics and forensic engineering: inverse ballistic trajectories and highway stopping distance reconstruction. Whereas forward ballistics projects where a shell lands when its launch angle ϑ is already known, the inverse ballistic problem fixes the target coordinates (x, y) on an elevated ridge at known muzzle speed v_0 and solves backward for the required launch angle ϑ. Substituting the trigonometric identity $\sec^2 \vartheta = 1 + \tan^2 \vartheta$ into Galileo's parabolic trajectory $y = x \tan \vartheta - (gx^2/2v_0^2) \sec^2 \vartheta$ converts the unknown launch slope $u = \tan \vartheta$ into a quadratic equation:

$$\left(\frac{gx^2}{2v_0^2}\right)u^2 - xu + \left(y + \frac{gx^2}{2v_0^2}\right) = 0 \qquad \textbf{(Eq. 7.5)}$$

Whenever the target lies within range, its two positive roots give the gunner two distinct paths to the same coordinates: a flat, fast low direct-fire trajectory ($\vartheta_1 = \arctan u_1$) and a high, arcing plunging mortar trajectory ($\vartheta_2 = \arctan u_2$) that descends from above. A similar quadratic governs highway braking and reaction stopping distance ($d = v_0\, t_r + v_0^2 / 2\mu\, g$). When a driver spots a hazard on the highway, the vehicle travels at its initial speed v_0 during the human perception-reaction time t_r ($d_{\text{react}} = v_0\, t_r$) before the brakes lock and tire-pavement friction μ decelerates the vehicle to a halt over a kinetic-energy braking distance $d_{\text{brake}} = v_0^2 / 2\mu\, g$. When crash investigators measure the total stopping distance d at the scene, reconstructing the vehicle's unknown initial speed v_0 requires extracting the positive root of the monic quadratic:

$$v_0^2 + (2\mu g t_r)\, v_0 - 2\mu g d = 0 \qquad \textbf{(Eq. 7.6)}$$

7.1.2 The 1535 Milanese Cubic Contest, Cardano's Formula & Casus Irreducibilis

For seven centuries after al-Khwarizmi, the *cubic equation* ($A\,x^3 + B\,x^2 + C\,x + D = 0$) stood unsolved. In 1494, the Franciscan friar Luca Pacioli (Leonardo da Vinci's mathematics collaborator) declared in his *Summa de arithmetica* that solving the general cubic was as impossible with existing mathematics as squaring the circle.

Pacioli was mistaken. Around 1515, Bologna mathematics professor Scipione del Ferro discovered how to solve the "depressed" cubic $t^3 + pt = q$ (a cubic with no t^2 term). In Renaissance Italy, however, university chairs were not protected by lifelong tenure; a mathematician could be publicly challenged to a high-stakes problem-solving contest in which the loser forfeited his salary, reputation, and professorship. Consequently, del Ferro kept his unpublished solution confidential until his deathbed in 1526, when he entrusted the manuscript to his student Antonio Maria Fior.

Possessing his late master's confidential formula, Fior began boasting that he alone could solve cubic equations, and in early 1535 he challenged Venetian mathematician Niccolò Fontana, born in Brescia and known as Tartaglia ("The Stammerer," after a French soldier's saber wounded his jaw and palate during the 1512 sack of Brescia and left him with a permanent speech impediment). Each contestant deposited thirty cubic problems with a notary in Milan and was given fifty days to solve the rival's list.

Fior had set all thirty of his problems as variations of del Ferro's single form $t^3 + pt = q$. Tartaglia, on the other hand, had posed several different families of cubics, including $t^3 + pt^2 = q$. On the night of February 12, 1535, eight days before the deadline, Tartaglia discovered the general reduction by dissecting a three-dimensional cube of side $(u - v)$ into smaller blocks:

$$(u-v)^3 + 3uv(u-v) = u^3 - v^3 \qquad \textbf{(Eq. 7.7)}$$

Tartaglia saw that setting $t = u - v$, $3uv = p$, and $u^3 - v^3 = -q$ reduces the cubic $t^3 + pt + q = 0$ to a *quadratic equation in* u^3. Within two hours Tartaglia solved all thirty of Fior's problems, whereas Fior failed to solve a single one of Tartaglia's.

News of Tartaglia's victory reached the Milanese physician and algebraist Gerolamo Cardano. In March 1539, Cardano invited Tartaglia to his house and requested the formula, pledging not to publish Tartaglia's method. Tartaglia relented and encoded his rule in a twenty-five-line Italian poem (*"Quando che'l cubo con le cose appresso…"*).

Cardano then combined Tartaglia's rule with his own discovery that *any* full cubic $A\,x^3 + B\,x^2 + C\,x + D = 0$ can be stripped of its quadratic x^2 term by translating the coordinate origin to the curve's point of inflection via $x = t - B/3A$:

$$t^3 + pt + q = 0, \qquad p = \frac{3AC - B^2}{3A^2}, \qquad q = \frac{27A^2D - 9ABC + 2B^3}{27A^3} \qquad \textbf{(Eq. 7.8)}$$

Cardano and his pupil Lodovico Ferrari, who soon discovered how to solve the fourth-degree *quartic equation* by reducing it to a resolvent cubic, now possessed the complete analytical solution to every polynomial up to degree four. In 1543, Cardano traveled to Bologna, inspected Scipione del Ferro's posthumous papers, and verified that del Ferro had discovered the formula twenty years *before* Tartaglia. Considering himself released from his pledge, Cardano published the complete cubic and quartic solutions in his 1545 treatise *Ars Magna* (*The Great Art*) *(Cardano, 1545)*, crediting both del Ferro and Tartaglia.

Inside Cardano's cubic formula appeared an algebraic paradox that led to the discovery of complex numbers:

$$t = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}}, \qquad \text{where } \Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3 \tag{Eq. 7.9}$$

Here the *cubic discriminant* Δ governs the nature of the three roots across three distinct regimes. When $\Delta > 0$, the S-shaped cubic crosses the horizontal axis only once and has *one real root* and *two complex conjugate roots*, formed by rotating $u = \sqrt[3]{(-q/2 + \sqrt{\Delta})}$ and $v = \sqrt[3]{(-q/2 - \sqrt{\Delta})}$ by the cube roots of unity $\omega = -1/2 + i(\sqrt{3}/2)$ and ω^2. When $\Delta = 0$, the cubic touches the horizontal axis tangentially and produces *three real roots with at least one repeated root*, as in the critically damped RLC filter polynomial $s^3 + 12s^2 + 45s + 50 = (s+2)(s+5)^2 = 0$ with a double pole at $s = -5$.

When $\Delta < 0$, the regime known as *casus irreducibilis*, the cubic crosses the horizontal axis three times to produce *three distinct real roots* (such as $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6 = 0$), yet the discriminant Δ under the square root is *strictly negative*. Extracting those three real numbers from Cardano's formula requires passing through the complex plane and taking the cube root of a complex number $r\,e^{i\vartheta}$ by De Moivre's angle trisection $r^{1/3}[\cos(\vartheta/3) + i\,\sin(\vartheta/3)]$, where the imaginary parts of the two conjugate cube roots cancel to zero.

7.1.3 Viète's Trigonometric Escape, Ferrari's Quartic & the Root-Coefficient Checksums

Sixteenth-century algebraists bypassed the complex radicals of *casus irreducibilis* through trigonometry and verified their roots through symmetric coefficient identities, methods that form the foundation of the SC-32's analytical solver.

On the SC-32, pressing [↗] [POLY] ([↗] + [XEQ]) opens the dedicated analytical polynomial workspace for $Ax^3 + Bx^2 + Cx + D = 0$, mapping the six LFU softkeys to [A], [B], [C/ON], [D], [EXEC], and [CLR]. Key each coefficient onto the stack and tap its softkey, setting **0** [D] whenever you want to solve a quadratic $Ax^2 + Bx + C = 0$. Pressing [EXEC] evaluates the discriminant Δ, places the primary root r_1 onto the stack in Gaussian complex format (Y = Real, X = Imag), and writes all three real or complex roots into numbered variable registers **1** through **6** (displayed as **R34**–**R39** in [↖] [REGS] → **STO**) and recalled on the keypad via indirect addressing **34** [STO] **i** [RCL] **(i)** through **39** [STO] **i** [RCL] **(i)**, written in shorthand below as [RCL] **1** through [RCL] **6**).

Behind that single [EXEC] keystroke stand three classical results of polynomial theory: François Viète's trigonometric trisection, Lodovico Ferrari's quartic resolvent, and Viète's root-coefficient relations. In 1593, the French cryptographer and algebraist François Viète noted the structural correspondence between the depressed cubic $t^3 + pt + q = 0$ (when $\Delta < 0$, so $p < 0$) and the trigonometric triple-angle identity for cosine:

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta \Rightarrow \cos^3\theta - \frac{3}{4}\cos\theta - \frac{1}{4}\cos(3\theta) = 0 \tag{Eq. 7.10}$$

By scaling the unknown as $t = 2\sqrt{(-p/3)}\,\cos\vartheta$, Viète matched the depressed cubic term-for-term against the triple-angle cosine identity and expressed all three real roots in closed form without complex square roots:

$$t_k = 2\sqrt{-\frac{p}{3}}\cos\left(\frac{1}{3}\arccos\left(\frac{3q}{2p}\sqrt{-\frac{3}{p}}\right) - \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2 \tag{Eq. 7.11}$$

Geometrically, solving a three-real-root cubic is equivalent to *trisecting an angle on the unit circle* and projecting the three 120°-spaced vertices of an inscribed equilateral triangle onto the horizontal real axis.

Right after Cardano mastered the cubic in 1539, Venetian mathematics teacher Zuanne da Coi challenged Cardano and Tartaglia with a fourth-degree *quartic equation*: $x^4 + 6x^2 + 36 = 60x$. In 1540, Cardano's pupil Lodovico Ferrari solved it by reducing the quartic to a resolvent cubic. After shifting x to remove the x^3 term ($x^4 + px^2 + qx + r = 0$), Ferrari moved $px^2 + qx + r$ to the right side, completed the square on the left as $(x^2 + p/2)^2$, and introduced an *auxiliary variable* y inside the square:

$$\left(x^2+\tfrac{p}{2}+y\right)^2=(p+2y)x^2-qx+\left(\tfrac{p^2}{4}-r+py+y^2\right) \qquad \textbf{(Eq. 7.12)}$$

The right-hand side is an ordinary *quadratic in x*, which becomes a perfect square $(mx + n)^2$—so that the square root can be taken on both sides—precisely when its quadratic discriminant $B^2 - 4AC$ equals zero:

$$q^2-4(p+2y)\left(y^2+py+\tfrac{p^2}{4}-r\right)=0 \qquad \textbf{(Eq. 7.13)}$$

That condition for the auxiliary variable y is a *third-degree cubic equation*, the *resolvent cubic*, which Cardano's [↷] [POLY] solver can solve directly. Once one root y_1 of the resolvent cubic is found, the quartic splits into two ordinary quadratic equations that yield all four roots.

Whenever [↷] [POLY] writes the roots of a cubic ($A\,x^3 + B\,x^2 + C\,x + D = 0$) into variables **1** through **6**, Viète's root-coefficient formulas verify the roots immediately without substituting each value back into the polynomial. Expanding $A(x - r_1)(x - r_2)(x - r_3) = A\,x^3 - A(r_1 + r_2 + r_3)x^2 + A(r_1\,r_2 + r_2\,r_3 + r_3\,r_1)x - A\,r_1\,r_2\,r_3$ and matching coefficients term by term yields the *sum of roots* (*trace*), *pairwise product sum*, and *product of roots* (*determinant*):

$$r_1+r_2+r_3=-\frac{B}{A}, \qquad r_1r_2+r_2r_3+r_3r_1=+\frac{C}{A}, \qquad r_1\cdot r_2\cdot r_3=-\frac{D}{A} \qquad \textbf{(Eq. 7.14)}$$

For our benchmark cubic $x^3 - 6x^2 + 11x - 6 = 0$ with roots 1, 2, 3, recalling and summing [RCL] **1** [RCL] **3** [+] [RCL] **5** [+] gives the trace $1 + 2 + 3 = 6 = -(-6)/1$. Likewise, their pairwise product sum is $1(2) + 2(3) + 3(1) = 11 = +11/1$, and their determinant product [RCL] **1** [RCL] **3** [×] [RCL] **5** [×] is $1 \times 2 \times 3 = 6 = -(-6)/1$. In any *depressed cubic* $t^3 + pt + q = 0$ (where $B = 0$), the three roots always sum identically to zero ($t_1 + t_2 + t_3 = 0$), because the coordinate shift $x = t - B/3A$ places the origin at the centroid of the three roots.

That chain of radical reductions stopped permanently after Ferrari's degree-4 quartic. For nearly three hundred years after *Ars Magna*, European mathematicians—including Ehrenfried Walther von Tschirnhaus, Leonhard Euler, and Joseph-Louis Lagrange—sought a general degree-5 quintic formula ($Ax^5 + Bx^4 + Cx^3 + Dx^2 + Ex + F = 0$). Lagrange identified the underlying obstacle: for a degree-2 quadratic, the resolvent has degree 1; for a degree-3 cubic, Tartaglia's resolvent has degree 2; for a degree-4 quartic, Ferrari's resolvent has degree 3. In each case, the auxiliary equation had *lower* degree than the original polynomial. Yet when Lagrange applied the same permutation method to a degree-5 quintic, the resolvent equation jumped *up* to degree 6 (with $5!/2 = 60$ alternating permutations).

In 1824, Norwegian mathematician Niels Henrik Abel built on Paolo Ruffini's 1799 work to prove that this barrier is absolute. Eight years later, on the night of May 29, 1832, French mathematician Évariste Galois wrote a testamentary letter summarizing his discoveries on the eve of his death at age twenty, founding *Group Theory* and explaining the structural cause. For degrees $n = 2$, 3, 4, the permutation group S_n of the roots decomposes into commutative cyclic quotients ("solvable groups") that correspond to extracting k-th roots $\sqrt[k]{(phantom x)}$. For $n \geq 5$, however, the alternating group A_5, which is isomorphic to the 60 rotational symmetries of a regular icosahedron, is a *simple non-abelian group* that cannot be factored into cyclic steps. No general radical formula in $+$, -, $\times$, $\div$, and $\sqrt[n]{(phantom x)}$ can exist for degree 5 or higher.

7.2 Unpeeling Polynomials from the Inside Out: Horner's Scheme

Before solving a polynomial equation $P(x) = 0$, one needs an efficient, numerically stable method to evaluate $P(x)$ at any argument $x = x_0$.

Consider the general fourth-degree quartic polynomial:

$$P(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 \qquad \textbf{(Eq. 7.15)}$$

Evaluating $P(x)$ term by term as written—computing x^4, multiplying by a_4, computing x^3, multiplying by a_3, computing x^2, and so on—repeats power computations unnecessarily ($4 + 3 + 2 + 1 = 10$ multiplications, or three separate exponentiations). Worse, near a root where large positive and negative power terms nearly cancel, computing $a_n\ x^n$ separately magnifies floating-point rounding error.

In 1819, the English schoolmaster William George Horner (rediscovering a method known to Isaac Newton in 1669 and the 13th-century Chinese algebraist Qin Jiushao in his 1247 *Mathematical Treatise in Nine Sections*) showed what happens when x is factored from the inside out:

$$P(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = (((a_4x + a_3)x + a_2)x + a_1)x + a_0 \qquad \textbf{(Eq. 7.16)}$$

In *Horner's nested scheme*, every explicit exponent vanishes. To evaluate any degree-n polynomial at x_0, you start with the leading coefficient $b_n = a_n$ and repeat a two-step linear recurrence $b_k = b_{k+1}\ x_0 + a_k$ exactly n times, multiplying the running value by x_0 ([×]) and adding the next coefficient a_k ([+] or [-]). Across the four nested stages of a quartic $P(x_0) = (((a_4\ x_0 + a_3)x_0 + a_2)x_0 + a_1)x_0 + a_0$, the running accumulator in Register X starts with $b_4 = a_4$, multiplies by x_0, and adds a_3 in the first stage to form $b_3 = a_4\ x_0 + a_3$; multiplies b_3 by x_0 and adds a_2 in the second stage to form $b_2 = b_3\ x_0 + a_2$; multiplies b_2 by x_0 and adds a_1 in the third stage to form $b_1 = b_2\ x_0 + a_1$; and finally multiplies b_1 by x_0 and adds a_0 to reach the remainder $b_0 = b_1\ x_0 + a_0 = P(x_0)$.

This recurrence requires four multiplications, four additions, zero powers, and zero temporary variables. On the SC-32, Horner's scheme runs even faster when all four stack registers are filled with x_0 at the outset via x_0 [ENTER] [ENTER] [ENTER]. Because the top register T replicates its value downward into Register Z every time the stack drops, a fresh copy of x_0 awaits in Register Y whenever a coefficient addition completes, so the four-register stack operates as an automatic Horner recurrence pipeline without re-typing x_0.

Furthermore, the intermediate values (b_4, b_3, b_2, b_1) produced at each step of Horner's recurrence have direct algebraic meaning: they are the exact coefficients of the cubic

quotient polynomial $Q(x)$ obtained by *Ruffini-Horner synthetic division* when dividing $P(x)$ by $(x - x_0)$:

$$P(x) = (x - x_0)\,Q(x) + b_0 = (x - x_0)(b_4x^3 + b_3x^2 + b_2x + b_1) + P(x_0) \qquad \textbf{(Eq. 7.17)}$$

This Polynomial Remainder Theorem identity $P(x) = (x - x_0)Q(x) + P(x_0)$ provides polynomial deflation, simultaneous derivative evaluation, and suppression of catastrophic cancellation near multiple roots. Once one exact root r_1 of a polynomial is found (where the remainder $P(r_1) = 0$), the intermediate numbers generated during Horner's evaluation at r_1 supply the deflated degree-(n-1) quotient polynomial $Q(x) = P(x)/(x - r_1)$, which reduces a fourth-degree quartic to a cubic or a cubic to a quadratic. Differentiating $P(x) = (x - x_0)Q(x) + P(x_0)$ with respect to x by the product rule yields:

$$P'(x) = Q(x) + (x - x_0)Q'(x) \Rightarrow P'(x_0) = Q(x_0) \qquad \textbf{(Eq. 7.18)}$$

Evaluating the quotient polynomial $Q(x)$ at x_0 by running a second Horner pass on the intermediate coefficients therefore gives the exact derivative $P'(x_0)$ without a separate symbolic derivative formula. In computational thermodynamics and flight-control engineering—where computers evaluate standard-atmosphere pressure-altitude curves $P(h) = a_3\,h^3 + a_2\,h^2 + a_1\,h + a_0$ and NASA Glenn molar heat-capacity fits $C_p(T)/R = a_0 + a_1\,T + a_2\,T^2 + a_3\,T^3 + a_4\,T^4$—nested Horner evaluation also suppresses cancellation near repeated roots such as $s = -5$ in the critically damped RLC filter by updating the local factor $(s + 5)$ at each step instead of subtracting large, independently rounded powers s^3 and $12s^2$.

That same nested-factoring approach simplifies rational formulas and symmetric high-degree equations across physics. In a silicon photovoltaic cell, the maximum power point satisfies a transcendental diode equation that cannot be inverted in closed form, but Australian solar physicist Martin Green (1982) established an accurate rational-logarithmic expression in the normalized open-circuit voltage $v_{oc} = V_{oc} / V_T$, evaluated in one pass on the stack by replicating v_{oc} across all four registers with [ENTER] [ENTER] [ENTER]:

$$\text{Fill Factor } FF = \frac{v_{oc} - \ln(v_{oc} + 0.72)}{v_{oc} + 1} \qquad \textbf{(Eq. 7.19)}$$

Similarly, when two coupled undamped spring-mass pendulums vibrate in their natural normal modes, the eigenvalue equation $\det(\boldsymbol{K} - \omega^2\boldsymbol{M}) = 0$ forms a biquadratic polynomial containing only even powers of frequency ($a\,x^4 + b\,x^2 + c = 0$), which reduces to a quadratic in $u = x^2$ whose positive square roots $\omega = \sqrt{u}$ give the two normal-mode frequencies.

7.3 Tangent Instabilities in Newton's Method and Brent's Hybrid Algorithm

Abel and Galois proved that $x^5 - x - 1 = 0$ cannot be solved with a radical formula, and in applied physics and engineering, governing laws rarely appear as pure polynomials. Physical conservation laws routinely combine algebraic powers x with sines, cosines, exponentials, and logarithms in *transcendental equations* where no symbolic manipulation can isolate the unknown on one side of the equals sign.

Isolating a root x^* where $f(x^*) = 0$ when symbolic algebra cannot invert $f(x)$ led numerical analysts to two contrasting approaches: interval bisection and local tangent or secant extrapolation. When a continuous curve $f(x)$ lies below the axis at $x = a$ with $f(a) < 0$ and above it at $x = b$ with $f(b) > 0$, so that $f(a) \cdot f(b) < 0$, Bernard Bolzano's 1817 *Intermediate*

Value Theorem guarantees that the curve crosses $y = 0$ at least once inside the *bracket* $[a, b]$. The *Bisection Method* evaluates the function at the midpoint $m = 1/2(a + b)$, inspects its sign, and discards whichever half does not contain the sign change. Every step halves the uncertainty and gains 1 binary bit of precision per step (linear convergence, order $p = 1.000$). After 40 bisections, the bracket shrinks by $2^{-40} \approx 10^{-12}$, though requiring 40 to 50 evaluations makes bisection slow.

To accelerate convergence, Isaac Newton in 1669 (testing John Wallis's cubic benchmark $x^3 - 2x - 5 = 0$ near $x_0 = 2$) and Joseph Raphson in 1690 proposed following the local tangent line of the curve:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \qquad \textbf{(Eq. 7.20)}$$

Near a simple root, the Newton-Raphson method is *quadratically convergent* ($p = 2.000$) and doubles the number of accurate decimal digits on every step ($1 \to 2 \to 4 \to 8 \to 16$ digits). When the analytic derivative $f'(x)$ is unavailable, replacing the tangent slope with the *Secant Chord* through the last two iterates yields superlinear convergence at the *golden-ratio exponent* $p = \varphi = (1+\sqrt{5}) / 2 \approx 1.618$. The vulnerability of Newton-Raphson and the Secant Method lies in the slope denominator $f'(x_n)$ (or $f(x_n) - f(x_{n-1})$). Whenever an iterate lands near a local extremum, an inflection plateau, or a horizontal asymptote where the slope flattens out ($f'(x_n) \approx 0$), the nearly horizontal tangent line projects the next estimate x_{n+1} far outside the region of interest or into an oscillatory cycle.

7.3.1 Richard Brent's 1973 Hybrid Root-Finding Algorithm

In 1973, building on earlier work by Theodorus Dekker, the Australian mathematician Richard P. Brent demonstrated in *Algorithms for Minimization Without Derivatives (Brent, 1973)* how to combine the guaranteed bracketing of Bisection with the superlinear convergence of Interpolation so that the solver achieves rapid convergence on smooth curves, requires no derivatives, and never diverges outside its bracket. Six years later, UC Berkeley numerical analyst William M. Kahan *(Kahan, 1979)* adapted this Dekker–Brent hybrid philosophy for Hewlett-Packard's pocket **SOLVE** key, which debuted on the 1979 HP-34C, continued into the 1982 HP-15C Voyager *(HP, 1982)*, and paired with symbolic equations on the 1991 HP-32SII Pioneer *(HP, 1991)*.

Brent's hybrid root-finding algorithm, which drives the SC-32's 64-bit [↗] [SOLVE] key ([↗] + [7], paired with the [↗] [EQN] equation editor), combines inverse quadratic interpolation with five acceptance tests. Given three distinct points $(a, f(a))$, $(b, f(b))$, and $(c, f(c))$ near the root, fitting an ordinary upright parabola $y = P(x)$ can fail whenever its vertex turns around before reaching the x-axis and yields no real roots. Brent instead fits x sideways as a *Lagrange quadratic polynomial of* y ($x = P(y)$) through $(f(a), a)$, $(f(b), b)$, and $(f(c), c)$, and then evaluates that sideways curve at $y = 0$:

$$x_{\text{IQI}} = \frac{af(b)f(c)}{(f(a)-f(b))(f(a)-f(c))} + \frac{bf(a)f(c)}{(f(b)-f(a))(f(b)-f(c))} + \frac{cf(a)f(b)}{(f(c)-f(a))(f(c)-f(b))} \qquad \textbf{(Eq. 7.21)}$$

Evaluating a sideways parabola at $y = 0$ always yields a single unique real number x_{IQI} and converges at order $p \approx 1.839$, nearly doubling the number of significant digits per step.

Rather than accepting every interpolated step unconditionally, the algorithm keeps the root enclosed inside a contracting sign-change bracket $[a, b]$ where $f(a)f(b) \leq 0$. Whenever Secant or Inverse Quadratic Interpolation proposes a trial step s, five acceptance conditions verify that s falls strictly inside the interior three-quarter window $((3a+b) / 4, b)$ and that consecutive steps contract the bracket by at least half every two iterations. If an

interpolated step attempts to leave the bracket or stalls on a flat plateau, the algorithm rejects the interpolation and executes a bisection step $m = 1/2(a + b)$.

Root-Finding Algorithm	Convergence Order (p)	Derivative $f'(x)$ Required	Bracket Guarantee	Behavior Near Flat Slopes ($f'(x) \approx 0$)
Bisection	1.000	No	Yes (Guaranteed)	Steady (1 bit/step), immune to slope traps
Newton-Raphson	2.000	Yes	No	Catastrophic divergence when $f'(x) \approx 0$
Secant Chord	1.618 (φ)	No	No	Can fling iterates far outside the search region
Brent's Hybrid Method	1.839	No	**Yes (Guaranteed)**	**Instant bisection fallback; zero divergence**

7.3.2 Where the Unknown Hides: Four Classic Transcendental Equations of Science and Engineering

Fundamental laws across physics and engineering routinely resist closed-form inversion because they couple two distinct physical mechanisms inside the same conservation balance and trap the unknown simultaneously inside two incompatible mathematical operations. In orbital mechanics, Johannes Kepler showed in his 1609 *Astronomia Nova* that as a planet sweeps out equal areas in equal times along an ellipse of eccentricity e, its uniform clock angle $M = 2\pi\ t\ /\ T$ (*mean anomaly*) and its geometric position angle E (*eccentric anomaly*) satisfy $E - e \sin E - M = 0$. Because the unknown angle E appears simultaneously as a linear algebraic arc outside the sine function and as a trigonometric phase inside sin E, Kepler observed that the equation *"cannot be solved a priori, on account of the different nature of the arc and the sine."*

A similar implicit coupling governs turbulent pipe flow. When water or rocket propellant flows through a rough pipe of diameter D at high Reynolds number Re, a viscous sublayer of thickness $\delta_v \propto D/(\mathrm{Re}\sqrt{f})$ cushions the pipe wall. Because the thickness of that sublayer depends on the Darcy–Weisbach friction factor f itself, Cyril Colebrook (1939) *(Colebrook, 1939)* showed that $\sqrt{f}$ appears both outside and inside a base-10 logarithm, which hydraulic engineers historically estimated from the logarithmic curves of the *Moody Chart*:

$$\frac{1}{\sqrt{f}} = -2.0\log_{10}\left(\frac{\epsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right) \qquad \textbf{(Eq. 7.22)}$$

In quantum radiation and semiconductor physics, exponential and linear terms couple in the same way. When Max Planck (1900) derived the blackbody spectrum $I(\lambda) \propto \lambda^{-5}/(e^{hc/\lambda\ k_B\ T} - 1)$, locating the Wien peak wavelength $\lambda_{\max}$ by setting the derivative to zero coupled the dimensionless photon energy $x = hc/(\lambda_{\max}\ k_B\ T)$ inside the exponential e^x and in the algebraic factor $(5 - x)$ outside it, $(5 - x)e^x = 5$ (or $5(1 - e^{-x}) - x = 0$), whose root $x^* = 4.9651142317$ sets the Wien displacement constant of stellar spectra. Likewise, when a DC voltage source V_s drives a semiconductor junction diode through a series load resistor R, Kirchhoff's voltage law $(V_s - V)/R = I_s(e^{V/V_T} - 1)$ balances a linear resistor voltage drop against William Shockley's exponential diode current and traps the junction voltage V both linearly and exponentially.

Even when a physical balance remains purely algebraic, three-dimensional geometry and conservation laws routinely produce cubic polynomials whose roots carry direct physical meaning. By Archimedes' buoyancy law, a sphere of radius R and density ρ_s sinks in water of density ρ_w until its submerged cap volume $(1/3)\pi\ h^2(3R - h)$ balances $(\rho_s/\rho_w)((4/3)\pi\ R^3)$, which forms the cubic $h^3 - 3Rh^2 + 4(\rho_s/\rho_w)R^3 = 0$ whose middle root (RCL 3 in → POLY) is

the physical waterline draft $h \in [0,\ 2R]$. In continuum mechanics and thermodynamics, twisting a brittle rod fractures it along a 45° helix governed by the three real eigenvalues of the Cauchy stress cubic $\lambda^3 - I_1\lambda^2 + I_2\lambda - I_3 = 0$, while compressing a real gas near its critical point follows Johannes Diderik van der Waals's cubic equation of state $(P + a/V^2)(V - b) = RT$.

7.3.3 Multiple-Root Conditioning and Polynomial Deflation Order

James H. Wilkinson, Alan Turing's colleague on the Pilot ACE computer and the founder of modern backward error analysis, established the conditioning contrast between simple crossings and repeated roots. At a *simple root* where the curve crosses the axis with non-zero slope ($f'(x^*) \neq 0$), a vertical rounding perturbation of size $\varepsilon_{\text{mach}} \approx 10^{-16}$ in evaluating $f(x)$ shifts the horizontal zero crossing by at most $\Delta x \approx \varepsilon_{\text{mach}} / |f'(x^*)|$. At an *m-fold repeated root* where the curve touches the axis tangentially as $f(x) \approx c(x - x^*)^m$, solving $|c(x - x^*)^m| \leq \varepsilon_{\text{mach}}$ yields *Wilkinson's Root Conditioning Law*:

$$\Delta x \approx \left(\frac{\varepsilon_{\text{mach}}}{|c|}\right)^{1/m} \qquad \textbf{(Eq. 7.23)}$$

For a double tangent root ($m = 2$, such as $(x - 3)^2 = 0$), the horizontal uncertainty widens to the square root of machine epsilon ($\varepsilon_{\text{mach}}^{1/2} \approx 10^{-8}$), where half the digits of precision are lost to floating-point rounding noise; for a triple root ($m = 3$), precision drops to $\varepsilon_{\text{mach}}^{1/3} \approx 10^{-5}$. Similarly, deflating a high-degree polynomial $P(x)$ to $Q(x) = P(x)/(x - r_1)$ by dividing out its *largest* root $|r_1|$ first contaminates the lower coefficients of $Q(x)$ with rounding errors and perturbs the remaining roots. Deflating roots in *increasing order of magnitude* ($|r_1| \leq |r_2| \leq \ldots$) preserves coefficient accuracy, and solving the derivative $f'(x) = 0$ at an even-multiplicity tangent minimum restores simple-root conditioning.

7.3.4 Inside the Symbolic Equation Bank (EQN): Pure Functions, Step Editing & Iverson Piecewise Logic

The SC-32's [→] [EQN] workspace ([→] + [STO]) provides a structured environment for entering, inspecting, and editing symbolic RPN formulas, including *piecewise physical laws* that change behavior across a threshold. Unlike keystroke-programmable calculators that mix arithmetic formulas with destructive memory writes and unconditional **GTO** jumps, the SC-32's 24-slot Equation Bank treats every stored entry (**A** through **Z**, up to 96 steps per equation) as a *pure, side-effect-free mathematical function.*

Pressing [→] [EQN] opens the Equation Catalog (**NEW**, **1:A**, **2:B**, ... on the top softkeys). Tapping **NEW** ([LFU0]) or pressing [↶] [LBL] (Left Shift on **+**) prompts **LBL _** and arms the **A–Z** Alpha layer so a single tap assigns the equation's letter (**00 LBL A**). As operations are keyed in, each step is recorded with a line number (**01 RCL X**, **02 x²**, ...). When typing a multi-digit constant such as **0.05**, pressing [ENTER] terminates the numeric literal without duplicating the stack, whereas a second tap inserts an explicit **ENTER** step. At any time, [↶] [▲] and [↶] [▼] (Left Shift on **8** and **7**) step the cursor backward and forward through the numbered instructions: typing any key inserts at the cursor, [⌫] deletes the preceding step, and [→] [RTN] (Right Shift on **+**) or [ON] saves the equation to non-volatile memory.

Inside an equation, pressing [RCL] followed by a letter (**A**–**Z**, or indirect **(i)**, where storing a slot number 1... 89 in the index pointer **i** via **STO .** lets **RCL x‖y** dereference **(i)** dynamically) reads that variable onto the stack, and **RCL + A** or **RCL × A** performs in-place recall arithmetic. By contrast, [STO] is blocked (**INVALID FOR EQN**) so that when [XEQ], [↶] [FN=], [→] [SOLVE], or [→] [∫] evaluates an equation repeatedly across an interval,

it cannot overwrite stored laboratory constants.

In 1919, the Cambridge mathematician William H. Macaulay introduced a compact notation for piecewise structural loads: when a bridge beam carries concentrated point loads or steps in cross-section at stations a_1, a_2, ..., writing a separate differential equation for every span is cumbersome. Macaulay defined the *singularity ramp bracket* $\langle\, x - a \,\rangle_+ = \max(0,\, x - a)$, which equals 0 to the left of $x = a$ and turns on as $(x - a)$ to the right of $x = a$. Four decades later, Kenneth E. Iverson (1962) generalized Macaulay's switch into the *Iverson Indicator Bracket* $[P] \in (1, 0)$, where any relational proposition P evaluates to **1** if True and **0** if False:

$$f(x) = f_1(x)\cdot[x \le c] + f_2(x)\cdot[x > c] \quad \text{or} \quad \langle u \rangle_+ = u\cdot[u > 0] = \max(0, u) \qquad \textbf{(Eq. 7.24)}$$

On the SC-32, the [→] [x?0] menu (Right Shift on [÷]) provides six unary zero comparisons (**x=0**, **x≠0**, **x>0**, **x<0**, **x≥0**, and **x≤0**), each replacing Register X in place with **1.0** for True or **0.0** for False. Its companion [↖] [x?y] menu (Left Shift on **÷**) offers the six binary two-register comparisons (**x=y**, **x≠y**, **x>y**, **x<y**, **x≥y**, and **x≤y**), which compare Register X against Register Y, drop the stack, and leave **1.0** or **0.0** in Register X. Consequently, on any expression u sitting in Register X, the three-keystroke *Macaulay ramp idiom* [ENTER] [x>0] [×] duplicates u into Y, converts X into $[u > 0] \in (0, 1)$, and multiplies them to produce $\langle\, u \,\rangle_+ = \max(0,\, u)$. This construction models bilinear contact springs, progressive tax or tariff brackets, Zener clippers, and piecewise fluid drag laws inside a single continuous **EQN** formula.

7.3.5 Brent's Two-Phase Convergence and Kahan's Stack Residual Audit

When [→] [SOLVE] evaluates an equation $f(x) = 0$ stored in [→] [EQN], it executes a two-phase bracketing state machine followed by William Kahan's post-convergence stack audit. In the first phase, if two initial guesses placed in Registers Y and X already straddle a sign change ($f(a)\cdot f(b) \le 0$), [→] [SOLVE] adopts the (Y, X) stack registers as the starting bracket $[a, b]$ immediately. When given only a single seed x_0 via [STO], or two guesses on the same side of the axis where $f(a)\cdot f(b) > 0$, an outward search generates a companion probe ($\Delta = 0.1|x_0|$) and advances outward for up to 50 steps using clamped secant steps, or expanding by the *golden ratio* $\varphi \approx 1.618034$ whenever the slope is flat. Along the way, the search retracts any probe that steps outside the domain of ln x or $\sqrt{x}$, and it detects even-multiplicity tangent roots such as $(x - 2)^2 = 0$ whenever $|b - a| < 10^{-12}$ and $|f(b)| < 10^{-7}$.

In the second phase, once a sign change $f(a)\cdot f(b) \le 0$ is bracketed, the solver attempts *Inverse Quadratic Interpolation* (IQI) whenever three distinct values ($f(a)$, $f(b)$, $f(c)$) are available, or a *Secant Chord* step otherwise. Before accepting the proposed step s, Brent's five acceptance tests verify that s lies strictly inside the three-quarter window ($(3a+b)$ / 4, b) and reduces the step width by at least 50% every two steps; if any check fails, the engine executes a *Bisection* step $m = 1/2(a + b)$ until $|f(b)| \le 10^{-13}$ or the half-bracket width reaches machine tolerance $\text{tol} = 2\varepsilon_{\text{mach}}|b| + (1/2)\cdot 10^{-12}$.

Following William Kahan's diagnostic solver design, the final stack state distinguishes a genuine root from a discontinuous pole. When [→] [SOLVE] finishes on the SC-32, it pushes the converged root x^* into Register X, lifts the prior contents of Register X into Register Y, and preserves the upper stack contents in Registers Z and T.

If an initial bracket straddles a *vertical asymptote* (*discontinuous pole*) where the curve jumps from $-\infty$ to $+\infty$, such as $f(x) = \tan x$ across $x = \pi/2$ or $f(x) = 1 / (x - 2)$ across $x = 2$, the opposite signs $f(a)\cdot f(b) < 0$ on either side of the discontinuity cause bisection to

contract around the pole $x = 2.0000000000$. Storing x^* from Register X into the equation's variable ([STO] **X**) and pressing [XEQ] followed by the equation's letter evaluates the residual $f(x^*)$ in Register X: at a *true root*, it returns $f(x^*) \approx 0$ ($|f(x^*)| \leq 10^{-12}$), whereas at a *false pole*, it reveals a large residual ($|f(x^*)| \gg 1$, or **DIVIDE BY 0**).

7.4 Horner Evaluation, Cardano's Cubic Solver & Brent's Root-Finder

On the SC-32 stack, root-finding separates polynomial problems of degree two or three from higher-degree and transcendental conservation laws.

When the governing relation is a quadratic ($ax^2 + bx + c = 0$) or cubic ($Ax^3 + Bx^2 + Cx + D = 0$) with explicit numerical coefficients, the [→] [POLY] workspace (above [XEQ]) bypasses symbolic equation entry altogether. Loading the coefficients from highest degree down onto the softkeys (**A** [A] **B** [B] **C** [C ON] **D** [D]) pushes all four coefficients onto Registers (T, Z, Y, X), and pressing [EXEC] dispatches directly to the Citardauq quadratic or Cardano–Viète cubic formulas; the solver returns the primary root in (Y = Real, X = Imag) and populates root registers **1** through **6** (indirect **R34**–**R39**).

When Abel and Galois's degree barrier or a transcendental coupling of algebraic, trigonometric, and exponential terms rules out closed-form extraction, the problem moves to [→] [EQN] ([→] + [STO]) and Brent's [→] [SOLVE] engine (above [7]). Defining the residual $f(X) =$ LHS - RHS $= 0$ in [→] [EQN], staging a two-guess bracket $[a, b]$ across Registers Y and X (or storing a single seed via [STO]), and invoking [→] [SOLVE] places the converged root x^* in Register X with its bracket companion in Register Y, ready for an immediate [XEQ] residual check.

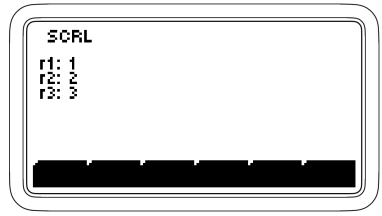


Figure: Cubic polynomial x^3 - 6x^2 + 11x - 6 = 0 solved via POLY

7.4.1 Horner's Nested Evaluation, T-Register Flooding & Biquadratic Reductions

For the fourth-degree polynomial $P(x) = 4x^4 - 3x^3 + 2x^2 - 5x + 7 = (((4x - 3)x + 2)x - 5)x + 7$ evaluated at $x_0 = 2.5$, where synthetic division yields $P(x) = (x - 2.5)(4x^3 + 7x^2 + 19.5x + 43.75) + 116.3750$, the four-register stack supports two execution patterns depending on whether $x_0 = 2.5$ is keyed at each stage or dispensed from Register T:

1. **Running Accumulator in Register X (Using Only Registers X and Y):** When $x = 2.5$ is short to key in, the Ruffini-Horner synthetic-division coefficients accumulate sequentially in Register X while Registers Z and T remain at zero:

- **First Horner Stage (b_3):** Multiplying the leading coefficient by x ($4 \times 2.5 = 10.0000$) and subtracting 3 forms the first synthetic-division quotient coefficient $b_3 = 4(2.5) - 3 =$ **7.0000** in Register X.
- **Second Horner Stage (b_2):** Multiplying by 2.5 (which lifts 7.0000 transiently into Register Y and forms 17.5000) and adding 2 produces the second quotient coefficient $b_2 = 7.0(2.5) + 2 =$ **19.5000** in Register X.
- **Third Horner Stage (b_1):** Multiplying by 2.5 and subtracting 5 yields the linear quotient coefficient $b_1 = 19.5(2.5) - 5 =$ **43.7500** in Register X.
- **Final Remainder ($b_0 = P(2.5)$):** One final multiply-and-add step ($43.75 \times 2.5 + 7$) via **2.5** [ENTER] **4** [×] **3** [-] **2.5** [×] **2** [+] **2.5** [×] **5** [-] **2.5** [×] **7** [+] completes the polynomial remainder $P(2.5) =$ **116.375** (**116.3750**).

2. **Flooding All Four Stack Registers (`2.5 ENTER ENTER ENTER`):** Rather than keying **2.5** four separate times—or re-typing a nine-digit evaluation point such as $x = 2.718281828$—filling all four registers (T, Z, Y, X) with x via **2.5** [ENTER] [ENTER] [ENTER] turns Register T into an automatic constant dispenser. Every time [×] consumes the copy of x in Register Y and drops the stack, Register T clones a fresh copy of $x = 2.5$ downward into Register Z. Starting with **4 × 3 -** ($\rightarrow$ 7.0000 in X, with 2.5 waiting in Y, Z, T) and streaming **× 2 + × 5 - × 7 +** reaches **116.3750** without re-typing **2.5**.

Applying that same T-register replication to Martin Green's rational-logarithmic solar-cell fill factor $FF = (v_{oc} - \ln(v_{oc} + 0.72)) / (v_{oc} + 1)$ at $v_{oc} = 25.145$ requires keying **25.145** only once: after flooding the stack with three taps of [ENTER], adding 0.72, taking [LN], and subtracting from the waiting copy of v_{oc} in Register Y forms the numerator 21.8921 in Register X, while a fresh copy of v_{oc} drops into Register Y ready for [x⇄y], + 1, and [÷] to resolve $FF =$ **0.837334**.

Similarly, on the coupled-oscillator biquadratic $x^4 - 10x^2 + 9 = 0$ ($u = x^2$), evaluating the quadratic radical $\sqrt{\Delta} = \sqrt{(10^2 - 4(9))} = 8.0000$ on the stack and pairing it with 10 via [↶] [LAST x] forms $10 \pm 8 = (18.0000, 2.0000)$ across Registers Y and X; halving both roots ($u_1 = 9$, $u_2 = 1$) and taking [√x] leaves both positive normal-mode frequencies side by side on the stack ($Y =$ **1.0000**, $X =$ **3.0000**).

7.4.2 Cardano's Analytical Cubic & Quadratic Solver

Inside [↗] [POLY] (Right Shift [↗] above [XEQ]), the softkeys and variable registers map the coefficients and extracted roots of $Ax^3 + Bx^2 + Cx + D = 0$ as follows:

Interface Element	Display Legend / Key Assignment	Functional Action / Internal Allocation
Menu Prompt	**`Ax^3 + Bx^2 + Cx + D = 0`**	Prompts for polynomial coefficient entry
LFU Softkeys	[A] [B] [C ON] [D] [EXEC] [CLR]	Softkey assignments for coefficient entry and execution
Internal Storage: Root 1	Variables **"1"** (Real), **"2"** (Imaginary)	Stores first calculated root (also placed in Registers Y and X)
Internal Storage: Root 2	Variables **"3"** (Real), **"4"** (Imaginary)	Stores second calculated root
Internal Storage: Root 3	Variables **"5"** (Real), **"6"** (Imaginary)	Stores third calculated root

Any unentered coefficient defaults to 0.0 (at least one of A or B must be non-zero, otherwise [EXEC] displays **A ≠ 0**). Evaluating representative cubics and quadratics shows how [↗] [POLY] dispatches across the three discriminant regimes ($\Delta < 0$, $\Delta > 0$, and $\Delta = 0$).

On the *casus irreducibilis* benchmark $(x - 1)(x - 2)(x - 3) = x^3 - 6x^2 + 11x - 6 = 0$ ($\Delta < 0$), [↗] [POLY] executes Viète's trigonometric cosine trisection.

Inside [↗] [POLY], loading the four coefficients $A = 1$, $B = -6$, $C = 11$, and $D = -6$ onto their top-row softkeys and pressing [EXEC] projects the three unit-circle vertices onto the real axis:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [A]	0.0000	0.0000	0.0000	1.0000	Coefficient $A = 1.0$
6 [+/−] [B]	0.0000	0.0000	1.0000	-6.0000	Coefficient $B = -6.0$
11 [C ON]	0.0000	1.0000	-6.0000	11.0000	Coefficient $C = 11.0$
6 [+/−] [D]	1.0000	-6.0000	11.0000	-6.0000	Coefficient $D = -6.0$
[EXEC]	1.0000	-6.0000	**1.000000**	**0.000000**	Primary root $r_1 = 1.0 + 0.0i$ in (Y, X); roots 1, 2, 3 in vars **1**–**6**

Here [EXEC] sorts all three real roots in ascending order ($r_1 = 1.0$ in variables **1**, **2**, $r_2 = 2.0$ in variables **3**, **4**, and $r_3 = 3.0$ in variables **5**, **6**), while placing the real part of r_1 in Register Y and its zero imaginary part in Register X.

In the positive-discriminant regime ($\Delta > 0$), a cubic with complex conjugate roots such as $x^3 - x^2 + x - 1 = (x - 1)(x^2 + 1) = 0$ crosses the horizontal axis only once at $r_1 = 1.0$, pairing that real intercept with the pure imaginary roots $r_{2,3} = 0.0 \pm 1.0i$. Loading its four alternating coefficients inside [↗] [POLY] separates the real crossing on the stack from the complex conjugate pair stored in the numbered registers:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [A]	0.0000	0.0000	0.0000	1.0000	Coefficient $A = 1.0$
1 [+/−] [B]	0.0000	0.0000	1.0000	-1.0000	Coefficient $B = -1.0$
1 [C ON]	0.0000	1.0000	-1.0000	1.0000	Coefficient $C = 1.0$
1 [+/−] [D]	1.0000	-1.0000	1.0000	-1.0000	Coefficient $D = -1.0$
[EXEC]	1.0000	-1.0000	**1.000000**	**0.000000**	Real r_1=1 in Y, Imag r_1=0 in X; conjugate pair $\pm\, i$ in vars **3**–**6**

Recalling [RCL] **3** and [RCL] **4** returns the real and imaginary parts 0.0000 and +1.0000 of $r_2 = +i$, while [RCL] **5** and [RCL] **6** return 0.0000 and -1.0000 for its conjugate $r_3 = -i$.

At the critical-damping boundary $\Delta = 0$—represented by the third-order RLC filter polynomial $s^3 + 12s^2 + 45s + 50 = (s+2)(s+5)^2 = 0$—the solver stores both copies of the double pole (-5.0, -5.0) and the simple pole (-2.0) across variables **1**, **3**, and **5**, while placing the first sorted real root in (Y = -5.000000, X = 0.000000) ([↗] [POLY] **1** [A] **12** [B] **45** [C ON] **50** [D] [EXEC] [x⇄y] → **-5**).

For the 3D Cauchy principal stress cubic ($\lambda^3 - 70\lambda^2 + 1175\lambda - 5750 = 0$) of a structural steel element with tensor invariants $I_1 = 70$ MPa, $I_2 = 1175$ MPa^2, and $I_3 = 5750$ MPa^3, the symmetry of Cauchy's stress tensor guarantees $\Delta \leq 0$ so all three eigenvalues $\sigma_3 \leq \sigma_2 \leq \sigma_1$ are real. Loading those invariants into [↗] [POLY] with [↗] [POLY] **1** [A] **190** [+/−] [B] **9900** [C ON] **135000** [+/−] [D] [EXEC] [RCL] **1** [RCL] **3** [RCL] **5** isolates the major principal stress **112.90729452** MPa in Variable **5**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [A] **70** [+/−] [B]	0.0000	0.0000	1.0000	-70.0000	Coefficients A, B = -I_1
1175 [C ON] **5750** [+/−] [D]	1.0000	-70.0000	1175.0000	-5750.0000	Coefficients $C = I_2$, D = -I_3
[EXEC]	1.0000	-70.0000	**10.000000**	**0.000000**	Minor stress $\sigma_3 = 10.0$ in Y; σ_2, σ_1 in vars **3**, **5**
[RCL] **5**	-70.0000	10.0000	0.0000	**48.027756**	Recall major principal stress σ_1 into Register X

Once $\sigma_1 = 48.0278$ MPa sits in Register X and $\sigma_3 = 10.0000$ MPa sits in Register Z, keying [R↓] [x⇄y] [-] **2** [÷] yields Henri Tresca's *maximum shear stress* $\tau_{max} = 1/2(\sigma_1 - \sigma_3) =$ **19.0139 MPa**.

Applying Archimedes' floating-sphere buoyancy cubic $h^3 - 3Rh^2 + 4(\rho_s/\rho_w)R^3 = 0$ to a solid oak marine buoy of radius $R = 0.50$ m (1.00 m diameter) and specific gravity $\rho_s/\rho_w = 0.65$ produces $h^3 - 1.5h^2 + 0.325 = 0$. Inside [↗] [POLY], the sequence [↗] [POLY] **1** [A] **1.5** [+/−] [B] **0** [C ON] **0.325** [D] [EXEC] [RCL] **3** isolates the physical waterline draft **0.6013896973** m in Variable **3**, which shows that 60.14% of the buoy's diameter sits submerged while the algebraic companion roots -0.4123 m and 1.3109 m lie outside the physical sphere [0, 1.00 m]:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [A] **1.5** [+/−] [B]	0.0000	0.0000	1.0000	-1.5000	Load cubic and quadratic coefficients $A = 1.0$, B = -1.5
0 [C ON] **0.325** [D]	1.0000	-1.5000	0.0000	0.3250	Load linear and constant coefficients $C = 0.0$, $D = 0.325$
[EXEC]	1.0000	-1.5000	-0.412257	0.0000	Solve cubic; real parts of 3 roots stored in Variables **1**, **3**, **5**
[RCL] **3**	-1.5000	-0.412257	0.0000	**0.601390**	Recall physical immersion depth $h \in [0, 2R]$ from Variable **3**

Evaluating the inverse ballistic trajectory quadratic ($360.3673\,u^2 - 1200u + 510.3673 = 0$) for an elevated target at horizontal range $x = 1{,}200$ m and height $y = 150$ m with muzzle speed $v_0 = 140$ m/s ($gx^2/2v_0^2 = 360.3673$ m) illustrates two ways to enter a quadratic $a\,u^2 + b\,u + c = 0$ inside [↗] [POLY]. Setting the cubic coefficient $A = 0$ (**0** [A]) and loading ($B = a$, $C = b$, $D = c$) solves $Bx^2 + Cx + D = 0$ directly, whereas mapping the coefficients into ($A = a$, $B = b$, $C = c$, $D = 0$) solves $u(a\,u^2 + b\,u + c) = 0$; this layout places a dummy root at $u = 0$ in Variable **1** and stores the two quadratic roots in Variables **3** and **5**. With the $D = 0$ layout, [↗] [POLY] **360.3673** [A] **1200** [+/−] [B] **510.3673** [C ON] **0** [D] [EXEC] [RCL] **3** [↖] [ATAN] recovers the low direct-fire angle **26.590115798**° before [RCL] **5** [↖] [ATAN] reveals the high mortar arc:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
360.3673 [A]	0.0000	0.0000	0.0000	360.3673	Quadratic coefficient A
1200 [+/−] [B]	0.0000	0.0000	360.3673	-1200.0000	Quadratic coefficient B
510.3673 [C ON]	0.0000	360.3673	-1200.0000	510.3673	Quadratic coefficient C
0 [D]	360.3673	-1200.0000	510.3673	0.0000	Depressed constant (D defaults to 0.0 if omitted)

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
EXEC	360.3673	-1200.0000	0.0000	0.0000	Solve polynomial; real roots (0, u_1, u_2) stored in Variables **1**, **3**, **5**
RCL **3** ↶ ATAN	-1200.0000	0.0000	0.0000	**26.590116**	Low direct-fire trajectory ϑ_1 in degrees
RCL **5** ↶ ATAN	0.0000	0.0000	26.590116	**70.532306**	High plunging mortar arc ϑ_2 in degrees

Similarly, on the forensic highway braking quadratic $v_0^2 + 20.594\ v_0 - 1166.99 = 0$ ($t_r = 1.5$ s, $\mu = 0.70$, $d = 85.0$ m), keying **1** A **20.594** B **1166.99** +/- C ON **0** D EXEC RCL **5** inside R/S POLY extracts the positive initial vehicle speed $v_0 = \mathbf{25.301\ m/s}$ (91.1 km/h).

In analytical chemistry, Wilhelm Ostwald's 1888 dilution law for a monoprotic weak acid ($HA + H_2O \rightleftharpoons H_3O^+ + A^-$) of initial molarity C_0 and dissociation constant K_a expands into the exact mass-action quadratic $K_a = (x^2 / (C_0 - x)) \Rightarrow x^2 + K_a\ x - K_a\ C_0 = 0$. For $C_0 = 0.010$ M chloroacetic acid ($K_a = 1.40 \times 10^{-3}$, $K_a\ C_0 = 1.40 \times 10^{-5}$), the textbook small-$x$ approximation $\sqrt{(K_a\ C_0)} = 0.003742$ M overstates ionization by 20%. Solving $x^2 + 0.0014x - 0.000014 = 0$ inside R/S POLY isolates the positive hydronium molarity $[H_3O^+] = \mathbf{0.003112\ M}$ in Variable **5**, and pressing ↶ LOG +/- converts that root in place into the exact solution pH = **2.5070**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 A **0.0014** B	0.0000	0.0000	1.0000	0.0014	Load quadratic coefficients 1.0 and $K_a = 0.0014$
0.000014 +/- C ON **0** D	1.0000	0.0014	-0.000014	0.0000	Load $-K_a\ C_0 = -1.4 \times 10^{-5}$ and depressed $D = 0$
EXEC	1.0000	0.0014	-0.004512	0.0000	Roots (-0.004512, 0, +0.003112) stored in Vars **1**, **3**, **5**
RCL **5**	0.0014	-0.004512	0.0000	0.003112	Recall physical hydronium molarity $[H_3O^+]$ (M)
↶ LOG +/-	0.0014	-0.004512	0.0000	**2.5070**	Transform $[H_3O^+]$ in place to exact pH = 2.5070

7.4.3 Hunting Transcendental Roots & Auditing Kahan's Stack with Brent's SOLVE

Comparing R/S SOLVE and R/S POLY side by side on two physical cubics shows how the symbolic equation solver complements closed-form extraction—noting that inside R/S EQN, recalling a variable letter automatically lifts the stack, whereas two consecutive numeric literals require ENTER between them so their digits do not concatenate.

On John Wallis and Isaac Newton's 1669 cubic benchmark ($x^3 - 2x - 5 = 0$), hand-evaluating Cardano's formula is cumbersome because $p = -2$ and $q = -5$ yield a positive discriminant $\Delta = (-5/2)^2 + (-2/3)^3 = 643/108 > 0$, requiring the sum of two nested cube roots $\sqrt[3]{(5/2 + \sqrt{(643/108)})} + \sqrt[3]{(5/2 - \sqrt{(643/108)})}$. In R/S EQN, the residual $f(x) = (x^2 - 2)x - 5$ enters directly in Horner nested form as **X x² 2 - X × 5 -**.

Because $f(2) = -1 < 0$ and $f(3) = +16 > 0$, either staging the two-guess bracket [2, 3] on (Y, X) via **2** ENTER **3** R/S SOLVE **X** or storing Newton's single seed **2** STO **X** and pressing **SOLVE X** converges to $x^* = \mathbf{2.0945514815}$ in Register X while lifting the starting seed 2.0000 into Register Y (with $Z = 0.0000$ and XEQ confirming a machine-zero residual $|f(x^*)| < 10^{-15}$).

Similarly, applying van der Waals's real-gas equation of state $(P + a/V^2)(V - b) = RT$ to carbon dioxide (CO_2, $a = 3.59$ L²·atm/mol², $b = 0.0427$ L/mol) compressed to $P = 50$ atm at $T = 300$ K ($RT = 24.618$ L·atm/mol) clears the V^2 denominator to form the expanded cubic $50V^3 - 26.753V^2 + 3.59V - 0.153293 = 0$. Inside [↗] [POLY], loading those four coefficients ([↗] [POLY] **50** [A] **26.753** [+/−] [B] **3.59** [C/ON] **0.153293** [+/−] [D] [EXEC] [x⇄y]) finds the real molar volume **0.3587362482** L/mol—27% denser than the ideal-gas estimate $V_{ideal} = RT/P = 0.4924$ L/mol and matching the V_m = **0.358736 L/mol** obtained by entering the RPN residual **3.59 V x² ÷ 50 + V 0.0427 - × 24.618 -** into [↗] [EQN] and running **0.49** [STO] **V** [↗] [SOLVE] **V**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
50 [A]	0.0000	0.0000	0.0000	50.0000	Coefficient $A = 50.0$
26.753 [+/−] [B]	0.0000	0.0000	50.0000	-26.7530	Coefficient $B = -26.753$
3.59 [C/ON]	0.0000	50.0000	-26.7530	3.5900	Coefficient $C = 3.59$
0.153293 [+/−] [D]	50.0000	-26.7530	3.5900	-0.153293	Coefficient $D = -0.153293$
[EXEC]	50.0000	-26.7530	**0.358736**	**0.000000**	Real root in Y, Imag in X
[x⇄y]	50.0000	-26.7530	0.000000	**0.358736**	Molar volume V_m in Register X

On coupled transcendental conservation laws, Brent's [↗] [SOLVE] engine resolves the unknowns directly from their [↗] [EQN] residuals, returning the converged root x^* in Register X, lifting the starting seed or bracket bound into Register Y, and preserving Registers Z and T for an immediate [XEQ] residual check ($|f(x^*)| \le 10^{-12}$):

- **Cyril Colebrook's Turbulent Pipe Friction Equation:** Evaluating the Colebrook–White turbulent pipe equation in commercial steel piping at Reynolds number Re = 100,000 and relative roughness $\varepsilon/D = 0.001$ brings both terms of the implicit balance onto one side of the equals sign:

$$g(f) = \frac{1}{\sqrt{f}} + 2.0\log_{10}\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\mathrm{Re}\sqrt{f}}\right) = 0 \qquad \textbf{(Eq. 7.25)}$$

Store **F √x 1/x 0.001 ENTER 3.7 ÷ 2.51 ENTER 100000 ÷ F √x ÷ + LOG 2 × +** in [↗] [EQN]. Because $1/\sqrt{f}$ diverges as $f \to 0^+$ and $\sqrt{f}$ is undefined for $f < 0$, seed **F** inside the physical Moody turbulent regime [0.008, 0.08] (or bracket **0.01** [ENTER] **0.05** on Y, X). Keying **0.02** [STO] **F** [↗] [SOLVE] **F** lifts 0.0200 into Register Y and converges to f^* = **0.022204** in Register X.

- **Johannes Kepler's Elliptical Orbital Anomaly ($E - e \sin E = M$):** Applying Kepler's equation $E - e \sin E - M = 0$ to an Earth-observation satellite with orbital eccentricity $e = 0.25$ at mean anomaly $M = 1.20$ rad in [RAD] mode yields $f(E) = E - 0.25 \sin E - 1.20 = 0$. Since $|\sin E| \le 1$, the physical root is guaranteed to lie inside $[M - e, M + e] = [0.95, 1.45]$ rad; storing **E 0.25 ENTER E SIN × - 1.20 -** in [↗] [EQN] and seeding the mean anomaly via **1.20** [STO] **E** [↗] [SOLVE] **E** converges to E^* = **1.488665 rad** (85.294°) in five iterations.

- **William Shockley's Silicon Diode Load Line:** For the DC diode load-line circuit ($V_s = 5.0$ V, saturation current $I_s = 10^{-12}$ A, thermal voltage $V_T = 0.02585$ V, and series load resistor $R = 1$ kΩ giving $I_s R = 10^{-9}$ V), Kirchhoff's loop residual is:

$$f(V) = 5.0 - 10^{-9}(e^{V/0.02585} - 1) - V = 0 \qquad \textbf{(Eq. 7.26)}$$

Whereas an unguarded Newton iteration from $V_0 = 0$ V shoots along the flat diode tangent to $V_1 = +5.0$ V ($e^{5.0\ /\ 0.02585} \approx 10^{84}$), Brent's bracket watchdogs keep the iterate inside the physical voltage drop. Storing **5 V 0.02585 ÷ eˣ 1 - 1E-9 × - V -** in [→] [EQN] and seeding near the silicon turn-on knee via **0.65** [STO] **V** [→] [SOLVE] **V** (or bracketing **[0.5, 0.8]** on Y, X) yields $V_D^* =$ **0.672944 V** (loop current $I_D = 4.3271$ mA).

- **Open-Channel Froude Critical Depth:** When $Q = 20.0$ m^3/s ($Q^2 = 400$) of floodwater rushes through a trapezoidal concrete spillway of bottom width $b = 3.0$ m and side slopes $z = 2.0$, the critical depth y_c where the surface wave speed matches the flow velocity (Fr = 1) satisfies $f(y) = Q^2\ T(y) - g\ A(y)^3 = 400(3 + 4y) - 9.81(3y + 2y^2)^3 = 0$—a sixth-degree polynomial that lies beyond Abel and Galois's degree-4 radical limit and beyond degree-3 [→] [POLY]. Recording the factored balance **3 ENTER 4 ENTER Y × + 400 × 3 ENTER Y × Y x² 2 × + 3 yˣ 9.81 × -** directly in [→] [EQN] and keying **1.0** [STO] **Y** [→] [SOLVE] **Y** converges to $y_c^* =$ **1.258641 m**.

- **Selecting Physical Branches in Multi-Root Equations (Nuclear Buckling & De Laval Nozzles):** Whenever a transcendental equation possesses multiple roots across separate physical regimes, your starting seed or (Y, X) bracket determines which branch [→] [SOLVE] locks onto:

  - **Nuclear Critical Buckling Radius:** For a spherical fission reactor core whose radial neutron flux vanishes at $\sin(0.045\ R) = 0$ in [RAD] mode, storing **R 0.045 × SIN** in [→] [EQN] and seeding **60** [STO] **R** [→] [SOLVE] **R** isolates the fundamental buckling mode $0.045\ R = \pi$ at $R_{\text{crit}} =$ **69.813170 cm** (avoiding the trivial origin $R = 0$).

  - **Aurel Stodola's De Laval Rocket Nozzle:** The area-Mach relation $(1/M)((1 + 0.2M^2)\ /\ 1.2)^3 - 2.50 = 0$ for area ratio $A/A^* = 2.50$ and $\gamma = 1.4$—entered in [→] [EQN] as **M x² 0.2 × 1 + 1.2 ÷ 3 yˣ M ÷ 2.50 -**—has two positive roots separated by the sonic throat $M = 1$. Seeding above the throat via **2.5** [STO] **M** [→] [SOLVE] **M** (or bracketing **1.1 ENTER 4.0** on Y, X) converges to the supersonic rocket exhaust branch $M_{\text{sup}} =$ **2.442804**, while seeding below the throat via **0.2** [STO] **M** [→] [SOLVE] **M** (or bracketing **0.05 ENTER 0.95**) isolates the subsonic diffuser branch $M_{\text{sub}} =$ **0.239541**.

7.4.4 Editing & Solving a Piecewise Equation in EQN Mode (x?0 & x?y)

Applying the Macaulay singularity ramp idiom ([ENTER] [x>0] [×]) to a spacecraft payload isolation mount—where a primary coil spring of stiffness $k_1 = 2{,}000$ N/m carries the load up to a clearance gap $x_0 = 0.05$ m (50 mm) before a secondary elastomer bump-stop of stiffness $k_2 = 8{,}000$ N/m engages—models the bilinear force law inside a single continuous equation:

$$F(x) = 2000x + 8000\langle x - 0.05\rangle_+ = 2000x + 8000(x - 0.05)\cdot[x - 0.05 > 0] \qquad \textbf{(Eq. 7.27)}$$

To find the deflection x^* under a static payload of $F_{\text{target}} = 350$ N, press [→] [EQN] (Right Shift [→] above [STO]), tap **NEW** ([LFU0]), press **A** for **00 LBL A**, and enter the residual $F(x)$ - 350 using [x>0] from [→] [x?0] (Right Shift [→] above [÷]):

[RCL] **X 2000** [×] [RCL] **X 0.05** [-] [ENTER] [x>0] [×] **8000** [×] [+] **350** [-] [→] [RTN]

Below the gap at $x = 0.03$ m, the indicator $[x - 0.05 > 0]$ evaluates to 0 so $F(0.03) = 60$ N. Above the gap at $x^* =$ **0.0750 m**—the root returned by **0.02** [ENTER] **0.10** [→] [SOLVE] **X**—the indicator flips to 1 and the bump-stop adds 200 N to reach $F(0.075) = 350$ N:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0.03 STO X	0.0000	0.0000	0.0000	0.0300	Test sub-gap deflection $x = 0.03$ m
2000 ×	0.0000	0.0000	0.0000	60.0000	Primary spring force $2000(0.03) = 60$ N
RCL X 0.05 -	0.0000	0.0000	60.0000	-0.0200	Gap clearance x - 0.05 = -0.02 m < 0
ENTER x>0 ×	0.0000	0.0000	60.0000	0.0000	Macaulay bracket $\langle -0.02 \rangle_+ = -0.02 \times 0 = 0$
8000 × +	0.0000	0.0000	0.0000	**60.0000**	Bump-stop inactive; total force $F(0.03) = 60$ N
0.075 STO X	0.0000	0.0000	60.0000	0.0750	Test post-gap root $x^* = 0.075$ m from **SOLVE X**
2000 ×	0.0000	0.0000	60.0000	150.0000	Primary spring force $2000(0.075) = 150$ N
RCL X 0.05 -	60.0000	60.0000	150.0000	0.0250	Gap overrun x - 0.05 = +0.025 m > 0
ENTER x>0 ×	60.0000	60.0000	150.0000	0.0250	Macaulay bracket $\langle +0.025 \rangle_+ = 0.025 \times 1 = 0.025$
8000 × +	60.0000	60.0000	60.0000	**350.0000**	Bump-stop adds 200 N; $F(0.075) =$ **350.0 N**

After g SOLVE terminates, the four-register stack preserves both the root estimate and the diagnostic context for William Kahan's residual check:

Stack / Action	Post-g SOLVE Contents	Engineering Diagnostic Interpretation
Register *X*	**Converged Root Estimate (x^*)**	The final root estimate where $f(x^*)$ crosses or touches zero, pushed onto Register X.
Register *Y*	**Starting Seed or Upper Bracket Bound**	Lifted from Register X when g SOLVE pushes the converged root x^* onto the stack.
Registers *Z*, *T*	**Prior Stack Context (Preserved)**	Any values previously in Registers Y and Z lift into Z and T undisturbed.
XEQ **Residual Audit**	**Evaluated Function Residual ($f(x^*)$)**	Storing x^* into your equation's target variable (STO X) and pressing XEQ evaluates $f(x^*)$: a true root yields $\lvert f(x^*) \rvert \leq 10^{-12}$, whereas an accidental vertical asymptote/pole exposes itself with $\lvert f(x^*) \rvert \gg 1$.

7.4.5 Practical Root-Finding Strategy: Bracket Seeding, Multiple Roots & Deflation on the Stack

Four stack techniques reinforce g POLY and g SOLVE when auditing nonlinear equations:

1. **Verifying g POLY Roots via Viète's Sum-and-Product Checksums:** Whenever g POLY writes the three roots of $Ax^3 + Bx^2 + Cx + D = 0$ into variables **1** (Real 1, indirect **R34**), **3** (Real 2, indirect **R36**), and **5** (Real 3, indirect **R38**), Viète's trace and determinant relations verify the roots directly:

- Summing the three real parts ([RCL] **1** [RCL] **3** [+] [RCL] **5** [+] via indirect registers **R34**, **R36**, **R38**) must equal -B/A (for the Cauchy stress cubic λ^3 - $70\lambda^2$ + 1175λ - $5750 = 0$, 10.0 + 11.972244 + 48.027756 = **70.000000**).
- Multiplying the three real roots ([RCL] **1** [RCL] **3** [×] [RCL] **5** [×]) must equal -D/A ($10.0 \times 11.972244 \times 48.027756$ = **5750.0000**).
- When a cubic has complex conjugate roots $r_{2,3} = \alpha \pm i\beta$ stored in variables **3, 4** (**R36, R37**) and **5, 6** (**R38, R39**), keying [RCL] **4** [RCL] **3** [↖] [▸θ,r] (Left Shift [↖] above [4]) converts (Re, Im) into the polar modulus $|r_2| = \sqrt{(\alpha^2 + \beta^2)}$ in Register X and phase angle ϑ in Register Y.

2. **Isolating Specific Branches of Multi-Root Transcendental Equations:** Whenever an equation has multiple roots across distinct physical regimes—such as low direct-fire vs. high mortar trajectories, subsonic vs. supersonic Mach branches across the $M^* = 1$ nozzle throat, or subcritical vs. supercritical conjugate depths in open-channel hydraulics—place two bracket bounds (a, b) in Registers Y and X that straddle only the desired branch (for example, **0.05** [ENTER] **0.95** for the subsonic nozzle branch, or **1.05** [ENTER] **4.0** for the supersonic branch) so Brent's solver remains locked inside that interval from iteration zero.

3. **Deflating Known Roots to Hunt Additional Roots ($(x - r_1)$ Division):** Once a first root r_1 of a transcendental or high-degree equation $f(x) = 0$ is isolated and stored in Variable **A** (deflating in increasing order of magnitude $|r_1| \leq |r_2|$ to preserve Wilkinson conditioning), appending **X** [RCL] **A** [-] [÷] to the end of the equation in [↗] [EQN] forms the deflated function $g(x) = f(x) \;/\; (x - r_1)$, dividing out the zero at r_1 so a second [↗] [SOLVE] pass cannot fall back into the first root's basin of attraction.

4. **Recognizing `NO ROOT FOUND` (Code 06) & Tangent Extrema:** If a curve $f(x)$ stays strictly positive everywhere—such as $f(x) = x^2 + 4 = 0$, which has only complex roots $\pm 2i$—the outward golden-ratio search terminates after 50 expansion steps and displays **NO ROOT FOUND** (Code 06, cleared via [C ON]). Conversely, when a curve touches the axis tangentially at an even-multiplicity root such as $f(x) = (x - 3)^2 = 0$, Brent's solver detects the tangent minimum once $|b - a| < 10^{-12}$ and $|f(b)| < 10^{-7}$, or solving the derivative $f'(x) = 0$ restores simple-root conditioning directly.

Together, Horner's nested recurrence, Cardano and Viète's closed-form [↗] [POLY] solver, and Richard Brent's bracketed [↗] [SOLVE] algorithm span both halves of algebraic and transcendental root-finding. Where radical formulas end at degree four, hybrid inverse-quadratic and bisection bracketing takes over, resolving implicit conservation laws to machine precision without sacrificing guaranteed convergence.

Chapter 8: Archimedes' Balance Beam, Central Differences & Romberg Quadrature

8.1 Symmetric Difference Quotients and Numerical Derivatives

At Launch Complex 39A at Cape Canaveral, when the five Rocketdyne F-1 engines ignite beneath the Apollo Saturn V moon rocket, the first-stage turbopumps deliver fifteen metric tons of kerosene and liquid oxygen per second into the combustion chambers. As hundreds of tons of propellant are expelled through the nozzles, the vehicle mass decreases while engine thrust increases in the thinning upper atmosphere, so acceleration climbs from 1.25 g at liftoff to 4.5 g just before staging.

In the seventeenth century, Isaac Newton and Gottfried Wilhelm Leibniz independently established that continuous motion is governed by two complementary operations—the derivative and the definite integral—linked as inverses by the *Fundamental Theorem of Calculus*. The derivative df/dt measures the instantaneous rate of change: if $x(t)$ is the altitude of the Saturn V at time t, its derivative $v(t) = dx/dt$ is the instantaneous vertical velocity, and the derivative of velocity $a(t) = dv/dt$ is the instantaneous acceleration. Conversely, the definite integral $\int_a^b f(t)\,dt$ measures cumulative change across an interval: integrating the area beneath the acceleration curve from ignition ($t = 0$) to first-stage burnout ($t = T$) recovers the total velocity gained by the spacecraft:

$$v(T) = v_0 + \int_0^T a(t)\,dt \tag{Eq. 8.1}$$

For the first pillar of calculus, a digital instrument evaluates the *instantaneous slope* (*derivative*) $f'(x)$ of a function at a point x directly from finite differences without symbolic algebra.

8.1.1 Symmetric Straddling and the Central Difference Formula

In an introductory calculus course, the derivative is defined by stepping a small distance h *forward* to $x + h$ and computing the difference quotient of the *one-sided forward difference*:

$$D_{\text{forward}}(h) = \frac{f(x+h) - f(x)}{h} \tag{Eq. 8.2}$$

A one-sided forward difference introduces systematic first-order bias on curved functions. On a curve that bends upward like a parabola ($f''(x) > 0$), the secant chord from x to $x + h$ is systematically *steeper* than the true tangent line at x because the curve rises more rapidly to the right of x. By Brook Taylor's 1715 series expansion:

$$f(x+h) = f(x) + f'(x)h + \tfrac{1}{2}f''(x)h^2 + \tfrac{1}{6}f'''(x)h^3 + \dots \tag{Eq. 8.3}$$

dividing $f(x + h) - f(x)$ by h leaves a one-sided curvature error of $+(1/2)f''(x)h$ that diminishes only linearly as $O(h)$. If one attempts to reduce that $O(h)$ truncation error by choosing a very small step size such as $h = 10^{-11}$, then $f(x + h)$ and $f(x)$ agree to their first eleven digits, and subtracting them destroys most of the calculator's precision through *catastrophic subtraction cancellation*.

Replacing the one-sided forward step with geometric symmetry eliminates that first-order bias. Instead of stepping only to the right, straddle the point x symmetrically by stepping backward to x - h and forward to $x + h$:

$$f(x+h)=f(x)+f'(x)h+\tfrac{1}{2}f''(x)h^2+\tfrac{1}{6}f'''(x)h^3+\ldots,\ f(x-h)=f(x)-f'(x)h+\tfrac{1}{2}f''(x)h^2-\tfrac{1}{6}f'''(x)h^3+\ldots \quad \textbf{(Eq. 8.4)}$$

Subtracting the second equation from the first and dividing by the total span $2h$ forms the *Central Difference Formula*:

$$f'(x)\approx\frac{f(x+h)-f(x-h)}{2h}=f'(x)+\tfrac{1}{6}f'''(x)h^2+\mathcal{O}(h^4) \quad \textbf{(Eq. 8.5)}$$

The $+(1/2)f''(x)h^2$ curvature terms on the right and left of x share the same sign and cancel to zero. Geometrically, the chord joining (x - h, $f(x$ - $h)$) and (x + h, $f(x$ + $h)$) runs nearly parallel to the true tangent line at x, and for any parabola $f(x) = ax^2 + bx + c$, where $f'''(x) = 0$, the central difference is exact for *any* step size h. Because the leading truncation error now scales as $O(h^2)$ while floating-point rounding error scales as $\varepsilon_{\text{mach}}/h$, the optimal step size lies near the cube root of machine epsilon, $h \approx \varepsilon_{\text{mach}}^{1/3} \approx 10^{-5}$. That balance delivers ten significant digits of derivative accuracy in two function evaluations.

On the SC-32, the four-register stack models this symmetric central difference directly: evaluate the forward point $f(x + h)$ in Register *X*, evaluate the backward point $f(x$ - $h)$ (which lifts $f(x+h)$ into Register *Y*), subtract via [-], and divide by $2h$ via [÷]. When differentiating $f(x) = e^{2x}$ at the origin $x_0 = 0$, where the exact slope is $f'(0) = 2$, [↶] [LASTx] and [x⇄y] streamline the sequence. Keying **2h** [ENTER] [e^x] [↶] [LASTx] [+/−] [e^x] [-] [x⇄y] [÷] reuses the step $2h$ from [↶] [LASTx] without re-typing it, or one can evaluate the hyperbolic sine identity (e^{2h} - e^{-2h}) / $2h$ = $\sinh(2h)/h$ via [↶] [HYP] [SIN]. Adding $f(x+h)$ and $f(x-h)$ instead of subtracting them cancels the odd terms and isolates the *second derivative* $f''(x)$ ≈ ($f(x+h)$ - $2f(x)$ + $f(x-h)$) / h^2.

In mechanical engineering, derivatives in polar coordinates govern James Watt's 1788 centrifugal flyball governor, the spinning brass-weight mechanism that regulated the speed of Industrial Revolution steam engines and prompted James Clerk Maxwell's 1868 formulation of negative-feedback control theory (*"On Governors"*). When a flyweight of mass m rotates at angular speed ω along a curved cam slot $r(\vartheta)$, the flyweight simultaneously moves tangentially at speed $v_\vartheta = r\omega$ and slides radially outward at speed $\dot{r} = \omega(dr/d\vartheta)$. Combining those two perpendicular velocity components via the Pythagorean theorem ($v^2 = \dot{r}^2 + (r\omega)^2$) yields its kinetic energy $E_k = (1/2)m\omega^2[(dr/d\vartheta)^2 + r(\vartheta)^2]$ at any cam angle ϑ.

8.1.2 Five-Point Stencil Acceleration via Richardson Extrapolation

Richardson's error-cancellation principle applies directly to the Taylor expansion of the central-difference derivative $D(h)$:

$$D(h)=\frac{f(x+h)-f(x-h)}{2h}=f'(x)+c_1h^2+c_2h^4+\ldots \qquad \left(\text{where } c_1=\tfrac{1}{6}f'''(x)\right) \quad \textbf{(Eq. 8.6)}$$

Because the leading error c_1 h^2 scales with the *square* of the step size h, evaluating the central difference at step h ($D(h)$) and at twice the step $2h$ ($D(2h)$, whose h^2 error is 4× larger) cancels the h^2 error completely via (4 $D(h)$ - $D(2h)$) / 3:

$$f'(x)\approx\frac{4D(h)-D(2h)}{3}=\frac{-f(x+2h)+8f(x+h)-8f(x-h)+f(x-2h)}{12h}+\mathcal{O}(h^4) \quad \textbf{(Eq. 8.7)}$$

This *five-point stencil* (-1, +8, 0, -8, +1 divided by $12h$) combines two central differences at h and $2h$ in the ratio 4/3 and -1/3, which eliminates the third-derivative error $(1/6)f'''(x)h^2$

and achieves fourth-order accuracy $O(h^4)$.

8.1.3 Leibniz's Integral Rule and Parametric Arc Length

Derivatives and integrals also combine directly in arc-length rectification, surfaces of revolution, mechanical work, and root-mean-square power. Christiaan Huygens and Jakob Bernoulli determined the exact length of a curved cable or orbit $y = f(x)$ from $x = a$ to $x = b$ by differential rectification. Over a differential horizontal step dx, the curve rises by $dy = f'(x)\,dx$, and by the Pythagorean theorem the diagonal arc element is $ds = \sqrt{(dx^2 + dy^2)} = \sqrt{(1 + [f'(x)]^2)}\,dx$. Integrating those differential elements across the interval yields the *Arc Length Formula*:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx \quad \text{or in polar coordinates:} \quad L = \int_{\theta_1}^{\theta_2} \sqrt{r(\theta)^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta \qquad \textbf{(Eq. 8.8)}$$

For a hanging catenary cable $y(x) = a\cosh(x/a)$, its derivative is $y'(x) = \sinh(x/a)$, and by the hyperbolic identity $1 + \sinh^2(x/a) = \cosh^2(x/a)$, the radical simplifies to $\sqrt{(1 + [y'(x)]^2)} = \cosh(x/a)$. Rotating that arc-length ribbon ds around the x-axis (extending Pappus's First Theorem to general curved profiles) sweeps out a frustum ring of radius $y(x)$ and circumference $2\pi\ y(x)$, which determines the curved surface area of any turned nosecone, parabolic dish, or cooling tower:

$$S = \int_a^b 2\pi y(x)\sqrt{1 + [y'(x)]^2}\,dx \qquad \textbf{(Eq. 8.9)}$$

In mechanics and electrical engineering, definite integrals accumulate variable forces and thermal power dissipation. When a constant force F displaces a mass through distance Δx, work is the product $W = F\Delta x$; when the force varies with position $F(x)$—as in a stiffening rally-car suspension spring, gas expanding against a piston ($W = \int_{V_1}^{V_2} P(V)\,dV$), or gravity diminishing with altitude—work is the definite integral $W = \int_{x_1}^{x_2} F(x)\,dx$. Similarly, the instantaneous heating power dissipated by a time-varying voltage $v(t)$ in a 1 Ω resistor is $v(t)^2$, and the *True Root-Mean-Square* (RMS) voltage is the equivalent DC voltage that delivers the same average thermal energy over a period T:

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T [v(t)]^2 dt} \qquad \textbf{(Eq. 8.10)}$$

For an undistorted sinusoid $v(t) = V_p \sin\vartheta$, the average of $\sin^2\vartheta$ is $1/2$, which yields $V_{\text{rms}} = V_p/\sqrt{2} \approx 0.7071\ V_p$. Whenever a varistor, thyristor dimmer, or rectifier clips the waveform, $V_p/\sqrt{2}$ no longer holds, and the integral $\sqrt{((1/T)\int_0^T v(t)^2\,dt)}$ is required.

8.2 Parabolic Quadrature and the Composite Trapezoidal Rule

The second pillar of calculus measures the exact area $\int_a^b f(x)\,dx$ enclosed underneath a curved boundary when no elementary geometric formula applies.

8.2.1 Archimedes' Mental Balance Beam and the 1 + 1/4 + 1/16 Parabolic Cascade

Around 240 BCE in Syracuse, Archimedes solved a problem that had resisted Greek geometers since Euclid: determining the exact area of the curved *parabolic segment* enclosed between a parabola and a straight secant chord.

In his treatise *Quadrature of the Parabola*—and in a companion manuscript titled *The Method of Mechanical Theorems*, which remained lost until it was discovered beneath the ink of a medieval prayer book (*The Archimedes Palimpsest*) in 1906—Archimedes combined mechanical equilibrium with geometric exhaustion. Before writing his formal geometric proof, Archimedes analyzed the problem using a mental balance beam, suspending thin vertical strips of the parabola from one end of a lever and counterbalancing them against strips of a known triangle at the other end. Comparing the lever arms at every horizontal station x revealed that the curved parabolic segment balanced at four-thirds (4/3) the weight of the largest inscribed triangle sharing the same base and apex.

To prove that 4/3 ratio rigorously by the *method of exhaustion*, Archimedes inscribed a central triangle of area A_0 inside the parabola, which left two smaller curved parabolic segments on the left and right. Inside each of those two segments he inscribed a new triangle, and proved from the quadratic law of the parabola that the combined area of the two new triangles equaled one-fourth ((1/4) A_0) of the first triangle. Repeating the construction inside the four remaining segments added $(1/4)^2$ A_0 = (1/16) A_0, then (1/64) A_0, and so on in a convergent geometric series:

$$\text{Area} = A_0\left(1 + \tfrac{1}{4} + \tfrac{1}{16} + \tfrac{1}{64} + \ \ldots\right) = A_0\left(\tfrac{1}{1 - 1/4}\right) = \mathbf{\tfrac{4}{3}A_0} \qquad \textbf{(Eq. 8.11)}$$

One scaling ratio governs Archimedes' geometric proof: every time the horizontal span of a parabolic arch is halved ($h \to h/2$), the remaining curved gap between the curve and the straight chord shrinks by a factor of one-fourth (1/4).

8.2.2 Composite Trapezoidal Quadrature and $O(h^2)$ Truncation Error

Archimedes' parabolic scaling ratio carries directly into modern numerical quadrature. To integrate a smooth function $f(x)$ from $x = a$ to $x = b$, one partitions the interval into N equal panels of width $h = (b - a) / N$. Capping each panel with a flat horizontal top at the left or right endpoint (a *Riemann step rectangle*) leaves a right-triangular gap of width h and height $\Delta y \approx f'(x)h$ on every sloping section of the curve; each panel incurs an $O(h^2)$ local error, and summing across $N = (b-a) / h$ panels produces an $O(h)$ first-order global error.

Connecting each sample point $(x_i, f(x_i))$ to the next point $(x_{i+1}, f(x_{i+1}))$ with a straight diagonal chord instead forms a *trapezoid* of average height $1/2[f(x_i) + f(x_i+1)]$ and area $(h/2)[f(x_i) + f(x_{i+1})]$. Every interior ordinate $f(x_i)$ is shared by two adjacent trapezoids (counted with weight 1/2 + 1/2 = 1), whereas the two outer endpoints $f(a)$ and $f(b)$ receive weight 1/2. Summing all N trapezoids yields the *Composite Trapezoidal Rule*:

$$T(h) = h\left[\tfrac{1}{2}f(a) + f(a+h) + f(a+2h) + \ \ldots\ + f(b-h) + \tfrac{1}{2}f(b)\right] \qquad \textbf{(Eq. 8.12)}$$

The truncation error of flat-topped trapezoids on a smooth curve follows from the geometry of a single narrow panel of width h. Over any small span, Taylor's theorem shows that a smooth curve bends locally like a parabola with curvature $f''(x)$, which encloses a thin crescent-shaped parabolic segment of width h between the curve and the straight secant chord of the trapezoid. Because parabolic deflection grows with the square of distance from the midpoint, the maximum vertical sag of the segment at the center of the panel is proportional to h^2 (specifically, $(1/8)f''(x)h^2$). By Archimedes' parabolic area theorem, the area of a parabolic segment is (2/3) × base × height, so the area of *one* panel's segment scales as (2/3) × h × $(1/8)f''(x)h^2 = (1/12)f''(x)h^3 \propto h^3$. Summing these segments across all $N = (b - a) / h$ panels spanning $[a, b]$ cancels one power of h against $1/h$:

$$\text{Total Trapezoidal Error } E(h) = T(h) - \int_a^b f(x)\,dx \approx \frac{b-a}{12} f''(\xi)\, h^2 \propto \mathcal{O}(h^2) \qquad \textbf{(Eq. 8.13)}$$

The sign of that error reflects the concavity of the integrand. When a curve is *concave up* ($f''(x) > 0$, such as e^x or x^2), the straight secant chord of each trapezoid lies *above* the curve, so $T(h)$ slightly *overestimates* the integral. When a curve is *concave down* ($f''(x) < 0$, such as $\sin x$ on $[0, \pi]$ or $\ln x$), the straight chord passes *beneath* the arch, so $T(h)$ slightly *underestimates* the integral.

The h^2 scaling law has a direct consequence: whenever an interval is partitioned into twice as many trapezoids, halving the step width ($h \to h/2$), the total error shrinks by a factor of $(1/2)^2 = \mathbf{1/4}$, matching Archimedes' 240 BCE ratio. Without extrapolation, that 1/4 ratio would require $2^{20} \approx 1{,}048{,}576$ trapezoids to reach twelve decimal digits of precision ($10^{-12} \approx 4^{-20}$). Richardson extrapolation eliminates this bottleneck in four or five halving steps.

8.3 The 4/3 Minus 1/3 Extrapolation: Richardson & Romberg Integration

In 1910–1911, the English physicist, meteorologist, and Quaker pacifist Lewis Fry Richardson *(Richardson, 1911)*—who formulated numerical weather prediction by solving fluid-dynamics equations by hand and later analyzed the fractal scaling of coastlines—recognized that the 1/4 error ratio could be used to cancel the leading truncation term algebraically.

Evaluating the area under a curve first with a coarse step size h produces the trapezoidal estimate $T_{\text{coarse}} = T(h)$, and halving the step size once to $h/2$ produces the fine trapezoidal estimate $T_{\text{fine}} = T(h/2)$.

First, that halving step reuses every previous ordinate. Refining N panels of width h into $2N$ panels of width $h/2$ keeps every sample point of the coarse grid as an even-numbered node of the fine grid:

$$T\left(\frac{h}{2}\right) = \frac{1}{2} T(h) + \frac{h}{2} \sum_{m=1}^{N} f\left(a + \left(m - \frac{1}{2}\right) h\right) \qquad \textbf{(Eq. 8.14)}$$

No function evaluation from the earlier pass is discarded; dividing the previous trapezoidal sum $T(h)$ by 2 and adding the newly sampled midpoints completes the refined sum.

Second, the trapezoidal error expands exclusively in *even* powers of h (h^2, h^4, h^6, ...) with all odd powers (h^3, h^5, ...) absent. In the 1730s, Leonhard Euler and Colin Maclaurin established this symmetry through the *Euler-Maclaurin Summation Formula*:

$$T(h) - \int_a^b f(x)\,dx = \sum_{m=1}^{K} \frac{B_{2m}}{(2m)!} h^{2m} \left[f^{(2m-1)}(b) - f^{(2m-1)}(a) \right] + \mathcal{O}(h^{2K+2}) \qquad \textbf{(Eq. 8.15)}$$

where $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$,... are Jakob Bernoulli's numbers. Every odd power of h vanishes identically because on every interior boundary x_i between adjacent panels, the derivative contribution from the right end of panel i ($+f^{(k)}(x_i)$) and the left end of panel $i+1$ ($-f^{(k)}(x_i)$) cancel in a *telescoping sum*, so only the odd-order derivatives at the outer endpoints $x = a$ and $x = b$ multiplied by even powers of h remain:

$$T(h) = I + c_1 h^2 + c_2 h^4 + c_3 h^6 + \ldots \qquad \textbf{(Eq. 8.16)}$$

Because halving the step size ($h \to h/2$) reduces the dominant $c_1\ h^2$ error term $E = c_1\ h^2$ to one-fourth ($(1/4)E$) of its previous value while reducing the $c_2\ h^4$ term to 1/16 of its value, the two estimates form a system of two linear equations:

$$T_{\text{coarse}} = I + E + c_2 h^4 + \mathcal{O}(h^6),\ T_{\text{fine}} = I + \tfrac{1}{4}E + \tfrac{1}{16}c_2 h^4 + \mathcal{O}(h^6) \qquad \textbf{(Eq. 8.17)}$$

Because the leading error in T_{fine} is one-fourth the error in T_{coarse}, multiplying the second equation by 4, subtracting the first equation, and dividing by 3 eliminates E:

$$\frac{4T_{\text{fine}} - T_{\text{coarse}}}{3} = \tfrac{4}{3}T_{\text{fine}} - \tfrac{1}{3}T_{\text{coarse}} = \left(\tfrac{4}{3} - \tfrac{1}{3}\right)I + \left(\tfrac{4}{3}\cdot\tfrac{1}{4}E - \tfrac{1}{3}E\right) - \tfrac{1}{4}c_2 h^4 = \mathbf{I + 0 \cdot E + \mathcal{O}(h^4)} \qquad \textbf{(Eq. 8.18)}$$

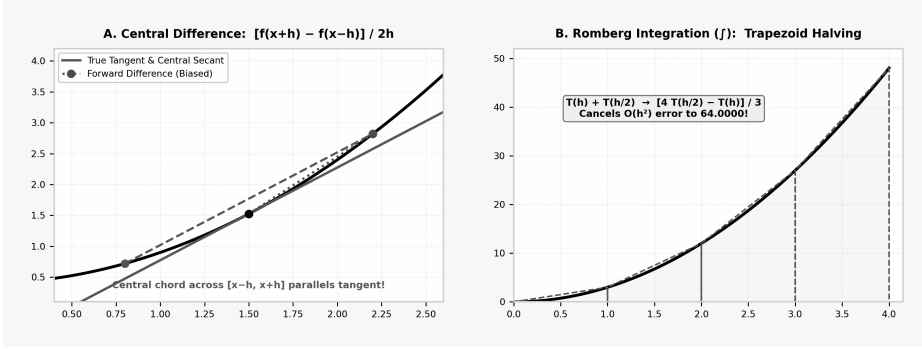


Figure: Symmetric central-difference secant slope cancelation and Romberg trapezoidal halving with Richardson 4^m extrapolation.

The leading h^2 error vanishes completely. Across a single coarse macro-panel of span $h = b$ - a (subdivided into two sub-steps of width $h/2$ at the midpoint m = (a+b) / 2), substituting $T(h)$ = $(h/2)[f(a)$ + $f(b)]$ and $T(h/2)$ = $(h/4)[f(a)$ + $2f(m)$ + $f(b)]$ into Richardson's 4/3 - 1/3 combination yields:

$$\tfrac{4}{3}\left(\tfrac{h}{4}[f(a) + 2f(m) + f(b)]\right) - \tfrac{1}{3}\left(\tfrac{h}{2}[f(a) + f(b)]\right) = \tfrac{h}{6}[f(a) + 4f(m) + f(b)] \qquad \textbf{(Eq. 8.19)}$$

Those 1–4–1 weights $(h/6)[f(a)$ + $4f(m)$ + $f(b)]$ (with $h = b$ - a, or Δx / 3 in terms of the half-step $\Delta x = h/2$) show that Richardson's 4/3 - 1/3 combination of two flat-topped trapezoid sums produces *Thomas Simpson's 1743 Parabolic Rule* ($O(h^4)$), the exact area under a parabola fitted through the left endpoint, the midpoint, and the right endpoint. Because the third derivative $f'''(x)$ of any cubic polynomial Ax^3 + Bx^2 + Cx + D is a constant (so $f'''(b)$ - $f'''(a)$ = 0 in the h^2 term and $f^{(4)}(x)$ = 0 in the h^4 term), Simpson's 1–4–1 rule is exact not only for parabolas, but for cubics as well.

In 1955, the German-Norwegian physicist Werner Romberg *(Romberg, 1955)* extended Richardson's elimination beyond the h^2 term. Once two Simpson parabolic estimates are available at step sizes h and $h/2$ across a macro-panel of span $h = b$ - a (subdivided into four sub-steps of width $h/4$ at nodes x_0, x_1, x_2, x_3, x_4), their leading $c_2\ h^4$ errors stand in the ratio $(1/2)^4$ = 1/16. Forming $(16/15)S_{\text{fine}}$ - $(1/15)S_{\text{coarse}}$ eliminates the h^4 error and produces *George Boole's 5-Point Rule* ($O(h^6)$) with its 7–32–12–32–7 weights (where $h/90$ = $2(h/4)/45$):

$$B = \frac{16S_{\text{fine}} - S_{\text{coarse}}}{15} = \frac{h}{90}[7f(x_0) + 32f(x_1) + 12f(x_2) + 32f(x_3) + 7f(x_4)] \qquad \textbf{(Eq. 8.20)}$$

Arranging each halving $h_k = (b\text{-}a)/2^k$ down a lower-triangular table $R(k, j)$ and sweeping across each row via the universal Richardson recurrence:

$$R(k,j) = R(k,j-1) + \frac{R(k,j-1) - R(k-1,j-1)}{4^j - 1}, \qquad \text{Error} = \mathcal{O}\left(h_k^{2j+2}\right), \qquad j = 1 \ldots \min(k, 5) \qquad \textbf{(Eq. 8.21)}$$

creates *Romberg Integration (Romberg, 1955)*:

Column Index	Equivalent Classical Quadrature Rule	Error Order	Extrapolation Multiplier
$j = 0$	Composite Trapezoidal Rule $T(h_k)$	$O(h^2)$	Base midpoint-reuse column
$j = 1$	Simpson's 1/3 Parabolic Rule	$O(h^4)$	(4 $R(k,0)$ - $R(k\text{-}1,0)$) / 3 (kills h^2 error)
$j = 2$	Boole's 5-Point Rule	$O(h^6)$	(16 $R(k,1)$ - $R(k\text{-}1,1)$) / 15 (kills h^4 error)
$j = 3$	9-Point Super-Extrapolation	$O(h^8)$	(64 $R(k,2)$ - $R(k\text{-}1,2)$) / 63 (kills h^6 error)
$j = 4, 5$	Capped High-Order Limit ($j_{\max} = 5$)	$O(h^{10}) \to O(h^{12})$	Prevents high-degree Runge ringing

To see the full triangular tableau $R(k, j)$ in action, trace how each row of $\int_0^2 e^x\,dx = e^2 - 1 \approx$ **6.3890560989** feeds the next column to the right:

Row k (Panels 2^k)	$j = 0$ (Trapezoid $O(h^2)$)	$j = 1$ (Simpson $O(h^4)$)	$j = 2$ (Boole $O(h^6)$)	$j = 3$ ($O(h^8)$)
$k = 0$ (1 panel)	8.38905610	—	—	—
$k = 1$ (2 panels)	6.91280988	6.42072780	—	—
$k = 2$ (4 panels)	6.52161011	6.39121019	6.38924235	—
$k = 3$ (8 panels)	6.42229781	6.38919370	6.38905927	6.38905637

On Row $k = 0$ (1 trapezoid, $h_0 = 2.0$), the single trapezoid $R(0,0) = 2/2(e^0 + e^2) = 8.38905610$ overshoots the exact integral by $+31.3\%$. On Row $k = 1$ (2 trapezoids, $h_1 = 1.0$), adding the single midpoint e^1 gives $R(1,0) = 6.91280988$, and applying (4 $R(1,0)$ - $R(0,0)$) / 3 produces the Simpson estimate $R(1,1) = 6.42072780$, which reduces the error tenfold. On Row $k = 2$ (4 trapezoids, $h_2 = 0.5$), adding the two quarter-points gives $R(2,0) = 6.52161011$, which extrapolates to $R(2,1) = 6.39121019$ and then $R(2,2) = (16(6.39121019) - 6.42072780) / 15 = 6.38924235$, within 0.0029% of the exact value using only 5 function evaluations. By Row $k = 3$ (8 trapezoids, $h_3 = 0.25$), using 9 function evaluations, $R(3,3) = 6.38905637$ is accurate to seven significant digits, and by Row $k = 4$ (17 points), the error drops below 10^{-12}.

On the SC-32, you run this entire Romberg tableau with three actions: enter the integrand expression into [R/S] [EQN], place the lower limit a in Register Y and the upper limit b in Register X, and press [R/S] [∫] (Right Shift on **8**) followed by the dummy variable softkey. The SC-32 caps the Romberg tableau at column $j_{\max} = 5$ ($O(h^{12})$) and splits wide intervals ($|b - a| > 20$) across logarithmic **1-2-5** decade boundaries because, as German mathematician Carl David Tolmé Runge proved in 1901, fitting a single high-degree polynomial across equally spaced points on a wide interval triggers boundary oscillations (*Runge's Phenomenon*).

When an integrand is a periodic wave such as $\cos^2 x$ or a bell curve whose odd derivatives $f^{(2m-1)}(b)$ - $f^{(2m-1)}(a)$ vanish at both endpoints in the Euler-Maclaurin expansion, the Composite Trapezoidal Rule converges exponentially fast on its own. Following William Kahan's architecture *(Kahan, 1979)*, the SC-32 couples its Romberg stopping tolerance to the active [←] [DISP] format (**FIX**, **SCI**, **ENG**, or **ALL**) and upon completion places the

certified integral I in Register X and the estimated truncation uncertainty $\pm\varepsilon$ in Register Y, while preserving whatever was previously in Registers Z and T so chained integrals never overwrite the upper stack registers.

8.4 Non-Elementary Integrals: Rockets, Dams & Large-Amplitude Pendulums

8.4.1 Joseph Liouville's Theorem and Non-Elementary Antiderivatives

In first-year calculus courses, nearly every exercise is constructed so that a closed-form symbolic antiderivative $F(x)$ exists for evaluating $F(b)$ - $F(a)$. Yet in 1835, the French mathematician Joseph Liouville proved that while any elementary combination of powers, roots, sines, exponentials, and logarithms can be *differentiated* by mechanical rules, most continuous curves encountered in physics possess no elementary antiderivative.

A simple pendulum of length L has the small-angle period $T_0 = 2\pi\sqrt{(L/g)}$. In his 1673 *Horologium Oscillatorium*, Christiaan Huygens demonstrated that a real pendulum swinging through a wide release angle ϑ_0 runs noticeably *slower* than T_0 because Galileo's linearized model replaces sin ϑ with ϑ. Retaining the exact restoring torque sin ϑ in conservation of energy leads to the *Complete Elliptic Integral of the First Kind* (with modulus $k = \sin(\vartheta_0/2)$):

$$\frac{T(\theta_0)}{T_0} = \frac{2}{\pi}\int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}} \qquad \textbf{(Eq. 8.22)}$$

Like the perimeter of an ellipse, no finite combination of elementary functions can express that integral, yet Romberg quadrature ([→] [∫]) evaluates it to twelve digits. The same barrier arises across quantum physics, optics, and civil engineering. In 1912, when Peter Debye explained why the heat capacity $C_V(T)$ of a diamond or copper crystal drops as T^3 near absolute zero, quantizing the lattice vibrations led to the non-elementary *Debye Integral*:

$$C_V(T) = 9Nk_B\left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2}\, dx \qquad \textbf{(Eq. 8.23)}$$

Similarly, integrating Max Planck's spectral radiance across a thermal camera's wavelength window $[\lambda_1, \lambda_2]$ or computing Carl Friedrich Gauss's normal error integral $(1 / \sqrt{2\pi})\int_a^b e^{-x^2/2}\, dx$ admits no closed-form antiderivative. When coherent light grazes a sharp edge—or when a civil engineer designs a highway exit ramp or roller-coaster loop whose curvature $\varkappa(s) \propto s$ increases linearly with arc length s to prevent a sudden lateral jerk—the trajectory follows *Augustin-Jean Fresnel's 1818 Diffraction Integrals*, tracing the double-coiled *Euler-Cornu clothoid spiral*:

$$C(u) = \int_0^u \cos\left(\frac{\pi t^2}{2}\right) dt, \qquad S(u) = \int_0^u \sin\left(\frac{\pi t^2}{2}\right) dt \qquad \textbf{(Eq. 8.24)}$$

Even when an integrand *does* admit a symbolic antiderivative, integrating the physical rate law in [→] [EQN] via [→] [∫] avoids algebraic transcription errors. This direct formulation applies to summing the hanging catenary arc length $\int_{-x_0}^{x_0} \cosh(x/a)\, dx$ of a transmission span, the voltage-dependent capacitance droop $C(V) = C_0(1 - \alpha\ V^2)$ of a ferroelectric $BaTiO_3$ capacitor *(Horowitz & Hill, 2015)*, or the cylindrical velocity rings $\int_0^R 2\pi\ r\ v(r)\ dr$ of Hagen-Poiseuille laminar pipe flow. In laminar pipe flow, because both the cross-sectional area $\pi\ R^2$ and the parabolic centerline speed $v_{\max} = (\Delta P\ /\ 4\mu\ L)R^2$ scale with R^2,

integrating the concentric shells yields *Poiseuille's fourth-power law*:

$$Q = \int_0^R 2\pi r\, v_{\max}\left(1 - \frac{r^2}{R^2}\right) dr = \tfrac{1}{2}\pi R^2 v_{\max} = \frac{\pi \Delta P}{8\mu L} R^4 \qquad \textbf{(Eq. 8.25)}$$

Consequently, a 16% plaque constriction to 0.84 R in a human coronary artery cuts blood supply in half (0.84^4 = **0.4979**), whereas a 19% smooth-muscle dilation to 1.19 R during exercise *doubles* blood flow (1.19^4 = **2.0053**), because laminar flow integrates across both cross-sectional area and velocity simultaneously.

8.4.2 Taming Improper Integrals & Singular Endpoints by Substitution

Physical integrals sometimes span an infinite upper limit $[a, \infty)$ or involve an integrand whose value or slope diverges at an endpoint, such as the hydrostatic pressure ribbon $(H - y)\sqrt{8y}$ at the floor $y = 0$ of a parabolic canyon dam or an inverse-square-root cusp $\int_0^b g(x)/\sqrt{x}\, dx$. If an endpoint triggers an explicit division by zero (such as $\sin x / x$ at $x = 0$), the SC-32 halts safely with **DIVIDE BY 0** (Code 01). As analyzed by William Kahan *(Kahan, 1979)* and in Press et al.'s *Numerical Recipes*, two classical variable substitutions inside [→] [EQN] eliminate endpoint singularities and infinite tails before pressing [→] [∫].

Whenever an integral has an integrable $1/\sqrt{x}$ singularity at $x = 0$ (as occurs when a pendulum reaches its turnaround point or light grazes a caustic), substituting $x = u^2$ introduces $dx = 2u\, du$ in the numerator, which cancels $\sqrt{x} = u$ in the denominator identically:

$$\int_0^b \frac{g(x)}{\sqrt{x}}\, dx = \int_0^{\sqrt{b}} \frac{g(u^2)}{u}\, 2u\, du = \int_0^{\sqrt{b}} 2\, g(u^2)\, du \qquad \textbf{(Eq. 8.26)}$$

For example, $\int_0^1 \cos x / \sqrt{x}\, dx$ diverges to $+\infty$ at $x = 0$, yet entering the smooth transformed integrand $2\cos(u^2)$ into [→] [EQN] as **U** [←] [x²] [COS] **2** [×] in [RAD] mode and integrating over **0** [ENTER] **1** [→] [∫] **U** converges in four Romberg rows to 1.8090484758.

Similarly, when integrating a rapidly decaying tail over $[a, \infty)$ with $a > 0$, the reciprocal substitution $x = 1/t$ ($dx = -dt/t^2$) maps the infinite horizon $x \to \infty$ onto the finite interval $[0, 1/a]$ as $t \to 0^+$:

$$\int_a^\infty f(x)\, dx = \int_0^{1/a} \frac{1}{t^2} f\left(\frac{1}{t}\right) dt \qquad \textbf{(Eq. 8.27)}$$

Displacing the lower limit from 0 to 10^{-12} (or truncating a Gaussian or exponential tail at 5 to 8 characteristic scale lengths where $e^{-x^2/2}$ has already fallen below machine epsilon) lets Romberg's tableau integrate improper tails to full precision.

8.5 Central Differences & Adaptive Romberg Quadrature

Both pillars of numerical calculus—central-difference differentiation and adaptive Romberg quadrature—map their two-point geometry onto the lower pair of stack registers (**Y** and **X**) while leaving the upper stack (**Z** and **T**) undisturbed.

For a general integrand $f(x)$ stored inside [→] [EQN] ([→] + [STO]), complementing direct 3-point Simpson evaluation $(1, 4, 1) \times h/6$ on the stack, the integration bounds $[a, b]$ occupy the lower two registers via **a** [ENTER] **b**. Pressing [→] [∫] (above [8]) and selecting the active dummy variable (**X**, **T**, **R**, or **V**) launches the midpoint-reuse trapezoidal cascade and Richardson's 4^j extrapolation across **Y** and **X** without disturbing any prior constants or

intermediate integrals stored in **Z** and **T**. When the active [↶] [DISP] precision is set to [FIX] **n**, [SCI] **n**, or [ENG] **n**, the stopping tolerance scales automatically with the displayed decimal unit while keeping all intermediate trapezoidal sums in full 64-bit IEEE-754 floating-point precision.

8.5.1 Rocket Burnout, Progressive Springs, Hydrostatic Dams & Poiseuille Flow

Unless degrees ([DEG]) are explicitly noted, trigonometric and calculus sequences use radian measure ([RAD]). Because a single Romberg integral consumes only the limits (a, b) in Registers Y and X, convergence replaces (Y, X) in place with the *Post-Integration Audit Frame* ($\pm\varepsilon$, I) while leaving Registers Z and T untouched for multi-integral chaining:

Stack Register	Post-[→] [∫] Contents	Engineering Purpose & Stack Chaining
Register *X*	**Certified Definite Integral ($I = \int_a^b f(x)\,dx$)**	The final Richardson-extrapolated quadrature result.
Register *Y*	**Estimated Quadrature Uncertainty ($\pm\varepsilon$)**	Upper bound on absolute truncation error. Swap via [x⇄y] to audit convergence.
Register *Z*	**Prior Register *Z* (Preserved)**	Holds whatever was in Z when [→] [∫] was pressed (for example, a first integral I_1 lifted into Z by typing **c** [ENTER] **d** for a second integral).
Register *T*	**Prior Register *T* (Preserved)**	Preserves the top of the stack undisturbed across quadrature passes.

Five mechanical and fluid accumulation laws show how the analytical smoothness of the integrand governs which column of Romberg's tableau delivers the final answer:

- **Cubic Rocket Thrust & Burnout Profiles (Exact in One Richardson Step at $R(1,1)$):** Because Richardson's first $(4/3)\ T(h/2)$ - $(1/3)\ T(h)$ column $R(1,1)$ is algebraically identical to Thomas Simpson's parabolic rule—whose fourth-order error $\propto f^{(4)}(\xi)\ h^4$ vanishes identically for every polynomial of degree 3 or less—any cubic velocity or acceleration law hits machine precision on the very first Romberg row.
  - **Sounding Rocket & Cubic Boost:** If a solid-propellant sounding rocket climbs at $v(t) = 30t$ - $0.5t^3$ m/s over $t \in [0, 6]$ s, one Simpson pass integrates the burnout altitude $\int_0^6 (30t - 0.5t^3)\ dt = [15t^2 - (1/8)t^4]_0^6 = 378.0$ m (202.5 m over $[0, 3]$ s for $45t$ - $0.5t^3$). For the cubic boost acceleration $a(t) = 4t^2$ - $0.5t^3$ m/s^2, store **T x² 4 × T 3 yˣ 0.5 × -** in [→] [EQN] and key **0** [ENTER] **6** [→] [∫] **T** to nail the burnout velocity $[(4/3)t^3 - (1/8)t^4]_0^6$ = **126.000000000 m/s** in Register X ($\pm\varepsilon = 1.24 \times 10^{-11}$ in Register Y).
  - **Liquid-Bipropellant Upper Stage:** When an upper stage expels propellant at a constant mass rate so thrust-to-mass acceleration climbs exponentially as $a(t) = 12.0e^{0.04t}$ - 9.80665 m/s^2 over a 25-second burn, integrating via **0** [ENTER] **25** [→] [∫] **T** yields **270.3183 m/s** in Register X (matching the analytical antiderivative $\Delta v = 300(e^{1.0}$ - $1)$ - 245.16625 = **270.3183 m/s**, where [↶] [LASTx] reuses **25** for the gravity-loss subtraction).
- **Progressive Rally-Car Suspension Springs:** That same exact-cubic property governs the variable-pitch *progressive suspension spring* ($F(x) = 2500x + 40000x^3$ N) of an off-road rally car. Over small ripples ($x \ll 0.05$ m), the 2,500 N/m linear rate gives a supple ride; on hard jump landings, the lower coils bind so the $40{,}000x^3$ stiffening prevents bottoming out (absorbing $[1250x^2 + 10000x^4]_0^{0.50} = 937.5$ J over a 0.50 m stroke). At $x = 0.15$ m, storing **X 2500 × X 3 yˣ 40000 × +** in [→] [EQN] and keying **0**

`ENTER` **`0.15`** `↷` `∫` **`X`** converges on the first Simpson column to W = **33.187500000 J** ($\pm\varepsilon = 3.55 \times 10^{-12}$ in Y), which evaluating $1250x^2 + 10000x^4$ via **`0.15`** `↶` `x²` **`1250`** `×` **`0.15`** `↶` `x²` `↶` `x²` **`10000`** `×` `+` confirms as **33.1875 Joules**—18% above a linear spring's 28.125 J.

- **Nonlinear Ferroelectric $BaTiO_3$ Ceramic Capacitors:** For a Class-II barium titanate ($BaTiO_3$) ceramic capacitor whose capacitance droops under high DC bias as $C(V) = C_0(1 - \alpha V^2)$ with $C_0 = 10$ μF and $\alpha = 10^{-6}$ V^{-2} *(Horowitz & Hill, 2015)*, integrating the differential energy $dW = C(V)\,V\,dV = 10^{-5}(V - 10^{-6}V^3)\,dV$ is again a degree-3 polynomial that Simpson's column $R(1,1)$ resolves exactly. Storing **`V 3 yˣ 1E-6 × V x↔y - 10E-6 ×`** in `↷` `EQN` and integrating over **`0`** `ENTER` **`500`** `↷` `∫` **`V`** reveals a true stored energy of **1.09375 Joules**—a 12.5% shortfall below the unadjusted linear estimate $(1/2)C_0\,V^2 = 1.2500$ J.

- **Parabolic Canyon Spillway Dam (Vertical Square-Root Tangents and Higher Romberg Columns):** Unlike those cubic polynomials, the hydrostatic pressure ribbon $(H - y)\sqrt{8y}$ across a parabolic canyon spillway of width $w(y) = \sqrt{8y}$ m and water depth $H = 4.0$ m ($\rho\,g = 9{,}810$ N/m^3) carries the singular endpoint derivative $(d/dy)\sqrt{8y} = \sqrt{2}/\sqrt{y} \to \infty$ at the canyon floor $y = 0$. Because the Euler-Maclaurin endpoint derivatives diverge there, $R(1,1)$ cannot cancel the curvature in one step; instead, Romberg's tableau subdivides the grid and marches to higher columns $R(k, j)$ until the singular boundary panel is pinched down to tolerance. Storing **`Y 8 × √x 4 Y - × 9810 ×`** in `↷` `EQN` and keying **`0`** `ENTER` **`4`** `↷` `∫` **`Y`** converges to **236,773.2915 N** ($\pm\varepsilon = 4.18 \times 10^{-10}$ in Register Y).

- **Hagen-Poiseuille Laminar Pipe Flow Across Concentric Rings:** Evaluating Poiseuille's concentric-shell integral $Q = \int_0^R 2\pi\, r\, v_{\max}(1 - r^2/R^2)\,dr$ for a pipe of radius $R = 0.05$ m ($R^2 = 0.0025$ m^2) and centerline speed $v_{\max} = 1.2$ m/s integrates the cubic ring flux $2\pi\, v_{\max}(r - r^3/R^2)$, whose peak contribution sits at $r = R/\sqrt{3} \approx 0.0289$ m where neither the wall shear ($v(R) = 0$) nor the vanishing centerline circumference ($2\pi(0) = 0$) suppresses the flow. Entering **`R x² 0.0025 ÷ 1 x↔y - 1.2 × R × 2 × π ×`** into `↷` `EQN`, keying **`0`** `ENTER` **`0.05`** `↷` `∫` **`R`** ($\to Q = 0.004712389$ m^3/s, $\pm\varepsilon = 1.45 \times 10^{-12}$ in Y), and scaling by **`1000`** `×` yields **4.712389 Liters/second**.

8.5.2 Transcendental Integrals, Piecewise RMS Waveforms & Chained Centroids

When we move from algebraic profiles to Gaussian bells, exponential lightning surges, hanging catenary cables, and chained integral ratios, both Euler-Maclaurin endpoint behavior and four-register stack preservation come to the fore.

Symmetry and exponential tail cutoffs resolve non-elementary improper integrals over $(-\infty, 0]$ or $[0, \infty)$ directly on (Y, X):

- **Gauss's Normal Bell & Vanishing Euler-Maclaurin Endpoint Derivatives ($\int_{-5}^{0} \varphi(x)\,dx$):** For the non-elementary standard normal density $\varphi(x) = (1 / \sqrt{2\pi})e^{-x^2/2}$, truncating the lower $(-\infty, 0]$ tail at five standard deviations ($x = -5$, where $\varphi(-5) \approx 1.487 \times 10^{-6}$) pairs an exponentially small left endpoint with the symmetry peak $x = 0$, where every odd derivative $\varphi'(0) = \varphi'''(0) = \varphi^{(5)}(0) = 0$ vanishes identically. Because the Euler-Maclaurin error series depends only on the odd-order endpoint derivatives $f^{(2m-1)}(b) - f^{(2m-1)}(a)$, storing **`X x² 2 ÷ +/- eˣ 2 π × √x ÷`** in `↷` `EQN` and integrating across **`5`** `+/-` `ENTER` **`0`** `↷` `∫` **`X`** converges rapidly to **0.4999997133** in Register X ($\pm\varepsilon = 2.15 \times 10^{-10}$ in Register Y)—rounding to exact **`0.500000`** in **`FIX 6`** and missing only the minuscule five-sigma tail $\Phi(-5) = 2.8665 \times 10^{-7}$.

- **Substation MOV Lightning Surge Arrester Across Five Time Constants:** When a lightning stroke drives an $i(t) = I_0\ e^{-t/\tau}$ current surge (I_0 = 10,000 A, τ = 5×10^{-5} s) through a zinc-oxide Metal-Oxide Varistor (MOV), the charge $Q = \int i(t)\ dt$ absorbed over $[0,\ 5\tau] = [0,\ 250\ \mu s]$ follows from integrating the exponential pulse. Storing **T 5E-5 ÷ +/- e^x 10000 ×** in [R↗] [EQN] and keying **0** [ENTER] **0.00025** [R↗] [∫] **T** integrates the pulse to **0.496631027 Coulombs** in Register X ($\pm\varepsilon = 1.12 \times 10^{-11}$ in Y), which cross-checking $I_0\tau(1 - e^{-5})$ via **1** [ENTER] **5** [+/-] [e^x] [-] **0.5** [×] confirms as **0.496631 C** ($1 - e^{-5}$ = **99.326%** of the total charge $I_0\tau$ = 0.5000 C).

- **Clinical Pharmacokinetics (AUC Bioavailability) & Civil Stormwater Hydrographs:** In clinical pharmacology, when an oral dose absorbs at k_a = 1.20 hr^{-1} and clears at k_e = 0.20 hr^{-1} along Harry Bateman's plasma curve $C(t) = 25.0(e^{-0.20t} - e^{-1.20t})$ mg/L (peaking at $t_{max} = \ln(k_a/k_e)\ /\ (k_a - k_e) = \ln(6)$ = **1.7918 hr**), storing **T 0.2 × +/- e^x T 1.2 × +/- e^x - 25 ×** in [R↗] [EQN] and keying **0** [ENTER] **24** [R↗] [∫] **T** integrates the 24-hour *Area Under the Curve* to $AUC_{0\rightarrow\ 24}$ = **103.1338 mg·hr/L** (99.01% of the infinite-horizon exposure $AUC_{0\rightarrow\infty}$ = 104.1667 mg·hr/L). Similarly, in civil hydrology, integrating a watershed's gamma storm hydrograph $Q(t) = 18.0\ t\ e^{-0.5t}$ m^3/s (**T 0.5 × +/- e^x T × 18 ×**) over **0** [ENTER] **8** [R↗] [∫] **T** ($\rightarrow$ 65.4079 m^3·hr/s) and scaling by **3600** [×] sizes a flood-detention basin at **235,468.3 m^3**.

All four stack registers (T, Z, Y, X) participate when chaining a Romberg integral into a subsequent calculation or combining two definite integrals on the stack. Applying the hyperbolic arc-length identity $\sqrt{(1 + [y'(x)]^2)} = \cosh(x/a)$ to a high-voltage transmission cable $y(x) = 800\ \cosh(x/800)$ spanning 200 m from x = -100 m to +100 m, integrating **X 800 ÷ HYP COS** via [R↗] [∫] **X** places the conductor arc length L = 200.5212 m in Register X and its uncertainty 8.55×10^{-11} in Register Y; as we then compute the mid-span sag $\Delta y = 800[\cosh(100/800) - 1]$ = 6.2581 m in Register X, the arc length 200.5212 m stays live in Register Y while the Romberg uncertainty rides up into Registers Z and T:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
100 [+/-] [ENTER] **100**	0.0000	0.0000	-100.0000	100.0000	Tower span interval [-100, +100]
[R↗] [∫] **X**	0.0000	0.0000	**8.55×10^{-11}**	**200.5212**	Conductor total arc length (meters)
100 [ENTER] **800** [÷] [↰] [HYP] [COS]	8.55×10^{-11}	8.55×10^{-11}	200.5212	1.007823	Dimensionless catenary height ratio
1 [-] **800** [×]	8.55×10^{-11}	8.55×10^{-11}	200.5212	**6.2581**	Conductor mid-span sag Δy (meters)

In nuclear engineering, a cylindrical heavy-water reactor core of radius R = 1.80 m, height H = 3.60 m, and peak volumetric power density $q_0''' = 120$ MW/m^3 has radial neutron flux governed by Bessel's drumhead mode $J_0(2.40483\ r/R) \approx 1 - (1/4)u^2 + (1/64)u^4$ (with $j_{0,1}/R \approx 1.3361\ m^{-1}$). Integrating the 3-term Horner-nested quartic radial ring integrand **R 1.3361 × x^2 ENTER ENTER 0.015625 × 0.25 - × 1 + R × 2 × π ×** over **0** [ENTER] **1.8** [R↗] [∫] **R** and scaling by the axial cosine factor $(2/\pi)\ q_0'''\ H = 864/\pi$ yields **1,263.13 MW_{th}**—within 4.5% of the exact Bessel J_0 thermal output **1,208.5866 MW_{th}** (or reaching **1,204.2 MW_{th}** when the next Horner term $-(1/2304)u^6$ is included).

Evaluating the True-RMS heating integral $V_{rms} = \sqrt{((1/T)\int_0^T v(t)^2\ dt)}$ on a clipped AC waveform shows why you should never integrate across a sharp slope corner in a single

Romberg pass. When a protection varistor clips a $V_p = 170$ V sine wave flat at $V_c = 136$ V, the derivative $v'(\vartheta)$ jumps discontinuously at the clipping knee $\vartheta_c = \arcsin(136/170) = \arcsin(0.80) = 0.927295$ rad, violating the smooth Euler-Maclaurin assumptions. Splitting the quarter-cycle $[0, \pi/2]$ right at ϑ_c—integrating the smooth rising sine squared $(170 \sin \vartheta)^2$ (**X** SIN **170** × ↶ x^2 in R/S EQN) over $[0, \vartheta_c]$ and adding the flat clamped plateau $136^2(\pi/2 - \vartheta_c)$ on the stack—computes the exact True-RMS voltage without corner ringing:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
RAD	0.0000	0.0000	0.0000	0.0000	Radian mode engaged for trig calculus
0 ENTER **0.8** ↶ **ASIN**	0.0000	0.0000	0.0000	0.927295	Integration bounds $[0, \vartheta_c]$ via $\vartheta_c = \arcsin(0.8)$
R/S ∫ **X**	0.0000	0.0000	**3.12 × 10^{-10}**	**6463.4190**	Integral of $(170 \sin \vartheta)^2$ over rising edge
0.8 ↶ **ASIN** R/S π **2** ÷ $x \rightleftarrows y$ −	6463.4190	6463.4190	6463.4190	0.643501	Clipped interval $(\pi/2 - \vartheta_c)$; rising integral stays in Y
136 ↶ x^2 ×	6463.4190	6463.4190	6463.4190	**11902.1965**	Flat clamped plateau contribution
+	6463.4190	6463.4190	6463.4190	**18365.6155**	Total quarter-cycle energy sum
2 × R/S π ÷ $\sqrt{x}$	6463.4190	6463.4190	6463.4190	**108.1292**	Mean square normalized and root taken (V_{rms})

Finally, many engineering quantities are defined as the ratio of two definite integrals. To locate the center of mass $\bar{\mathbf{x}} = \int_0^2 x\lambda(x)\,dx \,/\, \int_0^2 \lambda(x)\,dx$ of a tapered turbine spar with linear density $\lambda(x) = 1.5 + 0.8x^2$ kg/m, store the denominator mass $M = \int_0^2 \lambda(x)\,dx = 5.133333$ kg in Variable **A**, evaluate the first moment $M_x = \int_0^2 x\lambda(x)\,dx = 6.200000$ kg·m, and divide via RCL **A** ÷:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0 ENTER **2**	0.0000	0.0000	0.0000	2.0000	Integration limits [0, 2]
R/S ∫ **X**	0.0000	0.0000	**3.24 × 10^{-11}**	**5.133333**	Romberg evaluates rod mass M (kg)
STO **A**	0.0000	0.0000	3.24 × 10^{-11}	5.133333	Store total mass M in Variable A
0 ENTER **2**	3.24 × 10^{-11}	5.133333	0.0000	2.0000	Reload integration limits [0, 2]
R/S ∫ **X**	3.24 × 10^{-11}	5.133333	**1.87 × 10^{-11}**	**6.200000**	Romberg evaluates first moment M_x
RCL **A** ÷	5.133333	5.133333	1.87 × 10^{-11}	**1.207792**	Centroid position $\bar{\mathbf{x}} = M_x / M$ (meters)

The same two-pass chaining determines the *optical mode confinement factor* $\Gamma = \int_{-d}^{d} |E(x)|^2\,dx \,/\, \int_{-5d}^{5d} |E(x)|^2\,dx$ of a double-heterostructure gallium arsenide GaAs/AlGaAs quantum-well laser diode, where only the central active layer $[-d, +d]$ stimulates emission while the evanescent wings in $[-5d, +5d]$ remain passive. In normalized half-width units $d = 1$, we model the transverse waveguide intensity profile $|E(x)|^2 = 1.23\ \text{sech}^2(1.23\ x)$ by storing **1.23 X 1.23** × ↶ HYP COS ↶ x^2 ÷ in R/S EQN; as soon as the second limits **-5 ENTER 5** are keyed, the first integral **1.6852** rides up into Register Z:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [+/-] [ENTER] **1** [↻] [∫] **X**	0.0000	0.0000	1.25×10^{-11}	1.6852	Core field energy over [-d, d]
[STO] **A**	0.0000	0.0000	1.25×10^{-11}	1.6852	Core energy stored in register A
5 [+/-] [ENTER] **5** [↻] [∫] **X**	1.25×10^{-11}	1.6852	2.40×10^{-11}	2.0000	Total field energy over [-$5d$, $5d$]
[RCL] **A** [x⇄y] [÷]	1.6852	1.6852	2.40×10^{-11}	**0.8426**	Optical confinement factor $\Gamma = 84.26\%$

8.5.3 Central-Difference Derivatives & James Watt's Flyweight Governor

Turning from definite integrals to the symmetric central-difference formulas $f'(x_0) \approx (f(x_0 + h) - f(x_0 - h)) / 2h$ and $f''(x_0) \approx (f(x_0 + h) - 2f(x_0) + f(x_0 - h)) / h^2$, both operations execute directly on Registers X and Y without disturbing Registers Z or T.

First, verify the exact quadratic cancellation property on $f(x) = x^2$ at $x_0 = 3.0$ (where $f'(3.0) = 2(3.0) = 6.0$) using the $\varepsilon_{\text{mach}}^{1/3}$ step size $h = 10^{-5}$ (total span $2h = 0.00002$). Keying **3.00001** [↶] [x²] places the forward sample $f(x_0 + h) = 9.0000600001$ in Register X; keying **2.99999** [↶] [x²] [-] lifts the forward sample into Register Y, evaluates $f(x_0 - h) = 8.9999400001$ in Register X, and subtracts them so the $+(1/2)f''(x_0)h^2 = 10^{-10}$ curvature terms cancel to leave $\Delta f = 0.0001200000$ in Register X; and dividing by the span **0.00002** [÷] returns the exact derivative **6.0000000000** in Register X.

For the exponential benchmark $f(x) = e^{2x}$ at the origin $x_0 = 0$ with step $h = 10^{-5}$ (symmetric exponents $2x_{\pm} = \pm\, 2h = \pm\, 0.00002$ and span denominator $2h = 0.00002$), three stack sequences extract both the first slope $f'(0) = 2$ and the second curvature $f''(0) = 4$ across (Y, X):

- [↶] [LASTx] **Exponent Reuse:** Key **0.00002** [ENTER] [e^x] to place 0.00002 in Register Y and $e^{0.00002} = 1.0000200002$ in Register X while saving 0.00002 in **LASTX**. Then key [↶] [LASTx] [+/-] [e^x] [-] to form $e^{2h} - e^{-2h} = 0.0000400000$, and finish with [x⇄y] [÷] to obtain $f'(0) =$ **2.0000000003** while typing **0.00002** only once.
- **Hyperbolic Sine Identity:** Because $(e^{2h} - e^{-2h}) / 2h = 2 \sinh(2h)/2h$, keying **0.00002** [↶] [HYP] [SIN] [↶] [LASTx] [÷] **2** [×] evaluates $f'(0) =$ **2.0000000003** in seven keystrokes.
- **Second Derivative $f''(x_0) \approx (f(x_0+h) - 2f(x_0) + f(x_0-h)) / h^2$:** Adding $e^{2h} + e^{-2h} = 2 \cosh(2h)$ instead of subtracting them and dividing by $h^2 = 10^{-10}$ via **0.00002** [↶] [HYP] [COS] **1** [-] **2** [×] **1** [E] **10** [+/-] [÷] recovers the second derivative $f''(0) =$ **4.000000**.

Finally, evaluating the polar kinetic-energy equation $E_k = (1/2)m(\dot{r}^2 + v_\vartheta^2)$ for James Watt's centrifugal flyball governor on a brass flyweight of mass $m = 4.0$ kg whirling at $\omega = 40.0$ rad/s along the cam slot $r(\vartheta) = 0.15 + 0.05 \sin(2\vartheta)$ m at $\vartheta = \pi/6$ rad ([RAD] mode) pairs the radial and tangential velocity components directly across **Register Y** and **Register X**:

1. **Radial cam-sliding velocity in Register Y ($\dot{r} = \omega(dr/d\vartheta)$):** Storing $\vartheta = \pi/6 \approx 0.523599$ rad via [RAD] [↻] [π] **6** [÷] [STO] **A**, we evaluate the radial slope $dr/d\vartheta = 0.10 \cos(2\vartheta) = 0.050000$ m/rad either analytically via **2** [×] [COS] **0.1** [×] or numerically to $O(h^2)$ accuracy via the central difference $(r(\vartheta + 10^{-5}) - r(\vartheta - 10^{-5})) / (2 \times 10^{-5}) =$

0.0500000000 m/rad. Multiplying by $\omega = 40$ rad/s and squaring via **40** [×] [↰] [x²] places the squared radial sliding speed $\dot{r}^2 = (2.0000\ \text{m/s})^2 = \mathbf{4.000000\ m^2/s^2}$ in Register *X*.

2. **Tangential whirl velocity in Register *X* ($v_\vartheta = r\omega$) and Total Kinetic Energy:** Keying [RCL] **A 2** [×] [SIN] **0.05** [×] **0.15** [+] **40** [×] lifts $\dot{r}^2 = 4.000000$ into Register *Y* while evaluating the radius $r(\pi/6) = 0.193301$ m and tangential whirl speed $v_\vartheta = 40\ r = \mathbf{7.732051\ m/s}$ in Register *X*. Squaring v_ϑ, adding $\dot{r}^2$ from Register *Y*, and scaling by $(1/2)m = 2.0$ kg via [↰] [x²] [+] **2** [×] delivers the flyweight's exact kinetic energy $E_k = \mathbf{127.569226\ Joules}$—just as taking the second central difference $V''(\vartheta) \approx (V(\vartheta+h) - 2V(\vartheta) + V(\vartheta-h)) / h^2$ of the governor's effective potential $V(\vartheta) = mgL(1 - \cos\vartheta) - (1/2)m\omega^2 L^2 \sin^2\vartheta$ on (*Y*, *X*) certifies whether the equilibrium sleeve angle is dynamically stable ($V''(\vartheta^*) > 0$).

8.5.4 Display-Coupled Convergence, Chained Integrals & Endpoint Singularities

When integrating complex physical waveforms on the SC-32, four practical stack techniques let you control speed, accuracy, and boundary behavior:

1. **Controlling Quadrature Speed via Display Precision ([FIX], [SCI], [ENG]) & Register *Y* Error Inspection:** Following William Kahan's display-coupled stopping rule *(Kahan, 1979)*, the Romberg engine terminates once two successive diagonal extrapolations $|R(k, j) - R(k, j\text{-}1)|$ agree within half an ulp of the active display format ([FIX] **3** for a rapid 3-digit field check, or [SCI] **9** / [ALL] for full 11-to-12-digit certification), writing the converged definite integral into Register *X* and the final Richardson truncation uncertainty estimate $\pm\varepsilon$ directly into Register *Y*.

2. **Chaining Integrals via Register *Z*:** Because post-integration replaces (*Y*, *X*) with ($\pm\varepsilon$, *I*) while preserving *Z* and *T*, typing new limits **c** [ENTER] **d** lifts your first integral I_1 into Register *Z* (**1.6852** in our laser confinement example) so I_1 and I_2 sit ready for immediate ratio or sum chaining.

3. **Splitting Oscillatory or Sharp-Peak Integrands:** When an integrand contains a narrow spike (such as a high-*Q* resonance peak at $x = 10$ inside [0, 100]), splitting the interval at the peak ([0, 10] and [10, 100]) guarantees that the initial panel samples the apex directly.

4. **Handling Removable 0/0 Endpoints & Singular Cusp Substitutions:** Touching a division-by-zero pole such as $\int_0^1 1/x\ dx$ halts with **DIVIDE BY 0** (Code 01, cleared via [C ON]). For a removable 0/0 endpoint such as $\text{Si}(\pi) = \int_0^\pi \sin x\ /\ x\ dx \approx 1.8519370520$, displacing the lower limit to $a = 10^{-12}$ converges in four rows, while $x = u^2$ or $x = 1/t$ substitutions in [↻] [EQN] tame $1/\sqrt{x}$ cusps and semi-infinite tails.

Pairing symmetric central-difference derivatives with Werner Romberg's Richardson extrapolation turns the four-register stack into a numerical calculus laboratory. Whether extracting local rates of change where odd Taylor errors cancel or integrating non-elementary waveforms across recursive trapezoidal panels, both operations return certified floating-point bounds directly on Registers *Y* and *X*.

Chapter 9: Least Squares, Small Samples & the Geometry of Uncertainty

9.1 The Lost Planet Ceres, Galton's Quincunx & Bessel's n − 1 Correction

On the night of January 1, 1801, the Italian astronomer and Theatine priest Giuseppe Piazzi peered through his vertical Ramsden circle telescope at the Palermo Observatory in Sicily. While compiling a star catalog in the constellation Taurus, Piazzi noticed a faint, uncataloged eighth-magnitude point of light. The next night, it had shifted slightly against the background stars.

Across a forty-one-day span in January and early February 1801 (recording 24 nights of observations), Piazzi tracked the object before falling ill as it drifted into the daytime glare of the Sun. Piazzi had discovered Ceres, the largest body in the asteroid belt between Mars and Jupiter. Yet Piazzi had observed Ceres across barely three degrees of sky, less than one percent of its full orbit, and every telescope reading was affected by atmospheric refraction, clock drift, and human visual error. When European astronomers attempted to compute Ceres's orbit from three selected observations, their predictions diverged across the sky.

In Göttingen, Carl Friedrich Gauss, then twenty-four, read Piazzi's published sightings and approached the problem from a different premise. Instead of selecting three observations and treating them as exact, Gauss analyzed how random observational errors behave and how all of Piazzi's noisy sightings across that forty-one-day arc could be combined so that their random errors cancel one another.

Over three months in the autumn of 1801, Gauss combined his geometry of elliptical orbits with two foundational principles that established modern statistical inference *(Gauss, 1809)*: the *Method of Least Squares*, which identifies the most probable trajectory through a cloud of noisy measurements as the unique curve that minimizes the sum of the squared residuals (the squared vertical distances between the observed data points and the theoretical curve), and the *Gaussian Law of Error* (the normal distribution), under which small, independent random disturbances cluster symmetrically around the true value according to the exponential-square density $e^{z^2}/2$.

On December 31, 1801—after first re-sighting the faint object on December 7 and waiting through winter clouds to confirm its motion—the German astronomer Franz Xaver von Zach verified Ceres in the patch of sky in Virgo predicted by Gauss's least-squares calculation, within half a degree of Gauss's coordinates. Gauss had demonstrated that combining noisy measurements mathematically extracts a signal more accurate than any single instrument reading.

9.1.1 Francis Galton's 1873 Quincunx and the Central Limit Theorem

Gauss's bell curve recurs across the physical and biological sciences, governing astronomical telescope errors, the heights of forest trees, the thermal noise of resistors, and the diameters of machined steel shafts.

In 1873, during a Friday Evening Discourse at the Royal Institution in London, the Victorian polymath Francis Galton demonstrated a tabletop apparatus that made the

emergence of the normal curve directly visible: the Quincunx (now known as the Galton Board). Galton constructed a vertical wooden frame behind a pane of glass, with a funnel at the top, twelve staggered horizontal rows of brass pins in the middle (arranged like the five-spot *quincunx* pattern on a die), and a row of narrow vertical compartments at the bottom. When lead shot pellets were poured into the top funnel, each pellet cascaded through the brass pins. At each of the 12 rows of pins, a falling pellet encountered an independent 50/50 binary deflection: bounce Left (L) or bounce Right (R).

As the thousand pellets settled into the vertical compartments at the bottom, they neither formed a flat horizontal layer nor accumulated in an irregular heap; instead, the lead shot consistently formed the smooth, symmetric profile of Gauss's normal curve.

The shape follows from counting the combinatorial paths. For a pellet to land in Slot 0 at the far left edge, it must bounce Left twelve times in succession ($LLLLLLLLLLLL$), which represents only $C(12,\ 0) = 1$ path out of $2^{12} = 4{,}096$ possible trajectories (probability 0.024%). To land in Slot 1, there are $C(12,\ 1) = 12$ paths. By the time a pellet reaches Slot 6 in the center—six Left bounces and six Right bounces in any order—there are $C(12,\ 6) =$ **924** distinct paths leading to that central compartment.

Galton's Quincunx is a mechanical realization of the Central Limit Theorem (first proved for binomial trials by Abraham de Moivre in 1733 and generalized by Pierre-Simon Laplace in 1810): whenever a measurement is perturbed by a large number of small, independent random disturbances—lathe vibrations, air currents, thermal electron agitation—their sum converges to the Gaussian Normal Probability Density Function (PDF):

$$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),\qquad \phi(z)=\frac{1}{\sqrt{2\pi}}e^{-z^2/2}\quad\left(\text{where } z=\frac{x-\mu}{\sigma}\right)\qquad \textbf{(Eq. 9.1)}$$

The center of the distribution sits at the mean μ, while its dispersion is governed by the standard deviation σ (the horizontal distance from the peak μ to the two inflection points $x = \mu \pm \sigma$ where the curvature changes sign). Integrating the area under the density curve from $-\infty$ to z gives the cumulative probability $\Phi(z) = 1/2 + (1 / \sqrt{2\pi})\int_0^z e^{-t^2/2}\,dt$, which establishes the Empirical Rule (68.27%–95.45%–99.73%) for the fraction of observations falling within ± 1σ, ± 2σ, and ± 3σ of the mean.

9.1.2 Bessel's n - 1 Correction and Degrees of Freedom

When a sample of n parts is drawn from a production line, or n sensor readings x_1, x_2, ..., x_n are recorded, entering each reading into Register X (or a coordinate pair (y_k, x_k) into Registers Y and X) and pressing [Σ+] (with [↶] [Σ−] available to retract any mistyped reading) updates the SC-32's statistical accumulators. Opening the [↷] [x̄,ȳ] menu ([↷] + [yˣ]) returns the sample mean $\bar{\mathbf{x}} = (1/n)\sum_{i=1}{}^{n} x_i$ on softkey [x̄] (and, when each value x_k in X is paired with a weight w_k in Y, the weighted mean $\bar{\mathbf{x}}_w = \sum w_k\, x_k \,/\, \sum w_k$ on softkey [x̄w]).

Opening the [↷] [s,σ] menu ([↷] + [x⁻¹]) to quantify the dispersion of those n numbers around their mean places two standard deviation softkeys side by side: sample standard deviation [sx] (which divides the sum of squared deviations by n - 1) and population standard deviation [σx] (which divides by n):

$$\sigma_x=\sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n}},\qquad s_x=\sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}}=\sqrt{\frac{\Sigma x^2-\frac{(\Sigma x)^2}{n}}{n-1}}\qquad \textbf{(Eq. 9.2)}$$

Gauss's colleague, the astronomer Friedrich Wilhelm Bessel, demonstrated that dividing by n - 1 rather than n is required when estimating the variance of an underlying population

from a finite sample of size n.

Consider first the limiting case of a single observation: when a single shaft ($n = 1$) is pulled from a lathe and its diameter is measured as $x_1 = 25.02$ mm, the sample average $\bar{\mathbf{x}}$ of that single reading is 25.02 mm, and the deviation of the point from its own mean is identically zero (x_1 - $\bar{\mathbf{x}} = 0$). A single observation provides zero information about process spread, and Bessel's denominator 0 / (1 - 1) = 0/0 reflects this indeterminacy directly (raising **STAT ERROR** if [sx] is pressed at $n = 1$).

For any sample of n points drawn from a population whose *true* center is μ, the same algebraic mechanism applies. Because the true population mean μ is unknown, squared deviations must be measured around the sample's own average $\bar{\mathbf{x}}$. By Gauss's least-squares theorem, the sample mean $\bar{\mathbf{x}}$ is the unique value that minimizes the sum of squared deviations $\sum(x_i - c)^2$. Specifically, for any sample:

$$\sum_{i=1}^{n}(x_i-\bar{x})^2=\sum_{i=1}^{n}(x_i-\mu)^2-n(\bar{x}-\mu)^2\le\sum_{i=1}^{n}(x_i-\mu)^2 \qquad \textbf{(Eq. 9.3)}$$

Because the sample average $\bar{\mathbf{x}}$ lies at the center of mass of the n sampled points, those sample points always cluster closer to their own sample mean $\bar{\mathbf{x}}$ than to the true population center μ by an expected squared amount of σ^2, so the expectation of $\sum(x_i - \bar{\mathbf{x}})^2$ equals (n - 1)σ^2. Equivalently, because the n deviations (x_1 - $\bar{\mathbf{x}}$), ..., (x_n - $\bar{\mathbf{x}}$) must sum to zero:

$$\sum_{i=1}^{n}(x_i-\bar{x})=0 \qquad \textbf{(Eq. 9.4)}$$

only n - 1 of the deviations are independent (n - 1 degrees of freedom); once the first n - 1 deviations are specified, the final deviation is fixed. Dividing by n - 1 (Bessel's Correction) removes that downward bias.

In manufacturing and laboratory metrology, [x̄] and [sx] govern process capability indices, control charts, and the standard error of the mean. When a CNC lathe machines bearing shafts to a tolerance window [LSL, USL], the Six Sigma potential capability C_p = (USL - LSL) / 6 s_x measures how many times the natural $6s_x$ distribution span fits inside the specification limits, while the off-center capability index C_{pk} = min((USL - $\bar{\mathbf{x}}$) / $3s_x$, ($\bar{\mathbf{x}}$ - LSL) / $3s_x$) detects when tool wear shifts the sample mean $\bar{\mathbf{x}}$ toward the upper or lower limit.

In May 1924 at Bell Telephone Laboratories, physicist Walter A. Shewhart established statistical process control by proving that adjusting a machine after every random fluctuation ("tampering") doubles the output variance; plotting subgroup averages against $\pm$ $3\sigma_{\bar{\mathbf{x}}}$ control limits UCL, LCL = $\bar{\mathbf{X}}$ $\pm$ A_2 $\bar{\mathbf{R}}$ separates stationary random noise from assignable physical faults. Likewise, when astronomers and metrologists average $n = 36$ repeated readings, independent positive and negative errors partially cancel and reduce the standard deviation of the sample average—the Standard Error of the Mean—by the square root of the sample size:

$$\mathrm{SE}=\frac{s_x}{\sqrt{n}} \qquad \textbf{(Eq. 9.5)}$$

Averaging $n = 36$ readings reduces random uncertainty to $1/\sqrt{36} = 1/6$th of a single reading's noise.

9.2 Regression to the Mean, Least-Squares Lines & Linearized Curves

Twelve years after building his Quincunx, Francis Galton identified a second statistical phenomenon while analyzing the heights of 930 adult children and their 205 pairs of parents in 1886 (*"Regression towards Mediocrity in Hereditary Stature"*). Galton plotted average parent height on the horizontal x-axis and adult child height on the vertical y-axis, and fitted a straight line through the cloud of points.

Parents who were four inches taller than the population average had tall children, but on average their children were only about *two and a half inches* taller than the mean. Conversely, parents who were four inches shorter than average had children who were taller than their parents and closer to the center of the population. Whenever two variables are imperfectly correlated, extreme observations in x are accompanied on average by less extreme observations in y, closer to the population center ($\bar{x}$, $\bar{y}$). Galton named this phenomenon "regression toward the mean," and fitting a line or curve to noisy data has been termed regression ever since.

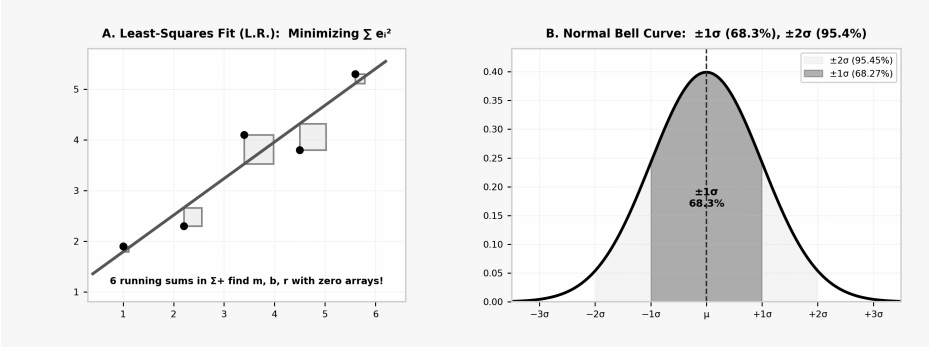


Figure: Gauss's method of least squares minimizing squared vertical residuals and the Gaussian normal probability distribution.

9.2.1 Streaming Regression Across Six Statistical Accumulators

Fitting Gauss's least-squares straight line $\hat{y} = m\,x + b$ through n noisy data pairs (x_1, y_1), (x_2, y_2), ..., (x_n, y_n) also determines Karl Pearson's 1895 Correlation Coefficient $r \in [-1, +1]$, which measures how closely the data cluster along the line: $r = +1$ indicates an exact positive linear relation, $r = -1$ an exact negative linear relation, and $r = 0$ zero linear correlation, while r^2 gives the fraction of variance in y explained by x. A pocket instrument evaluates all of these quantities across hundreds of points without storing an expanding array of coordinate pairs.

Minimizing the sum of squared vertical errors $\sum_{k=1}^{n} (y_k - (m\,x_k + b))^2$ by setting its partial derivatives with respect to m and b to zero compresses the entire dataset into six sufficient statistics *(Gauss, 1809)*:

$$\text{Slope } m = \frac{n\Sigma xy - \Sigma x\,\Sigma y}{n\Sigma x^2 - (\Sigma x)^2}, \qquad \text{Intercept } b = \bar{y} - m\bar{x} = \frac{\Sigma y - m\Sigma x}{n} \tag{Eq. 9.6}$$

$$\text{Pearson Correlation } r = \frac{n\Sigma xy - \Sigma x\,\Sigma y}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2]\,[n\Sigma y^2 - (\Sigma y)^2]}} \tag{Eq. 9.7}$$

In those three formulas, the individual data points (x_k, y_k) do not appear separately. The means, standard deviations, least-squares slope m, intercept b, and correlation r are determined entirely by six running totals, all six of which can be recalled at any moment on the SC-32 through the [→] [SUMS] menu ([→] + [Σ+], with softkeys [Σx], [Σy], [Σx²], [Σy²], [Σxy], and [n]) *(HP, 1982; HP, 1991)*:

$$\Sigma x = \sum x_k, \quad \Sigma y = \sum y_k, \quad \Sigma x^2 = \sum x_k^2, \quad \Sigma y^2 = \sum y_k^2, \quad \Sigma xy = \sum x_k y_k, \quad \text{and } n \qquad \textbf{(Eq. 9.8)}$$

Across 4 points or 4,000 points, updating those six registers on each [Σ+] observation retains the exact least-squares geometry of the dataset. If a mistyped entry or electromagnetic spike corrupts a data pair (x_k, y_k), keying that pair into Y and X and pressing [←] [Σ−] subtracts that single observation from the six running sums ($\sum x \leftarrow \sum x - x_k$, $\sum xy \leftarrow \sum xy - x_k\, y_k$, $n \leftarrow n - 1$) without disturbing the remaining readings.

Opening the [→] [L.R.] (Linear Regression) menu (Right Shift [→] above [LN]) exposes softkeys [ŷ], [x̂], [r], [m], and [b] for two-way calibration around $\hat{y} = mx + b$. Once a sensor is calibrated across reference pressures or temperatures, [ŷ] forecasts $\hat{y} = mx_0 + b$ at a known stimulus x_0, while [x̂] inverts the line $\hat{x} = (y_0 - b)/m$ to recover the unknown physical stimulus from a measured voltage y_0.

Because the least-squares line pivots through $(\bar{x}, \bar{y})$ and carries uncertainty in both height and slope, the pointwise Student-t confidence interval (and Working-Hotelling hyperbolic envelope) for the *mean regression line* $E(y \mid x_0)$ flares outward away from the centroid proportional to $s_e\sqrt{(1/n + (x_0 - \bar{x})^2 / \sum(x_i - \bar{x})^2)}$, with its minimum width at $x_0 = \bar{x}$. By contrast, the prediction band for a *single future measurement* adds $+1$ inside the radical, $\sqrt{(1 + 1/n + \ldots)}$, to account for the variance of an individual observation around the line.

When two variables rise monotonically together along a nonlinear curve—or are measured on an ordinal 1, 2, ..., n ranking scale (such as Mohs scratch hardness versus wear lifetime across ceramic coatings) where linear correlation r is distorted—replacing the raw measurements with their integer ranks and computing the rank differences $d_i = \text{rank}(x_i) - \text{rank}(y_i)$ reduces Pearson's formula to Charles Spearman's 1904 rank correlation:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)} \qquad \textbf{(Eq. 9.9)}$$

9.2.2 Bending Curves into Straight Lines: Moore's Law, Barometric Altitude & Wind Turbines

When the physical law governing a dataset is an exponential curve, a power law, or a saturation hyperbola rather than a straight line, the SC-32's RPN stack linearizes the coordinates directly before accumulation. Applying [LN] or [x^{-1}] to the y or x coordinates on the stack *before* pressing [Σ+] maps the nonlinear curve onto a straight line so the transformed parameters can be read from [m] and [b]:

Physical Model	Governing Equation	Coordinate Transformation	Linearized Straight Line	How Parameters Are Recovered
Linear	$y = m\,x + b$	$(x,\ y)$ unchanged	$y = m\,x + b$	Slope $= m$, Intercept $= b$
Exponential	$y = a\ e^{b\,x}$	$(x,\ \ln\,y)$ [Semi-Log]	$\ln\,y = b\,x + \ln\,a$	Rate $b = m$, Initial $a = e^{\text{Intercept}}$
Power Law	$y = a\ x^{b}$	$(\ln\,x,\ \ln\,y)$ [Log-Log]	$\ln\,y = b\,\ln\,x + \ln\,a$	Exponent $b = m$, Scale $a = e^{\text{Intercept}}$
Logarithmic	$y = a + b\,\ln\,x$	$(\ln\,x,\ y)$	$y = b\,\ln\,x + a$	Scale $b = m$, Offset $a =$ Intercept
Reciprocal	$1/y = b\,x + a$	$(x,\ 1/y)$	$1/y = b\,x + a$	Slope $= b$, Intercept $= a$
Hyperbolic	$y = x\ /\ (a + b\ x)$	$(1/x,\ 1/y)$ [Double-Reciprocal]	$1/y = a(1/x) + b$	Slope $= a$, Ceiling $1/b$

Three coordinate transformations—semi-logarithmic, log-log, and double-reciprocal—linearize common physical laws across science and engineering. Under the semi-logarithmic transformation ($\ln\,y$ versus x), plotting the natural logarithm of the response against an untransformed horizontal variable straightens exponential growth and decay. In 1965, Fairchild Semiconductor co-founder Gordon Moore *(Moore, 1965)* plotted the logarithm of the transistor count $\ln\,N(t)$ of integrated circuits against the calendar year t and observed the linear trend $\ln\,N(t) = \ln\,N_0 + r(t - t_0)$, corresponding to a doubling period $T_{\text{double}} = \ln\,2\ /\ r \approx 2$ years known as Moore's Law.

By Halley and Laplace's barometric formula, atmospheric pressure $P(h) = P_0\ e^{-h/H}$ decays exponentially with altitude h (scale height $H \approx 8.4$ km), so taking [LN] of pressure before pressing [Σ+] converts $\ln\,P(h) = \ln\,P_0 - (1/H)h$ into a straight line of slope [m] $= -1/H$. Similarly, in nuclear medicine, logging $\ln\,A(t)$ of Technetium-99m ($^{99\text{m}}\text{Tc}$) gamma-ray activity across time t converts Ernest Rutherford and Frederick Soddy's decay law $A(t) = A_0\ e^{-\lambda\ t}$ into a straight line of slope $m = -\lambda$, from which the half-life follows as $t_{1/2} = \ln\,2\ /\ \lambda$.

Under the log-log transformation ($\ln\,y$ versus $\ln\,x$), power laws become linear. Doubling the wind speed v multiplies a wind turbine's electrical output eightfold ($2^3 = 8$) because the mass of air crossing the rotor disk per second grows linearly with wind speed ($\dot{m} = \rho\ A\ v$) while the kinetic energy per unit mass grows with the square of wind speed ($(1/2)v^2$), so total wind power scales with the cube of wind speed: $P(v) = (1/2)\dot{m}v^2 = k\ v^3$. Taking [LN] of both power P and wind speed v before pressing [Σ+] transforms $P = k\ v^m$ into $\ln\,P = m\ \ln\,v + \ln\,k$, where [m] extracts the power-law exponent m and [b] [e^x] recovers the aerodynamic scale factor $k = e^b$.

Under the double-reciprocal transformation ($1/y$ versus $1/x$), saturating hyperbolas become linear. When biochemists Leonor Michaelis and Maud Menten (1913) measured the rate at which a catalytic enzyme processes substrate $[S]$, they found a rectangular hyperbola $v = V_{\max}[S]\ /\ (K_m + [S])$ that approaches a ceiling $V_{\max}$ as the active sites saturate. In 1934, Hans Lineweaver and Dean Burk showed that taking the reciprocal ([x^{-1}]) of both reaction velocity v and substrate concentration $[S]$ before [Σ+] linearizes the hyperbola:

$$\frac{1}{v} = \left(\frac{K_m}{V_{\max}}\right)\frac{1}{[S]} + \frac{1}{V_{\max}} \qquad \textbf{(Eq. 9.10)}$$

where the reciprocal of the intercept ([b] [x^{-1}]) yields the maximum turnover rate $V_{\max} = 1/b$ and multiplying by [m] yields the half-saturation constant $K_m = m \cdot V_{\max}$.

9.3 Student's t-Distribution and Small-Sample Inference

Around 1906, at St. James's Gate in Dublin, Arthur Guinness & Son, then the largest brewery in the world, confronted a fundamental question of small-sample inference. Guinness had recruited Oxford-educated chemist William Sealy Gosset as Head Experimental Brewer to apply quantitative quality control to the production of stout.

Gosset needed to determine whether a new variety of Irish barley yielded significantly more malt extract than an existing variety, or whether hop soft-resin content differed between two crop seasons. Classical nineteenth-century statistics, as developed by Carl Friedrich Gauss and Karl Pearson, assumed hundreds or thousands of observations ($n \to \infty$), so that the sample standard deviation s could be substituted for the exact population standard deviation σ without introducing additional uncertainty.

In industrial brewing—as in aerospace metallurgy or clinical trials—waiting years to cultivate and malt a thousand test plots of barley was impractical; Gosset typically worked with small experimental batches of $n = 4$ to 10. When n is small, the sample standard deviation s in the denominator of the standardized score fluctuates randomly from batch to batch alongside the sample mean $\bar{\mathbf{x}}$ in the numerator, which causes extreme ratios to occur more frequently than Gauss's normal curve predicts and produces a distribution with heavier tails.

Gosset derived the exact small-sample probability distribution of the ratio $t = (\bar{\mathbf{x}} - \mu) / (s / \sqrt{n})$. Because Guinness policy prohibited employees from publishing under their own names to protect trade secrets, Gosset obtained permission from the brewery board to publish his 1908 paper, *"The Probable Error of a Mean,"* in Karl Pearson's journal *Biometrika* under the pseudonym *"Student" (Student, 1908).*

To compare the means of two small experimental groups of sizes n_1 and n_2, such as tensile test coupons of two candidate aerospace titanium alloys, Student's Two-Sample t-Test pools the two sample variances weighted by their Bessel degrees of freedom (n_1 - 1 and n_2 - 1, with $df = n_1 + n_2$ - 2 total degrees of freedom):

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}, \qquad t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \tag{Eq. 9.11}$$

When $|t|$ exceeds the critical Student's t-threshold for $df = n_1 + n_2$ - 2, the difference between the sample means is statistically significant relative to the pooled small-sample variance.

9.3.1 Counting Rare Arrivals & Aging Machines: Poisson (1837) and Weibull (1951)

Two additional probability distributions model discrete random arrivals and the fatigue survival of physical hardware. Whenever independent discrete events occur randomly at a steady average rate λ per unit of time or space—photons from a distant nebula striking a detector pixel, cosmic-ray muons registering in a counter, or data packets arriving at an optical network router—the probability of observing k arrivals in a single interval follows the distribution derived in 1837 by French mathematician Siméon Denis Poisson. The Poisson Probability Mass Function (PMF) evaluates on the stack using [y^x], [e^x], and the factorial key [↶] [x!] via Left Shift [↶] above [x^{-1}] or inside the [↷] [PROB] menu:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{Eq. 9.12}$$

In 1898, Russian economist Ladislaus Bortkiewicz demonstrated the empirical accuracy of Poisson's law on rare events by fitting it to twenty years of Prussian cavalry records.

Whereas radioactive decay and Poisson packet arrivals are *memoryless*—a carbon-14 atom that has survived for five thousand years is no more likely to decay in the next minute than a newly formed atom—mechanical ball bearings, jet-engine turbine blades, and stressed steel shafts accumulate microscopic fatigue cracks as operating hours t increase. In 1951, Swedish engineer Waloddi Weibull *(Weibull, 1951)* introduced a survival distribution that encompasses all three regimes of the engineering bathtub reliability curve through a characteristic life scale η (the age at which $e^{-1} \approx 36.79\%$ of units survive) and a dimensionless shape parameter β:

$$R(t) = \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right] \qquad \textbf{(Eq. 9.13)}$$

When $\beta < 1$ (the infant mortality or burn-in regime), defective components fail early and the hazard rate *decreases* over time as weak units are eliminated. When $\beta = 1$ (memoryless random failures), the law reduces to the pure exponential $R(t) = e^{-t/\eta}$, where failures occur from random external shocks at a constant hazard rate $1/\eta$. When $\beta > 1$ (aging and fatigue wear-out), the failure hazard accelerates with age as bearings spall and metals fatigue. Taking the double natural logarithm $\ln[-\ln R(t)] = \beta \ln t - \beta \ln \eta$ via [LN] [+/−] [LN] transforms Weibull's survival curve into a straight line on the regression stack.

9.3.2 The Centroid Fulcrum, Lever-Arm Outliers & Residual Audits

Three geometric properties of a least-squares line govern outlier sensitivity, residual variance, and uncertainty propagation. Rewriting $b = \bar{y} - m\bar{x}$ as $\bar{y} = m\bar{x} + b$ shows that every least-squares regression line passes through the centroid $(\bar{x}, \bar{y})$ of the data cloud. Mechanically, the regression line behaves like a rigid beam pivoted on a fulcrum at $(\bar{x}, \bar{y})$ and linked by vertical springs to each data point (x_i, y_i). Because moment equals lever arm times force, any observation near the horizontal extremes where $|x_i - \bar{x}|$ is large acts with a long lever arm and exerts disproportionate leverage on the slope. A single transcription error at an extreme x_i can perturb the slope [m] and correlation [r] substantially, as illustrated by Francis Anscombe's 1973 Quartet, whereas retracting that point via [↶] [Σ−] restores the unperturbed fit.

Once $\hat{y} = mx + b$ is fitted through n data pairs, the vertical scatter remaining around the line is quantified by the residual standard error $s_e = \sqrt{((1 / (n-2))\sum(y_i - \hat{y}_i)^2)}$, where the denominator $n - 2$ reflects the two degrees of freedom consumed in estimating both slope m and intercept b. By the orthogonality of the least-squares projection, the total variance decomposes into the explained regression fraction r^2 and the unexplained residual fraction $(1 - r^2)$, so s_e can be evaluated directly from the accumulated statistics without re-scanning the n points:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = (n-1)s_y^2(1-r^2) \Rightarrow s_e = s_y\sqrt{\frac{n-1}{n-2}(1-r^2)} \qquad \textbf{(Eq. 9.14)}$$

Recalling [sy], [r], and [n] from the statistical menus determines the exact residual standard error s_e around the regression line.

Finally, when two independent random variables X and Y with standard uncertainties σ_X and σ_Y are combined linearly into $Z = aX \pm bY$, their variances add in quadrature ($\sigma_Z = \sqrt{(a^2\sigma_X^2 + b^2\sigma_Y^2)}$), which corresponds directly to the Pythagorean norm ([↶] [▸θ,r] on the stack). For a general differentiable function $Z = g(X, Y)$, Gauss's Law of Error Propagation combines the partial derivatives with the covariance $\mathrm{Cov}(X, Y) = r\, s_x\, s_y$:

$$\sigma_Z^2 \approx \left(\frac{\partial g}{\partial X}\right)^2 \sigma_X^2 + \left(\frac{\partial g}{\partial Y}\right)^2 \sigma_Y^2 + 2\left(\frac{\partial g}{\partial X}\right)\left(\frac{\partial g}{\partial Y}\right) r\,\sigma_X \sigma_Y \qquad \textbf{(Eq. 9.15)}$$

9.4 Statistical Accumulators, Probability Distributions & Regression

Because the six first- and second-order moments ($\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n) form a complete set of sufficient statistics for Gaussian means, Bessel variances, and linear least squares, the SC-32 connects the live four-register stack (**X, Y, Z, T**) directly to six 64-bit hardware accumulators (indirect registers **R28** through **R33**).

When logging a bivariate observation (x_k, y_k) after clearing prior runs with [CLΣ], keying **y_k** [ENTER] **x_k** [Σ+] lifts the vertical response y_k into Register Y and leaves the horizontal predictor x_k in Register X, which aligns the display registers (Y, X) with the (y, x) axes of the coordinate plane. For univariate samples x_k, keying **x_k** [Σ+] accumulates along X; for weighted averages, placing the weight w_k in Register Y and the reading x_k in Register X accumulates $\sum w_k\, x_k$ in $\sum xy$ and $\sum w_k$ in $\sum y$.

Because (y_k, x_k) sit live in the lower two stack registers prior to accumulation, applying [LN] or [x^{-1}] before pressing [Σ+] linearizes exponential, power-law, and Michaelis–Menten hyperbolic models in place before the four Right-Shift statistical menus query the six sums.



Figure: Standard Normal PDF f(1.0) = exp(−0.5)/√(2π) = 0.241970724519 on the SC-32.

9.4.1 The 6 Statistical Accumulators, Outlier Retraction & Softkey Menus

On the SC-32, pressing [Σ+] commits the coordinate pair ($Y = y_k$, $X = x_k$) into six non-volatile hardware registers while displaying the updated sample count n in Register X (and preserving **LASTX**):

Accumulator	Indirect Register	Mathematical Meaning	Primary Statistical Role
$\sum x$	**R29** (**LFU0**)	$\sum x_i$	Sample mean $\bar{x}$, regression slope m
$\sum y$	**R30** (**LFU1**)	$\sum y_i$	Sample mean $\bar{y}$, regression intercept b, total weight in $\bar{x}_w$
$\sum x^2$	**R31** (**LFU2**)	$\sum x_i^2$	x-variance and standard deviations (s_x, σ_x)
$\sum y^2$	**R32** (**LFU3**)	$\sum y_i^2$	y-variance and standard deviations (s_y, σ_y)
$\sum xy$	**R33** (**LFU4**)	$\sum (x_i \cdot y_i)$	Bivariate covariance, Pearson r, and weighted sum $\sum w_i\, x_i$
n	**R28** (**LFU5**)	Sample count	Normalization and Bessel degrees of freedom (n - 1)

Accessed via Right Shift [→] across the top scientific row, the four statistical menus map onto the six LFU dynamic softkeys ([LFU0] through [LFU5]):

Sub-Menu	Primary Invocation & Key Position	LFU0	LFU1	LFU2	LFU3	LFU4	LFU5
1. Sums Menu	[→] [SUMS] (above [Σ+])	Σx	Σy	Σx²	Σy²	Σxy	n
2. Means Menu	[→] [x̄,ȳ] (above [y^x])	x̄	ȳ	x̄w	—	—	—
3. Std Dev Menu	[→] [s,σ] (above [x^-1])	sx	sy	σx	σy	—	—
4. Regression Menu	[→] [L.R.] (above [LN])	ŷ	x̂	r	m	b	—

If a keystroke slip on the sixth data pair ($x = 15$, $y = 3.5$) transposes y into ($x = 15$, $y = 5.3$) when the sample count is $n = 6$, [←] [Σ−] excises the erroneous point and splices in the corrected reading without disturbing the first five points:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
5.3 [ENTER] 15	0.0000	0.0000	5.3000	15.0000	Key erroneous pair (x=15, y=5.3)
[←] [Σ−]	0.0000	0.0000	5.3000	**5.0000**	Subtract from accumulators; n decrements to 5
3.5 [ENTER] 15	0.0000	5.3000	3.5000	15.0000	Key corrected pair (x=15, y=3.5)
[Σ+]	0.0000	5.3000	3.5000	**6.0000**	Accumulate corrected pair; n restored to 6

9.4.2 Weighted Sensor Fusion, Six Sigma Capability (C_p, C_{pk}), Control Limits & Standard Error

Applying these quality-control and metrological indices ($\bar{\mathrm{x}}_w$, C_p, C_{pk}, Shewhart limits, and SE $= s/\sqrt{n}$) to shop-floor and observatory measurements shows how the six accumulators feed subsequent stack arithmetic:

For inverse-variance weighted sensor fusion ($\bar{\mathrm{x}}_w = \sum xy \,/\, \sum y$), when three observatory instruments measure an astrometric parallax angle—$x_1 = 10$ mas, $x_2 = 25$ mas, and $x_3 = 45$ mas—with inverse-variance precision weights $w_i = 1/\sigma_i^2$ equal to $w_1 = 40$, $w_2 = 60$, and $w_3 = 20$ mas^{-2}, loading each weight w_i into Register Y and measurement x_i into Register X before pressing [Σ+] accumulates $\sum(w_i\ x_i)$ inside **R33** ($\sum xy$) and $\sum w_i$ inside **R30** ($\sum y$). Opening [→] [x̄,ȳ] and tapping [x̄w] ([LFU2]) computes the minimum-variance fused estimate $\bar{\mathrm{x}}_w = \sum xy \,/\, \sum y = 2800/120 = 23.3333$ mas, while [→] [SUMS] → [Σy] ([LFU1]) recalls the combined precision weight 120.0 mas^{-2}:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.0000	0.0000	0.0000	0.0000	Clear all 6 statistical accumulators
40 [ENTER] 10 [Σ+]	0.0000	0.0000	40.0000	1.0000	Sensor 1 logged (w_1=40, x_1=10; n=1)
60 [ENTER] 25 [Σ+]	0.0000	40.0000	60.0000	2.0000	Sensor 2 logged (w_2=60, x_2=25; n=2)
20 [ENTER] 45 [Σ+]	40.0000	60.0000	20.0000	3.0000	Sensor 3 logged (w_3=20, x_3=45; n=3)
[x̄w]	40.0000	60.0000	20.0000	**23.3333**	Via [→] [x̄,ȳ] → [x̄w]: inverse-variance weighted mean $\bar{\mathrm{x}}_w$ (mas)
[Σy]	60.0000	20.0000	23.3333	**120.0000**	Via [→] [SUMS] → [Σy]: total precision weight $\sum y = \sum w_i$ (mas^{-2})

To evaluate the Six Sigma capability indices C_p and C_{pk} across $n = 10$ CNC-machined bearing shafts specified at 25.00 ± 0.10 mm (LSL = 24.90 mm, USL = 25.10 mm), keying each diameter into Register X and pressing [Σ+] yields sample mean $\bar{\mathrm{x}} =$ **25.0300 mm** and

Bessel sample spread s_x = **0.018257 mm**. Dividing the 0.20 mm tolerance width by $6s_x$ via **.20** ENTER **6** ÷ **.018257** ÷ yields potential capability C_p = **1.826**, while comparing $\bar{x}$ against USL = 25.10 mm via **25.10** ENTER **25.030** - **3** ÷ **.018257** ÷ yields C_{pk} = **1.278**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
25.02 Σ+ **25.04** Σ+ **25.01** Σ+ **25.03** Σ+ **25.05** Σ+ **25.02** Σ+ **25.00** Σ+ **25.04** Σ+ **25.03** Σ+ **25.06** Σ+	0.0000	0.0000	0.0000	10.0000	Accumulate 10 shaft diameter samples
R↓ x̄,ȳ x̂	0.0000	0.0000	0.0000	**25.0300**	Sample mean diameter $\bar{x}$ (mm)
R↓ s,σ sx	0.0000	0.0000	25.0300	**0.018257**	Sample standard deviation s_x (mm)
.20 ENTER **6** ÷ **.018257** ÷	25.0300	25.0300	0.0183	1.826	Process capability C_p = (USL-LSL) / $6s_x$
25.10 ENTER **25.030** - **3** ÷ **.018257** ÷	0.0183	0.0183	1.8258	1.278	Process capability index C_{pk} = (USL-$\bar{x}$) / $3s_x$

The pair (C_p = 1.8257, C_{pk} = 1.2780) provides a direct shop-floor diagnostic: because $C_p >$ 1.33, the lathe spindle itself is mechanically tight enough for Six Sigma production, but because C_{pk} = 1.2780 < 1.33, thermal tool growth has shifted the process center +0.030 mm toward the upper limit, which indicates a -0.030 mm tool-offset correction rather than a spindle rebuild.

- **Shift-Invariance Speed & Numerical Hygiene ($s_{x\,-\,c} = s_x$):** Translating an entire dataset by a constant datum c shifts the mean (x - $\bar{c}$ = $\bar{x}$ - c) without changing the distances between points ($s_{x\,-\,c} = s_x$). You can therefore subtract the 25.00 mm nominal diameter mentally and accumulate only the hundredths of a millimeter with Σ+: **2** Σ+ **4** Σ+ **1** Σ+ **3** Σ+ **5** Σ+ **2** Σ+ **0** Σ+ **4** Σ+ **3** Σ+ **6** Σ+. That shortcut cuts keystrokes in half—returning $\bar{d}$ = 3.0000 ⇒ $\bar{x}$ = 25.0300 mm and s_d = 1.8257 ⇒ s_x = 0.018257 mm—while shielding the one-pass variance sum $\sum x^2$ - $(\sum x)^2/n$ from catastrophic cancellation whenever raw readings cluster tightly around a large baseline offset.

Similarly, evaluating Walter Shewhart's ± $3\sigma_{\bar{x}}$ control limits UCL, LCL = $\bar{\bar{X}}$ ± $A_2\bar{R}$ around a centerline of $\bar{\bar{X}}$ = 50.00 mm with subgroup range $\bar{R}$ = 0.40 mm and n = 5 subgroup factor A_2 = 0.577 uses **LASTX** to compute both guardrails from a single centerline entry:

$$\mathrm{UCL} = \bar{\bar{X}} + A_2\bar{R}, \qquad \mathrm{LCL} = \bar{\bar{X}} - A_2\bar{R} \tag{Eq. 9.16}$$

The sequence **.40** ENTER **.577** × forms the control half-width $A_2\bar{R}$ = 0.2308 mm, and adding **50.00** + places the Upper Control Limit at UCL = **50.2308 mm**.

Subtracting that same half-width from the centerline via **.40** ENTER **.577** × **50.00** x⇄y - sets the Lower Control Limit at LCL = **49.7692 mm**.

Once $A_2\bar{R}$ = 0.2308 sits in Register X, keying **50.00** x⇄y - computes LCL = **49.7692 mm** while automatically capturing 0.2308 in **LASTX**; following at once with ↶ LASTx **2** × + adds 2($A_2\bar{R}$) to step straight across to UCL = **50.2308 mm** while typing **50.00** only once.

Applying the $\sqrt{n}$ averaging law (SE = $s/\sqrt{n}$) across $n = 36$ aluminum-alloy tensile yield specimens ($\bar{x} = 120.5$ MPa, $s = 8.4$ MPa) follows the same pattern. Dividing $s = 8.4$ MPa by $\sqrt{36} = 6$ with **8.4** [ENTER] **36** [√x] [÷] reduces the Standard Error of the Mean to SE = **1.4 MPa**, and scaling by the two-sided 95% normal multiplier **1.960** [×] gives the half-width margin $E =$ **2.7440 MPa**.

Adding E to $\bar{x}$ via **120.5** [ENTER] **2.744** [+] establishes the upper 95% confidence bound $\bar{x}$ + $E =$ **123.244 MPa**, while **120.5** [ENTER] **2.744** [-] gives the lower bound $\bar{x}$ - $E =$ **117.756 MPa**.

Alternatively, once $E = 2.7440$ sits in Register X, key **120.5** [x⇄y] [+] for the upper bound and [↶] [LASTx] **2** [×] [-] to recover E from [↶] [LASTx] and step straight to the lower bound.

9.4.3 Least-Squares Sensor Calibration, Confidence Bands & Spearman Rank Correlation

Streaming the four dead-weight calibration points of our piezoresistive pressure sensor—10 kPa at 1.25 V, 20 kPa at 2.45 V, 30 kPa at 3.80 V, and 40 kPa at 4.95 V—into the accumulators after [CLΣ] builds the regression model on the stack:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.00000000	0.00000000	0.00000000	0.00000000	Clear all 6 statistical accumulators ($\sum x$, $\sum y$, ... = 0)
1.25	0.00000000	0.00000000	0.00000000	1.25000000	Enter sensor output voltage $y_1 = 1.25$ V
[ENTER]	0.00000000	0.00000000	1.25000000	1.25000000	Stage $y_1 = 1.25$ V in Register Y
10	0.00000000	0.00000000	1.25000000	10.0000000	Enter reference pressure $x_1 = 10$ kPa in Register X
[Σ+]	0.00000000	0.00000000	1.25000000	1.00000000	Accumulate Point 1 (10, 1.25); display sample count $n = 1$
2.45 [ENTER]	0.00000000	1.25000000	2.45000000	2.45000000	Stage voltage $y_2 = 2.45$ V in Register Y
20 [Σ+]	0.00000000	1.25000000	2.45000000	2.00000000	Accumulate Point 2 (20, 2.45); sample count $n = 2$
3.80 [ENTER]	1.25000000	2.45000000	3.80000000	3.80000000	Stage voltage $y_3 = 3.80$ V in Register Y
30 [Σ+]	1.25000000	2.45000000	3.80000000	3.00000000	Accumulate Point 3 (30, 3.80); sample count $n = 3$
4.95 [ENTER]	2.45000000	3.80000000	4.95000000	4.95000000	Stage voltage $y_4 = 4.95$ V in Register Y
40 [Σ+]	2.45000000	3.80000000	4.95000000	4.00000000	Accumulate Point 4 (40, 4.95); sample count $n = 4$

Now inspect the accumulated state and calibration predictions across the softkey menus:

- **Raw Accumulator Audit ([→] [SUMS]):** [Σx] = 100.0, [Σy] = 12.45, [Σx²] = 3000.0, [Σy²] = 46.5075, [Σxy] = 373.50, and [n] = 4.
- **Regression Fit ([→] [L.R.]):** [m] = **0.1245000000 V/kPa**, [b] = **0.0000000000 V**, and [r] = **0.9995648074** (so $r^2 = 0.999130$, explaining 99.913% of variance).

- **Forward & Inverse Calibration (ŷ and x̂) & Two-Point Table Interpolation:** Keying **25** [R/S] [L.R.] [ŷ] forecasts **3.1125000000 V** while placing $r = 0.99956481$ in Register Y, and keying **4.00** [R/S] [L.R.] [x̂] inverts the calibration curve to recover **32.128514 kPa**. *(Corvallis Two-Point Table Interpolator: Whenever you need to interpolate between two adjacent rows (x_1, y_1) and (x_2, y_2) of a steam or atmosphere lookup table, log both points after* [CLΣ] *as* **y1 ENTER x1 Σ+ y2 ENTER x2 Σ+**. *Because two points determine an exact secant line with $r = \pm 1.000000$, querying* **ŷ** *at x_0 performs forward linear interpolation and querying* **x̂** *at y_0 performs inverse linear interpolation without manual difference quotients—and applying* [LN] *before* **Σ+** *performs log-linear interpolation across exponential tables.)*

- **Confidence vs. Prediction Margins at the Centroid ($x_0 = \bar{x} = 25$ kPa):** With residual standard error $s_e = 0.058095$ V and two-sided Student-t multiplier $t_{0.025,\ 2} = 4.303$, the $(x_0 - \bar{x})^2$ tilt term of the Working-Hotelling hyperbolic envelope vanishes at the centroid waist $x_0 = \bar{x}$. Evaluating $E = t_{0.025,2} \cdot s_e / \sqrt{4}$ via **4.303** [ENTER] **0.058095** [×] **4** [x^{-1}] [√x] [×] gives the mean-line confidence margin $E =$ **0.1250 V** (3.1125 ± 0.1250 V), whereas scaling by $\sqrt{(1 + 1/4)}$ via **4** [x^{-1}] **1** [+] [√x] **0.058095** [×] **4.303** [×] gives the single-observation prediction margin $E_{\text{pred}} =$ **0.2795 V** (3.1125 ± 0.2795 V).

In analytical and environmental chemistry, inverse regression calibration ([R/S] [L.R.] → [x̂]) is the standard tool for *Beer–Lambert UV-Vis spectrophotometry* ($A = -\log_{10}(I/I_0) = \varepsilon\ b\ C + A_{\text{blank}}$). When a water-quality chemist calibrates a $b = 1.00$ cm cuvette at $\lambda = 510$ nm across four iron(II)–phenanthroline standards—$C_1 = 1.00$ mg/L ($A_1 = 0.198$), $C_2 = 2.00$ mg/L ($A_2 = 0.395$), $C_3 = 3.00$ mg/L ($A_3 = 0.594$), and $C_4 = 4.00$ mg/L ($A_4 = 0.791$)—logging each (A_i, C_i) pair with [Σ+] and querying [r] and [m] confirms Beer's law linearity ($r =$ **0.999997**, absorptivity slope $m =$ **0.197700 L/mg**, blank offset $b =$ **0.000250**). Keying an unknown groundwater sample's absorbance $A_{\text{unk}} = 0.465$ and pressing [x̂] ([R/S] [L.R.] → [LFU1]) inverts the calibration line in place to yield $\hat{C} =$ **2.350784 mg/L**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.000000	0.000000	0.000000	0.000000	Clear all 6 statistical accumulators
0.198 [ENTER] **1** [Σ+]	0.000000	0.000000	0.198000	1.000000	Standard 1 (C_1=1.0 mg/L, A_1=0.198; n=1)
0.395 [ENTER] **2** [Σ+]	0.000000	0.198000	0.395000	2.000000	Standard 2 (C_2=2.0 mg/L, A_2=0.395; n=2)
0.594 [ENTER] **3** [Σ+]	0.198000	0.395000	0.594000	3.000000	Standard 3 (C_3=3.0 mg/L, A_3=0.594; n=3)
0.791 [ENTER] **4** [Σ+]	0.395000	0.594000	0.791000	4.000000	Standard 4 (C_4=4.0 mg/L, A_4=0.791; n=4)
[r]	0.395000	0.594000	0.791000	0.999997	Via [R/S] [L.R.] → [r]: Beer-Lambert correlation
[m]	0.594000	0.791000	0.999997	0.197700	Via [R/S] [L.R.] → [m]: slope $\varepsilon\ b$ (L/mg)
0.465 [x̂]	0.791000	0.999997	0.197700	**2.350784**	Via [R/S] [L.R.] → [x̂]: unknown concentration (mg/L)

Evaluating Charles Spearman's rank correlation $r_s = 1 - 6\sum d_i^2 / n(n^2-1)$ across five ceramic coatings ranked on Mohs scratch hardness versus wear lifetime with rank discrepancies $d = (1, -1, 2, -1, -1)$ runs on Registers X and Y while Registers Z and T remain at zero. Here the squared rank differences sum to $\sum d_i^2 = 1 + 1 + 4 + 1 + 1 = 8.0$—one-fifth of the antipodal reversal sum $n(n^2 - 1) / 3 = 40.0$ ($r_s = -1.0$) and well below

the uncorrelated expectation $n(n^2 - 1) / 6 = 20.0$ ($r_s = 0.0$). On the stack, form the numerator $6\sum d_i^2 = 48.0000$ in Register Y and the denominator $n(n^2 - 1) = 5(24) = 120.0000$ in Register X, divide to isolate the discordance penalty $48 / 120 = 0.4000$, and subtract from unity via **1** [↶] [x²] **1** [↶] [x²] [+] **2** [↶] [x²] [+] **1** [↶] [x²] [+] **1** [↶] [x²] [+] **6** [×] **5** [ENTER] **5** [↶] [x²] **1** [-] [×] [÷] **1** [x⇄y] [-] → **0.6000** ($r_s = +0.6000$).

9.4.4 Gaussian Normal Density, Cumulative Probability, Student's t-Test, Poisson & Weibull

Continuous and discrete probability laws—Gauss's normal distribution, Student's two-sample *t*-statistic, Poisson's arrival law, and Weibull's reliability model—evaluate directly on the stack:

Evaluating the Gaussian Normal PDF $f(x; \mu, \sigma)$ at $x = 115$ ($\mu = 100$, $\sigma = 15$), $f(115; 100, 15) = (1 / 15\sqrt{2\pi}) \exp(-(115\text{-}100)^2 / 2(15)^2)$, one standard deviation above the mean ($z = +1.0$) unfolds in two stack stages. First, square the deviation $(115 - 100)^2 = 225.0$ via **115** [ENTER] **100** [-] [↶] [x²], form the variance denominator $2\sigma^2 = 450.0$ in Register X with **15** [↶] [x²] **2** [×], and evaluate [÷] [+/-] [e^x] to leave the bell factor $e^{-0.5} = 0.60653066$ on the stack. Second, construct the normalization constant $\sigma\sqrt{2\pi} = 37.59942412$ via **15** [ENTER] **2** [↗] [π] [×] [√x] [×] while $e^{-0.5}$ rides in Register Y, and press [÷] to obtain **0.0161313817**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
115 [ENTER]	0.00000000	0.00000000	115.000000	115.000000	Stage observation $x = 115$ in Register Y
100 [-]	0.00000000	0.00000000	0.00000000	15.0000000	Subtract mean $\mu = 100$: deviation $x - \mu = 15.0$
[↶] [x²]	0.00000000	0.00000000	0.00000000	225.000000	Squared deviation $(x - \mu)^2 = 225.0$
15 [↶] [x²]	0.00000000	0.00000000	225.000000	225.000000	Variance $\sigma^2 = 15^2 = 225.0$ (lifts $(x\text{-}\mu)^2$ to Y)
2 [×]	0.00000000	0.00000000	225.000000	450.000000	Denominator $2\sigma^2 = 450.0$
[÷]	0.00000000	0.00000000	0.00000000	0.50000000	Normalized half-square $z^2/2 = 225/450 = 0.50$
[+/-] [e^x]	0.00000000	0.00000000	0.00000000	0.60653066	Gaussian exponential factor $\exp(-0.5) \approx 0.606531$
15 [ENTER]	0.00000000	0.60653066	15.0000000	15.0000000	Stage standard deviation $\sigma = 15$ (lifts $e^{-0.5}$ to Z)
2 [↗] [π]	0.60653066	15.0000000	2.00000000	3.14159265	Load 2 and π onto Registers Y and X
[×] [√x]	0.60653066	0.60653066	15.0000000	2.50662827	Evaluate $\sqrt{2\pi} \approx 2.506628$
[×]	0.60653066	0.60653066	0.60653066	37.5994241	Normalization constant $\sigma\sqrt{2\pi} \approx 37.599424$
[÷]	0.60653066	0.60653066	0.60653066	0.01613138	Gaussian density $f(115; 100, 15) =$ **0.0161313817**

*(Multiplying $f(115; 100, 15) = 0.0161313817$ by $\sigma = 15$ recovers the standard normal height $\varphi(1.0) = (1 / \sqrt{2\pi})e^{-0.5} =$ **0.2419707245** shown on the transflective LCD screenshot.)*

To evaluate the cumulative normal integral $\Phi(1.0) = 0.5 + (1 / \sqrt{2\pi})\int_0^{1.0} e^{-t^2/2}\,dt$ via Romberg quadrature ([↗] [∫]), recall that the left half of the symmetric bell curve from $-\infty$ to 0 encloses an area of 0.50000000. Finding $\Phi(z)$ for any $z > 0$ therefore requires only integrating the finite half-bell $\int_0^z e^{-t^2/2}\,dt$ with **X** [↶] [x²] **2** [÷] [+/-] [e^x] active in [↗] [EQN],

dividing by √2π, and adding 0.5:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
0 [ENTER] **1.0**	0.00000000	0.00000000	0.00000000	1.00000000	Push integration limits a=0, b=1.0
[R↓] [∫]	0.00000000	0.00000000	1.15×10^{-12}	0.85562439	Select variable **X**: Romberg quadrature $\int_0^1 e^{-t^2/2}\,dt$
2 [R↓] [π] [×] [√x] [÷]	0.00000000	0.00000000	0.00000000	0.34134475	Divide by √2π; area from mean to Z=1.0
0.5 [+]	0.00000000	0.00000000	0.00000000	**0.84134475**	Cumulative probability Φ(1.0) (84.13%)

Doubling the central slice 2 × 0.34134475 = **0.68268949** confirms the ± 1σ Empirical Rule (68.27%).

Applying Gosset's Student Two-Sample t-Test ($df = n_1 + n_2 - 2 = 16$) to Titanium Alloy A ($n_1 = 8$ coupons, $\bar{x}_1 = 450.0$ MPa, $s_1 = 12.5$ MPa) against Alloy B ($n_2 = 10$ coupons, $\bar{x}_2 = 468.0$ MPa, $s_2 = 14.0$ MPa), the pooled variance $s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2) / (n_1 + n_2 - 2)$ and t-statistic evaluate across Registers X and Y while Z and T remain at zero.

Adding Alloy A's sum of squares $(n_1 - 1)s_1^2 = 7(12.5^2) = 1{,}093.7500$ MPa² to Alloy B's $(n_2 - 1)s_2^2 = 9(14^2) = 1{,}764.0000$ MPa² forms 2,857.7500 MPa², and dividing by $df = 16$ before taking [√x] extracts the pooled standard deviation s_p = **13.364482 MPa** (weighted 9:7 toward the larger 10-coupon batch). Scaling s_p by $\sqrt{(1/8 + 1/10)}$ via [1/x] and **√x** gives the standard error SE = **6.339330 MPa**, and forming $\bar{x}_1 - \bar{x}_2$ = -18.0000 MPa before [x⇄y] [÷] yields Student's t = **-2.839422**—surpassing the two-sided 95% threshold $t_{0.025,\,16} = 2.120$.

Evaluating Poisson's rare-event PMF $P(X = k) = \lambda^k\, e^{-\lambda} / k!$ for $k = 6$ arrivals when an optical router receives $\lambda = 4.5$ packets/ms on average forms the arrival-rate power $\lambda^k = 4.5^6 = 8{,}303.765625$ in Register Y and the exponential survival factor $e^{-4.5} \approx 0.01110900$ in Register X, multiplies to collapse the numerator $\lambda^k\, e^{-\lambda} = 92.246538$, and divides by $k! = 6! = 720$ without touching Registers Z or T via **4.5** [ENTER] **6** [y^x] **4.5** [+/-] [e^x] [×] **6** [↶] [x!] [÷] → **0.128120** (12.81%).

Similarly, Waloddi Weibull's survival law $R(t) = \exp[-(t/\eta)^\beta]$ for a spindle bearing at t = 3,000 hr (η = 5,000 hr, wear-out exponent $\beta = 1.8 > 1$) normalizes service time ($t/\eta = 0.6000$), raises that ratio to $(0.6000)^{1.8} = 0.398660$, and evaluates exp(-0.398660) in place on Register X via **3000** [ENTER] **5000** [÷] **1.8** [y^x] [+/-] [e^x] → **0.6712** (67.12% survival, or 32.88% cumulative failure risk).

9.4.5 Linearized Curve Fitting: Log-Log Power Laws & Double-Reciprocal Kinetics

Applying the log-log transformation ($\ln P = m \ln v + \ln k$) to fit the wind turbine cubic power law $P = kv^m$ across four wind-tunnel test points (4, 16), (6, 54), (8, 128), (10, 250) transforms both output power P and wind speed v via [LN] right on the stack before tapping [Σ+]. Because unary keys like [LN] and [1/x] terminate digit entry and arm automatic stack lift, **16** [LN] **4** [LN] [Σ+] needs no [ENTER] between **16** [LN] and **4**—typing **4** lifts ln(16) into Register Y automatically, saving an [ENTER] tap on every data pair. Querying [R↓] [L.R.] [m] confirms the cubic exponent m = **3.000000**, [b] [e^x] recovers $k = e^b$ = **0.250000**, and forecasting output at 12 m/s with **12** [ENTER] **3** [y^x] **.25** [×] delivers **432** kW:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.00000000	0.00000000	0.00000000	0.00000000	Clear all 6 statistical accumulators
16 [LN] 4 [LN] [Σ+]	0.00000000	0.00000000	2.77258872	1.00000000	Point 1 logged (n=1 in X, ln P_1 in Y)
54 [LN] 6 [LN] [Σ+]	0.00000000	2.77258872	3.98898405	2.00000000	Point 2 logged (n=2 in X, ln P_2 in Y)
128 [LN] 8 [LN] [Σ+]	2.77258872	3.98898405	4.85203026	3.00000000	Point 3 logged (n=3 in X, ln P_3 in Y)
250 [LN] 10 [LN] [Σ+]	3.98898405	4.85203026	5.52146092	4.00000000	Point 4 logged (n=4 in X, ln P_4 in Y)
[↗] [L.R.] [m]	3.98898405	4.85203026	5.52146092	**3.00000000**	Regression slope $m = 3.0$ (cubic power law)
[b] [e^x]	4.85203026	5.52146092	3.00000000	**0.25000000**	Via [↗] [L.R.] → [b] then [e^x]: scale $k = e^b = 0.250$ kW/(m/s)^3
12 [ENTER] 3 [y^x] 0.25 [×]	3.00000000	3.00000000	0.25000000	**432.000000**	Predicted power $P(12) = 0.25 \times 12^3$ kW

Applying the double-reciprocal Lineweaver-Burk transformation ($1/v = (K_m/V_{\max})1/[S] + 1/V_{\max}$) to a Michaelis-Menten enzyme assay similarly taps [x^{-1}] on each reaction velocity v_k and substrate concentration $[S]_k$ before pressing [Σ+]. Streaming the four double-reciprocal pairs via [CLΣ] .182 [x^{-1}] 2.0 [x^{-1}] [Σ+] .286 [x^{-1}] 4.0 [x^{-1}] [Σ+] .400 [x^{-1}] 8.0 [x^{-1}] [Σ+] .471 [x^{-1}] 16.0 [x^{-1}] [Σ+] [b] [x^{-1}] [m] [×] determines the maximum turnover ceiling $V_{\max} = 1/b$ = **0.6346 μM/min** and the half-saturation Michaelis constant $K_m = m \cdot V_{\max}$ = **4.9484** mM:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[CLΣ]	0.0000	0.0000	0.0000	0.0000	Clear all 6 statistical accumulators
.182 [x^{-1}] 2.0 [x^{-1}] [Σ+]	0.0000	0.0000	5.4945	1.0000	Point 1 logged ($1/v_1$ in Y, $1/[S]_1$ in X; n=1)
.286 [x^{-1}] 4.0 [x^{-1}] [Σ+]	0.0000	5.4945	3.4965	2.0000	Point 2 logged ($1/v_2$ in Y, $1/[S]_2$ in X; n=2)
.400 [x^{-1}] 8.0 [x^{-1}] [Σ+]	5.4945	3.4965	2.5000	3.0000	Point 3 logged ($1/v_3$ in Y, $1/[S]_3$ in X; n=3)
.471 [x^{-1}] 16.0 [x^{-1}] [Σ+]	3.4965	2.5000	2.1231	4.0000	Point 4 logged ($1/v_4$ in Y, $1/[S]_4$ in X; n=4)
[b]	3.4965	2.5000	2.1231	**1.5759**	Via [↗] [L.R.] → [b]: double-reciprocal intercept $1/V_{\max}$
[x^{-1}]	3.4965	2.5000	2.1231	**0.6346**	Invert intercept for maximum velocity $V_{\max}$ (μM/min)
[m]	2.5000	2.1231	0.6346	**7.7981**	Via [↗] [L.R.] → [m]: slope K_m / $V_{\max}$ (lifts $V_{\max}$ into Y)
[×]	2.5000	2.5000	2.1231	**4.9484**	Multiply for Michaelis constant $K_m = m \times V_{\max}$ (mM)

Finally, the semi-log transformation (ln $A(t)$ = ln A_0 - λ t) applies [LN] solely to the vertical y-coordinate before pressing [Σ+]. For four Technetium-99m (^{99m}Tc) gamma-ray calibration readings—1000 MBq at t = 0 hr, 794 MBq at 2 hr, 630 MBq at 4 hr, and 500 MBq at 6 hr—clear with [CLΣ], then stream **1000 LN 0 Σ+**, **794 LN 2 Σ+**, **630 LN 4 Σ+**, and **500 LN 6 Σ+**; tapping [↗] [L.R.] [m] [+/−] reads the decay constant λ = **0.11552 hr^{-1}**, dividing ln 2 by λ yields the clinical half-life $t_{1/2}$ = ln 2 / λ = **6.000 hours**, and [b] [e^x] recovers A_0 =

1000.1 MBq (the same semi-log procedure weather-balloon radiosondes use on ln P_k vs. altitude h_k to calibrate sea-level pressure $P_0 = e^{b}$ and scale height $H = -1/m$).

In physical chemistry kinetics, pairing [LN] on the vertical rate constant k_i with [x^{-1}] on the horizontal absolute temperature T_i (**k_i** [LN] **T_i** [x^{-1}] [Σ+]) linearizes Svante Arrhenius's activation law $k = A \exp(-E_a / RT)$ into $\ln k = -(E_a/R)(1/T) + \ln A$ across multiple temperature runs without a single [ENTER] tap. Querying [↻] [L.R.] [m] [+/−] and multiplying by the molar gas constant $R = 8.31446$ J/(mol·K) extracts the least-squares activation barrier $E_a = -m\,R$ in J/mol, while [b] [e^x] recovers the pre-exponential collision frequency factor $A = e^{b}$ (and the identical ln P_i vs. $1/T_i$ pairing extracts the molar enthalpy of vaporization $\Delta H_{\text{vap}} = -m\,R$ from multi-point Clausius–Clapeyron vapor-pressure data).

For fatigue life testing across failure times t_k with empirical survival fractions R_k, applying the double-logarithm transform R_k (**LN +/- LN**) into Register Y and t_k (**LN**) into Register X before [Σ+] fits Waloddi Weibull's line $\ln[-\ln R(t)] = \beta \ln t - \beta \ln \eta$, where [m] reads the shape modulus β and $\eta = \exp(-b/\beta)$ recovers the characteristic life. Whenever you query Bessel's sample standard deviation [sx] or [sy] with fewer than two observations ($n \leq 1$), query [x̄] with $n = 0$, or query regression slope or correlation when all x-values are identical ($n\sum x^2 - (\sum x)^2 = 0$, a vertical line), the SC-32 raises **STAT ERROR** (Code 04), which clears non-destructively via [C/ON] without disturbing the six statistical registers.

Chapter 10: Boolean Logic, Two's Complement & the 36-Bit Word

10.1 Leibniz's Medallion, Boole's Logic & Claude Shannon's 1937 Master's Thesis

10.1.1 Leibniz's Binary Arithmetic and Shannon's Switching Relays

In the summer of 1937, MIT master's student Claude Elwood Shannon worked in the basement housing Vannevar Bush's Differential Analyzer alongside brass gears and electromechanical telephone relays. That hundred-ton mechanical analog computer used spinning steel shafts, torque amplifiers, and wheel-and-disc integrators to solve differential equations through physical rotation.

Every time a physicist or ballistician brought a new differential equation to the laboratory, Shannon configured two distinct subsystems: the continuous mechanical shafts of the analyzer, and a network of telephone relay circuits that controlled its switches, clutches, and safety interlocks.

Wiring those relays in series and in parallel, Shannon recognized the exact structure he had studied in an undergraduate logic course at the University of Michigan: the symbolic algebra of nineteenth-century English mathematician George Boole, published in 1854 as *An Investigation of the Laws of Thought*. Boole had formulated propositional reasoning as algebra. If a statement was true, Boole assigned it the number 1; if it was false, he assigned it the number 0. Combining two statements with the conjunction AND behaved like multiplication ($1 \times 1 = 1$, while $1 \times 0 = 0$); combining them with OR behaved like a saturating addition ($1 + 0 = 1$); and negating a statement with NOT inverted 1 to 0 and 0 to 1.

Boole had followed an even older trail. In 1703, the German polymath Gottfried Wilhelm Leibniz had presented a paper to the Royal Academy of Sciences in Paris titled *Explication de l'Arithmétique Binaire*. Leibniz noted that every integer could be represented using only two symbols, 1 and 0, and sketched a commemorative medallion showing light (1) emerging from the void (0) under the Latin motto *Omnibus ex nihilo ducendis sufficit unum* ("To derive everything from nothing, one suffices"). Yet for more than two centuries after Leibniz and eight decades after Boole, binary numbers and Boolean logic were treated as philosophical curiosities with no direct connection to machinery.

Observing the racks of telephone relays in the MIT basement, Shannon established the isomorphism between switching networks and Boolean algebra. Two switches wired in series allow electrical current to pass only if Switch A is closed AND Switch B is closed ($A \cdot B$). Two switches wired in parallel allow current to pass if Switch A is closed OR Switch B is closed ($A + B$). A normally closed relay contact that opens when its electromagnet is energized acts as a physical NOT inverter ($\overline{A}$). Every open or closed relay on a telephone switchboard was a physical realization of George Boole's two-valued algebra, and because Leibniz had shown that any number can be written in base-2 binary digits, circuits built from those logical switches could add, subtract, compare, and store numbers electronically.

In his 1937 MIT master's thesis, published in 1938 *(Shannon, 1938)*, Shannon proved that any logical decision or arithmetic operation can be synthesized from binary switching gates. Even the fundamental unit of digital information—the bit (*binary digit*)—was coined by

Shannon's Bell Labs colleague John Tukey and codified in Shannon's 1948 paper *A Mathematical Theory of Communication*. Every microprocessor, memory cell, router, and spacecraft guidance computer built over the past nine decades traces its architecture to that 1937 synthesis of symbolic logic and electrical switching.

10.1.2 Building Binary Addition from AND, OR, and XOR Gates

To see how Boolean switching gates perform arithmetic, consider how a silicon circuit adds two binary bits A and B. When adding $A + B$ in base 2, the circuit produces a two-bit sum consisting of a Carry bit (in the twos column, 2^1) and a Sum bit (in the ones column, 2^0). When $A = 0$ and $B = 0$, the result is 00_2 (Carry = 0, Sum = 0); when $A = 0$ and $B = 1$, or $A = 1$ and $B = 0$, the result is 01_2 (Carry = 0, Sum = 1); and when $A = 1$ and $B = 1$, the binary sum is $1 + 1 = 2_{10} = 10_2$ (Carry = 1, Sum = 0).

Those two output columns correspond directly to elementary Boolean gates:

Input Bit A	Input Bit B	Carry Bit (2^1 Column)	Sum Bit (2^0 Column)	Logical Meaning
`0`	`0`	`0`	`0`	Neither switch closed ⇒ 0
`0`	`1`	`0`	`1`	Exactly one switch closed ⇒ 1
`1`	`0`	`0`	`1`	Exactly one switch closed ⇒ 1
`1`	`1`	`1`	`0`	Both closed ⇒ Carry `1`, Sum wraps to `0`

The Carry column is `1` *if and only if* both A AND B are `1`, which is a single Boolean **`AND`** gate. Meanwhile, the Sum column is `1` *if and only if* A or B is `1`, but not both, matching the truth table of the Exclusive-OR (**`XOR`**, ⊕) gate. Wiring one **`XOR`** gate and one **`AND`** gate side by side across the same two input wires forms a physical Half-Adder, a circuit of two-state logic switches that performs binary addition. Chaining thirty-six adders together so that each column's carry feeds into the next column to its left enables a silicon register to add ten-digit integers in a single clock cycle.

10.2 Positional Radices and the 36-Bit Word Architecture

10.2.1 Positional Radices: Decimal, Binary, Octal, and Hexadecimal

The integer **`255`** takes four distinct written forms across the standard positional bases—**`255`** in decimal, **`11111111b`** in binary, **`377o`** in octal, and **`FFh`** in hexadecimal—even though all four strings represent the identical quantity.

When humans learned to count in antiquity, they grouped objects in tens (Decimal, base $b = 10$) corresponding to the ten fingers of the hands. Writing the decimal number $1{,}042_{10}$ expresses a polynomial in powers of 10 from the inside out:

$$1,042_{10} = 1\times10^3 + 0\times10^2 + 4\times10^1 + 2\times10^0 = ((1\times10+0)\times10+4)\times10+2 \qquad \textbf{(Eq. 10.1)}$$

The choice of radix 10 is a convention of human anatomy rather than a requirement of arithmetic. Building an electronic circuit that distinguishes ten distinct voltage levels (0 V, 0.5 V, 1.0 V, ..., 4.5 V) is vulnerable to thermal drift and noise margins. Instead, digital circuits operate with two states: OFF (0) and ON (1). In Binary (base $b = 2$), each column from right to left represents the next power of two ($2^0 = 1$, $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, ...):

$$1042_{10} = 1024 + 16 + 2 = 1\times2^{10} + 1\times2^4 + 1\times2^1 = 10000010010_2 \qquad \textbf{(Eq. 10.2)}$$

Although binary matches two-state hardware, long strings of thirty-six binary digits such as **000011000000101010000000101000101101b** exceed the four-to-seven-item span of human working memory identified by George Miller. Computer scientists compress binary words into compact human-readable notation without dividing by 10 by using radices that are exact powers of two: octal (**OCT**, base $b = 8 = 2^3$) and hexadecimal (**HEX**, base $b = 16 = 2^4$).

Because $8 = 2^3$, each octal digit (**0** through **7**) corresponds to an exact 3-bit triad (**000** through **111**), so converting between binary and octal requires only partitioning the bit string into 3-bit groups from right to left. Similarly, because $16 = 2^4$, each hexadecimal digit corresponds to an exact 4-bit nibble (**0000** through **1111**). For values from 0 to 9, hexadecimal uses the standard digits **0**–**9**; for the remaining six 4-bit patterns (10 through 15), it uses the letters **A** (= 10), **B** (= 11), **C** (= 12), **D** (= 13), **E** (= 14), and **F** (= 15). Two hexadecimal digits side by side (**00h** through **FFh**) pack an exact 8-bit byte (an octet, $2^8 = 256$ values from 0 to 255).

Decimal (DEC)	4-Bit Binary Nibble (BIN)	Octal (OCT)	Hexadecimal (HEX)	Positional Power Decomposition
0	0000b	0o	0h	0
5	0101b	5o	5h	$4 + 1 = 2^2 + 2^0$
7	0111b	7o	7h	$4 + 2 + 1 = 2^3 - 1$
8	1000b	10o	8h	2^3 (one full octal triad carry)
10	1010b	12o	Ah	$8 + 2 = 2^3 + 2^1$
12	1100b	14o	Ch	$8 + 4 = 2^3 + 2^2$
15	1111b	17o	Fh	$8 + 4 + 2 + 1 = 2^4 - 1$ (all 4 nibble bits ON)

Hexadecimal reduces binary inspection to direct nibble translation. When inspecting the 16-bit hexadecimal word **3C6Ah**, the four nibbles expand from left to right as **3h** $= 2 + 1 =$ 0011, **Ch** $(12_{10}) = 8 + 4 = 1100$, **6h** $= 4 + 2 = 0110$, and **Ah** $(10_{10}) = 8 + 2 = 1010$. Concatenating those four nibbles yields **0011 1100 0110 1010b**, compressing sixteen binary digits into four hexadecimal characters. On the SC-32, pressing the [↖] [BASE] menu via Left Shift [↖] + [×] places those four radix modes—[HEX], [DEC], [OCT], and [BIN]—onto the top softkeys and re-renders all four stack registers (**T, Z, Y, X**) on screen without altering their underlying 36-bit values. Whenever [HEX] is active, the top row of scientific keys from [√x] through [Σ+] operates as an **A**–**F** hexadecimal keypad.

10.2.2 The 36-Bit Word Architecture of the IBM 7090 and DEC PDP-10

Modern general-purpose processors organize their registers in multiples of 8-bit bytes, typically 32 or 64 bits wide. By contrast, the scientific and artificial-intelligence mainframes of the 1950s through the 1970s standardized on the 36-bit word.

When Katherine Johnson, Dorothy Vaughan, and NASA's Flight Research Division computed the orbital trajectories for John Glenn's *Friendship 7* Mercury flight and the Apollo lunar rendezvous studies, chronicled in Margot Lee Shetterly's *Hidden Figures*, the mainframes at Langley and Goddard were IBM 7090 and IBM 7094 systems built around 36-bit registers. When John McCarthy and Marvin Minsky founded the MIT Artificial Intelligence Laboratory and created the LISP programming language (**CAR** and **CDR** stood for *"Contents of the Address part of Register"* and *"Contents of the Decrement part of Register"* on the 36-bit IBM 704), they designed around 36-bit words. Likewise, when researchers at MIT, Stanford, CMU, and BBN built the initial nodes of the ARPANET,

their primary time-sharing platform was Digital Equipment Corporation's DEC PDP-10, another 36-bit architecture.

Architects of mid-twentieth-century scientific mainframes selected the 36-bit word for its divisibility and decimal range. In number theory, 36 is an abundant, highly composite integer divisible by 1, 2, 3, 4, 6, 9, 12, and 18. Whereas a 32-bit word divides evenly into eight 4-bit hexadecimal nibbles ($32 = 8 \times 4$) but leaves a 2-bit remainder when grouped into 3-bit octal triads ($32 = 10 \times 3 + 2$), a 36-bit word is the smallest practical computer word that divides with zero remainder into both twelve 3-bit octal digits ($36 = 12 \times 3$) and nine 4-bit hexadecimal nibbles ($36 = 9 \times 4$).

Furthermore, whereas a signed 32-bit integer overflows at +2,147,483,647 (fewer than ten full decimal digits, insufficient for a 10-digit telephone number or an unsigned 32-bit IPv4 broadcast address), $2^{36} = 68{,}719{,}476{,}736 \approx 6.87 \times 10^{10}$ provides ten full decimal digits of integer capacity. Thirty-six bits also partition into two symmetric 18-bit half-words ($36 = 2 \times 18$, which allowed LISP and the PDP-10 to pack two memory pointers into a single register) or six 6-bit alphanumeric characters ($36 = 6 \times 6$).

When IBM introduced the System/360 in 1964, Gene Amdahl and Fred Brooks standardized commercial computing on 8-bit bytes and 32-bit words. On a signed 32-bit calculator, however, entering an unsigned 32-bit hexadecimal value such as **`0xFFFFFFFF`** (4,294,967,295) encounters a sign-bit conflict: because Bit 31 is the sign bit on a 32-bit word, **`0xFFFFFFFF`** is either rejected as an overflow or interpreted as -1.

By adopting a 36-bit word across all four stack registers (T, Z, Y, X), the SC-32 places its sign bit four positions higher at Bit 35. Every 32-bit unsigned memory address, IPv4 network mask (**`0xFFFFFFFF`**), or microcontroller register sits in Bits 31–0 as a strictly positive integer, with four additional bits above the 32-bit boundary: three positive magnitude headroom bits (**`Bits 34-32`**, expanding positive range 8×) and the dedicated sign bit at **`Bit 35`** to accommodate overflow carries, checksum folds, and bit-shifts without truncation.

10.3 Mechanical Odometers, Two's Complement Arithmetic & Bitwise Masks

10.3.1 Two's-Complement Modular Addition

Because a silicon register contains only binary switches that are **`1`** (ON) or **`0`** (OFF), representing signed negative integers requires encoding the sign within the bit pattern itself.

Early computer designers in the 1940s tried the obvious idea first: use the leftmost bit as a "sign flag" (**`0`** for +, **`1`** for -) and leave the remaining bits as the ordinary magnitude (Sign-Magnitude representation). That sounds simple until you try to build the adder circuit: to add +5 and -3, the hardware first has to compare which number is larger, switch from an adder circuit to a subtractor circuit, and resolve two distinct representations of zero (**`+0`** and **`-0`**).

Modular arithmetic provides a direct way to represent negative numbers so that an ordinary binary adder can add positive and negative integers together without knowing which operands are negative. A 4-digit mechanical bicycle odometer set to **`0000`** miles illustrates how modular wrap-around encodes negative values. Riding forward 1 mile advances the rightmost wheel upward to **`0001`**, whereas starting at **`0000`** and rolling the wheel backward one mile (0 - 1 = -1) wraps all four mechanical number wheels backward one click from **`0`** to **`9`** so the odometer window reads **`9999`**. Rolling backward a second mile

(0 - 2 = -2) yields **9998**, and rolling backward three miles (-3) yields **9997**.

Starting with the odometer at **9997** (the four-digit modular representation of -3) and riding forward 5 miles (+5) advances the wheels five clicks in sequence through **9998**, **9999**, **0000** (where the carry spills off the left end of the 4-digit window), **0001**, and finally **0002**. By representing -3 as 10,000 - 3 = 9997, ordinary forward addition (9997 + 0005 = 10002 → **0002**) automatically computes (-3) + 5 = +2 without separate subtraction circuitry.

In binary hardware, that odometer wrap-around is called Two's Complement arithmetic. On a 36-bit binary odometer (**000000000h** in nine-nibble hexadecimal, the entry bound enforced when switching from [DEC] into [HEX], [OCT], or [BIN]), rolling forward from **000000000h** (0) counts upward through all positive 36-bit integers up to **7FFFFFFFFh** ($+2^{35}$ - 1 = +34,359,738,367), where Bit 35 (the 36-bit sign boundary) is **0** and all 35 lower bits are **1**. Rolling backward one click from **000000000h** (0 - 1 = -1) turns every bit to **1** (**FFFFFFFFFh** across a 36-bit mask, or **FFFFFFFFFFFFFFFFh** across the SC-32's 64-bit internal two's-complement register). The Two's Complement negative (-X) of any integer X follows directly from bitwise inversion: on a 36-bit odometer, X plus its bitwise logical inverse $\overline{X}$ (where every **0** flips to **1** and every **1** flips to **0**) always equals thirty-six **1**s (**FFFFFFFFFh**), which is one click below zero (-1):

$$X + \overline{X} = \text{FFFFFFFFFh} \equiv -1 \pmod{2^{36}} \implies -X = \overline{X} + 1 \qquad \textbf{(Eq. 10.3)}$$

Flipping every bit (**NOT**) and adding 1 produces the exact additive inverse. That single rule ($\overline{X}$ + 1) is how modern microprocessors and the [+/-] change-sign key on the SC-32 negate integers and reduce subtraction (Y - X = Y + $\overline{X}$ + 1) to pure addition.

Bit Position	Field Designation	Meaning in 36-Bit Two's Complement
Bit 35	**MSB** / Sign Bit	**0** = Positive integer (0 to $+2^{35}$-1); **1** = Negative integer (-2^{35} to -1)
Bits 34–32	Upper Headroom	Provides 3 full bits of positive magnitude headroom (8× capacity) below Sign Bit 35
Bits 31–0	Standard 32-Bit Word	Holds full 32-bit unsigned addresses, IPv4 netmasks, and GPIO bitfields

10.3.2 Cleaving Words at Bit Boundaries via Euclidean Division (÷R)

In elementary arithmetic, dividing 12,345 by 16 produces a quotient of 771 with a remainder of 9, because:

$$12,345 = 771 \times 16 + 9 \qquad \textbf{(Eq. 10.4)}$$

Number theorists call this Euclidean Division ($N = Q \cdot D + R$, with $0 \le R < D$). In algebra class, we often abandon remainders in favor of decimal fractions (12,345 / 16 = 771.5625). Yet in computer science, keeping the integer quotient $Q = \lfloor N / D \rfloor$ and the integer remainder $R = N \bmod D$ separate is one of the most versatile operations in mathematics.

Dividing a binary integer N by a power of two $D = 2^k$ via Euclidean division cleaves the binary word in two at bit position k. The integer quotient $Q = \lfloor N / 2^k \rfloor$ performs a right bit-shift ($N \gg k$), sliding all upper bits from Bit k upward down by k positions into the ones column and discarding the lower k bits. Simultaneously, the remainder $R = N \bmod 2^k$ forms a bitwise **AND** low-bit mask (N & (2^k - 1)) that preserves the exact, unshifted value of the lower k bits (**Bits [k-1 : 0]**).

On the SC-32, the dedicated Euclidean division key [→] [÷R] ([→] + [E]) models this cleavage directly across the two lower registers of the stack: placing any integer in Register Y and a power-of-two divisor 2^k in Register X and pressing [→] [÷R] deposits the right-shifted upper quotient $\lfloor\ Y\ /\ X\ \rfloor$ into Register X and the masked lower remainder Y mod X into Register Y in a single keystroke. In the 12,345 ÷ 16 example expressed in hexadecimal, $12{,}345_{10}$ = 3039h and $16_{10} = 2^4$ = 10h. Dividing 3039h by 10h (2^4) via [→] [÷R] places the quotient Q = 303h (771_{10}, the upper nibbles shifted right by 4 bits) in Register X and the remainder R = 9h (9_{10}, the bottom 4-bit nibble masked off) in Register Y.

That single mathematical relationship unifies six classic computer-engineering operations without dedicated bit-shift submenus. Shifting left by k bits ($X \ll k$) multiplies by $M = 2^k$, whereas shifting right by k bits ($X \gg k$) takes the integer quotient on dividing by $M = 2^k$. Masking the lowest k bits (X & (2^k - 1)) extracts the remainder X mod 2^k, and snapping an address down to a 2^k-aligned block (X & (2^k - 1)) takes the quotient $\lfloor\ X\ /\ 2^k\ \rfloor$ and multiplies back by 2^k to zero out the bottom k bits. Extracting an interior bit-field (**`Bits [m : k]`**) takes the remainder modulo 2^{m+1} to strip bits above bit m and divides by 2^k to shift bit k down to bit 0. Finally, adding $+2^k$ to a register whose k-th bit is **`0`** sets Bit k to **`1`** (**`OR`**), while subtracting 2^k when Bit k is **`1`** clears Bit k to **`0`** (**`AND NOT`**) without affecting neighboring bits.

10.3.3 Euclid's Remainder Ladder: Greatest Common Divisor (GCD) and Least Common Multiple (LCM)

Around 300 BCE in Alexandria, Euclid recorded in Book VII, Proposition 2 of his *Elements* what Donald Knuth called the oldest nontrivial algorithm in mathematics: the Euclidean Algorithm for the Greatest Common Divisor GCD(a, b) of two integers. When a digital signal processor runs one clock domain at a = 1,071 MHz and another at b = 462 MHz, finding the largest common reference frequency that divides both evenly does not require prime factorization. Euclid noted that any divisor of both a and b must also divide the remainder $R = a$ mod b from Euclidean division ([→] [÷R]). Replacing (a, b) with (b, a mod b) and pressing [→] [÷R] repeatedly steps down Euclid's Remainder Ladder: 1071 = 2 × 462 + 147 → 462 = 3 × 147 + 21 → 147 = 7 × 21 + 0. The last non-zero remainder before **`0`** is GCD(1071, 462) = **21**.

On the SC-32, you can either step down Euclid's ladder manually with [→] [÷R] or execute the entire ladder in a single keystroke inside the [→] [PARTS] menu (Right Shift [→] above [√x]), which places [GCD] ([LFU4]) and [LCM] ([LFU5]) side by side alongside [IP], [FP], [ABS], and [SGN]. Once GCD(a, b) is known, dividing the product $|a \cdot b|$ by GCD(a, b) gives the Least Common Multiple:

$$\mathrm{LCM}(a,b) = \frac{|a \cdot b|}{\mathrm{GCD}(a,b)} \qquad \textbf{(Eq. 10.5)}$$

Together, [GCD] and [LCM] solve three everyday problems across computer engineering and mechanics: reducing gear-tooth or aspect ratios (1920 × 1080 divided by GCD(1920, 1080) = 120 gives the 16:9 display ratio), finding the exact cycle period LCM(T_1, T_2) when two asynchronous timer buses realign in phase, and verifying that a public exponent e is coprime to Euler's totient (GCD(e, $\varphi(n)$) = 1) in RSA public-key cryptography.

10.3.4 Real-World Bitwise Engineering: Subnet Masks, RGB Pixels, Base64, Endianness & IEEE-754 Floats

Reading hexadecimal nibbles and cleaving binary words at powers of two (2^k) clarifies five foundational operations in digital systems. In IPv4 address slicing and **CIDR** subnetting, every IPv4 address A on the Internet is a single unsigned 32-bit integer written for human convenience in dotted-quad decimal notation $B_3.B_2.B_1.B_0$ as four 8-bit bytes (octets, each ranging from **0** to **255**, or **00h** to **FFh**), packed via Horner's radix-256 polynomial $A = ((B_3 \cdot 256 + B_2) \cdot 256 + B_1) \cdot 256 + B_0$:

Address Component	Octet 1 (**Bits 31–24**)	Octet 2 (**Bits 23–16**)	Octet 3 (**Bits 15–8**)	Octet 4 (**Bits 7–0**)
32-Bit Host IP (*A*)	B_3 ($\times 2^{24}$)	B_2 ($\times 2^{16}$)	B_1 ($\times 2^{8}$)	B_0 ($\times 2^{0}$)
/24 Netmask (FFFFFF00h)	**11111111** (Network)	**11111111** (Network)	**11111111** (Network)	**00000000** (Host)
Aligned /24 Network Base	B_3	B_2	B_1	**00000000** (**0**)

In Classless Inter-Domain Routing (**CIDR**), a **/p** prefix notation specifies that the upper p bits (**Bits [31 : 32-p]**) form the shared Network Prefix, while the remaining $k = 32 - p$ lower bits (**Bits [k-1 : 0]**, spanning $M = 2^k$ addresses per subnet block) identify the individual Host Offset. Dividing a 32-bit address A (or its active boundary octet) by the block stride $M = 2^{32-p}$ via [MOD] [÷R] cleaves the address right at the subnet boundary: the integer remainder $R = A \bmod 2^k$ isolates the host offset within the block, while multiplying the integer quotient $Q = \lfloor A \,/\, 2^k \rfloor$ back by 2^k zeroes out the k host bits to snap directly to the aligned network base address $A_{\text{net}} = \lfloor A \,/\, 2^k \rfloor \times 2^k$. Adding the inverted host wildcard mask $W = 2^k - 1$ = FFFFFFFFh - Netmask then populates all k host bits with **1**s to form the directed broadcast address $A_{\text{bcast}} = A_{\text{net}} + (2^k - 1)$.

A second application is packing color channels and Base64 characters by Horner's method. A smartwatch or microcontroller color display packs three separate Red, Green, and Blue brightness values into a single 16-bit memory word (**RGB565**). Because the cones in the human retina are most sensitive to subtle shades of green, **RGB565** allocates 5 bits ($R_5 \in [0, 31]$) to Red in **Bits [15:11]**, 6 bits ($G_6 \in [0, 63]$) to Green in **Bits [10:5]**, and 5 bits ($B_5 \in [0, 31]$) to Blue in **Bits [4:0]**:

Channel Allocation	Bit Positions	Bit Width & Range	Positional Shift Multiplier
Red (R_5)	Bits **15–11**	5 bits (0... 31, **00h–1Fh**)	$R_5 \times 2^{11}$ (**× 800h**)
Green (G_6)	Bits **10–5**	6 bits (0... 63, **00h–3Fh**)	$G_6 \times 2^{5}$ (**× 20h**)
Blue (B_5)	Bits **4–0**	5 bits (0... 31, **00h–1Fh**)	$B_5 \times 2^{0}$ (unshifted)
Packed RGB565 Word	Bits **15–0**	16-bit unsigned pixel (**0000h–FFFFh**)	$(R_5 \times 800\text{h}) + (G_6 \times 20\text{h}) + B_5$

Packing the three non-overlapping channels into a single 16-bit integer evaluates $(R_5 \times 2^{11}) + (G_6 \times 2^5) + B_5$, whereas unpacking any 16-bit pixel back into its three color channels requires only two Euclidean divisions ([MOD] [÷R]) at $2^{11} = 800\text{h}$ and $2^5 = 20\text{h}$. That same bit-slicing geometry powers Base64 encoding across the web. To transmit binary data over text-only protocols, Base64 welds each group of three 8-bit ASCII bytes (B_2, B_1, B_0) into a single 24-bit integer ($3 \times 8 = 24$ bits) and slices it at $2^{18} = 40000\text{h}$, $2^{12} = 1000\text{h}$, and $2^6 = 40\text{h}$ into four 6-bit sextet indices ($4 \times 6 = 24$ bits, each ranging from 0 to 63) that map onto the 64 printable characters **A–Z**, **a–z**, **0–9**, **+**, and **/**.

A fourth operation concerns Big-Endian versus Little-Endian byte order and Internet checksums. In 1980, computer scientist Danny Cohen borrowed the terms Big-Endian and

Little-Endian from Jonathan Swift's *Gulliver's Travels* (where two empires warred over whether to crack a soft-boiled egg at the big or little end) to describe multi-byte memory ordering. Network cables transmit bytes Big-Endian (**12345678h**, most-significant byte **12h** first), whereas x86 and ARM processors store bytes Little-Endian (**78563412h**, least-significant byte **78h** first). When routers verify packet integrity in flight, they use RFC 1071 16-bit one's-complement addition with end-around carry *(Knuth, 1997)*, dividing the 32-bit accumulated sum by 2^{16} = 10000h and adding the upper carry quotient right back into the lower 16-bit remainder—turning the 16-bit register into a circular loop where high-byte carries wrap into the low byte, so Big-Endian and Little-Endian machines compute the exact same checksum without byte-swapping.

Finally, just as decimal scientific notation writes -625 as -6.25 × 10^2, William Kahan's IEEE-754 floating-point standard *(IEEE, 2019; Golub & Van Loan, 1996; Parker, 2019)* writes -6.25 in binary scientific notation as -1.5625 × 2^2 = -1.1001_2 × 2^2, factoring every normal non-zero value into a base-2 exponent and a binary significand with an implicit leading **1.** bit before the binary point:

$$x = (-1)^S \times 2^{E-\text{bias}} \times \left(1 + \frac{M}{2^p}\right) \qquad \textbf{(Eq. 10.6)}$$

Because every non-zero normalized binary significand begins with **1.**, IEEE-754 omits that hidden bit from storage to gain one extra bit of precision. A 32-bit single-precision float (**binary32**, p = 23, bias = 127) packs 1 Sign bit S in **Bit 31**, an 8-bit Biased Exponent E = e + 127 in **Bits [30:23]** (reserving E = 0 for ± 0 and subnormals and E = FFh for ±∞ and **NaN**), and the 23-bit Fractional Mantissa M in **Bits [22:0]**, so cleaving the 32-bit word at 2^{23} = 800000h and 2^8 = 100h via [→] [÷R] unpacks M, E, and S onto the stack in two operations (while a 64-bit **binary64** double uses p = 52 and an 11-bit exponent with bias = 1023 and E = 7FFh reserved, cleaved in its upper 32-bit half at 2^{20} = 100000h and 2^{11} = 800h).

10.4 Combinatorics (PROB), Cryptographic Keyspaces, and the Birthday Collision Paradox

Sizing the state space of a digital register, a network topology, or a cryptographic cipher, and determining the collision threshold of random hashes, rests on the five functions inside the SC-32's [→] [PROB] menu: combinations [Cn,r], permutations [Pn,r], factorials [n!] ([←] [x!]), the uniform pseudo-random generator [RAND], and the PRNG seed command [SD].

10.4.1 Ordered Permutations, Unordered Combinations & Pascal's Symmetry

When drawing r items without replacement from n distinct items (such as n = 16 hexadecimal digits **0**–**F** or n = 12 server racks), the counting law depends on whether the order of selection is significant. When the order of selection matters (such as assigning r = 4 distinct hexadecimal nibbles to a 4-stage shift register, or ordering r = 5 packet routes out of n = 12), there are n choices for the first slot, n - 1 for the second, down to n - r + 1 for the r-th slot, which defines the ordered permutation [Pn,r]:

$$P(n,r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!} \qquad \textbf{(Eq. 10.7)}$$

Conversely, when order does not matter (such as choosing any subset of r = 5 storage drives out of n = 12 for a RAID parity group, or wiring every pair of r = 2 routers among n = 7 ARPANET nodes), dividing $P(n, r)$ by the $r!$ ways to rearrange the chosen subset

yields Blaise Pascal's binomial coefficient [Cn,r]:

$$C(n,r) = \binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!\,(n-r)!} \qquad \textbf{(Eq. 10.8)}$$

10.4.2 Richard von Mises's Birthday Paradox & Cryptographic Hash Collisions

That ratio between $P(N, k)$ and N^k governs a classic result in probability and cybersecurity: Richard von Mises's Birthday Paradox (1939). When k independent items (people in a room, or random packets in a network) are each assigned one of N equally likely labels (such as $N = 365$ calendar days, or $N = 2^b$ possible values of a b-bit hash function), first-order intuition suggests that a collision becomes 50% likely only when k approaches $N/2$.

That estimate is far too large because collisions can occur across any of the $C(k, 2) = k(k\text{-}1)/2$ distinct pairs among the k items. The exact probability P_{unique} that all k items receive *distinct* labels is the ordered permutation $P(N, k) = N(N\text{-}1)\cdots(N\text{-}k\text{+}1)$ divided by the N^k total unrestricted assignments, and using the first-order Taylor bound $1 - x \approx e^{-x}$ yields the Gaussian collision law:

$$P(\text{at least one collision}) = 1 - \frac{P(N,k)}{N^k} \approx 1 - \exp\left(-\frac{k(k-1)}{2N}\right) \qquad \textbf{(Eq. 10.9)}$$

Setting $\exp(-k^2 / 2N) = 1/2$ shows that the 50% collision threshold scales not with N, but with its square root: $k_{50\%} \approx \sqrt{((2 \ln 2)\ N)} \approx 1.1774\sqrt{N}$. In computer science, that $\sqrt{N}$ scaling is known as the Birthday Bound: when a b-bit hash function or database UUID spans $N = 2^b$ possible values, random collisions become likely after roughly $\sqrt{(2^b)} = 2^{b/2}$ operations, which is $2^{16} = 65{,}536$ packets for a 32-bit CRC or $2^{32} \approx 4.29 \times 10^9$ blocks for a 64-bit cipher. Modern cryptographic hashes such as SHA-256 therefore use $b = 256$ bits, which raises the birthday collision threshold to $2^{128} \approx 3.4 \times 10^{38}$ operations.

10.5 36-Bit Radix Representations, Boolean Masks & Bitwise Arithmetic

Because the four positional radices (**DEC**, **HEX**, **OCT**, **BIN**), modular two's-complement arithmetic ($Z/2^{36}Z$), and Euclidean division ($N = Q \cdot D + R$) operate on the same underlying 36-bit integer words across Registers **T, Z, Y, X**, base conversions and bit-field extractions execute directly on the lower two stack registers (**Y** and **X**) without disturbing values staged in **Z** and **T**.

Switching the active radix mode via [↶] [BASE] ([↶] + [×]) re-formats all four visible registers on screen without altering their binary contents, while [↷] [+R] ([↷] + [E]) cleaves any word N in Register **Y** at a power-of-two boundary $D = 2^k$ in Register **X** into its right-shifted upper quotient $Q = \lfloor\ Y\ /\ 2^k\ \rfloor$ in **X** and its masked lower remainder $R = Y \bmod 2^k$ in **Y** (where a single tap of [x⇄y] swaps **R** into **X** and **Q** into **Y**).

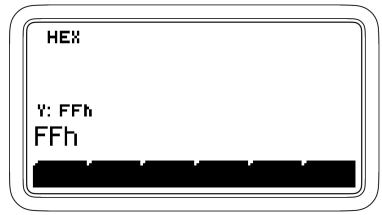


Figure: Decimal 255 converted in place to hexadecimal FFh on the SC-32 transflective display while preserving all four stack registers.

10.5.1 Radix Modes (DEC, HEX, OCT, BIN) & Direct Hex Entry

Following the classic HP-32SII design, the [↰] [BASE] menu (Left Shift [↰] on [×], Row 5, Column 3) selects the active display and entry radix across all four stack registers:

Base System	Keystroke	Screen Suffix	Allowed Character Digits	Example Formatted Display
Decimal	[↰] [BASE] [DEC]	*(none)*	**0** through **9**	**1042**
Hexadecimal	[↰] [BASE] [HEX]	**h**	**0** through **9**, **A** through **F**	**412h**
Octal	[↰] [BASE] [OCT]	**o**	**0** through **7**	**2022o**
Binary	[↰] [BASE] [BIN]	**b**	**0** and **1**	**10000010010b**

If Register X holds **255** in Decimal mode, pressing [↰] [BASE] [HEX] re-renders the register as **FFh**, pressing [↰] [BASE] [OCT] shows **377o**, and pressing [↰] [BASE] [BIN] shows all eight asserted bits as **11111111b**.

When Hexadecimal mode ([HEX]) is active, the top row of primary scientific keys across Row 1 maps onto the printed alpha legends A through F: [√x] enters $\boldsymbol{A} = 10_{10}$, [e^x] enters $\boldsymbol{B} = 11_{10}$, [LN] enters $\boldsymbol{C} = 12_{10}$, [y^x] enters $\boldsymbol{D} = 13_{10}$, [x^{-1}] enters $\boldsymbol{E} = 14_{10}$, and [Σ+] enters $\boldsymbol{F} = 15_{10}$. Entering **0xDEADBEEF** thus uses the top-row sequence **y^x 1/x √x y^x e^x 1/x 1/x Σ+**. In stack tables, hexadecimal literals such as **C0A80A2D** or **1F** are written in compact hex notation for readability; on the keypad in **HEX** mode, enter **A**–**F** using Row 1 (**√x** through **Σ+**). To apply scientific functions such as square root, natural logarithm, or powers to a hexadecimal value, switch back to Decimal mode ([↰] [BASE] [DEC]) first so the top-row keys resume their mathematical roles.

Because the SC-32 preserves all four stack registers across base changes, you can also mix radices in the middle of a calculation without jotting down a single intermediate number. To add a hexadecimal base offset $3A7_{16}$ to an octal permission mask 452_8 and read the sum in decimal, enter **3** [√x] **7** [ENTER] (935_{10}) in [HEX] mode, switch to [OCT] mode to add **452** [+] (2261_8), and switch to [DEC] mode to read the combined total $\mathbf{1{,}201}_{10}$ in Register X.

10.5.2 Euclidean Division (÷R): Flash Memory Pages, IPv4 Octets & Clock Bus Alignment

Applying Euclidean division [↷] [÷R] across Registers Y and X while Registers Z and T remain untouched executes byte-address and IPv4 subnet splits directly on the bottom two registers:

- **16-Byte Cache-Line Row & Offset (`12345 ÷R 16`):** For the byte address 12,345 = 771 × 16 + 9 (**0x3039**), keying **12345** [ENTER] **16** [↷] [÷R] in [DEC] mode cleaves at the $2^4 = 16$ bit boundary, placing the 16-byte row quotient Q = **771** in Register X and the intra-row byte remainder R = **9** in Register Y (where a single tap of [x⇄y] parks the

quotient **771** in Register Y and brings the remainder **9** into Register X).

- **256-Byte Flash Page & Offset (`12345 ÷R 256`):** Mapping that same byte address 12,345 (**0x3039**) onto a SPI NOR flash chip organized into $2^8 = 256$-byte programmable pages, keying **12345** [ENTER] **256** [R↓] [÷R] (followed by [x⇄y] when inspecting the offset in X and page in Y) cleaves the word right at Bit 8 into Page 48 ($\lfloor$ 12,345 / 256 $\rfloor = 48$, upper byte **0x30**) and Offset 57 (12,345 mod 256 = 57, lower byte **0x39**), since $48 \times 256 + 57 = 12{,}345$, while Registers Z and T remain at zero.

That same two-register [R↓] [÷R] cleavage decomposes IPv4 network addresses in base 256 (2^8) and aligns **/28** and **/24** Classless Inter-Domain Routing (**CIDR**) subnets:

- **Base-256 IPv4 Octet Unpacking (`3232235777 → 192.168.1.1`):** An unsigned 32-bit IPv4 socket address such as **3232235777** (**0xC0A80101**) is a four-digit integer written in base 256. Keying **3232235777** [ENTER] **256** [R↓] [÷R] in [DEC] mode strips off the lowest 8-bit octet **1** into Register Y and leaves the upper 24-bit prefix **12625921** in Register X. Repeating **256** [R↓] [÷R] peels off the third octet **1** into Y (with **49320** in X), and a third **256** [R↓] [÷R] splits the remaining 16-bit prefix into the first octet **192** in Register X and the second octet **168** in Register Y, which displays **192.168.1.1** across the four stack registers (**1, 1, 168, 192**).

- **Aligning a `/28` VLAN Interface (`205 → Block 192, Offset 13`):** On a **/28** office subnet (**255.255.255.240**, where 32 - 28 = 4 host bits give a block stride of $2^4 = 16$ addresses), dividing host octet **205** by **16** via [R↓] [÷R] places the block index **12** in Register X and the host offset **13** in Register Y ($205 = 12 \times 16 + 13$, with Z and T at zero), and multiplying Register X back by **16** zeroes out the low 4 host bits: keying **205** [ENTER] **16** [R↓] [÷R] **16** [×] snaps directly to the aligned **/28** network base ID **192** (**12 × 16 = 192**).

For a packed 32-bit **/24** workstation address **192.168.10.45** (**C0A80A2Dh**, where **192** = C0h, **168** = A8h, **10** = 0Ah, and **45** = 2Dh), working in [HEX] mode slices at the 8-bit host boundary ($2^8 = 100$h) while Registers Z and T rest at **0h**:

- **Align the `/24` Network Base (`192.168.10.0`):** Cleaving **C0A80A2Dh** by **100h** separates the 24-bit network prefix **C0A80Ah** in Register X from the 8-bit host ID **2Dh** (45_{10}) in Register Y, and multiplying Register X back by **100h** zeros out the host byte. Keying **C0A80A2D** [ENTER] **100** [R↓] [÷R] **100** [×] produces the aligned **/24** network base **C0A80A00h** (**192.168.10.0**).

- **Isolate the Inverted Host Wildcard Mask (`0.0.0.255`):** Subtracting the **/24** netmask **FFFFFF00h** from the 32-bit all-ones broadcast word **FFFFFFFFh** flips every network bit to **0** and every host bit to **1**. Keying **FFFFFFFF** [ENTER] **FFFFFF00** [−] isolates the 8-bit host wildcard mask **FFh**.

- **Populate the Directed `/24` Broadcast Address (`192.168.10.255`):** Because the aligned network base (**C0A80A00h**) and the host wildcard (**000000FFh**) occupy non-overlapping bit lanes, adding them performs an exact carry-free bitwise **OR** across the bottom 8 bits. Keying **C0A80A00** [ENTER] **FF** [+] delivers the directed **/24** broadcast address **C0A80AFFh**.

10.5.3 36-Bit Two's Complement Odometer Seams & Hardware Bit-Fields

The two boundary seams of the 36-bit two's-complement word (**7FFFFFFFFh** and **0 - 1**) appear directly in [HEX] mode on Registers *Y* and *X (Parker, 2019)*:

- **Signed 36-Bit Overflow Seam (7FFFFFFFFh + 1 → 800000000h):** At the highest positive 36-bit integer **7FFFFFFFFh** ($+2^{35}$ - 1 = +34,359,738,367$_{10}$), Bit 35 is **0** while all 35 lower magnitude bits (Bits 34–0) are **1**. Adding **1** propagates a ripple carry through all 35 lower bits into Bit 35 (**800000000h**), triggering signed two's-complement overflow (**V = 1**): keying **7FFFFFFFF** [ENTER] **1** [+] advances across the 36-bit boundary to **800000000h** (which represents -2^{35} = -34,359,738,368$_{10}$ in a wrapped 36-bit register, or +34,359,738,368$_{10}$ = 800000000h in the SC-32's unclipped 64-bit accumulator).

- **Zero Underflow Seam & Sign Extension (0 - 1 → -1):** Decrementing the register by one from **0** borrows across every bit position simultaneously and sets all two's-complement bits ON (**FFFFFFFFFh** masked to 36 bits, or **FFFFFFFFFFFFFFFFh** with hardware *sign extension* replicating Bit 35 across Bits 63–36 of the 64-bit register): keying **0** [ENTER] **1** [-] produces **FFFFFFFFFh** (-1$_{10}$).

Every bit-field shift and mask operation maps directly to a power of two $M = 2^k$ *(Shannon, 1938)*:

Bit-Field / Masking Operation	Mathematical Equivalent ($M = 2^k$)	SC-32 RPN Keystrokes (**HEX** / **DEC**)	Result Location
Left Shift ($X \ll k$)	$X \times 2^k$	**X** [ENTER] **M** [×]	Shifted word in **X**
Right Shift ($X \gg k$)	$\lfloor X / 2^k \rfloor$	**X** [ENTER] **M** [R/S] [÷R]	Upper bits in **X**
Low-Bit Mask (X & (2^k - 1))	$X \bmod 2^k$	**X** [ENTER] **M** [R/S] [÷R] [x≷y]	Low k bits in **X**
High-Bit Alignment (X & ~ (2^k - 1))	$\lfloor X / 2^k \rfloor \times 2^k$	**X** [ENTER] **M** [R/S] [÷R] **M** [×]	Aligned base in **X**
Set Disjoint Bit k	$X + 2^k$	**X** [ENTER] **M** [+]	Updated word in **X**
Clear Set Bit k	X - 2^k	**X** [ENTER] **M** [-]	Updated word in **X**
32-Bit Unsigned Complement	FFFFFFFF$_{16}$ - X	**FFFFFFFF** [ENTER] **X** [-]	Inverted mask in **X**

(For right bit-shifts in [DEC] mode, [R/S] [÷R] is preferred over [÷] because [÷] retains fractional decimal digits in Decimal mode, whereas [R/S] [÷R] always truncates to the integer quotient $\lfloor X / 2^k \rfloor$ in every base.)

Memory-mapped microcontroller peripherals pack independent hardware control lines into a single register word so firmware can toggle GPIO pins, isolate telemetry fault masks, and extract ADC bit-fields via positional masks 2^k without disturbing neighboring lines:

- **Asserting an Unset GPIO Direction Pin (GPIO_DIR | (1 << 5)):** When a 32-bit GPIO direction register holds **0x00001804**—meaning Pins 2 (2^2 = 4h), 11 (2^{11} = 800h), and 12 (2^{12} = 1000h) are already configured as outputs while Pin 5 is currently an input (**0**)—adding the unasserted Bit 5 positional weight $2^5 = 32_{10}$ = 20h in [HEX] mode performs a carry-free bitwise **OR**: keying **1804** [ENTER] **20** [+] asserts Pin 5 to produce **1824h** (**1824h**).

- **Clearing and Re-Asserting an Active Heartbeat Pin (PORT & ~(1 << 2)):** If a port output register holds **0x00000017** (0001 0111$_2$) with an active heartbeat line on Bit 2 (2^2 = 4h), subtracting the bit weight in [HEX] mode performs a borrow-free **AND NOT**: keying **17** [ENTER] **4** [-] clears Bit 2 to **13h**, and adding **4h** back via **13** [ENTER] **4** [+] (or [↶] [LASTx] [+]) re-asserts the pin to **17h** without disturbing neighboring pins.

- **Isolating Tripped Telemetry Fault Flags (`Status ^ Baseline`):** Whenever a baseline register's asserted bits are a subset of an active status word, borrow-free subtraction acts as an exact bitwise **XOR** difference mask. Comparing a spacecraft downlink status word **0x5F5A** against its nominal baseline **0x5A50** via **5F5A** [ENTER] **5A50** [-] strips away the nominal bits to isolate the newly tripped fault mask **50Ah**, which renders as **10100001010b** in [BIN] with a Hamming weight of 4 fault bits at positions 1, 3, 8, and 10.

- **Extracting an Interior Multi-Bit Field (`Bits [7:4] of 0x3C6A`):** For the 16-bit word **0x3C6A**, when an ADC control register packs a 4-bit oversampling selector in Bits [7:4] (**6h**), extracting that interior field combines a modulo mask (mod 2^8 = 100h) with an integer right-shift (÷ 2^4 = 10h):

$$\text{Bits}\,[7:4] = \left\lfloor \frac{3\text{C6Ah} \bmod 2^8}{2^4} \right\rfloor = \left\lfloor \frac{6\text{Ah}}{10\text{h}} \right\rfloor = 6\text{h} \qquad \textbf{(Eq. 10.10)}$$

First, cleaving at 2^8 = 100h via **3C6A** [ENTER] **100** [↻] [÷R] [x⇄y] masks off the upper configuration byte (**3Ch**) and brings the lower 8-bit byte **6Ah** into Register X; second, right-shifting by 4 bits (÷ 2^4 = 10h) via **6A** [ENTER] **10** [÷] discards the bottom nibble (**Ah**) so Bits [7:4] land right-justified at Bit 0 as **6h**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3C6A [ENTER]	0h	0h	3C6Ah	3C6Ah	Stage 16-bit ADC register word **3C6Ah**
100 [↻] [÷R]	0h	0h	6Ah	3Ch	Split upper byte **3Ch** into X and lower byte **6Ah** into Y
[x⇄y]	0h	0h	3Ch	6Ah	Swap lower byte **6Ah** (Bits [7:0]) into Register X
10 [÷]	0h	0h	3Ch	6h	Shift right by 4 bits (**÷ 10h**) to isolate Bits [7:4] = **6h**

10.5.4 Packing RGB565 Pixels, Slicing Base64 Sextets & Reversing Endianness

Executing **RGB565** pixel packing ($R_5 \times 2^{11}$) + ($G_6 \times 2^5$) + B_5 for 5-bit Red (**1Fh** = 31/31), 6-bit Green (**15h** = 21/63), and 5-bit Blue (**06h** = 6/31) in [HEX] mode shifts and merges the three non-overlapping bit lanes across Registers Y and X:

- **Shift Red into `Bits [15:11]` (`× 800h`):** Multiplying Red (**1Fh**) by 2^{11} = 800h left-shifts those 5 bits across the Green and Blue lanes: keying **1F** [ENTER] **800** [×] places **F800h** in Register X.

- **Shift Green into `Bits [10:5]` (`× 20h`) and Merge:** Multiplying Green (**15h**) by 2^5 = 20h via **15** [ENTER] **20** [×] left-shifts the 6 green bits into Bits [10:5] as **2A0h** while **F800h** waits in Register Y.

- **Combine All Three Lanes (`FAA6h`):** Merging Red and Green (**`FAA0h`**) and adding the unshifted 5-bit Blue channel (**`06h`**) in Bits [4:0] via **`FAA0`** [ENTER] **`6`** [+] completes the packed 16-bit framebuffer pixel **`FAA6h`**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`1F` [ENTER]	**`0h`**	**`0h`**	**`1Fh`**	**`1Fh`**	Stage 5-bit Red channel **`1Fh`** (31_{10})
`800` [×]	**`0h`**	**`0h`**	**`0h`**	**`F800h`**	Shift Red left 11 bits (**`× 800h`**) into Bits [15:11]
`15` [ENTER]	**`0h`**	**`F800h`**	**`15h`**	**`15h`**	Stage 6-bit Green channel **`15h`** (21_{10}); Red lifts to Z
`20` [×]	**`0h`**	**`0h`**	**`F800h`**	**`2A0h`**	Shift Green left 5 bits (**`× 20h`**) into Bits [10:5]
[+]	**`0h`**	**`0h`**	**`0h`**	**`FAA0h`**	Combine shifted Red and Green channels (**`FAA0h`**)
`6` [+]	**`0h`**	**`0h`**	**`0h`**	**`FAA6h`**	Add 5-bit Blue channel **`06h`** for packed **`RGB565`** word **`FAA6h`**

Conversely, given a packed 16-bit RGB565 pixel **`0xF38E`** in [HEX] mode, keying **`F38E`** [ENTER] **`800`** [↷] [÷R] [x⇄y] **`20`** [↷] [÷R] unpacks all three color channels onto the visible stack in one pass: Red in Register Z (**`1Eh`** $= 30_{10}$), Green in Register X (**`1Ch`** $= 28_{10}$), and Blue in Register Y (**`Eh`** $= 14_{10}$).

Similarly, slicing the 24-bit ASCII word **`"Man"`** (**`'M' = 4Dh`**, **`'a' = 61h`**, **`'n' = 6Eh`** ⇒ 4D616Eh) into its four 6-bit Base64 indices cleaves at $2^{18} = 40000$h, $2^{12} = 1000$h, and $2^6 = 40$h in [HEX] mode:

- **Sextet 1 (`Bits [23:18] → 'T'`):** Right-shifting by 18 bits (**`÷ 40000h`**) via **`4D616E`** [ENTER] **`40000`** [÷] extracts **`13h`** (19_{10} ⇒ Base64 **`'T'`**).
- **Sextet 2 (`Bits [17:12] → 'W'`):** Masking the 18-bit remainder and right-shifting by 12 bits (**`÷ 1000h`**) via **`4D616E`** [ENTER] **`40000`** [↷] [÷R] [x⇄y] **`1000`** [÷] isolates **`16h`** (22_{10} ⇒ **`'W'`**).
- **Sextet 3 (`Bits [11:6] → 'F'`):** Cleaving the 12-bit remainder at 6 bits (**`÷ 40h`**) via **`4D616E`** [ENTER] **`1000`** [↷] [÷R] [x⇄y] **`40`** [÷] extracts **`5h`** (5_{10} ⇒ **`'F'`**).
- **Sextet 4 (`Bits [5:0] → 'u'`):** Masking the final 6-bit remainder via **`4D616E`** [ENTER] **`40`** [↷] [÷R] [x⇄y] leaves **`2Eh`** (46_{10} ⇒ **`'u'`**), spelling out **`TWFu`**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`4D616E` [ENTER] **`40000`** [÷]	**`0h`**	**`0h`**	**`0h`**	13h	Sextet 1: **`4D616Eh >> 18 = 13h`** (19_{10} ⇒ ***T***)
`4D616E` [ENTER] **`40000`** [↷] [÷R] [x⇄y] **`1000`** [÷]	**`13h`**	**`13h`**	**`13h`**	16h	Sextet 2: remainder **`>> 12 = 16h`** (22_{10} ⇒ ***W***)
`4D616E` [ENTER] **`1000`** [↷] [÷R] [x⇄y] **`40`** [÷]	**`16h`**	**`16h`**	**`4D6h`**	5h	Sextet 3: remainder **`>> 6 = 5h`** (5_{10} ⇒ ***F***)
`4D616E` [ENTER] **`40`** [↷] [÷R] [x⇄y]	**`4D6h`**	**`5h`**	**`13585h`**	2Eh	Sextet 4: lower 6-bit remainder = **`2Eh`** (46_{10} ⇒ ***u***)

(Single-Entry **`÷R`** *Cascade: Chaining* **`4D616E`** [ENTER] **`40000`** [↷] [÷R] [x⇄y] **`1000`** [↷] [÷R] [x⇄y] **`40`** [↷] [÷R] *unpacks all four Base64 indices across the four stack registers—***`13h`** *in T,* **`16h`** *in Z,* **`2Eh`** *in Y, and* **`5h`** *in X—with* **`4D616E`** *keyed only once.)*

Reversing the 32-bit Big-Endian network header **`0x12345678`** into its Little-Endian x86/ARM memory representation **`0x78563412`** (**`BSWAP`** / **`REV`**) in [HEX] mode folds each shifted byte into a running sum on Register X across Registers Y and X while Registers Z and T stay at **`0h`**:

- **Promote Byte 0 (78h ≪ 24) into Byte 3:** Multiplying **78h** by 2^{24} = 1000000h via **12345678** [ENTER] **100** [R↓] [x⇄y] **1000000** [×] yields **78000000h** (**78000000h**).
- **Promote Byte 1 (56h ≪ 16) into Byte 2:** Multiplying **56h** by 2^{16} = 10000h via **12345678** [ENTER] **10000** [R↓] [x⇄y] **100** [R↓] [R↓] **10000** [×] positions **560000h** (**560000h**) and merges them into **78560000h**.
- **Promote Byte 2 (34h ≪ 8) into Byte 1:** Multiplying **34h** by 2^8 = 100h via **12345678** [ENTER] **1000000** [R↓] [R↓] [x⇄y] **10000** [R↓] [R↓] **100** [×] places **3400h** (**3400h**) into Byte 1 (**78563400h**).
- **Demote Byte 3 (12h ≫ 24) into Byte 0:** Right-shifting the original most-significant byte via **12345678** [ENTER] **1000000** [÷] isolates **12h** (**12h**), so adding **12h** completes the byte-swapped Little-Endian word **78563412h**.

10.5.5 RFC 1071 Checksums, Circular Rotations (ROL), UART Dividers & IEEE-754 Unpacking

Executing the RFC 1071 16-bit one's-complement end-around carry checksum on an IPv4 header whose ten 16-bit words sum to **2653Dh** in [HEX] mode cleaves at the 16-bit boundary 2^{16} = 10000h via **2653D** [ENTER] **10000** [R↓] [R↓], separating the upper carry quotient **2h** in Register X from the lower 16-bit sum **653Dh** in Register Y (with $Z = T = 0$h). Pressing [+] folds that end-around carry directly into Bit 0 (**653Fh**), and subtracting from the all-ones 16-bit mask **FFFFh** via **FFFF** [x⇄y] [-] inverts every bit to form the one's-complement wire checksum **9AC0h** ($39{,}616_{10}$ in [DEC]).

In cryptographic block ciphers and cyclic redundancy checks, that same two-register end-around carry implements a circular 1-bit left rotation (**ROL 1**) across an 8-bit register boundary (2^8 = 100h), where the bit shifted out of Bit 7 wraps around to enter Bit 0 (**0x85** = $10000101_2 \Rightarrow 00001011_2$ = 0Bh):

- **Left-Shift All 8 Bits (× 2):** Doubling **85h** via **85** [ENTER] **2** [×] shifts the byte left by one position to **10Ah** (**10Ah**), pushing the old MSB into Bit 8.
- **Cleave at the 8-Bit Boundary (÷R 100h):** Dividing by **100h** via **10A** [ENTER] **100** [R↓] [R↓] separates the low 8-bit remainder (**Ah** in Register Y) from the Bit 8 carry (**1h** = 1h in Register X).
- **Fold the Carry into Bit 0 (+):** Adding the carry back into the low byte via **10A** [ENTER] **100** [R↓] [R↓] [+] completes the circularly rotated byte **Bh** (**Bh**).

When configuring an asynchronous serial **UART** transceiver from a f_{clk} = 150.0 MHz system bus at 115,200 baud with 8× oversampling, hardware fractional baud-rate generators split the ideal divisor Divider = 150,000,000 / (8 × 115,200) ≈ 162.760417 into a 16-bit integer mantissa register (**IBRD**) and a 6-bit fixed-point sixty-fourths register (**FBRD** = ⌊ 64 × FP + 0.5 ⌋). Switching to [DEC] mode and keying **150000000** [ENTER] **8** [÷] **115200** [÷] yields the exact divisor **162.760416667** in Register X. Duplicating that divisor and extracting [IP] isolates **IBRD = 162** while swapping to take the fractional part [FP] isolates 0.76041667; multiplying by 64 and rounding via [←] [RND] converts 48.6667 to **FBRD = 49** (49/64 = 0.765625) in Register X right beside **IBRD = 162** in Register Y.

To unpack the 32-bit IEEE-754 single-precision (**binary32**) word **0xC0C80000**—encoding $-6.25_{10} = -110.01_2 = (-1)^1 \times 2^2 \times 1.1001_2$ as $(-1)^S \times 2^{E\,-\,127} \times (1 + M\,/\,2^{23})$—chain two [R↓] [R↓] cleavages in [HEX] mode at the 23-bit mantissa boundary (2^{23} = 800000h) and the 8-bit exponent boundary (2^8 = 100h). This places the 23-bit significand **480000h** (4,718,592

/ 2^{23} = 0.5625, or 1.5625 with the hidden leading **1.**) in Register Z, the biased exponent **81h** (129_{10}, so e = 129 - 127 = 2) in Register Y, and the sign bit **1h** in Register X to verify $(-1)^1 \times 2^2 \times 1.5625$ = -6.25:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
C0C80000 ENTER	**0h**	**0h**	**C0C80000h**	**C0C80000h**	Stage 32-bit IEEE-754 float word **C0C80000h**
800000 R↓	**0h**	**0h**	**480000h**	**181h**	Cleave at Bit 23: mantissa **480000h** in Y, upper 9 bits **181h** in X
100 R↓	**0h**	**480000h**	**81h**	**1h**	Cleave at Bit 8: Sign **1h** in X, Exponent **81h** in Y, Mantissa in Z
← BASE DEC	0	4718592	129	1	Switch to Decimal: S = 1, E = 129, M = 4,718,592
x⇄y **127** -	4718592	4718592	1	2	Subtract bias 127 to reveal true binary exponent e = 2

Similarly, for the upper 32-bit word **0xC02AC000** of -13.375$_{10}$ (**0xC02AC00000000000**, **binary64**) in HEX mode, chaining **C02AC000** ENTER **100000** R↓ **800** R↓ cleaves at 2^{20} = 100000h and 2^{11} = 800h to leave Sign S = **1** in X, Biased Exponent E = **402h** (1026_{10}, e = 1026 - 1023 = **3**) in Y, and significand **AC000h** (704,512 / 2^{20} = 0.671875) in Z for $(-1)^1 \times 2^3 \times 1.671875$ = **-13.375**.

10.5.6 Combinatorics (PROB) & Birthday Paradox Collision Probability

In DEC mode, the PROB menu (Right Shift above e^x) evaluates discrete state spaces on the stack. For a 12-drive array selecting 5 drives for an unordered RAID group (C(12, 5)) versus an ordered failover chain (P(12, 5)), **12** ENTER **5** PROB Cn,r counts **792** subsets, while **12** ENTER **5** PROB Pn,r spans **95,040** sequences:

- **4-Stage Non-Repeating Hex Shift Register (P(16, 4)):** Assigning an ordered 4-nibble sequence without repetition from 16 hexadecimal digits via **16** ENTER **4** PROB Pn,r yields **43,680** distinct register states.
- **7-Router Full-Mesh Point-to-Point Links (C(7, 2)):** Wiring every unordered pair of routers across a 7-node network via **7** ENTER **2** PROB Cn,r requires **21** cables.
- **15-Packet Priority Queue Permutations (15!):** Ordering an entire 15-packet queue via **15** PROB ← x! spans **1,307,674,368,000** schedules (12! = 479,001,600 for 12 items).

Evaluating Richard von Mises's Birthday Collision probability 1 - $P(365, k)/365^k$ across N = 365 days for k = 23 people (C(23, 2) = 253 pairs) via **365** ENTER **23** PROB Pn,r gives 4.220082 × 10^{58} collision-free assignments; dividing by **365** ENTER **23** y^x ÷ yields P_{unique} = **0.492703**, and **1** x⇄y - confirms a shared-birthday probability of **0.507297** (50.73%).

Chapter 11: Compound Interest, Amortization & the Time Value of Money

11.1 Fibonacci's Merchants, Percentage Asymmetry & The Rule of 72

In 1202, Leonardo Pisano, known to posterity as Fibonacci, returned to Pisa from the commercial ports of Bugia (in modern-day Algeria), Alexandria, and Constantinople. His father, Guglielmo Bonacci, had served as a customs official representing Pisan traders across North Africa, and young Leonardo had grown up watching Arab and North African mathematicians calculate with the nine Hindu-Arabic numerals (**1** through **9**) and the symbol **zephirum** (**0**), while European counting houses were still struggling to multiply Roman numerals (**MCCIV**) on counting boards.

When Fibonacci completed his treatise *Liber Abaci* (*The Book of Calculation*) in 1202, popular history remembered the work chiefly for its exercise on breeding rabbits. In fact, that rabbit problem occupied barely half a page. Fibonacci filled the longest, most ambitious chapters of *Liber Abaci* with the real mathematical problem that kept twelfth-century Venetian, Genovese, and Pisan sea captains awake at night: comparing the worth of a gold bezant in hand today against a stream of payments arriving across different years in the future.

Four centuries later, after Simon Stevin printed the first public compound-interest tables in Leiden (*Tafelen van Interest*, 1582), English mathematician Richard Witt published *Arithmeticall Questions, Touching the Buying or Exchange of Annuities* in 1613 London, just one year before John Napier unveiled his logarithms in Edinburgh. Witt demonstrated to London merchants, landlords, and actuaries that time acts as a geometric operator on capital: every period that a principal PV is invested at periodic rate i, it scales by the growth factor $(1 + i)$. Conversely, every period one must wait to receive a promised future sum FV, its present value is discounted through division by $(1 + i)$.

Across personal savings, amortizing residential mortgages, fixed-income securities, and corporate capital budgeting, compound growth and present-value discounting translate cash flows occurring at different dates into a single common measure of value.

11.1.1 Multiplicative Asymmetry in Percentage Gains and Losses

Even over a two-period horizon, percentage changes exhibit a fundamental asymmetry between gains and losses. When a portfolio of initial value V_0 suffers a fractional drawdown of δ during an autumn sell-off, and then rallies by an equal fractional gain of $+\delta$ the following spring, linear intuition—trained on symmetric addition and subtraction ($-\delta + \delta = 0$)—expects the portfolio to return to its starting balance V_0.

Tracing the capital across the two periods reveals the multiplicative asymmetry. Losing a fraction δ first reduces the initial capital V_0 to a diminished base of $V_1 = V_0(1 - \delta)$; for a 20% drop on \$100.00, the capital shrinks to \$80.00. When the market subsequently rebounds by $+\delta$, that percentage gain acts on the shrunken base V_1 rather than on V_0, bringing the terminal value to $V_2 = V_1(1 + \delta) = V_0(1 - \delta)(1 + \delta)$. On the \$80.00 base, a $+20\%$ gain adds only $+\$16.00$ and ends at \$96.00, so the portfolio remains down by -4.0%. More dramatically, when an asset loses half its value ($\delta = 50\%$), a subsequent $+50\%$ gain restores only three-quarters of the initial principal; full recovery to V_0 requires solving $V_0(1$

- δ)(1 + g) = V_0, which demands a hyperbolic recovery gain of g = δ / (1 - δ) (or +100% after a 50% loss, and +25% after a 20% loss).

Equal percentage losses and gains never cancel out because percentages compose multiplicatively rather than additively. By the algebraic identity for the difference of two squares, a fractional drop -δ followed by an equal fractional gain +δ produces:

$$(1-\delta)(1+\delta) = 1-\delta^2 < 1 \qquad \textbf{(Eq. 11.1)}$$

Whenever δ > 0, the product (1 - δ)(1 + δ) falls short of unity by the quadratic volatility drag δ^2 (for δ = 0.20, the drag is $\delta^2 = 0.04 = 4\%$).

That same multiplicative structure governs successive chain markdowns and commercial gross-margin pricing. When a merchant applies a primary fractional markdown d_1 followed by an additional promotional discount d_2 at checkout, the two concessions do not sum to d_1 + d_2. Instead, the net retention factor is the product (1 - d_1)(1 - d_2), so the effective combined discount is d_{eff} = 1 - (1 - d_1)(1 - d_2) = d_1 + d_2 - d_1 d_2. A 50% clearance markdown followed by an extra 40% off retains 0.50 × 0.60 = 0.30 of list price, an effective discount of 70% rather than 90%.

Similarly, in commercial accounting, a markup rate r is measured as a fraction of unit landed cost C to produce a selling price of $P_{\text{markup}} = C(1 + r)$, whereas a gross margin requirement m specifies the fraction of the final selling price P retained as gross profit, (P - C)/P = m. Applying a cost markup of $r = m$ realizes a gross margin of only m / (1 + m). To achieve a true target gross margin m, the unit cost C must represent the complement (1 - m) of the selling price:

$$P_{\text{margin}} = \frac{C}{1-m} \qquad \textbf{(Eq. 11.2)}$$

11.1.2 Base-Preserving Percentages and the Logarithmic Rule of 72

Everyday percentage calculations frequently go astray because every percentage only has meaning relative to its reference base.

In the 1970s, Hewlett-Packard's RPN calculator architects and UC Berkeley numerical analyst William M. Kahan, the chief architect of the IEEE-754 floating-point standard and financial solver algorithms *(Kahan, 1979)*, examined why conventional pocket calculators make percentage math so error-prone. On an algebraic calculator, computing a tax or discount rate r% on an invoice base B via **B × r %** overwrites the display with the adjustment $B \cdot r$ / 100 and destroys the original base B, so the operator must re-enter the base from scratch to obtain the net total.

Hewlett-Packard's architects and Kahan recognized that in an inside-out stack architecture, the percentage operator must be a two-register state transformation (Y, X) *mapsto* (Y, $Y \cdot X$ / 100) that preserves the reference base in Register Y. On the SC-32, [→] [%] (Right Shift [→] above [COS]) and [→] [%CHG] (Right Shift [→] above [TAN]) enforce this base-preserving invariant. Staging the base amount in Register Y and the percentage rate in Register X lets [→] [%] place the exact currency adjustment in Register X while holding the original base untouched in Register Y, ready for [+] to add a tax or [-] to deduct a markdown in a single keystroke.

When a periodic growth rate $r\%$ compounds across N consecutive periods, each period multiplies the balance by $(1 + r/100)$ and compounds additive rates into exponential growth:

$$FV = PV\left(1 + \frac{r}{100}\right)^N \quad \textbf{(Eq. 11.3)}$$

To find the number of periods N required for compound growth at rate $r\%$ to double an initial principal ($FV / PV = 2$), take the natural logarithm of both sides:

$$\left(1 + \frac{r}{100}\right)^N = 2 \implies N\ln\left(1 + \frac{r}{100}\right) = \ln 2 \approx 0.693147 \quad \textbf{(Eq. 11.4)}$$

For moderate rates ($r \leq 12\%$), the Mercator-Taylor expansion $\ln(1 + u) = u - (1/2)u^2 + \cdots$ with $u = r/100$ gives $N \approx (69.31/r)(1 + r/200)$. Franciscan friar Luca Pacioli, who first printed the rule in his monumental *Summa de arithmetica* (Venice, 1494), and Renaissance merchants across Venice and Florence adjusted the numerator from 69.3 to 72 (matching the exact discrete-compounding curvature around $r \approx 8\%$) because 72 is evenly divisible by 1, 2, 3, 4, 6, 8, 9, and 12 to establish the Rule of 72:

$$\text{Doubling Time (Years)} \approx \frac{72}{\text{Annual Interest Rate } r\%} \quad \textbf{(Eq. 11.5)}$$

At 6% annual return, capital doubles in approximately 72 / 6 = 12 years; at 8%, in 72 / 8 = 9 years; at 9%, in 72 / 9 = 8 years. Conversely, when price inflation runs at an annual rate π over N years, the real purchasing power of uninvested cash decays exponentially according to $PV_{\text{real}} = FV \cdot (1 + \pi)^{-N}$ and cuts purchasing power in half every $72 / (100\pi)$ years.

11.2 The Five-Variable Balance Law and Mortgage Amortization

11.2.1 Deriving the Five-Variable TVM Balance Equation

In practice, financial contracts rarely involve a single lump sum left untouched for decades. Instead, three distinct cash components interact across N compounding periods at periodic interest rate i: the present value PV exchanged at time $t = 0$ (such as loan principal disbursed at closing or an opening investment balance), the uniform periodic payment stream PMT transferred at each period $k = 1, 2, \ldots, N$ (such as a periodic loan installment, lease payment, or sinking-fund contribution), and the terminal future value FV remaining at the end of period N (0 for a fully amortizing loan, a contractual balloon payoff, or an accumulated savings target). All five variables—N, i, PV, PMT, and FV—are unified by Francis Baily's (1808) and Augustus De Morgan's (1838) principle of actuarial equivalence, which discounts every cash flow to a common reference epoch.

Taking the present epoch ($t = 0$) as the valuation fulcrum, each periodic payment PMT occurring at period k is discounted to $t = 0$ through multiplication by $(1 + i)^{-k}$, and the terminal balance FV at period N is discounted by $(1 + i)^{-N}$:

$$PV + \frac{PMT}{(1+i)^1} + \frac{PMT}{(1+i)^2} + \cdots + \frac{PMT}{(1+i)^N} + \frac{FV}{(1+i)^N} = 0 \quad \textbf{(Eq. 11.6)}$$

The central summation of N discounted payments is a finite geometric series with common ratio $q = (1 + i)^{-1}$. Applying Euclid's closed-form summation identity collapses those N terms into the standard actuarial Annuity Present-Value Factor ($a_{\overline{N}|i}$):

$$\sum_{k=1}^{N}(1+i)^{-k}=\frac{1-(1+i)^{-N}}{i}=a_{\overline{N}|i} \qquad \text{(Eq. 11.7)}$$

Substituting $a_{\mathrm{N}|i}$ yields the Master Five-Variable Time Value of Money (**TVM**) Equation:

$$PV+PMT\left[\frac{1-(1+i)^{-N}}{i}\right]+FV(1+i)^{-N}=0 \qquad \text{(Eq. 11.8)}$$

Payment timing within each period modifies the annuity factor. In an Ordinary Annuity (**END** mode), standard for residential mortgages, term loans, and coupon bonds, each payment PMT occurs at the end of its compounding period after one interval of interest has accrued. In an Annuity Due (**BEG** mode), standard for real-estate leases, equipment rentals, and insurance premiums, each payment occurs at the beginning of the period, so it compounds for one extra interval and multiplies the PMT bracket by $(1 + i)$.

11.2.2 Cash-Flow Conservation and the Sign Convention

The right-hand side of the Master TVM Equation equals zero because every financial contract conserves discounted value across time. Financial mathematics models an individual wallet or corporate treasury as a thermodynamic control volume across time, governed by a strict directional sign convention. Cash inflows entering the operator's hands, such as loan proceeds received at closing, coupon checks collected, or maturity redemptions paid out, carry a positive sign (+). Cash outflows leaving the operator's hands, such as periodic loan repayments, savings deposits transferred into an external account, or the purchase price paid to acquire a security, carry a negative sign (-).

When all inflows and outflows along the timeline are discounted to $t = 0$ at periodic rate i, their algebraic sum vanishes. This sign discipline makes the direction of every transaction unambiguous: when a borrower receives loan proceeds $PV > 0$ at $t = 0$, the N periodic repayments must carry the opposite sign ($PMT < 0$) to reduce the remaining debt to $FV = 0$.

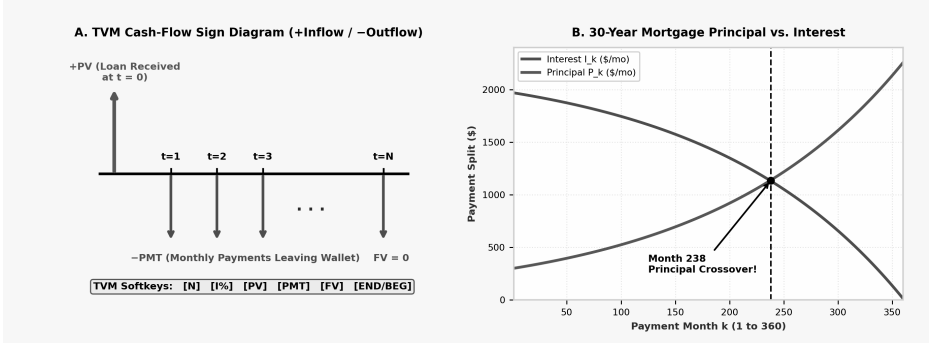


Figure: Time Value of Money (TVM) cash-flow sign conservation timeline and 30-year mortgage principal-versus-interest amortization crossover.

11.2.3 Principal-Interest Decomposition and Amortization Crossover

For a fully amortizing level-payment loan of principal $PV > 0$ repaid across N periods (**`FV = 0`**) at periodic interest rate i, solving the Master TVM Equation fixes the constant periodic payment magnitude $|PMT|$:

$$|PMT| = PV\left[\frac{i}{1-(1+i)^{-N}}\right] \qquad \textbf{(Eq. 11.9)}$$

Although the total check $|PMT|$ remains constant at every period $k = 1, 2, \ldots, N$, its internal allocation between interest charge I_k and principal reduction P_k shifts exponentially over the life of the contract. At the end of period k - 1, the remaining loan balance B_{k-1} equals the present value of the N - $k + 1$ payments still remaining:

$$B_{k-1} = |PMT|\left[\frac{1-(1+i)^{-(N-k+1)}}{i}\right] \qquad \textbf{(Eq. 11.10)}$$

During period k, the lender first assesses periodic interest $I_k = i \cdot B_{k-1}$ on that outstanding balance, and only the residual portion $P_k = |PMT|$ - I_k reduces the principal:

$$I_k = |PMT|\left[1-(1+i)^{-(N-k+1)}\right], \qquad P_k = |PMT|(1+i)^{-(N-k+1)} \qquad \textbf{(Eq. 11.11)}$$

These two identities govern the structural dynamics of every amortizing loan. Because $P_{k+1} / P_k = 1 + i$, the principal portion of each payment follows a geometric progression that grows exponentially at rate i from the first period to the last. On long-duration loans where $N \cdot i \gg 1$, the factor $(1 + i)^{-N}$ is small, so I_1 consumes the overwhelming majority of the opening payments while P_1 retires only a small fraction of principal. The amortization crossover period k_{cross} at which principal reduction P_k finally overtakes interest I_k occurs when $P_k = 1/2|PMT|$, which simplifies directly to $(1 + i)^{N - k + 1} = 2$. Taking logarithms reveals that the crossover point depends only on N and i, completely independent of the loan size PV:

$$k_{\text{cross}} = N + 1 - \frac{\ln 2}{\ln(1+i)} \qquad \textbf{(Eq. 11.12)}$$

Thus, the amortization crossover always occurs exactly one Rule-of-72 doubling time prior to maturity. Furthermore, because early principal reduction P_k is small, sending an unscheduled periodic principal curtailment ΔP on top of $|PMT|$ accelerates the payoff horizon from N down to N^*, derived in closed form by inverting the annuity factor:

$$N^* = -\frac{\ln\left(1-\frac{i \cdot PV}{|PMT|+\Delta P}\right)}{\ln(1+i)} \qquad \textbf{(Eq. 11.13)}$$

Each extra dollar directed to principal in early periods bypasses all compound interest that would otherwise accrue on that dollar across the remaining N - k periods of the schedule.

11.3 Valuing Future Cash Streams: Bond Pricing, Net Present Value & Internal Rate of Return

11.3.1 Coupon Discounting, Inverse Yield Dynamics, and Macaulay Duration

A fixed-rate bond is a standardized loan contract in which an issuer promises to pay a sequence of periodic coupon inflows $PMT = +C$ across N compounding periods together with a par redemption inflow $FV = +M$ at maturity. When the market requires a periodic yield to maturity y on securities of comparable credit risk, the fair purchase price P on a coupon date (entered as an acquisition outflow PV = -P; between coupon dates, the settlement "dirty" invoice price adds accrued interest to the quoted "clean" price) is the sum of the discounted coupons and discounted par redemption:

$$P(y) = C\left[\frac{1-(1+y)^{-N}}{y}\right] + M(1+y)^{-N} \qquad \textbf{(Eq. 11.14)}$$

Because the contractual cash promises C and M are fixed at issuance, the market price $P(y)$ moves inversely to the prevailing yield y. When the market yield y equals the coupon rate $c = C / M$, the bond trades at par ($P = M$); when prevailing yields rise above the coupon rate ($y > c$), the bond must trade at a discount ($P < M$) so that the capital gain M - P at maturity compensates the buyer for the below-market coupons; when yields fall below the coupon rate ($y < c$), the bond trades at a premium ($P > M$).

In 1938, economist Frederick Macaulay introduced a closed-form measure of a bond's effective maturity and interest-rate sensitivity by treating the discounted cash flows $w_t = C_t\ (1 + y)^{-t}$ at epochs $t = 1, 2, \ldots, N$ as point masses distributed along a timeline. The center-of-mass fulcrum where that timeline balances—identical in form to Archimedes' weighted centroid $\bar{t} = \sum t\ w_t\ /\ \sum w_t$—is Macaulay Duration ($D$, measured in periods, or $D_{\text{yr}} = D/m$ in years for m coupons per year). Differentiating $P(y)$ with respect to y and dividing D_{yr} by $(1 + y/m)$ yields Modified Duration (D^*), the first-order elasticity linking percentage price changes to annual yield shifts Δy:

$$D = \frac{1}{P}\sum_{t=1}^{N}\frac{t\cdot C_t}{(1+y)^t}, \qquad \frac{\Delta P}{P} \approx -D^{*}\cdot\Delta y = -\frac{D}{1+y}\cdot\Delta y \qquad \textbf{(Eq. 11.15)}$$

This duration lever explains why short-dated notes with small D^* exhibit minimal price volatility when central-bank rates move, whereas long-dated bonds—whose cash-flow centroid sits far out along the time axis—experience large percentage price swings for every 1% shift in yield (if a bond fund has $D^* = 7.0$ yr and market interest rates rise by 1%, the fund's price drops by roughly 7%, before second-order convexity curvature $(1/2)C(\Delta y)^2$).

11.3.2 Net Present Value (NPV) as a Horner Polynomial and IRR Root-Finding

Unlike level-coupon bonds, capital projects, industrial plants, and operating businesses generate uneven cash flows $C_1, C_2, \ldots, C_N$ across successive years. Discounting each cash flow at the firm's periodic cost of capital r defines the present value of the stream (and, after subtracting the initial outlay $|C_0|$, the Net Present Value):

$$\text{PV}(r) = \sum_{t=1}^{N}\frac{C_t}{(1+r)^t}, \qquad \text{NPV}(r) = C_0 + \sum_{t=1}^{N}\frac{C_t}{(1+r)^t} \qquad \textbf{(Eq. 11.16)}$$

Letting $x = 1 + r$ represent the gross periodic discount factor, the discounted sum is a degree-N polynomial in the reciprocal factor $u = 1/x$, which folds from the terminal horizon $t = N$ backward to $t = 0$ via Horner's nested recurrence:

$$\mathrm{NPV}(x) = C_0 + (\cdots((C_N \div x + C_{N-1}) \div x + C_{N-2}) \div x + \cdots + C_1) \div x \qquad \textbf{(Eq. 11.17)}$$

This nested polynomial representation directly governs the inverse valuation problem: given an initial investment outlay $C_0 < 0$ followed by future cash flows C_1, ..., C_N, the Internal Rate of Return (**IRR**) is the critical discount rate r^* at which the Net Present Value vanishes, $\mathrm{NPV}(r^*) = 0$.

Determining **IRR**—or solving the Master TVM Equation for an unknown periodic rate **I%** ([I%] on [LFU1])—requires finding the root of a degree-N polynomial. As Niels Henrik Abel and Évariste Galois proved in the nineteenth century, no closed-form radical formula exists for general polynomial equations of degree $N \geq 5$. Moreover, as William Kahan demonstrated in his 1979 analysis of financial calculators, direct evaluation of $(1 - (1 + i)^{-N}) / i$ suffers catastrophic cancellation when $i \to 0$, requiring evaluation through $\ln(1 + i)$ and $\exp(x) - 1$ primitives during iterative root-finding:

$$a_{\overline{N}|i} = \frac{1 - \exp(-N \ln(1+i))}{i} \qquad \textbf{(Eq. 11.18)}$$

Two structural properties govern numerical root-finding on discounted cash-flow polynomials. First, **NPV** and **IRR** can rank mutually exclusive projects differently when their scales or cash-flow timings diverge: **IRR** measures the constant compounding rate earned on capital while it remains invested inside the project (whereas Modified **MIRR** explicitly compounds intermediate inflows to the horizon at the firm's external reinvestment rate before taking the N-th root), while **NPV** discounts every cash flow at the external opportunity cost of capital r.

Second, by René Descartes's 1637 Rule of Signs, a conventional investment whose cash-flow sequence changes sign only once ($C_0 < 0$ followed by $C_t \geq 0$) possesses a unique positive root $x^* = 1 + r^*$. By contrast, a project with a large terminal decommissioning or reclamation outflow (-, +, +, -) changes sign twice and can exhibit two distinct internal rates of return. To handle both regimes cleanly, the SC-32 combines Horner stack evaluation with Richard Brent's [↻] [SOLVE] root-finder and William Kahan's financial convergence safeguards *(Kahan, 1979; Brent, 1973)* across **N**, **I%**, **PV**, **PMT**, **FV**, and **END**/**BEG**.

11.4 Percentages, Mortgage Amortization & Discounted Cash Flows

Because a unit of capital today compounds at periodic rate i into $(1 + i)^N$ units N periods hence, every commercial invoice, amortizing loan, bond, or capital budget is governed by discounted cash-flow conservation. On the StackCalc SC-32, financial computation relies on a two-register discipline: Register **Y** holds the baseline principal or subtotal untouched while Register **X** evaluates [↻] [%] or [↻] [%CHG], and the five-variable [↖] [TVM] softkey bar ([N], [I%], [PV], [PMT], [FV]) reads known inputs from and writes solved roots directly to Register **X** without disturbing the upper stack.

Pressing [↖] [TVM] (Left Shift [↖] above [π‖y]) maps the five financial variables plus **END/BEG** across the six top softkeys ([LFU0]–[LFU5]), where two operational invariants govern every entry. Under the wallet reference frame, cash flowing into your pocket is entered as positive (**+**), whereas cash flowing out of your pocket is negated with [+/−] (**-**). To solve for an unknown quantity from four known parameters, keying a number and tapping a softkey

stores that parameter from Register **X**, while tapping a fifth softkey without prior numeric entry solves for that unknown variable.

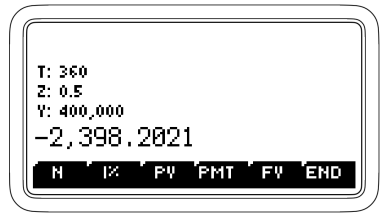


Figure: The SC-32 transflective display executing multi-variable financial calculations with continuous stack preservation.

11.4.1 Base-Preserving Percentages (%), Chain Discounts & Gross Margin Pricing

To format currency outputs rounded to cents while retaining full 64-bit internal precision, set the display mode to two decimal places ([FIX] **2**) via [↶] [DISP] → [FIX] → **2** *(HP, 1982)*. On the StackCalc SC-32, the Percentage Function ([→] [%]) adheres strictly to William Kahan's non-destructive specification *(Kahan, 1979)*:

$$\text{Calculation}: \quad X \leftarrow \frac{Y \times X}{100}, \quad Y \leftarrow Y\,(\text{Preserved}), \quad \text{Lift} \leftarrow \text{TRUE} \qquad \textbf{(Eq. 11.19)}$$

Because [→] [%] leaves your original base amount intact in Register Y while placing the computed dollar adjustment directly beneath it in Register X (with Registers Z and T at zero), everyday commercial calculations flow in two natural moves:

- **Consulting Invoice Tax (`$150.00 + 8.25%`):** To add 8.25% sales tax to a $150.00 invoice, keying **150** [ENTER] **8.25** [→] [%] places the $12.38 tax in Register X while holding $150.00 in Register Y, so pressing [+] completes the $162.38 total in one stroke. Likewise, running **100** [ENTER] **8** [→] [%] on a $100.00 baseline leaves the exact dollar adjustment (**8**) in Register X while freezing 100.00 in Register Y.

- **Enterprise Server Purchase Order (`$18,450.00 + 8.25%`):** On an enterprise server order subtotaling $18,450.00 with 8.25% state sales tax ([FIX] **2**), keying **18450** [ENTER] **8.25** [→] [%] stages $18,450.00 in Register Y, computes the $1,522.13 sales tax in Register X, and sums the two registers to **$19,972.13** without retyping the base.

- **Single Volume Discount (`$85,000.00 - 17.5%`):** On an $85,000.00 software license carrying a 17.5% volume discount, **85000** [ENTER] **17.5** [→] [%] places the $14,875.00 concession in Register X directly beneath the $85,000.00 list price in Register Y, so subtracting settles the net invoice payable at **$70,125.00**.

- **Stacked Chain Markdowns (`$280.00 - 30% - 15%`):** When a 30% seasonal sale is followed by an extra 15% VIP coupon on a $280.00 field jacket, evaluating **280** [ENTER] **30** [→] [%] and subtracting the initial $84.00 markdown leaves the intermediate $196.00 sale price in Register X; applying the second 15% discount ($29.40) via **15** [→] [%] [-] then lands at $166.60, an effective 40.5% reduction rather than 45%.

- **Shared vs. Itemized Bill Splitting:** When 8 diners split a $480.00 restaurant tab where 3 guests shared $96.00 of wine while all 8 split the food equally, deducting the wine tab via **480** [ENTER] **96** [-] isolates the $384.00 food subtotal and divides by 8 to establish the $48.00 per-person food share in Register X; chaining **96** [ENTER] **3** [÷]

then stages the $32.00 per-drinker wine share in Register X beneath $48.00 in Y and adds them to finish the wine-drinker total ($80.00).

By contrast, comparing the Markup Trap against true Gross Margin Pricing on a product with landed cost C = $60.00 and target 40.0% gross margin activates all four stack registers so you can hold the flawed markup price in the upper registers while computing the true margin price below: adding a 40% markup onto $60.00 (**60** [ENTER] **40** [→] [%] [+] → **$84.00**) yields only $24 / $84 = 28.57% margin, whereas dividing cost by the margin complement Price = Cost / (1 - Margin) via **60** [ENTER] **1** [ENTER] **0.40** [-] [÷] resolves the true retail price **$100.00**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
60 [ENTER] **40** [→] [%]	0.00	0.00	60.00	24.00	40% markup on $60.00 cost: $24.00
[+]	0.00	0.00	0.00	84.00	Flawed markup price: $84.00 (28.57% margin)
60 [ENTER] **1** [ENTER]	84.00	60.00	1.00	1.00	True margin pricing: stage cost $60 in Z and 1.00 in Y
.40 [-]	84.00	84.00	60.00	0.60	1 - 0.40 = 0.60 margin complement; $60.00 drops to Y
[÷]	84.00	84.00	84.00	100.00	True gross margin retail price: $100.00

11.4.2 Percent Change (%CHG), Cross-Metric Elasticity, CAGR & Inflation

The Percent Change Function ([→] [%CHG]) is executed via Right Shift [→] followed by [%CHG] using the chronological order Old Value [ENTER] New Value [→] [%CHG], preserving the Old Value in Register Y and the New Value in **LASTX**.

In a quarterly SaaS operating review where Annual Recurring Revenue (**ARR**) grew from $12.0M to $15.6M, **12** [ENTER] **15.6** [→] [%CHG] returns +30.00%. Over the same period, Customer Acquisition Cost (**CAC**) expanded from $2.4M to $2.88M, where **2.4** [ENTER] **2.88** [→] [%CHG] returns +20.00%. Dropping the old baseline out of the way via [x⇄y] [R↓] aligns **ARR** growth in Register Y (30.00) and **CAC** growth in Register X (20.00) side-by-side on the stack. Pressing [-] then gives the **+10.00%** growth spread, while **30** [ENTER] **20** [÷] delivers the **1.5** bang-for-the-buck growth elasticity multiplier (1.50× revenue growth per 1× marketing spend growth):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
12 [ENTER] **15.6** [→] [%CHG]	0.00	0.00	12.00	30.00	ARR growth +30.00% in X; baseline 12.00 in Y
2.4 [ENTER] **2.88** [→] [%CHG]	12.00	30.00	2.40	20.00	CAC growth +20.00% in X; ARR growth lifts to Z
[x⇄y]	12.00	30.00	20.00	2.40	Swap CAC baseline 2.40 into X
[R↓]	2.40	12.00	30.00	20.00	Roll stack down to align 30.00% in Y and 20.00% in X
[-]	2.40	2.40	12.00	10.00	ARR vs. CAC growth spread +10.00 percentage points

To compute the n-year Compound Annual Growth Rate (**CAGR**), CAGR = $(FV\ /\ PV)^{1/n}$ - 1, the entire inside-out chain executes across Registers X and Y alone while Registers Z and T remain at zero. For a venture investment that expands from PV = $14.0M to FV = $88.5M over n = 5 years, keying **88.5** [ENTER] **14.0** [÷] **5** [x^{-1}] [y^x] **1** [-] **100** [×] forms the terminal wealth multiple 88.5 / 14.0 = 6.3214× in Register X, raises it to the reciprocal power 1/5 = 0.20 via [x^{-1}] and [y^x] to obtain the annualized growth factor 1.4460, and

subtracts unity before scaling by 100 to reveal **44.60%** (44.60%/yr; by the same five-year root, an equity share rising from \$4.20 to \$11.45 compounds at $(11.45 / 4.20)^{1/5} - 1 = 22.21\%$/yr).

Conversely, deflating \$100,000 of uninvested cash by 20 years of 3.2% annual CPI inflation ($PV = 100{,}000 / (1.032)^{20}$) via **100000** [ENTER] **1.032** [ENTER] **20** [y^x] [÷] reveals that real purchasing power erodes to **53260.60** (\$53,260.60). By the same exponential logic, translating a nominal 18.5% APR compounded monthly ($m = 12$) into its true Effective Annual Rate, EAR = $(1 + 0.185/12)^{12} - 1$, needs **12** typed only once when you recover the compounding frequency from [↶] [LASTx] right after dividing 0.185 / 12, which compounds the monthly factor $1.015417^{12} - 1$ to an effective annual yield of 20.15%.

11.4.3 The Five-Variable TVM Engine: Mortgages, Amortization & Balloon Notes

The onboard Five-Variable Time Value of Money ([↶] [TVM]) Engine is accessed via Left Shift [↶] above [x⇄y] (Row 7, Column 3):

TVM Variable	LFU Softkey	Actuarial Description	Computational Units
N	[N] ([LFU0])	Total number of compounding periods	Integer or fractional periods
***I*%**	[I%] ([LFU1])	Periodic interest rate per compounding period (i%)	Percent per period (e.g. **6.5** for 6.5%/yr or **6.75/12 = 0.5625** for monthly)
PV	[PV] ([LFU2])	Present value (initial loan principal or investment)	Currency (\$)
PMT	[PMT] ([LFU3])	Periodic annuity payment amount	Currency (\$ per period)
FV	[FV] ([LFU4])	Future value (final balloon balance or accumulated savings)	Currency (\$)
Mode	**END/BEG** ([LFU5])	Payment timing within each period	**END** (Ordinary Annuity) or **BEG** (Annuity Due)

Four hardware state conventions govern the [↶] [TVM] softkey interface:

1. **Store versus Solve Dispatch:** Tapping any [↶] [TVM] softkey immediately after numeric entry or an arithmetic operation stores Register X into that financial register; tapping a softkey immediately after another [↶] [TVM] softkey solves for that unknown variable into Register X.

2. **Periodic Interest Rate Scaling:** Softkey [I%] ([LFU1]) stores the periodic interest rate i% per compounding interval. For a monthly contract quoted at nominal annual rate APR, divide by 12 (**6.75** [ENTER] **12** [÷] → 0.5625%) before storing.

3. **Payment Timing Mode (END/BEG):** Softkey [LFU5] toggles between Ordinary Annuity (**END**, end-of-period cash flows) and Annuity Due (**BEG**, beginning-of-period cash flows).

4. **Register Persistence Across Scenarios:** All five [↶] [TVM] registers retain their stored parameters across calculations for sensitivity analysis; explicitly zero any unused variable (such as **0** [FV] or [↶] [CLEAR] → [CLVARS] on [LFU1]) before starting an unrelated contract.

Applying this five-variable sequence to a 30-year fixed residential mortgage of $PV = +\$350{,}000.00$ at 6.75% APR establishes the monthly payment and lifetime interest burden:

1. **Total Periods (`N = 360`):** Thirty years of monthly payments span $30 \times 12 = 360$ compounding periods → **`360`** [N].
2. **Monthly Periodic Rate (`I% = 6.75 / 12 = 0.5625`):** Divide the 6.75% annual rate by 12 months to get 0.5625% per month → **`.5625`** [I%].
3. **Present Value Loan Inflow (`PV = +350000`):** At closing, the lender wires +$350,000.00 into your transaction → **`350000`** [PV].
4. **Target Future Payoff Balance (`FV = 0`):** A fully amortizing mortgage pays the balance down to zero at Month 360 → **`0`** [FV].
5. **Solve for the Unknown Monthly Outflow (`PMT`):** With four variables stored, tap [PMT] without typing a number to solve for the monthly check (-$2,270.09):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
`360` [N]	0.00	0.00	0.00	360.00	Store term $N = 360$ months
`.5625` [I%]	0.00	0.00	360.00	0.56	Store monthly periodic rate $i = 0.5625\%$
`350000` [PV]	0.00	360.00	0.56	350000.00	Store loan principal inflow $PV = +\$350{,}000.00$
`0` [FV]	360.00	0.56	350000.00	0.00	Zero unused balloon register $FV = \$0.00$ before solving
[PMT]	360.00	0.56	350000.00	-2270.09	Solve monthly mortgage outflow $PMT =$ -$2,270.09
`360` [×]	0.56	0.56	350000.00	-817233.60	Total 30-year payments -$817,233.60
`350000` [+]	0.56	0.56	350000.00	-467233.60	Total lifetime interest paid -$467,233.60

Dissecting the Month 1 interest/principal split requires only Registers X and Y while Registers Z and T stay at zero (noting that Truth-in-Lending cent-rounding from -$2,270.0934 adjusts the Month 360 payoff by a few cents): in Month 1, the bank assesses one periodic rate ($i = 0.0675 \,/\, 12 = 0.005625$) against the $350,000.00 balance. Keying **`350000`** [ENTER] **`0.005625`** [×] computes Month 1 interest $I_1 = \$1{,}968.75$ in Register X—consuming 86.7% of the $2,270.09 check before a single cent touches the debt. Deducting that interest via **`2270.09`** [x⇄y] [-] isolates the Month 1 principal reduction $P_1 = \$301.34$ (13.3%), and subtracting P_1 via **`350000`** [x⇄y] [-] leaves $349,698.66 owed after Month 1. By the crossover formula $k_{\text{cross}} = 361 - \ln 2 \,/\, \ln(1.005625) = 237.43$, Month 238 is the first integer month where principal overtakes interest:

Month	Remaining Balance	Monthly Payment	Interest (I_k)	Principal (P_k)	Equity %	Financial Milestone
1	$349,698.66	$2,270.09	$1,968.75	$301.34	0.09%	86.7% of check goes straight to interest
Month 238 *(Principal Crossover)*	$199,998.44	$2,270.09	$1,131.40	$1,138.70	42.86%	**Amortization Crossover:** P_k finally overtakes I_k
360	$0.00	$2,270.09	$12.70	$2,257.39	100.00%	Final payoff; $467,233.60 total lifetime interest

Because the **TVM** registers retain your loan parameters, testing an extra $250.00/month principal curtailment means storing the augmented outflow $PMT = -\$2{,}520.09$/mo and tapping [N] (**`6.75`** [ENTER] **`12`** [÷] [I%YR] **`350000`** [PV] **`0`** [FV] **`2520.09`** [+/−] [PMT] [N]), which

compresses the repayment horizon to **270.93** months (22.6 years, saving 89 payments and $134,474.66 in interest):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
6.75 ENTER **12** ÷ I%	0.00	0.00	0.00	0.56	Monthly rate i = 0.5625%
350000 PV **0** FV	0.00	0.56	350000.00	0.00	Store PV = $350,000.00 and zero FV = $0.00 before solving
2520.09 +/- PMT	0.56	350000.00	0.00	-2520.09	Augmented payment: -$2,520.09/mo
N	0.56	350000.00	0.00	270.93	Accelerated repayment: 270.93 months

Next, take a commercial 5-year balloon note lending $500,000 at 7.5% APR ($i = 0.625\%$/month), amortized on a 25-year schedule ($N = 300$ months) with a balloon maturity at Month 60. Because the monthly check is sized as if the loan ran for 300 months to $FV = 0$, first solve for PMT at $N = 300$ via **300** N **7.5** ENTER **12** ÷ I%YR **500000** PV **0** FV PMT → **-3,694.95589**, and then shorten the horizon to **60** N and tap FV to read off the remaining principal payoff due at Month 60 (**-$458,662.75**):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
300 N	0.00	0.00	0.00	300.00	Store 25-year amortization schedule $N = 300$ months
.625 I%	0.00	0.00	300.00	0.63	Store monthly periodic rate 7.5% / 12 = 0.625%
500000 PV	0.00	300.00	0.63	500000.00	Store commercial loan principal PV = +$500,000.00
0 FV	300.00	0.63	500000.00	0.00	Target $FV = 0$ over full 300-month amortization
PMT	300.00	0.63	500000.00	-3694.96	Monthly payment PMT = -$3,694.96
60 N	0.63	500000.00	-3694.96	60.00	Set horizon to 5-year maturity (N = 60 months)
FV	0.63	500000.00	-3694.96	-458662.75	Remaining 5-year balloon payoff FV = -$458,662.75

Every ↶ TVM calculation can also be cross-checked from the closed-form annuity identities on the 4-level stack:

Financial Instrument	Analytical Closed-Form Formula	Native RPN Inside-Out Recipe
Ordinary Annuity PV	$PV = PMT\,[\,(1 - (1+i)^{-N}) \,/\, i\,]$	**1 ENTER i + N +/- y^x 1 x↔y - i ÷ PMT ×**
Ordinary Annuity FV	$FV = PMT\,[\,((1+i)^{N} - 1) \,/\, i\,]$	**1 ENTER i + N y^x 1 - i ÷ PMT ×**
Amortization Payment	$PMT = PV\,[\,i \,/\, (1 - (1+i)^{-N})\,]$	**1 ENTER i + N +/- y^x 1 x↔y - i x↔y ÷ PV ×**
Sinking Fund Payment	$PMT = FV\,[\,i \,/\, ((1+i)^{N} - 1)\,]$	**1 ENTER i + N y^x 1 - i x↔y ÷ FV ×**

11.4.4 Bond Yield to Maturity (YTM), Duration, NPV, IRR & Portfolio Risk

For a 10-year corporate bond (N = 20 semi-annual periods) with par value FV = +\$1,000.00 and a 5.5% annual coupon paid semi-annually (PMT = +\$27.50) purchased at PV = -\$925.00, storing N, PV, PMT, FV and pressing [I%] finds the semi-annual yield 3.2666%, which doubles (**2** [×]) to the Wall Street Bond-Equivalent Yield BEY = $2i$ = 6.53% (or compounds via $(1 + i)^2$ - 1 = 6.64% as an Effective Annual Yield EAY):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
20 [N]	0.00	0.00	0.00	20.00	Store N = 20 semi-annual coupon periods
925 [+/−] [PV]	0.00	0.00	20.00	-925.00	Store bond purchase price outflow PV = -\$925.00
27.50 [PMT]	0.00	20.00	-925.00	27.50	Store semi-annual coupon inflow PMT = +\$27.50
1000 [FV]	20.00	-925.00	27.50	1000.00	Store maturity par redemption FV = +\$1,000.00
[I%]	20.00	-925.00	27.50	3.27	Solve semi-annual periodic yield i = 3.2666%
2 [×]	-925.00	-925.00	27.50	6.53	Annualized bond-equivalent YTM = 6.53%

To evaluate Macaulay Duration D and Modified Duration D^* for a 3-year 8.0% annual coupon bond trading at par of \$1,000.00 with yield y = 8.0%, accumulate the time-weighted present values $\sum t \cdot C_t / (1.08)^t$ and divide by price and $(1 + y)$:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
80 [ENTER] **1.08** [÷]	0.00	0.00	0.00	74.0741	Year 1 weighted PV: 1 × 80 / 1.08 = \$74.0741
80 [ENTER] **1.08** [ENTER] **2** [y^x] [÷] **2** [×] [+]	74.07	74.07	74.07	211.2483	Add Year 2 weighted PV (2 × 80 / 1.08^2 = \$137.1742)
1080 [ENTER] **1.08** [ENTER] **3** [y^x] [÷] **3** [×] [+]	211.25	211.25	211.25	2783.2648	Add Year 3 weighted PV (3 × 1080 / 1.08^3 = \$2,572.02)
1000 [÷]	211.25	211.25	211.25	2.7833	Divide by \$1,000 price for Macaulay Duration D = 2.7833 yr
1.08 [÷]	211.25	211.25	211.25	2.5771	Divide by 1.08 for Modified Duration D^* = 2.5771 yr

To discount a 4-year corporate free-cash-flow stream of \$10M, \$12M, \$15M, and \$234M at a 10% cost of capital (Enterprise Value = $10/1.1 + 12/1.1^2 + 15/1.1^3 + 234/1.1^4$), running **10** [ENTER] **1.1** [÷] **12** [ENTER] **1.1** [↶] [x^2] [÷] [+] **15** [ENTER] **1.1** [ENTER] **3** [y^x] [÷] [+] **234** [ENTER] **1.1** [ENTER] **4** [y^x] [÷] [+] accumulates the four present values to **190.10313503** (\$190.10M):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
10 [ENTER] **1.1** [÷]	0.00	0.00	0.00	9.09	Year 1 discounted FCF: 10 / 1.10 = \$9.09M
12 [ENTER] **1.1** [↶] [x^2] [÷]	0.00	0.00	9.09	9.92	Year 2 discounted FCF: 12 / 1.10^2 = \$9.92M
[+]	0.00	0.00	0.00	19.01	Cumulative PV through Year 2: \$19.01M
15 [ENTER] **1.1** [ENTER] **3** [y^x] [÷]	19.01	19.01	19.01	11.27	Year 3 discounted FCF: 15 / 1.10^3 = \$11.27M

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[+]	19.01	19.01	19.01	30.28	Cumulative PV through Year 3: $30.28M
234 [ENTER] **1.1** [ENTER] **4** [y^x] [÷]	30.28	30.28	30.28	159.83	Year 4 discounted FCF + terminal value: $159.83M
[+]	30.28	30.28	30.28	190.10	Total Enterprise Value PV = $190.10M

(Horner Nested Discounting via Register T: Pre-filling the stack with **1.1** [ENTER] [ENTER] [ENTER] *and alternating* [x⇄y] [÷] *with each cash-flow addition from Year 4 (234) down to Year 1 (10) evaluates $190.10M with* **1.1** *typed only once and zero* [y^x] *exponentiations.)*

To solve for the Internal Rate of Return (**IRR**) of an upfront outlay C_0 = -$100,000 returning C_1 = $30,000, C_2 = $40,000, and C_3 = $50,000, store the Net Present Value polynomial ((50000 / X + 40000) / X + 30000) / X - 100000 in [↗] [EQN] mode using Horner RPN tokens **50000 X ÷ 40000 + X ÷ 30000 + X ÷ 100000 -**, where $X = 1 + r$ is the gross annual discount factor. Because Richard Brent's root-finder evaluates the stored equation internally, passing a two-guess bracket **1.0** [ENTER] **1.2** (or seeding **1.1** [STO] **X**) and invoking [↗] [SOLVE] pushes the converged zero-NPV root $X = 1.088963$ into Register X (lifting the prior contents of Register X into Y and preserving the upper stack in Z and T). Subtracting unity and multiplying by 100 then extracts the project's exact internal rate of return, IRR = 8.90%/yr (8.8963%).

Finally, two classic metrics quantify risk-adjusted return and position sizing on the stack. William Sharpe's Ratio $(R_p - R_f)/\sigma_p$ measures the excess return earned above a risk-free Treasury bill per unit of volatility: for $R_p = 18.5\%$ against $R_f = 4.5\%$ with volatility $\sigma_p = 11.2\%$, **18.5** [ENTER] **4.5** [-] **11.2** [÷] yields **1.25** ((18.5 - 4.5) / 11.2 = 1.25). John Kelly's 1956 Criterion $f^* = (bp - q) / b = p - (1-p) / b$ determines the fraction of capital to wager on a favorable repeat bet to maximize long-run logarithmic growth without risking ruin: for win probability $p = 0.56$ ($q = 0.44$) and odds $b = 1.50$, **1.5** [ENTER] **0.56** [×] **0.44** [-] **1.5** [÷] **100** [×] yields **26.67%** (26.67%, with [↶] [LASTx] recalling 0.56; in practice, portfolio managers often deploy "Half-Kelly" $(1/2)f^* \approx 13.3\%$ to halve drawdown volatility while retaining three-quarters of the compound growth rate).

Chapter 12: Analytic Geometry, Composite Areas & Integral Mensuration

12.1 René Descartes's Coordinate Plane, Gauss's Shoelace Theorem & Analytic Loci

12.1.1 Algebraic Equations as Geometric Loci

During his years as a boarding student at the College of La Flèche in Anjou, René Descartes was granted an unusual privilege by the rector, Father Charlet: while his classmates rose at five o'clock every morning to recite Latin prayers in unheated stone halls, young René was allowed to remain warm under his blankets until late morning, thinking quietly with his eyes open.

Descartes kept that habit of morning meditation for the rest of his life. Observing the orthogonal grid of ceiling tiles and wall joints in his room—and investigating Pappus of Alexandria's classical locus problem—Descartes considered how to specify the exact position of a moving point on a flat plane at any instant, and how to measure the geometric region enclosed by its path.

He realized that if he measured the perpendicular distance x from a horizontal reference axis and the perpendicular distance y from a second reference axis, every position on the plane corresponded to a unique pair of numbers (x, y). If the point moved along a straight diagonal line, a polygonal boundary, or a smooth parabolic arch, its two distances x and y obeyed a strict algebraic equation; conversely, *any* algebraic equation connecting x and y defined a geometric boundary across the plane whose intercepts, peaks, enclosed area, and surface of revolution could be calculated directly from its coordinates.

When Descartes published this discovery in 1637 as *La Géométrie (Descartes, 1637)*, an appendix to his *Discourse on the Method*, he ended a two-thousand-year separation that had divided mathematical thought since antiquity. Ancient Greek geometry, as practiced by Euclid, Archimedes, and Apollonius, was rich in spatial intuition with circles, triangles, parabolas, and tangent lines drawn by straightedge and compass, yet it possessed almost no symbolic algebra, so proving a new mensuration theorem required inventing a custom geometric dissection from scratch. Conversely, medieval and Renaissance algebra, developed by al-Khwarizmi, Tartaglia, Cardano, and Viète, provided a symbolic calculus for balancing unknowns (x^2, x^3), yet an equation sat on the page as an abstract string of characters divorced from spatial area and volume.

Descartes proved that algebra and geometry are two languages describing one underlying reality. John Stuart Mill later called analytic geometry *"the greatest single step ever made in the progress of the exact sciences,"* because it allowed every geometric boundary, cross-sectional area, and three-dimensional shell to be analyzed with the full precision of numerical algebra.

12.1.2 Reading the Anatomy of a Boundary Curve: Roots, Peaks, Valleys & Inflection Points

Once a physical boundary or structural profile $y = f(x)$ is expressed across a horizontal abscissa x and a vertical ordinate y, stored on the SC-32 inside the [→] [EQN] bank ([→] + [STO]) *(HP, 1991)*, every geometric landmark that bounds an enclosed area corresponds to an exact algebraic condition on $f(x)$ and its derivatives.

Hunting for the roots or zeros of a boundary equation means solving for the abscissas x where $f(x) = 0$ via [→] [SOLVE] *(Brent, 1973)*. In planar mensuration and integral calculus, a root is a shoreline where the profile crosses the reference axis ($y = 0$) and switches from negative elevation ($y < 0$, below the datum) to positive elevation ($y > 0$, above the datum). Whenever an engineer computes the total unsigned area between a wave or beam profile and the x-axis, locating the exact zero crossings first via [→] [SOLVE] is essential so that positive lobes above the axis and negative lobes below the axis do not cancel each other out during integration.

Along a smooth boundary curve, the first derivative $f'(x)$ measures the local tangent slope: along a rising ramp, $f'(x) > 0$, and along a descending slope, $f'(x) < 0$. At the exact crown of an arch (*local maximum*) or the bottom of a sag trough (*local minimum*), the tangent line is momentarily horizontal ($f'(x) = 0$).

An *inflection point* marks a reversal in the sign of curvature along a structural profile, such as an S-shaped highway transition ramp or a reverse-curve viaduct arch modeled by a general cubic polynomial:

$$y(x) = ax^3 + bx^2 + cx + d, \qquad y''(x) = 6ax + 2b \tag{Eq. 12.1}$$

Wherever the second derivative is negative ($y''(x) < 0$), the tangent slope decreases with x and the arch bends downward (*concave down*); wherever the second derivative is positive ($y''(x) > 0$), the tangent slope increases with x and the trough bends upward (*concave up*). At the unique abscissa $x_{\text{infl}} = -b/3a$ where $y''(x_{\text{infl}}) = 0$, the bending moment passes through zero as the profile reverses its direction of curvature, forming the exact geometric center of symmetry between the cubic's local peak and local valley.

Vertical and horizontal asymptotes govern the bounding envelopes of rational and exponential profiles. When a rational denominator approaches zero, such as the thin-lens magnification law $f(x) = 1 / (x - 2)$ as an object crosses the focal point $x = 2$, the ordinate diverges toward $\pm\infty$ along a *vertical asymptote*. Conversely, when a falling body approaches terminal velocity $v(t) = v_{\text{term}}\tanh(gt/v_{\text{term}})$, or when an uncharged capacitor charges through a resistor toward supply voltage $V_{\max}$ with RC time constant τ:

$$V(t) = V_{\max}(1 - e^{-t/\tau}) \tag{Eq. 12.2}$$

the profile approaches a flat horizontal ceiling—a *horizontal asymptote* at $V = V_{\max}$ as $t \to \infty$—and the area trapped *between* the charging curve and that horizontal ceiling, $\int_0^{\infty} [V_{\max} - V(t)]\, dt = V_{\max}\tau$, measures the total magnetic flux-linkage or charge deficit required to reach steady state.

12.1.3 Carl Friedrich Gauss's Shoelace Theorem & the Trapezoidal Surveyor's Ledger

Before turning to curved boundaries that require integral calculus, Descartes's coordinate plane provides an exact algebraic method for computing the area of *any* flat

polygon—however irregular—directly from the (x_k, y_k) coordinates of its corners.

Suppose a civil engineer or land surveyor records the n boundary monuments (x_1, y_1), (x_2, y_2), ..., (x_n, y_n) in order around the perimeter of a polygonal parcel, closing the loop back to $(x_{n+1}, y_{n+1}) = (x_1, y_1)$ (whether counterclockwise in math coordinates (x, y) or clockwise in surveyor coordinates (N, E), since taking the absolute value $|A|$ at the end makes the area invariant to traversal direction). As Carl Friedrich Gauss formalized in his geodetic survey lectures, dropping vertical plumb lines from each boundary segment $(x_k, y_k) \rightarrow (x_{k+1}, y_{k+1})$ down to the x-axis partitions the parcel into a signed ring of vertical trapezoids of width $\Delta x_k = x_k - x_{k+1}$ and mean altitude $\bar{y}_k = 1/2(y_k + y_{k+1})$. Expanding and canceling the symmetric $x_k\, y_k$ terms yields *Gauss's Shoelace Formula* (also known as Meister's 1769 determinant formula):

$$A = \frac{1}{2}\left|\sum_{k=1}^{n}(x_k y_{k+1} - x_{k+1} y_k)\right| = \frac{1}{2}\left|\sum_{k=1}^{n}(x_k - x_{k+1})(y_k + y_{k+1})\right| \qquad \textbf{(Eq. 12.3)}$$

Both expressions evaluate cleanly on the SC-32 by coupling the 4-register stack to the statistical accumulator [Σ+] (and [↶] [Σ−]). For each directed edge from vertex k to vertex k+1, you form the 2 × 2 cross-product $x_k\, y_{k+1} - x_{k+1}\, y_k$ (or the doubled trapezoid $(x_k - x_{k+1})(y_k + y_{k+1})$) on Registers Y and X and press [Σ+]. Register X increments the edge counter $n = 1, 2, ..., N$ so you never lose your place along the survey traverse, and pressing [↷] [SUMS] → [Σ×] followed by **2** [÷] [ABS] delivers the exact enclosed area of the polygon.

12.2 Composite Shapes, Cutout Apertures & Surface Area Takeoffs in STAT Mode

12.2.1 Additive and Subtractive Dissection of Composite Cross-Sections (Σ+ and Σ−)

In structural, mechanical, and architectural engineering, real-world cross-sections and three-dimensional enclosures are rarely single textbook primitives. A bridge gusset plate, an aircraft bulkhead rib, or a cast-aluminum gearbox housing is a *composite region* built by welding or extruding simple outer shapes (rectangles, right triangles, trapezoids, circular sectors, and parabolic spandrels) and then milling out interior lightening holes, bolt bores, and cooling slots.

Rather than writing down intermediate sub-areas on scrap paper—which invites transcription errors and rounding drift—the SC-32's statistical registers ([Σ+], [↶] [Σ−], and [↷] [SUMS]) act as an dedicated *additive and subtractive mensuration ledger*:

1. **Zero the Ledger ([CLΣ]):** Press [↶] + [Σ+] ([CLΣ]) to clear all six statistical registers ($\sum x = \sum y = \sum x^2 = \sum y^2 = \sum xy = n = 0$).

2. **Accumulate Solid Sub-Regions ([Σ+]):** Evaluate the area A_k of each positive sub-shape (or the surface area of each exterior wall group) in Register X and press [Σ+]. The calculator adds A_k into $\sum x$ at full 15-digit internal precision and displays the running feature count $n = 1, 2, 3, ...$ in Register X.

3. **Deduct Cutouts, Bores & Apertures ([↶] [Σ−]):** Whenever a blueprint specifies a drilled bolt circle, a rectangular inspection window, or a chamfered corner cutout, evaluate its positive area A_{cutout} in Register X and press Left Shift [↶] + [Σ+] ([↶] [Σ−] via the primary Σ**-** shifted key, or negate with [+/−] before [Σ+]). Using [↶] [Σ−] subtracts A_{cutout} directly from $\sum x$ (and decrements n), whereas using [+/−] [Σ+] subtracts the cutout area from $\sum x$ while *incrementing* the audited feature count n so that n

equals the total number of blueprint features verified.

4. **Recall Net Area & Convert to Mass or Coating Volume ([→] [SUMS] → [Σx]):** Press Right Shift [→] + [Σ+] ([→] [SUMS]) and tap softkey [Σx] ([LFU0]) to place the net composite area $A_{\text{net}} = \sum A_{\text{solid}} - \sum A_{\text{cutout}}$ onto Register X, ready to multiply by plate thickness t and material density ρ for structural mass, or by paint/anodizing film thickness for surface treatment takeoffs.

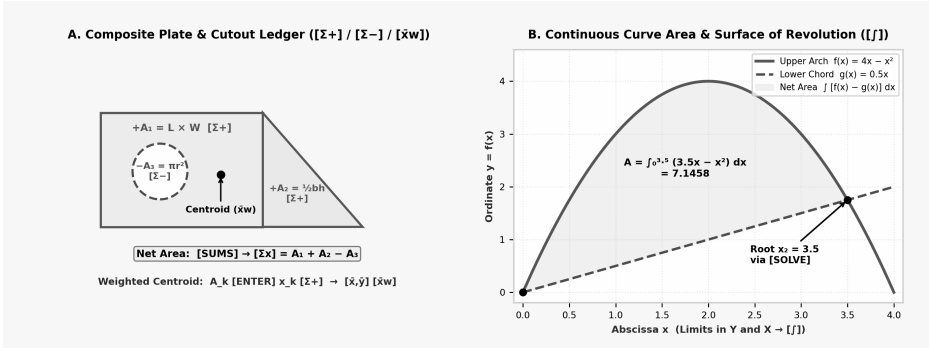


Figure: Composite plate and cutout dissection via Σ+ / Σ− and x̄w paired with continuous definite-integral area between curves via SOLVE and ∫.

12.2.2 Simultaneous Net Area & Composite Centroid Auditing (Σy and x̄w)

When designing an eccentric machine bracket, a retaining wall cross-section, or a ship's hull frame, knowing the total cross-sectional area A_{net} is only half the problem: you must also locate the *area centroid* $(\bar{x}, \bar{y})$ through which the resultant axial load or hydrostatic force acts, and about which the section's bending stiffness (second moment of area $I = \sum (I_{c,k} + A_k\, d_k^2)$) is referenced.

Because [Σ+] is a *bivariate* accumulator that processes Register Y and Register X together, the SC-32 can compute **both the net composite area and the exact composite centroid in a single pass** without typing a single sub-area twice:

$$\bar{x} = \frac{\sum_{k=1}^{m} A_k x_k}{\sum_{k=1}^{m} A_k} = \frac{\Sigma xy}{\Sigma y}, \qquad A_{\text{net}} = \sum_{k=1}^{m} A_k = \Sigma y \tag{Eq. 12.4}$$

Here is how the two-register mapping works on the stack:

- For each solid sub-shape k, place its **area weight A_k in Register Y** and its **centroid coordinate x_k in Register X** (A_k [ENTER] x_k [Σ+]), since the SC-32's weighted mean [x̄w] divides $\sum xy$ / $\sum y$ using Register Y as the weight and Register X as the coordinate.
- For a cutout hole or slot of area A_{hole} centered at x_{hole}, either place the negative area $-A_{\text{hole}}$ ([+/−]) in Register Y and x_{hole} in Register X before pressing [Σ+], or place $+A_{\text{hole}}$ in Register Y and x_{hole} in Register X and press [↶] [Σ−]! Both paths subtract A_{hole} from $\sum y$ and subtract the cutout's first moment $A_{\text{hole}}\, x_{\text{hole}}$ from $\sum xy$.

- When all sub-shapes and cutouts have been logged:

1. Open [↷] [SUMS] ([↷] + [Σ+]) and tap [Σy] ([LFU1]) to recall the **net composite area** $A_{\text{net}} = \sum A_k$ into Register X.
2. Open [↷] [SUMS] and tap [Σxy] ([LFU4]) if you need the raw **first moment of area** $Q = \sum A_k\, x_k$ (used in beam shear-flow formula $\tau = VQ/It$).
3. Open [↷] [x̄,ȳ] ([↷] + [$y^{\hat{}}$]) and tap [x̄w] ([LFU2]) to compute the **composite centroid** $\bar{x} = \sum xy \,/\, \sum y$ in a single keystroke!

12.2.3 Classical Composite Primitives & Archimedes' Parabolic Spandrels

To dissect curved as well as polygonal plates using [Σ+] and [↶] [Σ−], Table 12.1 catalogs the exact area A and centroid coordinates $(\bar{x}, \bar{y})$ of the fundamental engineering sub-regions (including Archimedes' $(2/3)bh$ parabolic arch and its complementary $(1/3)bh$ parabolic spandrel, which allow curved architectural fillets and streamlined fairings to be audited inside **STAT** mode alongside rectangles and circles):

Sub-Region Primitive	Enclosed Area A	Centroid Location $(\bar{x}, \bar{y})$	Mensuration Rule & Stack Entry Idiom
Rectangle ($b \times h$)	$A = b\,h$	$(b/2,\ h/2)$	From corner origin; **b ENTER h ×**
Right Triangle ($b \times h$)	$A = (1/2)\ b\,h$	$(b/3,\ h/3)$	One-third of leg lengths from the 90° corner
Trapezoid ($a \parallel b$, height h)	$A = ((a + b)\ /\ 2)\ h$	$\bar{y} = (h/3)((2a + b)\ /\ (a + b))$	Height above base b (where a is top width)
Full Circle / Bore (r)	$A = \pi\ r^2$	$(x_c,\ y_c)$	Deduct drilled bores via **r x² π × Σ-**
Semicircle (radius r)	$A = (1/2)\pi\ r^2$	$\bar{y} = 4r/3\pi \approx 0.424413\ r$	Measured perpendicular from flat diameter base
Quarter-Circle Fillet (r)	$A = (1/4)\pi\ r^2$	$\bar{x} = \bar{y} = 4r/3\pi \approx 0.424413\ r$	Measured from the two orthogonal straight radii
Circular Sector (angle 2α rad)	$A = \alpha\ r^2$	$\bar{x} = 2r \sin\alpha\ /\ 3\alpha$	Along symmetry bisector from vertex (α in rad)
Parabolic Arch ($y = h(1 - x^2/b^2)$, $x \in [-b,b]$)	$A = 2/3\ (2b)\ h$	$\bar{y} = (2/5)\ h = 0.4000\ h$	Archimedes' 2/3 rule across full span $2b$
Parabolic Spandrel ($y = (h/b^2)x^2$)	$A = (1/3)\ b\,h$	$((3/4)b,\ (3/10)h)$	Complementary corner fillet outside a parabola

12.2.4 Francis Anscombe's 1973 Quartet: Auditing Σ+ Residuals on the Stack

The same six statistical registers ($\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n) that accumulate composite areas and centroids also drive Carl Friedrich Gauss's method of least squares ([↷] [L.R.] above [LN]). When an engineer feeds $(x_i,\ y_i)$ calibration pairs—such as surface area versus convective heat dissipation, or pressure versus transducer voltage—into [Σ+], the [↷] [L.R.] menu distills the cloud into the linear regression slope m ([m]), vertical intercept b ([b]), and Karl Pearson's correlation coefficient r ([r]).

In 1973, British statistician Francis John Anscombe, founder of Yale University's Department of Statistics, published a landmark paper in *The American Statistician* titled *"Graphs in Statistical Analysis"* demonstrating why an analyst must never accept m, b, and r blindly without auditing the individual data residuals. Anscombe constructed four small

datasets (*Anscombe's Quartet*), each consisting of $n = 11$ coordinate pairs (x_i, y_i), whose summary statistics are identical to three decimal places:

Statistical Metric	Dataset I	Dataset II	Dataset III	Dataset IV
Sample Size (n)	11	11	11	11
Mean of x ($\bar{x}$) & Variance (s_x^2)	$\bar{x} = 9.0$, $s_x^2 = 11.0$	$\bar{x} = 9.0$, $s_x^2 = 11.0$	$\bar{x} = 9.0$, $s_x^2 = 11.0$	$\bar{x} = 9.0$, $s_x^2 = 11.0$
Mean of y ($\bar{y}$) & Variance (s_y^2)	$\bar{y} = 7.50$, $s_y^2 = 4.125$	$\bar{y} = 7.50$, $s_y^2 = 4.125$	$\bar{y} = 7.50$, $s_y^2 = 4.125$	$\bar{y} = 7.50$, $s_y^2 = 4.125$
Least-Squares Line ($\hat{y} = m\,x + b$)	$\hat{y} = 0.500x + 3.00$	$\hat{y} = 0.500x + 3.00$	$\hat{y} = 0.500x + 3.00$	$\hat{y} = 0.500x + 3.00$
Correlation (r) & Determination (R^2)	$r = 0.816$, $R^2 = 0.667$	$r = 0.816$, $R^2 = 0.667$	$r = 0.816$, $R^2 = 0.667$	$r = 0.816$, $R^2 = 0.667$

Despite sharing $\hat{y} = 0.500x + 3.00$ and $r = 0.816$, the four datasets represent completely different physical mechanisms:

- **Dataset I (Genuine Noisy Linear Trend):** Eleven measurements scattered naturally around a straight line, with small, unpatterned residuals $e_i = y_i - \hat{y}(x_i)$ of alternating sign across the domain.
- **Dataset II (Smooth Quadratic Parabola):** A noiseless parabolic arch $y = -0.125x^2 + 2.75x - 6.00$ that rises to a peak at $x = 11.0$ ($y = 9.13$) and bends back down at $x = 14.0$ ($y = 8.10$).
- **Dataset III (Exact Line Plus One Corrupted Outlier):** Ten points lying on a tight straight line of slope $m = 0.345$, plus a single corrupted sensor reading at (13.0, 12.74) that wrenches the least-squares slope up to 0.500.
- **Dataset IV (Vertical Cluster Plus One Leverage Point):** Ten points stacked vertically at $x = 8.0$ with zero correlation, plus one isolated point at (19.0, 12.50) that single-handedly fabricates $r = 0.816$.

On a non-graphing scientific instrument like the SC-32, you unmask all three pathologies in seconds using a **Three-Point Stack Residual Audit** via [↷] [L.R.] → [ŷ] ([LFU0]) and [↶] [REGS] ([↶] + [0], **DATA** tab):

1. **Check the Endpoints and Midpoint via [ŷ]:** After accumulating your pairs with [Σ+], evaluate the linear prediction $\hat{y}$ at the left endpoint ($x = 4$), the midpoint ($x = 11$), and the right endpoint ($x = 14$) by keying **x** [↷] [L.R.] [ŷ] and subtracting from the measured y_i. In Dataset II, the residuals $e_i = y_i - \hat{y}_i$ at $x = (4, 11, 14)$ follow a systematic **negative–positive–negative arch** (-1.89, +0.63, -1.90), immediately diagnosing concave-down quadratic curvature ($y'' < 0$) without drawing a single pixel!
2. **Test Log/Power Linearization or Remove a Suspect Outlier via [↶] [Σ−]:** In Dataset III, inspecting the raw pairs in the **DATA** tab of [↶] [REGS] (indirect slots **40**–**89**) spots the spike at (13.0, 12.74). Keying **12.74** [ENTER] **13** [↶] [Σ−] removes that single corrupted pair from the accumulators in place, and tapping [↷] [L.R.] → [r] watches the correlation jump from $r = 0.816$ to $\boldsymbol{r} = \mathbf{0.99999}$, proving that the remaining ten measurements obey a pure linear law.

12.3 Tying Definite Integration (∫) to Planar Area, Cross-Sections & Surface Area of Revolution

12.3.1 From Discrete Σ+ Sums to Continuous Romberg Quadrature (∫)

Whereas [Σ+] and [↶] [Σ−] sum the areas of discrete polygons, circles, and parabolic spandrels, continuous boundaries—such as hydrofoil ribs, sinusoidal corrugated roofing, Gaussian probability bells, and exponentially flared horn reflectors—require summing infinitely many infinitesimal vertical strips of width *dx*.

As established in Chapter 8, Werner Romberg's adaptive quadrature engine ([→] [∫], Right Shift [→] above [8]) *(Romberg, 1955; Kahan, 1979)* automates this transition from discrete summation to continuous mensuration:

1. Store the boundary profile $f(x)$ in the [→] [EQN] bank ([→] + [STO]) as a pure RPN token stream, and press [ENTER] (or [XEQ] / **FN=** if selecting from multiple equations).
2. Stage the lower abscissa bound a in Register Y and the upper abscissa bound b in Register X via **a** [ENTER] **b**.
3. Press [→] [∫] and tap the dummy variable softkey (**X**, **T**, etc.). Upon convergence, the SC-32 places the **certified area $A = \int_a^b f(x)\,dx$ in Register X** and the **uncertainty bound $\pm\varepsilon$ in Register Y**, while leaving Registers Z and T completely untouched!

Because Registers Z and T are preserved across every [→] [∫] execution, you can **combine discrete geometric areas and continuous integrals directly on the stack or inside** [Σ+]: evaluate a curved fairing or arch area via **a** [ENTER] **b** [→] [∫] **X**, press [Σ+] to log that curved area into $\sum x$, and then add rectangular base plates or deduct circular bolt holes via [↶] [Σ−] in the very same statistical ledger!

12.3.2 Area Between Intersecting Curves & Root-Bracketed Mensuration (SOLVE → ∫)

When a planar cross-section is bounded above by an upper curve $y_{\text{upper}} = f(x)$ and below by a lower chord or curve $y_{\text{lower}} = g(x)$—such as a crescent-shaped turbine blade profile, a meniscus lens cross-section, or an arched masonry rib over a sloped ramp—the vertical height of the strip at any abscissa x is the difference $h(x) = f(x) - g(x)$. The net enclosed area between their intersection points x_1 and x_2 is:

$$A = \int_{x_1}^{x_2} [f(x) - g(x)]\,dx \qquad \textbf{(Eq. 12.5)}$$

Notice that the integrand $h(x) = f(x) - g(x)$ serves a **dual role** on the SC-32:

1. **The Intersection Condition ($h(x) = 0$):** At the left and right corners where the upper and lower curves meet, their vertical separation vanishes: $f(x) - g(x) = 0$.
2. **The Area Integrand ($\int h(x)\,dx$):** Between those two corners, integrating that *exact same equation* $h(x) = f(x) - g(x)$ accumulates the enclosed cross-sectional area!

This duality creates a closed **Solve-then-Integrate workflow** using a single [→] [EQN] entry:

- **Step 1 (Define the Difference Profile once in [→] [EQN]):** Enter $h(x) = f(x) - g(x)$ into [→] [EQN]. For example, to find the area trapped between the parabolic arch $f(x) = 4x - x^2$ and the inclined chord $g(x) = 0.5x$ (shown in Panel B of the concept diagram above), enter $h(x) = 4x - x^2 - 0.5x = 3.5x - x^2$ in Horner RPN form as **3.5 X - X ×**.
- **Step 2 (Find the Intersection Bounds via [→] [SOLVE]):** The left intersection sits at the origin $x_1 = 0$. To find the right intersection x_2 near **[3, 4]**, key **3** [ENTER] **4** [→] [SOLVE] **X**, which refines the exact intersection root $x_2 = \mathbf{3.500000}$ into Register X.
- **Step 3 (Integrate Across the Exact Root via [x⇄y] and [→] [∫]):** With $x_2 = 3.500000$ already sitting in Register X, simply key **0** [x⇄y] [→] [∫] **X** to place ($Y = 0$, $X = 3.5$) on the stack and integrate the *same* active equation $3.5x - x^2$ from 0 to 3.5. Romberg's first Simpson column $R(1,1)$ returns the exact enclosed area:

$$A = \int_0^{3.5} (3.5x - x^2)\,dx = \left[\tfrac{3.5}{2}x^2 - \tfrac{1}{3}x^3\right]_0^{3.5} = \tfrac{3.5^3}{6} = \tfrac{343}{48} \approx \mathbf{7.145833} \qquad \textbf{(Eq. 12.6)}$$

And if you press [↖] [FDISP] ([↖] + [•]), the SC-32 reduces **7.1458333333** into the exact rational fraction `7 7/48` (343/48)!

12.3.3 Arc Length & Surface Area of Revolution via ∫

Integral calculus also extends Chapter 6's solid mensuration formulas to the **arc length *L*** and **3D surface area of revolution *S*** of any smooth profile $y = f(x)$ over $x \in [a, b]$.

When a curve $y = f(x)$ steps horizontally by dx and rises vertically by $dy = f'(x)\,dx$, the Pythagorean theorem gives the infinitesimal chord length $ds = \sqrt{(dx^2 + dy^2)} = \sqrt{(1 + [f'(x)]^2)}\,dx$. Revolving that infinitesimal arc segment ds around the x-axis sweeps out a thin conical frustum ribbon of radius $y = f(x)$, circumference $2\pi\,f(x)$, and slant width ds, whose lateral surface area is $dS = 2\pi\,f(x)\,ds$. Integrating from $x = a$ to $x = b$ yields the two classic integral mensuration formulas of Huygens, Leibniz, and Euler:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\,dx, \qquad S_{\text{rev}} = 2\pi \int_a^b f(x)\sqrt{1 + [f'(x)]^2}\,dx \qquad \textbf{(Eq. 12.7)}$$

Similarly, if a planar profile $y = f(x)$ is revolved around the *vertical* y-axis over radius $x \in [a, b]$ (such as a parabolic microwave dish antenna $z(r) = r^2/4F$ of focal length F and aperture radius R, where $z'(r) = r/2F$), each concentric ring at radius r has circumference $2\pi\,r$ and slant width $ds = \sqrt{(1 + (r/2F)^2)}\,dr$:

$$S_{\text{dish}} = 2\pi \int_0^R r\sqrt{1 + \left(\tfrac{r}{2F}\right)^2}\,dr = \tfrac{8\pi F^2}{3}\left[\left(1 + \tfrac{R^2}{4F^2}\right)^{3/2} - 1\right] \qquad \textbf{(Eq. 12.8)}$$

By storing the integrand $r\sqrt{(1 + [z'(r)]^2)}$ in [→] [EQN], running **a** [ENTER] **b** [→] [∫] **R**, and multiplying the resulting integral in Register X by **2** [×] [→] [π] [×], you can compute the exact 3D surface area of arbitrary aerodynamic nose cones, hyperbolic cooling towers, bell nozzles, and catenary domes that have no elementary closed-form formula.

12.4 Worked Stack Traces: Composite Area Ledgers, Curve Profiles & Integral Mensuration

12.4.1 Composite Structural Gusset Plate with Cutout Bore (Σ+, Σ−, and x̄w)

Consider the composite steel structural gusset plate illustrated in Panel A of our concept diagram:

- **Sub-region 1 (Base Rectangle):** Width $b_1 = 120.0$ mm, height $h_1 = 80.0$ mm ($x \in [0, 120]$, $y \in [0, 80]$), with area $A_1 = 120 \times 80 = 9{,}600.0$ mm^2 and centroid at $x_1 = 60.0$ mm.
- **Sub-region 2 (Triangular Nose):** Right triangle extending from $x = 120.0$ mm to $x = 180.0$ mm (base $b_2 = 60.0$ mm, height $h_2 = 80.0$ mm), with area $A_2 = 1/2(60)(80) = 2{,}400.0$ mm^2 and centroid one-third of its base from the $x = 120$ mm weld line: $x_2 = 120 + 60/3 = 140.0$ mm.
- **Sub-region 3 (Drilled Bolt Bore Cutout):** Circular hole of radius $r = 20.0$ mm centered at $x_3 = 50.0$ mm, removing area $A_3 = \pi\,(20^2) = 400\pi \approx 1{,}256.637061$ mm^2.

To tally the **net surface area of one face** ($A_{\text{net}} = A_1 + A_2 - A_3 = 12{,}000 - 400\pi =$ **10,743.362939 mm²**) using the univariate [Σ+] / [↶] [Σ−] ledger, clear the statistical registers with [CLΣ], add the rectangle and triangle with [Σ+], deduct the circular bore with [↶] [Σ−], and recall [↷] [SUMS] → [Σx]. Because [Σ+] and [↶] [Σ−] disarm stack lift after displaying the count n in Register X, typing the first dimension of the next sub-shape overwrites n in place so Registers Y, Z, and T stay at zero throughout the entire takeoff:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↶] [CLEAR] [CLΣ]	0.000000	0.000000	0.000000	0.000000	Zero all 6 statistical accumulators ($\sum x = 0$, $n = 0$)
120 [ENTER] 80 [×] [Σ+]	0.000000	0.000000	0.000000	1.000000	Add rectangle $A_1 = 9{,}600$ mm^2 into $\sum x$; $n = 1$ (lift disarmed)
60 [ENTER] 80 [×] 2 [÷] [Σ+]	0.000000	0.000000	0.000000	2.000000	Overwrite n=1; add triangle $A_2 = 2{,}400$ mm^2 ($\sum x = 12{,}000$); $n = 2$
20 [↶] [x²] [↷] [π] [×] [↶] [Σ−]	0.000000	0.000000	0.000000	1.000000	Overwrite n=2; deduct bore $A_3 = 400\pi = 1{,}256.637061$ mm^2 from $\sum x$
[↷] [SUMS] [Σx]	0.000000	0.000000	0.000000	**10743.362939**	Recall net composite plate area $A_{\text{net}} = 10{,}743.3629$ mm^2

Now watch how the **bivariate weighted-centroid method** (**A_k** [ENTER] **x_k** [Σ+]) computes *both* that same net area $A_{\text{net}} = 10{,}743.3629$ mm^2 (in $\sum y$) and the exact horizontal balance point $\bar{x} = (9600(60) + 2400(140) - 400\pi(50)) / 10743.362939 = 849{,}168.1469 / 10{,}743.362939 =$ **79.041186 mm** (via [↷] [x̄,ȳ] → [x̄w]):

1. Clear [CLΣ], then log the rectangle: **9600** [ENTER] **60** [Σ+] ($n = 1$).
2. Log the triangular nose: **2400** [ENTER] **140** [Σ+] ($n = 2$).
3. Log the circular bore cutout centered at $x_3 = 50$ mm: **20** [↶] [x²] [↷] [π] [×] **50** [↶] [Σ−].

4. Tap [R/S] [x̄,ȳ] → [x̄w] ([LFU2]) to read the exact composite centroid $\bar{x} =$ **79.041186 mm**, and tap [R/S] [SUMS] → [Σy] ([LFU1]) to read the net area $A_{net} =$ **10,743.362939 mm²**!

12.4.2 Evaluating Cubic Arch Inflections, Harmonic Series, Damped Envelopes & sinc Lobes

When analyzing continuous engineering profiles—whether computing ordinances along an arch, verifying integrands before pressing [R/S] [∫], or evaluating peak heights at critical points—Horner's nested form and [←] [LASTx] keep every calculation inside the 4-register stack without scratchpad variables.

First, when laying out equally spaced suspension-bridge hanger stations, CNC toolpath passes, or trapezoidal integration nodes across an interval $[X_{min}, X_{max}]$ partitioned into N - 1 equal panels:

$$x_k = X_{min} + k\left(\frac{X_{max} - X_{min}}{N-1}\right), \qquad k = 0, 1, \ldots, N-1 \qquad \textbf{(Eq. 12.9)}$$

Because each arithmetic step in $x_k = X_{min} + k((X_{max} - X_{min}) / (N - 1))$ transforms a single scalar in place—from total span to panel width to horizontal displacement to world coordinate—the entire calculation executes inside Register X while Registers Y, Z, and T remain at zero. Across a 64-point (N - 1 = 63) domain spanning [-3.0, +3.0] at station k = 21 (one-third of the span from the left boundary), keying **3** [ENTER] **3** [+/−] [-] computes the total span X_{max} - X_{min} = 6.000000 in Register X. Dividing by **63** [÷] forms the panel width Δx = 6.0 / 63 ≈ 0.095238, scaling by **21** [×] advances $k \cdot \Delta x$ = 2.000000 units rightward, and adding the left boundary via **3** [+/−] [+] lands at the exact abscissa x_{21} = -1.000000.

Second, take the cubic highway S-curve profile $y(x) = x^3 - 6x^2 + 9x + 2$ spanning $x \in [0, 4]$: its local maximum rises to (1, 6), its local minimum dips to (3, 2), and its inflection point ($y''(x) = 6x - 12 = 0$) sits midway between them at $x = 2$. Because a cubic polynomial is odd-symmetric around its inflection point $(x_{infl}, y_{infl}) = (2, 4)$, the area under $y(x)$ across any interval symmetric about $x = 2$ (such as [0, 4]) equals the base width 4.0 multiplied by the inflection ordinate $y(2) = 4.0$ ($\int_0^4 (x^3 - 6x^2 + 9x + 2)\,dx = 4 \times 4 = 16.000000$). Writing $y(x)$ in nested Horner form $y(x) = ((x - 6)x + 9)x + 2$ and pre-filling the 4-level stack evaluates that inflection ordinate ($X = 4.000000$) with **2** typed only once:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
2 [ENTER] [ENTER] [ENTER]	2.000000	2.000000	2.000000	2.000000	Flood all four stack registers with inflection abscissa $x = 2.0$
6 [-]	2.000000	2.000000	2.000000	-4.000000	Innermost Horner factor x - 6 = -4.0; Register T replicates 2.0
[×] **9** [+]	2.000000	2.000000	2.000000	1.000000	Second Horner stage $(x - 6)x + 9$ = 1.0; $x = 2.0$ drops into Y
[×] **2** [+]	2.000000	2.000000	2.000000	4.000000	Inflection point ordinate $y(2)$ = $((x - 6)x + 9)x + 2 = 4.000000$

Third, four classical periodic and decaying waveforms illustrate how ordinate verification on the stack pairs with definite-integral area ([R/S] [∫]):

- **Joseph Fourier's Odd-Harmonic Square Wave Synthesis:** When Jean-Baptiste Joseph Fourier announced in 1807 that a flat-topped square wave of height π/4 could be synthesized by summing smooth odd-harmonic sine waves:

$$y_K(t) = \sum_{m=1}^{K} \frac{\sin((2m-1)t)}{2m-1} = \sin(t) + \tfrac{1}{3}\sin(3t) + \tfrac{1}{5}\sin(5t) + \cdots \qquad \textbf{(Eq. 12.10)}$$

each odd harmonic (1 / (2*m*-1)) sin((2*m*-1)t) contributes positive half-cycle area $\int_0^{\pi}$ sin((2*m*-1)t) / (2*m*-1) dt = 2/(2*m*-1)^2, summing across all odd integers to Euler's $\sum_{m=1}^{\infty}$ 2/(2*m*-1)^2 = $\pi^2/4$ (matching the rectangle $\pi \times \pi/4$). At the midpoint of the plateau (t = 90.0° = $\pi/2$), successive odd harmonics alternate in sign (+1 - 1/3 + 1/5 - ⋯ → $\pi/4$). Evaluating the three-harmonic plateau ordinate at t = 90.0° on the stack via **1** [ENTER] **3** [x^{-1}] [-] **5** [x^{-1}] [+] needs only Registers Y and X while Z and T stay at zero: the fundamental crest sin(90°) = +1.000000 sits in Register Y while the third-harmonic trough (1/3) sin(270°) = -0.333333 forms in Register X and folds down to 0.666667, after which adding +(1/5) sin(450°) = +0.200000 lifts the plateau center to **0.866667**.

- **Underdamped Mechanical Ringdown Envelope:** When an underdamped sports suspension $x(t) = e^{-\zeta\omega_n t}\cos(\omega_d t) = e^{-0.60\ t}\cos(3.9546\ t)$ (ζ = 0.15, ω_n = 4.0 rad/s, ω_d = 3.9546 rad/s) responds to a step impulse in Radians mode ([RAD]), evaluating the instantaneous displacement at t = 1.0 s via [RAD] **0.6** [+/-] [e^x] **3.9546** [COS] [×] yields **-0.377160**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 [ENTER] **.60** [×]	0.00	0.00	0.00	0.600000	Damping exponent $\zeta\,\omega_n\,t$ = 0.600000 at t = 1.0 s
[+/-] [e^x]	0.00	0.00	0.00	0.548812	Exponential decay envelope $e^{-0.60}$ = 0.548812
1 [ENTER]	0.00	0.548812	1.000000	1.000000	Stage time t = 1.0 s for oscillatory phase; envelope lifts to Z
3.9546 [×]	0.00	0.00	0.548812	3.954600	Damped phase angle $\omega_d\,t$ = 3.954600 rad; envelope drops to Y
[COS]	0.00	0.00	0.548812	-0.687229	Cosine carrier cos(3.9546 rad) = -0.687229
[×]	0.00	0.00	0.00	-0.377160	Instantaneous suspension displacement x(1.0) = -0.377160 m

- **Fraunhofer Single-Slit Diffraction `sinc` Function ($y(x)$ = sin x / x):** In physical optics, coherent light passing through a narrow slit produces the amplitude profile $y(x)$ = sin x / x, with a central principal maximum $\lim_{x \to 0}$sin x / x = 1 and side lobes that cross zero at every non-zero multiple of π (enclosing the Sine Integral area Si(π) = $\int_0^{\pi}$ sin x / x dx ≈ 1.851937052, evaluated via **0.000001** [ENTER] [↗] [π] [↗] [∫] **X**). Evaluating the half-lobe ordinate at $x = \pi/2$ in Radians mode ([RAD]) via [RAD] [↗] [π] **2** [÷] [SIN] [↖] [LASTx] [÷] yields **0.636620** (2/π ≈ 0.636620, where recalling $x = \pi/2$ from [↖] [LASTx] right after evaluating sin(π/2) divides sin(x)/x without retyping the angle).

12.4.3 Biochemical Saturation, Cantilever Beam Deflection & Piecewise Iverson Profiles

Biochemical reaction kinetics, structural beam deflection curves, and piecewise voltage clippers likewise pair single-register [↖] [LASTx] evaluation on the stack with [↗] [EQN] integration:

- **Michaelis–Menten Enzyme Saturation Hyperbola (v = $V_{\max}[S]$ / (K_m + [S])):** For an enzyme with maximum turnover rate $V_{\max}$ = 100.0 μmol/min and half-saturation constant K_m = 15.0 mM, the reaction curve climbs steeply near the

origin where active sites are abundant, then bends toward the horizontal asymptote $V_{\max} = 100$. Using [↶] [LASTx] verifies the two reference points along that hyperbola with $[S]$ typed only once while leaving Registers Z and T at zero: at half-saturation $[S] = K_m = 15.0$ mM, multiplying 100 × 15 and recalling $[S] = 15.0$ mM from [↶] [LASTx] to form $K_m + [S] = 30.0$ mM confirms $v = 50.000000$ μmol/min ($(1/2)\,V_{\max}$); at $[S] = 3K_m = 45.0$ mM, the same [↶] [LASTx] recall divides 100 × 45 by 45 + 15 = 60.0 mM to confirm $v = 75.000000$ μmol/min ($(3/4)\,V_{\max}$).

- **Euler–Bernoulli Cantilevered Beam Deflection Curve ($y(x) = (Px^2/6EI)(3L - x)$):** Across an $L = 4.0$ m cantilever span ($P/6EI = 3.0625 \times 10^{-5}$ m^{-2}, or in normalized form $0.5x^2(12\text{-}x)$ entered in [→] [EQN] as **12 X - X x² × 0.5 ×**), the elastic curve emerges horizontally from the clamped wall at $x = 0$ ($y(0) = y'(0) = 0$) before bowing cubically toward the free tip at $x = 4.0$ m. Because [↶] [LASTx] retains $x = 4.0$ m after forming $3L = 3 \times 4.0 = 12.0$ m, the tip-deflection verification stays inside Register X ($Y = Z = T = 0$): recalling [↶] [LASTx] twice subtracts $x = 4.0$ for $3L - x = 8.000000$ m and squares $x^2 = 16.0$ m^2 via [↶] [x²] to form $x^2(3L - x) = 128.000000$ m^3, which scales by 3.0625×10^{-5} m^{-2} to resolve the tip sag $\delta_{\max} = 0.003920$ m (3.92 mm).

- **Piecewise Zener-Diode Limiter & Trapezoidal Area via Iverson Booleans ([x>0]):** Inside [→] [EQN], the relational test menus [↶] [x?y] and [→] [x?0] evaluate to Kenneth Iverson's Boolean indicator 1.0 (True) or 0.0 (False) in Register X while preserving Registers Y and Z. Combining a linear ramp with the three-keystroke Macaulay bracket idiom ([ENTER] [x>0] [×]) models a piecewise Zener-diode voltage limiter—acting like an electronic pressure-relief valve or surge protector that passes normal voltages untouched and clips any spike above 5.0 V—via $v_{\text{out}}(x) = \min(x, 5.0\text{ V}) = x - \langle\, x - 5 \,\rangle_+$ over $x \in [0, 8]$ V ([RCL] X [RCL] X 5 [-] [ENTER] [x>0] [×] [-], whose enclosed area over $[0, 8]$ is the $[0, 5]$ triangle 12.5 plus the $[5, 8]$ rectangle 15.0, or $A = 27.500000$ V^2). Below the 5.0 V knee (at $x = 3.2$ V), [x>0] evaluates to 0 so $v_{\text{out}} = 3.2$ V; above the knee (at $x = 7.4$ V), [x>0] evaluates to 1 and subtracts the 2.4 V excess to clamp the output flat at 5.000000 V:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3.2 [STO] X	0.000000	0.000000	0.000000	3.200000	Sample linear ramp below knee ($x = 3.2$ V)
[RCL] X 5 [-]	0.000000	0.000000	3.200000	-1.800000	Excess above Zener knee $x - 5 = -1.8$ V < 0
[ENTER] [x>0] [×] [-]	0.000000	0.000000	0.000000	**3.200000**	Macaulay ramp is 0; linear output $v_{\text{out}}(3.2) = 3.2$ V
7.4 [STO] X	0.000000	0.000000	3.200000	7.400000	Sample clamped plateau above knee ($x = 7.4$ V)
[RCL] X 5 [-]	3.200000	3.200000	7.400000	2.400000	Excess above Zener knee $x - 5 = +2.4$ V > 0
[ENTER] [x>0] [×] [-]	3.200000	3.200000	3.200000	**5.000000**	Subtract 2.4 × 1 = 2.4 V; clamped $v_{\text{out}}(7.4) =$ **5.0 V**

12.4.4 Parabolic Dish Surface Area of Revolution & Bivariate Calibration Auditing

To see how surface area of revolution ([→] [∫]) and closed-form mensuration cross-check each other on the stack, consider a parabolic solar concentrator dish $z(r) = r^2/4F$ with focal length $F = 1.0$ m and outer rim radius $R = 1.5$ m (where $z'(r) = 0.5r$ and the radial slant factor is $\sqrt{(1 + 0.25r^2)}$):

1. **Continuous Integral Evaluation via [→] [EQN] and [→] [∫]:** Store the radial surface-area integrand $r\sqrt{(1 + 0.25r^2)}$ in [→] [EQN] as **R x² 0.25 × 1 + √x R ×**. Stage the radial bounds **0** [ENTER] **1.5** and press [→] [∫] **R** to obtain the radial integral $I = \int_0^{1.5} r\sqrt{(1 + 0.25r^2)}\, dr$ = **1.270833333 m²** in Register X, then scale by **2** [×] [→] [π] [×] to obtain the exact 3D reflective dish surface area S_{dish} = **7.984881 m²**.

2. **Closed-Form Verification on the Stack:** Evaluating $(8\pi\ F^2\ /\ 3)[(1 + R^2/4F^2)^{3/2} - 1]$ for $F = 1.0$ m and $R = 1.5$ m forms $1 + 2.25/4 = 1.5625 = (5/4)^2$, whose 3/2 power (**1.5** [y^x]) is $(5/4)^3 = 125/64 = 1.953125$. Subtracting 1 leaves $61/64 = 0.953125$, and multiplying by $8\pi/3$ (**8** [×] [→] [π] [×] **3** [÷]) confirms $S_{\text{dish}} = 61\pi/24$ = **7.984881 m²**—an 12.96% surface-area increase over the flat circular aperture $\pi\ R^2 = 7.068583$ m²!

Finally, when auditing experimental $(x_k,\ y_k)$ calibration data accumulated via [Σ+]—where up to 25 bivariate pairs reside in indirect slots **40**–**89** (**DATA0**–**DATA49** under [↖] [REGS])—combining [→] [L.R.] slope/correlation checks ([m], [r]) with predicted-value residual checks ([ŷ]) verifies that a sensor response is genuinely linear across its operating span. Take four piezoresistive pressure sensor calibration pairs: (10 kPa, 1.25 V), (20 kPa, 2.45 V), (30 kPa, 3.80 V), (40 kPa, 4.95 V). Accumulating the pairs via [Σ+] and querying [→] [L.R.] → [m] yields the least-squares sensitivity $m = 0.124500$ V/kPa ($y = 0.1245x + 0.0000$, $r = 0.999565$, $R^2 = 0.999130$), while checking **20** [→] [L.R.] [ŷ] (→ 2.4900 V) and **30** [→] [L.R.] [ŷ] (→ 3.7350 V) confirms that the residuals alternate cleanly without parabolic drift:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↖] [CLEAR] [CLΣ]	0.00	0.00	0.00	0.00	Clear statistical accumulators
1.25 [ENTER] **10** [Σ+]	0.00	0.00	1.25	1.00	Pair 1: (10 kPa, 1.25 V); count n = 1
2.45 [ENTER] **20** [Σ+]	0.00	1.25	2.45	2.00	Pair 2: (20 kPa, 2.45 V); count n = 2
3.80 [ENTER] **30** [Σ+]	1.25	2.45	3.80	3.00	Pair 3: (30 kPa, 3.80 V); count n = 3
4.95 [ENTER] **40** [Σ+]	2.45	3.80	4.95	4.00	Pair 4: (40 kPa, 4.95 V); count n = 4
[m]	2.45	3.80	4.95	0.1245	Regression slope m = 0.124500 V/kPa

By uniting Descartes's analytic coordinate geometry, Gauss's Shoelace polygon theorem, additive and subtractive composite area ledgers ([Σ+], [↖] [Σ−], [x̄w]), and Romberg's definite integral ([→] [∫]), the SC-32 turns every planar cross-section and three-dimensional surface of revolution into a direct, verifiable four-register stack calculation.

Chapter 13: Coupled Laws of Classical and Modern Engineering

13.1 Bending Light and Stacking Tolerances: Compound Microscopes & Pythagorean Error

Across optical design, mechanical assembly, fluid hydraulics, structural statics, thermodynamics, acoustics, and orbital mechanics, physical laws couple across multiple variables and orders of magnitude. While academic curricula divide science into separate courses labeled *Optics*, *Manufacturing*, *Fluid Mechanics*, *Structures*, *Thermodynamics*, and *Astrodynamics*, a space telescope, a jet turbine, or a deep-sea submersible subjects the same structural assembly to all of these regimes simultaneously.

Across these physical domains, a small set of recurring mathematical structures—harmonic reciprocals, Pythagorean sums of squares, logarithmic ratios, and fractional power laws—governs the underlying equations *(HP, 1982; Horowitz & Hill, 2015)*. Evaluating these coupled laws from the inside out isolates the dimensionless or geometric core of each relationship before scaling by material or environmental constants.

13.1.1 Thin-Lens Conjugates, Two-Stage Magnification & Abbe's Diffraction Limit

In the 1670s, Dutch microscopist Antonie van Leeuwenhoek demonstrated to the Royal Society of London a previously unseen microscopic domain—"little animalcules" swimming inside a single drop of pond water. Yet Leeuwenhoek's instruments were simple single-lens magnifiers: he melted a tiny soda-lime glass thread into a spherical bead barely 1.5 millimeters across and clamped it between two brass plates. To observe specimens at 250× magnification, Leeuwenhoek had to hold that tiny glass sphere at an extremely short working distance right in front of his eye.

Reaching 400× magnification with a single magnifying glass runs into a severe optical limit.

Every single converging lens of focal length f obeys Carl Friedrich Gauss's Thin-Lens Conjugates Law: when an illuminated specimen sits at object distance d_o in front of the lens and focuses a sharp image at distance d_i behind the lens, the wavefront curvatures (optical vergences $1/d_o$ and $1/d_i$) add in reciprocal space:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \quad \Longleftrightarrow \quad f = \frac{d_o d_i}{d_o + d_i} = d_o \| d_i \tag{Eq. 13.1}$$

Because the human eye cannot focus comfortably on anything closer than the near point $N \approx 250$ mm (25 cm, about ten inches), a single lens of focal length f used as a magnifying glass provides an angular magnification of $M = N / f$. Achieving 400× magnification with a single glass singlet would require a focal length of $f = 250 / 400 = 0.625$ millimeters—smaller than a poppy seed, with severe spherical and chromatic aberrations.

Galileo Galilei (who built his *occhiolino* compound microscope in 1609), Robert Hooke (whose 1665 *Micrographia* coined the biological word *"cell"* after inspecting cork bark), and German physicist Ernst Abbe (who partnered with instrument maker Carl Zeiss in Jena in the 1870s to establish the diffraction theory of modern microscopy) overcame this single-lens barrier. Their compound optical architecture refracts light twice in series across

a brass optical tube of standardized mechanical tube length L. In the first stage, a precision short-focal-length objective lens of focal length f_{obj} sits just above the specimen slide and projects an enlarged, inverted real intermediate image into the focal plane near the top of the tube, producing a linear lateral magnification of:

$$M_{\text{obj}} = \frac{L}{f_{\text{obj}}} \qquad \textbf{(Eq. 13.2)}$$

In the second stage, a wide ocular eyepiece of focal length f_{eye} at the top of the tube acts as a virtual angular magnifier that enlarges the floating intermediate image a second time relative to the observer's near-point distance N:

$$M_{\text{eye}} = \frac{N}{f_{\text{eye}}} \qquad \textbf{(Eq. 13.3)}$$

Because the second lens magnifies an image that has already been magnified by the first lens, the total compound magnification is the product of two moderate lens ratios:

$$M_{\text{total}} = M_{\text{obj}} \times M_{\text{eye}} = \left(\frac{L}{f_{\text{obj}}}\right) \times \left(\frac{N}{f_{\text{eye}}}\right) \qquad \textbf{(Eq. 13.4)}$$

No 0.6 mm glass bead is required, only two modest focal lengths working in series (or $f_{\text{tube}}/f_{\text{obj}}$ on modern infinity-corrected microscopes, up to Ernst Abbe's 1873 wave-optics diffraction resolution limit $d_{\text{min}} = \lambda \,/\, 2\ \text{NA} \approx 200$ nm for visible light). On the SC-32, the dedicated parallel operator [≠‖∀] resolves thin-lens conjugate focal lengths $f = d_o \parallel d_i$ in a single tap, while the four-register stack holds the objective ratio M_{obj} in Register Y as you form M_{eye} in Register X and multiply via [×].

13.1.2 Orthogonal Variance Addition and Root-Sum-Square (RSS) Tolerance Stacking

In the machine shop where a brass microscope barrel or a multi-stage turbine rotor is turned on a lathe, no cutting tool produces an exact mathematical dimension; every machined feature carries a manufacturing tolerance $\pm\ \sigma_i$. When n independent machined components, each carrying an identical bilateral tolerance $\pm\ \sigma$, are stacked end-to-end inside a housing, adding all n tolerances linearly yields $\pm\ n\sigma$. In manufacturing engineering, that arithmetic sum is called Worst-Case Linear Stackup, and it overstates the probable assembly variation whenever n is large.

That simultaneous worst case is statistically improbable. Expecting all n independent components from an unbiased lathe to arrive simultaneously at their positive extreme $+\sigma$ is analogous to flipping n fair coins and observing *Heads* on every toss ($1/2^n$, or less than 0.1% for $n = 10$). Because individual machining variations fluctuate independently above and below nominal size, positive and negative departures partially cancel across the assembly.

Carl Friedrich Gauss and Pierre-Simon Laplace proved that for independent random variations, variances (σ_i^2) add rather than standard deviations (σ_i). Geometrically, two independent random errors σ_1 and σ_2 are uncorrelated and act orthogonal (90°) to each other in probability space, combining along the Pythagorean hypotenuse. For any assembly of n independent machined dimensions with bilateral 3σ tolerances σ_1, σ_2, ..., σ_n, the statistical Root-Sum-Square (**RSS**) Tolerance Stackup is the Euclidean norm of the n-dimensional tolerance vector:

$$\sigma_{\text{RSS}} = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2} \qquad \textbf{(Eq. 13.5)}$$

For n equal tolerances $\pm\ \sigma$, the statistical assembly envelope grows only with the square root of the part count ($\sigma_{\text{RSS}} = \sigma\sqrt{n}$ rather than $n\sigma$). On the SC-32, you can accumulate $\sum \sigma_i^2$ with [↖] [x²], [+], and [√x], or chain the polar hypotenuse key [↖] [▸θ,r] (Left Shift [↖] above [4]) across each tolerance in turn to compute the multi-dimensional **RSS** norm without a single squaring stroke.

13.2 Osborne Reynolds's Dye Ribbon, Continuity & Leonhard Euler's Buckling Column

13.2.1 Osborne Reynolds's 1883 Dye Experiment, Continuity & Venturi Pressure Drops

Smoke rising from a blown-out candle ascends for three inches as a smooth, glassy silver ribbon before erupting into a swirling cloud of curls, and water flows smoothly from a partially opened kitchen faucet yet roars when the tap is opened wide. For two centuries after Isaac Newton, physicists could write equations for ideal, frictionless water, yet they could not explain why smooth flow suddenly turns violent. In 1883, Osborne Reynolds, the first professor of engineering at Owens College in Manchester, built a transparent glass-tube apparatus that isolated the governing dimensionless parameter.

Reynolds filled a large elevated glass tank with calm water and fitted a horizontal glass pipe with a polished bell-shaped brass mouth inside it. Just in front of the brass trumpet, he positioned a fine glass needle fed with crimson aniline dye. In the subcritical laminar regime, when Reynolds cracked the outlet valve slightly so water glided slowly through the pipe, the crimson dye stretched down the center of the clear glass tube like a motionless silk thread; adjacent layers (*laminae*) of water slid past one another without mixing. In the supercritical regime, as Reynolds slowly opened the valve to increase the flow velocity v, the red thread remained sharp until the speed reached a critical threshold, at which moment, about two feet down the tube, the ribbon burst into curling, eddying vortices that tinted the entire pipe pink (*turbulent flow*).

Reynolds tested pipes of different diameters D and warmed or cooled the water to vary its viscosity μ and density ρ. He discovered that the transition from laminar to turbulent flow never happens at a fixed speed v. Instead, it always occurs at the same critical value of a dimensionless four-variable ratio now known throughout physics as the Reynolds Number (Re):

$$\text{Re} = \frac{\rho v D}{\mu} = \frac{\text{Inertial Momentum Forces}}{\text{Viscous Friction Forces}} \qquad \textbf{(Eq. 13.6)}$$

Reynolds's fraction balances two competing physical mechanisms. The inertial numerator $\rho\ v\ D$ measures fluid inertia, the tendency of a moving parcel of fluid of density ρ and speed v across diameter D to keep advancing and roll up into swirling eddies. The viscous denominator μ measures dynamic viscosity, the internal shear friction between fluid layers that damps out perturbations and keeps the flow orderly. In any circular pipe, when $\text{Re} < 2{,}300$, viscous friction dominates and the flow remains smoothly laminar (as in blood gliding through a capillary, lubricating oil in a bearing clearance, or melted polymer moving through a 0.40 mm micro-extrusion nozzle). When $\text{Re} > 4{,}000$, inertia overwhelms viscosity and the flow becomes fully turbulent (as in high-speed water mains or air rushing over an airliner wing at $\text{Re} \approx 10^7$).

Once an incompressible fluid is flowing steadily through a conduit, two partner laws govern its speed and pressure at every cross-section. First, by the Law of Continuity ($Q = A_1\ v_1 = A_2\ v_2$), because fluid mass cannot vanish or accumulate inside a closed pipe, the volumetric

flow rate Q passing a cross-section each second remains constant. When a solid circular polymer filament of radius r and area $A_f = \pi r^2$ is fed at slow speed v_f into a heated liquefier and squeezed through a fine nozzle into a ribbon of width w and height h ($A_{\text{bead}} = w \times h$), narrowing the cross-sectional area more than twenty-five-fold causes the exiting melt velocity $v = \text{Q} / (w \times h)$ to increase by that same area ratio.

Second, in 1738 Swiss mathematician Daniel Bernoulli formulated the energy balance of moving fluids, and in 1797 Italian physicist Giovanni Battista Venturi demonstrated its throat-constriction effect: where fluid accelerates through a narrow throat ($v_2 > v_1$), its kinetic energy per unit volume $(1/2)\rho v^2$ rises, so by conservation of energy its static wall pressure drops according to $P_1 - P_2 = (1/2)\rho(v_2^2 - v_1^2)$. That Venturi pressure drop governs how carburetor jets atomize fuel, how Pitot tubes measure airspeed on aircraft, and how wind blowing over a curved roof generates upward suction.

13.2.2 Elastic Column Buckling and Euler's 1757 Critical Load ($P_{\text{cr}} = \pi^2 EI/L^2$)

When a slender vertical strut, such as a flexible ruler stood upright on a desk, is loaded axially at its top end, it exhibits two distinct mechanical regimes. Under a small axial force (1 Newton), the strut remains straight and carries the load in pure compression. Increasing the load slightly (2 Newtons) produces only tiny axial shortening. Yet the instant the applied force reaches a specific critical threshold P_{cr}, the slender member suddenly snaps sideways into a bowed arch.

The plastic did not crush; the compressive stress in the ruler was less than one-hundredth of the plastic's crushing yield strength. Instead, the ruler failed by elastic instability, a phenomenon analyzed in 1757 by Swiss mathematician Leonhard Euler while working at the Berlin Academy.

Euler determined the exact axial load P_{cr} at which a straight, slender column of length L, material stiffness (Young's modulus) E, and cross-sectional bending moment of inertia I bifurcates into a laterally bowed curve $y(x)$ rather than remaining straight. Setting the internal elastic restoring moment $E\,I\,y''(x)$ equal to the overturning bending moment $-P\,y(x)$ gives the exact differential equation of a harmonic oscillator:

$$EI\frac{d^2y}{dx^2} + Py = 0 \implies y(x) = A\sin\left(\sqrt{\frac{P}{EI}}\,x\right) \qquad \textbf{(Eq. 13.7)}$$

For a column pinned at both ends ($y(0) = 0$ and $y(L) = 0$), the lowest non-zero bowed shape occurs when the sine wave completes one half-arch across the length L, where $\sqrt{(P/EI)}\ L = \pi$. Squaring both sides and solving for P delivers Euler's Critical Column Buckling Formula:

$$P_{\text{cr}} = \frac{\pi^2 EI}{(KL)^2} \qquad \textbf{(Eq. 13.8)}$$

(where K is the effective-length factor: $K = 1.0$ for a column pinned at both ends, $K = 0.5$ when clamped rigidly at both ends, and $K = 2.0$ for a flagpole free at the top).

Two structural scaling laws follow directly from Euler's equation. First, although there are no circles in a straight column, the half-sine curvature factor $\pi^2 \approx 9.8696$ appears because the buckled column bows into a half-sine wave $\sin(\pi\ x\ /\ L)$, and differentiating that sine wave twice (once for slope, once for curvature) extracts two factors of $\pi\ /\ L$. Second, the material's yield strength does not appear in Euler's formula at all: replacing mild structural steel with ultra-high-strength alloy steel barely changes $E \approx 200$ GPa, so it does not

prevent a slender column from buckling. Instead, buckling strength is governed by the inverse-square length penalty I / L^2, where doubling the unsupported length L of a column cuts its buckling capacity to one-fourth ($1/2^2 = 1/4$). When that same steel member is laid horizontally as a pin-supported floor beam of span L carrying a center load P, its midspan sag grows with the cube of the span:

$$\delta_{\max} = \frac{PL^3}{48EI} \quad \textbf{(Eq. 13.9)}$$

Doubling the span of a bridge or balcony beam multiplies its sag by $2^3 = 8$. On the SC-32, both circular pipe continuity ($A_f = \pi\ r^2$) and Euler column buckling ($P_{\text{cr}} = \pi^2\ E\ I\ /\ L^2$) flow directly from [↷] [π], [↶] [x²], and scientific exponent entry [E], holding the flexural rigidity $\pi^2\ E\ I$ in Register Y while you square the column length L^2 in Register X and divide via [÷].

13.3 Sadi Carnot's Heat Engine Limit & Counter-Flow Heat Exchangers

13.3.1 Sadi Carnot's 1824 Heat-Engine Efficiency Limit

In 1824, French military engineer Nicolas Léonard Sadi Carnot examined how British coal-fired steam engines had transformed industrial production following the Napoleonic Wars. Engineers such as Richard Trevithick and Arthur Woolf had doubled and tripled the efficiency of James Watt's early steam engines by raising boiler pressures and adding condensers. Engineers across Europe sought to determine whether eliminating all mechanical friction in the pistons and insulating every pipe so zero heat leaked through the walls would allow a steam engine to convert 100% of the heat energy of burning coal into mechanical work.

In *Réflexions sur la puissance motrice du feu* (*Reflections on the Motive Power of Fire*, 1824), Carnot proved that an upper thermodynamic bound prevents total conversion of heat into work by comparing a heat engine to a hydraulic waterwheel. In hydraulics, water does not turn a millwheel merely by sitting in a millpond; it turns the wheel because it falls from a high elevation to a lower elevation, and if the tailrace stream below the millwheel sits at the same height as the millpond above it, no water flows and the wheel cannot turn regardless of the pond's depth. By the same physical logic, heat cannot be converted into work merely by sitting inside a hot furnace at absolute temperature T_H. To extract mechanical work from thermal motion, heat must flow from a hot source at absolute temperature T_H into a colder sink at absolute temperature T_C, such as condenser cooling water or outside air, and that cold sink must carry away the rejected thermal entropy.

Carnot proved that even for a theoretically reversible heat engine operating between T_H and T_C (measured in absolute Kelvin, $T_{\text{K}} = T_{^\circ\text{C}} + 273.15$, where 0 K = -273.15°C is absolute zero where molecular thermal motion ceases), the maximum fraction of heat that can become mechanical work is:

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \quad \textbf{(Eq. 13.10)}$$

Carnot's ratio (1 - T_C / T_H) imposes two design constraints on every thermal power cycle. First, because molecular thermal kinetic energy vanishes only at absolute zero (0 K = -273.15°C), substituting raw Celsius temperatures t_C / t_H into Carnot's equation commits a severe origin-shift error. Even for moderate-temperature geothermal or ocean-thermal plants where the Celsius ratio t_C / t_H appears small, adding the 273.15 K offset to both reservoirs

shows that T_C / T_H remains a large fraction of unity, bounding theoretical work recovery before mechanical friction is even considered.

Second, modern jet turbines and rocket engines run their combustion chambers as hot as single-crystal nickel superalloys allow (T_H > 1,700 K) because raising T_H shrinks the rejected fraction T_C / T_H and raises the Carnot ceiling. On the SC-32, translating T_C and T_H to Kelvin in Registers Y and X, dividing via [÷], and completing the complement 1 - T_C/ T_H via [+/−] 1 [+] (or 1 [x⇄y] [-]) evaluates the Carnot limit in one continuous pass, while [x^{-1}] forms the reciprocal temperature differences $1/T_1$ - $1/T_2$ in Arrhenius activation-energy studies.

13.3.2 Counter-Flow Heat Exchangers and the Log-Mean Temperature Difference (LMTD)

Once a power plant, chemical reactor, or computer liquid-cooling loop generates thermal energy, heat exchangers and insulated barriers govern how efficiently that heat transfers from one fluid to another or stays inside a building in winter.

In 1822, two years before Sadi Carnot's memoir, French mathematician Jean-Baptiste Joseph Fourier published *Théorie analytique de la chaleur* (*The Analytical Theory of Heat*), formulating the law of thermal conduction *(Incropera et al., 2007).* When a flat wall or window pane of area A, thickness d, and thermal conductivity k separates a warm room at T_{in} from a cold exterior at T_{out}, the heat-flow rate Q in Watts across the barrier is:

$$\dot{Q} = \frac{kA}{d}(T_{\text{in}} - T_{\text{out}}) = \frac{T_{\text{in}} - T_{\text{out}}}{R_{\text{th}}}, \qquad \text{where } R_{\text{th}} = \frac{d}{kA} \tag{Eq. 13.11}$$

Architects and physicists write Fourier's law in terms of thermal resistance $R_{\text{th}} = d$ / k A (K/W) because it turns heat flow into the thermal counterpart of Georg Simon Ohm's electrical law ($I = \Delta V$ / R). Solid soda-lime window glass has a thermal conductivity of $k_{\text{glass}} \approx 0.96$ W/(m·K), whereas still dry air has a conductivity of only $k_{\text{air}} \approx 0.025$ W/(m·K), nearly forty times more insulating than glass. A single 4 mm pane of glass offers minimal barrier to winter cold, whereas sandwiching a 12 mm dead-air or argon gap between two 4 mm glass panes places three thermal resistances in series ($R_{\text{total}} = R_{\text{glass}} + R_{\text{air}} + R_{\text{glass}}$) and reduces conductive heat loss by more than 90%.

In a flowing shell-and-tube heat exchanger that cools a hot process stream from inlet temperature $T_{h,\text{in}}$ to outlet temperature $T_{h,\text{out}}$ using a coolant stream rising from $T_{c,\text{in}}$ to $T_{c,\text{out}}$, the internal thermal profile depends on whether the two streams travel in the same direction (parallel flow) or in opposite directions (counterflow). In parallel flow, both streams enter at End 1 with a large initial temperature difference $\Delta T_1 = T_{h,\text{in}} - T_{c,\text{in}}$, but their temperatures converge exponentially along the tube length toward a common asymptote, so the hot outlet $T_{h,\text{out}}$ can never be cooled below the cold outlet $T_{c,\text{out}}$.

In counterflow, with the hot fluid flowing left-to-right while the cold fluid flows right-to-left, the hottest incoming fluid $T_{h,\text{in}}$ meets the warmest exiting coolant $T_{c,\text{out}}$ at End 1 ($\Delta T_1 = T_{h,\text{in}} - T_{c,\text{out}}$), while the cooled exiting fluid $T_{h,\text{out}}$ meets the coldest incoming coolant $T_{c,\text{in}}$ at End 2 ($\Delta T_2 = T_{h,\text{out}} - T_{c,\text{in}}$). This countercurrent arrangement maintains a nearly uniform driving gradient along the entire heat-transfer surface and permits $T_{h,\text{out}}$ to drop below $T_{c,\text{out}}$.

Nature evolved counterflow heat exchangers in the vascular bundles (*rete mirabile*) of emperor penguins, wading birds, and Arctic mammals: warm arterial blood flowing down from the body core runs in countercurrent contact with cold venous blood returning from the ice, transferring heat from artery to vein so the extremities remain just above freezing

without draining core body heat. Because the local temperature difference $\Delta T(x)$ varies exponentially between ΔT_1 at End 1 and ΔT_2 at End 2, integrating Fourier's differential conduction law $dQ = U\,\Delta T(x)\,dA$ along the exchanger proves that the effective driving potential is not the arithmetic mean $1/2(\Delta T_1 + \Delta T_2)$, but the Logarithmic Mean Temperature Difference (**LMTD**):

$$\mathrm{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)} \qquad \textbf{(Eq. 13.12)}$$

On the SC-32, the 4-register stack models **LMTD** directly across the four registers. By duplicating ΔT_1 across Registers Z, Y, and X via [ENTER] [ENTER] before subtracting ΔT_2, the linear difference ΔT_1 - ΔT_2 forms in Register X while [↶] [LASTx] preserves ΔT_2. Swapping via [x⇄y], recalling [↶] [LASTx], and pressing [÷] [LN] [÷] then divides that linear difference by the natural logarithm of the ratio $\ln(\Delta T_1 \,/\, \Delta T_2)$ in one clean sequence.

13.4 Acoustic Reverberation, Rocket Staging & Geostationary Orbits

13.4.1 Wallace Sabine's 1898 Fogg Lecture Hall Experiments and Reverberation Time (RT60)

In the autumn of 1895, Harvard University finished constructing a grand semicircular lecture room inside the new Fogg Art Museum. Architecturally, the room looked magnificent. Acoustically, it was a failure: when a professor stood at the lectern and spoke at a normal pace of three syllables per second, the plaster dome and bare walls kept every word reverberating for 5.52 seconds, so fifteen previous syllables were still echoing around the room on top of the word being spoken.

Harvard President Charles W. Eliot assigned the acoustic investigation to assistant professor of physics Wallace Clement Sabine. At that moment in history, architectural acoustics did not exist as a quantitative science; architects designed concert halls by copying older rooms or stringing wires across the ceiling.

Sabine decided to measure the room from first principles. Because streetcars clattering down Broadway rattled his sensitive instruments during the day, Sabine conducted his measurements during the quiet overnight hours. Using an organ pipe tuned to 512 Hz (one octave above middle C) and an electro-pneumatic chronograph, he measured how long it took the sound intensity in the empty room to decay by 60 decibels (one-millionth of its initial energy, the threshold where a loud organ note fades into silence), a universal metric now called RT_{60}.

To measure how sound absorption changes RT_{60}, Sabine and two assistants spent months hauling more than 400 velvet-upholstered seat cushions on hand trucks from nearby Sanders Theatre into the Fogg lecture hall a few dozen at a time. With 0 cushions in the room, the reverberation lasted 5.52 seconds; with 50 linear meters of Sanders Theatre cushions installed, it dropped to 3.94 seconds; and with 200 meters of cushions covering the seats and aisles, it dropped to 1.78 seconds.

In the autumn of 1898, plotting the reciprocals of his acoustic decay measurements, Sabine discovered that the product of room reverberation time RT_{60} and total acoustic absorption A_{total} was an exact constant proportional to the room's air volume V, which established Sabine's Reverberation Law (in metric SI units):

$$RT_{60} = \frac{0.161V}{A_{\text{total}}} \qquad \textbf{(Eq. 13.13)}$$

where V is the room's interior volume in cubic meters (m^3) and $A_{\text{total}} = \sum \alpha_i S_i$ is the total effective open-window absorption in metric Sabins (m^2; for heavily damped broadcast studios where $\bar{\alpha} > 0.3$, Carl Eyring's logarithmic form replaces $\bar{\alpha}$ with $-\ln(1 - \bar{\alpha})$). Applying that hyperbolic relation, Sabine was commissioned in 1898 to design Boston's new 2,600-seat concert hall (targeting $RT_{60} \approx 1.85$ s for orchestral warmth, whereas a university lecture hall targets a much shorter $RT_{60} \approx 0.7$ s so spoken consonants stay crisp). Over a century later, that 1900 Boston auditorium remains a benchmark of architectural acoustics.

13.4.2 Flight & Orbital Physics: Breguet's Aircraft Range, Tsiolkovsky's Rocket & Clarke's Geostationary Ring

Three foundational laws govern long-range atmospheric flight, spacecraft propulsion, and synchronous orbital geometry. In aerodynamics, French aviation pioneer Louis Charles Breguet showed that because a jet aircraft in level cruise supports its instantaneous weight W via wing lift ($L = W$) and overcomes drag $D = W / (L/D)$ at weight-specific fuel consumption c_T (hr^{-1}), integrating true airspeed V against fuel burn $dW/dt = -c_T W / (L/D)$ from initial gross weight W_0 to final landing weight W_1 yields the logarithmic *Breguet Range Equation*:

$$R = \frac{V}{c_T}\left(\frac{L}{D}\right)\ln\left(\frac{W_0}{W_1}\right) \qquad \textbf{(Eq. 13.14)}$$

In 1903, Russian theorist Konstantin Eduardovich Tsiolkovsky, working in Kaluga, integrated Newton's second law for a body whose mass changes continuously in a vacuum as it ejects exhaust gas at velocity v_e, deriving the identical logarithmic structure for rocket propulsion: $\Delta v = v_e \ln(m_0 / m_f)$. Because both a transoceanic aircraft and an orbital rocket must carry and accelerate their own unburned fuel before consuming it, range and velocity grow with the natural logarithm ([LN]) of the starting-to-final mass ratio (m_0 / m_f). That logarithmic law explains why no single-stage chemical rocket can easily carry a heavy payload to high orbit, and why multi-stage rockets jettison their heavy empty Stage 1 tank mid-flight ($\Delta v_1 = v_{e1}\ln(m_{01}/m_{f1})$) before igniting a lighter Stage 2 upper stage ($\Delta v_2 = v_{e2}\ln(m_{02}/m_{f2})$).

Once a launch vehicle reaches orbit, Johannes Kepler's Third Law governs its period as a function of orbital radius. In October 1945, twelve years before *Sputnik 1*, Royal Air Force radar officer Arthur C. Clarke published a three-page paper in *Wireless World* on extra-terrestrial relays for worldwide radio coverage. Clarke applied Kepler's Third Law, which equates the ratio of orbital radius cubed to orbital period squared with Earth's standard gravitational parameter $\mu = GM_{\oplus}$: $4\pi^2 a^3 / \mu = T^2$.

In low Earth orbit (300 km altitude), a satellite circles the globe every 90 minutes; at the Moon's distance (384,400 km), it takes 27.3 days. Between those two extremes lies a unique synchronous radius a where the satellite's orbital period T matches Earth's sidereal rotation period relative to the fixed stars (roughly 3 minutes, 56 seconds shorter than the 24-hour solar day because Earth advances nearly 1° along its solar orbit during each rotation). Isolating a from Kepler's Third Law by taking the cube root (1/3 power via **3** [↶] [x^y] or **3** [x^{-1}] [y^x]) gives the synchronous semi-major axis:

$$a = \left(\frac{\mu T^2}{4\pi^2}\right)^{1/3} \qquad \textbf{(Eq. 13.15)}$$

Subtracting Earth's equatorial radius $R_\oplus$ from a yields the geostationary altitude $h_{\text{GEO}} = a - R_\oplus$ above sea level. A communications or weather relay placed in a circular equatorial orbit at this Clarke Belt radius revolves at the exact angular velocity of the planet below and remains stationary above a fixed meridian on the ground.

13.5 Multi-Domain Engineering Calculations

In a multi-stage engineering calculation, every intermediate number sitting on the stack represents a physical quantity with explicit units that can be verified before the next operation executes. Keeping all four operational registers (T, Z, Y, X) visible exposes intermediate subtotals—such as a 40× objective magnification in Register Y, a 0.09 mm² nozzle cross-section in Register X, a 295.15 K cold-sink temperature, or a 4,491.14 m/s first-stage velocity increment—so unit mismatches and decimal slips are caught immediately. Hardware operators like [x‖y] and [↶] [▸θ,r] further compress harmonic reciprocals and multidimensional Euclidean norms into single keystrokes.



Figure: Thin-lens focal length 1/f = 1/30 + 1/60 solved in a single keystroke via [x‖y] → 20.0000 cm on the SC-32.

13.5.1 Optical Lens Conjugates, Microscope Cascades & Parallel Acoustic Loads (x‖y)

Comparing single-stage harmonic combinations against two-stage series cascades on Ernst Abbe's optical bench illustrates why some physical laws collapse inside a single stack register while others require multi-register staging:

- **Single-Lens Conjugate Calibration ($1/f = 1/d_o + 1/d_i$):** When an optical technician calibrates a converging test lens by placing an illuminated reticle at object distance $d_o = 30.0$ cm and capturing a sharp real image on a ground-glass screen at $d_i = 60.0$ cm, Gauss's conjugate equation $f = d_o\ d_i\ /\ (d_o + d_i) = d_o \parallel d_i$ has the exact harmonic symmetry of the SC-32's hardware [x‖y] operator. Staging 30.0 cm in Register Y and 60.0 cm in Register X (**30** [ENTER] **60** [x‖y]) → **20** collapses both vergences in a single tap while leaving Registers Y, Z, and T at zero.

- **Ernst Abbe's Two-Stage Compound Microscope ($M_{\text{total}} = L\ /\ f_{\text{obj}} \times N\ /\ f_{\text{eye}}$):** Now apply the two-stage compound architecture ($L = 160.0$ mm, $N = 250.0$ mm) to an optical bench mounting an $f_{\text{obj}} = 4.0$ mm objective in series with an $f_{\text{eye}} = 25.0$ mm ocular eyepiece. Because the objective stage ($M_{\text{obj}} = 160\ /\ 4 = 40.0\times$) and the eyepiece stage ($M_{\text{eye}} = 250\ /\ 25 = 10.0\times$) are two independent physical ratios that multiply in series, the inside-out plan computes $M_{\text{obj}} = 40.0\times$ first, lets it ride up to Register Z while $M_{\text{eye}} = 10.0\times$ forms in X, and multiplies them when M_{obj} drops into Register Y: **160** [ENTER] **4** [÷] **250** [ENTER] **25** [÷] [×] → **400**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
160 ENTER	0.00	0.00	160.0000	160.0000	Tube length $L = 160.0$ mm entered
4 ÷	0.00	0.00	0.00	40.000000	Objective magnification $M_{obj} = 40.0$x
250 ENTER	0.00	40.000000	250.0000	250.0000	Near point $N = 250.0$ mm entered, M_{obj} lifts to Z
25 ÷	0.00	0.00	40.000000	10.000000	Eyepiece magnification $M_{eye} = 10.0$x; M_{obj} drops to Y
×	0.00	0.00	0.00	400.0000	Compound microscope magnification $M_{total} = 400.0$x

- **Three-Driver Parallel Acoustic Bus (8 Ω ∥ 8 Ω ∥ 4 Ω):** When two 8.0 Ω mid-bass loudspeaker drivers and a 4.0 Ω subwoofer share an amplifier bus in parallel, harmonic addition is associative, so you never need to hold all three impedances in separate registers at once. Fold the twin 8.0 Ω drivers into a 4.000000 Ω subtotal in Register X ($Y = Z = T = 0$) and combine that subtotal with the 4.0 Ω subwoofer: **8** ENTER **8** x∥y **4** x∥y → **2** (2.000000 Ω). *(Similarly, to pad down a studio audio line by 10.0 dB without altering its $Z_0 = 600$ Ω impedance using a Bridged-T resistor network with voltage ratio $K = 10^{10/20} \approx 3.1623$, duplicating 600 Ω across Z and Y before forming K - 1 = 2.1623 multiplies for the bridge resistor $R_1 = Z_0(K\text{-}1) = 1{,}297.37$ Ω in Y and recalls K - 1 from* ↶ **LAST**x *to divide for the shunt resistor $R_2 = Z_0/(K\text{-}1) = 277.49$ Ω in X.)*

13.5.2 Statistical Root-Sum-Square (RSS) Mechanical Tolerance Stacking

When a precision bearing shaft stacks three independent machined shoulders with bilateral 3σ tolerances $\sigma_1 = \pm 0.04$ mm, $\sigma_2 = \pm 0.03$ mm, and $\sigma_3 = \pm 0.12$ mm, our inside-out plan treats the statistical Root-Sum-Square (**RSS**) assembly envelope $\sigma_{RSS} = \sqrt{(\sigma_1{}^2 + \sigma_2{}^2 + \sigma_3{}^2)}$ as a running variance accumulator across Registers X and Y while Registers Z and T stay at zero:

- **Step-by-Step Variance Accumulation:** Square the first shoulder tolerance ($0.04^2 = 0.001600$ mm^2), add the second shoulder's variance ($0.03^2 = 0.000900$ mm^2) to form the two-shoulder variance 0.002500 mm^2 (a 3–4–5 right-triangle sub-hypotenuse of 0.05 mm), add the third shoulder's variance ($0.12^2 = 0.014400$ mm^2 → 0.016900 mm^2, a 5–12–13 Pythagorean triple), and take the square root: **0.04** ↶ x² **0.03** ↶ x² + **0.12** ↶ x² + √x → **±0.130 mm** (± 0.130000 mm, a 31.6% reduction below the ± 0.190 mm worst-case linear sum, which shows that standard CNC turning meets print without secondary grinding).
- **Five-Token Polar Hypotenuse Shortcut (↶ →θ,r):** Better still, chaining the SC-32's polar hypotenuse operator via **0.04** ENTER **0.03** ↶ →θ,r **0.12** ↶ →θ,r resolves that exact 3D Euclidean norm 0.130000 mm in five tokens without a single ↶ x², +, or √x keystroke: the first ↶ →θ,r leaves the planar hypotenuse $r_{xy} = \sqrt{(0.04^2 + 0.03^2)} = 0.050000$ mm in Register X with stack lift armed, so typing **0.12** lifts 0.050000 mm into Register Y ready for the second ↶ →θ,r tap.

13.5.3 Osborne Reynolds's Pipe Flow, Continuity & Euler Column Buckling

Single-branch multiplicative chains in fluid mechanics and structural mechanics likewise execute as streamlined two-register sequences (**Registers X** and **Y** only, with **Z = T = 0**) because each physical factor scales the running product or quotient in place:

- **Osborne Reynolds's Turbulent Pipe Flow ($\mathrm{Re} = \rho\, v\, D \,/\, \mu$):** Inside a heat exchanger's copper tubes ($D = 0.025$ m internal diameter), cooling water at 20°C ($\rho = 998.2$ kg/m^3, dynamic viscosity $\mu = 1.002 \times 10^{-3}$ Pa·s) circulates at bulk velocity $v = 1.85$ m/s. To verify that the flow is turbulent, form the mass flux $\rho\, v = 1{,}846.67$ kg/(m^2·s) in Register X, multiply by diameter $D = 0.025$ m to get the inertial momentum numerator $\rho\, v\, D = 46.16675$ kg/(m·s), and divide by viscosity μ: keying **998.2** [ENTER] **1.85** [×] **0.025** [×] **1.002** [E] **3** [+/−] [÷] yields $\mathrm{Re} = 46{,}074.60$, more than twenty times the critical laminar-to-turbulent transition ($\mathrm{Re}_{\mathrm{crit}} \approx 2{,}300$), which confirms vigorous eddy mixing along the copper wall.

- **Microfluidic Flow-Cytometer Channel & Capillary Continuity ($Q = w\, h\, v_{\mathrm{chan}} = \pi\, r_c^2\, v_{\mathrm{cap}}$):** Inside a lab-on-a-chip flow cytometer where a rectangular optical interrogation channel of width $w = 0.45$ mm and depth $h = 0.20$ mm carries a laminar sheath stream at $v_{\mathrm{chan}} = 180.0$ mm/s, form the rectangular cross-section $A_{\mathrm{chan}} = 0.45 \times 0.20 = 0.090000$ mm^2 and scale by 180.0 mm/s: **.45** [ENTER] **.20** [×] **180** [×] → **16.2** ($Q = 16.200000$ mm^3/s). Conversely, from $Q = 16.2$ mm^3/s, dividing by $w = 0.45$ mm and $h = 0.20$ mm via **16.2** [ENTER] **.45** [÷] **.20** [÷] recovers $v_{\mathrm{chan}} =$ **180** (180.0000 mm/s), while dividing Q by the circular area $A_c = \pi(0.875)^2 = 2.405282$ mm^2 of the 1.75 mm supply tubing (**.875** [↶] [x²] [↷] [π] [×] [÷]) yields the upstream syringe-pump feed velocity $v_{\mathrm{cap}} = 6.735177$ mm/s.

- **Euler's Critical Column Buckling Load ($P_{\mathrm{cr}} = \pi^2\, E\, I \,/\, L^2$):** For a pin-ended structural steel column of unsupported length $L = 6.5$ m, Young's modulus $E = 200.0$ GPa (200×10^9 Pa), and weak-axis moment of inertia $I = 4.85 \times 10^{-6}$ m^4 under a safety factor SF = 2.5, build the flexural rigidity numerator $\pi^2\, E\, I = 9.573516 \times 10^6$ N·m^2 first via [↷] [π] [↶] [x²] **200** [E] **9** [×] **4.85** [E] **6** [+/−] [×]. Squaring the column length (**6.5** [↶] [x²]) lifts that rigidity into Register Y while forming $L^2 = 42.250000$ m^2 in X, so dividing via [÷] yields the critical Euler buckling load $P_{\mathrm{cr}} = 226{,}592.10$ N (226.59 kN), and **2.5** [÷] establishes the allowable working load $P_{\mathrm{allow}} = 90{,}636.84$ N (90.64 kN).

- **Simply Supported Beam Midspan Sag ($\delta_{\max} = PL^3/48EI$):** For a horizontal pin-supported steel beam of span $L = 4.0$ m ($E = 200$ GPa, $I = 8.5 \times 10^{-6}$ m^4) carrying a midspan point load $P = 5{,}000$ N, cube the span ($4.0^3 = 64.0$ m^3) via **4** [ENTER] **3** [y^x], multiply by $P = 5{,}000$ N to form $PL^3 = 3.20 \times 10^5$ N·m^3, and divide sequentially by 48, E, and I to resolve the midspan deflection $\delta_{\max} = 0.003922$ m (3.92 mm).

13.5.4 Sadi Carnot's Heat Engine Limit & Counter-Flow LMTD

Thermodynamics offers an even sharper contrast between a two-register temperature ratio and a genuine multi-register heat-exchanger calculation:

- **Sadi Carnot's Efficiency Ceiling ($\eta_{\mathrm{Carnot}} = 1 - T_C/T_H$):** In a geothermal binary power plant operating between an underground brine reservoir at $T_H = 185$°C and river cooling water at $T_C = 22$°C (where an unshifted Celsius ratio 1 - 22/185 would falsely suggest 88.1% efficiency), first translate the cold sink to absolute Kelvin ($T_C = 22 + 273.15 = 295.150000$ K in Register X), lift T_C into Register Y while translating the hot source to Kelvin ($T_H = 185 + 273.15 = 458.150000$ K in Register X), divide to form the entropy-rejection fraction $T_C \,/\, T_H = 0.644221$, and subtract from 1 via **22** [ENTER] **273.15** [+] **185** [ENTER] **273.15** [+] [÷] **1** [x⇄y] [−] to obtain **0.355778** ($\eta_{\mathrm{Carnot}} = 35.58\%$).

- **Counter-Flow Heat Exchanger `LMTD` ($(\Delta T_1 - \Delta T_2) \,/\, \ln(\Delta T_1 \,/\, \Delta T_2)$):** In a counterflow oil cooler where oil cools from $T_{h,\mathrm{in}} = 120$°C to $T_{h,\mathrm{out}} = 65$°C against water

rising from $T_{c,\text{in}} = 20°\text{C}$ to $T_{c,\text{out}} = 45°\text{C}$, the terminal approaches are $\Delta T_1 = 120°\text{C} - 45°\text{C} = 75°\text{C}$ and $\Delta T_2 = 65°\text{C} - 20°\text{C} = 45°\text{C}$. Duplicating $\Delta T_1 = 75°\text{C}$ across Registers Z, Y, and X (**75** ENTER ENTER) before subtracting $\Delta T_2 = 45°\text{C}$ forms $\Delta T_1 - \Delta T_2 = 30.0°\text{C}$ in Register X while holding $\Delta T_1 = 75.0°\text{C}$ in Register Y and archiving $\Delta T_2 = 45.0°\text{C}$ in ↶ LASTx, evaluating LMTD = 58.7285°C with **75** and **45** each typed only once:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
75 ENTER ENTER	0.000000	75.000000	75.000000	75.000000	Stage and duplicate terminal temperature approach $\Delta T_1 = 75.0°\text{C}$
45 -	0.000000	0.000000	75.000000	30.000000	Terminal difference $\Delta T_1 - \Delta T_2 = 30.0°\text{C}$; **LASTX** saves 45.0
x⇄y ↶ LASTx ÷	0.000000	0.000000	30.000000	1.666667	Swap $\Delta T_1 = 75.0$ to X and divide by **LASTX** (45.0) for ratio 1.666667
LN ÷	0.000000	0.000000	0.000000	58.728496	Divide $\Delta T_1 - \Delta T_2$ by ln(1.666667); LMTD = 58.7285°C

13.5.5 Sabine Reverberation (RT60), Breguet Range, Tsiolkovsky Staging & Clarke's Orbit

Acoustic reverberation decay, transoceanic aircraft range, geodetic EDM baseline reduction, and Keplerian orbit sizing contrast single-product scaling against multi-stage logarithmic and trigonometric reductions:

- **Wallace Sabine's Reverberation Law ($RT_{60} = 0.161\ V\ /\ A_{\text{total}}$):** For a lecture hall of volume $V = 675.0\ \text{m}^3$ and total absorption $A_{\text{total}} = 152.75$ metric Sabins, form $0.161 \times 675 = 108.67500\ \text{s·m}^2$ in Register X and divide by 152.75: **.161** ENTER **675** × **152.75** ÷ → **0.7115** (0.7115 s).

- **Louis Breguet's Aircraft Range & Geodetic EDM Slope-and-Refraction Reduction:** For a transport aircraft departing at gross weight $W_0 = 78{,}000$ kg and landing at $W_1 = 56{,}000$ kg with lift-to-drag ratio $L/D = 18.5$ and specific range factor $V/c_T = 1{,}420$ km, dividing 78,000 / 56,000, pressing LN (→ 0.331357), and scaling by 18.5 and 1,420 km yields a maximum still-air range of $R =$ **8,704.7513 km**. In geodetic land surveying, when an infrared Electronic Distance Measurement (EDM) total station measures a slope distance $S = 845.620$ m at zenith angle $z = 84°\ 15'\ 00''$ (**84.15** ↶ ▸HR → 84.2500°) under warm ambient air requiring a +18.5 ppm ($1 + 18.5 \times 10^{-6}$) meteorological refractive-index correction, scaling S by ($1 + 18.5 \times 10^{-6}$) and projecting by $\sin(z)$ reduces the sightline to the local tangent-plane horizontal baseline $H =$ **841.3802 m** (prior to multi-kilometer Earth curvature and refraction corrections):

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
78000 ENTER **56000** ÷ LN	0.0000	0.0000	0.0000	0.331357	Logarithmic weight ratio $\ln(W_0 \,/\, W_1)$
18.5 × **1420** ×	0.0000	0.0000	0.0000	8704.7513	Breguet cruise range R = 8,704.75 km
845.620 ENTER **1** ENTER **18.5** E **6** +/− + ×	8704.7513	8704.7513	8704.7513	845.6356	Apply +18.5 ppm meteorological EDM scale
84.15 ↶ ▸HR SIN ×	8704.7513	8704.7513	8704.7513	**841.3802**	Project by sin(84° 15') for horizontal baseline (m)

- **Konstantin Tsiolkovsky's Two-Stage Rocket ($\Delta v_{\text{net}} = v_{e1}\ln(m_{01}/m_{f1}) + v_{e2}\ln(m_{02}/m_{f2})$ - Δv_{loss}):** Stage 1 (v_{e1} = 2,950 m/s, mass ratio 4.5833) and Stage 2 (v_{e2} = 3,420 m/s, mass ratio 6.1290) sum across Registers Y and X before deducting gravity and drag losses (Δv_{loss} = 1,350 m/s), with $Z = T = 0$. Evaluate Stage 1 (Δv_1 = 2950 ln(4.5833) = 4,491.1368 m/s) via **4.5833** LN **2950** ×, evaluate Stage 2 (Δv_2 = 3420 ln(6.1290) = 6,200.5681 m/s) via **6.1290** LN **3420** × while Δv_1 waits in Y, and combine via + **1350** - to obtain v_{net} = 9,341.7049 m/s.
- **Arthur C. Clarke's Geostationary Altitude ($h_{\text{GEO}} = (\mu\ T^2 \,/\, 4\pi^2)^{1/3}$ - $R_\oplus$):** With $\mu = 3.986004 \times 10^{14}$ m^3/s^2, sidereal period T = 86,164.09 s, and equatorial radius $R_\oplus = 6.37814 \times 10^6$ m, square T^2, multiply by μ, divide by 4 and π^2, and raise to the 1/3 power to obtain the synchronous semi-major axis a = 42,164,140 m (42,164.14 km), then subtract $R_\oplus$ via **86164.09** ↶ x^2 **3.986004** E **14** × **4** ÷ R↓ π ↶ x^2 ÷ **3** x^{-1} y^x **6.37814** E **6** - to find the geostationary altitude (X = 35,786,000 m = 35,786.0 km).

Across optical instruments, precision bearing assemblies, turbulent heat exchangers, acoustic enclosures, and orbital launch vehicles, multi-domain engineering analysis rests on a remarkably small set of recurring algebraic structures. Harmonic reciprocals ($x \parallel y$), orthogonal variance norms (↶ ▸θ,r), logarithmic temperature ratios (LN), and fractional power laws (y^x) evaluate across the same four registers regardless of whether the physical units are micrometers, Pascals, Sabins, or kilometers per second.

Chapter 14: Fermi Estimation, Algebraic Symmetry & High-Speed Calculation

14.1 Enrico Fermi's Torn Paper Scraps and the Art of Back-of-the-Envelope Estimation

At 5:29 a.m. on July 16, 1945, in the Jornada del Muerto desert of New Mexico, the physicist Enrico Fermi stood at Base Camp ten miles ($r \approx 16$ km) south of Ground Zero as the world's first nuclear detonation turned the pre-dawn sky white. While instruments whirred and oscillographs traced pressure curves that would take weeks to reduce, Fermi held a handful of torn notebook-paper scraps above his head. Forty seconds after the flash, when the shockwave reached Base Camp, he released the scraps into the sudden gust of air. They fluttered backward and landed on the desert floor about two and a half meters ($\Delta r \approx 2.5$ m) behind him.

Pacing off that displacement, Fermi mentally estimated the mechanical work $W \approx \Delta P \, (2\pi r^2) \Delta r$ done by a hemispherical blast shell of area $2\pi r^2 \approx 1.6 \times 10^9$ m^2 at weak-shock overpressure $\Delta P \approx 0.1 \, P_0 \approx 10^4$ Pa pushing the air outward by $\Delta r \approx 2.5$ m ($W \approx 4 \times 10^{13}$ J). Within seconds, he announced to his colleagues that the blast equaled roughly 10 kilotons of TNT (4.184×10^{13} J). Weeks later, exhaustive radiochemical analysis of the fused desert glass settled on 18.6 kilotons—well within a factor of two of Fermi's torn-paper estimate.

Fermi trained his University of Chicago students to cultivate that same numerical poise before touching a slide rule or desk calculator. His classic oral-exam problem, estimating the number of piano tuners working in the city of Chicago, seemed intractable without a telephone directory, yet collapsed in thirty seconds when decomposed into a chain of bounded factors. Starting from a city population of roughly 3×10^6 residents at about 4 people per household gives 7.5×10^5 homes. If roughly 1 household in 15 owns a piano (bounded between 1 in 10 and 1 in 20), the city contains about 5×10^4 pianos, which tuned once a year require 5×10^4 annual service visits. Because a full-time technician servicing 4 pianos a day across 250 working days completes 1,000 tunings per year, dividing 50,000 by 1,000 estimates 50 piano tuners in Chicago.

Multiplying five bounded estimates produces a reliable answer instead of compounding into error *(Munroe, 2014)* because independent overestimates and underestimates cancel in logarithmic space. Whenever you are unsure of a factor in a Fermi chain, bracket it between a plausible lower bound L (too small) and a plausible upper bound H (too large) and take their geometric mean, modeled on the SC-32 in three strokes via **L** [ENTER] **H** [×] [√x]:

$$x_{\text{est}} = \sqrt{L \times H} \tag{Eq. 14.1}$$

Taking the base-10 logarithm reveals why the geometric mean is the natural center of an order-of-magnitude interval:

$$\log_{10}\left(\sqrt{L \times H}\right) = \frac{\log_{10} L + \log_{10} H}{2} \tag{Eq. 14.2}$$

If you bound a quantity between 20 and 80, the arithmetic mean (50) overweights the upper end in relative terms, whereas the geometric mean $\sqrt{(20 \times 80)} = \sqrt{(1{,}600)} = 40$ sits at

an exact factor of 2 above 20 and a factor of 2 below 80.

14.1.1 Random Log-Error Cancellation and the $\sqrt{N}$ Law

A common objection to Fermi estimation is that breaking one unknown number into five separate estimates appears to introduce five opportunities for error instead of one.

The mathematics of logarithms and statistics proves that the exact opposite is true. Try guessing the number of piano tuners in Chicago in a single blind leap: because an unanchored guess has no everyday human scale to tether it, you might easily guess 5 or 5,000, an error of two orders of magnitude ($10^{\pm 2}$).

Decomposing that unknown quantity into a product of $k = 5$ everyday factors $X = f_1 \times f_2 \times f_3 \times f_4 \times f_5$ (population, household size, piano ownership fraction, tunings per year, and tunings per day) anchors each step in familiar daily life. Because a family household has around 3 to 5 people rather than 400, and a technician tunes 3 to 5 pianos a day rather than 300, your relative error on each individual factor f_i is tightly bounded (say within $\pm$ 30%, or $\pm$ 0.12 in base-10 log units).

When you multiply k independent factors together, their base-10 logarithms add:

$$\log_{10} X = \log_{10} f_1 + \log_{10} f_2 + \log_{10} f_3 + \log_{10} f_4 + \log_{10} f_5 \qquad \textbf{(Eq. 14.3)}$$

If your individual log-errors ε_i are unbiased (some slightly high, like guessing 1 in 15 households own a piano when the true fraction is 1 in 20; others slightly low, like guessing 3 million residents when the metro area has 4 million), they do not add up in the worst-case direction. Instead, just like random machining tolerances, independent logarithmic errors combine via the Root-Sum-Square (**RSS**) random-walk law ($\sigma_{\text{total}} = \sqrt{k}\ \sigma_{\text{step}}$ rather than $k\ \sigma_{\text{step}}$). Overestimating two links and underestimating three links causes the errors to cancel each other out across the multiplication chain.

14.1.2 Four Everyday Sanity Benchmarks Worth Knowing by Heart

To run a fast Fermi check on any physics, chemistry, finance, or engineering problem before your fingers touch the keys, four compact physical anchors establish immediate order-of-magnitude bounds. First, one solar year of 365.2422 days $\times$ 86,400 s/day equals 3.1557×10^7 seconds, which lies within half a percent of $\pi \times 10^7$ seconds; whenever a problem converts annual power consumption, radioactive decay, or satellite drift into per-second rates, $\pi \times 10^7$ sets the order of magnitude at a glance. Second, liquid water has a density of 1 g/cm^3 = 1 kg/L = 1,000 kg/m^3 (one metric ton per cubic meter), whereas air at sea level is roughly 1/800 as dense ($\approx$ 1.225 kg/m^3), aluminum is 2.7$\times$ water, and structural steel is $\approx$ 7.8$\times$ water.

Third, standard sea-level atmospheric pressure (101.325 kPa $\approx$ 14.7 psi) equals the hydrostatic weight of a 10.3-meter (34-foot) column of fresh water, so every 10 meters deeper a submarine or dam face descends adds roughly 1 atmosphere (100 kPa) of water pressure. Fourth, at room temperature ($T \approx 298$ K = 25°C), the Nernst and Shockley thermal voltage $k_B\ T\ /\ q$ governing both silicon computer chips and human nerve cells is roughly 25.7 mV, which produces 25.7 mV $\times$ ln 10 $\approx$ 59.2 mV per tenfold decade of ion concentration or diode current, while sound travels through room-temperature air at roughly 343 m/s, or about 1 kilometer every 3 seconds and 1 mile every 5 seconds.

In multi-step technical calculation, pairing every keypad evaluation with an order-of-magnitude Fermi bound catches slipped decimal points and transposed digits before recording the result. Whenever you combine two positive rates or impedances via the

harmonic operator $x \parallel y = xy / (x+y)$, the result must lie strictly below the smaller input and at least half of that smaller input:

$$\tfrac{1}{2}\min(x,y) \le x \parallel y < \min(x,y) \qquad \textbf{(Eq. 14.4)}$$

In a bottom-up continued fraction $a_0 + 1 / (a_1 + 1 / (a_2 + ...))$, the leading integer plus the first reciprocal rung $(a_0 + 1/a_1)$ brackets the exact convergent within $\pm 1/a_1^2$; similarly, in a nested radical pyramid $\sqrt{(a + \sqrt{(b + \sqrt{c})})}$, adding the integer floor of the inner root gives a two-digit mental preview of the outer square root. Finally, by Luca Pacioli's Rule of 72, for compound interest or exponential growth $FV = P(1+r)^n$, dividing 72 by the percentage rate $r_{\%}$ gives the doubling time $T_{\text{double}} \approx 72/r_{\%}$. Over an n-year horizon, capital undergoes $d = n / T_{\text{double}}$ doublings, bounding the terminal balance near $P \times 2^d$ before a single key is pressed.

14.2 Structural Algebraic Symmetries and Mental Calculation

Legend holds that when young Carl Friedrich Gauss sat in J. G. Büttner's arithmetic classroom in Brunswick in 1784 *(Sartorius von Waltershausen, 1856)*, the schoolmaster assigned the class the task of summing the integers from 1 to 100. While his classmates scribbled column additions on their slates, Gauss laid his slate on the teacher's desk after a few moments with a single number written on it: 5,050. Rather than adding from left to right, Gauss had folded the ascending staircase of numbers onto a reversed copy of itself, pairing $1 + 100 = 101$, $2 + 99 = 101$, ..., $50 + 51 = 101$ into 50 identical pairs ($S_n = n(n+1)/2 = 50 \times 101 = 5{,}050$). Almost two centuries later, the Indian mental calculator Shakuntala Devi demonstrated that same structural vision on stage, out-computing mainframe computers by spotting algebraic symmetries that collapse ten arithmetic operations into two.

Four geometric symmetries turn multi-step algebraic expressions into rapid mental or three-keystroke reflexes.

14.2.1 Difference-of-Squares Symmetry

An $a \times a$ square with a smaller $b \times b$ square removed from its upper-right corner leaves an L-shaped gnomon of area $a^2 - b^2$. Slicing off the narrow vertical leg of dimensions $b \times (a - b)$ and rotating it 90° onto the top edge locks the two rectangles together into a single $(a - b) \times (a + b)$ rectangle:

$$a^2 - b^2 = (a-b)(a+b) \qquad \textbf{(Eq. 14.5)}$$

Read from left to right, this identity factors the difference of two squares without squaring either number: for $68^2 - 32^2$, multiplying their difference $(68 - 32 = 36)$ by their sum $(68 + 32 = 100)$ produces 3,600 immediately. Read from right to left, the same identity turns any product of two numbers equidistant from a round midpoint a into a one-second mental subtraction $(a - b)(a + b) = a^2 - b^2$. For 47×53, because both factors sit at distance $b = 3$ from $a = 50$:

$$47 \times 53 = (50-3)(50+3) = 50^2 - 3^2 = 2{,}500 - 9 = 2{,}491 \qquad \textbf{(Eq. 14.6)}$$

14.2.2 Squaring Any Number Ending in 5

Setting the offset $b = 5$ in the difference-of-squares rectangle rewrites $a^2 - 5^2 = (a - 5)(a + 5)$ as $a^2 = (a - 5)(a + 5) + 25$, showing why squaring any number ending in 5 requires zero multiplication of the full number. For a two-digit number $10n + 5$ (such as 35 or 85, or a half-integer like 8.5), subtracting 5 gives $10n$ and adding 5 gives $10(n + 1)$, so:

$$(10n+5)^2 = 100n(n+1) + 25 \quad \textbf{(Eq. 14.7)}$$

Multiplying the leading digit n by the next integer $n + 1$ and appending 25 evaluates the square in place: for 35^2, multiplying $3 \times 4 = 12$ and appending 25 gives 1,225, while for 8.5^2, multiplying $8 \times 9 = 72$ and appending .25 gives 72.25 (the exact lower-base square R^2 of a conical frustum).

14.2.3 Harmonic Round-Trip Speeds

When a vehicle climbs a mountain pass of length d at speed v_1 and descends that same distance d at a faster speed v_2, intuition often suggests the arithmetic mean $1/2(v_1 + v_2)$, overlooking the unequal duration spent on each leg. Because the vehicle spends a longer time $t_1 = d / v_1$ ascending at the slower speed than descending ($t_2 = d / v_2$), the round-trip average speed over total distance $2d$ is weighted by travel time and produces the harmonic mean:

$$v_{\text{avg}} = \frac{2d}{\frac{d}{v_1} + \frac{d}{v_2}} = \frac{2v_1v_2}{v_1 + v_2} = 2(v_1 \parallel v_2) \quad \textbf{(Eq. 14.8)}$$

Because the single-leg factor $v_1 v_2 / (v_1 + v_2)$ is identical to the parallel harmonic operator $v_1 \parallel v_2$, any equal-distance round-trip average speed reduces to twice the parallel combination of the two leg speeds, and by the Cauchy-Schwarz inequality satisfies $v_{\text{avg}} < 1/2(v_1 + v_2)$ whenever $v_1 \neq v_2$.

14.2.4 Right-Triangle Inradius via Corner-Square Folding

When a geometry problem asks for the inscribed circle radius (inradius) r of a right triangle with legs a and b and hypotenuse c, the general Heron-semiperimeter formula computes the area $A = (1/2)ab$ and semiperimeter $s = (a+b+c) / 2$ to divide $r = A/s$. When the inscribed circle is tucked into the 90° corner of the right triangle, the two radii perpendicular to the legs form an $r \times r$ square in the right-angle vertex, while the remaining segments of the legs ($a - r$ and $b - r$) fold onto the hypotenuse c as equal tangent segments from the two acute vertices. Therefore $(a - r) + (b - r) = c$, which solves directly for the right-triangle inradius identity:

$$r = \frac{a+b-c}{2} \quad \textbf{(Eq. 14.9)}$$

Because the $r \times r$ corner square must fit inside the shorter leg of the triangle, this tangent decomposition also provides the immediate upper bound $r < (1/2) \min(a, b)$.

14.2.5 Replacing Divisions by 5, 25, and 125 with Doubling Cascades

Mental division of a multi-digit number by 5, 25, or 125 is slow when carried out by long division with remainders, whereas multiplying by 2, 4, or 8 reduces to repeated doubling. Because 5, 25, and 125 are the exact powers of 5 that pair with 2, 4, and 8 to form powers of 10:

$$\frac{x}{5}=\frac{2x}{10},\qquad \frac{x}{25}=\frac{4x}{100},\qquad \frac{x}{125}=\frac{8x}{1,000} \tag{Eq. 14.10}$$

Whenever a problem asks you to divide by 5, 25, or 125 (or multiply by 0.25, 2.5, 12.5, or 25), replace the division with a left-to-right doubling cascade and shift the decimal point. To divide 430 ÷ 5, double 430 → 860 and shift one decimal place left to obtain 86. To divide 1,700 ÷ 25, quadruple 17 × 4 = 68 (double 17 → 34 → 68) to obtain 68. Conversely, to multiply 36 × 25, divide 36 by 4 (→ 9) and append two zeros to obtain 900.

14.2.6 Nicomachus's Sum-of-Cubes Identity ($1^3 + 2^3 + \ldots + n^3$)

Once you know Gauss's triangular formula for the sum of the first n integers, the n-th triangular number $T_n = 1 + 2 + \ldots + n = n(n+1)/2$, two higher power-sum formulas follow directly for moments of inertia, Riemann integral limits, and mathematical competitions. The sum of the first n squares equals $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1) / 6 = (2n+1) / 3\ T_n$, while Nicomachus of Gerasa's sum-of-cubes theorem (c. 100 CE) shows that adding the first n cubes ($1^3 + 2^3 + 3^3 + \ldots + n^3$) produces the exact square of Gauss's triangular number T_n:

$$1^3+2^3+3^3+\ldots+n^3=(1+2+3+\ldots+n)^2=\left[\frac{n(n+1)}{2}\right]^2 \tag{Eq. 14.11}$$

For $n = 4$: $1 + 2 + 3 + 4 = 10$, and $1^3 + 2^3 + 3^3 + 4^3 = 1 + 8 + 27 + 64 = 100 = 10^2$. To sum the cubes from 1^3 to 20^3, never cube twenty numbers: compute $T_{20} = 20 \times 21 / 2 = 210$ and square it ($210^2 = 44{,}100$) in three seconds.

14.3 Reading Giant Expressions from the Inside Out: Visual Chunking & Horner Rhythm

A four-tier fraction or nested radical pyramid appears intricate on a printed page because traditional mathematical typesetting is written from the outside in, whereas numerical evaluation happens from the inside out.

14.3.1 Visual Chunking: Finding the Innermost Seed

When a practiced problem-solver glances at a complex equation, their eye ignores the outer fraction bars and radical roofs and locks onto the innermost seed. In a nested radical pyramid such as $\sqrt{(17 + \sqrt{(29 + \sqrt{83})})}$, rather than reading from left to right and counting brackets, start at the right-hand core 83 and take its square root ≈ 9.11, add 29 to get 38.11, take the root ≈ 6.17, add 17 to get 23.17, and take the outer root ≈ 4.81, checking each intermediate value as it appears.

Similarly, a continued-fraction ladder such as $3 + 1 / (7 + 1 / (15 + 1 / (1 + 1/292)))$ is an inverted staircase of unit fractions: starting at the bottom denominator (292) and alternating reciprocals with integer additions climbs all five tiers without a single parenthesis or scratchpad note. The same visual chunking governs single-entry polynomial and mensuration cores such as the truncated-cone (frustum) volume formula from the 1850 BCE Moscow Mathematical Papyrus:

$$V=\frac{1}{3}\pi h(R^2+Rr+r^2) \tag{Eq. 14.12}$$

In the symmetric quadratic core $R^2 + Rr + r^2$, as well as in the Law of Cosines core $a^2 + b^2 - 2ab \cos C$ or the depressed cubic $f(x) = x^3 - cx$ in a central-difference derivative $f'(x) \approx (f(x+h) - f(x-h)) / 2h$, chaining the calculator's automatic last-argument memory ([↶]

[LAST*x*]) as a single-entry spine evaluates R^2, recalls R to multiply by r, and recalls r to form r^2 while typing each multi-digit dimension only once.

14.3.2 Single-Entry Evaluation and Transcription Error Prevention

Experienced calculators rely on single-entry spines, entering a parameter such as $x = 3.847$ or $R = 8.5$ only once when evaluating $P(x) = 2x^3 - 5x^2 + 4x - 7$ or $V = (1/3)\pi\, h(R^2 + Rr + r^2)$. Retyping a five-keystroke decimal number (**3 . 8 4 7**) three separate times inside a polynomial carries three distinct costs. First, typing **3.847** three times consumes 15 keystrokes instead of 5, tripling the manual entry overhead on a single evaluation. Second, during rapid entry, every manual digit sequence carries a non-zero probability of a finger slip (such as typing **3.874** on the quadratic term after typing **3.847** on the cubic term), and a small transposition error in a secondary term often produces an answer that still passes a rough order-of-magnitude check, so the mistake remains hidden.

Third, on a four-register stack, once a number is keyed into Register X, it is an exact IEEE-754 floating-point object. Tapping [ENTER] duplicates it into Register Y in one tap, filling the stack via [ENTER] [ENTER] [ENTER] uses top-of-stack replication in Register T to supply exact copies of x for Horner's scheme, and tapping [↶] [LAST*x*] after [↶] [x^2] or [×] restores the exact pre-operation operand without pressing a single digit key.

14.3.3 Structural Principles of Inside-Out Expression Evaluation

Whenever you face an unfamiliar multi-tier formula in physics, chemistry, finance, or engineering, four inside-out principles *(Pólya, 1945)* determine the evaluation order. First, start at the deepest asymmetric leaf by scanning inside the innermost parentheses, radical roof, or exponent for the operation that requires two numbers (such as a/b or $c + d$), and evaluate that seed first so every subsequent unary function (**LN**, **SIN**, **√x**, $\mathbf{e}^x$, **1/x**) executes in place on Register X without lifting the stack. Second, never press **ENTER** after a completed calculation: under the Stack-Lift State Machine, every arithmetic key (**+**, **-**, **×**, **÷**) and every function key (**√x**, $\mathbf{x}^2$, **LN**, **SIN**, **1/x**) automatically leaves Stack Lift enabled, so typing the next constant pushes your finished subtotal up into Register Y, whereas pressing **ENTER** after **+** or **√x** duplicates the subtotal into both Y and X and wastes a stack register.

Third, let [$x \leftrightarrow y$] and [x^{-1}] replace parentheses in subtracted or divided denominators: once a complicated denominator D sits in Register X and you need N/D or $A - D$, tap [x^{-1}] and multiply by N ($N \times D^{-1}$), or type N (which lifts D to Register Y), tap [$x \leftrightarrow y$] to put N in Y and D in X, and press [÷] (or [-]). Fourth, park finished branches in Register Z while building on Y and X: any expression of the form (A op B) ± (C op D) needs only three registers (X, Y, Z), because computing (A op B) first lets the first branch ride up to Register Z while C and D are entered, and when (C op D) collapses into Register X, the first branch drops from Z to Y ready for the final dyadic operator.

14.3.4 Beyond Four Chunks: Cognitive Horizon, Unbounded Stacks & the Clean-Slate Ethos

Harvard psychologist George A. Miller's working-memory principle *(Miller, 1956)* and later cognitive studies show that human attention holds roughly four active chunks at once, the exact cognitive horizon mirrored by the classic four-register stack (X, Y, Z, T). By induction on the branching depth of an expression tree (its *Strahler number*), four registers suffice to evaluate any binary tree of branching depth 4 (including every standard closed-form formula in physics and finance) from the inside out without touching a named memory register, while top-of-stack T-replication turns [ENTER] [ENTER] [ENTER] into a self-replenishing constant supply for Horner's scheme.

Two practical considerations govern stack depth expansion beyond four registers and examination memory-state enforcement. When an engineering problem branches into five or six parallel sub-expressions—such as a five-branch parallel impedance network ($R_1 \parallel R_2 \parallel R_3 \parallel R_4 \parallel R_5$), a four-mass rotational inertia sum $\sum_{k=1}^{4} m_k\ r_k^2$ where each $m_k\ r_k^2$ term is inspected before addition, or a five-term rational polynomial quotient whose numerator and denominator each require three active registers—a strict 4-register ceiling forces a trade-off.

Pushing a fifth value onto a 4-level stack silently drops the oldest subtotal off Register T, and subsequent binary operations replicate whatever sits in T downward. Expanding the stack into an 8-register ([8-LVL]) or unbounded Last-In, First-Out ([INF]) push-down heap via [↗] [FLAGS] → **STACK** removes that register-depth bookkeeping: you can load five or ten given numbers onto the stack in a single left-to-right pass, or build five independent sub-branches without overwriting an outer subtotal at the top of the stack.

In practice, three tactical rules govern stack-mode selection:

- `4-LVL` **(Classic 4-Register Default):** Keep **4-LVL** active whenever you rely on x_0 [ENTER] [ENTER] [ENTER] top-of-stack T-replication to supply an automatic constant for Horner polynomial evaluation or repeated scaling.
- `8-LVL` **(Extended 8-Register / 4-Complex-Level Stack):** Switch to **8-LVL** for multi-branch complex phasor calculations—where 8 real registers provide a full 4-level complex stack (z_1 ... z_4) so two-branch networks like $(Z_A \parallel Z_C) + (Z_B \parallel Z_D)$ evaluate without **STO**/**RCL**—or for 5-to-8-register real expressions when you still want top-register replication at R_8.
- `INF` **(999-Level Unbounded LIFO Heap):** Switch to **INF** for five-or-more-branch parallel networks ($R_1 \parallel R_2 \parallel R_3 \parallel R_4 \parallel R_5$), multi-term product sums ($\sum_{k=1}^{6} m_k\ r_k^2$), and deep rational expressions where a dyadic drop must pull cleanly from stack history without duplicating the top register.

At the same time, quantitative examinations and professional licensure assessments test live mathematical reasoning rather than pre-stored equation libraries in non-volatile memory. Because proctors cannot manually inspect twenty-six lettered registers and scroll through stored equations on every desk, examination rules require a hardware-enforced clean-slate state: an irreversible one-action purge that zeroes every stored equation, variable, financial register, and statistical accumulator, suppresses help overlays, and illuminates a tamper-evident status badge, while leaving the live stack and numerical solvers running at full precision.

14.3.5 Expression Complexity & Register Footprint Classification

Across number theory, planar and solid mensuration, coordinate geometry, and applied physics, quantitative problems stratify naturally by the depth of their expression trees (their Strahler number) and the number of active stack registers required to evaluate them from the inside out without external variable storage:

Complexity Tier	Mathematical Structure	Active Registers	Representative Operations
Tier 1	Single-branch arithmetic & percentages	1–2 (X, Y)	Chained products/quotients, [↗] [%], [↗] [%CHG]
Tier 2	Nested unary cascades & continued fractions	1–2 (X, Y)	Radical pyramids ([$\sqrt{x}$]), reciprocal ladders ([x^{-1}])

Complexity Tier	Mathematical Structure	Active Registers	Representative Operations
Tier 3	Planar right-triangle & vector geometry	2 (X, Y)	Pythagorean norms (↶ ▸θ,r), polar/rectangular conversions (↷ ▸y,x), inradii
Tier 4	Oblique trigonometry & two-branch quotients	2–3 (X, Y, Z)	Law of Sines/Cosines, compound fractions $(A+B)/(C-D)$, secant slopes
Tier 5	Solid mensuration & harmonic networks	2–3 (X, Y, Z)	Conical frustums (R^2+Rr+r^2 via ↶ LASTx), thin lenses & parallel loads ($x\|\|y$)
Tier 6	Transcendental modeling & numerical calculus	2–3 (X, Y, Z)	Exponential decay (e^x), central differences, Brent root-finding (↷ SOLVE), Romberg (↷ ∫)
Tier 7	Coupled multi-branch & astrodynamic systems	3–4+ (X ... T or **INF**)	Hohmann transfer impulses, **LMTD** exchangers, multi-branch **INF** networks

Recognizing a problem's structural tier before pressing a key determines the evaluation path immediately: Tiers 1 and 2 fold in place inside Register X with automatic stack lift, Tiers 3 through 6 stage at most one or two intermediate subtotals in Registers Y and Z while reusing shared arguments from ↶ LASTx, and Tier 7 either parks an outer branch in Register Z or expands the stack to INF so five or more parallel terms can accumulate in a single pass.

14.4 High-Speed Estimation & Multi-Stage Evaluations



Figure: High-speed inside-out evaluation of a nested radical pyramid sqrt(11 + sqrt(21 + sqrt(16))) = 4 on the StackCalc SC-32, producing the exact integer result 4 with zero parentheses.

14.4.1 Verification Discipline, Error Recovery & Examination Mode

Four operational practices maintain numerical accuracy and stack hygiene across multi-stage calculations:

1. **Significant-Figure Precision & Angular-Mode Verification:** Standard engineering and scientific evaluations report results to three or four significant figures (**14.2**, **0.0482**, **1.48E5**, or **3.14**), using E for scientific-notation mantissa-exponent entry, and always verify that the **DEG** or **RAD** annunciator matches the problem formulation before evaluating trigonometric expressions (↶ MODES DEG).

2. **Instant Error Recovery via ↶ LASTx:** If you accidentally press + instead of - on step 8 of a 10-step calculation, never clear the stack and start over. Because **LASTX** holds the operand that was in Register X just before the mistake, pressing ↶ LASTx - undoes the addition and restores your prior subtotal, and a second ↶ LASTx - performs the intended subtraction.

3. **Diagram Sketching vs. Intermediate Stack Retention:** Reserve scratch paper for sketching free-body diagrams, geometric orientations, or boundary conditions rather than transcribing intermediate floating-point numbers that already reside at full 64-bit

precision in Registers X, Y, Z, and T.

4. **Examination Mode ([EXAM]) & Stack Depth Selection (4-LVL, 8-LVL & INF):** Opening [↗] [FLAGS] exposes **STACK** (softkey 4) and [EXAM] (softkey 5). Tapping **EXAM** executes an irreversible **CLALL** memory wipe that zeroes all stack levels, **LASTX**, all six statistical accumulators, all 26 variables A–Z, indirect registers i and (i), all five **TVM** financial registers, and every stored **EQN** formula, while resetting the calculator to **DEC**/**ALL** and illuminating the persistent **EXAM** status badge. Selecting **STACK** opens the **2-LVL | 4-LVL | 8-LVL | INF | VIEW** softkey bar, where you can choose [4-LVL] for the classic 4-register stack with T-replication, [8-LVL] for 8 registers (X ... R_8), [INF] for an unbounded 999-level LIFO heap stack (System Flag 5) without top-of-stack loss or T-duplication, and [↗] [VIEW] for multi-line stack inspection (System Flag 4).

14.4.2 Rational Number Crunchers, Reciprocals & Compound Growth

Before writing down any result, pair your RPN stack evaluation with a fast mental bound. For example, with [x⇄y], if **30** [ENTER] **60** [x⇄y] displays **20.0**, your eye verifies at a glance that $15 \leq 20.0 < 30$; and for a right triangle with legs a, b, obtaining the hypotenuse via **a** [ENTER] **b** [↶] [▸θ,r] lets you check $r = (a+b-c) \,/\, 2 < (1/2)\,\min(a,b)$.

Mental bounding and automatic stack lift work together on compound rational fractions:

$$Q = \frac{148.6 + 92.4}{314.2 - 188.7} \qquad \textbf{(Eq. 14.13)}$$

Mentally, the numerator is ≈ 150 + 90 = 240 and the denominator is ≈ 314 - 189 = 125, so the quotient must lie just under 240 / 120 = 2.0. On the SC-32, automatic stack lift acts as a natural fraction bar: once [+] finishes the numerator (241.0), keying the denominator's first number (**314.2**) automatically lifts 241.0 into Register Y, and pressing [ENTER] lifts it again into Register Z while the denominator difference (125.5) forms in X and Y. When [-] collapses the denominator into X, the numerator drops right back into Y, so keying **148.6** [ENTER] **92.4** [+] **314.2** [ENTER] **188.7** [-] [÷] delivers **1.9203187251**:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
148.6 [ENTER]	0.0000	0.0000	148.6000	148.6000	Lift first numerator term to Y
92.4 [+]	0.0000	0.0000	0.0000	241.0000	Numerator sum 148.6 + 92.4 = 241.0
314.2 [ENTER]	0.0000	241.0000	314.2000	314.2000	Auto-lift numerator to Z, enter minuend
188.7 [-]	0.0000	0.0000	241.0000	125.5000	Denominator difference 314.2 - 188.7 = 125.5
[÷]	0.0000	0.0000	0.0000	**1.9203**	Quotient 241.0 / 125.5 = 1.920319

That same automatic stack lift accelerates differences of reciprocals:

$$Q = \frac{1}{0.0482} - \frac{1}{0.0895} \qquad \textbf{(Eq. 14.14)}$$

Mentally, 1/0.05 - 1/0.09 ≈ 20 - 11 = 9. Because every unary function on the SC-32—including [x^{-1}]—terminates digit entry and arms stack lift, this reciprocal difference requires zero [ENTER] strokes. Keying **.0482** [x^{-1}] **.0895** [x^{-1}] [-] computes the first reciprocal (20.7469) and arms the stack, auto-lifts 20.7469 into Register Y when **.0895** is typed, inverts 0.0895 in place (11.1732), and subtracts via [-] to deliver **9.5737036093** in five operations.

In analytic geometry, keeping signed coordinate differences organized on the stack prevents sign flips when computing the secant slope between two survey benchmarks $(x_1,\ y_1) = (12.4,\ 8.5)$ and $(x_2,\ y_2) = (34.8,\ -14.2)$:

$$m = \frac{y_2 - y_1}{x_2 - x_1} \qquad \textbf{(Eq. 14.15)}$$

Evaluating the vertical rise Δy = -14.2 - 8.5 = -22.7 first and then computing the horizontal run Δx = 34.8 - 12.4 = 22.4 automatically lifts Δy into Register Y for [÷]. Keying **14.2** [+/−] [ENTER] **8.5** [-] **34.8** [ENTER] **12.4** [-] [÷] subtracts the ordinates (Δy = -22.7000), forms the abscissa run (Δx = 22.4000 in X while Δy waits in Y), and divides to return **-1.013392857**.

For multi-period exponential growth, when a principal of $5,000.00 is invested in a college endowment fund earning 6.5% annual interest compounded annually for $n = 18$ years ($FV = P(1+r)^n$, where Pacioli's Rule of 72 estimates 72 / 6.5 ≈ 11 yr per doubling, or 1.6 doublings to ≈ $15,000), forming the growth factor mentally (1 + 0.065 = 1.065), raising it to the 18th power, and scaling by principal via **1.065** [ENTER] **18** [y^x] **5000** [×] determines the terminal balance **$15,533.2719** in three operations.

14.4.3 Radical Pyramid Cascades & Bottom-Up Continued Fractions

Multi-tier radical pyramids and continued-fraction ladders require zero upper-stack registers when evaluated from the inside out. On an algebraic keypad, an expression like

$$Q = \sqrt{17 + \sqrt{29 + \sqrt{83}}} \quad \text{or} \quad Q = 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1 + \frac{1}{292}}}} \qquad \textbf{(Eq. 14.16)}$$

forces you to open three or four nested parentheses from the outside in, holding every outer constant in pending memory until the final closing bracket. On an RPN stack starting at the deepest inner seed, the exact opposite happens: at the end of every single step, Registers T, Z, and Y remain 0.0000, because each unary operation ([$\sqrt{x}$] or [x^{-1}]) transforms Register X in place and each immediate integer offset (**29** [+] or **15** [+]) folds straight back into Register X.

To unpeel the radical pyramid (**T = Z = Y = 0** throughout), lock your eye on the innermost radicand (83) and work outward via **83** [$\sqrt{x}$] **29** [+] [$\sqrt{x}$] **17** [+] [$\sqrt{x}$]. Evaluating the inner core $\sqrt{83}$ = 9.11043358 in Register X arms stack lift; adding 29 lifts 9.11043358 momentarily to Y and immediately collapses the middle radicand 29 + 9.11043358 = 38.1104336 back into Register X (returning Y to 0.0000); extracting the middle root via [$\sqrt{x}$] yields 6.17336485 in place; adding 17 forms the outer radicand 23.1733649 in X; and taking the final square root delivers **4.8138721265** (mentally bounded by $\sqrt{(17 + \sqrt{(29 + 9)})} \approx \sqrt{(23.2)} \approx 4.8$) without a single [ENTER] tap.

To climb the continued-fraction ladder (**T = Z = Y = 0** throughout), that identical single-register rhythm evaluates the 5-tier continued-fraction convergent for π (355/113 extended by 292) from the ground floor up via **292** [x^{-1}] **1** [+] [x^{-1}] **15** [+] [x^{-1}] **7** [+] [x^{-1}] **3** [+]. Starting at the deepest denominator, inverting 292 via [x^{-1}] places 1/292 = 0.00342466 in Register X and arms stack lift; adding 1 and inverting flips the fourth-tier rung (1.00342466 → 0.99658703) in place; adding 15 and inverting steps up the third rung (15.9965870 → 0.06251333); adding 7 and inverting steps up the second rung (7.06251333 → 0.14159265); and adding the leading integer 3 lands on **3.141592653** (≈ π) with zero [ENTER] presses and zero parentheses.

14.4.4 Inside-Out Mensuration & Multistage Physics Across Twenty Orders of Magnitude

In multi-stage geometry and physics problems, isolating a self-contained dimensionless core inside the formula and evaluating that core first in Register X allows all outer scaling constants to multiply or divide in a straight line without needing Registers Z or T.

For circular segment wetted area ($A = (1/2)r^2(\vartheta_{\text{rad}} - \sin\vartheta)$), when liquid fills a horizontal cylindrical pipe of radius $r = 12.0$ cm until the wetted perimeter subtends a central angle of $\vartheta = 65.0°$, the cross-sectional wetted area subtracts the isosceles triangle $(1/2)r^2 \sin\vartheta$ from the pie-slice sector $(1/2)r^2\vartheta_{\text{rad}}$:

$$A = \tfrac{1}{2}r^2\left(\frac{\pi\theta}{180} - \sin\theta\right) \qquad \textbf{(Eq. 14.17)}$$

Rather than computing $(1/2)r^2$ first and parking it on the upper stack while evaluating the mixed radian-and-degree parenthesis, evaluate the dimensionless segment shape factor ($\pi\vartheta/180$ - sin ϑ) first in [DEG] mode via **65** [R↓] [π] [×] **180** [÷] **65** [SIN] [-] **12** [R↑] [x^2] [×] **2** [÷]. Converting $\vartheta = 65°$ to radians places $\vartheta_{\text{rad}} = 65\pi/180 = 1.134464$ in Register X, evaluating $\sin(65°) = 0.906308$ automatically lifts ϑ_{rad} to Register Y, subtracting collapses their difference into the pure shape factor 0.228156 in Register X (leaving $Y = Z = T = 0$), and scaling in place by $(1/2)r^2 = 1/2(12)^2 = 72$ cm^2 yields **16.427248327** cm^2.

Alternatively, evaluating $\sin(65°)$ first and recalling $\vartheta = 65°$ straight from [R↑] [LASTx] before multiplying by π / 180 and swapping via [$x\rightleftarrows y$] lets you type the angle only once.

For Lord Kelvin's resonant tank circuit ($f_0 = 1 / 2\pi\sqrt{LC}$), in 1853 William Thomson (Lord Kelvin) proved that discharging a capacitor C through an inductor L produces a sinusoidal electromagnetic oscillation at natural frequency $f_0 = 1 / 2\pi\sqrt{LC}$. For an RF receiver tank circuit with inductance $L = 2.5$ mH (2.5×10^{-3} H) and capacitance $C = 4.7$ nF (4.7×10^{-9} F), never compute 1 first: build the denominator from the inside out in Register X via **2.5** [E] **3** [+/-] [ENTER] **4.7** [E] **9** [+/-] [×] [$\sqrt{x}$] **2** [×] [R↓] [π] [×] [x^{-1}]—forming the reactive time constant $\sqrt{LC} = 3.4278 \times 10^{-6}$ s, scaling by 2π to get the oscillation period $T_0 = 2.1538 \times 10^{-5}$ s, and flipping once at the very end via [x^{-1}] to read off **46,430.26884** Hz without a single parenthesis.

For Walter Hohmann's 1925 geostationary transfer impulse (Δv_1), that same inside-out factoring streamlines orbital mechanics at 10^{14} scale. To boost a spacecraft from a circular low Earth parking orbit ($r_1 = 6.678 \times 10^6$ m, 300 km altitude) onto an elliptical geostationary transfer orbit ($r_2 = 4.2164 \times 10^7$ m) in Earth's gravitational field ($\mu = 3.986 \times 10^{14}$ m^3/s^2), the perigee velocity impulse is:

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}}\left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1\right) \qquad \textbf{(Eq. 14.18)}$$

In this expression, the right-hand parenthesis ($\sqrt{(2r_2 / (r_1 + r_2))}$ - 1) is the dimensionless fractional velocity boost above circular speed (≈ 31.4%). Evaluating that dimensionless impulse factor *first* via **2** [ENTER] **4.2164** [E] **7** [×] **6.678** [E] **6** [ENTER] **4.2164** [E] **7** [+] [÷] [$\sqrt{x}$] **1** [-] **3.986** [E] **14** [ENTER] **6.678** [E] **6** [÷] [$\sqrt{x}$] [×] keeps Registers Z and T completely empty: forming $2r_2 = 8.4328 \times 10^7$ m in Register X and then summing the major axis $r_1 + r_2 = 4.8842 \times 10^7$ m automatically lifts $2r_2$ into Register Y; dividing, taking [$\sqrt{x}$], and subtracting 1 collapses their ratio into the dimensionless impulse factor 0.313988 in Register X (returning Y to zero); and computing the circular parking speed $v_{c1} = \sqrt{(\mu/r_1)} = 7{,}725.84$ m/s lifts 0.313988 into Register Y so a final multiplication delivers **2,425.767684** m/s.

14.4.5 Single-Entry Quadratic & Cubic Cores via LASTX

For a conical frustum of height $h = 14.0$ cm, lower radius $R = 8.5$ cm, and upper radius $r = 4.2$ cm ($V = (1/3)\pi\, h(R^2 + Rr + r^2)$), the symmetric quadratic core $R^2 + Rr + r^2$ folds into Register X across (Y, X) while Z and T stay at zero. Keying **8.5** [↶] [x²] **8.5** [ENTER] **4.2** [×] [+] **4.2** [↶] [x²] [+] **14** [×] **3** [÷] [↻] [π] [×] squares $R^2 = 8.5^2 = 72.2500$ cm^2 ($8 \times 9 = 72 \Rightarrow 72.25$), adds $Rr = 35.70$ cm^2 ($\rightarrow 107.9500$ cm^2), adds $r^2 = 17.64$ cm^2 ($\rightarrow 125.5900$ cm^2), and scales by $(1/3)\pi\, h$ to yield **1,841.245566** cm^3.

Chaining [↶] [LASTx] after squaring R recalls R to multiply by r and then recalls r to square r^2, evaluating $R^2 + Rr + r^2$ with each radius typed once (just as factoring a closed cylinder's area $A_{\text{total}} = 2\pi\, r\, h + 2\pi\, r^2 = 2\pi\, r(h + r)$ for $h = 84.0$ mm and $r = 18.5$ mm sums 84.0 + 18.5 = 102.5 mm and recalls $r = 18.5$ mm from [↶] [LASTx] to scale by 2π for 11,914.49 mm^2).

When a surveyor measures two sides of a triangular plot ($a = 18.5$ m and $b = 24.2$ m) and the included angle ($C = 52.0°$), the Law of Cosines $c = \sqrt{(a^2 + b^2 - 2ab \cos C)}$ leaves $a^2 + b^2$ in Register Y while building $2ab \cos C$ in Register X ($Z = T = 0$): keying **18.5** [↶] [x²] **24.2** [↶] [x²] [+] **2** [ENTER] **18.5** [×] **24.2** [×] **52** [COS] [×] [-] [√x] sums $18.5^2 + 24.2^2 = 927.8900$ m^2, forms $2(18.5)(24.2)\cos(52°) = 551.2636$ m^2, and subtracts before [√x] to yield **19.40687288** m (or chains [↶] [LASTx] across a^2, ab, and b^2 so neither side is retyped).

To approximate the derivative of $f(x) = x^3 - 2x$ at $x = 3.0$ via a central difference with step $h = 0.001$, $f'(3) \approx (f(3.001) - f(2.999)) / 0.002 = 25.000$, keying **3.001** [ENTER] **3** [y^x] **2** [ENTER] **3.001** [×] [-] **2.999** [ENTER] **3** [y^x] **2** [ENTER] **2.999** [×] [-] [-] **.002** [÷] evaluates $f(3.001) = 21.025009$, forms $f(2.999) = 20.975009$ in X while $f(3.001)$ waits in Y, and divides their difference by 0.002 to yield **25.000001**. Inside that sequence, squaring 3.001 via [↶] [x²] saves **3.001** in **LASTX**, multiplying by [↶] [LASTx] forms 3.001^3 while keeping **3.001** in **LASTX**, and a second recall subtracts $2(3.001)$ with zero [ENTER] strokes.

Likewise, in Newton's Law of Cooling ($T(15) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$ for $T_0 = 95.0°$C, $T_{\text{env}} = 22.0°$C, $k = 0.045$ min^{-1}, $t = 15.0$ min), keying **0.045** [+/-] [ENTER] **15** [×] [e^x] **95** [ENTER] **22** [-] [×] **22** [+] scales $e^{-0.675} = 0.5092$ by $T_0 - T_{\text{env}} = 73.0°$C and adds 22.0°C to resolve **59.168418704** °C (or recalls $T_{\text{env}} = 22°$C from [↶] [LASTx] after forming 95 - 22).

14.4.6 Two-Register Harmonic, Vector & Geometric Identities (x‖y, →y,x & Inradius)

Four two-register identities resolve harmonic round-trip speeds, inverse parallel branches, inclined-plane forces, and right-triangle inradii in Registers X and Y ($Z = T = 0$). For a round trip at outbound speed $v_1 = 40.0$ mph and return speed $v_2 = 60.0$ mph, $v_{\text{avg}} = 2v_1\, v_2 / (v_1 + v_2) = 2(v_1 \parallel v_2)$, so keying **40** [ENTER] **60** [x‖y] **2** [×] collapses $40 \parallel 60 = 24.0000$ mph and doubles it to **48.0 mph** mph in four operations. Conversely, when the combined parallel resistance $R_{\text{eq}} = 20.0\ \Omega$ (or focal length $f = 20.0$ cm) and one branch $R_1 = 30.0\ \Omega$ are given, negating the known branch before pressing [x‖y] evaluates the inverse harmonic identity $R_2 = R_{\text{eq}} \parallel (-R_1) = 20(-30) / (20 - 30) =$ **60.0000 Ω** via **20** [ENTER] **30** [+/-] [x‖y] without three separate [x^{-1}] inversions.

Similarly, for a block of weight $W = mg = 250.0$ N on a $\vartheta = 28.0°$ incline, keying **28** [ENTER] **250** [↻] [▸y,x] resolves both orthogonal components simultaneously, placing normal force $F_perp = W \cos 28° =$ **220.7369 N** in Register X and downhill ramp force $F_{\parallel} = W \sin 28° =$ **117.3679 N** in Register Y. And for the inscribed circle radius of a right triangle with legs $a = 8$, $b = 15$ and hypotenuse $c = 17$ ($r = (a + b - c) / 2$), keying **8** [ENTER] **15**

[+] **17** [-] **2** [÷] sums the legs (23.0000), subtracts 17, and divides by 2 to resolve **3.00**.

14.4.7 Unlimited Stack Mode (INF) on Multi-Branch Networks

With the 999-level heap stack active ([↷] [FLAGS] [INF]), five-branch and multi-term evaluations collapse in a single left-to-right pass. Take a five-branch parallel telemetry shunt network with resistances $R_1 = 20\ \Omega$, $R_2 = 30\ \Omega$, $R_3 = 60\ \Omega$, $R_4 = 40\ \Omega$, and $R_5 = 120\ \Omega$:

$$R_{\text{eq}} = 20 \| 30 \| 60 \| 40 \| 120 = 7.5000\,\Omega \qquad \textbf{(Eq. 14.19)}$$

In [INF] mode, you can enter all five resistor values sequentially onto the stack (**20** [ENTER] **30** [ENTER] **60** [ENTER] **40** [ENTER] **120**) and then collapse all four parallel pairs in a four-tap [x‖y] cascade ([x‖y] [x‖y] [x‖y] [x‖y]). Across rows 2 through 6 in **Register T** below, when **120** is entered on row 2, **20.0000** is not lost—it rides up into Level 5 above **Register T** (30.0000), and on the opening [x‖y] tap (row 3), **20.0000** drops right back from Level 5 into **Register T** while true LIFO history pulls **0.0000** (never duplicating T) on rows 4, 5, and 6:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[↷] [FLAGS] [INF] **20** [ENTER] **30** [ENTER] **60** [ENTER] **40**	20.0000	30.0000	60.0000	40.0000	Enable 999-level **INF** stack; load first 4 branches (R_1 ... R_4)
[ENTER] **120**	30.0000	60.0000	40.0000	120.0000	Enter $R_5 = 120\ \Omega$; R_1 = 20.0000 rides safely up to Level 5
[x‖y]	20.0000	30.0000	60.0000	30.0000	$40 \| 120 = 30\ \Omega$; $R_1 = 20.0000$ drops from Level 5 back into T
[x‖y]	0.0000	20.0000	30.0000	20.0000	$60 \| 30 = 20\ \Omega$; true LIFO drop pulls 0.0000 into T (no T-smearing)
[x‖y]	0.0000	0.0000	20.0000	12.0000	$30 \| 20 = 12\ \Omega$; $R_1 = 20.0000$ drops to Y
[x‖y]	0.0000	0.0000	0.0000	**7.5000**	Final collapse $20 \| 12 =$ **7.5000 Ω**

That same [INF] depth accelerates multi-term product sums and five-term rational expressions. Take the total rotational moment of inertia $I = \sum_{k=1}^{4} m_k\, r_k^2$ of a four-mass rotor assembly with $m_1 = 1.5$ kg at $r_1 = 0.20$ m, $m_2 = 2.0$ kg at $r_2 = 0.30$ m, $m_3 = 2.5$ kg at $r_3 = 0.40$ m, and $m_4 = 3.0$ kg at $r_4 = 0.50$ m. Evaluating each $m_k\, r_k^2$ term via **m** [ENTER] **r** [↶] [x²] [×] in succession pushes all four individual inertias—0.06, 0.18, 0.40, and 0.75 kg·m^2—onto the unbounded stack simultaneously; collapsing them via three [+] taps then yields 1.3900 kg·m^2, or lets you divide by a fifth denominator branch via [÷] without touching [STO] or [RCL].

14.4.8 Summary of Inside-Out Evaluation Patterns & Postfix Keystroke Economy

Mathematical Expression	Result	Postfix Keystroke Sequence	Infix Brackets	Postfix Brackets
1. Triple Harmonic Network: $12 \| 20 \| 30$	6.00	**12** [ENTER] **20** [x‖y] **30** [x‖y]	2 (**(**, **)**)	0
2. Inverse Harmonic Partner: $(20^{-1} - 30^{-1})^{-1}$	60.0	**20** [ENTER] **30** [+/−] [x‖y]	2 (**(**, **)**)	0

Mathematical Expression	Result	Postfix Keystroke Sequence	Infix Brackets	Postfix Brackets
3. Nested Continued Fraction: 1 / (2 + 1/(3 + 1/4))	0.433	4 x^{-1} 3 + x^{-1} 2 + x^{-1}	4 ((,))	0
4. Radical Nested Pyramid: √(17 + √(29 + √83))	4.81	83 $\sqrt{x}$ 29 + $\sqrt{x}$ 17 + $\sqrt{x}$	6 (((,)))	0
5. 3D Euclidean Norm: ‖(3, 4, 12)‖	13.0	3 ENTER 4 ↶ ▸θ,r 12 ↶ ▸θ,r	2 (,)	0
6. Sequential Markup & Discount: $250 + 15% - 12%	$253	250 ENTER 15 ↷ % + 12 ↷ % -	2 (,)	0
7. Quadratic Roots: x^2 - 7x + 12 = 0	3.00, 4.00	↷ POLY: 0 A 1 b 7 +/− C ON 12 D EXEC	4 (,)	0
8. Sexagesimal Time Addition: 3h 45m + 2h 35m	6h 20m	3.45 ↶ ▸HR 2.35 ↶ ▸HR + ↷ ▸HMS	4 (,)	0
9. Simultaneous Ramp Forces: 250 N at 28°	*Y*: 117.4, *X*: 220.7	28 ENTER 250 ↷ ▸y,x	4 (2× entry)	0
10. Affine Temperature Scaling: 37.0°C → °F	98.6°F	37 ENTER 1.8 × 32 +	2 (,)	0

14.5 Epilogue: The Mind and the Instrument

Conventional infix notation writes algebraic expressions from the outside in, deferring operations inside nested parentheses until the closing delimiter is reached. Physical and mathematical dependencies, by contrast, propagate from the inside out: an inner quantity is evaluated at the core of an expression, and each successive operator, scaling ratio, or functional mapping transforms the value already present. That structural correspondence—formalized in Jan Łukasiewicz's 1924 parenthesis-free logic and realized in Charles Hamblin's and Bill Hewlett's hardware evaluation stacks *(Łukasiewicz, 1924; Hamblin, 1957; Whitney, Rodé & Tung, 1972)*—aligns the order of keystrokes with the order of mathematical construction.

Across scientific and engineering computation, a single architectural principle connects elementary arithmetic, transcendental analysis, linear regression, vector phasors, numerical calculus, and coupled system design. Logarithms convert multiplicative products into additive shifts; complex exponentials unify circular rotation with hyperbolic growth; orthogonal variance projections govern least-squares curve fitting and mechanical tolerance stacks; and Richard Brent's root-finder and Werner Romberg's quadrature engine refine continuous models to machine precision. On the SC-32, these methods share a common four-register workspace where operands and operators interact directly without modal syntax trees.

When calculation proceeds in postfix order, the instrument operates not as an opaque black box that accepts an entire formula string at once, but as an inspectable register machine. Because Registers *X*, *Y*, *Z*, and *T* expose every intermediate quantity together with its physical units and order of magnitude, each step of an evaluation can be bounded by Fermi estimation and verified against the governing law before the next operator is applied. Cultivating that habit—estimating before computing, factoring before keying, and evaluating from the inner core outward—unites mathematical insight with disciplined quantitative practice.

Appendix A: Formal Stack-Lift State Machine

A.1 State-Machine Architecture & Register Transitions

The four operational registers (X, Y, Z, T) of a Reverse Polish Notation (RPN) processor form a four-register last-in, first-out (LIFO) evaluation stack, with X as the active bottom display register, Y as the second operative register, Z as the third, and T as the top constant register.

Across the four-register stack, every completed result resides in the primary working register, Register X, while intermediate subtotals wait in the upper registers (Y, Z, T). In an ordinary algebraic calculator, the display register is passive: if you want to compute (14 × 25) + (36 × 48), you must either open nested parentheses or copy **350** onto scratch paper before typing **36**. On the SC-32, Register X is governed by a deterministic George H. Mealy (1955) finite-state automaton (Q, $\sum$, Δ, δ, λ) called the Stack Lift Automaton. Its internal state set Q = (EDIT, ARMED, DISABLED) and keystroke alphabet $\sum$ jointly determine both the next state $\delta(q, \sigma)$ and the immediate register transformation $\lambda(q, \sigma)$ in $O(1)$ time:

- **Building a Number (`EDIT` / Digit Entry Active):** While you are tapping digits ([3], [6], [•], [5]), characters append directly to the active input buffer in Register X. Each digit appends to the right of the previous digit in Register X without disturbing Registers Y, Z, or T.
- **Stack Lift Armed (`ARMED` / `Lift = TRUE`):** The moment you press any mathematical operator ([+], [×], [√x], [SIN], [LN]), the calculator terminates digit entry, writes the evaluated result into Register X, and arms the stack-lift state (**`Lift = TRUE`**). You do not need to press [ENTER] to protect that result. Pressing the next digit key ([0]–[9] or [•]) triggers an automatic stack-lift transition, pushing the finished result in Register X up to Register Y ($Y \leftarrow X$, $Z \leftarrow Y$, $T \leftarrow Z$) before starting your new number in X.
- **Stack Lift Neutral (`DISABLED` / `Lift = FALSE`):** We press [ENTER] only to separate two numbers typed *back-to-back* with no operator between them (such as **14** followed by **25**). Pressing [ENTER] pushes **14** up to Register Y, leaves a duplicate copy in X, and disables stack lift (**`Lift = FALSE`**) so the first digit of **25** overwrites X in place without pushing a second **14** into Z.

Rooted in the Corvallis engineering specifications of the HP-15C and HP-32SII *(HP, 1982; HP, 1991)*, this Mealy automaton executes every register transition deterministically without operator-precedence buffers or parenthesis stacks.

A.1.1 Stack Lift Transition Diagram

Current State	Triggering Keystroke Class	Next State	Stack Register Effect
LIFT ARMED (**Lift = TRUE**)	First digit ([0]–[9], [•])	**EDIT BUFFER**	Pushes stack up (**X → Y → Z → T**) before writing new digit
EDIT BUFFER (In Progress)	Any function/operator ([+], [×], [√x])	**LIFT ARMED** (**Lift = TRUE**)	Terminates digit entry, evaluates result into **X**, and arms stack lift
EDIT BUFFER or **LIFT ARMED**	[ENTER], [CLx], [Σ+]	**LIFT DISABLED** (**Lift = FALSE**)	Next digit typed will overwrite **X** in place without lifting **Y**
LIFT DISABLED (**Lift = FALSE**)	[x⇄y], [R↓], [f] [R↑], or operator	**LIFT ARMED** (**Lift = TRUE**)	Re-arms stack lift for subsequent numeric entry

A.1.2 The Three Keystroke Classes

Every key on the 43-button matrix belongs to one of three deterministic state-transition classes:

1. **Enables Stack Lift (`Lift = TRUE`):** Nearly all mathematical operators (**+**, **-**, **×**, **÷**, **√x**, **SIN**, **LN**, etc.), constants ([f] [π]), recalls ([RCL], [←] [LASTx]), and stack permutations ([R↓], [f] [R↑], [x⇄y]) set **Lift = TRUE**. The next digit key pushes $T \leftarrow Z$, $Z \leftarrow Y$, $Y \leftarrow X$ (discarding the old T in **4-LVL** mode) before writing the new digit into X.

2. **Disables Stack Lift (`Lift = FALSE`):** Established exclusively by [ENTER] (the Stack Entry Invariant, which duplicates $Y \leftarrow X$ and disarms lift so the second operand overwrites X), [g] [ON] / [⌫] (**CLx**), and statistical accumulation ([Σ+], [←] [Σ−]). The next digit key overwrites Register X in place while leaving Y, Z, and T untouched.

3. **Preserves Stack Lift (State Unchanged):** Prefix and modal keys ([←], [f], [STO], [E], softkey menu openers, and [+/−] during active **EDIT** digit entry—whereas pressing [+/−] on a finalized register enables stack lift) leave the active stack-lift flag unaltered.

Because every operator sets the **Lift** flag deterministically upon completion, intermediate results never require explicit parentheses or scratchpad storage; tracking whether an operation enables or disables stack lift prevents accidental operand duplication.

A.1.3 Definitive 43-Key Physical Matrix Audit Table

All 43 physical switch positions of the SC-32 keypad are indexed by electrical scan coordinate (**Row, Col**), primary legend, and post-execution Stack Lift state:

Key Position	Primary Legend	Stack Lift Status	Updates LASTX	Register Delta Transformation	Operational Description
Row 0, Col 0	[LFU0]	Preserves	No	Context dependent	Softkey Menu Item 1 / Dynamic LFU Slot 1
Row 0, Col 1	[LFU1]	Preserves	No	Context dependent	Softkey Menu Item 2 / Dynamic LFU Slot 2
Row 0, Col 2	[LFU2]	Preserves	No	Context dependent	Softkey Menu Item 3 / Dynamic LFU Slot 3
Row 0, Col 3	[LFU3]	Preserves	No	Context dependent	Softkey Menu Item 4 / Dynamic LFU Slot 4

Key Position	Primary Legend	Stack Lift Status	Updates LASTX	Register Delta Transformation	Operational Description
Row 0, Col 4	LFU4	Preserves	No	Context dependent	Softkey Menu Item 5 / Dynamic LFU Slot 5
Row 0, Col 5	LFU5	Preserves	No	Context dependent	Softkey Menu Item 6 / Dynamic LFU Slot 6
Row 1, Col 0	$\sqrt{x}$	**Enables**	**Yes**	$X \leftarrow \sqrt{X}$; Y, Z, T unch	Unary square root; captures prior X
Row 1, Col 1	e^x	**Enables**	**Yes**	$X \leftarrow e^X$; Y, Z, T unch	Unary natural exponential; captures prior X
Row 1, Col 2	LN	**Enables**	**Yes**	$X \leftarrow \ln X$; Y, Z, T unch	Unary natural logarithm; captures prior X
Row 1, Col 3	y^x	**Enables**	**Yes**	$X \leftarrow Y^X$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic power; stack drops; T replicated
Row 1, Col 4	x^{-1}	**Enables**	**Yes**	$X \leftarrow 1/X$; Y, Z, T unch	Unary reciprocal; captures prior X
Row 1, Col 5	$\Sigma+$	**Disables**	No (Preserved)	$X \leftarrow n$; Y, Z, T unch; $\sum$ accums updated	Statistical data accumulation; disables lift
Row 2, Col 0	STO	Preserves	No	$X \rightarrow \text{Var}[k]$; Stack unch	Store X into designated variable
Row 2, Col 1	RCL	**Enables**	No	$T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow \text{Var}[k]$	Recall variable k (pushes stack upward)
Row 2, Col 2	R↓	**Enables**	No	$X \leftarrow Y$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow X$	Roll stack down (circular rotation)
Row 2, Col 3	SIN	**Enables**	**Yes**	$X \leftarrow \sin X$; Y, Z, T unch	Trigonometric sine in active angular mode
Row 2, Col 4	COS	**Enables**	**Yes**	$X \leftarrow \cos X$; Y, Z, T unch	Trigonometric cosine in active angular mode
Row 2, Col 5	TAN	**Enables**	**Yes**	$X \leftarrow \tan X$; Y, Z, T unch	Trigonometric tangent in active angular mode
Row 3, Col 0-1	ENTER	**Disables**	No	$T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow X$	Duplicate $X \rightarrow Y$; disable stack lift
Row 3, Col 2	$x \leftrightarrow y$	**Enables**	No	$X \leftarrow Y$; $Y \leftarrow X$; Z, T unch	Exchange Registers X and Y
Row 3, Col 3	+/-	Preserves (in **EDIT**) / **Enables**	**Yes** (finalized) / No (**EDIT**)	$X \leftarrow -X$; Y, Z, T unch	Toggle arithmetic sign of active number
Row 3, Col 4	E	Preserves	No	Arms scientific exponent entry buffer	Exponent entry separator
Row 3, Col 5	⌫	Preserves (in **EDIT**) / **Disables** (as **CLx**)	No	Deletes trailing digit in **EDIT**; sets $X \leftarrow 0$ as **CLx**	Dual-mode backspace / Clear X
Row 4, Col 0	XEQ	Preserves	No	Context dependent	Execute designated program or equation
Row 4, Col 1	7	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 7$	Digit input 7

Key Position	Primary Legend	Stack Lift Status	Updates LASTX	Register Delta Transformation	Operational Description
Row 4, Col 2	8	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 8$	Digit input 8
Row 4, Col 3	9	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 9$	Digit input 9
Row 4, Col 4	÷	**Enables**	**Yes**	$X \leftarrow Y \div X$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic division; stack drops; T replicated
Row 5, Col 0	↰ (Left Shift)	Preserves	No	Arms left-shifted alternate functions	Primary shift prefix modifier
Row 5, Col 1	4	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 4$	Digit input 4
Row 5, Col 2	5	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 5$	Digit input 5
Row 5, Col 3	6	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 6$	Digit input 6
Row 5, Col 4	×	**Enables**	**Yes**	$X \leftarrow Y \times X$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic multiplication; stack drops
Row 6, Col 0	↱ (Right Shift)	Preserves	No	Arms right-shifted alternate functions	Secondary shift prefix modifier
Row 6, Col 1	1	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 1$	Digit input 1
Row 6, Col 2	2	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 2$	Digit input 2
Row 6, Col 3	3	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 3$	Digit input 3
Row 6, Col 4	-	**Enables**	**Yes**	$X \leftarrow Y - X$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic subtraction; stack drops
Row 7, Col 0	ON (ON)	**Disables**	No	Clears active entry or dismisses error	Clear entry / Hardware power wake
Row 7, Col 1	0	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 0$	Digit input 0
Row 7, Col 2	•	(Consumes Lift)	No	If Lift: $T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow 0.$	Decimal radix point input
Row 7, Col 3	$x \parallel y$	**Enables**	**Yes**	$X \leftarrow Y \times X \,/\, (Y + X)$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic parallel harmonic operator; drops
Row 7, Col 4	+	**Enables**	**Yes**	$X \leftarrow Y + X$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic addition; stack drops; T replicated

A.2 Shifted Operations & Formal Register Delta Algebra

Every shifted key on the SC-32 adheres strictly to the Stack Lift Automaton. Tables A.1 and A.2 catalog the mathematical, stack-control, and modal left-shifted (↰, Primary Shift) and right-shifted (↱, Secondary Shift) operations—while the ten physical unit conversion keys (↰ ▶km/↱ ▶mi, ↰ ▶kg/↱ ▶lb, ↰ ▶°C/↱ ▶°F, ↰ ▶cm/↱ ▶in, ↰ ▶l/↱ ▶gal) and angle/time conversions (↰ ▶HR/↱ ▶HMS, ↰ ▶DEG/↱ ▶RAD) act as unary in-place transformations ($X \leftarrow f(X)$, LASTX $\leftarrow X$) that arm stack lift—followed by the formal vector algebra of register transformations:

A.2.1 Table A.1: Left-Shifted Operations (Primary Shift)

Shifted Combination	Function Name	Stack Lift Status	Updates LASTX	Register Delta Transformation	Operational Notes
$\sqrt{x}$	**x²**	**Enables**	**Yes**	$X \leftarrow X^2$; Y, Z, T unch	Unary square
e^x	**10ˣ**	**Enables**	**Yes**	$X \leftarrow 10^X$; Y, Z, T unch	Unary common exponential
LN	**LOG**	**Enables**	**Yes**	$X \leftarrow \log_{10} X$; Y, Z, T unch	Unary common logarithm
y^x	**ˣ√y**	**Enables**	**Yes**	$X \leftarrow Y^{1/X}$; $Y \leftarrow Z$; $Z \leftarrow T$; $T \leftarrow T$	Dyadic x-th root of y; stack drops
x^{-1}	**x!**	**Enables**	**Yes**	$X \leftarrow \Gamma(X{+}1)$; Y, Z, T unch	Factorial / Gamma function
Σ+	**Σ-**	**Disables**	No (Preserved)	$X \leftarrow n{-}1$; Y, Z, T unch; $\sum$ decremented	Statistical correction; disables lift
STO	**CMPLX**	Preserves	No	Sets one-shot complex evaluator prefix	Arms Gaussian plane arithmetic
R↓	**HYP**	Preserves	No	Sets hyperbolic prefix flag	Arms hyperbolic evaluation
SIN	**ASIN**	**Enables**	**Yes**	$X \leftarrow \arcsin X$; Y, Z, T unch	Inverse sine in active angular mode
COS	**ACOS**	**Enables**	**Yes**	$X \leftarrow \arccos X$; Y, Z, T unch	Inverse cosine in active angular mode
TAN	**ATAN**	**Enables**	**Yes**	$X \leftarrow \arctan X$; Y, Z, T unch	Inverse tangent in active angular mode
ENTER	**LASTX**	**Enables**	No	$T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow$ LASTX	Recalls prior X; pushes stack upward
+/−	**MODES**	Preserves	No	Menu: **DEG**, **RAD**, **GRAD**, **.**, **,**	Modal angular and radix settings
E	**DISP**	Preserves	No	Menu: **FIX n**, **SCI n**, **ENG n**, **ALL**	Non-destructive display precision
⌫	**CLEAR**	Preserves	No	Menu: **CLEQN**, **CLVARS**, **CLALL**, **CLΣ**, **CLREGS**, **CLSTK**	Selective memory wipe menu
÷	**x?y**	Preserves	No	Comparison menu: **x=y**, **x≠y**, **x>y**, etc.	Conditional branching tests
4	**→ϑ,r**	**Enables**	**Yes**	$X' \leftarrow \sqrt{(X^2+Y^2)}$; $Y' \leftarrow \text{atan2}(Y, X)$; $L \leftarrow X$	Rectangular to polar conversion
×	**BASE**	Preserves	No	Menu: **HEX**, **DEC**, **OCT**, **BIN**	Integer base switching menu
0	**REGS**	Preserves	No	Displays register viewer table	Full-screen register browser (**STACK**, **STO**, **STATS**, **DATA**)
•	**FDISP**	Preserves	No	Toggles fraction display mode (Flag 7)	Rational fraction representation
$x \parallel y$	**TVM**	Preserves	No	Opens 5-variable TVM workspace	5-variable Time Value of Money solver
+	**LBL**	Preserves	No	Prompts for equation label letter	Inserts equation label token in **EQN** mode

A.2.2 Table A.2: Right-Shifted Operations (Secondary Shift)

Shifted Combination	Function Name	Stack Lift Status	Updates LASTX	Register Delta Transformation	Operational Notes
$\sqrt{x}$	**PARTS**	Preserves	No	Menu: **IP**, **FP**, **ABS**, **SGN**, **GCD**, **LCM**	Number theory and decomposition
e^x	**PROB**	Preserves	No	Menu: **Cn,r**, **Pn,r**, **x!**, **RAND**, **SD**	Combinatorics and probability
LN	**L.R.**	Preserves	No	Menu: $\hat{y}$, $\hat{x}$, **r**, **m**, **b**	Linear regression estimates & slope
y^x	$\bar{x}$,$\bar{y}$	Preserves	No	Menu: $\bar{x}$, $\bar{y}$, $\bar{x}_w$	Sample and weighted means
x^{-1}	**s**,σ	Preserves	No	Menu: **sx**, **sy**, σ**x**, σ**y**	Sample and population std dev
$\Sigma+$	**SUMS**	Preserves	No	Menu: Σ**x**, Σ**y**, Σ**x**^{2}, Σ**y**^{2}, Σ**xy**, **n**	Statistical accumulator viewer
STO	**EQN**	Preserves	No	Enters postfix Equation Bank editor	User-defined RPN equations
R↓	**R↑**	**Enables**	No	$X \leftarrow T$; $Y \leftarrow X$; $Z \leftarrow Y$; $T \leftarrow Z$	Circular roll stack upward
SIN	π	**Enables**	No	$T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow \pi$	Pushes π constant onto stack
COS	**%**	**Enables**	**Yes**	$X \leftarrow (Y \times X)/100$; Y, Z, T unch	Kahan base-preserving percentage
TAN	**%CHG**	**Enables**	**Yes**	$X \leftarrow 100(X - Y)/Y$; Y, Z, T unch	Base-preserving percent change
ENTER	**SHOW**	Preserves	No	Renders 15-digit internal mantissa	Non-destructive temporary peek
+/−	**ABS**	**Enables**	**Yes**	$X \leftarrow \lvert X \rvert$; Y, Z, T unch	Unary absolute value magnitude
E	**÷R**	**Enables**	**Yes**	$X \leftarrow \text{trunc}(Y/X)$; $Y \leftarrow \text{rem}(Y, X)$; Z, T unch	Integer quotient (X) & remainder (Y)
⌫	*(None)*	Preserves	No	Unassigned	Intentionally unassigned to prevent accidental clear
XEQ	**POLY**	Preserves	No	Dynamic cubic solver workspace (**A B C D EXEC CLR**)	Closed-form quadratic and cubic root workspace
7	**SOLVE**	**Enables**	No	$T \leftarrow Z$; $Z \leftarrow Y$; $Y \leftarrow X$; $X \leftarrow x^*$	Richard Brent hybrid root solver
8	$\int$	**Enables**	No	$X \leftarrow \int_Y^X f(t)\,dt$; $Y \leftarrow \lvert\varepsilon\rvert$	Adaptive Romberg quadrature
÷	**x?0**	Preserves	No	Comparison menu: **x=0**, **x≠0**, **x>0**, etc.	Zero-relative comparison tests
4	**→y,x**	**Enables**	**Yes**	$X' \leftarrow X \cos Y$; $Y' \leftarrow X \sin Y$; $L \leftarrow X$	Polar to rectangular conversion
×	**FLAGS**	Preserves	No	Menu: **SF n**, **CF n**, **FS? n**, **STACK** (**2/4/8-LVL**, **INF**), **EXAM**	System flags & stack size
•	**/c**	**Enables**	No	Pops $c \in [2, 4095]$ from X; drops stack	Fixed fraction denominator ($1/c$)
$x \lVert y$	*(None)*	Preserves	No	Unassigned on physical non-graphing hardware	Intentionally unassigned (**TVM** sits on ↖)

A.2.3 The Formal Register Delta Algebra

To verify calculation sequences with mathematical certainty, the effect of any operator Ω on the four-register stack $\boldsymbol{S} = [X,\ Y,\ Z,\ T]^T$ is expressed as a matrix or functional transformation $\boldsymbol{S}' = \boldsymbol{f}_\Omega(\boldsymbol{S})$:

- **Unary Operators ($f(X)$):** Functions such as [√x], [1/x], [eˣ], [LN], [SIN], [COS], [TAN], [↶] [x²], [↶] [10ˣ], [↶] [LOG], [↶] [x!], and [↷] [|x|] transform X in place while leaving Y, Z, T undisturbed:

$$[X',\ Y',\ Z',\ T'] = [f(X),\ Y,\ Z,\ T],\quad \text{LASTX}' = X,\quad \text{Lift}' = \text{TRUE} \qquad \textbf{(Eq. A.1)}$$

- **Dyadic Operators ($f(Y, X)$):** Binary operators ([+], [−], [×], [÷], [yˣ], [↶] [x\y], and [x‖y]) combine Y and X into X, drop $Z \to Y$, and replicate $T \to Z$:

$$[X',\ Y',\ Z',\ T'] = [f(Y,X),\ Z,\ T,\ T],\quad \text{LASTX}' = X,\quad \text{Lift}' = \text{TRUE} \qquad \textbf{(Eq. A.2)}$$

- **Circular Stack Rotations ([R↓], [↷] [R↑], [x⇄y]):** Rearrange registers without altering numerical values or updating **LASTX**, while arming stack lift (Lift' = TRUE):

$$\mathbf{S}'_{\mathrm{R}\downarrow} = [Y,\ Z,\ T,\ X],\qquad \mathbf{S}'_{\mathrm{R}\uparrow} = [T,\ X,\ Y,\ Z],\qquad \mathbf{S}'_{x\leftrightarrow y} = [Y,\ X,\ Z,\ T] \qquad \textbf{(Eq. A.3)}$$

- **Stack Duplication ([ENTER]):** Pushes the stack upward while replicating X into Y and disarming stack lift:

$$[X',\ Y',\ Z',\ T'] = [X,\ X,\ Y,\ Z],\quad \text{LASTX}' = \text{LASTX},\quad \text{Lift}' = \text{FALSE} \qquad \textbf{(Eq. A.4)}$$

During any dyadic operation in **4-LVL** mode, Register T drops into Z while T retains its original value ($T' \leftarrow T$). This top-register constant replication invariant allows a constant stored in T to be reused across an unbounded series of operations without re-entry.

A.3 Comprehensive Stack Lift Traces

Twelve operational traces illustrate the Stack Lift Automaton across representative engineering calculations. Multi-register expression traces display the full four-register stack (**Register T, Register Z, Register Y, Register X**), focused state-machine traces highlight the active **Stack Lift Flag** alongside the working and auxiliary registers (**LASTX**, **Register Z**, or **Stat Register n**), and compact dyadic traces display the lower three registers (**Register X, Register Y, Register Z**).

A.3.1 Chained Arithmetic Without Intermediary ENTER

Calculate: $(14 \times 25) + (36 \times 48)$

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
[1] [4]	0	0	0	14	First operand keyed (**EDIT**)
[ENTER]	0	0	14	14	Stack pushed; X copied to Y (**DISABLED**)
[2] [5]	0	0	14	25	Overwrites X in place (**EDIT**); Y preserved
[×]	0	0	0	350	$14 \times 25 = 350$; stack drops (**ARMED**)
[3] [6]	0	0	350	36	**No ENTER needed:** **ARMED** lift pushes **350** into Y (**EDIT**)
[ENTER]	0	350	36	36	Pushes **350** into Z (**DISABLED**)
[4] [8]	0	350	36	48	Overwrites X (**EDIT**); registers Y and Z preserved

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
×	0	0	350	1728	$36 \times 48 = 1728$; Z drops to Y (**ARMED**)
+	0	0	0	2078	$350 + 1728 = 2078$; final sum in X (**ARMED**)

A.3.2 The ENTER Lift-Disable Invariant

Demonstrating why pressing ENTER prevents accidental stack duplication.

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
5 0	0	0	0	50	Base value entered (**EDIT**)
ENTER	0	0	50	50	**DISABLED**; X duplicated into Y
8	0	0	50	8	Digit **8** replaces X in place (**EDIT**) without pushing stack
+	0	0	0	58	Addition yields **58**; stack lift **ARMED**

A.3.3 LASTX Dyadic Operator Correction

The user intends to compute $100 \times 8 = 800$, but accidentally hits ÷, producing 12.5. The error is corrected without re-keying 100 or 8:

#	Keystroke	Stack Lift Flag	Reg X	Reg Y	LASTX	Operational Commentary
1	1 0 0 ENTER	**DISABLED**	100	100	0	Operands staged
2	8 ÷	**ARMED**	12.5000	0	8	Error: Divided instead of multiplied
3	↶ LASTx	**ARMED**	8	12.5000	8	Recalls operand **8** that caused the operation
4	×	**ARMED**	100.0000	0	8	Multiplies 12.5×8 to recover original numerator 100
5	↶ LASTx	**ARMED**	8	100.0000	8	Recalls operand **8** once more
6	×	**ARMED**	800.0000	0	8	Multiplies 100×8 to achieve intended target 800

A.3.4 CLx Stack Lift Disabling

Demonstrating how CLx clears an erroneous entry without collapsing or dropping upper registers:

Keystroke	Stack Lift Flag	Reg X	Reg Y	Reg Z	Operational Commentary
1 5 0 ENTER	**DISABLED**	150	150	0	Operand staged
9 9 9	**EDIT**	999	150	0	User mistypes second operand
CLx	DISABLED	0	150	0	CLx zeroes X and sets state to **DISABLED** (**Lift = FALSE**)
2 5	**EDIT**	25	150	0	Typing **25** overwrites zero in place
-	**ARMED**	125	0	0	150 - 25 = 125

A.3.5 Unary Function Lift Re-Arming

Unary functions evaluate in place on Register X and re-arm stack lift so subsequent inputs push automatically:

#	Keystroke	Stack Lift Flag	Reg X	Reg Y	Operational Commentary
1	1 6	**EDIT**	**16**	**0**	Raw value entered
2	√x	ARMED	**4**	**0**	$\sqrt{16} = 4$; Lift flag **ARMED** (**Lift = TRUE**)
3	9	**EDIT**	**9**	**4**	Inputting **9** triggers lift push; $4 \rightarrow Y$
4	+	**ARMED**	**13**	**0**	$4 + 9 = 13$

A.3.6 Constant Factor Calculation via T-Replication

Compute sales tax (8.25%, multiplier 1.0825) on three amounts: \$15.00, \$24.50, and \$80.00:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
1 . 0 8 2 5	**0**	**0**	**0**	**1.0825**	Tax factor keyed into X
ENTER ENTER ENTER	**1.0825**	**1.0825**	**1.0825**	**1.0825**	Stack flooded with constant factor
1 5 ×	1.0825	**1.0825**	**1.0825**	**16.2375**	15×1.0825; T replicates into Z and Y
ON C 2 4 . 5 ×	1.0825	**1.0825**	**1.0825**	**26.5213**	**C** clears X (disabling lift); 24.5×1.0825; T replicates
ON C 8 0 ×	1.0825	**1.0825**	**1.0825**	**86.6000**	**C** clears X; 80×1.0825; constant remains active

A.3.7 Statistical Accumulation and Non-Destructive Stack Isolation

Entering pairs $(x, y) = (10, 25)$ into statistical memory:

#	Keystroke	Stack Lift Flag	Reg X	Reg Y	Stat Register n	Operational Commentary
1	2 5 ENTER	**DISABLED**	**25**	**25**	**0**	y-coordinate pushed to Y
2	1 0	**EDIT**	**10**	**25**	**0**	x-coordinate keyed into X
3	Σ+	DISABLED	**1**	**25**	**1**	Register X shows sample count n=1; lift **DISABLED**
4	3 5	**EDIT**	**35**	**25**	**1**	Next y overwrites X directly without stack lift

A.3.8 Complex Number Evaluation via CMPLX

Compute $(3 + 4i) + (5 - 2i)$:

Keystroke	Reg T	Reg Z	Reg Y	Reg X	What Happens on the Stack
3 ENTER 4	**0**	**0**	**3** (Real)	**4** (Imag)	First complex number staged in (Y, X)
ENTER	**0**	**3**	**4**	**4**	Single **ENTER** lifts (3, 4) to (Z, Y) and disables lift
5 ENTER 2 +/−	**3**	**4**	**5**	**-2**	**5** overwrites X; **ENTER 2 +/-** lifts (3, 4) to (T, Z)
↶ CMPLX +	**3**	**3**	**8** (Real)	**2** (Imag)	Vector sum: $(3+5) + (4-2)i = 8 + 2i$; T replicates

A.3.9 Parallel Resistance via the Harmonic Operator

Calculate parallel equivalent of 30 Ω and 60 Ω:

Keystroke	Reg X	Reg Y	Reg Z	Operational Commentary
3 0 ENTER	30	30	0	First resistor staged
6 0	60	30	0	Second resistor keyed
x‖y	20.0000	0	0	Single-keystroke harmonic calculation: 30 × 60 / (30 + 60) = 20 Ω

A.3.10 Integer Division with Remainder

Divide 29 by 6 to find quotient and integer remainder:

Keystroke	Reg X	Reg Y	Reg Z	Operational Commentary
2 9 ENTER	29	29	0	Dividend staged in Y
6	6	29	0	Divisor in X
g ÷R	4 (Quotient)	5 (Remainder)	0	$29 = 4 \times 6 + 5$; quotient in X, remainder in Y

A.3.11 Absolute Value Evaluation

Compute |-45.75|:

Step	Keystroke	Reg X	Reg Y	Operational Commentary
1	4 5 • 7 5 +/-	-45.75	0	Negative operand entered
2	ABS	45.75	0	Instant absolute value; stack lift armed
3	1 0 ×	457.50	0	Multiplies without needing ENTER

A.3.12 Dynamic Stack Depth Expansion (4-LVL vs. 8-LVL vs. INF)

Under extended system flags (g FLAGS), the user can expand the stack beyond 4 levels:

- `4-LVL` **(Classic HP):** 4 registers (X, Y, Z, T). T replicated on drop; values pushed off T are discarded.
- `8-LVL` **(Extended STEM):** 8 registers (X, Y, Z, T, R_5, R_6, R_7, R_8). Deeper parenthetical evaluations can occur without variable storage. R_8 replicated on drop.
- `INF` **(Infinite Dynamic):** Dynamic LIFO stack scaling up to 999 levels (**`RPNCore`** ceiling). Stack drop pulls upper elements down without top-register replication.

Stack Depth Architecture Comparison (After Keying 2 ENTER 3 ENTER 4 ENTER 5 ENTER 6 ENTER 7)

Level	4-Level Stack (`4-LVL`)	8-Level Stack (`8-LVL`)	Infinite Stack (`INF`)
Top / Level 8	— *(No R_8: ceiling at T)*	**`R8 = 0`** *(Top Replicator)*	Dynamic Stack Top (up to Level 999)
Level 7	—	**`R7 = 0`**	*... (Empty until pushed)*
Level 6	— (**2** *pushed off T)*	**`R6 = 2`** *(Preserved above T)*	**`Level 6 = 2`** *(Preserved in LIFO heap)*
Level 5	— (**3** *pushed off T)*	**`R5 = 3`** *(Preserved above T)*	**`Level 5 = 3`** *(Preserved in LIFO heap)*
Level 4	**`T = 4`** *(Constant Replicator)*	**`T (R4) = 4`**	**`Level 4 (T) = 4`**
Level 3	**`Z = 5`** *(Third Operative)*	**`Z (R3) = 5`**	**`Level 3 (Z) = 5`**

Level	4-Level Stack (`4-LVL`)	8-Level Stack (`8-LVL`)	Infinite Stack (`INF`)
Level 2	`Y = 6` *(Second Operative)*	`Y (R2) = 6`	`Level 2 (Y) = 6`
Level 1 (Display)	`X = 7` *(Primary Display)*	`X (R1) = 7` *(Primary Display)*	`Level 1 (X) = 7` *(Primary Display)*

Appendix B: Error Catalog & Singularity Analysis



Figure: Exploded hardware assembly showing the internal keypad matrix, display module, and continuous memory architecture.

B.1 The Non-Destructive Error Recovery Principle & Domain Singularities

On an expression-evaluating calculator without pre-execution domain guards, encountering an arithmetic singularity—such as dividing by zero, taking a real logarithm of a negative number, or exceeding the floating-point exponent ceiling—aborts the pending syntax tree and discards intermediate subtotals.

The StackCalc SC-32 implements non-destructive error recovery across all arithmetic, transcendental, and iterative subsystems. Because every operation in Reverse Polish Notation evaluates one step at a time on the operands sitting in Registers Y and X, the processor validates the mathematical domain *before* mutating the stack.

B.1.1 The Non-Destructive Error Recovery Guarantee

When any of the 12 hardware error conditions occurs:

1. **Diagnostic Banner:** The descriptive error message is posted to the status line of the transflective LCD.
2. **Pre-Mutation Trap:** The operation causing the exception is halted before mutating operational memory.
3. **Register Preservation:** All four operational stack registers (X, Y, Z, T) and **LASTX** remain frozen intact.
4. **Single-Key Dismissal:** Pressing [C ON] or [⌫] dismisses the error banner and restores the exact prior stack state.

You can then inspect your operands, swap values via [x⇄y], change a sign via [+/−], or re-key the intended operator without losing a single intermediate calculation.

Feature / Behavior	StackCalc SC-32	Vintage HP-32SII	Conventional Algebraic Calculator (Infix)
Stack Preservation	100% frozen; zero register mutation	Frozen stack	Stack cleared or lost in syntax tree
Recovery Keystroke	Single [ON] or [⌫] restores display	[ON] or [⌫] restores	Requires quitting or clearing editor
Operand Inspection	Operands remain visible in X and Y	Operands remain	Operands often must be retyped
LASTX Protection	Prior valid X retained safely	Prior valid X retained	No dedicated **LASTX** register
Shift/Prefix Clearance	[ON] cancels pending prefix harmlessly	[ON] cancels	Keypad state unclear

When a division by zero occurs during a multi-step calculation, there is no need to clear the stack or restart from the beginning: pressing [ON] dismisses the banner while leaving the numerator intact in Register Y and the zero divisor in Register X, ready to be overwritten or swapped on the spot.

B.1.2 Hardware Error Code Summary

Every hardware error code on the SC-32 corresponds to a deterministic mathematical domain singularity or hardware boundary:

Error Code	Designation	Error Code	Designation	Error Code	Designation
Code 01	**DIVIDE BY 0**	Code 05	**STAT ERROR**	Code 09	**MEMORY FULL**
Code 02	**INVALID DATA**	Code 06	**NO ROOT FOUND**	Code 10	**TIMEOUT**
Code 03	**OVERFLOW**	Code 07	**A ≠ 0**	Code 11	**INTERRUPTED**
Code 04	**TOO BIG**	Code 08	**ERROR 5** *(HP-12C TVM)*	Code 12	**NO NEG** / **EQN** / **INT** Guards

B.1.3 Code 01: DIVIDE BY 0 (Division by Zero & Pole Singularities)

- **Why the Singularity Happens:** Division asks how many times a divisor x fits into a dividend y. As x shrinks toward zero (0.1, 0.001, 10^{-12}), the quotient y/x explodes without bound toward $+\infty$ from one side and $-\infty$ from the other, meaning $y/0$ has no finite value on the real number line. The same vertical asymptote (a *pole singularity*) appears inside disguised quotients across algebra and trigonometry:
  - **Direct Division & Reciprocals:** $y \div 0$ ([÷]), integer remainder division y rem 0 ([SHIFT] [÷R]), or multiplicative reciprocal $1/0$ ([x^{-1}]).
  - **Logarithmic Origin Pole:** Evaluating $\ln(0)$ or $\log_{10}(0)$ asks what power of e or 10 equals zero; as $x \to 0^+$, the logarithm plunges vertically to $-\infty$.
  - **Zero Base with Negative Exponent:** Evaluating 0^y for $y < 0$ (such as $0^{-2} = 1/0^2$) places zero in the denominator (whereas 0^0 is an indeterminate power form guarded under domain rules).
- **IEEE-754 Protection:** Rather than silently polluting your stack with **NaN** or **Inf** symbols that ruin subsequent calculations, the hardware trap intercepts the operation before mutating any register.

- **Stack State & Recovery Procedure:**

1. Register X holds **0.0000** (the offending divisor or argument), Register Y holds your original numerator untouched, and **LASTX** preserves the X value from your previous completed calculation.
2. Press [ON] to dismiss the **DIVIDE BY 0** banner.
3. Key the intended non-zero divisor (e.g., [1] [5])—since stack lift remains disabled or overwrites in place after [ON]—and press [÷].

When computing 450/15 and mistakenly keying 0 as the divisor:

Keystroke	Display Output	Reg X	Reg Y	Reg Z
[4] [5] [0] [ENTER]	450.0000	450.0000	450.0000	0.0000
[0] [÷]	DIVIDE BY 0	0.0000	450.0000	0.0000
[ON]	0.0000	0.0000	450.0000	0.0000
[1] [5] [÷]	30.0000	30.0000	0.0000	0.0000

B.1.4 Code 02: INVALID DATA (Real-Valued Domain Violations)

- **Why the Singularity Happens:** Many mathematical functions are defined only on a specific interval of the real number line R, and stepping outside that interval asks a question that has no real-number answer:
  - **Even Roots of Negative Numbers ($\sqrt{x}$ for $x < 0$):** Because the square of any real number—positive or negative—is always non-negative ($x^2 \geq 0$), no point on the real line squares to -16.
  - **Logarithms of Negative Numbers ($\ln x$ or $\log_{10} x$ for $x < 0$):** Because $e \approx 2.718$ and 10 are strictly positive bases, raising them to any real exponent y always produces a strictly positive result ($e^y > 0$). Thus $\ln x$ exists on the real line only for $x > 0$.
  - **Inverse Sine & Cosine Outside [-1, +1] (arcsin x or arccos x for $|x| > 1$) and Orthogonal Tangent Poles (tan 90°, tan 270°):** On the unit circle of radius 1, the vertical leg $\sin \vartheta$ and horizontal leg $\cos \vartheta$ can never exceed the hypotenuse of length 1, while tan(90°) and tan(270°) sit on vertical asymptotes where $\cos \vartheta = 0$.
  - **Factorial Poles & Combinatorial Boundaries:** Real factorials [↶] [x!] evaluate $\Gamma(x+1)$ across real arguments except at the negative-integer poles $x \in$ (-1, -2, -3, ...) where $\Gamma(x+1)$ diverges, and discrete combinations $C(n, r)$ ([nCr]) or permutations $P(n,r)$ ([nPr]) cannot choose $r > n$ items or negative integer counts.
- **Stack State & Recovery Procedure:**

1. All four registers remain frozen with the offending argument in Register X.
2. Press [ON] to clear the error banner.
3. **If you intended a complex-plane calculation:** Arm the complex prefix by pressing [↶] [CMPLX], then re-execute the function. For example, with **-16** in X, pressing [↶] [CMPLX] [$\sqrt{x}$] steps off the real line into the Gaussian plane and returns $0 + 4i$ (**0** in Y, **4** in X).

4. **If you made a sign error:** Press [+/−] to flip the sign in X and press the function key again.

B.1.5 Code 03: OVERFLOW (Floating-Point Magnitude Ceiling)

- **Why the Singularity Happens:** In 64-bit IEEE-754 double-precision arithmetic, 11 bits are allocated to the binary exponent and establish a maximum finite magnitude of $(2 - 2^{-52}) \times 2^{1023} \approx 1.7976931348623157 \times 10^{308}$. Exponential and factorial functions grow so rapidly that modest inputs cross this threshold:
  - **Factorials ([↶] [x!]):** $170! \approx 7.2574 \times 10^{306}$ fits in Register X, whereas $171! \approx 1.2410 \times 10^{309}$ exceeds 10^{308}.
  - **Natural & Common Exponentials:** e^x overflows for $x > 709.782712893384$ ($\ln(1.7977 \times 10^{308})$), and 10^x overflows for $x > 308.2547155599167$.
- **Stack State & Recovery Procedure:**

1. Press [ON] to dismiss the **OVERFLOW** banner; your unscaled operands remain intact in Y and X.
2. Work in logarithmic space (for example, using Stirling's approximation $\ln(n!) \approx n \ln n - n + (1/2) \ln(2\pi n)$ or adding logarithms $\log_{10}(a \times b) = \log_{10} a + \log_{10} b$) so huge exponents never overflow the hardware mantissa.

B.1.6 Code 04: TOO BIG (36-Bit Two's-Complement Word Overflow)

- **Why the Singularity Happens:** When operating in Computer Science integer base modes (**HEX**, **OCT**, **BIN**), the SC-32 models a classic 36-bit signed two's-complement architecture, where 1 bit represents the sign and 35 bits represent magnitude:

$$\text{Signed 36-Bit Range: } -2^{35} = -34,359,738,368 \text{ (0x800000000) to } +2^{35}-1 = +34,359,738,367 \text{ (0x7FFFFFFF)} \quad \textbf{(Eq. B.1)}$$

Attempting to switch from **DEC** into **HEX**, **OCT**, or **BIN** while Register X holds a floating-point number outside $[-2^{35}, 2^{35}-1]$ triggers **TOO BIG**.

- **Stack State & Recovery Procedure:**

1. Press [ON] to dismiss the message; your full floating-point value remains intact in Register X.
2. Remain in [↶] [BASE] [DEC] for arbitrary floating-point magnitudes, or reduce the integer modulo 2^{36} before converting to hexadecimal or binary.

B.2 Statistical, Iterative Solver & Financial Singularities

While Codes 01–04 protect single-step arithmetic and transcendental functions, Codes 05–08 guard multi-variable statistical accumulators, numerical root-finders, polynomial solvers, and compound-interest financial registers.

B.2.1 Code 05: STAT ERROR (Degrees-of-Freedom & Vertical-Line Singularities)

- **Why the Singularity Happens:** Statistical summaries condense a cloud of n data points into a center and a spread, which requires enough independent observations to define those geometric quantities:
  - **Sample Mean with Empty Data ($n = 0$):** Computing $\bar{x} = \sum x_i \,/\, n$ when no points have been accumulated via [Σ+] divides 0/0.
  - **Sample Standard Deviation with a Single Point ($n < 2$):** Sample standard deviation uses Friedrich Bessel's correction in the denominator:

$$s_x = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \qquad \textbf{(Eq. B.2)}$$

The denominator uses n - 1 rather than n because the sample mean $\bar{x}$ is calculated from the data itself, which uses up 1 degree of freedom. If you have only $n = 1$ data point, that point sits on its own mean ($x_1 = \bar{x}$) and produces zero apparent spread with a denominator of n - 1 = 0. You need at least $n = 2$ points to measure a distance between measurements.

  - **Linear Regression with Identical x-Coordinates ($\sigma_x = 0$):** Fitting a line $\hat{y} = a + bx$ requires at least two distinct x-values. If all entered (x_i, y_i) pairs share the same x-coordinate, the points stack in a vertical column of infinite slope ($b = \Delta y \,/\, 0$), and Pearson's correlation $r = s_{xy} \,/\, s_x\, s_y$ divides by $s_x = 0$.

- **Stack State & Recovery Procedure:**

1. Press [C ON] to dismiss; all six statistical accumulators ($\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n) and the stack remain intact.
2. Inspect your current sample count via [→] [SUMS] (**n** on [LFU5]) and add at least a second distinct data point with [Σ+], or use population standard deviation σ_x ([→] [s, σ] → **σx**) if a single-point population divisor $n = 1$ is intended.

B.2.2 Code 06: NO ROOT FOUND (Unbracketed Zero-Crossing in Brent's Solver)

- **Why the Singularity Happens:** Richard Brent's hybrid root-finding algorithm ([→] [SOLVE]) combines the guaranteed convergence of bisection with the rapid superlinear speed of secant and inverse quadratic interpolation. By the Intermediate Value Theorem, a continuous function $f(x)$ is guaranteed to cross the horizontal axis ($f(x^*) = 0$) inside an interval $[a, b]$ if and only if $f(a)$ and $f(b)$ have *opposite algebraic signs* ($f(a) \cdot f(b) < 0$—one point above the axis, one point below). **NO ROOT FOUND** occurs when:
  - The curve never crosses the real axis at all (for example, $f(x) = x^2 + 4 > 0$ for every real x).
  - Your two starting guesses in Registers Y and X sit on the same side of the root ($f(a) \cdot f(b) > 0$), and 100 secant steps fail to locate a sign reversal.

- **Stack State & Recovery Procedure:**

1. Press [C ON] to clear the banner; the solver deposits its closest candidate x in Register X while preserving your equation.

2. Sample the active **FN=** equation at trial endpoints via [XEQ] (for example, storing trial abscissas in your equation variable and inspecting the sign of $f(x)$ in Register X) to locate an interval $[a, b]$ where $f(a)$ and $f(b)$ have opposite algebraic signs.

3. Enter the two verified bracket guesses a ([ENTER]) and b that straddle opposite sides of the zero crossing, and press [↗] [SOLVE].

B.2.3 Code 07: A ≠ 0 (Degenerate Leading Coefficient in POLY Solver)

- **Why the Singularity Happens:** In the characteristic polynomial workspace ([↗] [POLY]), a cubic equation $A\,x^3 + B\,x^2 + C\,x + D = 0$ is normalized by dividing through by its highest non-zero leading coefficient (A for a cubic, or B when $A = 0$ for a quadratic). If both $A = 0$ and $B = 0$ (or when a cubic normalization requires a non-zero leading coefficient), dividing by zero is undefined because the polynomial degree collapses.

- **Stack State & Recovery Procedure:**

1. Press [C ON] to dismiss; your stored coefficients in **A**, **B**, **C**, and **D** remain untouched.

2. Store a non-zero leading coefficient and press [LFU4] (**EXEC**). The [↗] [POLY] workspace solves quadratics $bx^2 + cx + d = 0$ either natively by setting cubic coefficient $A = 0$ (**0** [A]) and entering (B, C, D), or by entering (A, B, C) with $D = 0$ (**0** [D], factoring $x(Ax^2 + Bx + C) = 0$). All three analytical roots (r_1, r_2, r_3)—including complex conjugate pairs—are written to indirect registers **R34**–**R39** (**34**–**35** for $\mathrm{Re}(r_1)$, $\mathrm{Im}(r_1)$, **36**–**37** for r_2, and **38**–**39** for r_3) and can be browsed in the [↖] [REGS] ([0] or [↖] [MEM] → **REGS**, **STO** tab) catalog or recalled via indirect addressing (**34** [STO] **i**, [RCL] **(i)**).

B.2.4 Code 08: ERROR 5 (Cash-Flow Sign Violation in Time Value of Money)

- **Why the Singularity Happens:** The Time Value of Money ([↖] [TVM]) equation balances present value (PV), periodic payments (PMT), and future value (FV) discounted at periodic rate i. The display message **ERROR 5** intentionally preserves the classic HP-12C actuarial error designation for TVM cash-flow sign violations where **PV**, **PMT**, and **FV** all share the same sign:

$$PV + PMT\left[\frac{1-(1+i)^{-N}}{i}\right] + FV(1+i)^{-N} = 0 \qquad \text{(Eq. B.3)}$$

Here the right-hand side is zero. If you enter $PV > 0$, $PMT > 0$, and $FV \geq 0$ (all positive), you are asking the solver to find an interest rate where money flows *only toward you* and never leaves your pocket—a sum of strictly positive numbers can never add up to zero for any real interest rate $i > -100\%$.

- **Stack State & Recovery Procedure:**

1. Press [C ON] to dismiss; all five financial registers (**N**, **I%**, **PV**, **PMT**, **FV**) remain intact.

2. Enforce the cash-flow sign convention: cash *received* (such as a loan principal PV) is positive, and cash *paid out* (such as monthly mortgage payments PMT or a savings deposit) is negative ([+/−]). Flip the sign of the outflow variable, store it back into its TVM softkey, and solve again.

B.3 Memory, Quadrature & Hardware Diagnostic Recovery Protocols

Codes 09 through 12 govern non-volatile equation bank capacity, numerical integration convergence limits, user interrupts, and unsigned arithmetic guards.

B.3.1 Code 09: MEMORY FULL (Equation Bank Capacity Reached)

- **Triggering Keystrokes:** Pressing [↗] [EQN] (Right Shift [↗] above [STO]) and attempting to create a 25th stored equation or exceed the flash equation token budget.
- **Why the Singularity Happens:** The SC-32 stores up to 24 persistent user-defined postfix equations in continuous flash memory ([↗] [EQN]). Attempting to allocate a 25th equation slot or exceed the flash sector step budget triggers **MEMORY FULL**.
- **Non-Destructive Recovery:**

1. Press [C/ON] to dismiss; all existing equations, variables **A**–**Z**, and the stack remain 100% intact.
2. Open the equation bank ([↗] [EQN]), scroll to a temporary or obsolete equation, and delete it via [←] [CLEAR] → **CLEQN** ([LFUθ]).

B.3.2 Code 10: TIMEOUT (Romberg Quadrature Subdivision Exhaustion)

- **Triggering Keystrokes:** Invoking [↗] [∫] (Right Shift [↗] above [8]) on an integrand that fails to converge within the maximum Romberg refinement depth.
- **Why the Singularity Happens:** Adaptive Romberg quadrature ([↗] [∫]) repeatedly halves the trapezoid width $h_k = (b\text{-}a)/2^k$ and applies Richardson $(4^m\ T_{k,m\text{-}1}\ \text{-}\ T_{k\text{-}1,m\text{-}1})\ /\ (4^m\ \text{-}\ 1)$ extrapolation until successive diagonal estimates agree to within $\varepsilon = 10^{\text{-}9}$. If an integrand has a vertical asymptote inside $[a,\ b]$, a sharp step discontinuity, or violent high-frequency oscillation (such as $\sin(1/x)$ near $x = 0$), the error estimate may not settle within 100,000 function evaluations.
- **Non-Destructive Recovery:**

1. Press [C/ON] to dismiss; the finest-grid numerical approximation achieved before timeout is preserved in Register *X*.
2. Split the integration interval $[a,\ b]$ at the discontinuity or narrow the bounds around smooth sub-intervals, and sum the piecewise integrals on the stack.

B.3.3 Code 11: INTERRUPTED (Clean User Abort of Iterative Engine)

- **Triggering Keystrokes:** Pressing [C ON] during an active [↗] [SOLVE], [↗] [∫], or [↗] [POLY] sweep.
- **Why the Singularity Happens:** Pressing [C ON] while the calculator is actively running a multi-step Romberg integral ([↗] [∫]), a Brent root search ([↗] [SOLVE]), or a cubic solve ([↗] [POLY]) halts the iterative loop.
- **Non-Destructive Recovery:**

1. Temporary solver scratch variables are discarded, and your four operational stack registers (X, Y, Z, T) are restored to their pre-execution state.
2. Press [C ON] once more to clear the **INTERRUPTED** notice and resume calculation.

B.3.4 Code 12: Unsigned Flag 6 (NO NEG), Equation & Integer Guard Banners

- **Triggering Keystrokes:** Subtracting $Y < X$ or pressing [+/−] while System Flag 6 (**NO NEG**) is active (**INVALID DATA** on the SC-32; **CANNOT BE < 0** on the elementary SC-6), invoking [XEQ], [↗] [SOLVE], or [↗] [∫] without an active equation (**NO EQN**) or with a malformed token stream (**INVALID FOR EQN**), or invoking [GCD] / [LCM] in **PARTS** on a fractional operand (**INTEGER REQUIRED**).
- **Why the Singularity Happens:** When Flag 6 (**NO NEG**, the unsigned whole-number arithmetic profile) is active, negative inputs and results (Y - $X < 0$) are blocked at zero. Likewise, equation engines require a valid stored RPN formula in [↗] [EQN], and number-theoretic greatest common divisor ([GCD]) and least common multiple ([LCM]) are defined strictly over the ring of integers Z.
- **Non-Destructive Recovery:**

1. Press [C ON] to dismiss the diagnostic banner; both Y and X remain untouched.
2. For **NO NEG**, press [x⇄y] to swap Y and X or clear Flag 6 ([↗] [FLAGS] → **CF 6**); for **NO EQN** / **INVALID FOR EQN**, enter or select a valid equation in [↗] [EQN]; and for **INTEGER REQUIRED**, strip fractional parts via [↗] [PARTS] → [IP] before computing [GCD] or [LCM].

B.3.5 SC-32 Hardware Recovery Protocols

For collegiate and professional engineering examinations, laboratory baseline resets, or power transitions, the SC-32 provides three graduated reset levels:

1. **Soft Status Clear (Non-Destructive):** Press [ON] twice. Cancels any active menu prompt, dismisses error banners, and clears pending [←] or [→] shift flags without erasing variables, statistical accumulators, or the stack.

2. **Selective Workspace Clear ([←] [CLEAR] Menu):** Press [←] [CLEAR] (Left Shift [←] above [⌫]) to choose a targeted reset on the six top LFU softkeys:

   - **CLEQN** ([LFU0]): Deletes the selected equation in the Equation Bank.
   - **CLVARS** ([LFU1]) / **CLREGS** ([LFU4]): Zeroes letter storage registers **A**–**Z** (and **CLVARS** also zeroes the five **TVM** financial registers **N**, **I%**, **PV**, **PMT**, **FV**).
   - **CL**Σ ([LFU3]): Zeroes the six statistical accumulators ($\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n).
   - **CLSTK** ([LFU5]): Zeroes stack registers T, Z, Y, X while keeping stored variables intact.

3. **Full Factory Baseline Reset (CLALL):** Hold [ON] while powering on, or press [←] [CLEAR] $\rightarrow$ **CLALL** ([LFU2]). Erases all variables **A**–**Z**, statistical accumulators, TVM financial registers, stored equations, and the stack, returning angular mode to **DEG** and display format to **ALL**. On every power-up, the firmware also verifies the cyclic redundancy checksum of its dual non-volatile state slots, automatically falling back to the last verified atomic snapshot if a battery swap occurred mid-write.

Appendix C: Physical Keycap Directory & Engineering Architecture

C.1 Physical Keypad Layout & the 43-Key Matrix

A pocket scientific instrument faces a fundamental human-factors trade-off: it must expose dozens of transcendental functions, statistical estimators, coordinate transformations, and numerical solvers without burying everyday arithmetic inside scrolling multi-page menus or crowding the faceplate with tiny, closely spaced keys. The SC-32 addresses this trade-off through a 43-key tactile matrix divided into three ergonomic zones, paired with a three-layer shift grammar that groups functions by mathematical association rather than arbitrary menu trees.



Figure: StackCalc SC-32 Scientific RPN Calculator physical faceplate render, showing the 43-key layout, upper dynamic LFU softkeys, double-wide ENTER bar, and high-contrast primary and secondary shift legends.

C.1.1 The Three-Layer Ergonomic Grammar

Below the top row of six dynamic softkeys, each of the 37 main physical keys carries three coordinated operational meanings—plus an automatic alpha character for variable and equation naming—organized by a strict conceptual rule:

- **Primary Legend (White Center Label — 1 Tap):** Every key's unshifted tap executes the foundational, highest-frequency operation for that position on the faceplate. These include core transcendental functions ([√x], [eˣ], [LN], [yˣ], [x⁻¹], [SIN], [COS], [TAN]) and statistical accumulation ([Σ+]), alongside stack and memory control ([STO], [RCL], [R↓], [ENTER], [x⇄y]) and immediate postfix arithmetic ([÷], [×], [-], [+], [x‖y]).
- **Left Shift ([↰] — Mathematical Inverse or Direct Companion):** Tapping [↰] accesses the mathematical inverse or companion printed above the upper-left corner of the key: [√x] ↔ [↰] [x²], [eˣ] ↔ [↰] [10ˣ], [LN] ↔ [↰] [LOG], [yˣ] ↔ [↰] [ˣ√y], [x⁻¹] ↔ [↰] [x!], [Σ+] ↔ [↰] [Σ−], [SIN] ↔ [↰] [ASIN], [COS] ↔ [↰] [ACOS], [TAN] ↔ [↰] [ATAN], and [ENTER] ↔ [↰] [LASTx] to recover the pre-operation operand. Across the lower number pad, [↰] converts imperial inputs into metric (SI) units ([↰] [▸km], [↰] [▸kg], [↰] [▸°C], [↰] [▸cm], [↰] [▸1]) and polar/decimal formats ([↰] [▸θ,r], [↰] [▸HR], [↰] [▸DEG]).

- **Right Shift ([→] — Modal Workspaces, Solvers & Softkey Menus):** Tapping [→] opens the analytical workspace or specialized operator printed above the upper-right corner. On the top rows, [→] opens the softkey menus associated with each key's domain: **PARTS** above [√x], **PROB** above [eˣ], **L.R.** above [LN], x̄,ȳ above [yˣ], **s,**σ above [x⁻¹], and **SUMS** above [Σ+]. Elsewhere on the keypad, [→] launches the numerical engines ([→] [POLY], [→] [SOLVE], [→] [∫], **EQN**) and invokes specialized operators (π, [→] [%], [→] [%CHG], [ABS], [→] [1/x], [→] [SHOW]). On the digit pad, [→] converts metric values into US/Imperial units ([→] [▸mi], [→] [▸lb], [→] [▸°F], [→] [▸in], [→] [▸gal]) and rectangular/sexagesimal formats ([→] [▸y,x], [→] [▸HMS], [→] [▸RAD]).

- **Alpha Labels (A–Z, i, (i)):** Printed to the lower-right of the main keys, the 26 alphabetical register labels **A** through **Z** plus the indirect-addressing indicators **i** and **(i)** activate automatically whenever a command prompts for a register or program label, such as after pressing [STO], [RCL], [XEQ], **FN=**, **LBL**, [→] [SOLVE], or [→] [∫]. You never need to toggle a dedicated alpha-lock mode to store a physical constant in Register **G** or solve an equation for variable **T**.

C.1.2 Upper Function Zone (Rows 0–2, 6 Columns)

The top three rows of the keypad form a compact 6-column scientific command deck positioned beneath the transflective display so your eyes can scan the stack and the top keys in a single field of view. Row 0 ([LFU0] through [LFU5]) is a dedicated, context-aware LFU (Least Frequently Used) dynamic softkey strip: in normal state, it dynamically surfaces the six functions tracked by the LFU algorithm, and whenever you open a menu or workspace such as **PARTS**, **PROB**, **L.R.**, **BASE**, **MODES**, **DISP**, **FLAGS**, **TVM**, or **POLY**, it remaps to that menu's softkey labels along the bottom pixel row of the LCD. Pressing any [LFU0]–[LFU5] button executes the corresponding on-screen command in a single tap without hiding the active stack registers or hijacking the fixed scientific keys below.

Below the softkey bar, Row 1 ([√x], [eˣ], [LN], [yˣ], [x⁻¹], [Σ+]) groups every power, root, exponential, logarithmic, reciprocal, and statistical accumulation operator along a single horizontal row. Four of these keys ([√x], [eˣ], [LN], [x⁻¹]) and their [←] companions ([←] [x²], [←] [10ˣ], [←] [LOG], [←] [x!]) are immediate unary transformations that replace X in place while preserving Y, Z, T. By contrast, [yˣ] and [←] [x√y] combine Y and X dyadically as Y^X and $Y^{1/X}$. At the right end of Row 1, [Σ+] and [←] [Σ−] feed (x, y) data pairs into the six statistical accumulators while updating X with the sample count n and disarming stack lift. The right shifts ([→]) across Row 1 turn those six keys into a statistical and number-theory toolbar: **PARTS**, **PROB**, **L.R.**, x̄,ȳ, **s,**σ, and **SUMS**.

Row 2 ([STO], [RCL], [R↓], [SIN], [COS], [TAN]) pairs named register management and stack rotation on the left with planar trigonometry on the right. Pressing [STO] or [RCL] followed by any letter key **A**–**Z** stores or recalls a value, and inserting **+**, **-**, **×**, or **÷** before the letter performs non-destructive Storage or Recall Arithmetic without cluttering the stack. Their shifted legends open two primary workspaces: Left Shift [←] above [STO] (**CMPLX**) arms the four-register Gaussian complex phasor engine on (Y, X) and (T, Z), while Right Shift [→] above [STO] (**EQN**) opens the symbolic Equation Bank.

Next to [RCL], [R↓] and its right shift [→] [R↑] rotate all four stack registers circularly so you can inspect or reposition any intermediate value. Left Shift [←] above [R↓] ([←] [HYP]) serves as the hyperbolic prefix for [SIN], [COS], [TAN], and their [←] inverse functions. Finally, the right shifts of the three trig keys place π above [SIN], Kahan's base-preserving percentage [→] [%] above [COS], and percent change [→] [%CHG] above [TAN] within thumb reach.

Row	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5
Row 0	LFU0	LFU1	LFU2	LFU3	LFU4	LFU5
Row 1	$\sqrt{x}$	e^x	LN	y^x	x^{-1}	$\Sigma+$
Row 2	STO	RCL	R↓	SIN	COS	TAN

C.1.3 Transition Zone (Row 3)

Row 3 forms the structural and ergonomic bridge between the upper scientific function array and the lower number pad. Anchoring Columns 0 and 1 is the double-wide ENTER bar—the physical fulcrum of Reverse Polish Notation. When you finish typing the first number of a dyadic pair, pressing ENTER terminates digit entry, pushes a copy of X up into Y ($T \leftarrow Z$, $Z \leftarrow Y$, $Y \leftarrow X$), and disables stack lift so the next number you type overwrites X in place. Its shifted legends above ENTER provide operand recovery and full-precision inspection: Left Shift ($\leftarrow$ LASTx) recalls the exact value that was in Register X before the last mathematical operation, while Right Shift ($\rightarrow$ SHOW) temporarily reveals the full 15-digit internal mantissa and exponent of Register X without altering your active **FIX**, **SCI**, or **ENG** display setting.

To the right of ENTER sits $x \rightleftarrows y$ at Row 3, Col 2, the register-exchange key that gives RPN its algebraic freedom. Whenever you evaluate a complex denominator or subtrahend *first* and then push the numerator or minuend onto the stack, a single tap of $x \rightleftarrows y$ swaps X and Y so ÷, -, or y^x executes in the proper order without parentheses. Its shifted functions inspect available memory ($\leftarrow$ MEM) or swap X with any letter register ($\rightarrow$ $x \rightleftarrows ?$). Beside it at Row 3, Col 3, +/- toggles the sign of the number currently in X—or negates the power-of-ten exponent if pressed after E—while hosting the angular/radix **MODES** menu on $\leftarrow$ and absolute value ABS ($|x|$) on $\rightarrow$.

Completing the transition bridge are E at Row 3, Col 4 and ⌫ at Row 3, Col 5. Pressing E during number entry transitions from the mantissa to the base-10 exponent ($m \times 10^n$) so you can key physical constants like Planck's constant or Avogadro's number without multiplying by 10^x. Its left shift opens the **DISP** precision menu (**FIX**, **SCI**, **ENG**, **ALL**), and its right shift executes Euclidean integer division with remainder ($\rightarrow$ $\div_R$), placing the quotient $\lfloor Y/X \rfloor$ in X and the remainder in Y. At the far right, the backspace key ⌫ adapts to your state: while you are actively typing digits, it erases one character at a time from the right. Once a number is finalized on the stack, pressing ⌫ acts as **CLx**, zeroing Register X and disabling stack lift without dropping Y, Z, or T, while Left Shift $\leftarrow$ opens the selective **CLEAR** menu.

Row	Col 0 & 1 (Double-Wide)	Col 2	Col 3	Col 4	Col 5
Row 3	ENTER	$x \rightleftarrows y$	+/-	E	⌫

C.1.4 Lower Arithmetic Zone (Rows 4–7, 5 Columns)

The lower four rows widen into a generous 5-column pitch engineered for rapid, eyes-free numeric entry and postfix arithmetic. Along the left margin in Column 0, four dedicated control keys govern execution and shift state. XEQ executes stored equations by label, while its shifts designate the target equation **FN=** on $\leftarrow$ and open the quadratic/cubic **POLY** root workspace on $\rightarrow$. Below it, $\leftarrow$ and $\rightarrow$ act as latching, toggleable left- and right-shift prefixes. At the bottom-left corner, ON ($\leftarrow$ **ON**) wakes the hardware, cancels active prompts, or clears Register X, with $\leftarrow$? on $\leftarrow$ for the interactive guided tutorial and **OFF** on $\rightarrow$ for deep sleep.

Along the right margin in Column 4, the four foundational dyadic operators—[÷], [×], [-], and [+]—fuse Y and X, drop the stack ($Y \leftarrow Z$, $Z \leftarrow T$), replicate T, and leave stack lift armed. Their shifted legends provide relational tests (**x?y**, **x?0**), computer-science integer **BASE** modes and system **FLAGS**, liter/gallon conversions ([◀] [▶l]/[▶] [▶gal]), and equation labels ([LBL]/[RTN]).

Across the central **0**–**9** and **.** digit pad in Columns 1–3, the shifted legends are organized into five engineering clusters so you can model physical problems on the stack:

- **Calculus, Root-Finding & Equation Stepping (7 and 8):** Right-shifting [7] ([▶] [SOLVE]) and [8] ([▶] [∫]) invokes Richard Brent's root finder and Werner Romberg's adaptive quadrature on the active **FN=** equation using your bounds in Y and X. Left-shifting [7] ([◀] [▼]) and [8] ([◀] [▲]) steps vertically through multi-equation banks and register catalogs.
- **2D Vector, Sexagesimal Time & Angular Conversions (4, 5, and 6):** Key [4] converts between rectangular coordinates (x, y) in (X, Y) and polar phasors (r, ϑ) via [◀] [▶θ,r] and [▶] [▶y,x]. Key [5] translates between **DD.MMSS** sexagesimal angles/timestamps and decimal hours/degrees ([◀] [▶HR] ↔ [▶] [▶HMS]), and Key [6] converts between radians and degrees ([◀] [▶DEG] ↔ [▶] [▶RAD]).
- **Physical SI ↔ Imperial Unit Conversions (9, 1, 2, 3, and -):** Every paired conversion puts the metric unit on [◀] and the imperial unit on [▶]. Distance sits on [9] ([◀] [▶km] ↔ [▶] [▶mi]), mass on [1] ([◀] [▶kg] ↔ [▶] [▶lb]), and temperature on [2] ([◀] [▶°C] ↔ [▶] [▶°F]), followed by length on [3] ([◀] [▶CM] ↔ [▶] [▶in]) and fluid volume on [-] ([◀] [▶l] ↔ [▶] [▶gal]).
- **Register Inspection, Fractions & Indirect Addressing (0, ., i, (i)):** Key [0] opens the full-screen **REGS** table ([◀]) or single-register **VIEW** ([▶]). Next to it, the radix point [•] toggles exact fraction display [◀] [FDISP] ([◀]), sets the fixed fraction denominator **/c** ([▶]), and carries the Alpha label **i** for the index register. The Alpha label **(i)** on [x‖y] dereferences the integer pointer in **i** across all 89 slots of the memory map: **1**–**26** = **A**–**Z**, **27** = **i**, **28**–**33** = statistical sums, **34**–**39** = **POLY** roots, and **40**–**89** = raw statistical data points (**DATA0**–**DATA49**).
- **The Harmonic Hub (x‖y) & Relational Tests (x?y, x?0):** At Row 7, Col 3 sits the [x‖y] key: a single unshifted tap evaluates the parallel harmonic product $Y \times X / (Y + X)$, while its left shift ([◀]) launches the 5-variable **TVM** financial solver (**N**, **I%**, **PV**, **PMT**, **FV**, **END/BEG**) and its Alpha label **(i)** provides indirect register dereferencing. Meanwhile, the division key [÷] at Row 4, Col 4 hosts the two six-operator relational comparison menus [◀] [x?y] ([◀]) and [▶] [x?0] ([▶]), which flash **YES**/**NO** non-destructively on the live stack and act as 1/0 Iverson Boolean indicator switches inside [▶] [EQN] to model piecewise functions.

Row	Col 0	Col 1	Col 2	Col 3	Col 4
Row 4	[XEQ]	[7]	[8]	[9]	[÷]
Row 5	[◀] (Left Shift)	[4]	[5]	[6]	[×]
Row 6	[▶] (Right Shift)	[1]	[2]	[3]	[-]
Row 7	[ON] (ON)	[0]	[•]	[x‖y]	[+]

C.1.5 The 43-Key Function Directory

Each key position is indexed by matrix coordinate (Row, Col), primary legend, left [◀] and right [▶] shift functions, alpha register letter, and operational behavior:

Key #	Matrix (R, C)	Primary Legend	Left Shift [↖]	Right Shift [↗]	Alpha	Functional Summary
1	R0, C0	LFU0	Softkey 1	Softkey 1	—	Dynamic LFU slot 1 / active softkey menu item 1
2	R0, C1	LFU1	Softkey 2	Softkey 2	—	Dynamic LFU slot 2 / active softkey menu item 2
3	R0, C2	LFU2	Softkey 3	Softkey 3	—	Dynamic LFU slot 3 / active softkey menu item 3
4	R0, C3	LFU3	Softkey 4	Softkey 4	—	Dynamic LFU slot 4 / active softkey menu item 4
5	R0, C4	LFU4	Softkey 5	Softkey 5	—	Dynamic LFU slot 5 / active softkey menu item 5
6	R0, C5	LFU5	Softkey 6	Softkey 6	—	Dynamic LFU slot 6 / active softkey menu item 6
7	R1, C0	$\sqrt{x}$	[↖] x^2	**PARTS** Menu	**A**	Square root √x, square x^2, and number-theory PARTS menu (IP, FP, ABS, SGN, GCD, LCM)
8	R1, C1	e^x	[↖] 10^x	**PROB** Menu	**B**	Natural exponential e^x, common antilog 10^x, and combinatorics PROB menu (Cn,r, Pn,r, x!, RAND, SD)
9	R1, C2	LN	[↖] LOG	**L.R.** Menu	**C**	Natural logarithm LN, base-10 logarithm LOG, and linear regression L.R. menu ($\hat{y}$, $\hat{x}$, r, m, b)
10	R1, C3	y^x	[↖] $\sqrt[x]{y}$	$\bar{x}$,$\bar{y}$ Menu	**D**	Dyadic power y^x, x-th root $\sqrt[x]{y}$, and sample/weighted means menu ($\bar{x}$, $\bar{y}$, $\bar{x}$w)
11	R1, C4	x^{-1}	[↖] x!	**s**,σ Menu	**E**	Reciprocal 1/x, factorial/Gamma x!, and sample/population standard deviation menu (sx, sy, σx, σy)
12	R1, C5	Σ+	[↖] Σ−	**SUMS** Menu	**F**	Statistical data accumulation Σ+, removal Σ-, and summation register menu (Σx, Σy, Σx^2, Σy^2, Σxy, n)
13	R2, C0	STO	**CMPLX** Mode	**EQN** Mode	**G**	Store register STO, complex phasor prefix CMPLX, and symbolic RPN Equation Bank EQN
14	R2, C1	RCL	[↖] RND Round	SCRL Scroll	**H**	Recall register RCL, round mantissa to display precision RND, and scroll long equations
15	R2, C2	R↓	[↖] HYP Prefix	[↗] R↑ Roll Up	**I**	Roll stack down R↓, hyperbolic function prefix HYP, and roll stack up R↑
16	R2, C3	SIN	[↖] ASIN	π Constant	**J**	Sine SIN, inverse sine ASIN, and 15-digit π constant
17	R2, C4	COS	[↖] ACOS	[↗] % Base Pct	**K**	Cosine COS, inverse cosine ACOS, and Kahan base-preserving percentage %
18	R2, C5	TAN	[↖] ATAN	[↗] %CHG Pct Chg	**L**	Tangent TAN, inverse tangent ATAN, and base-preserving percent change %CHG

Key #	Matrix (R, C)	Primary Legend	Left Shift [↖]	Right Shift [↗]	Alpha	Functional Summary
19	R3, C0-1	ENTER	[↖] LASTx	[↗] SHOW	**M**	Duplicate X to Y and disable lift ENTER, recall prior operand LASTX, and full-precision peek SHOW
20	R3, C2	x⇄y	**MEM** Specs	x⇄? Swap Reg	**N**	Swap X and Y registers x↔y, memory status summary MEM, and exchange X with named register x↔?
21	R3, C3	+/−	**MODES** Menu	ABS Abs	**O**	Toggle sign +/-, angular/radix MODES menu (DEG, RAD, GRAD, ., ,), and absolute value |x|
22	R3, C4	E	**DISP** Menu	[↗] ÷R Div/Rem	**P**	Scientific exponent entry E, display format DISP menu (FIX, SCI, ENG, ALL), and Euclidean quotient/remainder ÷R
23	R3, C5	⌫	**CLEAR** Menu	*(Unassigned)*	—	Digit backspace / clear X register CLx, and selective workspace CLEAR menu
24	R4, C0	XEQ	**FN=** Define	**POLY** Solver	—	Execute equation XEQ, select target function FN=, and closed-form quadratic/cubic POLY root solver
25	R4, C1	7	[↖] [▼] Step Down	[↗] SOLVE Brent	**Q**	Digit 7, step down equation/register list ↓, and Richard Brent hybrid root solver SOLVE
26	R4, C2	8	[↖] [▲] Step Up	[↗] ∫ Quadrature	**R**	Digit 8, step up equation/register list ↑, and adaptive Romberg numerical integration ∫
27	R4, C3	9	[↖] ▶km Conv	[↗] ▶mi Conv	**S**	Digit 9, miles-to-kilometers conversion →km, and kilometers-to-miles conversion →mi
28	R4, C4	÷	**x?y** Menu	**x?0** Menu	—	Dyadic division ÷, relational comparison menu x?y, and zero-comparison menu x?0
29	R5, C0	[↖] (Left Shift)	*(Left Shift Latch)*	*(Left Shift Latch)*	—	Primary Left Shift modifier prefix (toggles upper-left key legends)
30	R5, C1	4	[↖] ▶θ,r Polar	[↗] ▶y,x Rect	**T**	Digit 4, rectangular-to-polar conversion →ϑ,r, and polar-to-rectangular conversion →y,x
31	R5, C2	5	[↖] ▶HR Hours	[↗] ▶HMS Time	**U**	Digit 5, sexagesimal DD.MMSS to decimal hours/degrees →HR, and decimal to sexagesimal →HMS
32	R5, C3	6	[↖] ▶DEG Angle	[↗] ▶RAD Angle	**V**	Digit 6, radians-to-degrees conversion →DEG, and degrees-to-radians conversion →RAD
33	R5, C4	×	**BASE** Menu	**FLAGS** Menu	—	Dyadic multiplication ×, radix BASE menu (HEX, DEC, OCT, BIN), and system FLAGS / stack-depth menu

Key #	Matrix (R, C)	Primary Legend	Left Shift ↰	Right Shift ↱	Alpha	Functional Summary
34	R6, C0	↱ (Right Shift)	*(Right Shift Latch)*	*(Right Shift Latch)*	—	Secondary Right Shift modifier prefix (toggles upper-right key legends)
35	R6, C1	1	↰ ▸kg Mass	↱ ▸lb Mass	W	Digit 1, pounds-to-kilograms conversion →kg, and kilograms-to-pounds conversion →lb
36	R6, C2	2	↰ ▸°C Temp	↱ ▸°F Temp	X	Digit 2, Fahrenheit-to-Celsius conversion →°C, and Celsius-to-Fahrenheit conversion →°F
37	R6, C3	3	↰ ▶CM Length	↱ ▸in Length	Y	Digit 3, inches-to-centimeters conversion →cm, and centimeters-to-inches conversion →in
38	R6, C4	-	↰ ▸l Volume	↱ ▸gal Volume	—	Dyadic subtraction -, US gallons-to-liters conversion →l, and liters-to-US gallons conversion →gal
39	R7, C0	ON (ON)	↰ ? Tutorial	**OFF** Sleep	—	Clear entry / dismiss error / wake hardware C, interactive guided tutorial ?, and power sleep OFF
40	R7, C1	0	**REGS** Viewer	**VIEW** Single	Z	Digit 0, full-screen register browser REGS (STACK, STO, STATS, DATA), and single-register inspect VIEW
41	R7, C2	•	↰ FDISP Frac	↱ /c Denom	i	Decimal radix point ., toggle rational fraction display FDISP, and set fixed fraction denominator /c
42	R7, C3	x‖y	**TVM** Solver	*(Unassigned)*	(i)	Parallel harmonic operator x‖y, 5-variable TVM financial solver, and indirect address pointer (i)
43	R7, C4	+	LBL Label	RTN Return	—	Dyadic addition +, equation label token LBL, and subroutine return token RTN

C.2 Architectural Refinements, Dynamic LFU Menus & Complete HP-32SII / HP-15C / HP-12C Comparison

The SC-32 keypad layout follows the Dvorak Principle applied to calculators: whereas computer terminals and menu-driven graphing calculators place common operations inside nested menu trees, routine scientific and engineering calculations on a handheld instrument should execute from direct tactile keystrokes.

C.2.1 The Corvallis HP-32SII Structural Foundation

The 1991 HP-32SII Pioneer provides an ergonomic and symbolic baseline for a portrait RPN scientific instrument. Its one-handed geometry keeps the digit pad, ENTER, and arithmetic column inside the natural sweep of a single thumb. At the same time, its 26 automatic alpha variables (**A**–**Z**) and symbolic Equation Bank (↱ EQN) replace anonymous register numbers (**R0**–**R9**) and blind keystroke subroutines (**LBL 0..9**), while eight SI ↔ Imperial unit conversions (expanded to ten on the SC-32) and native rational fractions (↰ FDISP, ↱ /c) sit directly on the number pad.

The SC-32 preserves that 1991 Pioneer matrix while replacing its 1-line display and infix-equation mode with a 4-line stack display and pure RPN equation steps. It also integrates the analytical engines of the 1982 HP-15C Voyager ([↗] SOLVE, [↗] ∫, [↖] CMPLX), a closed-form quadratic/cubic [↗] POLY engine, and the 1981 HP-12C Voyager financial functions ([↖] TVM, [↗] %, [↗] %CHG) onto shifted keys:

Architectural Dimension	HP-12C Voyager (1981)	HP-15C Voyager (1982)	HP-32SII Pioneer (1991)	StackCalc SC-32
Form Factor & Ergonomics	Landscape (two-handed)	Landscape (two-handed)	Portrait (one-handed thumb arc)	Portrait 43-key Pioneer geometry (one-handed thumb arc)
Display & Stack Visibility	1-line, 10-digit 7-seg (**X** only)	1-line, 10-digit 7-seg (**X** only)	1-line, 12-char dot-matrix (**X** only)	4-line 132 × 65 transflective LCD (*T*, *Z*, *Y*, *X* visible simultaneously)
Softkey Menu Bar	None	None	Remaps top row (**√x**..**Σ+**) with 1-line text	Dedicated 6-key Dynamic LFU Bar ([LFUθ]–[LFU5]) above **√x**..**Σ+**
Operational Stack Depth	4-level (*X*, *Y*, *Z*, *T*)	4-level (*X*, *Y*, *Z*, *T*)	4-level (*X*, *Y*, *Z*, *T*)	Selectable [4-LVL] (classic Corvallis), [8-LVL], or [INF] (999-level LIFO)
Storage Registers	**R0**–**R9**, **.0**–**.9** (numbered)	**R0**–**R9**, **.0**–**.9**, **I** (numbered)	**A**–**Z** (26 named letters) + **i**, **(i)**	**A**–**Z** (26 named letters) + **i**, **(i)** across 89 indirect slots (**R1**–**R89**)
Equation Bank (EQN)	None (keystroke **R/S** only)	None (keystroke **LBL 0..9** only)	Algebraic infix **EQN** (**(**, **)**, **=**)	Pure RPN [↗] [EQN] Bank with Iverson Boolean [x?0]/[x?y] piecewise switches
Root Finder & Calculus	**IRR** / **NPV** cash-flow only	Brent **SOLVE** & Romberg **∫f(x)** via **LBL**	Brent **SOLVE** & Romberg ∫ via **EQN**	Brent [↗] [SOLVE] & Romberg [↗] [∫] via [↗] [EQN] + Dedicated Analytic [↗] [POLY] (**R34**–**R39**)
Complex Numbers (CMPLX)	None	Parallel complex stack via **I** / **Re↔Im**	4-register (*Y*, *X*) & (*T*, *Z*) via **CMPLX** prefix	4-register (*Y*, *X*) & (*T*, *Z*) via [↖] [CMPLX] prefix (including complex [x⇌y] & [↗] [POLY] roots)
Time Value of Money (TVM)	Dedicated **n**, **i**, **PV**, **PMT**, **FV** keys	None	None (requires manual keystroke program)	Built-in 5-Variable [↖] [TVM] (**N**, **I%**, **PV**, **PMT**, **FV**, **END/BEG**) on [x⇌y]
Physical Unit Conversions	None	**→H.MS**, **→RAD** only	8 SI ↔ US keys + polar/HMS/angle	10 SI ([↖]) ↔ US ([↗]) keys (adds **km** ↔ **mi** on **9**) + polar/HMS/angle
Exact Fractions & Number Theory	None	None	**a b/c**, **FDISP**, **/c**; **PARTS** menu	Exact fractions ([↖] [FDISP], [↗] [/c]), direct [↗] [1/R] & [ABS] keys (**GCD**/**LCM** in **PARTS** softkey menu)
Parallel / Harmonic Operator	None	None	None	Dedicated unshifted [x⇌y] key (*Y* × *X* / (*Y* + *X*) for resistors, lenses, capacitors)
Exam Mode & Base Word	Non-programmable	Programmable (10-digit)	Programmable (32-bit unsigned **BASE**)	Non-graphing scientific + proctored [EXAM] lockdown & 36-bit signed **BASE**

C.2.2 Complete Key-by-Key & Menu Differences: HP-32SII vs. SC-32

Every primary key, left shift ([↖]), corresponding to the HP-32SII orange shift), right shift ([↗]), corresponding to the HP-32SII blue shift), and alpha legend on the SC-32 behaves identically to the 1991 HP-32SII except for the intentional hardware and menu additions listed below:

Position / Menu	Key Face (SC-32 vs. HP-32SII)	Left Shift [↖] (SC-32 vs. HP-32SII)	Right Shift [↗] (SC-32 vs. HP-32SII)	Engineering Rationale for the SC-32 Design
R0, C0–C5	[LFU0]–[LFU5] *(New dedicated row)*	Dynamic LFU / Menu softkeys	Dynamic LFU / Menu softkeys	Reserves 6 dedicated physical softkeys beneath the 4-line LCD (showing 6 dynamic LFU functions in normal state and menu labels in menus) so menus never hijack [√x]..[Σ+] (**Row 1**).
R3, C3	[+/−] *(Same)*	**MODES** *(Adds* **.** */* **,** *radix)*	[ABS] (**|x|**) *(HP-32SII:* **(** *open paren)*	Parentheses are unnecessary in RPN mode; places Absolute Value (**|x|**) on a direct right-shifted key while also retaining **ABS** inside the **PARTS** softkey menu.
R3, C4	[E] *(Same)*	**DISP** *(Same)*	[↗] [÷R] *(HP-32SII:* **)** *close paren)*	Parentheses are unnecessary in RPN mode; adds Euclidean integer division (**÷R**: quotient in *X*, remainder in *Y*) as a direct key.
R3, C5	[⌫] *(Same)*	**CLEAR** *(Expanded menu)*	*(Unassigned) (HP-32SII:* **=** *equals)*	Infix **=** is not needed in RPN mode; leaving Right-Shift [⌫] unassigned prevents accidental clears during fast entry.
R4, C0	[XEQ] *(Same)*	**FN=** *(Same)*	[↗] [POLY] *(HP-32SII: no direct key)*	Adds the closed-form quadratic/cubic [↗] [POLY] coefficient entry and root solver (**A**, **B**, **C**, **D**, **EXEC**, **CLR**) on **BLUE + XEQ**.
R4, C3	[9] *(Same)*	[↖] [▶km] *(HP-32SII: blank)*	[↗] [▶mi] *(HP-32SII: blank)*	Adds direct kilometer ↔ mile unit conversions ([↖] [▶km] on [↖], [↗] [▶mi] on [↗]) on [9].
R7, C1	[0] *(Same)*	**REGS** *(HP-32SII:* **INPUT***)*	**VIEW** *(Same)*	Repurposes programming-only **INPUT** to the **REGS** register viewer while keeping single-variable **VIEW** on [↗].
R7, C3	[x⇄y] *(HP-32SII:* **R/S***)*	**TVM** *(HP-32SII:* **PRGM***)*	*(Unassigned) (HP-32SII:* **PSE***)*	Repurposes **R/S** (**PRGM**/**PSE**) to Primary [x⇄y], Left-Shift **TVM** financial solver, and Alpha **(i)** indirect pointer.
R7, C4	[+] *(Same)*	[LBL] *(Restored for* **EQN***)*	[RTN] *(Same)*	Retains [LBL] on Left-Shift [+] so stored RPN equations can begin with an explicit single-letter label.

Position / Menu	Key Face (SC-32 vs. HP-32SII)	Left Shift [↖] (SC-32 vs. HP-32SII)	Right Shift [↗] (SC-32 vs. HP-32SII)	Engineering Rationale for the SC-32 Design
PARTS Menu	Trigger: **Right-Shift** [√x]	HP-32SII: **IP**, **FP**, **ABS**	SC-32 adds **SGN**, **GCD**, **LCM**	Places **SGN**, **GCD**, and **LCM** inside the 6-item **PARTS** softkey menu (**IP**, **FP**, **ABS**, **SGN**, **GCD**, **LCM**; only the elementary SC-6 has physical **GCF**/**LCM** buttons).
CLEAR Menu	Trigger: **Left-Shift** [⌫]	HP-32SII: **CLx**, **CLVARS**, **CLΣ**, **ALL**	SC-32: **CLEQN**, **CLVARS**, **CLALL**, **CLΣ**, **CLREGS**, **CLSTK**	Expands **CLEAR** to six softkeys (plus **FIN** inside **TVM** to zero **N**, **I%**, **PV**, **PMT**, **FV** while preserving **END**/**BEG**).
FLAGS Menu	Trigger: **Right-Shift** [×]	HP-32SII: **SF n**, **CF n**, **FS? n**, **FC? n**	SC-32 adds **4-LVL**, **8-LVL**, **INF**, **EXAM**	Extends **FLAGS** with selectable stack depth (**4-LVL**, **8-LVL**, **INF**) and proctored **EXAM** lockdown.
EQN Mode	Trigger: **Right-Shift** [STO]	HP-32SII: Infix strings (**SIN(X)+1**)	SC-32: RPN step entry + **EQN_NEW**/**EQN_EDIT**	Stores and edits equations as pure RPN instruction steps with LFU softkeys (**EQN_NEW**, **EQN_EDIT**).

C.2.3 Seven Architectural Enhancements Over the Vintage HP-32SII

Beyond the physical keycap legends, the SC-32 expands seven core subsystems while preserving 100% compatibility with classic HP-32SII keystroke sequences:

- **Four-Line Stack Display & Dedicated 6-Key LFU Bar (R0, C0–C5):** Replaces the HP-32SII's single-line 12-character LCD with a 132 × 65 px transflective 4-line stack view (T, Z, Y, X) paired with six physical LFU softkeys ([LFU0]–[LFU5]), so active submenus never hijack [√x]..[Σ+].

- **Pure RPN Equations & Iverson Piecewise Booleans ([↗] [EQN]):** Evaluates stored equations as pure, side-effect-free RPN token streams rather than infix strings. Relational tests from [↖] [x?y] and [↗] [x?0] (**R4C4**) act as non-destructive Iverson indicator brackets $[P] \in (1, 0)$ on Register X (preserving Y and Z) to support branchless piecewise formulas across [XEQ], [↗] [SOLVE], and [↗] [∫].

- **Direct |x| & ÷R Keys and Expanded PARTS Softkey Menu (R1C0, R3C3–C4):** Reclaiming infix **(** and **)** on Row 3 places Absolute Value [ABS] (**|x|**) and Euclidean quotient-remainder division [↗] [÷R] on direct right-shifted keys, while the **PARTS** softkey menu (**Right-Shift** [√x]) retains the HP-32SII's **IP**, **FP**, and **ABS** softkeys (**LFU0**–**LFU2**) and adds Signum (**SGN**), Greatest Common Divisor (**GCD**), and Least Common Multiple (**LCM**) on **LFU3**–**LFU5** (unlike the elementary SC-6, where **GCF** and **LCM** are physical keys).

- **Analytic POLY Root Workspace & TVM Financial Engine (R4C0, R7C3):** Adds a closed-form quadratic and cubic polynomial solver ([↗] [POLY]: **A**, **B**, **C**, **D**, **EXEC**, **CLR**) that writes all real and complex conjugate roots to **R34**–**R39**, alongside a 5-variable HP-12C [↖] [TVM] solver (**N**, **I%**, **PV**, **PMT**, **FV**, **END/BEG**).

- **Dedicated Parallel-Harmonic Operator ([x‖y] at R7C3):** Evaluates the reciprocal product-over-sum law $Y \times X / (Y + X)$ in a single unshifted keystroke across real and complex ([↖] [CMPLX]) operands for parallel resistors, series capacitors, thin lenses, and combined flow rates.

- **Configurable Stack Depth (4-LVL, 8-LVL, INF) & Proctored EXAM Mode (R5C4):** The **FLAGS** menu preserves user Flags **0**–**9** (**SF**, **CF**, **FS?**) and adds the **STACK** submenu (**2-LVL**, **4-LVL**, **8-LVL**, **INF**, **VIEW**) plus proctored **EXAM** lockdown. Default **4-LVL** mode preserves exact Corvallis 4-register physics (including top-of-stack T-replication on drop), while **8-LVL** and **INF** (999-level LIFO) eliminate register overflow during deeply nested evaluations.
- **36-Bit Signed Two's-Complement BASE Arithmetic & 89-Slot Indirect Memory Map (i, (i)):** Expands integer **BASE** operations (**DEC**, **HEX**, **OCT**, **BIN**) from 32-bit unsigned truncation to a 36-bit signed two's-complement word (9 hex nibbles, 12 octal digits). Integer pointers $k \in (1 \ldots 89)$ stored in **i** (**STO** [•]) and dereferenced via **(i)** (**RCL** [x⇄y] or **STO** [x⇄y]) address:
  - **1**–**26**: Named variables **A** through **Z** (**1 = A**, **2 = B**, ..., **26 = Z**), plus **27** for the index register **i** itself.
  - **28**–**33**: Live statistical accumulators (Σ**n**, Σ**x**, Σ**y**, Σ**x**^{2}, Σ**y**^{2}, Σ**xy** at indirect slots **28**–**33**; arranged as Σ**x**, Σ**y**, Σ**x**^{2}, Σ**y**^{2}, Σ**xy**, **n** on the **SUMS** softkey bar).
  - **34**–**39**: Analytic polynomial roots from [↗] [POLY] ($\mathrm{Re}(r_1)$, $\mathrm{Im}(r_1)$, $\mathrm{Re}(r_2)$, $\mathrm{Im}(r_2)$, $\mathrm{Re}(r_3)$, $\mathrm{Im}(r_3)$).
  - **40**–**89**: Up to 50 raw statistical data values (**DATA0**–**DATA49**, storing up to 25 bivariate (x_k, y_k) pairs) accumulated via [Σ+] for residual and regression auditing (**DATA** tab in [↖] [REGS]).

C.2.4 Dynamic LFU Softkey Menus Directory

The six top keys ([LFU0] through [LFU5]) automatically remap depending on the active menu state:

Menu	LFU0	LFU1	LFU2	LFU3	LFU4	LFU5
DISP	FIX n	SCI n	ENG n	ALL	(None)	(None)
MODES	DEG	RAD	GRAD	.	,	(None)
BASE	HEX	DEC	OCT	BIN	(None)	(None)
CLEAR	CLEQN	CLVARS	CLALL	CLSTAT (CLΣ)	CLREGS	CLSTK
FLAGS	SF n	CF n	FS? n	STACK	EXAM	(More →)
STACK	2-LVL	4-LVL	8-LVL	INF	VIEW	(None)
PARTS	IP	FP	ABS	SGN	GCD	LCM
PROB	C n, r	P n, r	n! (x!)	RAND	SD	(None)
SUMS	Σx	Σy	Σx²	Σy²	Σxy	n
$\bar{x}$,$\bar{y}$	$\bar{x}$	$\bar{y}$	$\bar{x}$w	(None)	(None)	(None)
s,σ	sx	sy	σx	σy	(None)	(None)
L.R.	$\hat{y}$	$\hat{x}$	r	m	b	(None)
x?y	x=y	x≠y	x>y	x<y	x≥y	x≤y
x?0	x=0	x≠0	x>0	x<0	x≥0	x≤0
TVM	N	I%	PV	PMT	FV	END/BEG
POLY	A	B	C	D	EXEC	CLR
REGS	STACK	STO	STATS	DATA	▲	▼

C.3 Open-Source Hardware & Firmware Architecture

Live schematics, Gerber fabrication files, and firmware sources are maintained at **stackcalc.io/hardware** and in our public engineering build logs on **Hackaday**.

C.3.1 Deterministic State-Machine Execution

The SC-32 firmware is structured around a pure, side-effect-free transition function $\delta(S, k) \rightarrow S'$ that separates mathematical state execution from display rendering and matrix scanning. Rather than dispersing stack-lift rules, edit-buffer termination, and error trapping across user-interface handlers, the entire calculator state—comprising the active stack registers, **LASTX**, the 26 named variables (**A**–**Z**), indirect index register **i**, statistical accumulators, **TVM** financial registers, and the Stack Lift flag—resides in a single deterministic value structure.

Maintaining behavioral parity between the physical microcontroller firmware and the desktop/web hardware simulator requires strict adherence to edge-case invariants: how [ENTER] disarms stack lift, how an in-progress digit entry interacts with [x⇄y], how [↶] [LASTx] captures pre-operation operands differently across unary and dyadic operators, and how pre-mutation domain guards trap tan(90°) or ln(0) without corrupting upper registers. Every physical row-and-column switch closure on the handheld PCB steps the formal four-register automaton defined in Appendix A before updating the 132 × 65 display buffer.

C.3.2 Mathematical Rigor in Silicon: Lanczos, Brent, and Romberg

Every transcendental function, statistical estimator, root solver, and definite integral on the SC-32 is implemented using established algorithms from numerical analysis:

- **Continuous Factorials via Cornelius Lanczos's Gamma Approximation:** Rather than limiting [↶] [x!] to a lookup table of whole numbers, the engine evaluates $\Gamma(x+1)$ across the real line using Cornelius Lanczos's Chebyshev rational approximation paired with Euler's reflection formula $\Gamma(z)\Gamma(1\text{-}z) = \pi \,/\, \sin(\pi\, z)$, maintaining 15-digit internal precision from fractional half-integers up to 170! ≈ 7.2574×10^{306}.
- **Guaranteed Root-Finding via Richard Brent's Hybrid Algorithm:** Pure Newton–Raphson iteration can diverge when a curve flattens near a local extremum, while pure bisection converges at one binary bit per step. The [↷] [SOLVE] engine implements Richard Brent's 1973 hybrid method, which uses inverse quadratic and secant interpolation whenever the iterate converges rapidly and falls back to bisection whenever an iterate would step outside the bracket $[a, b]$ or stall.
- **Adaptive Calculus via Werner Romberg's Richardson Tableau:** When you press [↷] [∫], the calculator constructs a triangular Romberg tableau, halving the trapezoidal step size and applying Lewis Fry Richardson's $(4^m\ T_{k,m\text{-}1} - T_{k\text{-}1,m\text{-}1}) \,/\, (4^m - 1)$ error-cancellation cascade to eliminate successive powers of h^2, h^4, h^6 until the integral converges to machine tolerance.

Each of these numerical engines—alongside William Kahan's compensated summation and base-preserving percentages, Gaussian complex phasor arithmetic, and 36-bit two's-complement word operations—is verified against reference mathematical test vectors across the full domain.

C.3.3 Four-Layer Gold-Immersion Hardware, Daylight Optics & Tactile Key-Feel

The physical hardware integrates the PCB switch matrix, reflective optics, and mechanical enclosure as a unified assembly:

- **Four-Layer ENIG Circuit Architecture & Continuous Memory:** The custom four-layer printed circuit board uses an electroless nickel immersion gold (ENIG) surface finish across its switch matrix pads for corrosion-free contact longevity over hundreds of thousands of actuations. Between calculations, the firmware writes atomic, FNV-1a checksum-verified state snapshots—preserving all four stack registers (T, Z, Y, X), **`LASTX`**, the 26 letter registers (**`A`**–**`Z`**), all six statistical accumulators, TVM financial registers, and the 24-slot Equation Bank—into dual alternating 32 KiB non-volatile flash slots.
- **Reflective Daylight Readability & Tactile Dome-Switch Mechanics:** Rather than an emissive smartphone screen that washes out in direct sunlight, the SC-32 pairs a 132 × 65 transflective liquid-crystal display with a laser-micro-foamed thermoplastic faceplate and a unified TPU 95A keypad membrane over 43 discrete tactile dome switches. Ambient light passes through the liquid-crystal layer and reflects off the internal metallic reflector, increasing contrast as ambient illumination rises, while the 43 tactile dome switches beneath the sculpted keycaps provide crisp mechanical feedback on each actuation.
- **Simulating Silicon & Board-Aware Parametric CAD:** During board fabrication, a browser-based hardware simulator wired to a cycle-accurate microcontroller emulator verified matrix debouncing, sleep wake-up timing, and pixel typography before physical bring-up. As documented in the Hackaday engineering logs, the parametric OpenSCAD enclosure scripts read mounting and switch coordinates directly from the KiCad layout—separating mechanical alignment, capture, stop, and release geometries so the tapered chassis, faceplate, TPU keypad membrane, and snap-retained cap assemble without screws or adhesive. Complete KiCad schematics, parametric CAD sources, printable **`.stl`**/**`.3mf`** files, and firmware build logs are published at **stackcalc.io/downloads** for students, educators, and engineers.

Appendix D: Mathematical Constants, SI Units & STEM Equations

D.1 Mathematical Constants, BIPM SI Prefixes & Exact Unit Conversions

One hundred core equations across classical mechanics, electromagnetism, thermodynamics, optics, chemistry, astronomy, and finance pair each formula with its physical constants and step-by-step RPN stack execution on the StackCalc SC-32.

D.1.1 SC-32 Scientific Register Hierarchy & Data Flow

Register / Bank	Architectural Level	Operational Role & Data Flow
Register T	Top / Constant Buffer	Retains constant factor and replicates on dyadic stack drop
Register Z	Third Stack Level	Holds intermediate sub-expressions and vector/matrix terms
Register Y	Second Stack Level	Stores waiting dyadic base, dividend, or coordinate component
Register X	Active / Display	Direct keyboard entry and live computed result readout
LASTX	Recovery Buffer	Automatically captures previous X before arithmetic/transcendentals
STATS Bank	6 Accumulators	Hardware registers $\sum x$, $\sum y$, $\sum x^2$, $\sum y^2$, $\sum xy$, n
Memory A–Z	26 Global Registers	Persistent variable storage via STO and RCL

D.1.2 Scientific Notation, Register Telemetry, and Inside-Out Evaluation

Physical constants spanning dozens of orders of magnitude (such as Planck's constant $6.62607015 \times 10^{-34}$ J·s) are entered directly via the exponent key (**6.62607015 E 34 +/-**). As multi-step equations execute, intermediate physical quantities (such as angular frequency ω in Register Y and capacitance C in Register X) remain continuously visible on the 132 × 65 transflective LCD, so deeply nested blackbody radiation, thermodynamic, and relativistic formulas collapse inside-out without parentheses or scratchpad registers.

D.1.3 Fundamental Mathematical & Exact 2019 SI Physical Constants

Under the May 20, 2019 SI Redefinition (26th CGPM), seven defining constants of nature are fixed as exact numerical invariants with zero measurement uncertainty: the vacuum speed of light (c = 299,792,458 m/s), Planck's constant ($h = 6.62607015 \times 10^{-34}$ J·s), elementary charge ($e = 1.602176634 \times 10^{-19}$ C), Boltzmann's constant ($k_B = 1.380649 \times 10^{-23}$ J/K), Avogadro's constant ($N_A = 6.02214076 \times 10^{23}$ mol^{-1}), the molar gas constant ($R = N_A\, k_B = 8.314462618$ J/(mol·K)), and the Stefan–Boltzmann constant ($\sigma = 2\pi^5\, {k_B}^4 / 15\, c^2\, h^3 = 5.670374419 \times 10^{-8}$ W/(m^2·K^4)).

Alongside these exact invariants stand the CODATA 2022 experimentally measured constants—the Newtonian gravitational constant ($G = 6.67430 \times 10^{-11}$ N·m^2/kg^2), vacuum permittivity ($\varepsilon_0 = 8.8541878188 \times 10^{-12}$ F/m), and vacuum permeability ($\mu_0 = 1.25663706127 \times 10^{-6}$ N/A^2)—and the fundamental mathematical constants with their SC-32 RPN generation sequences:

Constant Name	Symbol	Exact Value (15-Digit Reference)	SC-32 RPN Entry Sequence
Archimedes' Constant	π	**3.14159265358979**	[↗] [π]
Euler's Number (Natural Base)	e	**2.71828182845905**	**1** [e^x]
Golden Ratio ((1+√5) / 2)	φ	**1.61803398874989**	**5** [$\sqrt{x}$] **1** [+] **2** [÷]
Euler–Mascheroni Constant	γ	**0.57721566490153**	**0.57721566490153**
Natural Logarithm of 2	ln 2	**0.69314718055995**	**2** [LN]
Natural Logarithm of 10	ln 10	**2.30258509299405**	**10** [LN]
Common Logarithm of e	$\log_{10} e$	**0.43429448190325**	**1** [e^x] [↶] [LOG]
Pythagorean Constant	√2	**1.41421356237310**	**2** [$\sqrt{x}$]
Theodorus' Constant	√3	**1.73205080756888**	**3** [$\sqrt{x}$]
Diagonal of Unit Golden Rectangle	√5	**2.23606797749979**	**5** [$\sqrt{x}$]
Radians-to-Degrees Factor (180/π)	rad→deg	**57.2957795130823**	**1** [↶] [▸DEG]
Degrees-to-Radians Factor (π/180)	deg→rad	**0.01745329251994**	**1** [↗] [▸RAD]

D.1.4 International Standard Unit Conversions & BIPM SI Prefixes

The International Bureau of Weights and Measures (BIPM) defines the complete set of SI prefixes—including the **ronna** (10^{27}), **quetta** (10^{30}), **ronto** (10^{-27}), and **quecto** (10^{-30}) prefixes adopted at the 27th CGPM—together with their direct [E] exponent keystrokes:

Macro Prefix	Symbol	Factor	SC-32 Keystroke	Micro Prefix	Symbol	Factor	SC-32 Keystroke
quetta	**Q**	10^{30}	**1** [E] **30**	deci	**d**	10^{-1}	**1** [E] **1** [+/−]
ronna	**R**	10^{27}	**1** [E] **27**	centi	**c**	10^{-2}	**1** [E] **2** [+/−]
yotta	**Y**	10^{24}	**1** [E] **24**	milli	**m**	10^{-3}	**1** [E] **3** [+/−]
zetta	**Z**	10^{21}	**1** [E] **21**	micro	μ	10^{-6}	**1** [E] **6** [+/−]
exa	**E**	10^{18}	**1** [E] **18**	nano	**n**	10^{-9}	**1** [E] **9** [+/−]
peta	**P**	10^{15}	**1** [E] **15**	pico	**p**	10^{-12}	**1** [E] **12** [+/−]
tera	**T**	10^{12}	**1** [E] **12**	femto	**f**	10^{-15}	**1** [E] **15** [+/−]
giga	**G**	10^{9}	**1** [E] **9**	atto	**a**	10^{-18}	**1** [E] **18** [+/−]
mega	**M**	10^{6}	**1** [E] **6**	zepto	**z**	10^{-21}	**1** [E] **21** [+/−]
kilo	**k**	10^{3}	**1** [E] **3**	yocto	**y**	10^{-24}	**1** [E] **24** [+/−]
hecto	**h**	10^{2}	**1** [E] **2**	ronto	**r**	10^{-27}	**1** [E] **27** [+/−]
deka	**da**	10^{1}	**1** [E] **1**	quecto	**q**	10^{-30}	**1** [E] **30** [+/−]

D.1.5 Exact International Unit Definitions & Hardware Conversion Keys

Physical Quantity	From Unit	To SI Unit	Exact Defining Factor / Formula	Dedicated SC-32 Key
Length	Inch (**in**)	Centimeter (**cm**) / Meter (**m**)	1 in ≡ 2.54 cm ≡ 0.0254 m (exact)	↶ ▶CM / ↷ ▶in
Length	Foot (**ft**)	Meter (**m**)	1 ft ≡ 0.3048 m (exact)	**0.3048** ×
Length	Yard (**yd**)	Meter (**m**)	1 yd ≡ 0.9144 m (exact)	**0.9144** ×
Length	Statute Mile (**mi**)	Kilometer (**km**)	1 mi ≡ 1.609344 km (exact)	↶ ▶km / ↷ ▶mi
Length	Nautical Mile (**nmi**)	Meter (**m**)	1 nmi ≡ 1852 m (exact)	**1852** ×
Mass	Avoirdupois Pound (**lb**)	Kilogram (**kg**)	1 lb ≡ 0.45359237 kg (exact)	↶ ▶kg / ↷ ▶lb
Volume	U.S. Liquid Gallon (**gal**)	Liter (**L**)	1 gal ≡ 3.785411784 L (exact, 231 in^3)	↶ ▶l / ↷ ▶gal
Pressure	Standard Atmosphere (**atm**)	Pascal (**Pa**)	1 atm ≡ 101,325 Pa (exact)	**101325** ×
Pressure	Bar (**bar**)	Pascal (**Pa**)	1 bar ≡ 100,000 Pa (exact)	**1** E **5** ×
Energy	Thermochemical Calorie (**cal**)	Joule (**J**)	1 cal_{th} ≡ 4.184 J (exact)	**4.184** ×
Energy	Kilowatt-hour (**kWh**)	Joule (**J**)	1 kWh ≡ 3.6 × 10^6 J (exact)	**3.6** E **6** ×
Temperature	Fahrenheit (**°F**)	Celsius (**°C**)	$T_C = (T_F - 32) \times 5/9$ (exact)	↶ ▶°C / ↷ ▶°F
Temperature	Celsius (**°C**)	Kelvin (**K**)	$T_K = T_C + 273.15$ (exact)	**273.15** +
Time / Angle	Sexagesimal (**H.MMSS**)	Decimal (**HR** / **DEG**)	$H + M/60 + S/3600$	↶ ▶HR / ↷ ▶HMS

D.2 Fundamental Equations of Science and Engineering

D.2.1 Classical Mechanics, Orbital Dynamics & Statics

Kinematic Velocity-Displacement Relation

$$v_f = \sqrt{v_i^2 + 2a\Delta x} \qquad \text{(Eq. D.1)}$$

Derived by eliminating elapsed time t between the velocity-time and position-time equations, this work-energy identity equates the specific work $a\ \Delta x$ performed by a constant acceleration a (m/s^2) over a displacement Δx (m) to the gain in kinetic energy per unit mass $1/2(v_f^2 - v_i^2)$ from initial speed v_i (m/s). Because v_f^2 grows linearly with stopping or takeoff distance Δx, doubling a vehicle's initial velocity requires four times the braking distance under fixed deceleration.

On the SC-32 stack, squaring v_i in place first and allowing automatic stack lift to hold v_i^2 in Register Y while forming $2a\Delta x$ in Register X (**v_i** ↶ x² **2** ENTER **a** × **Δx** × + √x) evaluates the benchmark takeoff roll (v_i = 12.0 m/s, a = 4.5 m/s^2, Δx = 25.0 m, where v_i^2 = 144.0 m^2/s^2 and $2a\Delta x$ = 225.0 m^2/s^2) to **19.209373** m/s.

Projectile Motion Maximum Range

$$R = \frac{v_0^2 \sin(2\theta)}{g} \qquad \textbf{(Eq. D.2)}$$

For a projectile launched over level ground in a vacuum with muzzle speed v_0 (m/s) and elevation angle ϑ (degrees) under standard gravity g = 9.80665 m/s², the horizontal range multiplies the vertical flight time $2v_0 \sin \vartheta \,/\, g$ by the constant horizontal ground speed $v_0 \cos \vartheta$. Collapsing 2 sin ϑ cos ϑ through the trigonometric double-angle identity sin(2ϑ) proves both that maximum range occurs at ϑ = 45° (where sin 90° = 1) and that complementary launch angles ϑ and 90° - ϑ strike the exact same downrange point. In [DEG] mode, squaring v_0 first so it waits in Register Y while doubling and taking the sine of ϑ (**v₀** [↶] [x²] **θ** [ENTER] **2** [×] [SIN] [×] **9.80665** [÷]) resolves v_0 = 45.0 m/s and ϑ = 35.0° (v_0^2 sin 70° = 1902.8776 m²/s²) to **194.0395** m.

Centripetal Acceleration & Radial Force

$$F_c = \frac{mv^2}{r} \qquad \textbf{(Eq. D.3)}$$

Christian Huygens proved in *Horologium Oscillatorium* (1673) that deflecting a body of mass m (kg) along a circular arc of radius r (m) at constant tangential speed v (m/s) requires a continuous inward radial acceleration $a_c = v^2 \,/\, r$ perpendicular to the velocity vector. Because lateral force scales quadratically with speed v^2 and inversely with turn radius r, doubling cornering velocity quadruples tire lateral shear or turbine blade root tension, whereas doubling the turn radius halves the load. Squaring v immediately in Register X before multiplying by mass and dividing by radius (**v** [↶] [x²] **m** [×] **r** [÷]) resolves a cornering vehicle (m = 1250 kg, v = 24.0 m/s, r = 60.0 m) via **24** [↶] [x²] **1250** [×] **60** [÷] to **12000.0** N (12.0 kN).

Newton's Law of Universal Gravitation

$$F_g = G\frac{m_1 m_2}{r^2} \qquad \textbf{(Eq. D.4)}$$

Isaac Newton's *Principia* (1687) unified terrestrial weight and celestial mechanics by demonstrating that any two spherically symmetric masses m_1 and m_2 (kg) separated by center-to-center distance r (m) attract each other with a force proportional to the universal gravitational constant G = 6.67430 × 10^{-11} N·m²/kg². The inverse-square $1/r^2$ geometric dilution reflects conservation of gravitational flux across concentric spherical shells of surface area $4\pi\, r^2$ in three-dimensional space. Entering the numerator product $G\, m_1\, m_2$ via the scientific exponent key [E] and dividing by r^2 (**6.67430** [E] **11** [+/-] [ENTER] **m₁** [×] **m₂** [×] **r** [↶] [x²] [÷]) for an m_2 = 1500 kg spacecraft orbiting Earth (m_1 = 5.9722 × 10^{24} kg) at r = 6.771 × 10^6 m yields an orbital attraction of **13041.44** N.

Circular Orbital Velocity

$$v = \sqrt{\frac{\mu}{r}} \qquad \textbf{(Eq. D.5)}$$

Equating the inward gravitational pull μ / r^2—where $\mu = GM_{\oplus} = 3.986004418 \times 10^{14}$ m^3/s^2 is Earth's standard geocentric gravitational parameter—to the centripetal acceleration v^2 / r required to maintain a circular orbit of radius r (m) yields the local circular orbital speed. Because r appears in the denominator under the radical, orbital mechanics exhibits the paradoxical property that spacecraft in higher-altitude orbits travel at slower linear velocities than those in low Earth orbit. Dividing μ by r and extracting the square root in place (**3.986004418** [E] **14** [ENTER] **r** [÷] [√x]) for a 400 km Low Earth Orbit ($r = 6.771 \times 10^6$ m) returns **7672.60** m/s.

Planetary Escape Velocity

$$v_{esc} = \sqrt{\frac{2\mu}{r}} = v_{orbital} \times \sqrt{2} \qquad \textbf{(Eq. D.6)}$$

A spacecraft escapes a central gravitational field of parameter μ (m^3/s^2) from radial distance r (m) onto a zero-total-energy parabolic trajectory when its specific kinetic energy $(1/2)v_{esc}^2$ exactly balances the local gravitational potential well depth μ / r. Regardless of the planet's mass or radius, the local escape speed is always exactly $\sqrt{2} \approx 1.41421$ times the circular orbital velocity at that same altitude. Evaluating the quotient under the radical directly on the stack (**2** [ENTER] μ [×] **r** [÷] [√x]) from Earth's mean surface radius ($r = 6.371 \times 10^6$ m, $\mu = 3.986004418 \times 10^{14}$ m^3/s^2) yields **11186.1** m/s (11.19 km/s).

Tsiolkovsky Rocket Burnout Equation

$$\Delta v = v_e \ln\left(\frac{m_0}{m_f}\right) \qquad \textbf{(Eq. D.7)}$$

Konstantin Tsiolkovsky's 1903 differential momentum integration governs a rocket expelling propellant at effective exhaust velocity $v_e = I_{sp}\, g_0$ (m/s) as vehicle mass decreases from initial wet mass m_0 to final dry burnout mass m_f. Because ideal velocity change Δv grows only with the natural logarithm of the mass ratio m_0 / m_f, achieving orbital or interplanetary velocities requires either high-specific-impulse propulsion (v_e) or staging to jettison empty structural mass along the ascent. Forming the mass ratio in Register X, taking [LN], and scaling by v_e (**m0** [ENTER] **mf** [÷] [LN] **ve** [×]) for $v_e = 3100$ m/s, $m_0 = 45000$ kg, $m_f = 6500$ kg yields **5998.07** m/s.

Rotational Kinetic Energy of Rigid Body

$$E_{rot} = \tfrac{1}{2} I \omega^2 \qquad \textbf{(Eq. D.8)}$$

Summing the translational kinetic energy $(1/2)\ dm\ (\omega\ r)^2$ of every differential mass element across a rigid rotor replaces linear mass m with the mass moment of inertia $I = \int r^2\ dm$ (kg·m^2) and linear speed v with angular velocity ω (rad/s). Because geometric mass distribution scales I with the square of radial distance r^2 while spin rate enters as ω^2, energy-storage flywheels and satellite reaction wheels concentrate rim mass at the outer periphery and spin at high angular speeds. Squaring ω first on Register X and multiplying by $I/2$ (ω [↰] [x²] **I** [×] **2** [÷]) for a rotor with $I = 4.25$ kg·m^2 at $\omega = 150.0$ rad/s gives **47812.5** J (47.81 kJ).

Simple Gravity Pendulum Period

$$T = 2\pi\sqrt{\frac{L}{g}} \qquad \textbf{(Eq. D.9)}$$

Under the small-angle linearization $\sin\vartheta \approx \vartheta$, the restoring gravitational torque on a simple pendulum of pivot-to-bob length L (m) in local gravity $g = 9.80665$ m/s^2 forms a harmonic oscillator of angular frequency $\omega_0 = \sqrt{(g/L)}$. As Galileo observed and Christiaan Huygens formalized, the bob mass cancels out of both the gravitational restoring force and the inertial acceleration, so the round-trip period T depends solely on L and g. Computing the inner ratio L/g, taking its square root, and scaling by 2π (**L** [ENTER] **9.80665** [÷] [√x] **2** [×] [↗] [π] [×]) for a Huygens seconds pendulum ($L = 0.99362$ m, beating 1 s per half-swing) confirms $T =$ **2.00000** s.

Euler Critical Buckling Load for Slender Column

$$P_{\mathrm{cr}} = \frac{\pi^2 E I}{K^2 L^2} \qquad \textbf{(Eq. D.10)}$$

Leonhard Euler's 1744 eigenvalue analysis of the elastic elastica shows that a slender compression strut of Young's modulus E (Pa), minimum weak-axis area moment of inertia I (m^4), unsupported length L (m), and effective end-restraint factor K (where $K = 1.0$ for pinned-pinned ends and $K = 0.5$ for fixed-fixed ends) bifurcates into lateral bending long before reaching the material's compressive yield stress. Because the critical load P_{cr} depends solely on flexural rigidity EI and drops inversely with the square of effective length $(KL)^2$, doubling a column's unbraced span cuts its buckling capacity fourfold.

Forming the numerator $\pi^2 E I$ in Register Y and dividing by $(KL)^2$ ([↗] [π] [↖] [x²] **E** [×] **I** [×] **K** [ENTER] **L** [×] [↖] [x²] [÷]) for a steel column ($E = 200 \times 10^9$ Pa, $I = 6.4 \times 10^{-6}$ m^4, $K = 1.0$, $L = 4.5$ m) yields **623856.5** N (623.86 kN).

D.2.2 Fluid Mechanics, Hydraulics & Aerodynamics

Torricelli Efflux Jet Velocity

$$v = \sqrt{2gh} \qquad \textbf{(Eq. D.11)}$$

Evangelista Torricelli demonstrated in 1643—a result later recognized as a direct application of Daniel Bernoulli's streamline energy conservation between a free reservoir surface and an open atmospheric discharge jet—that liquid under hydrostatic head h (m) exits a sharp orifice at the exact speed acquired by a body falling freely from height h under gravity $g = 9.80665$ m/s^2. Because efflux velocity grows only with the square root of head $\sqrt{h}$, quadrupling the liquid depth in a gravity-fed tank is required to double the nozzle discharge speed. Multiplying $2g$ by head h and extracting the square root in place (**2** [ENTER] **9.80665** [×] **h** [×] [√x]) for a tank head of $h = 5.0$ m ($2gh = 98.0665$ m^2/s^2) yields **9.90285** m/s.

Volumetric Flow Continuity Equation

$$Q = A_1 v_1 = \frac{\pi D_1^2}{4} v_1 \qquad \textbf{(Eq. D.12)}$$

For an incompressible fluid moving through a circular conduit of internal diameter D_1 (m) at mean cross-sectional velocity v_1 (m/s), conservation of mass requires the volumetric flow rate Q (m³/s) to remain invariant across every cross-section ($A_1\ v_1 = A_2\ v_2$). Because circular cross-sectional area $A_1 = (\pi/4)D_1^2$ scales with the square of diameter, tapering a nozzle to half its upstream diameter reduces the flow area fourfold and forces the fluid velocity to quadruple. Squaring D_1 first on Register X and scaling by $(\pi/4)\ v_1$ (**D** [↶] [x²] [↷] [π] [×] **4** [÷] **v** [×]) for a $D_1 = 0.15$ m supply main flowing at $v_1 = 2.4$ m/s yields **0.0424115** m³/s (42.41 L/s).

Reynolds Number for Pipe Flow

$$Re = \frac{\rho v D}{\mu} \qquad \textbf{(Eq. D.13)}$$

Introduced by George Gabriel Stokes and popularized by Osborne Reynolds's 1883 dye-filament pipe experiments, the dimensionless Reynolds number measures the ratio of convective inertial momentum ($\rho\ v^2\ /\ D$) to Newtonian viscous shear stress ($\mu\ v\ /\ D^2$) for a fluid of density ρ (kg/m³) and dynamic viscosity μ (Pa·s) flowing at speed v (m/s) through hydraulic diameter D (m). Internal pipe flow remains smooth and laminar below Re ≈ 2300, passes through an intermittent transitional regime, and becomes fully turbulent with chaotic eddy mixing above Re ≈ 4000. Chaining the inertial numerator and dividing by viscosity (ρ [ENTER] **v** [×] **D** [×] μ [÷]) for water at 20°C ($\rho = 998$ kg/m³, $v = 1.8$ m/s, $D = 0.05$ m, $\mu = 0.001002$ Pa·s) gives Re = **89640.7** (fully turbulent).

Darcy-Weisbach Friction Head Loss

$$h_f = f \frac{L}{D} \frac{v^2}{2g} \qquad \textbf{(Eq. D.14)}$$

Formulated by Henry Darcy and Julius Weisbach, this foundational hydraulic loss equation expresses the irreversible pressure head h_f (m of fluid column) dissipated by wall shear across a pipe of length L (m) and diameter D (m) carrying flow at velocity v (m/s) under gravity $g = 9.80665$ m/s², where f is the dimensionless Darcy friction factor from the Colebrook-White or Moody chart. Because head loss scales with the aspect ratio L/D and the velocity head $v^2\ /\ (2g)$, doubling volumetric throughput in a fixed conduit quadruples frictional pumping head. Evaluating the dimensionless geometric factor $f\ L\ /\ D$ first and multiplying by $v^2\ /\ (2g)$ (**f** [ENTER] **L** [×] **D** [÷] **v** [↶] [x²] [×] **2** [÷] **9.80665** [÷]) for $f = 0.024$, $L = 120.0$ m, $D = 0.20$ m, $v = 2.5$ m/s yields $h_f =$ **4.5887** m.

Manning Formula for Open Channel Uniform Flow

$$v = \frac{1}{n} R_h^{2/3} S^{1/2} \qquad \textbf{(Eq. D.15)}$$

Irish engineer Robert Manning's 1889 empirical correlation governs steady, uniform, free-surface gravity flow in aqueducts, storm culverts, and natural riverbeds, balancing the downslope gravitational component $S = \Delta z \,/\, L$ against boundary roughness n across a channel of hydraulic radius $R_h = A \,/\, P_{\text{wet}}$ (m, the cross-sectional flow area divided by wetted perimeter, with SI unit factor $k_n = 1.0$ $\text{m}^{1/3}/\text{s}$ or U.S. customary factor 1.4859 $\text{ft}^{1/3}/\text{s}$). Deep, compact channels with a large hydraulic radius R_h minimize wall drag per unit flow area, accelerating mean velocity as $R_h^{2/3}$. Raising R_h to the 2/3 power via [y^x], multiplying by $\sqrt{S}$, and dividing by Gauckler–Manning roughness n (**R_h** [ENTER] **2** [ENTER] **3** [÷] [y^x] **S** [$\sqrt{x}$] [×] **n** [÷]) for a concrete-lined canal ($n = 0.015$, $R_h = 0.85$ m, $S = 0.0025$) produces **2.9911** m/s.

Poiseuille Laminar Capillary Flow Rate

$$Q = \frac{\pi r^4 \Delta P}{8 \mu L} \qquad \textbf{(Eq. D.16)}$$

Derived independently by Jean Léonard Marie Poiseuille and Gotthilf Hagen from the exact parabolic velocity profile $u(r)$ of steady laminar flow (Re $<$ 2300), this law relates volumetric throughput Q (m^3/s) to pressure drop ΔP (Pa) across a cylindrical capillary tube of inner radius r (m) and length L (m) filled with fluid of viscosity μ (Pa·s). Because both the cross-sectional area ($\propto r^2$) and the peak centerline velocity ($\propto r^2$) scale quadratically with radius, laminar conductance varies with the fourth power r^4—meaning a 16% constriction in a microfluidic channel or coronary artery cuts volumetric flow in half.

Raising r to the fourth power first and chaining the remaining factors (**r** [ENTER] **4** [y^x] [↻] [π] [×] **ΔP** [×] **8** [÷] **μ** [÷] **L** [÷]) for $r = 0.002$ m, $\Delta P = 5000$ Pa, $\mu = 0.0035$ Pa·s, $L = 0.50$ m yields 1.7952×10^{-5} m^3/s.

Aerodynamic Dynamic Pressure

$$q = \frac{1}{2} \rho v^2 \qquad \textbf{(Eq. D.17)}$$

In subsonic aerodynamics, dynamic pressure q (Pa) represents the kinetic energy per unit volume of an airstream of density ρ (kg/m^3) moving at true airspeed v (m/s), measured physically as the differential pressure between a forward-facing Pitot stagnation port and a flush static pressure port. Aerodynamic lift $L = q\,S\,C_L$ and parasitic drag $D = q\,S\,C_D$ scale directly with q, explaining why aircraft flying in thin high-altitude air ($\rho \ll \rho_0$) must maintain a much higher true airspeed v to preserve the same indicated dynamic pressure and stall margin. Squaring airspeed v in Register X and multiplying by ρ / 2 (**v** [↰] [x^2] **ρ** [×] **2** [÷]) at standard sea level ($\rho = 1.225$ kg/m^3, $v = 65.0$ m/s) returns **2587.81** Pa.

Compressible Flow Mach Number

$$M = \frac{v}{\sqrt{\gamma R_{\text{specific}} T}} \quad \textbf{(Eq. D.18)}$$

Named after Austrian physicist Ernst Mach, the Mach number M normalizes true flight speed v (m/s) by the local speed of sound $a = \sqrt{(\gamma\ R_{\text{specific}}\ T)}$ in an ideal gas of adiabatic index $\gamma = 1.40$, specific gas constant $R_{\text{specific}} = 287.058$ J/(kg·K) for dry air, and absolute static temperature T (K). Because acoustic wave propagation speed depends exclusively on the square root of absolute temperature $\sqrt{T}$, the speed of sound drops from 340.3 m/s at sea level (288.15 K) to 313.8 m/s in the cold stratosphere (245.0 K), bringing cruising airliners near the transonic compressibility threshold ($M \approx 0.8$).

Computing the acoustic denominator $\sqrt{(\gamma\ R\ T)}$ first in Register X, entering v, and swapping via [x⇄y] (**1.4** [ENTER] **287.058** [×] **T** [×] [√x] **v** [x⇄y] [÷]) for $v = 240.0$ m/s at $T = 245.0$ K yields Mach **0.7649**.

D.2.3 Thermodynamics, Gas Laws & Heat Transfer

Ideal Gas State Equation

$$P = \frac{nRT}{V} \quad \textbf{(Eq. D.19)}$$

By synthesizing Robert Boyle's isothermal compression law, Jacques Charles's isobaric expansion law, and Amedeo Avogadro's molar volume hypothesis, Émile Clapeyron's ideal gas equation relates the absolute pressure P (Pa) exerted on a vessel of volume V (m^3) to the molar quantity n (mol), the exact 2019 SI molar gas constant $R = N_A\ k_B = 8.314462618$ J/(mol·K), and the absolute temperature T (K). Microscopically, the numerator nRT measures two-thirds of the total translational kinetic energy of the enclosed gas molecules colliding elastically against the container walls. Chaining the thermal energy product nRT in Register X before dividing by volume V (**n** [ENTER] **8.314462618** [×] **T** [×] **V** [÷]) for $n = 4.5$ mol at $T = 350.0$ K inside $V = 0.035$ m^3 ($nRT = 13{,}095.28$ J) yields **374150.8** Pa (374.15 kPa).

Carnot Thermodynamic Cycle Maximum Efficiency

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \quad \textbf{(Eq. D.20)}$$

Sadi Carnot's 1824 treatise *Réflexions sur la puissance motrice du feu* established that no heat engine operating between a hot thermal reservoir at absolute temperature T_H (K) and a cold rejection sink at T_C (K) can exceed the reversible isothermal-isentropic cycle efficiency η_{Carnot}. Because the Second Law of Thermodynamics mandates that entropy extracted from the hot source ($Q_H\ /\ T_H$) must be discharged into the cold sink ($Q_C\ /\ T_C \geq Q_H\ /\ T_H$), achieving 100% thermal efficiency would require either an infinitely hot heat source or an unachievable absolute-zero sink ($T_C = 0$ K). Computing the absolute temperature ratio $T_C\ /\ T_H$ first and subtracting from unity via [x⇄y] (**T_C** [ENTER] **T_H** [÷] **1** [x⇄y] [−]) for $T_C = 295.0$ K and $T_H = 850.0$ K yields **0.65294** (65.29%).

Stefan-Boltzmann Total Radiative Power

$$P = \sigma \varepsilon A T^4 \qquad \textbf{(Eq. D.21)}$$

Derived by Jozef Stefan (1879) and Ludwig Boltzmann (1884) and later anchored to quantum mechanics by integrating Max Planck's spectral blackbody distribution over all frequencies and solid angles, this law expresses the total electromagnetic thermal power P (W) radiated by a body of surface area A (m^2), gray-body emissivity $\varepsilon \in (0, 1]$, and absolute temperature T (K), where $\sigma = 5.670374419 \times 10^{-8}$ W/($m^2 \cdot K^4$). Because radiative emission scales with the fourth power of absolute temperature (T^4), doubling the temperature of a vacuum furnace filament or deep-space radiator panel multiplies its radiant heat rejection sixteenfold.

Raising T to the fourth power first and multiplying by A, ε, and σ (**T** ENTER **4** y^x **A** × **ε** × **5.670374419** **E** **8** +/- ×) for T = 1200.0 K, A = 1.8 m^2, ε = 0.85 gives **179898.7** W (179.90 kW).

Fourier Conduction Heat Transfer Rate

$$q = \frac{kA(T_1 - T_2)}{L} \qquad \textbf{(Eq. D.22)}$$

Jean-Baptiste Joseph Fourier's 1822 *Théorie analytique de la chaleur* established that steady-state conductive heat flow q (W) across a planar slab of thermal conductivity k (W/(m·K)), cross-sectional area A (m^2), and thickness L (m) is directly proportional to the face-to-face temperature difference T_1 - T_2 (K). Writing the relation in thermal-circuit form $q = (T_1 - T_2) / R_{th}$ with thermal resistance $R_{th} = L / (kA)$ provides an exact mathematical parallel to Ohm's law for multi-layer building envelopes, semiconductor package baseplates, and cryogenic Dewar insulation. Forming the temperature drop T_1 - T_2 first and scaling by k A / L (**T_1** ENTER **T_2** - **k** × **A** × **L** ÷) for a masonry wall (k = 0.75 W/(m·K), A = 15.0 m^2, L = 0.20 m) spanning T_1 = 295.15 K to T_2 = 263.15 K yields **1800.0** W.

Newton's Law of Cooling Transient Temperature

$$T(t) = T_\infty + (T_0 - T_\infty)e^{-kt} \qquad \textbf{(Eq. D.23)}$$

When a thermally conductive body with low internal Biot number (Bi = h L_c / k ≪ 0.1) cools or warms from initial temperature T_0 in a convective bath of ambient temperature T_∞, its temperature excess $T(t)$ - T_∞ decays exponentially at rate $k = hA$ / (m c_p) (s^{-1}) over elapsed time t (s). Because the instantaneous cooling rate dT/dt = -k(T - T_∞) is proportional to the remaining thermal excess, each elapsed time constant $\tau = 1/k$ reduces the temperature gap to e^{-1} ≈ 36.79% of its initial value. Evaluating inside-out from the decay factor e^{-kt} outward (**k** +/- ENTER **t** × e^x **T_0** ENTER **T_∞** - × **T_∞** + , or **k** +/- ENTER **t** × e^x **N_0** × when T_∞ = 0) for T_∞ = 20.0°C, T_0 = 90.0°C, k = 0.005 s^{-1}, t = 180.0 s resolves T(180) = **48.46** °C.

Logarithmic Mean Temperature Difference (LMTD)

$$\mathrm{LMTD} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)} \qquad \textbf{(Eq. D.24)}$$

In a parallel-flow or counterflow heat exchanger where the local fluid-to-fluid temperature approach $\Delta T(x)$ varies from ΔT_1 at the inlet header to ΔT_2 at the outlet header (K or °C), the local heat flux $dq = U\ \Delta T(x)\ dA$ causes $\Delta T(x)$ to decay exponentially along the exchanger length. Integrating that exponential profile yields the logarithmic mean temperature difference LMTD for $q = U\ A \cdot \mathrm{LMTD}$, which is always strictly less than the simple arithmetic average $(\Delta T_1 + \Delta T_2)/2$ whenever $\Delta T_1 \neq \Delta T_2$. Computing the terminal difference ΔT_1 - ΔT_2 in Register Y and dividing by the natural logarithm of the terminal ratio $\ln(\Delta T_1 / \Delta T_2)$ (**ΔT₁** ENTER **ΔT₂** - **ΔT₁** ENTER **ΔT₂** ÷ LN ÷) for $\Delta T_1 = 45.0$ K and $\Delta T_2 = 18.0$ K gives **29.4665** K.

Ideal Brayton Gas Turbine Thermal Efficiency

$$\eta = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}} \qquad \textbf{(Eq. D.25)}$$

Named after George Brayton, the ideal open or closed gas-turbine power cycle compresses intake gas isentropically across pressure ratio $r_p = P_2 / P_1$, adds heat at constant pressure in the combustor, and expands the hot gas isentropically through a power turbine, where $\gamma = c_p / c_v$ is the specific-heat ratio ($\gamma = 1.40$ for diatomic air). Because the isentropic temperature ratio across the compressor equals $T_2 / T_1 = r_p^{(\gamma-1)/\gamma}$, raising the overall compressor pressure ratio r_p directly increases thermal efficiency and explains why modern turbofan cores operate at pressure ratios exceeding 40:1. Forming the exponent $(\gamma - 1)/\gamma$ on the stack, raising r_p via y^x, taking the reciprocal, and subtracting from unity (**rₚ** ENTER **1.4** ENTER **1** - **1.4** ÷ y^x x^{-1} **1** $x \rightleftarrows y$ -) at $r_p = 16.0$ and $\gamma = 1.40$ yields **0.5471** (54.71%).

Latent Heat Phase Transition Energy

$$Q = mL_v \qquad \textbf{(Eq. D.26)}$$

Discovered by Joseph Black in 1761, latent heat Q (J) represents the isothermal enthalpy required to break or reform intermolecular bonds across a first-order phase transition of mass m (kg) with specific latent heat of vaporization (or fusion) L_v (J/kg) while the thermodynamic temperature remains pinned at the boiling or melting point. Water's exceptionally high hydrogen-bond enthalpy of vaporization ($L_v = 2.26 \times 10^6$ J/kg at 100°C—more than five times the sensible heat needed to warm liquid water from 0°C to 100°C) underpins industrial steam Rankine boilers, heat pipes, and evaporative cooling towers. Multiplying mass m by L_v (**m** ENTER **Lᵥ** ×) for $m = 2.5$ kg of boiling water at 100°C yields **5650000.0** J (5.65 MJ).

D.2.4 Electricity, Magnetism & AC Circuit Analysis

Ohm's Law and DC Power Dissipation

$$P = I^2R = \frac{V^2}{R} = VI \qquad \textbf{(Eq. D.27)}$$

Combining Georg Simon Ohm's linear conduction law ($V = IR$) with James Prescott Joule's calorimetric heating law ($P = VI$) expresses the thermal power P (W) dissipated in a resistive load R (Ω) under terminal voltage V (V) and current I (A). Because $P = V^2 / R$ scales quadratically with applied voltage across a fixed resistance (and $P = I^2 R$ scales quadratically with line current along a transmission cable), doubling supply voltage quadruples resistive heating, whereas stepping transmission voltage up tenfold reduces $I^2 R$ grid losses a hundredfold for the same delivered power. Squaring V in Register X and dividing by R (**V** [↰] [x²] **R** [÷]) for a $V = 120.0$ V heating element of resistance $R = 15.0$ Ω yields **960.0** W.

Equivalent Resistance of Parallel Networks via x‖y

$$R_{\text{eq}} = R_1 \| R_2 \| R_3 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \qquad \textbf{(Eq. D.28)}$$

When resistive branches R_1, R_2, R_3 (Ω) share a common terminal node pair, Kirchhoff's current law dictates that their individual conductances $G_i = 1 / R_i$ (S) sum linearly, so the combined equivalent resistance R_{eq} is always strictly smaller than the smallest individual branch resistor. Rather than keying three reciprocals, two additions, and a final reciprocal, the SC-32's dedicated hardware parallel operator [x‖y] evaluates $X \cdot Y / (X + Y)$ in a single keystroke and associatively chains across any number of branches. Keying $\boldsymbol{R_1}$ [ENTER] $\boldsymbol{R_2}$ [x‖y] $\boldsymbol{R_3}$ [x‖y] for $R_1 = 100.0$ Ω, $R_2 = 150.0$ Ω, $R_3 = 300.0$ Ω collapses $100 \parallel 150 = 60.0$ Ω on the first press and $60 \parallel 300 =$ **50.0** Ω on the second.

Electrostatic Energy Stored in Capacitor

$$U_C = \frac{1}{2} C V^2 \qquad \textbf{(Eq. D.29)}$$

Charging a capacitor of capacitance C (F) to terminal voltage V (V) requires moving incremental charge $dq = C\,dv$ against the rising potential v, so integrating $\int_0^V C\,v\,dv$ yields the stored electrostatic field energy U_C (J) with its characteristic factor of $1/2$ (the remaining half of the energy $C\,V^2$ supplied by a fixed-voltage DC source is dissipated as heat in the charging resistance). Because stored energy scales quadratically with voltage V^2, high-voltage pulsed-laser flashlamp banks, medical defibrillators, and EV traction-inverter DC-link capacitors pack substantial energy into compact dielectric volumes. Squaring V first on Register X and multiplying by $C/2$ (**V** [↰] [x²] **C** [×] **2** [÷]) for a $C = 47.0 \times 10^{-6}$ F bus capacitor charged to $V = 450.0$ V gives **4.75875** J.

Magnetic Energy Stored in Inductor

$$U_L = \tfrac{1}{2} L I^2 \qquad \textbf{(Eq. D.30)}$$

As the electromagnetic dual of a capacitor's electric field energy, establishing a current I (A) through an inductor of self-inductance L (H) requires doing work against Faraday's counter-EMF $v_L = L\ di/dt$, integrating $\int_0^I L\ i\ di$ into magnetic flux energy U_L (J) stored within the core air gap and winding field. Because magnetic energy scales with I^2 and cannot change discontinuously without generating a destructive voltage spike ($v_L = L\ di/dt$), buck/boost switching regulators and solenoid drivers require freewheeling diodes or synchronous switches to commutate inductor current safely. Squaring I in place and scaling by $L/2$ (**I** [↶] [x²] **L** [×] **2** [÷]) for an $L = 0.25$ H filter choke carrying $I = 6.4$ A yields **5.12** J.

RC Transient Charging Voltage

$$V_C(t) = V_0(1 - e^{-t/(RC)}) \qquad \textbf{(Eq. D.31)}$$

Applying a step supply voltage V_0 (V) through series resistance R (Ω) into an initially uncharged capacitor C (F) produces a first-order differential relaxation governed by the time constant $\tau = R\ C$ (s). As the capacitor voltage $V_C(t)$ rises, the voltage drop V_0 - $V_C(t)$ driving charging current through R decays exponentially, bringing $V_C(t)$ to 63.21% of V_0 at $t = 1\tau$, 95.02% at $t = 3\tau$, and 99.33% at $t = 5\tau$. Evaluating from the dimensionless exponent $-t\ /\ (RC)$ outward (**t** [+/-] [ENTER] **R** [÷] **C** [÷] [e^x] [+/-] **1** [+] ***V₀*** [×]) for $V_0 = 12.0$ V, $R = 10000.0$ Ω, $C = 220.0 \times 10^{-6}$ F ($\tau = 2.2$ s) at $t = 3.5$ s gives **9.5551** V.

RL Inductive Current Decay

$$I(t) = I_0 e^{-t(R/L)} \qquad \textbf{(Eq. D.32)}$$

When an inductor L (H) carrying steady initial current I_0 (A) is switched across a dissipative loop resistance R (Ω) at $t = 0$, its collapsing magnetic flux sustains current that decays exponentially with inductive time constant $\tau = L\ /\ R$ (s). Unlike an RC circuit where larger resistance slows the transient ($\tau = RC$), increasing loop resistance R in an RL flyback path dissipates stored magnetic energy $(1/2)LI_0^2$ faster and shortens the decay time constant $\tau = L/R$. Forming the negative exponent $-t\ R\ /\ L$ first and scaling by I_0 (**t** [+/-] [ENTER] **R** [×] **L** [÷] [e^x] ***I₀*** [×], or **t** [+/-] [ENTER] **R** [×] **L** [÷] [e^x] [+/-] **1** [+] **V** [×] **R** [÷] for step energization) for $I_0 = 5.0$ A, $R = 24.0$ Ω, $L = 0.80$ H at $t = 0.05$ s yields **1.1157** A.

LC Resonant Frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \qquad \textbf{(Eq. D.33)}$$

William Thomson (Lord Kelvin) derived in 1853 that an undamped tank circuit of inductance L (H) and capacitance C (F) exchanges energy harmonically between its magnetic and electric fields at the natural frequency f_0 (Hz) where inductive reactance $X_L = 2\pi\ f_0\ L$ and capacitive reactance $X_C = 1\ /\ (2\pi\ f_0\ C)$ are equal in magnitude and opposite in phase. Because f_0 varies inversely with $\sqrt{LC}$, sweeping a variable tuning capacitor across a four-to-one capacitance ratio tunes an AM or HF radio receiver across a two-to-one frequency octave. Multiplying L by C, taking [$\sqrt{x}$], scaling by 2π, and inverting via [x^{-1}] (**L** [ENTER] **C** [×] [$\sqrt{x}$] **2** [×] [→] [π] [×] [x^{-1}]) for $L = 10.0 \times 10^{-3}$ H and $C = 100.0 \times 10^{-12}$ F ($\sqrt{LC} = 1.0 \times 10^{-6}$ s) yields **159154.9** Hz (159.15 kHz).

RLC Series AC Total Impedance

$$Z = \sqrt{R^2 + \left(2\pi f L - \frac{1}{2\pi f C}\right)^2} \qquad \textbf{(Eq. D.34)}$$

In steady-state sinusoidal AC analysis at frequency f (Hz), a series branch of resistance R (Ω), inductance L (H), and capacitance C (F) presents complex impedance $\boldsymbol{Z} = R + j(X_L - X_C)$, where inductive reactance $X_L = 2\pi\, f\, L$ leads current by +90° and capacitive reactance $X_C = 1/(2\pi\, f\, C)$ lags by -90°. Because the two reactive components lie 180° apart along the imaginary axis, they partially or completely cancel each other before combining in quadrature with real resistance R. Storing angular frequency $\omega = 2\pi\, f$ in register **A** to evaluate $X_L - X_C$ without re-keying $2\pi\, f$ (**2** [↷] [π] [×] **f** [×] [STO] **A L** [×] [RCL] **A C** [×] [x^{-1}] [−] [↶] [x^2] **R** [↶] [x^2] [+] [$\sqrt{x}$]) for R = 50.0 Ω, L = 0.05 H, C = 10.0 × 10^{-6} F, f = 120.0 Hz gives **107.293** Ω.

Quality Factor of Resonant Circuit

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} \qquad \textbf{(Eq. D.35)}$$

Introduced by K. S. Johnson at Western Electric in 1914, the dimensionless quality factor Q of a series RLC resonator measures 2π times the ratio of peak energy stored in the reactive tank to the energy dissipated per cycle in series resistance R (Ω), expressed directly as the characteristic impedance $\sqrt{(L/C)}$ divided by R. High-Q resonators exhibit both sharp spectral selectivity—with half-power (-3 dB) fractional bandwidth $\Delta f \,/\, f_0 = 1/Q$—and a resonant voltage magnification across L and C equal to Q times the source excitation voltage. Computing the characteristic impedance $\sqrt{(L/C)}$ first and dividing by R (**L** [ENTER] **C** [÷] [$\sqrt{x}$] **R** [÷]) for R = 8.0 Ω, L = 2.5 × 10^{-3} H, C = 47.0 × 10^{-9} F yields Q = **28.829**.

Decibel Power Gain Ratio

$$\mathrm{dB} = 10\log_{10}\left(\frac{P_2}{P_1}\right) \qquad \textbf{(Eq. D.36)}$$

Originated at Bell Telephone Laboratories in the 1920s to quantify signal loss per mile of standard telephone cable, the decibel (dB, one-tenth of a bel) expresses the ratio of output power P_2 (W) to reference input power P_1 (W) on a base-10 logarithmic scale. Mapping power ratios through 10 $\log_{10}(\cdot)$ converts long chains of cascaded RF amplifier gains, mixer conversion losses, and waveguide attenuations from multi-step multiplications into simple algebraic additions (+3.0103 dB per power doubling and +10 dB per decade). Dividing P_2 by P_1, pressing [↶] [LOG], and scaling by 10 (***P*₂** [ENTER] ***P*₁** [÷] [↶] [LOG] **10** [×]) for P_1 = 0.025 W amplified to P_2 = 120.0 W gives **36.8124** dB.

D.2.5 Wave Mechanics, Acoustics & Physical Optics

Snell's Law of Refraction

$$\theta_2 = \arcsin\left(\frac{n_1 \sin\theta_1}{n_2}\right) \quad \textbf{(Eq. D.37)}$$

First discovered by Ibn Sahl in 984 CE and rediscovered by Willebrord Snellius in 1621, Snell's law follows from Pierre de Fermat's principle of least optical time across a boundary separating dielectric media of refractive indices $n_1 = c \,/\, v_1$ and $n_2 = c \,/\, v_2$. Continuity of tangential phase velocity along the interface forces an incident ray at angle ϑ_1 (measured from the surface normal) to bend toward the normal when entering a denser medium ($n_2 > n_1$) and causes total internal reflection when propagating from a denser to a rarer medium at angles where ($n_1 \,/\, n_2$) sin $\vartheta_1 > 1$. In [DEG] mode, taking sin ϑ_1, scaling by the index ratio $n_1 \,/\, n_2$, and applying [↶] [ASIN] (θ_1 [SIN] n_1 [×] n_2 [÷] [↶] [ASIN]) for a beam entering crown glass ($n_2 = 1.52$) from air ($n_1 = 1.0003$) at $\vartheta_1 = 45.0°$ gives **27.7297** °.

Gaussian Thin Lens Equation via x‖y

$$f = \frac{d_o \cdot d_i}{d_o + d_i} = d_o \,\|\, d_i \quad \textbf{(Eq. D.38)}$$

In Carl Friedrich Gauss's paraxial optics approximation, a thin converging lens of focal length f adds optical power $1/f$ to the diverging spherical wavefront curvature $1/d_o$ from an object at distance d_o to focus a real conjugate image at distance d_i according to $1/f = 1/d_o + 1/d_i$. Because the reciprocal sum of conjugate distances is algebraically identical to two parallel resistors, the SC-32's hardware [x‖y] operator solves thin-lens and spherical-mirror imaging in a single keystroke without reciprocals. Entering d_o [ENTER] d_i [x‖y] for an object at $d_o = 30.0$ cm projecting a real image at $d_i = 60.0$ cm directly returns the focal length $f =$ **20.0** cm.

Compound Optical Microscope Magnification

$$M = -\left(\frac{L_{\text{tube}}}{f_{\text{obj}}}\right)\left(\frac{d_{\text{near}}}{f_{\text{eye}}}\right) \quad \textbf{(Eq. D.39)}$$

A standard compound microscope achieves high overall angular magnification M by cascading two positive lens stages separated by the mechanical tube length $L_{\text{tube}} = 160$ mm: a short-focal-length objective lens (f_{obj}, mm) projects a real, inverted intermediate image magnified laterally by $-L_{\text{tube}} \,/\, f_{\text{obj}}$, and an eyepiece ($f_{\text{eye}}$, mm) acts as a simple magnifier referenced to the human eye's relaxed near point $d_{\text{near}} = 250$ mm. Multiplying the objective's lateral magnification by the ocular's angular magnification (**160** [ENTER] f_{obj} [÷] **250** [ENTER] f_{eye} [÷] [×] [+/−]) for an $f_{\text{obj}} = 4.0$ mm (40×) objective paired with an $f_{\text{eye}} = 25.0$ mm (10×) ocular yields **-400.0** (× 400 inverted angular magnification).

Acoustic Doppler Frequency Shift (Moving Source)

$$f' = f_0\left(\frac{v}{v - v_s}\right) \quad \textbf{(Eq. D.40)}$$

Christian Doppler's 1842 principle explains why an acoustic source emitting at rest frequency f_0 (Hz) while advancing toward a stationary observer at speed v_s (m/s) through air of sound speed $v = 343.0$ m/s crowds successive wave crests closer together in space. Because the emitter chases its own outgoing wavefronts, the compressed wavelength ahead of the source shrinks to $\lambda' = (v - v_s)/f_0$, raising the observed pitch $f' = v \,/\, \lambda'$ hyperbolically until wavefronts pile up into a Mach-1 shock cone as $v_s \to v$. Forming the relative velocity $v - v_s$ first, dividing into v via [x⇄y], and scaling by f_0 (**343** [ENTER] v_s [−] **343** [x⇄y] [÷] f_0 [×]) for an $f_0 = 500.0$ Hz train horn approaching at $v_s = 35.0$ m/s returns **556.818** Hz.

Sound Pressure Decibel Level

$$L_p = 20\log_{10}\left(\frac{P_{\text{rms}}}{P_0}\right) \qquad \textbf{(Eq. D.41)}$$

Acoustic Sound Pressure Level (L_p, in dB SPL) compares the root-mean-square pressure fluctuation P_{rms} (Pa) of a sound wave against the internationally standardized threshold of human hearing at 1 kHz, $P_0 = 20.0 \times 10^{-6}$ Pa (20 μPa). Because acoustic power intensity $I = P_{\text{rms}}^2 / (\rho\ c)$ is proportional to the square of pressure amplitude, the 10 $\log_{10}(I / I_0)$ power-ratio definition pulls the exponent of 2 out front to become 20 $\log_{10}(P_{\text{rms}} / P_0)$, where every tenfold increase in pressure corresponds to +20 dB. Dividing P_{rms} by 20 μPa, pressing [↶] [LOG], and multiplying by 20 (***P*rms** [ENTER] **20** [E] **6** [+/−] [÷] [↶] [LOG] **20** [×]) at $P_{\text{rms}} = 0.632$ Pa gives **90.0** dB SPL.

Sabine Reverberation Time for Architectural Enclosure

$$RT_{60} = \frac{0.161V}{A_{\text{total}}} \qquad \textbf{(Eq. D.42)}$$

Pioneering architectural acoustician Wallace Clement Sabine established in 1898 that the reverberation time RT_{60} (s)—the duration required for diffuse reverberant sound energy in a room of volume V (m^3) to decay by 60 dB (one-millionth of its initial intensity) after a source stops—is directly proportional to room volume and inversely proportional to the effective absorption area $A_{\text{total}} = \sum S_i\ \alpha_i$ (in metric sabins, m^2). Physically, a larger room volume increases the mean free path $4V/S$ between wall reflections, whereas acoustic panels of high absorption coefficient α_i drain energy faster at each bounce (with empirical constant 0.161 s/m at 20°C). Multiplying 0.161 by V and dividing by A_{total} (**0.161** [ENTER] **V** [×] ***A*tot** [÷]) for a recital hall of $V = 850.0$ m^3 and $A_{\text{total}} = 110.0$ sabins yields **1.244** s.

Optical Diffraction Grating Wavelength

$$\lambda = \frac{d\sin\theta}{m} \qquad \textbf{(Eq. D.43)}$$

When collimated light illuminates a diffraction grating ruled with periodic slit spacing d (m), each groove acts as a coherent Huygens secondary wavelet emitter across all outgoing angles ϑ (degrees). Adjacent wavelets interfere constructively only at discrete angles where their geometric path-length difference $d \sin \vartheta$ equals an integer multiple $m\lambda$ ($m = \pm 1, \pm 2,$ ...) of the optical wavelength λ (m), dispersing polychromatic light into separated spectral orders. In [DEG] mode, evaluating sin ϑ first, multiplying by grating pitch d, and dividing by order m (θ [SIN] **d** [×] **m** [÷]) for a 600 lines/mm grating ($d = 1.6667 \times 10^{-6}$ m) at first-order angle $\vartheta = 22.45°$ ($m = 1$) determines $\lambda = 6.365 \times 10^{-7}$ m (636.5 nm).

Numerical Aperture of Optical Fiber

$$NA = \sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2} \qquad \textbf{(Eq. D.44)}$$

For a step-index optical fiber with core refractive index n_{core} surrounded by lower-index cladding n_{clad}, Snell's law at the entrance face combined with the critical angle $\sin \vartheta_c = n_{\text{clad}} / n_{\text{core}}$ at the core-cladding wall defines the dimensionless Numerical Aperture $NA = n_0 \sin \vartheta_{\max}$. Any ray entering the fiber end-face within the acceptance cone half-angle $\vartheta_{\max} = \arcsin(NA)$ strikes the internal core wall shallower than the critical angle and propagates down the waveguide via lossless total internal reflection. Squaring both refractive indices, subtracting, and taking the square root (***n*core** [↶] [x²] ***n*clad** [↶] [x²] [-] [√x]) for $n_{\text{core}} = 1.482$ and $n_{\text{clad}} = 1.450$ yields $NA =$ **0.3063** ($\vartheta_{\max} = 17.84°$).

D.2.6 Nuclear, Atomic & Relativistic Physics

Einstein Mass-Energy Equivalence

$$E = mc^2 \quad \textbf{(Eq. D.45)}$$

Albert Einstein's September 1905 paper demonstrated that rest mass m (kg) is a concentrated reservoir of internal energy E (J) scaled by the square of the exact vacuum speed of light c = 299,792,458 m/s, representing the zero-momentum case of the relativistic energy-momentum invariant $E^2 = (pc)^2 + (mc^2)^2$. Because the conversion factor c^2 = $8.98755179 \times 10^{16}$ J/kg is immense, a nuclear fission or fusion reaction that converts even a tiny mass defect Δm of 1.0 gram (1.0×10^{-3} kg) into kinetic and radiative energy releases macroscopic power on the scale of a city's daily electrical consumption. Squaring c in Register X and multiplying by m (**299792458** [↶] [x²] **m** [×]) for $m = 1.0 \times 10^{-3}$ kg yields 8.98755×10^{13} J (89.88 TJ).

Planck-Einstein Photon Energy

$$E_{\text{photon}} = \frac{hc}{\lambda} \quad \textbf{(Eq. D.46)}$$

Max Planck's 1900 resolution of the ultraviolet blackbody catastrophe and Albert Einstein's 1905 explanation of the photoelectric effect established that electromagnetic radiation of wavelength λ (m) and frequency ν = c / λ is emitted and absorbed in discrete quanta of energy $E_{\text{photon}} = h\nu$, where $h = 6.62607015 \times 10^{-34}$ J·s and c = 299,792,458 m/s. Because photon energy varies inversely with wavelength λ, a single short-wavelength ultraviolet or X-ray photon can eject a bound inner-shell electron that no intensity of long-wavelength infrared illumination could ever dislodge.

Forming the universal constant product $hc = 1.98644586 \times 10^{-25}$ J·m on the stack and dividing by λ (**6.62607015** [E] **34** [+/−] [ENTER] **299792458** [×] **λ** [÷]) for a green photon ($\lambda = 500.0 \times 10^{-9}$ m) yields 3.9729×10^{-19} J (2.48 eV).

de Broglie Particle Wavelength

$$\lambda = \frac{h}{mv} \quad \textbf{(Eq. D.47)}$$

In his 1924 doctoral thesis, Louis de Broglie extended wave-particle duality from massless photons ($p = h/\lambda$) to massive matter particles, proposing that any particle of mass m (kg) moving at non-relativistic speed v (m/s) with momentum $p = mv$ possesses an intrinsic matter-wave wavelength $\lambda = h / p$, where $h = 6.62607015 \times 10^{-34}$ J·s. While macroscopic objects have immeasurably small wavelengths, electrons accelerated across kilovolt potentials exhibit sub-nanometer de Broglie wavelengths shorter than interatomic crystal spacings and support atomic-scale diffraction and transmission electron microscopy. Dividing Planck's constant h sequentially by mass m and velocity v (**6.62607015** [E] **34** [+/−] [ENTER] **m** [÷] **v** [÷]) for an electron ($m = 9.1093837 \times 10^{-31}$ kg) traveling at $v = 2.0 \times 10^6$ m/s gives 3.6370×10^{-10} m (0.3637 nm).

Relativistic Lorentz Factor (γ)

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{1}{\sqrt{1-(v/c)^2}} \qquad \textbf{(Eq. D.48)}$$

Hendrik Lorentz and Albert Einstein showed that invariance of the spacetime interval $ds^2 = (c\ dt)^2 - dx^2$ across inertial reference frames moving at relative velocity v requires scaling time and space coordinates by the Lorentz factor γ, parameterized by the dimensionless speed ratio $\beta = v / c \in [0, 1)$. Near everyday speeds ($\beta \ll 1$), $\gamma \approx 1 + (1/2)\beta^2$ is indistinguishable from unity, but as $\beta \to 1$, γ diverges toward infinity—dilating the proper lifetime $\Delta t = \gamma\ \Delta\tau$ of high-energy atmospheric muons and contracting longitudinal lengths $L = L_0 / \gamma$. Squaring β in Register *X*, subtracting from 1 via [x⇄y], taking [√x], and inverting via [1/x] (β [↶] [x²] **1** [x⇄y] [-] [√x] [1/x]) at $\beta = 0.95$ ($1 - \beta^2 = 0.0975$) computes $\gamma =$ **3.20256**.

Relativistic Momentum

$$p = \gamma m v = \frac{m v}{\sqrt{1-(v/c)^2}} \qquad \textbf{(Eq. D.49)}$$

In special relativity, conservation of four-momentum across collisions requires the spatial momentum of a particle of rest mass m (kg) moving at speed v (m/s) to scale with the Lorentz factor as $p = \gamma\ m\ v$ rather than the Newtonian mv. As v approaches the speed of light c = 299,792,458 m/s, continuous force application ($F = dp/dt$) pumps energy into the diverging γ factor rather than pushing v past c, which directly determines the magnetic bending rigidity $B\rho = p / q$ required to steer relativistic protons inside a synchrotron ring. Computing the radical $\sqrt{(1 - (v/c)^2)}$ first, forming mv, and dividing via [x⇄y] (**v** [ENTER] **299792458** [÷] [↶] [x²] **1** [x⇄y] [-] [√x] **m** [ENTER] **v** [×] [x⇄y] [÷]) for a proton ($m = 1.67262 \times 10^{-27}$ kg) at $v = 0.85\ c = 2.548236 \times 10^8$ m/s yields 8.0906×10^{-19} kg·m/s.

Radioactive Decay Exponential Law

$$N(t) = N_0 e^{-\lambda t} = N_0 \left(\tfrac{1}{2}\right)^{t/t_{1/2}} \qquad \textbf{(Eq. D.50)}$$

Ernest Rutherford and Frederick Soddy established in 1902 that radioactive transmutation is a memoryless quantum tunneling process in which each unstable nucleus has a constant probability $\lambda = \ln(2) / t_{1/2}$ per unit time of decaying, independent of its age. Integrating $dN/dt = -\lambda\ N$ from an initial population or activity N_0 over elapsed time t shows that every elapsed half-life $t_{1/2}$ reduces the surviving radioisotope inventory by an exact factor of two. Raising 0.5 to the elapsed half-life count $t / t_{1/2}$ and scaling by N_0 (**0.5** [ENTER] **t** [ENTER] $t_{1/2}$ [÷] [y^x] N_0 [×], or **k** [+/-] [ENTER] **t** [×] [e^x] N_0 [×] when given decay rate $k = \lambda$) for a Cobalt-60 source (N_0 = 10000.0 Bq, $t_{1/2}$ = 5.27 years) after t = 15.0 years returns **1390.5** Bq.

Rydberg Equation for Hydrogen Spectral Series

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \qquad \textbf{(Eq. D.51)}$$

Formulated empirically by Johannes Rydberg in 1888 and derived from first principles by Niels Bohr's 1913 quantized angular-momentum model of the hydrogen atom, this relation gives the vacuum wavelength λ (m) of photons emitted or absorbed when an electron transitions between principal quantum shells n_1 and n_2 ($n_1 < n_2$), where $R_\infty = m_e\ e^4 / 8\ \varepsilon_0{}^2\ h^3\ c = 1.0973731568 \times 10^7$ m^{-1}. Because bound orbital energies scale inversely with the square of the shell index (E_n = -13.6 eV/n^2), transitions terminating on n_1 = 1, 2, 3 generate the ultraviolet Lyman, visible Balmer, and infrared Paschen series respectively.

Computing the reciprocal-square difference $1/n_1^2$ - $1/n_2^2$, multiplying by R_∞, and pressing [x^{-1}] (n_1 [↶] [x^2] [x^{-1}] n_2 [↶] [x^2] [x^{-1}] [-] **1.0973731568** [E] **7** [×] [x^{-1}]) for the Balmer H_α red line ($n_1 = 2$, $n_2 = 3$) gives 6.5611×10^{-7} m (656.11 nm with R_∞, or 656.47 nm with reduced-mass R_H).

D.2.7 Euclidean Mensuration, Trigonometry & Coordinate Geometry

Volume and Surface Area of Sphere

$$V = \tfrac{4}{3}\pi r^3, \quad A = 4\pi r^2 \tag{Eq. D.52}$$

In *On the Sphere and Cylinder* (c. 225 BCE)—the geometric ratio Archimedes regarded as his crowning achievement—he proved that a sphere of radius r occupies exactly two-thirds of the volume ($\pi\, r^2 \cdot 2r = 2\pi\, r^3$) and two-thirds of the total closed surface area ($2\pi\, r^2 + 2\pi\, r \cdot 2r = 6\pi\, r^2$) of its circumscribing right circular cylinder. Differentiating $V(r) = (4/3)\pi\, r^3$ with respect to r recovers $dV/dr = 4\pi\, r^2 = A$, because an infinitesimal radial expansion dr adds a spherical shell of volume $A\ dr$. Cubing r on the stack via **r** [ENTER] **3** [y^x] [↷] [π] [×] **4** [×] **3** [÷] for volume and squaring via **r** [↶] [x^2] [↷] [π] [×] **4** [×] for surface area at $r = 7.50$ cm ($r^3 = 421.875$ cm^3) yields Volume = **1767.146** cm^3 and Area = **706.858** cm^2.

Right Circular Cone Volume & Lateral Surface Area

$$V = \tfrac{1}{3}\pi r^2 h, \quad A_{\text{lat}} = \pi r\sqrt{r^2 + h^2} \tag{Eq. D.53}$$

First established by Eudoxus of Cnidus via the method of exhaustion, a right circular cone of base radius r and perpendicular altitude h encloses exactly one-third the volume of its bounding cylinder $\pi\, r^2\, h$ because horizontal cross-sectional disks shrink quadratically with distance from the apex ($\int_0^h (z/h)^2\, dz = h/3$). Unrolling the conical sheet along a generator of Pythagorean slant height $\ell = \sqrt{(r^2 + h^2)}$ produces a planar circular sector of radius ℓ and arc length $2\pi\, r$, whose area is $1/2(2\pi\, r)\ell = \pi\, r\sqrt{(r^2 + h^2)}$. Evaluating volume via **r** [↶] [x^2] **h** [×] [↷] [π] [×] **3** [÷] and lateral area via **r** [↶] [x^2] **h** [↶] [x^2] [+] [$\sqrt{x}$] **r** [×] [↷] [π] [×] for $r = 6.0$ cm and $h = 14.0$ cm returns Volume = **527.788** cm^3 and Lateral Area = **287.108** cm^2.

Frustum of Right Circular Cone Volume

$$V = \tfrac{1}{3}\pi h(R^2 + Rr + r^2) \tag{Eq. D.54}$$

Slicing a right circular cone parallel to its base at height h leaves a conical frustum bounded by lower radius R and upper radius r—the circular counterpart of the ancient Egyptian Moscow Mathematical Papyrus (Problem 14, c. 1850 BCE) pyramidal frustum. Rather than multiplying h by the arithmetic mean of the two end disks, exact integration over the linearly tapering radius $r(z)$ averages three equally weighted disks: the bottom base $\pi\, R^2$, the top base $\pi\, r^2$, and their geometric mean $\pi\, Rr$. Accumulating the quadratic sum $R^2 + Rr + r^2$ in Register X before scaling by $(1/3)\pi\, h$ (**R** [↶] [x^2] **R** [ENTER] **r** [×] [+] **r** [↶] [x^2] [+] **h** [×] **3** [÷] [↷] [π] [×]) for $R = 8.0$ cm, $r = 3.0$ cm, $h = 12.0$ cm ($R^2 + Rr + r^2 = 97.0$ cm^2) yields **1218.938** cm^3.

Torus Volume and Total Surface Area

$$V = 2\pi^2 R r^2, \quad A = 4\pi^2 R r \qquad \textbf{(Eq. D.55)}$$

By Pappus of Alexandria's centroid theorems (Book VII of the *Collection*, c. 340 CE), revolving a planar circular cross-section of minor tube radius r (area $\pi\ r^2$, perimeter $2\pi\ r$) around an external coplanar axis at major centroid radius R ($R \geq r$) sweeps out a volume equal to the cross-sectional area times the centroid path length $2\pi\ R$, and a surface area equal to the perimeter times $2\pi\ R$. Because inward curvature on the inner equator (R - r) exactly cancels outward curvature on the outer equator (R + r), the torus behaves metrically like a straight cylinder of radius r and length $2\pi\ R$, governing toroidal inductors, O-ring elastomer seals, and tokamak fusion vessels.

Evaluating **r** [↶] [x²] **R** [×] **2** [×] [↷] [π] [↶] [x²] [×] for volume and **R** [ENTER] **r** [×] **4** [×] [↷] [π] [↶] [x²] [×] for surface area at R = 12.0 cm and r = 2.5 cm yields Volume = **1480.44** cm^3 and Area = **1184.35** cm^2.

Heron's Formula for Arbitrary Triangle Area

$$A = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2} \qquad \textbf{(Eq. D.56)}$$

Recorded by Heron of Alexandria in Book I of his *Metrica* (c. 60 CE), this intrinsic distance formula determines the planar area A of any triangle directly from its three edge lengths a, b, c and semiperimeter $s = (a + b + c)/2$ without requiring an interior angle or dropped altitude. Each factor (s - a), (s - b), (s - c) represents the tangent segment from a vertex to the inscribed circle of inradius $r = A / s$, and their product vanishes whenever the triangle degenerates into a collinear segment ($a + b = c$). Computing semiperimeter s once, storing it in register **S**, and forming the four-term product under the radical (**a** [ENTER] **b** [+] **c** [+] **2** [÷] [STO] **S** **a** [-] [RCL] **S** **b** [-] [×] [RCL] **S** **c** [-] [×] [RCL] **S** [×] [√x]) for the classic (13, 14, 15) Heronian triangle (s = 21.0, factors 21 × 8 × 7 × 6 = 7056) gives **84.000**.

Law of Cosines for Arbitrary Triangle

$$c = \sqrt{a^2 + b^2 - 2ab\cos C} \qquad \textbf{(Eq. D.57)}$$

Generalized from Book II, Propositions 12–13 of Euclid's *Elements* by the Persian astronomer Jamshīd al-Kāshī in 1427, the Law of Cosines resolves the third side c of any oblique triangle from two adjacent sides a, b and their included angle C (degrees). Subtracting the Euclidean inner-product projection $2ab\cos C$ from the Pythagorean sum $a^2 + b^2$ accounts for acute ($C < 90°$, where $\cos C > 0$ shortens c) and obtuse ($C > 90°$, where $\cos C < 0$ lengthens c) vertex geometry, collapsing to $c = \sqrt{(a^2 + b^2)}$ at $C = 90°$. In [DEG] mode, summing $a^2 + b^2$ in Register Y, subtracting $2ab\cos C$, and taking [√x] (**a** [↶] [x²] **b** [↶] [x²] [+] **2** [ENTER] **a** [×] **b** [×] **C** [COS] [×] [-] [√x]) for a = 18.0, b = 25.0, C = 62.0° returns **22.945**.

Great Circle Spherical Distance (Haversine Formula)

$$d = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)}\right) \quad \textbf{(Eq. D.58)}$$

Introduced by James Inman in his 1835 *Navigation and Nautical Astronomy*, the haversine formula computes the shortest geodesic arc d across a sphere of mean Earth radius R = 6371.0 km between coordinates (φ_1, λ_1) and (φ_2, λ_2) using half-versed sines $\text{hav}(\vartheta) = \sin^2(\vartheta/2)$ of the latitude difference $\Delta\varphi = \varphi_2 - \varphi_1$ and longitude difference $\Delta\lambda = \lambda_2 - \lambda_1$. Expressing central angle entirely in terms of $\sin^2(\Delta/2)$ avoids the severe floating-point cancellation that plagues the spherical law of cosines $\arccos(\sin\varphi_1 \sin\varphi_2 + \cos\varphi_1 \cos\varphi_2 \cos\Delta\lambda)$ when two waypoints lie only meters or kilometers apart ($\cos\vartheta \approx 0.99999999$).

In [DEG] mode, evaluating the two haversine terms from the inside out and converting the half-central angle to radians via [↷] [▸RAD] (φ₂ [ENTER] φ₁ [-] 2 [÷] [SIN] [↶] [x²] φ₁ [COS] φ₂ [COS] [×] λ₂ [ENTER] λ₁ [-] 2 [÷] [SIN] [↶] [x²] [×] [+] [√x] [↶] [ASIN] [↷] [▸RAD] 2 [×] 6371.0 [×]) from New York (40.7128°N, 74.0060°W) to London (51.5074°N, 0.1278°W) yields **5570.2** km (3461.2 miles).

Circular Sector Area and Arc Length

$$s = r\theta, \quad A = \tfrac{1}{2} r^2 \theta \quad (\theta \text{ in radians}) \quad \textbf{(Eq. D.59)}$$

Because radian measure defines a central angle ϑ (in rad) as the dimensionless ratio of subtended arc length s to radius r, a circular sector occupying fraction ϑ / (2π) of a complete turn has arc length $s = (\vartheta / 2\pi)(2\pi r) = r\vartheta$ and enclosed pie-slice area $A = (\vartheta / 2\pi)(\pi r^2) = (1/2)r^2\vartheta$. By a direct structural parallel to linear kinematics ($v = at$ and $x = (1/2)at^2$), integrating circumferential arc length $r\vartheta$ radially from 0 to r produces $\int_0^r \rho\vartheta \, d\rho = (1/2)r^2\vartheta$. Multiplying **r** [ENTER] θ [×] for arc length or **r** [↶] [x²] θ [×] 2 [÷] for sector area at r = 12.0 cm and ϑ = 1.047198 rad (60.0°, convert via **60** [↷] [▸RAD]) yields Arc Length = **12.566** cm and Sector Area = **75.398** cm².

Circular Segment Area

$$A = \tfrac{1}{2} r^2 (\theta - \sin\theta) \quad (\theta \text{ in radians}) \quad \textbf{(Eq. D.60)}$$

A circular segment—the region bounded by a secant chord and its subtended arc of central angle ϑ (in radians) on a circle of radius r—is obtained by subtracting the area of the central isosceles triangle $(1/2)r^2 \sin\vartheta$ from the full circular sector $(1/2)r^2\vartheta$. Civil and process engineers rely on this difference to calibrate the liquid volume inside partially filled horizontal cylindrical storage tanks and storm sewers as a function of wetted angle $\vartheta = 2\arccos((r - h)/r)$. In [RAD] mode, duplicating ϑ with [ENTER] to form ϑ - sin ϑ before multiplying by r^2 / 2 (θ [ENTER] [SIN] [-] **r** [↶] [x²] [×] 2 [÷]) for r = 8.0 m and ϑ = 1.250 rad gives **9.633** m².

Circular Annulus & Annular Sector Area

$$A = \pi(R^2 - r^2) = \pi(R - r)(R + r), \quad A_{\text{sector}} = \frac{\theta}{2}(R - r)(R + r) \qquad \textbf{(Eq. D.61)}$$

The planar ring (annulus) between two concentric circles of outer radius R and inner bore radius r has area $\pi(R^2 - r^2)$, while a sector subtending central angle ϑ (rad) encloses $A_{\text{sector}} = (\vartheta/2)(R^2 - r^2)$. Factoring the difference of squares as $(R - r)(R + r)$ reveals that the annulus area equals the radial wall thickness $w = R - r$ multiplied by the mean circumference $2\pi\, r_{\text{mean}} = \pi(R + r)$, and prevents loss of significant digits when evaluating thin-walled pipe cross-sections where $R \approx r$. Forming $(R - r)$ and $(R + r)$ on the stack and scaling by π (**R** ENTER **r** − **R** ENTER **r** + × 1/x π ×) for $R = 15.0$ cm and $r = 10.0$ cm evaluates to **392.699** cm^2.

Ellipse Area & Ramanujan's Second Perimeter Approximation

$$A = \pi ab, \quad h = \frac{(a-b)^2}{(a+b)^2}, \quad P \approx \pi(a+b)\left[1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right] \qquad \textbf{(Eq. D.62)}$$

Because an ellipse of semimajor axis a and semiminor axis b ($a \geq b$) is a uniform vertical scaling of a circle of radius a by the aspect ratio b/a, its enclosed area scales exact circle area $\pi\, a^2$ to $A = \pi\, a\, b$. By contrast, its arc-length perimeter $P = 4a\, E(e)$ requires the complete elliptic integral of the second kind and admits no elementary closed form; however, Srinivasa Ramanujan's 1914 second rational approximation in terms of the symmetric oblateness parameter $h = (a - b)^2 / (a + b)^2$ matches the exact elliptic integral up to the h^5 Maclaurin term (accurate to better than 0.0004% even at aspect ratios of 3:1).

Evaluating area via **a** ENTER **b** × 1/x π × and storing h in **A** to unpeel Ramanujan's fraction inside-out (**a** ENTER **b** − **a** ENTER **b** + ÷ √x x² STO **A** **3** × **4** x⇄y − √x **10** + 1/x **3** × RCL **A** × **1** + **a** ENTER **b** + × 1/x π ×) for $a = 10.0$ cm, $b = 6.0$ cm ($h = 0.0625$) yields Area = **188.496** cm^2 and Perimeter = **51.054** cm.

Parabolic Segment Area (Archimedes' Quadrature)

$$A = \frac{2}{3}bh \qquad \textbf{(Eq. D.63)}$$

In *Quadrature of the Parabola* (c. 240 BCE)—the earliest known summation of an infinite geometric series ($1 + 1/4 + 1/16 + \ldots = 4/3$)—Archimedes proved that the planar region bounded by a parabola and a chord of width b with perpendicular vertex sagitta h encloses exactly four-thirds the area of the inscribed triangle ($(1/2)bh$), or two-thirds the area of the circumscribing rectangle (bh). Structural engineers use this exact $(2/3)bh$ factor daily when integrating parabolic shear-stress distributions across rectangular beams and parabolic bending-moment diagrams under uniform loading. Multiplying base b by height h and scaling by 2/3 (**b** ENTER **h** × **2** × **3** ÷) for an arch of span $b = 18.0$ m and rise $h = 6.0$ m gives **72.000** m^2.

Regular *n*-gon Area, Inradius & Circumradius

$$A = \tfrac{1}{4} n s^2 \cot\left(\tfrac{\pi}{n}\right), \quad r = \frac{s}{2\tan(\pi/n)}, \quad R = \frac{s}{2\sin(\pi/n)} \qquad \textbf{(Eq. D.64)}$$

Connecting the center of a regular n-sided polygon of edge length s to its n vertices partitions the figure into n congruent isosceles triangles, each with central apex angle 2π / n (rad), base s, apothem (inradius) $r = (s/2)\ \cot(\pi/n)$, and circumradius $R = (s/2)\ \csc(\pi/n)$. Summing the n sub-triangles yields $A = n((1/2)\ s\ r) = (1/4)\ n\ s^2\ \cot(\pi/n)$, which converges smoothly to $\pi\ r^2$ as $n \to \infty$ and sizes hexagonal bolt flats ($n = 6$), octagonal architectural piers ($n = 8$), and milled polygonal shafts. In [RAD] mode, evaluating $\cot(\pi\ /\ n)$ via the reciprocal of tangent ([RAD] [↻] [π] **n** [÷] [TAN] [x^{-1}] **s** [↰] [x^2] [×] **n** [×] **4** [÷]) for a regular hexagon ($n = 6$, $s = 5.0$ cm) yields **64.952** cm^2.

Brahmagupta's Formula & Ptolemy's Theorem for Cyclic Quadrilaterals

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d)}, \quad s = \frac{a+b+c+d}{2}, \quad ac + bd = ef \qquad \textbf{(Eq. D.65)}$$

Derived by the Indian mathematician Brahmagupta in his *Brāhmasphuṭasiddhānta* (628 CE), this formula gives the exact area A of any cyclic quadrilateral—a four-sided polygon of side lengths a, b, c, d inscribed within a circle—in terms of its semiperimeter $s = (a\ +\ b\ +\ c\ +\ d)/2$, while Claudius Ptolemy's theorem (*Almagest*, c. 150 CE) relates the opposite-side products to the interior diagonals e, f via $ac\ +\ bd\ =\ ef$. By Bretschneider's identity, a cyclic quadrilateral encloses the strictly maximum possible area of any quadrilateral formed from four fixed edge lengths, and setting $d = 0$ immediately recovers Heron's triangle formula.

Storing semiperimeter s in register **S** and multiplying the four residual factors (**a** [ENTER] **b** [+] **c** [+] **d** [+] **2** [÷] [STO] **S a** [-] [RCL] **S b** [-] [×] [RCL] **S c** [-] [×] [RCL] **S d** [-] [×] [$\sqrt{x}$]) for $a = 3.0$, $b = 4.0$, $c = 5.0$, $d = 6.0$ ($s = 9.0$) evaluates to **18.974**.

Stewart's Theorem for Triangle Cevians

$$man + d^2a = b^2m + c^2n \Rightarrow d = \sqrt{\frac{b^2m + c^2n - man}{a}} \qquad \textbf{(Eq. D.66)}$$

Published by Matthew Stewart in 1746 (and memorized across generations by the Gaelic-flavored mnemonic *"A man and his dad put a bomb in the sink"*: $m\ a\ n\ +\ d\ a\ d\ =\ b\ m\ b\ +\ c\ n\ c$), this theorem gives the length d of any internal cevian segment from a triangle vertex to a point dividing the opposite base a into adjacent sub-segments m and n ($m\ +\ n\ =\ a$). Applying the Law of Cosines to the two supplementary sub-triangles sharing the cevian foot eliminates the supplementary cosines $\cos\vartheta\ +\ \cos(\pi\ -\ \vartheta)\ =\ 0$ and yields a single master formula for triangle medians ($m\ =\ n\ =\ a/2$), angle bisectors ($m/n\ =\ c/b$), and internal structural truss web members.

Forming $b^2\ m\ +\ c^2\ n\ -\ m\ a\ n$ on the stack, dividing by a, and taking [$\sqrt{x}$] (**b** [↰] [x^2] **m** [×] **c** [↰] [x^2] **n** [×] [+] **m** [ENTER] **a** [×] **n** [×] [-] **a** [÷] [$\sqrt{x}$]) for the median to the hypotenuse of a (6, 8, 10) right triangle ($a = 10.0$, $b = 8.0$, $c = 6.0$, $m = 5.0$, $n = 5.0$) confirms $d =$ **5.000**.

D.2.8 Three-Dimensional Solid Geometry & Mensuration

Spherical Cap (Dome) & Spherical Zone Volume

$$V_{\text{cap}} = \tfrac{1}{3}\pi h^2(3R - h), \quad A_{\text{dome}} = 2\pi Rh \tag{Eq. D.67}$$

$$V_{\text{zone}} = \frac{\pi h}{6}(3r_1^2 + 3r_2^2 + h^2) \tag{Eq. D.68}$$

Integrating horizontal disk cross-sections $\pi(R^2 - z^2)\,dz$ from $z = R - h$ to the pole $z = R$ of a sphere of radius R yields the volume V_{cap} of a spherical cap of sagitta height h, while Archimedes' cylinder-projection theorem proves that the curved dome surface area $A_{\text{dome}} = 2\pi\,R\,h$ equals the lateral area of a slice of height h on the circumscribing cylinder of radius R. When a spherical shell is truncated between two parallel planes separated by thickness h with cut radii r_1 and r_2, the enclosed spherical zone volume V_{zone} sizes cryogenic rocket tank bulkheads, architectural domes, and thick optical meniscus lenses.

Evaluating cap volume from the inside out via **3** [ENTER] **R** [×] **h** [−] **h** [↶] [x²] [×] [↷] [π] [×] **3** [÷] (and dome area via **2** [↷] [π] [×] **R** [×] **h** [×]) for $R = 10.0$ cm and $h = 3.0$ cm returns Volume = **254.469** cm^3 and Dome Area = **188.496** cm^2.

Frustum of Regular Pyramid Volume

$$V = \frac{h}{3}(A_1 + A_2 + \sqrt{A_1 A_2}) \tag{Eq. D.69}$$

For any pyramidal or conical frustum of perpendicular altitude h bounded by parallel, geometrically similar lower and upper base polygons of areas A_1 and A_2, cross-sectional area varies quadratically with elevation z, so exact integration yields h times the Heronian mean $1/3(A_1 + A_2 + \sqrt{(A_1\,A_2)})$. Setting $A_1 = \pi\,R^2$ and $A_2 = \pi\,r^2$ immediately recovers the circular cone frustum, whereas setting $A_2 = 0$ recovers the full pyramid volume $(1/3)\,A_1\,h$ and furnishes a single unified formula for tapered concrete bridge piers, pedestal footings, and bulk grain hoppers. Computing the geometric mean $\sqrt{(A_1\,A_2)}$ first on the stack and adding A_1 and A_2 before scaling by $h/3$ (**A₁** [ENTER] **A₂** [×] [√x] **A₁** [+] **A₂** [+] **h** [×] **3** [÷]) for $A_1 = 36.0$ m^2, $A_2 = 16.0$ m^2, $h = 9.0$ m yields **228.000** m^3.

Ellipsoid Volume

$$V = \frac{4}{3}\pi abc \tag{Eq. D.70}$$

A general triaxial ellipsoid defined by $(x/a)^2 + (y/b)^2 + (z/c)^2 \le 1$ is obtained by applying a three-dimensional linear transformation with Jacobian determinant $\det(\boldsymbol{J}) = abc$ to the unit sphere $x'^2 + y'^2 + z'^2 \le 1$ of volume $(4/3)\pi$. Scaling the unit sphere along the three orthogonal semiaxes a, b, c yields $V = (4/3)\pi\,a\,b\,c$, which collapses to an oblate or prolate spheroid when two axes are equal ($a = b$, as in the WGS-84 reference geoid of Earth) and to a sphere when $a = b = c = r$. Chaining the product of the three semiaxes and scaling by $(4/3)\pi$ (**a** [ENTER] **b** [×] **c** [×] [↷] [π] [×] **4** [×] **3** [÷]) for $a = 6.0$ cm, $b = 4.0$ cm, $c = 3.0$ cm gives **301.593** cm^3.

Paraboloid of Revolution Volume

$$V = \tfrac{1}{2}\pi r^2 h \qquad \textbf{(Eq. D.71)}$$

In *On Conoids and Spheroids*, Archimedes demonstrated that revolving a parabola around its symmetry axis to form a circular paraboloid of rim radius r and axial depth h encloses exactly one-half the volume of its circumscribing cylinder ($\pi\ r^2\ h$). The simple factor of $1/2$ arises because the cross-sectional disk area $A(z) = \pi\ [r(z)]^2 = \pi\ (r^2\ /\ h)\ z$ grows strictly *linearly* with axial distance z from the vertex ($0 \le z \le h$), so its average cross-section is just the midpoint disk $(1/2)\pi\ r^2$. Squaring rim radius r in Register X and multiplying by $(1/2)\pi\ h$ (**r** [↶] [x²] **h** [×] [↷] [π] [×] **2** [÷]) for a parabolic reflector or nosecone of $r =$ 4.0 m and $h =$ 3.0 m yields **75.398** m^3.

Cylindrical Ungula (Hoof) Volume

$$V = \tfrac{2}{3} r^3 \tan\alpha \qquad \textbf{(Eq. D.72)}$$

When a right circular cylinder of radius r is severed by an oblique plane that passes directly through a diameter of the circular base at inclination angle α (degrees), the wedge-shaped solid cut off is known in classical solid mensuration as a *cylindrical ungula* (from the Latin for a horse's hoof). Slicing the ungula perpendicular to the base diameter produces a family of similar right triangles of base $x = \sqrt{(r^2 - y^2)}$ and height $z = x \tan\alpha$, whose integral $\int_{-r}^{r} 1/2(r^2 - y^2) \tan\alpha\ dy = (2/3) r^3 \tan\alpha$ is completely free of π. In [DEG] mode, cubing r and multiplying by $(2/3) \tan\alpha$ (**r** [ENTER] **3** [y^x] **α** [TAN] [×] **2** [×] **3** [÷]) for $r =$ 6.0 cm and $\alpha = 45.0°$ evaluates to exactly **144.000** cm^3.

Universal Prismoidal Formula (Simpson's 3D Rule)

$$V = \frac{h}{6}(A_{\text{top}} + 4A_{\text{mid}} + A_{\text{bot}}) \qquad \textbf{(Eq. D.73)}$$

Because Thomas Simpson's 1:4:1 parabolic quadrature rule integrates any polynomial of degree three or less with zero truncation error, the prismoidal formula gives the *exact* volume of any solid of altitude h—including spheres, spherical segments, cones, pyramids, frustums, wedges, paraboloids, hyperboloids, and civil highway earthwork prismoids—whose horizontal cross-sectional area $A(z)$ is a quadratic or cubic function of height z, using only the top area A_{top}, bottom area A_{bot}, and exact mid-plane area A_{mid} at $z = h/2$. Here A_{mid} is the true geometric cross-section at mid-height (not the average of A_{top} and A_{bot}).

Scaling A_{mid} by 4 first, adding A_{top} and A_{bot}, and multiplying by $h/6$ (**A_{mid}** [ENTER] **4** [×] **A_{top}** [+] **A_{bot}** [+] **h** [×] **6** [÷]) for a square frustum with $A_{\text{top}} = 16.0$, $A_{\text{mid}} = 36.0$, $A_{\text{bot}} = 64.0$, $h = 6.0$ yields **224.000**.

D.2.9 Financial Mathematics, Amortization & Time Value of Money

Present Value of an Ordinary Annuity

$$PV = PMT\left[\frac{1-(1+i)^{-N}}{i}\right] \quad \textbf{(Eq. D.74)}$$

Summing the finite geometric progression $\sum_{k=1}^{N} PMT\ (1 + i)^{-k}$ of N equal end-of-period cash flows PMT discounted at periodic compound interest rate i yields the closed-form present-value annuity factor $[1 - (1+i)^{-N}] \ / \ i$. Financially, the numerator $1 - (1+i)^{-N}$ represents the difference between a perpetual stream of payments ($PMT \ / \ i$) and the present value of an identical perpetuity deferred by N periods, pricing fixed-rate mortgages, equipment leases, and level-coupon bond streams. Evaluating the discount factor $(1 + i)^{-N}$ first on the stack, subtracting from unity via [x⇄y], dividing by i, and scaling by PMT (**1** [ENTER] **i** [+] **N** [+/−] [y^x] **1** [x⇄y] [−] **i** [÷] **PMT** [×]) for $PMT = 500$ USD, $i = 0.005$ (6% APR compounded monthly), and $N = 36$ months yields **16,435.51** USD.

Future Value of an Ordinary Annuity

$$FV = PMT\left[\frac{(1+i)^{N}-1}{i}\right] \quad \textbf{(Eq. D.75)}$$

Compounding N regular end-of-period deposits PMT forward at periodic interest rate i sums the geometric series $\sum_{k=0}^{N-1} PMT(1 + i)^{k}$ into the accumulated terminal balance $FV = PV\ (1 + i)^{N}$. Because the compound growth factor $(1 + i)^{N}$ grows exponentially with horizon N, interest earned on prior interest eventually dominates the linear sum of raw principal contributions $N \cdot PMT$ in long-term retirement accounts and capital reserve funds. Forming $(1 + i)^{N}$, subtracting 1, dividing by i, and multiplying by PMT (**1** [ENTER] **i** [+] **N** [y^x] **1** [−] **i** [÷] **PMT** [×]) for $PMT = 250$ USD, $i = 0.006$ (7.2% APR monthly), and $N = 60$ months yields **17,991.18** USD.

Loan Amortization Periodic Payment

$$PMT = PV\left[\frac{i}{1-(1+i)^{-N}}\right] \quad \textbf{(Eq. D.76)}$$

Inverting the present-value annuity relation multiplies the initial borrowed principal PV by the capital-recovery factor $i \ / \ [1 - (1+i)^{-N}]$ to determine the level periodic installment PMT that retires both accruing interest and principal in exactly N compounding periods. Early in an amortization schedule, the interest charge $i \cdot B_{k-1}$ consumes the majority of each payment PMT, whereas the principal reduction $PMT - i \cdot B_{k-1}$ accelerates exponentially toward the end of the loan term. Computing the denominator $1 - (1+i)^{-N}$ first, dividing into i via [x⇄y], and multiplying by PV (**1** [ENTER] **i** [+] **N** [+/−] [y^x] **1** [x⇄y] [−] **i** [x⇄y] [÷] **PV** [×]) for $PV = 25{,}000$ USD, $i = 0.0045$ (5.4% APR monthly), and $N = 48$ months gives **580.27** USD.

Sinking Fund Periodic Savings Payment

$$PMT = FV\left[\frac{i}{(1+i)^{N}-1}\right] \qquad \textbf{(Eq. D.77)}$$

Inverting the future-value annuity relation determines the uniform end-of-period contribution PMT required to accumulate a scheduled target sum FV over N periods at periodic compound rate i. An algebraic identity links the capital-recovery and sinking-fund factors, $i \,/\, [1 - (1+i)^{-N}] = i + i \,/\, [(1+i)^{N} - 1]$, showing that amortizing a loan PV is mathematically equivalent to paying interest-only $i \cdot PV$ each period while depositing into a sinking fund that accumulates $FV = PV$ at maturity. Forming $(1 + i)^{N} - 1$ in Register X, dividing into i via [x⇄y], and scaling by FV (**1** [ENTER] **i** [+] **N** [y^x] **1** [-] **i** [x⇄y] [÷] **FV** [×]) for a target $FV = 50{,}000$ USD at $i = 0.005$ (6% APR monthly) over $N = 120$ months yields **305.10** USD.

Flat & Growing Perpetuities

$$PV_{\text{flat}} = \frac{PMT}{i}, \quad PV_{\text{growing}} = \frac{PMT}{i-g} \qquad \textbf{(Eq. D.78)}$$

As the horizon $N \to \infty$, the discount term $(1 + i)^{-N}$ vanishes for any positive discount rate $i > 0$, collapsing the present value of an infinite stream of level cash flows PMT to $PV_{\text{flat}} = PMT \,/\, i$. When the cash flow grows geometrically at rate g per period ($PMT_k = PMT(1 + g)^{k-1}$ with $i > g$), Myron Gordon's 1956 dividend-discount summation of $\sum_{k=1}^{\infty} PMT\,(1+g)^{k-1}/(1+i)^{k}$ subtracts g directly from the discount denominator to give $PV_{\text{growing}} = PMT \,/\, (i - g)$, valuing university endowments, preferred shares, and capitalized real estate. Dividing **PMT** [ENTER] **i** [÷] for a flat perpetuity or **PMT** [ENTER] **i** [ENTER] **g** [-] [÷] for a growing perpetuity at $PMT = 1{,}200$ USD/yr, $i = 0.06$, $g = 0.02$ returns Flat Perpetuity = **20,000.00** USD and Growing Perpetuity = **30,000.00** USD.

D.2.10 Chemistry, Electrochemistry, Chemical Kinetics & Biochemistry

pH & Hydronium Ion Activity

$$\mathrm{pH} = -\log_{10}[\mathrm{H_3O^+}], \quad [\mathrm{H_3O^+}] = 10^{-\mathrm{pH}} \qquad \textbf{(Eq. D.79)}$$

Introduced by Danish biochemist Søren Peder Lauritz Sørensen at the Carlsberg Laboratory in 1909, the pH scale (*pondus hydrogenii*) maps the aqueous molar activity of solvated hydronium ions $[\mathrm{H_3O^+}]$ (mol/L) onto a negative base-10 logarithm. Because aqueous proton concentrations span more than fourteen orders of magnitude from 1 M strong acids (pH 0) to 1 M strong bases (pH 14, anchored by water's autoionization product $K_w = [\mathrm{H_3O^+}][\mathrm{OH^-}] = 10^{-14}$ at 25°C), every unit decrease in pH represents a tenfold increase in acidity. Keying **[$\mathbf{H_3O^+}$]** [↶] [LOG] [+/−] (or inverting via **pH** [+/−] [↶] [10^x]) for $[\mathrm{H_3O^+}] = 3.50 \times 10^{-5}$ mol/L (**3.5** [E] **5** [+/−] [↶] [LOG] [+/−]) yields pH = **4.4559**.

Henderson–Hasselbalch Buffer Equilibrium

$$\mathrm{pH} = \mathrm{p}K_a + \log_{10}\left(\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}\right) \qquad \textbf{(Eq. D.80)}$$

Derived by Lawrence Joseph Henderson (1908) and Karl Albert Hasselbalch (1916) by taking the negative common logarithm of the weak-acid dissociation equilibrium $K_a = [\mathrm{H_3O^+}][\mathrm{A^-}] \,/\, [\mathrm{HA}]$, this equation relates solution pH to the acid's $\mathrm{p}K_a = -\log_{10} K_a$ and the molar ratio of conjugate base $[\mathrm{A^-}]$ to undissociated weak acid [HA]. Because $\log_{10}([\mathrm{A^-}]/[\mathrm{HA}])$ has its flattest slope when $[\mathrm{A^-}] = [\mathrm{HA}]$, a buffer resists pH excursions most strongly within $\pm$ 1 unit of $\mathrm{pH} = \mathrm{p}K_a$, governing both laboratory Tris/acetate buffers and human arterial blood bicarbonate homeostasis (pH 7.40).

Dividing conjugate base molarity by weak acid molarity, taking [↶] [LOG], and adding $\mathrm{p}K_a$ (**A_base** [ENTER] **HA** [÷] [↶] [LOG] **pKa** [+]) for an acetate buffer ($\mathrm{p}K_a = 4.76$, $[\mathrm{A^-}] = 0.25$ M, [HA] = 0.15 M) gives **4.9818**.

Nernst Electrochemical Cell Potential

$$E_{\mathrm{cell}} = E^{\circ}_{\mathrm{cell}} - \frac{RT}{nF}\ln Q \qquad \textbf{(Eq. D.81)}$$

Walther Nernst derived in 1889 how the electromotive force E_{cell} (V) of a galvanic cell or ion-selective membrane departs from its standard-state potential $E^{\circ}_{\mathrm{cell}}$ (V) as a function of absolute temperature T (K), the stoichiometric number of electrons transferred n, the Faraday constant $F = N_A\, e$ = 96,485.3 C/mol, the gas constant R = 8.31446 J/(mol·K), and the reaction quotient Q of product-to-reactant activities. Derived directly from the Gibbs free energy identity $\Delta G = \Delta G^{\circ} + RT \ln Q$ via $\Delta G = -n\,F\,E_{\mathrm{cell}}$, the Nernst equation shows that cell voltage drops to zero ($E_{\mathrm{cell}} = 0$) once the redox reaction reaches chemical equilibrium ($Q = K_{\mathrm{eq}}$).

Forming the thermal voltage factor $RT \,/\, (nF)$ first, multiplying by ln Q, and subtracting from $E^{\circ}_{\mathrm{cell}}$ via [x⇄y] (**8.31446** [ENTER] **T** [×] **n** [÷] **96485.3** [÷] **Q** [LN] [×] **E0** [x⇄y] [-]) for a Daniell Zn/Cu cell ($E^{\circ}_{\mathrm{cell}} = 1.100$ V, $T = 298.15$ K, $n = 2$, $Q = 0.050$) yields **1.1385** V.

Arrhenius Chemical Reaction Rate Constant

$$k = A\exp\left(-\frac{E_a}{RT}\right) \qquad \textbf{(Eq. D.82)}$$

Svante Arrhenius demonstrated in 1889 that the rate constant k ($\mathrm{s^{-1}}$ for a first-order reaction) of a thermally activated chemical process equals the pre-exponential collision-and-orientation frequency factor A ($\mathrm{s^{-1}}$) weighted by the Boltzmann factor $\exp[-E_a \,/\, (RT)]$, where E_a (J/mol) is the molar activation energy barrier, R = 8.314462618 J/(mol·K), and T (K) is absolute temperature. Because T sits in the denominator of a negative exponent, even a modest 10 K rise near room temperature for a typical $E_a \approx$ 50–80 kJ/mol barrier roughly doubles or triples the fraction of molecular collisions energetic enough to cross the transition state.

Evaluating the dimensionless barrier ratio $-E_a \,/\, (RT)$ first, exponentiating via [e^x], and scaling by A (**Ea** [ENTER] **8.314462618** [÷] **T** [÷] [+/-] [e^x] **A** [×]) for $A = 1.25 \times 10^{11}$ $\mathrm{s^{-1}}$, E_a = 75,000 J/mol, T = 350.0 K gives $k =$ **0.8017** $\mathrm{s^{-1}}$.

Michaelis–Menten Enzyme Saturation Kinetics

$$v_0 = \frac{V_{\max}[S]}{K_m + [S]} \qquad \textbf{(Eq. D.83)}$$

Formulated by Leonor Michaelis and Maud Menten in 1913 from the quasi-steady-state formation and breakdown of an enzyme–substrate complex $E + S \rightleftharpoons ES \rightarrow E + P$, this rectangular hyperbola relates initial catalytic reaction velocity v_0 to substrate concentration $[S]$, maximum turnover velocity $V_{\max} = k_{\text{cat}}[E]_{\text{total}}$, and the Michaelis constant K_m (the substrate concentration at which $v_0 = (1/2)\,V_{\max}$). At trace substrate levels ($[S] \ll K_m$), active sites are mostly empty and reaction rate scales linearly as first-order kinetics $v_0 \approx (V_{\max}/K_m)[S]$; at high substrate excess ($[S] \gg K_m$), every catalytic site is saturated and v_0 plateaus asymptotically at zero-order ceiling $V_{\max}$.

Forming the numerator $V_{\max}[S]$ in Register Y and dividing by $K_m + [S]$ (**Vmax** ENTER **S** × **Km** ENTER **S** + ÷) for $V_{\max}$ = 140.0 μM/min, $[S]$ = 15.0 mM, K_m = 5.0 mM yields **105.0000** μM/min.

Beer–Lambert Spectrophotometric Absorbance

$$A = \varepsilon \ell c = -\log_{10}\left(\frac{I}{I_0}\right) \qquad \textbf{(Eq. D.84)}$$

Developed successively by Pierre Bouguer (1729), Johann Heinrich Lambert (1760), and August Beer (1852), the Beer–Lambert law states that monochromatic light of incident intensity I_0 passing through a homogeneous absorbing solution of molar concentration c (mol/L) and cuvette path length ℓ (cm) suffers differential attenuation $dI \,/\, I = -\ln(10)\varepsilon\, c\, dx$, where ε (L·mol^{-1}·cm^{-1}) is the wavelength-specific molar absorptivity. Because transmitted intensity $I = I_0\ 10^{-\varepsilon\,\ell\,c}$ decays exponentially with concentration, defining spectrophotometric absorbance as $A = -\log_{10}(I \,/\, I_0)$ produces a strictly linear calibration curve $A = \varepsilon\,\ell\,c$ for quantitative chemical assays.

Multiplying **ε** ENTER **1** × **c** × (and recovering transmittance $T = I/I_0$ via **A** +/− ↶ 10ˣ) for an NADH enzymatic assay at 340 nm (ε = 6,220 L·mol^{-1}·cm^{-1}, ℓ = 1.00 cm, c = 8.50 × 10^{-5} M) gives A = **0.5287** (T = 29.60%).

Clausius–Clapeyron Vapor Pressure Scaling

$$P_2 = P_1 \exp\left[\frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] \qquad \textbf{(Eq. D.85)}$$

Rudolf Clausius and Émile Clapeyron showed that along a liquid–vapor phase-equilibrium boundary where molar vapor volume vastly exceeds liquid volume ($\Delta V_{\text{vap}} \approx RT/P$), the thermodynamic slope $dP \,/\, dT = \Delta H_{\text{vap}} \,/\, (T\Delta V_{\text{vap}})$ integrates assuming constant molar enthalpy of vaporization ΔH_{vap} (J/mol) and R = 8.314462618 J/(mol·K) to link saturation vapor pressures P_1, P_2 at absolute temperatures T_1, T_2 (K). Because saturation vapor pressure rises exponentially with $-1/T$, this relation predicts both the depressed boiling point of water at high mountain elevations and the vacuum pressure required for low-temperature solvent distillation.

Subtracting the reciprocal temperatures $1/T_1$ - $1/T_2$ via 1/x, scaling by $\Delta H_{\text{vap}} \,/\, R$, exponentiating, and multiplying by P_1 (**T1** 1/x **T2** 1/x − **ΔH** × **8.314462618** ÷ eˣ **P1** ×) for water (P_1 = 23.76 Torr at T_1 = 298.15 K, T_2 = 323.15 K, ΔH_{vap} = 43,990 J/mol) yields P_2 = **93.7702** Torr.

Debye–Hückel Solution Ionic Strength

$$I = \frac{1}{2}\sum_{i=1}^{n} c_i z_i^2 \qquad \textbf{(Eq. D.86)}$$

Introduced by Gilbert N. Lewis and Merle Randall in 1921 and grounded in electrostatic statistical mechanics by Peter Debye and Erich Hückel in 1923, the ionic strength I (mol/L) of an electrolyte solution sums the molar concentration c_i (mol/L) of every dissolved cationic and anionic species weighted by the square of its integer charge valency z_i (divided by 2 because both positive cations and negative anions contribute to the screening cloud). Because both the Coulomb force on an ion and the Boltzmann clustering of counter-ions scale linearly with charge z_i, multivalent ions ($z_i = \pm 2, \pm 3$) contract the Debye screening length $\varkappa^{-1} \propto 1/\sqrt{I}$ four to nine times as strongly as monovalent salts at equal molarity.

Squaring each ion charge z_i before multiplying by concentration c_i (**z₁** [↶] [x²] **c₁** [×] **z₂** [↶] [x²] **c₂** [×] [+] **2** [÷]) for 0.10 M $MgCl_2$ ($c_{Mg^{2+}} = 0.10$ M, $z_1 = +2$; $c_{Cl^-} = 0.20$ M, $z_2 = -1$) evaluates to **0.3000** mol/L.

D.2.11 Probability, Statistics, Quality Control & Information Theory

Binomial Probability Mass Function

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \qquad \textbf{(Eq. D.87)}$$

Derived by Jacob Bernoulli in *Ars Conjectandi* (1713), the binomial probability mass function gives the exact probability of observing k successes across n independent, identically distributed Bernoulli trials when each trial succeeds with probability $p \in [0, 1]$ and fails with probability $q = 1 - p$. Any specific ordered sequence of k successes and $n - k$ failures occurs with product probability $p^k (1 - p)^{n-k}$, and the binomial coefficient $C(n, k) = n!/k!(n-k)!$ (computed directly via the **PROB** menu [Cn,r] key without overflowing intermediate factorials) counts how many distinct permutations share that exact tally. Evaluating $C(n, k)$ first and multiplying by p^k and $(1 - p)^{n-k}$ (**n** [ENTER] **k** [Cn,r] **p** [ENTER] **k** [y^x] [×] **1** [ENTER] **p** [−] **n** [ENTER] **k** [−] [y^x] [×]) for $n = 10$ wafer dies, $k = 3$ defects, $p = 0.25$ yields **0.2503** (25.0282%).

Poisson Rare-Event Arrival Probability

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \qquad \textbf{(Eq. D.88)}$$

Siméon Denis Poisson derived in 1837 that as the number of trials $n \to \infty$ and the per-trial probability $p \to 0$ with expected count $\lambda = np$ held constant, the binomial distribution converges to this single-parameter law for the probability of observing $k \in (0, 1, 2, \ldots)$ independent random events within a fixed time or space interval of mean arrival rate λ. Because a memoryless Poisson process has variance equal to its mean ($\sigma^2 = \mu = \lambda$), this distribution governs shot-noise photon arrivals on a photodiode, radioactive disintegration counts, cosmic-ray bit flips, and unscheduled packet arrivals at a network switch. Raising λ to power k, multiplying by $e^{-\lambda}$, and dividing by $k!$ (**λ** [ENTER] **k** [y^x] **λ** [+/−] [e^x] [×] **k** [↶] [x!] [÷]) for $\lambda = 4.2$ photons/ms and $k = 3$ photons gives **0.1852** (18.5165%).

Gaussian Standard *z*-Score & Confidence Interval Half-Width

$$z = \frac{x - \mu}{\sigma}, \quad E = z_{\alpha/2}\frac{s}{\sqrt{n}} \tag{Eq. D.89}$$

Standardizing a normally distributed observation x by subtracting the population mean μ and dividing by standard deviation σ maps any Gaussian variable onto the dimensionless unit normal distribution $N(0, 1)$, while the Central Limit Theorem guarantees that the sample mean $\bar{x}$ of n independent measurements with sample standard deviation s ([sx]) has standard error SE = s / √n. Multiplying s / √n by the two-sided critical quantile $z_{\alpha/2}$ (1.9600 for a 95% confidence interval or 2.5758 for 99%) yields the confidence interval half-width E, where quadrupling the sample size n is required to halve the margin of error. Computing **x** [ENTER] μ [-] σ [÷] for a z-score or **s** [ENTER] **n** [√x] [÷] **1.96** [×] for the 95% confidence half-width with s = 8.40 MPa, n = 49 tensile specimens, $z_{\alpha/2}$ = 1.96 returns E = **2.3520** MPa.

Bayes' Theorem Diagnostic Posterior Probability

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A) + P(B \mid A')P(A')} \tag{Eq. D.90}$$

Communicated posthumously by Richard Price in 1763, Thomas Bayes's theorem inverts conditional probabilities by dividing the joint true-positive probability $P(B \mid A)P(A)$ by the total probability $P(B) = P(B \mid A)P(A) + P(B \mid A')P(A')$ of observing a positive test or sensor alarm B across both the true state A (prior prevalence $P(A)$) and its complement A' ($P(A') = 1 - P(A)$). When screening for a low-prevalence defect or condition ($P(A) \ll 1$), even a small false-positive rate $P(B \mid A')$ applied to the vast healthy population $P(A')$ can outnumber true positives $P(B \mid A)P(A)$ in the denominator.

Duplicating the true-positive product $P(B \mid A)P(A)$ on the stack with [ENTER] so it serves in both numerator and denominator (***P*(*B*|*A*)** [ENTER] ***P*(*A*)** [×] [ENTER] ***P*(*B*|¬*A*)** [ENTER] ***P*(¬*A*)** [×] [+] [÷]) for $P(A)$ = 0.01, sensitivity $P(B \mid A)$ = 0.99, and false-positive rate $P(B \mid A')$ = 0.05 ($P(A')$ = 0.99) yields **0.1667** (16.67% posterior probability).

Six Sigma Process Capability Index (C_{pk})

$$C_{pk} = \min\left(\frac{\mathrm{USL} - \bar{x}}{3s}, \frac{\bar{x} - \mathrm{LSL}}{3s}\right) \tag{Eq. D.91}$$

In statistical process control (SPC) and semiconductor/mechanical manufacturing, the process capability index C_{pk} compares the distance from the sample process mean $\bar{x}$ ([x̄]) to the nearer engineering tolerance boundary—either the Upper Specification Limit (USL) or Lower Specification Limit (LSL)—against the 3s one-sided natural process spread, where s ([sx]) is the sample standard deviation. Unlike C_p = (USL - LSL)/(6s), which assumes a perfectly centered process, C_{pk} penalizes off-center mean drift: C_{pk} = 1.00 corresponds to a ± 3σ (99.73% yield) boundary, C_{pk} = 1.33 is the standard ± 4σ industrial acceptance benchmark, and C_{pk} = 2.00 defines centered ± 6σ capability.

Evaluating the closer upper limit via **USL** [ENTER] $\bar{x}$ [-] **3** [÷] **s** [÷] for a CNC shaft (USL = 10.050 mm, LSL = 9.950 mm, $\bar{x}$ = 10.010 mm, s = 0.010 mm) gives C_{pk} = **1.3333**.

Shannon Information Entropy (Base-2 Bits)

$$H(X) = -\sum_{i=1}^{m} p_i \log_2 p_i = -\frac{1}{\ln 2} \sum_{i=1}^{m} p_i \ln p_i \qquad \textbf{(Eq. D.92)}$$

In his 1948 monograph *A Mathematical Theory of Communication*, Claude Elwood Shannon defined the entropy $H(X)$ (in bits/symbol when using base-2 logarithms) of a discrete random source emitting symbols x_1, ..., x_m with probabilities p_1, ..., p_m ($\sum p_i = 1$) as the expected self-information $E[-\log_2 p_i]$. By Shannon's source coding theorem, $H(X)$ sets the exact theoretical lower bound on the average number of bits per symbol required for lossless compression (approached by Huffman and arithmetic coding), peaking at $\log_2 m$ bits when all m symbols are equiprobable and vanishing when one outcome is certain ($p_i = 1$). Summing $p_i \ln p_i$ on the stack and dividing by -ln 2 (**p** ENTER LN × **1** ENTER **p** - ENTER LN × + **2** LN ÷ +/-) for a biased binary telemetry source ($p_1 = 0.75$, $p_2 = 0.25$) yields **0.8113** bits/symbol.

Shannon–Hartley Channel Capacity Law

$$C = B \log_2 \left(1 + \frac{S}{N}\right) = \frac{B \ln(1 + S/N)}{\ln 2} \qquad \textbf{(Eq. D.93)}$$

Building on Ralph Hartley's pulse-level signaling rate and Nyquist's $2B$ sampling theorem, Claude Shannon proved that a communication channel of analog bandwidth B (Hz) corrupted by additive white Gaussian noise (AWGN) of mean noise power N can transmit information with arbitrarily low bit-error probability at any rate up to the capacity ceiling C (bits/s), where S/N is the linear signal-to-noise power ratio. Because bandwidth B scales capacity linearly whereas signal-to-noise ratio S/N sits inside a base-2 logarithm, power-starved deep-space transponders and spread-spectrum links trade wider bandwidth for exponentially lower transmit power.

Converting S/N via ln(1 + S/N) / ln 2 and multiplying by B (***SNR*** ENTER **1** + LN **2** LN ÷ **B** ×, or from decibels via $\boldsymbol{SNR}_{\mathbf{dB}}$ ENTER **10** ÷ ↶ 10^x **1** + LN **2** LN ÷ **B** ×) for a B = 20.0 MHz channel at linear S/N = 31.0 (14.91 dB) gives **100.0000** Mbps.

M/M/1 Queueing Utilization, Queue Length & Latency

$$\rho = \frac{\lambda}{\mu}, \quad L = \frac{\lambda}{\mu - \lambda}, \quad W = \frac{1}{\mu - \lambda} \qquad \textbf{(Eq. D.94)}$$

In Agner Krarup Erlang's and David George Kendall's Markovian $M/M/1$ single-server queueing model, independent Poisson requests arriving at mean rate λ (req/s) are processed by a server with exponentially distributed service times of mean completion rate μ (req/s, requiring server utilization ρ = λ / μ < 1 for steady-state stability). Solving the birth–death balance equations yields the mean number of requests in the system L = ρ / (1 - ρ) = λ / (μ - λ) and—via John Little's universal conservation law L = λ W—the mean end-to-end response latency W = 1 / (μ - λ) (queuing wait plus service time), which diverges hyperbolically as utilization ρ approaches 100%.

Evaluating λ ENTER μ ENTER λ - ÷ for L and μ ENTER λ - x^{-1} for W at λ = 45.0 req/s and μ = 60.0 req/s (ρ = 75%) yields L = **3.0000** requests and W = **0.0667** s (66.7 ms).

D.2.12 Structural Mechanics, Semiconductor Electronics, RF & Computer Engineering

Cylindrical Shaft Torsional Shear Stress & Angle of Twist

$$\tau_{\max} = \frac{Tr}{J}, \qquad \theta = \frac{TL}{GJ}, \qquad J = \frac{\pi r^4}{2} \tag{Eq. D.95}$$

Charles-Augustin de Coulomb's torsion theory shows that twisting a solid circular shaft of radius r (m), length L (m), and shear modulus G (Pa) under torque T (N·m) produces a shear strain distribution that grows linearly from zero along the neutral axis to a peak outer-fiber shear stress $\tau_{\max} = T\,r\,/\,J$, where $J = \int_0^r \rho^2\,(2\pi\rho\,d\rho) = (1/2)\pi\,r^4$ (m^4) is the polar moment of inertia. Because J scales with the fourth power of radius (r^4), doubling a drive shaft's diameter reduces peak shear stress eightfold ($\propto 1/r^3$) and cuts total elastic twist angle $\vartheta = TL/(GJ)$ (rad) sixteenfold ($\propto 1/r^4$).

Computing $J = (1/2)\pi\,r^4$ first to obtain $\tau_{\max}$ (**r** [↶] [x²] [↶] [x²] [↷] [π] [×] **2** [÷] **T** [x⇄y] [÷] **r** [×]) and recovering T/J via [↶] [LASTx] for twist angle ([↶] [LASTx] [÷] **L** [×] **G** [÷]) at $T = 450$ N·m, $r = 0.025$ m, $L = 1.20$ m, $G = 79.3 \times 10^9$ Pa ($J = 6.1359 \times 10^{-7}$ m^4) yields $\tau_{\max} = 1.8335 \times 10^7$ Pa (18.33 MPa) and $\vartheta =$ **0.01110** rad (0.636°).

Cantilever & Simply Supported Beam Tip Deflection

$$\delta_{\text{cantilever}} = \frac{FL^3}{3EI}, \quad \delta_{\text{simple}} = \frac{FL^3}{48EI} \tag{Eq. D.96}$$

Integrating the Euler–Bernoulli elastic beam differential equation $EI\,d^2w/dx^2 = M(x)$ twice across a span L (m) of Young's modulus E (Pa) and cross-sectional area moment of inertia I (m^4) subjected to a concentrated point load F (N) shows that maximum transverse deflection δ (m) grows with the *cube* of span length (L^3) and inversely with flexural rigidity EI. Supporting the beam at both ends with a mid-span load reduces the peak bending moment from FL to $FL/4$ and halves the effective half-span integration length, so a simply supported beam is sixteen times stiffer (1/48 vs. 1/3) than an end-loaded cantilever of identical span.

Cubing L on the stack and scaling by $F\,/\,(3EI)$ (**L** [ENTER] **3** [y^x] **F** [×] **3** [÷] **E** [÷] **I** [÷]) for a cantilever with $F = 5{,}000$ N, $L = 2.0$ m, $E = 200 \times 10^9$ Pa, $I = 4.0 \times 10^{-6}$ m^4 yields **0.0167** m (16.67 mm).

Hooke's Law Axial Stress, Strain & Elastic Elongation

$$\sigma = \frac{F}{A}, \quad \varepsilon = \frac{\Delta L}{L_0}, \quad \Delta L = \frac{FL_0}{AE} \tag{Eq. D.97}$$

Combining Robert Hooke's 1678 linear spring law (*"ut tensio, sic vis"*) with Thomas Young's 1807 material modulus $E = \sigma\,/\,\varepsilon$ (Pa) relates engineering axial stress $\sigma = F\,/\,A$ (Pa) and dimensionless engineering strain $\varepsilon = \Delta L\,/\,L_0$ across a prismatic bar of unstrained length L_0 (m) and cross-sectional area A (m^2) under axial load F (N). Writing $\Delta L = F\,/\,(AE\,/\,L_0)$ identifies $k = AE\,/\,L_0$ (N/m) as the exact axial spring stiffness of structural tension rods, suspension cables, and preloaded bolted joints below the proportional elastic limit.

Multiplying F by L_0 and dividing sequentially by A and E (**F** [ENTER] L_0 [×] **A** [÷] **E** [÷]) for a steel tie rod ($F = 45{,}000$ N, $L_0 = 2.50$ m, $A = 3.00 \times 10^{-4}$ m^2, $E = 200 \times 10^9$ Pa) yields $\Delta L =$ **0.001875** m (1.875 mm, at axial stress $\sigma = 150.0$ MPa).

Shockley Semiconductor Diode Current-Voltage Law

$$I_D = I_S\left[\exp\left(\frac{V_D}{nV_T}\right) - 1\right] \qquad \textbf{(Eq. D.98)}$$

Derived by William Shockley in 1949 from minority-carrier injection and diffusion across a semiconductor p-n depletion region, this equation relates diode current I_D (A) to applied junction voltage V_D (V), reverse saturation current I_S (A), emission ideality factor $n \in [1, 2]$, and the thermal voltage $V_T = k_B\ T\ /\ q$ ($\approx$ 25.85 mV at T = 300 K). Under forward bias ($V_D \gg V_T$), the exponential term dominates unity and produces a slope of ln(10) $n\ V_T$ $\approx$ 59.5 mV per decade of current at room temperature (n = 1), whereas under reverse bias ($V_D < -4V_T$), current saturates at the tiny leakage floor -I_S.

Dividing V_D by n and V_T, exponentiating via [e^x], subtracting 1, and multiplying by I_S (**V_D** [ENTER] **n** [÷] **V_T** [÷] [e^x] **1** [-] **I_S** [×]) for a silicon junction (I_S = 1.0 × 10^{-12} A, V_D = 0.650 V, n = 1.0, V_T = 0.02585 V) gives **0.0832** A (83.25 mA).

Operational Amplifier Closed-Loop Gain & -3 dB Bandwidth

$$A_{v,\text{non}} = 1 + \frac{R_f}{R_g}, \quad A_{v,\text{inv}} = -\frac{R_f}{R_{\text{in}}}, \quad f_{-3\text{dB}} = \frac{\text{GBWP}}{A_{v,\text{non}}} \qquad \textbf{(Eq. D.99)}$$

By applying Harold Stephen Black's 1927 negative-feedback principle to an operational amplifier with high open-loop differential gain $A_{\text{OL}} \gg 10^5$, a resistive voltage divider of feedback resistor R_f (Ω) and ground or input resistor R_g, R_{in} (Ω) forces the inverting input to track the non-inverting input, pinning closed-loop voltage gain solely to precision passive resistor ratios. Because dominant-pole internal compensation rolls off open-loop gain at -20 dB/decade above the first pole, the product of closed-loop noise gain $A_{v,\text{non}}$ and -3 dB bandwidth $f_{\text{-3dB}}$ remains constant at the amplifier's unity-gain bandwidth product GBWP (Hz).

Computing $A_{v,\text{non}}$ = 1 + R_f / R_g first and dividing into GBWP via [x⇄y] (**R_f** [ENTER] **R_g** [÷] **1** [+] **GBWP** [x⇄y] [÷]) for R_f = 90 kΩ, R_g = 10 kΩ, GBWP = 10.0 MHz yields $A_{v,\text{non}}$ = **10.0000** (20.0 dB) and $f_{\text{-3dB}}$ = **1.0000** MHz.

Friis Free-Space RF Path Loss (FSPL)

$$\text{FSPL}_{\text{dB}} = 20\log_{10}\left(\frac{4\pi df}{c}\right) \qquad \textbf{(Eq. D.100)}$$

Derived by Harald Trap Friis at Bell Labs in 1946, the free-space path loss FSPL_{dB} quantifies the ratio of transmitted to received power between two isotropic antennas separated by unobstructed line-of-sight distance d (m) at carrier frequency f (Hz) with vacuum light speed c = 2.99792458 × 10^8 m/s ($\lambda = c/f$). Physically, $(4\pi\ d\ /\ \lambda)^2$ combines two distinct factors: spherical wavefront spreading over area $4\pi\ d^2$ and the frequency-dependent effective capture aperture $A_e = \lambda^2\ /\ (4\pi)$ of an isotropic reference dipole—explaining why doubling either link distance d or carrier frequency f adds 6.02 dB of attenuation between isotropic antennas.

Multiplying 4π d f, dividing by c, taking [↰] [LOG], and scaling by 20 (**4** [↷] [π] [×] **d** [×] **f** [×] **2.99792458** [E] **8** [÷] [↰] [LOG] **20** [×]) for d = 1,000 m (1 km) at f = 2.40 × 10^9 Hz (2.4 GHz) yields **100.0520** dB.

Amdahl's Law Multiprocessor Speedup & CMOS Dynamic Power

$$S(N) = \frac{1}{(1-p) + \frac{p}{N}}, \quad P_{\text{dyn}} = \alpha C V_{DD}^2 f \qquad \textbf{(Eq. D.101)}$$

Gene Amdahl demonstrated in 1967 that if a fraction $p \in [0, 1]$ of a program's execution time can be parallelized across N processor cores while the remaining serial fraction (1 - p) must execute sequentially, overall speedup $S(N)$ is bounded above by the serial ceiling $S(\infty)$ = 1 / (1 - p) regardless of how many cores are added. Simultaneously, because synchronous CMOS gates charge and discharge load capacitance C (F) at clock frequency f (Hz) and switching activity factor α, dynamic power $P_{\text{dyn}} = \alpha\ C\ V_{DD}^2\ f$ grows quadratically with supply voltage V_{DD} (V)—driving modern architectures to scale performance via parallel cores at reduced V_{DD} and f rather than pushing single-core clock rates.

Evaluating **p** [ENTER] **N** [÷] **1** [ENTER] **p** [-] [+] [x^{-1}] for Amdahl speedup (or $\boldsymbol{V_{DD}}$ [↶] [x^2] **α** [×] **C** [×] **f** [×] for CMOS power) at p = 0.90 (90% parallelizable) and N = 16 cores gives $S(16)$ = **6.4000** × speedup (asymptotic ceiling $S(\infty)$ = 10.0×).

D.3 Stack Optimization Idioms, Solver Templates & Boundary Guards

Fluency on the SC-32 comes from recognizing recurring mathematical forms and applying dedicated hardware keys to minimize button presses and eliminate redundant variable stores.

D.3.1 The Harmonic Mean Idiom (Parallel Operator)

The SC-32 provides a dedicated primary key [$x \parallel y$] which calculates:

$$X = \frac{X \cdot Y}{X + Y} = \frac{1}{\frac{1}{X} + \frac{1}{Y}} \qquad \textbf{(Eq. D.102)}$$

In algebraic calculators, entering a reciprocal sum requires 12 to 16 keystrokes with fractions and parentheses. On the SC-32, it takes one keystroke:

- **Parallel Resistors:** $R_{\text{eq}} = R_1 \parallel R_2 \parallel R_3 \rightarrow$ $\boldsymbol{R_1}$ [ENTER] $\boldsymbol{R_2}$ [$x \parallel y$] $\boldsymbol{R_3}$ [$x \parallel y$].
- **Series Capacitors:** $C_{\text{eq}} = C_1 \parallel C_2 \rightarrow$ $\boldsymbol{C_1}$ [ENTER] $\boldsymbol{C_2}$ [$x \parallel y$].
- **Thin Lens Optics:** Focal length $f = d_o \parallel d_i \rightarrow$ $\boldsymbol{d_o}$ [ENTER] $\boldsymbol{d_i}$ [$x \parallel y$].
- **Spring Series Compliance:** $k_{\text{eq}} = k_1 \parallel k_2 \rightarrow$ $\boldsymbol{k_1}$ [ENTER] $\boldsymbol{k_2}$ [$x \parallel y$].
- **Reduced Mass in Two-Body Problem:** $\mu = m_1 \parallel m_2 \rightarrow$ $\boldsymbol{m_1}$ [ENTER] $\boldsymbol{m_2}$ [$x \parallel y$].

D.3.2 The Difference-of-Squares Idiom

When evaluating expressions of the form $\sqrt{(c^2 - a^2)}$ (such as Pythagorean legs $b = \sqrt{(c^2 - a^2)}$ or numerical aperture $NA = \sqrt{(n_1^2 - n_2^2)}$):

- **Traditional RPN:** **c** [↶] [x^2] **a** [↶] [x^2] [-] [$\sqrt{x}$] (6 keystrokes).
- **Single-Entry Stack Factoring Idiom:** **c** [ENTER] [ENTER] **a** [-] [$x \rightleftarrows y$] [↶] [LASTx] [+] [×] [$\sqrt{x}$]. Entering multi-digit operands c and a only once and factoring $\sqrt{((c-a)(c+a))}$ via [$x \rightleftarrows y$] and [↶] [LASTx] avoids re-typing and prevents catastrophic cancellation when $c \approx a$.

D.3.3 Constant Factor Replication in Register T

When evaluating a formula with a repeated scalar factor K across multiple inputs (such as converting feet to meters with 0.3048, or applying a 1.0825 sales-tax factor):

1. Fill all four stack registers: **1.0825** ENTER ENTER ENTER (X, Y, Z, T).
2. Price 1: **10** × → 10.825; Register T drops into Z while duplicating 1.0825 in T.
3. Price 2: CLx **25** × → 27.0625; CLx disables stack lift so **25** overwrites X and T feeds 1.0825 downward on every × without STO/RCL.

D.3.4 Numerical Root-Finding and Quadrature Function Selector

The SC-32 provides root-finding via Richard Brent's hybrid algorithm (↗ SOLVE) and adaptive Romberg quadrature (↗ ∫) across equations stored in the 24-slot Equation Bank (↗ EQN, labeled **A**–**Z**):

1. **User Equations (↗ EQN):** Enter or select an RPN expression in ↗ EQN, stage initial guesses or integration limits $[a, b]$ in Registers Y and X (**a** ENTER **b**, or seed the target variable via STO), and press ↗ SOLVE or ↗ ∫ followed by the equation label (**A**–**Z**) and variable softkey (**X**, **T**, etc.).
2. **Closed-Form Polynomials (↗ POLY):** For quadratics $Ax^2 + Bx + C = 0$ (**0** D) and cubics $Ax^3 + Bx^2 + Cx + D = 0$, press ↗ POLY (↗ + XEQ), load coefficients **A**–**D**, and tap **EXEC** to compute all real and complex conjugate roots in closed form (stored in indirect registers **R34**–**R39**).
3. **Brent Solver (↗ SOLVE):** Pushes the converged root x^* onto Register X while lifting prior stack registers into Y, Z, T (STO **X** followed by XEQ audits the residual $f(x^*)$).
4. **Romberg Quadrature (↗ ∫):** With integration limits $[a, b]$ in (Y, X), returns the definite integral $\int_a^b f(x)\,dx$ in Register X and the Richardson error estimate $|\varepsilon|$ in Register Y.

D.3.5 Storing Physical Constants & Numerical Boundary Guards

Mathematical invariants (π via ↗ π and e via **1** e^x) are hardcoded in firmware. Exact 2019 SI defined constants (h = 6.62607015 × 10^{-34} J·s, c = 2.99792458 × 10^{8} m/s, e = 1.602176634 × 10^{-19} C, k_B = 1.380649 × 10^{-23} J/K, R = 8.314462618 J/(mol·K), N_A = 6.02214076 × 10^{23} mol^{-1}) and CODATA 2022 constants (G, ε_0, μ_0) are keyed via E or stored into the 26 non-volatile registers (STO **A**–**Z**) for recall (RCL **C**, RCL **H**, RCL **R**).

- **Scientific Exponent Syntax:** Use E (**1.602176634** E **19** +/− enters one atomic value in X), never × **10** y^x.
- **Angular Modes:** Surveying, navigation, and Snell's law require Degrees (↖ MODES DEG); calculus and Fourier formulas require Radians (↖ MODES RAD).
- **Singularity Guards:** Division by zero ($x/0$ or ln 0) shows **DIVIDE BY 0** (Code 01), and $\sqrt{(-x)}$ in Real mode shows **INVALID DATA** (Code 02; in ↖ CMPLX, $\sqrt{(-x)}$ returns $i\sqrt{x}$ across Y, X). Press ON or ⌫ to dismiss without losing stack state.

Appendix E: Primary Sources & Annotated Bibliography

The architecture of the StackCalc SC-32 draws on four centuries of mathematical discovery and a century of research across mathematical logic, stack-oriented computer organization, numerical error analysis, human-factors engineering, and scientific instrument design. From Jan Łukasiewicz's 1924 parenthesis-free notation in Warsaw and John Napier's 1614 logarithmic tables at Merchiston Castle to Carl Friedrich Gauss's method of Least Squares, Claude Shannon's symbolic relay algebra, and Bill Hewlett's 1971 pocket-calculator specification in Palo Alto, these primary works trace the lineage of direct numerical computation.

E.1 Primary Historical & Technical Sources

E.1.1 Foundations of Logic & Computer Architecture

Jan Łukasiewicz (1929) — *Elementy logiki matematycznej (Elements of Mathematical Logic; parenthesis-free notation devised 1924). Komisja Wydawnicza Koła Matematyczno-Fizycznego Słuchaczów Uniwersytetu Warszawskiego (Warsaw; English trans. Pergamon Press, 1963).*

> Introduces parenthesis-free prefix and postfix notation (devised in 1924 and published in his 1929 Warsaw logic treatise), demonstrating that operator position alone uniquely determines evaluation order without grouping symbols.
>
> *Cited in: Ch 02, Ch 14, Foreword*

Claude E. Shannon (1938) — *A Symbolic Analysis of Relay and Switching Circuits. Transactions of the American Institute of Electrical Engineers, Vol. 57, No. 12, pp. 713–723.*

> Proves that Boolean algebra directly models physical electromechanical relays, establishing the foundation of binary arithmetic and digital logic gates.
>
> *Cited in: Ch 10*

Arthur W. Burks, Don W. Warren & Jesse B. Wright (1954) — *An Analysis of a Logical Machine Using Parenthesis-Free Notation. Mathematical Tables and Other Aids to Computation, Vol. 8, No. 46, pp. 53–57.*

> Formalizes Polish parenthesis-free postfix notation for electronic digital computers, laying the theoretical foundation for hardware-evaluated operational stacks.
>
> *Cited in: Ch 02, Foreword*

Charles L. Hamblin (1957) — *Computer Languages. The Australian Journal of Science, Vol. 20, No. 5, pp. 135–139.*

Pioneers the push-down operational stack in computer hardware (alongside his June 1957 WRE Salisbury conference paper "An Addressless Coding Scheme Based on Mathematical Notation" and 1962 The Computer Journal paper "Translation to and from Polish Notation"), demonstrating that a zero-address stack architecture eliminates memory traffic and register allocation overhead.

Cited in: Ch 02, Ch 14, Foreword

Edsger W. Dijkstra (1961) — *Making a Translator for ALGOL 60. Mathematisch Centrum, Amsterdam, Netherlands.*

Details the Shunting-Yard algorithm, proving the hidden buffering debt and parenthesis parsing overhead imposed by infix notation.

Cited in: Ch 02

Donald E. Knuth (1997) — *The Art of Computer Programming, Volume 1: Fundamental Algorithms. Addison-Wesley, Reading, MA.*

The monumental reference on sequential stack allocation, LIFO queue dynamics, combinatorics, and exact algorithm analysis.

Cited in: Ch 02, Ch 10

E.1.2 Corvallis Silicon & Handheld Architecture

David S. Cochran (1972) — *Algorithms and Accuracy in the HP-35. Hewlett-Packard Journal, Vol. 23, No. 10, pp. 10–11.*

Documents the 4-register XYZT stack, serial bit-slice silicon architecture, CORDIC transcendental algorithms, and T-register constant replication.

Cited in: Ch 01, Ch 02, Foreword

Thomas M. Whitney, France Rodé & Chung C. Tung (1972) — *The 'Powerful Pocketful': An Electronic Calculator Challenges the Slide Rule. Hewlett-Packard Journal, Vol. 23, No. 10, pp. 2–9.*

Documents William R. Hewlett's shirt-pocket challenge to HP's Palo Alto engineering team and the architectural design of the HP-35 as an electronic slide rule.

Cited in: Ch 01, Ch 02, Ch 14, Foreword

Richard J. Nelson (1974) — *PPC Journal (formerly 65 Notes). Personal Programmable Calculator Club, Santa Ana, CA.*

Pioneering user-group forensic engineering publications demanding absolute register transparency and synthetic programming capabilities.

Cited in: Ch 01

Bernard E. Musch, John J. Wong & David R. Conklin (1980) — *Powerful Personal Calculator System Sets New Standards. Hewlett-Packard Journal, Vol. 31, No. 3, pp. 3–12.*

Introduces the HP-41C alphanumeric handheld computer system with continuous CMOS memory, alphanumeric display segments, user key reassignment, and expandable peripheral architectures.

Cited in: Ch 01

Eric A. Evett, Paul J. McClellan & Joseph P. Tanzini (1983) — *Scientific Pocket Calculator Extends Range of Built-In Functions. Hewlett-Packard Journal, Vol. 34, No. 5, pp. 25–35.*

Establishes the split-stack complex number convention, matrix solver algorithms, root finding, numerical integration, and continuous memory architecture of the HP-15C Voyager.

Cited in: Ch 01

Kathy B. Hamrick (1996) — *The History of the Hand-Held Electronic Calculator. The American Mathematical Monthly, Vol. 103, No. 8, pp. 633–639.*

Traces the historical evolution of Polish notation from abstract Warsaw mathematical logic to Corvallis handheld scientific computing.

Cited in: Ch 01

E.1.3 Numerical Stability & Scientific Algorithms

Gerolamo Cardano (1545) — *Artis Magnae, Sive de Regulis Algebraicis (The Great Art, or The Rules of Algebra). Johannes Petreius, Nuremberg.*

Publishes the depressed cubic reduction and radical root formulas discovered by Scipione del Ferro and Niccolò Tartaglia, along with the first appearance of casus irreducibilis complex intermediates.

Cited in: Ch 07

John Napier (1614) — *Mirifici Logarithmorum Canonis Descriptio. Andrew Hart, Edinburgh.*

The invention of logarithms, transforming laborious multi-digit multiplication into addition and doubling the working lifespan of astronomers.

Cited in: Ch 03

René Descartes (1637) — *La Géométrie (Appendix to Discours de la méthode). Jan Maire, Leiden.*

Unifies Euclidean geometry and symbolic algebra via coordinate axes, laying the foundation for analytic geometry and function plotting.

Cited in: Ch 05, Ch 12

Galileo Galilei (1638) — *Discorsi e Dimostrazioni Matematiche, intorno à due nuoue scienze (Dialogues Concerning Two New Sciences). Louis Elsevier, Leiden.*

Establishes the geometric Square-Cube Law governing surface-area-to-volume scaling in structural mechanics and biological thermoregulation, as well as the isochronous law of the pendulum.

Cited in: Ch 04, Ch 06

Carl Friedrich Gauss (1809) — *Theoria Motus Corporum Coelestium in Sectionibus Conicis Solem Ambientium. Perthes and Besser, Hamburg.*

Introduces the Method of Least Squares and the Gaussian Normal error distribution, recovering the lost celestial orbit of Ceres.

Cited in: Ch 09

Charles Proteus Steinmetz (1893) — *Complex Quantities and Their Use in Electrical Engineering. Proceedings of the International Electrical Congress, Chicago, August 21–25, 1893, pp. 33–74, New York.*

Landmark paper introducing complex phasor representation (Z = R + jX) to alternating-current analysis, reducing differential equations to complex arithmetic.

Cited in: Ch 05

William Sealy Gosset ("Student") (1908) — *The Probable Error of a Mean. Biometrika, Vol. 6, No. 1, pp. 1–25.*

Introduces Student's t-distribution for small-sample statistical inference, accounting for the sampling variance of Bessel-corrected standard deviation s.

Cited in: Ch 09

Lewis Fry Richardson (1911) — *The Approximate Arithmetical Solution by Finite Differences of Physical Problems Involving Differential Equations. Philosophical Transactions of the Royal Society of London, Series A, Vol. 210, pp. 307–357.*

Formulates Richardson extrapolation (deferred approach to the limit), canceling lowest-order error terms in numerical approximations.

Cited in: Ch 08

Cyril F. Colebrook (1939) — *Turbulent Flow in Pipes, with Particular Reference to the Transition Region between the Smooth and Rough Pipe Laws. Journal of the Institution of Civil Engineers, Vol. 11, No. 4, pp. 133–156.*

Establishes the implicit non-linear Colebrook-White Darcy friction factor equation for turbulent pipe hydraulics, requiring numerical root finding.

Cited in: Ch 07

Waloddi Weibull (1951) — *A Statistical Distribution Function of Wide Applicability. Journal of Applied Mechanics, Vol. 18, No. 3, pp. 293–297.*

Establishes the Weibull survival and reliability distribution unifying infant-mortality, random-shock, and fatigue wear-out failure regimes.

Cited in: Ch 09

Werner Romberg (1955) — *Vereinfachte numerische Integration. Det Kongelige Norske Videnskabers Selskab Forhandlinger, Vol. 28, No. 7, pp. 30–36.*

Applies triangular Richardson extrapolation to composite trapezoidal approximations, canceling successive Euler-Maclaurin error terms for rapid convergence.

Cited in: Ch 08, Ch 12

Richard Bellman (1957) — *Dynamic Programming. Princeton University Press, Princeton, NJ.*

Introduces the principle of optimality and dynamic programming, structuring complex multi-stage computational problems.

Cited in: Foundational reference & curriculum background

Gordon E. Moore (1965) — *Cramming More Components onto Integrated Circuits. Electronics, Vol. 38, No. 8, pp. 114–117.*

Establishes the semi-logarithmic exponential scaling trajectory of integrated circuit transistor density (Moore's Law).

Cited in: Ch 09

Richard P. Brent (1973) — *Algorithms for Minimization without Derivatives. Prentice-Hall, Englewood Cliffs, NJ.*

Synthesizes bisection, secant, and inverse quadratic interpolation into an unconditionally stable numerical root finder converging to 10^{-12} precision.

Cited in: Ch 07, Ch 11, Ch 12

William M. Kahan (1979) — *Personal Calculator Has Key to Solve Any Equation f(x) = 0. Hewlett-Packard Journal, Vol. 30, No. 12, pp. 20–26.*

Establishes the non-destructive base-preserving percentage invariant and foundational floating-point error bounds for handheld solvers.

Cited in: Ch 07, Ch 08, Ch 11, Ch 12, Foreword

William J. Cody & William Waite (1980) — *Software Manual for the Elementary Functions. Prentice-Hall, Englewood Cliffs, NJ.*

The definitive treatise on extended-precision range reduction for transcendentals, preserving 53 bits of IEEE-754 mantissa across large arguments.

Cited in: Ch 03

Gene H. Golub & Charles F. Van Loan (1996) — *Matrix Computations. Johns Hopkins University Press, Baltimore, MD.*

The standard treatise on matrix decomposition, singular value decomposition, least-squares solvers, and floating-point stability.

Cited in: Ch 10

E.1.4 Breakout Mathematics & Scientific Nonfiction

Wolfgang Sartorius von Waltershausen (1856) — *Gauss zum Gedächtniss. S. Hirzel, pp. 12–13, Leipzig.*

> Primary biographical source recording young Carl Friedrich Gauss's arithmetic series solution in J.G. Büttner's Brunswick classroom and his Low German exclamation 'Ligget se!' ('There it lies!').
>
> *Cited in: Ch 14*

Martin Gardner (1959) — *The Scientific American Book of Mathematical Puzzles and Diversions. Simon & Schuster, New York.*

> Pioneered popular mathematical recreations, flexagons, hexaflexagons, and spatial puzzles that celebrate mathematical play.
>
> *Cited in: Foundational reference & curriculum background*

Simon Singh (1997) — *Fermat's Enigma: The Epic Quest to Solve the World's Greatest Mathematical Problem. Walker & Company, New York.*

> Captures the three-century human quest to prove Fermat's Last Theorem, conveying the passion and stakes of mathematical discovery.
>
> *Cited in: Foundational reference & curriculum background*

Georges Ifrah (2000) — *The Universal History of Numbers: From Prehistory to the Invention of the Computer. John Wiley & Sons, New York.*

> Comprehensive historical account of tally stones, Mesopotamian sexagesimal accounting, Egyptian hieroglyphic numerals, and the independent invention of zero and positional place value in India and Mesoamerica.
>
> *Cited in: Foundational reference & curriculum background*

Cleomedes (trans. Alan C. Bowen & Robert B. Todd) (2004) — *Cleomedes' Lectures on Astronomy: A Translation of The Heavens. University of California Press, Berkeley, CA.*

> Preserves the primary classical account (Book I, Chapter 7) of Eratosthenes of Cyrene's geometric measurement of the Earth's circumference using solstice shadows at Syene and Alexandria.
>
> *Cited in: Ch 04*

Steven Strogatz (2012) — *The Joy of x: A Guided Tour of Math, from One to Infinity. Eamon Dolan / Houghton Mifflin Harcourt, Boston.*

> Opens every mathematical section with everyday curiosity and intuitive aha-moments, showing how mathematical beauty connects to human experience.
>
> *Cited in: Ch 02, Foreword*

Jordan Ellenberg (2014) — *How Not to Be Wrong: The Power of Mathematical Thinking. Penguin Press, New York.*

Exposes common cognitive biases, non-linear scaling traps, and statistical survivorship fallacies with conversational wit.

Cited in: Ch 02, Foreword

Randall Munroe (2014) — *What If?: Serious Scientific Answers to Absurd Hypothetical Questions. Houghton Mifflin Harcourt, Boston.*

Applies order-of-magnitude dimensional analysis, physics Fermi approximations, and creative mathematical modeling to wild scenarios.

Cited in: Ch 14

Matt Parker (2019) — *Humble Pi: A Comedy of Maths Errors. Riverhead Books, New York.*

Explores famous engineering catastrophes and numerical bugs caused by round-off errors, integer overflows, and unit conversion failures.

Cited in: Ch 10

Steven Strogatz (2019) — *Infinite Powers: How Calculus Reveals the Secrets of the Universe. Eamon Dolan / Houghton Mifflin Harcourt, Boston.*

Illuminates numerical integration, derivatives, and continuous change from Archimedes to modern spacecraft trajectory navigation.

Cited in: Foreword

E.1.5 Cognitive Science, Problem Solving & Mathematical Intuition

George Pólya (1945) — *How to Solve It: A New Aspect of Mathematical Method. Princeton University Press, Princeton, NJ.*

Defines the heuristic problem-solving framework (Understand, Plan, Carry Out, Look Back), aligned with step-by-step RPN stack inspection.

Cited in: Ch 14

George A. Miller (1956) — *The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information. Psychological Review, Vol. 63, No. 2, pp. 81–97.*

Identifies the 7 ± 2 item capacity limit of human short-term working memory, validating the 4-register stack as the optimal cognitive span.

Cited in: Ch 01, Ch 02, Ch 14, Foreword

John Sweller (1988) — *Cognitive Load During Problem Solving: Effects on Learning. Cognitive Science, Vol. 12, No. 2, pp. 257–285.*

Establishes Cognitive Load Theory, demonstrating that parentheses and operator precedence buffering in algebraic calculators impose extraneous cognitive load.

Cited in: Ch 01, Ch 02, Foreword

E.1.6 Standards & Hardware Handbooks

Hewlett-Packard Company (1982) — *HP-15C Advanced Functions Handbook. Hewlett-Packard Corvallis Division, Corvallis, OR.*

> The pinnacle reference manual for handheld scientific computation, detailing complex matrix inversion, numerical integration, and root finding.
>
> *Cited in: App A, Ch 01, Ch 02, Ch 05, Ch 07, Ch 09, Ch 11, Ch 13, Foreword*

Hewlett-Packard Company (1991) — *HP 32S II RPN Scientific Calculator Owner's Manual. Hewlett-Packard Corvallis Division, Corvallis, OR.*

> The classic owner's manual for collegiate STEM practitioners, integrating two-variable statistical regression, algebraic equation solving, and RPN.
>
> *Cited in: App A, Ch 01, Ch 07, Ch 09, Ch 12, Foreword*

Frank P. Incropera, David P. DeWitt, Theodore L. Bergman & Adrienne S. Lavine (2007) — *Fundamentals of Heat and Mass Transfer. John Wiley & Sons, Hoboken, NJ.*

> Standard engineering reference on Fourier thermal conduction, Newton's law of convective cooling, extended fin surface area efficiency, and electronics thermal management.
>
> *Cited in: Ch 06, Ch 13*

Paul Horowitz & Winfield Hill (2015) — *The Art of Electronics. Cambridge University Press, Cambridge, UK.*

> The definitive hardware design treatise, establishing the master-practitioner voice, rule-of-thumb heuristics, and parallel resistance shortcuts.
>
> *Cited in: Ch 05, Ch 08, Ch 13, Foreword*

IEEE Computer Society (2019) — *IEEE Standard for Floating-Point Arithmetic. Institute of Electrical and Electronics Engineers, Piscataway, NJ.*

> The international standard governing 64-bit double-precision floating-point arithmetic, subnormal representation, and arithmetic exceptions.
>
> *Cited in: Ch 10*

E.2 Thematic Reading Pathways

E.2.1 Foundations of Problem-Solving, Calculus & Physical Intuition

- **Problem Formulation, Calculus & Estimation:** George Pólya's *How to Solve It* (1945) formalizes heuristic reasoning—isolating unknowns, working backward, and testing limiting cases before computing—while Sanjoy Mahajan's *Street-Fighting Mathematics* (2010) extends Enrico Fermi's order-of-magnitude estimation and dimensional analysis. Steven Strogatz's *The Joy of x* (2012) and *Infinite Powers* (2019) trace the conceptual arc from elementary arithmetic to the infinitesimal calculus of Archimedes, Newton, and Leibniz.

- **Physical Scaling, Thermodynamics & Complex Rotation:** Richard Feynman's *Lectures on Physics* (Volume I, Chapter 22: *"Algebra"*) progresses in ten pages from integer counting to Euler's formula, $e^{i\vartheta} = \cos\vartheta + i\sin\vartheta$, and shows why multiplication by $\sqrt{(-1)}$ performs a 90° rotation in the plane. Galileo Galilei's *Two New Sciences* (1638), Sadi Carnot's *Reflections on the Motive Power of Fire* (1824), and Wallace Sabine's *Collected Papers on Acoustics* (1922) establish dimensional scaling laws, thermodynamic efficiency bounds, and architectural reverberation physics.

- **Transcendentals, Imaginary Units & Small Samples:** Eli Maor's *e: The Story of a Number* (1994) and Paul Nahin's *An Imaginary Tale* (1998) chronicle the analytic emergence of natural exponentials and complex numbers, while David Salsburg's *The Lady Tasting Tea* (2001) and Francis Anscombe's *"Graphs in Statistical Analysis"* (1973) illuminate the statistical methods of Carl Friedrich Gauss, Francis Galton, and William Sealy Gosset.

E.2.2 History of Science, Navigation & Information Theory

- **Logarithms, Shadows & Navigation:** John Napier's *Mirifici Logarithmorum Canonis Descriptio* (1614, translated into English by Edward Wright) and Dava Sobel's *Longitude* (1995) show how logarithmic tables, spherical trigonometry, and marine chronometers reduced oceanic navigation to arithmetic addition and subtraction.

- **Symbolic Logic & Information Theory:** Claude Shannon's 1938 master's thesis (*"A Symbolic Analysis of Relay and Switching Circuits"*) and his 1948 monograph (*"A Mathematical Theory of Communication"*) establish the direct correspondence between Boolean switching operators (**`AND`**, **`OR`**, **`XOR`**) and logarithmic information entropy ($\log_2 p$).

E.2.3 Computer Architecture, Numerical Analysis & Floating-Point Silicon

- **Postfix Stacks & Pocket Computation:** Jan Łukasiewicz's *Elementy logiki matematycznej* (1929), Charles Hamblin's 1957 and 1962 papers on reverse Polish hardware stacks, Edsger Dijkstra's 1961 shunting-yard algorithm, and the June 1972 *Hewlett-Packard Journal* articles by Thomas M. Whitney, France Rodé, Chung C. Tung, and David S. Cochran on the HP-35 four-register stack and CORDIC algorithms form the foundation of stack-machine evaluation, analyzed systematically in Donald Knuth's *The Art of Computer Programming* (Volumes 1 and 2).

- **Floating-Point Arithmetic & Numerical Analysis:** William Kahan's monographs on IEEE-754 floating-point design and principal branch cuts (including *"Mathematics Written in Sand"*) explain why guard digits and compensated summation prevent catastrophic cancellation, while Richard Brent's *Algorithms for Minimization Without Derivatives* (1973), Werner Romberg's 1955 quadrature paper, and William Press et al.'s *Numerical Recipes* document the root-finding and integration algorithms implemented in the SC-32 firmware.

Interactive web simulators, firmware sources, and open-source hardware schematics are available at **stackcalc.io**.

Alphabetical Subject Index

3

36-Bit Word Architecture (IBM 7090 & DEC PDP-10) 174, 175, 176

4

4-Level XYZT Operational Stack 3, 5, 14, 35, 36

43-Key Tactile Matrix Keypad 245, 264, 267

A

Abel, Niels Henrik & Évariste Galois (Quintic Insolvability) 126, 127, 128, 133, 196

AC Phasors, Impedance & Charles Proteus Steinmetz 12, 14, 19, 29, 36

Adaptive Romberg Quadrature ([∫]) & Werner Romberg 3, 14, 18, 69, 142

Al-Khwarizmi & Completing the Square 122, 123

Amortization Schedules & Mortgage Crossover 190, 192, 194, 196

Anscombe's Quartet & Visual Data Inspection 164, 208

Archimedes (Sphere-Cylinder Ratio, Parabola & Buoyancy) 15, 34, 35, 42, 74

Arrhenius Chemical Kinetics & Activation Energy 53, 173, 222, 304

B

Bessel's Correction (Sample Variance Degrees of Freedom n - 1) 51, 128, 129, 153, 157, 158

Binary ([BIN]), Octal ([OCT]), Decimal ([DEC]) & Hexadecimal ([HEX]) 2, 17, 18

Bitwise Logic ([AND], [OR], [XOR], [NOT]) & Subnet Masks 14, 174, 175

Brent's Hybrid Root-Finding Algorithm ([SOLVE]) 3, 122, 128, 129

Bresenham's Line Algorithm & Affine Pixel Mapping 38

C

Carbon-14 Radiometric Dating & Willard Libby 52, 53, 162

Cardano, Gerolamo & Tartaglia (1535 Milan Cubic Contest) 3, 14, 18, 19, 22, 83

Carnot Heat-Engine Efficiency & Counter-Flow Heat Exchangers 221

Catastrophic Cancellation & Numerical Stability (William Kahan) 3, 14, 54, 88, 94

Catenary Curve (cosh x), Hanging Chains & Gateway Arch 74

Cavalieri's Principle & 3D Solid Mensuration 103, 106, 107

Center of Mass, Moment of Inertia & Parallel-Axis Theorem 108, 110, 111

Central Difference Derivative & Taylor Error Cancellation 105, 142, 143

Claude Shannon & Boolean Switching Circuits (1937 Thesis) 147, 174

Complex Numbers ([CMPLX]) & 90-Degree Rotation (i = √-1) 3, 83, 84

Composite Cross-Sections, Parallel-Axis Theorem & Singularity Functions . . 72, 120, 132, 139, 150
Compound Microscopes & Thin-Lens Optics 7, 205, 211, 215, 217
Continued Fractions (Golden Ratio φ & Pi 355/113) 33, 34

D

De Moivre's Theorem & Roots of Unity 55, 85, 87
Descartes, René & the Cartesian Coordinate Plane 72, 83, 84, 106, 120
Difference of Two Squares & Mental Multiplication 104, 191, 232, 233
Dijkstra's Shunting-Yard Algorithm & Charles Hamblin 2, 3, 30

E

Eratosthenes of Cyrene & Earth's Circumference 65, 82
Error Catalog & Mathematical Singularities 55, 93, 177, 258
Euler's Column Buckling Law ($Pcr = \pi^2EI / L^2$) 220, 221, 226, 282
Euler's Identity ($e^{i\pi} + 1 = 0$) & Complex Exponential 83, 86, 96
Euler's Natural Base e ([e^x], [LN]) & Continuous Growth 20, 47, 52, 53, 88

F

Factorials ([x!]), Combinatorics & Lanczos Gamma Approximation 52, 54, 55
Fermi Estimation & Enrico Fermi's Trinity Paper-Drop Test 230
Fibonacci (Leonardo of Pisa) & Present Value in Liber Abaci 34, 36, 42, 190

G

Galileo Galilei & the Square-Cube Scaling Law 17, 68, 74, 102
Galton, Sir Francis (Quincunx Bean Machine & Regression to the Mean) 157, 158, 160
Gauss, Carl Friedrich (1801 Ceres Orbit, Least Squares & Bell Curve) 14, 87, 93, 96, 101, 125
Great-Circle Navigation, Surveying Traverses & Crosswind Vectors 71, 72, 73

H

Heron's Triangle Formula & Ramanujan's Ellipse Approximation 34, 103, 104, 105
Hewlett, Bill, Dave Packard & the 1972 HP-35 Pocket Challenge 3, 9
Horner's Nested Polynomial Scheme & Single-Entry LASTX Spine 45, 127
Hyperbolic Functions ([HYP], sinh, cosh, tanh) & Ocean Waves 54, 65, 74

I

Inside-Out Thinking vs. Outside-In Paper Notation 1, 7, 8, 30

K

Kepler's Orbital Laws & Transcendental Anomaly Equation 46, 49, 66, 130, 224
Keycap [1/x] (Reciprocal) 7, 11, 14, 19, 24, 42

Keycap [∫] (Adaptive Romberg Numerical Quadrature) 11, 13, 15, 69, 148
Keycap [CMPLX] (Complex Prefix) 11, 14, 84
Keycap [ENTER] (Stack Lift & Duplication) 5, 6, 7, 8, 11
Keycap [E] (Scientific Exponent Entry) 11, 14, 17, 19, 24
Keycap [HYP] (Hyperbolic Prefix) 75, 76, 79, 143, 154
Keycap [LASTX] (Argument Recovery & Horner Verification) 11, 15, 26, 36, 37
Keycap [LN] & [e^x] (Natural Log & Exponential) 8, 11, 14, 20, 52
Keycap [LOG] & [10^x] (Common Base-10 Log & Antilog) 11, 48, 52
Keycap [m,b], [$\bar{x}$] & [sx] (Linear Regression & Sample Statistics) 112, 113, 120, 158, 159
Keycap [POLY] (Polynomial Root Solver) 3, 11, 14, 125, 126
Keycap [R↓] (Roll Stack Down) 11, 36, 75, 111
Keycap [SIN], [COS], [TAN] & Inverse Trig 11, 41, 68, 69
Keycap [SOLVE] (Brent Numerical Root Solver) 11, 13, 15, 129, 132
Keycap [STO] & [RCL] (Memory Storage & Recall) 11, 13, 14, 37, 84
Keycap [x↔y] (Register X-Y Swap) 8, 11, 26, 36, 41, 43
Keycap [x‖y] (Universal Harmonic / Parallel Operator) 3, 7, 11, 15, 18
Keycap [y^x] & [$\sqrt[x]{y}$] (Powers & Roots) 11, 14, 19, 24, 49
Keycap [÷R] (Euclidean Integer Quotient & Remainder) 3, 11, 13, 14, 26, 179
Keycap [Σ+] (Statistical & Geometric Accumulator) 14, 72, 76, 80, 103, 110
Keycap [→ϑ,r] & [→y,x] (Polar-Rectangular Vector Conversion) 11, 67, 71, 72
Kleiber's 3/4-Power Metabolic Scaling Law 49, 50, 52, 102

L

Leibniz, Gottfried Wilhelm (Binary Arithmetic & Calculus) 74, 106, 142, 144, 174
Linear Regression, Karl Pearson's Correlation (r) & Linearization 160, 161, 162, 163
Loudspeaker Butterworth Crossovers & Standing-Wave Ratio (VSWR) 19, 92
Lukasiewicz, Jan & 1924 Warsaw Parenthesis-Free Logic 1, 2, 3

M

Miller, George A. & Working-Memory Limits (1956 Paper) 2, 10, 11, 30

N

Napier, John & Henry Briggs (1614 Invention of Logarithms) 9, 20, 46
Net Present Value (NPV), Internal Rate of Return (IRR) & Bonds 195, 196
Newton's Law of Cooling & Thermal Relaxation 52, 53, 103, 241, 286
Normal Bell Curve, Central Limit Theorem & Cumulative Φ(z) 157, 158, 170

P

Percentages ([%], [%CHG]), Multiplicative Asymmetry & Rule of 72 . . . . . 61, 190, 191, 192

Q

Quartz Crystal Oscillator (32,768 Hz = 2^{15} Hz) & Reflective LCD 3, 9, 14, 15, 16, 17

R

Relativistic Time Dilation (Cosmic-Ray Muons & Lorentz Factor γ) 49, 50, 163, 294

Reynolds Number, Pipe Turbulence & Venturi Pressure Drop 130, 150, 219

Richardson Extrapolation (4/3 - 1/3 Error Cancellation) 143, 146, 156, 316

Root-Sum-Square (RSS) Mechanical Tolerance Stacking 218, 226, 231

S

Sabine's Reverberation Law (Boston Symphony Hall) & Acoustics 48, 223

Stack Lift State Machine & Register T Replication 5, 6, 7, 23, 31, 32

Student's t-Distribution & William Sealy Gosset (Guinness Brewery) 163, 170

T

Time Value of Money ([TVM]: [N], [I%], [PV], [PMT], [FV]) 14, 190, 192, 193, 194

Tsiolkovsky Rocket Equation, Saturn V & Clarke Geostationary Orbit 142, 223, 224

Two's Complement Negative Numbers & Odometer Arithmetic 174, 177, 178

U

Unit Circle Volvelle, Astrolabes & Law of Cosines 35, 65, 66, 68

V

Van der Waals Real Gas Cubic & 3D Principal Stress Eigenvalues 50, 135, 138, 285

Viète's Nested Square-Root Cascade 1, 8, 33, 35

W

Weber-Fechner Law (Decibels, Richter Scale, Stellar Magnitudes & pH) 46, 48, 223, 290

Weibull Reliability Distribution & Poisson Rare Events 51, 163, 164